Birational Geometry Seminar

2026-10-10

Contents

1. Ample and Nef Line Bundle **6**

1.1. Cartier divisors, Line Bundles and Linear Series 7

1.1A. Cartier divisors and Line Bundles 7

1.1B. Linear Series 12

1.1C. Intersection Numbers via Cohomological Approach 15

1.2. The Classical Theory 26

1.2A. Cohomological Properties 27

1.2B. Numerical Properties 36

1.3. $\mathbb{Q}$-Cartier divisors and $\mathbb{R}$-Cartier divisors 44

1.3A. Definition of $\mathbb{Q}$-Cartier divisors. 44

1.3B. $\mathbb{R}$-Cartier divisors and Their ampleness 48

1.4. Nef line bundle and Cartier divisors 51

1.4A. Definitions and Formal Properties 51

1.4B. Kleiman's Theorem 53

1.4C. Cones 62

1.4D. Applications of Cones on Examples 69

1.4E. Fujita's Vanishing Theorem 78

1.5. Ampleness in Relative Version 84

1.5A. Relative Ampleness 84

1.5B. Intersections in Relative Version 87

1.5C. Nefness in Relative Version 91

1.5D. Kleiman's Criterion in Relative Version 92

2. Rational Curves & Bend and Break **97**

2.1. Parametrizing Morphisms 98

2.1A. Rational Curves on Exceptional locus 98

2.1B. Parametrizing Rational Curves 109

2.1C. Parametrizing Morphisms . . . 111
2.1D. Morphisms With Fixed Points . . . 121
2.1E. Morphism From a Curve . . . 122
2.2. Bend-and-Break Lemmas . . . 124
2.3. Application: Finding Rational Curves . . . 132
2.3A. Rational Curves On Fano Varieties . . . 132
2.3B. Finding Rational Curves when K_X is not Nef . . . 142
2.4. Mori's Cone Theorem . . . 146
2.4A. Cone Theorem on the Nonsingular Case . . . 146
2.4B. Application: Minimal Models of Surfaces . . . 152

3. Linear Series **158**
3.1. Asymptotic Theory . . . 159
3.1A. Basic Definitions . . . 159
3.1B. Semiample Line Bundles . . . 169
3.1C. Iitaka Fibration . . . 175
3.2. Big Line Bundles and Cartier divisors . . . 190
3.2A. Basic Properties of Big Cartier divisors . . . 190
3.2B. Pseudo-effective and Big Cones . . . 198
3.2C. Volume of a Big Cartier divisor . . . 200
3.3. Bigness in Relative Version . . . 213
3.4. Examples . . . 220
3.4A. Zariski's Construction . . . 220
3.4B. Cutkosky's Construction . . . 220
3.4C. Okounkov's Construction . . . 220

4. Vanishing Theorems **221**
4.1. Preliminaries . . . 222
4.1A. Simple Normal Crossings and Resolution of Singularities . . . 222
4.1B. Covering Trick . . . 228
4.2. Kodaira and Nakano Vanishing Theorems . . . 241
4.2A. Kodaira Vanishing Theorem . . . 242
4.2B. Differential Forms with Logarithmic Poles . . . 243
4.2C. Nakano Vanishing Theorem . . . 246
4.3. Vanishing for Big and Nef Line Bundles . . . 248
4.3A. Statement and Proof of the Theorem . . . 248
4.3B. Some Applications . . . 253

4.4. Kawamata-Viehweg vanishing theorem 254
4.4A. Statement and Proof of the Theorem 255
4.4B. Generalizations of the Vanishings in Relative Version 257

5. Introduction to the Minimal Model Program **269**
5.1. Introduction to the Mori's program 270
5.2. Singularities in the Minimal Model Program 286
5.2A. Definition of Discrepancies 287
5.2B. Computation of Discrepancies 291
5.2C. Singularities for the Pair 309

6. Cone Theorems **336**
6.1. Introduction to the Proof of the Cone Theorem 337
6.1A. Basepoint-free Theorem 341
6.1B. The Cone Theorem 347
6.1C. The Rationality Theorem 373
6.1D. The Non-vanishing Theorem 386
6.2. Relative versions 392
6.2A. The Relative Non-vanishing Theorem 392
6.2B. The Relative Basepoint-free & Effective Basepoint-free Theorem 400
6.2C. The Relative Rationality Theorem 417
6.2D. The Relative Cone Theorem 425
6.3. Running the MMP 428

A. Relatively Linearly Equivalent **429**

B. Local Properties of the Algebraic Scheme **431**
B.1. Bertini's Theorem 431
B.2. Differential Forms in Algebraic Geometry 433
B.3. Implicit Function Theorem in Algebraic Geometry 446
B.3A. Proof of the Main Theorem 458
B.4. Blowing-up Property and Infinitesimal Lifting 459

C. Chevalley's Nullstellensatz **467**

D. Spectral Sequence towards to Algebraic Geometry **471**
D.1. Bicomplexes 471
D.2. Filtrations and Exact Couples 476

D.3. Convergence 483
D.4. Cartan-Eilenberg Resolution 487
D.5. Cohomology of The Total Complex 495
D.6. Grothendieck Spectral Sequence 502

Bibliography **511**

Chapter 1

Ample and Nef Line Bundle

In this chapter all schemes will be considered over $\mathbb{C}$, but if we don't mention the characteristic, things can go through arbitrary field.

1.1 Cartier divisors, Line Bundles and Linear Series

1.1A Cartier divisors and Line Bundles

> Cartier divisors are important for studying maps from a given variety to a projective space.
>
> *Robin Hartshorne*

Definition 1.1.1. (Cartier divisor) A *Cartier divisor* on X is identified as a global section of the quotient sheaf $\mathscr{K}_X^*/\mathscr{O}_X^*$, expressed as:

$$\mathrm{Div}(X) := \Gamma(X, \mathscr{K}_X^*/\mathscr{O}_X^*)$$

Concretely, $D \in \mathrm{Div}(X)$ is represented by the collection $\{(U_i, f_i)\}$, where X is covered by U_i, and they are compatible on $U_i \cap U_j$ such that:

$$f_i = g_{ij} f_j \text{ for some } g_{ij} \in \Gamma(U_{ij}, \mathscr{O}_X^*)$$

The function f_i is called *local equation* for D at any point $x \in U_i$. Now for D and D' two such Cartier divisors, they are the same if they have a common refinement V_k such that

$$f_k = h_k f'_k \text{ on } V_k, \text{ for some } h_k \in \Gamma(V_k, \mathscr{O}_X^*)$$

Although $Div(X)$ operates as a multiplicative group, it is expressed using additive notation:

$$D = \big\{(U_i, f_i)\big\},\ D' = \big\{(U_i, f'_i)\big\},\ D + D' = \big\{(U_i, f_i f'_i)\big\}$$

where U_i represents the chosen good refinement. *Support* of $D = \{(U_i, f_i)\}$ is the set of points $x \in X$ such that f_i at x is **not** unit in $\mathscr{O}_{X,x}$. D is *effective* if $f_i \in \Gamma(U_i, \mathscr{O}_X)$, i.e. regular on U_i (has no poles), written by $D \geq 0$, and $D \geq D'$ if $D - D' \geq 0$.

Remark 1.1.2. For non-reduced, general schemes, using the "sheaf of total rings":

$$\Gamma(U, \mathscr{K}_X) = \Gamma(U, \mathscr{O}_X)_{tot} = A_{tot}$$

$$\mathscr{K}_{X,x} = (\mathscr{O}_{X,x})_{tot}$$

Definition 1.1.3. (Cycles) X is a variety (or scheme) pure of dimension n.

A *k-cycle* on X is a $\mathbb{Z}$-linear combination of irreducible subvarieties of codimension $n-k$.

A *Weil divisor* on X is $(n-1)$-cycle, written by $\mathrm{WDiv}(X) = Z_{n-1}(X)$.

Remark 1.1.4. Cycle map:

$$\begin{aligned} \mathrm{Div}(X) &\longrightarrow \mathrm{WDiv}(X) \\ D &\longmapsto [D] = \sum \mathrm{ord}_V(D) \cdot [V] \end{aligned}$$

where the sum is formal sum and $\mathrm{ord}_V(D) = v_V(f_i)$ for $V \cap U_i \neq \emptyset$ is independent of i since compatible on intersection. And if X is **normal**, then this map is injective; if **non-singular** we get an isomorphism (or just need locally factorial).

Now if it's not emphasized on purpose, "Cartier divisor" will always means the Cartier divisor.

Definition 1.1.5.

$$\begin{aligned} &\textit{Principal Cartier divisor}: \ \mathrm{div}(f) = D \in \mathrm{Div}(X) \ , \ f \in \Gamma(X, \mathscr{M}_X^*) \\ &\textit{Linearly equivalent}: \ D_1 \sim_{\mathrm{lin}} D_2 \text{ if } D_1 - D_2 = \mathrm{div}(f) \ , \ f \in \Gamma(X, \mathscr{M}_X^*) \end{aligned}$$

Remark 1.1.6. (Pullback of Cartier divisors) Let D be a Cartier divisor on X. Given $f : Y \to X$, one would like to define a Cartier divisor f^*D on Y by pulling back the generators for D, i.e.

$$D = \{(U_i, f_i)\} \overset{f^*}{\longmapsto} f^*D = \{(f^{-1}U_i, f_i \circ f)\}$$

we need the condition below to ensure pullback is meaningful:

> Let $V \subseteq Y$ be any associated subvariety, can easily regard V as an irreducible component of Y, then $f(V)$ should NOT $\subseteq \mathrm{Supp}(D)$.

Since if it is, for example we set D is effective, then $D|_{U_i} = \mathrm{div}(f_i)$ for which f_i has zeros on $\mathrm{Supp}(D)$, so if $f(V) \subseteq \mathrm{Supp}(D)$, then $f_i \circ f \equiv 0$ on $f^{-1}U_i \subseteq V$, that is, cannot define $\mathrm{div}(f_i \circ f)$, because we need sections in $\mathscr{K}_X^*$.

Now we discuss the case in general, following the [GW20, 11.16], but it is not commonly used.

Inverse Image of Cartier divisors:

In general we can always pull back the line bundles, but can't pullback the Cartier divisors. That is, we always deal with some particular Cartier divisors that behave

well under the support chosen, corresponding to line bundle up to linear equivalent relation this is a sort of "moving lemma" philosophy.

Key Question:

$f : X \to Y$ is a morphism of schemes, when $\mathscr{O}_Y \to f_*\mathscr{O}_X$ given by f can extend to a homomorphism $\mathscr{K}_Y \to f_*\mathscr{K}_X$, then induce to

$$\mathscr{K}_Y^*/\mathscr{O}_Y^* \to f_*\mathscr{K}_X^*/f_*\mathscr{O}_X^* \to f_*(\mathscr{K}_X^*/\mathscr{O}_X^*)$$

$$\rightsquigarrow f^* : \mathrm{Div}(Y) = \Gamma(Y, \mathscr{K}_Y^*/\mathscr{O}_Y^*) \to \Gamma(Y, f_*(\mathscr{K}_X^*/\mathscr{O}_X^*) = \Gamma(X, \mathscr{K}_X^*/\mathscr{O}_X^*) = \mathrm{Div}(X)$$

Definition 1.1.7. (Pullback) $f : X \to Y$ scheme map such that $\mathscr{O}_Y \to f_*\mathscr{O}_X$ can be extended to $\mathscr{K}_Y \to f_*\mathscr{K}_X$. Let D be Cartier divisor on Y, its image f^*D is called the *pullback of* D.

Proposition 1.1.8. *The pulling back has the following properties:*

(a) $f^*(D_1 + D_2) = f^*D_1 + f^*D_2$

(b) $f^*\mathscr{O}_Y(D) \cong \mathscr{O}_X(f^*D)$ *i.e. also has* $f^* : Pic(Y) \to Pic(X)$

Proof. We give a sketch of proof:

(a) $(f_1 \cdot f_2) \circ f = (f_1 \circ f) \cdot (f_2 \circ f)$,

(b) locally they're both $\mathscr{O}_{U_i} \cdot (f_i \circ f)^{-1}$.

□

Theorem 1.1.9. (Pullback Condition) *Let $f : X \to Y$ be a morphism of schemes, and assume the one of the following conditions holds:*

(a) f is flat, or

(b) X is reduced and locally noetherian, each of irreducible components of X dominates an irreducible component of Y.

Then the homomorphism $f^\flat : \mathscr{O}_Y \to f_*\mathscr{O}_X$ extends to $\mathscr{K}_Y \to f_*\mathscr{K}_X$

Proof. Recall that $\mathscr{K}_X$ is the sheaf associated to $U \mapsto S_U^{-1}\Gamma(U, \mathscr{O}_X)$, where $S_U^{-1} \subseteq \Gamma(U, \mathscr{O}_X)$ is a multiplicative subset of regular sections.

So enough to show $\mathscr{O}_Y \to f_*\mathscr{O}_X$ can extends to a morphism between these presheaves, i.e. on $\forall\ V \underset{open}{\subseteq} Y$, the map $f_V^\flat : \Gamma(V, \mathscr{O}_Y) \to \Gamma(f^{-1}V, \mathscr{O}_X)$ induces $S_V^{-1}\Gamma(V, \mathscr{O}_Y) \to S_{f^{-1}V}^{-1}\Gamma(f^{-1}V, \mathscr{O}_X)$, that is, $f_V^\flat$ sends regular sections to regular sections.

Case(a): OK, since flat keeps injective.

Case(b): $\mathscr{O}_{X,x}$ is reduced, so zeroCartier divisors live in minimal primes, i.e. regular sections are outside minimal primes. Since irreducible components corresponding to minimal primes which are generic points of these irreducible components, and dominant map sends generic points to generic points, so sends regular sections to regular sections. □

Remark 1.1.10. (divisor of zeros) Let $s \in \Gamma(X, \mathscr{L})$ be a global section of $\mathscr{L}$, and assume that s does not vanish on any associated subvariety of X: e.g. if X is reduced, this means s doesn't vanish identically on ANY irreducible components of $X \rightsquigarrow (s)_0 = div(s) \in \mathrm{Div}(X)$ is an effective Cartier divisor.

Line Bundles:

A Cartier divisor D on $X \rightsquigarrow$ line bundle $\mathscr{O}_X(D)$: $\mathscr{O}_X(D)|_{U_i} \cong \mathscr{O}_{U_i} \cdot f_i^{-1}$, i.e.

$$\begin{aligned}\mathscr{O}_X(D)(U_i) &= \left\{ f \in \Gamma(X, \mathscr{K}_X^*) \middle| f \cdot f_i = s_i \in \mathscr{O}_{U_i} \right\} \\ &= \left\{ f \in \Gamma(X, \mathscr{K}_X^*) \middle|\ \mathrm{div}(f) + D|_{U_i} \geq 0 \right\}\end{aligned}$$

in which for $f_i = g_{ij} f_j \Rightarrow$ transition functions of $\mathscr{O}_X(D)$ are g_{ij} on $U_i \cap U_j$.

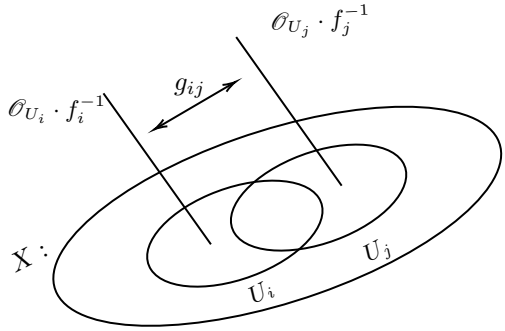


Figure 1.1.: Line Bundle

Moreover:

$$0 \to \mathscr{O}_X^* \to \mathscr{K}_X^* \to \mathscr{K}_X^*/\mathscr{O}_X^* \to 0$$

induces

$$0 \to \Gamma(X, \mathscr{O}_X^*) \to \Gamma(X, \mathscr{K}_X^*) \to \Gamma(X, \mathscr{K}_X^*/\mathscr{O}_X^*) \to H^1(X, \mathscr{O}_X^*) \cong \mathrm{Pic}(X)$$

we get $\mathscr{O}_X(D_1) \cong \mathscr{O}_X(D_2) \Leftrightarrow D_1 \sim_{\mathrm{lin}} D_2$.

Notice that only for Cartier divisors we can get an line bundle, but for Weil divisors D the notation $\mathscr{O}_X(D)$ is called by *divisorial sheaf* used in Kawamata's book.

Example 1.1.11. (Define the Cartier divisor of a section of line bundle) Let X be an irreducible variety, L a line bundle on X.

Picking $s \in H^0(X, L)$ a non-zero global section, we define $\mathrm{div}(s) = (s)_0$ the *divisor of zeros* as follows: $\phi : L|_{U_i} \longrightarrow \mathscr{O}_{U_i}$ is the local trivialization, then

$$\mathrm{div}(s) = \{(U_i, \phi(s|_{U_i}))\}$$

is an effective Cartier divisor.

If $L = \mathscr{O}_X(D)$, then for $D = \{(U_i, f_i)\}$, the local trivialization

$$\phi : \mathscr{O}_X(D)|_{U_i} \cong \mathscr{O}_{U_i} \cdot \frac{1}{f_i} \longrightarrow \mathscr{O}_{U_i}$$

is just the multiplication by f_i, so for non-zero $s \in H^0(X, \mathscr{O}_X(D))$,

$$\mathrm{div}(s) = \big\{(U_i, s|_{U_i} \cdot f_i)\big\}$$

Moreover, if D is an effective Cartier divisor represented by (U_i, f_i), then there is an canonical global section $s_D := s \in H^0\big(X, \mathscr{O}_X(D)\big)$ with $s|_{U_i} = 1 \in \mathscr{O}_X(D)|_{U_i} \cong \mathscr{O}_{U_i} \cdot \frac{1}{f_i}$ such that

$$\mathrm{div}(s) = \big\{(U_i, s|_{U_i} \cdot f_i)\big\} = \big\{(U_i, 1 \cdot f_i)\big\} = \big\{(U_i, f_i)\big\} = D,$$

which recover the original information of that Cartier divisor.

Now we give some admissible facts in algebraic geometry.

Canonical Bundle and Cartier divisor:

X smooth projective variety over $\mathbb{C}$, $\dim X = n$.

$$\omega_X := \bigwedge^n \Omega_X = \Omega_X^n$$

the *canonical line bundle*, and the corresponding Cartier divisor K_X is *canonical Cartier divisor*

such that $\mathscr{O}_X(K_X) \cong \omega_X$.

Kodaira Vanishing Theorem:

Let L be an ample Cartier divisor on a smooth projective variety X which is of dimension n.

Then we have

$$H^i\big(X, \mathscr{O}_X(K_X + L)\big) = 0, \text{ for every } i > 0.$$

We will prove it in Chapter 4, by using the covering trick.

1.1B Linear Series

Let X be a variety, and L is a line bundle on X with $V \subseteq H^0(X, L)$ a non-zero subspace of finite dimension. Then we define

$$|V| := \mathbb{P}_{\text{sub}}(V) \text{ the projective space of 1-dimensional subspaces of V.}$$

that is, these 1-dimensional subspaces form the points of $\mathbb{P}_{\text{sub}}(V)$.

When X is a *complete variety*, $|V|$ is the linear series of Cartier divisors of sections of V.

$\Rightarrow H^0(X, L)$ is a finite dimensional k-vector space, see [Har77, III, Rem8.8.1].

$\Rightarrow$ for $V = H^0(X, L)$, yields a complete linear series.

D Cartier divisor $\rightsquigarrow |D|$: complete linear series related to $\mathscr{O}_X(D)$.

Morphism to a Projective Space:

Evaluation of sections in V gives a morphism:

$$\text{eval}_V : V \underset{\mathbb{C}}{\otimes} \mathscr{O}_X \longrightarrow L$$

where $- \underset{\mathbb{C}}{\otimes} \mathscr{O}_X$ is equipped with an $\mathscr{O}_X$-module structure by the line bundle L is an $\mathscr{O}_X$-module, so elements of $V \subseteq H^0(X, L)$ has $\mathscr{O}_X$-elements as scalars:

$$\text{On } U \underset{\text{open}}{\subset} X : \sum v_i \otimes f_i \longmapsto \sum f_i \cdot v_i|_U \ , \ f_i \in \mathscr{O}_X(U) \ , \ \forall i$$

where $\sum f_i \cdot v_i|_U : x \mapsto \sum f_i(x) \cdot v_i(x)$ in $k(x) = \mathscr{O}_{X,x}/m_x$.

Definition 1.1.12. (Base ideal) The *base ideal* of $|V|$ is written by

$$\mathfrak{b}(|V|) = \mathfrak{b}(X, |V|) \subseteq \mathscr{O}_X$$

is the image of the map $V \underset{\mathbb{C}}{\otimes} L^\vee \to \mathscr{O}_X$ determined by eval_V.

$$\text{Where } V \underset{\mathbb{C}}{\otimes} L^\vee \to \mathscr{O}_X \text{ is induced by } L \underset{\mathbb{C}}{\otimes} L^\vee \cong \mathscr{O}_X,\ L^\vee =: \mathscr{H}om_{\mathscr{O}_X}(L, \mathscr{O}_X)$$
$$v \otimes f \mapsto f(v), \text{ on } U\text{: } f|_U(v|_U) \in \mathscr{O}_X(U) \text{ and at } x\text{: } f_x(v_x) \in \mathscr{O}_{X,x}$$

Definition 1.1.13. (Scheme-theoretic Base locus) The *base locus* of $|V|$ is the closed subset of X cut out by the base ideal $\mathfrak{b}(|V|)$, i.e. $\text{Bs}(|V|) := \text{Supp}\, \mathscr{O}_X/\mathfrak{b}(|V|)$, equipped with the scheme structure, we say the *base scheme*.

When $V = H^0(X, L)$ or $H^0(X, \mathscr{O}_X(D))$ are finite dimensional, [Har77, III, 5.2]:

$$\rightsquigarrow \mathfrak{b}(|L|) \text{ or } \mathfrak{b}(|D|)$$

Remark 1.1.14. $\mathrm{Bs}(|V|) = \{x \in X | \forall s \in V,\ s(x) = 0 \text{ in } k(x)\}$.

Example 1.1.15. Suppose that X is a complete variety with a Cartier divisor D.

Then $\forall m, l \geq 1$ integers, has inclusion:

$$\mathfrak{b}(|lD|) \cdot \mathfrak{b}(|mD|) \subseteq \mathfrak{b}(|(l+m)D|)$$

via

$$\begin{aligned} &H^0(X, \mathcal{O}_X(\ell D)) \otimes H^0(X, \mathcal{O}_X(mD)) \otimes \mathcal{O}_X(-(\ell+m)D) \\ &\quad \xrightarrow{\mu \otimes 1} H^0(X, \mathcal{O}_X((\ell+m)D)) \otimes \mathcal{O}_X(-(\ell+m)D) \xrightarrow{\text{eval}} \mathcal{O}_X, \end{aligned}$$

where μ comes from

$$\begin{aligned} \mu : H^0\big(X, \mathscr{O}_X(lD)\big) \otimes H^0\big(X, \mathscr{O}_X(mD)\big) &\longrightarrow H^0\Big(X, \mathscr{O}_X\big((l+m)D\big)\Big) \\ s \otimes t &\longmapsto s \cdot t. \end{aligned}$$

Definition 1.1.16. (Free linear series) Say $|V|$ is *free*, or *base-point free*, if the base locus is $\emptyset$, i.e. $\mathfrak{b}(|V|) = \mathscr{O}_X$.

A Cartier divisor D or line bundle is *free* if the corresponding complete linear series is so. Say that L is *generated by its global sections* or *globally generated*.

Proposition 1.1.17.

$|V|$ is free $\Longleftrightarrow$ for each point $x \in X$, can find a section $s \in V$ such that $s(x) \neq 0$

Proof.

$$\begin{aligned} \mathfrak{b}(|V|) = \mathscr{O}_X &\Longleftrightarrow \forall x \in X,\ \mathfrak{b}(|V|)_x = \mathscr{O}_{X,x} \\ &\Longleftrightarrow V \otimes L_x^\vee \to \mathscr{O}_{X,x} \text{ is surjective.} \\ &\qquad \text{i.e. has elements mapping to units of } \mathscr{O}_{X.x}, \text{ i.e. not in } m_x. \\ &\Longleftrightarrow \exists s(x) \neq 0 \text{ equivalent to } s_x \notin m_x L_x = m_x \\ &\qquad \text{where } s_x \otimes f_x \text{ can't be mapped into } m_x, \text{ for } f_x \in L_x^\vee = \mathscr{O}_{X,x}^\vee. \end{aligned}$$

□

Assume now that $\dim V \geq 2$. Set $B = \mathrm{Bs}(|V|)$, then $|V|$ determines a morphism:

$$\phi = \phi_{|V|} : X - B \longrightarrow \mathbb{P}(V)$$

where the coordinates of $\mathbb{P}(V)$ is a basis of vector subspace V.

Given $x \in X$, $\phi(x)$ is the hyperplane in V consisting of the sections might vanish at x, but do **not** vanish simultaneously, which are points in $\mathrm{Bs}(|V|)$.

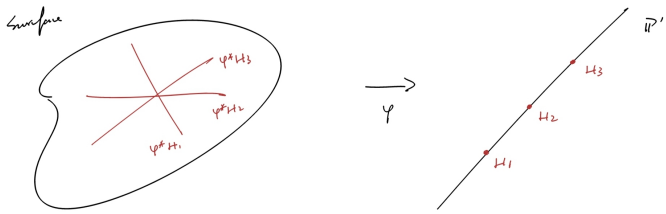


Figure 1.2.: Morphism undefined on base locus

So if we choose a basis $s_0, \ldots, s_r \in V$, therefore ϕ is given in homogeneous coordinates by the expression:

$$\phi : x \longmapsto [s_0(x), \ldots, s_r(x)] \in \mathbb{P}(V).$$

When X is an irreducible variety, $X - B$ is an open subset of X hence dense, so can view $\phi_{|V|}$ as a rational map $X \dashrightarrow \mathbb{P}(V)$.

If $|V|$ is free, then $\phi_{|V|} : X \to \mathbb{P}(V)$ is indeed a morphism. We get that when $B = \emptyset$, the construction can be reversed, i.e. relate a morphism to a **free** linear series.[Har77, II, Rem7.8.1]

Now gives a morphism to a projective space:

$$\phi : X \longrightarrow \mathbb{P} = \mathbb{P}(V),$$

and assume thay $\phi(X)$ doesn't lies on **any** hyperplane, where $V = H^0(\mathbb{P}, \mathscr{O}_{\mathbb{P}(1)}$ corresponding to hyperplane section H of $\mathbb{P}(V) \Rightarrow |V|$ is a free linear series (on X as below).

So if $\exists s \in V$ such that $\phi^*(s) = 0$, then $s \circ \phi(x) = 0$ for $\forall x \in X$, get a contradiction.

$\Rightarrow \phi^* : V = H^0(\mathbb{P}, \mathscr{O}_{\mathbb{P}}(1)) \to H^0(X, \phi^* \mathscr{O}_{\mathbb{P}}(1)) = H^0(X, \mathscr{O}_{\mathbb{P}}(\phi^* H))$ is injective, realizing V as a subspace of $H^0(X, \phi^* \mathscr{O}_{\mathbb{X}}(1))$.

$\Rightarrow \phi$ is identified with the corresponding morphism $\phi_{|V|}$.

Example 1.1.18. (Projection) Suppose that $W \subseteq V$ is a subspace of dim≥ 2.

Then $\mathrm{Bs}(|V|) \subseteq \mathrm{Bs}(|W|)$ by:

$$\begin{aligned} W \subseteq V \Rightarrow & W \underset{\mathbb{C}}{\otimes} L^* \subseteq V \underset{\mathbb{C}}{\otimes} L^* \\ \Rightarrow & \mathfrak{b}(|W|) \subseteq \mathfrak{b}(|V|) \\ \Rightarrow & \mathrm{Supp}\ {}^{\mathscr{O}_X}/_{\mathfrak{b}(|V|)} \subseteq \mathrm{Supp}\ {}^{\mathscr{O}_X}/_{\mathfrak{b}(|W|)} \end{aligned}$$

so that $\phi_{|V|}$ and $\phi_{|W|}$are both defined on $X - \mathrm{Bs}(|W|)$.

Viewed as a morphism on this set, one has the relation:

$$\phi_{|W|} = \pi \circ \phi_{|V|}$$

where

$$\pi : \mathbb{P}(V) - \mathbb{P}(V/W) \longrightarrow \mathbb{P}(W)$$

is linear projection centered along the subspace $\mathbb{P}(V/W) \subseteq \mathbb{P}(V)$.

Assume $|W|$ is free and X is complete, then $|V|$ is free by $\mathrm{Bs}(|V|) \subseteq \mathrm{Bs}(|W|) = \emptyset$, so $\phi_{|V|}$ is a morphism.

X complete [Har77, II, ex4.4] $\Rightarrow \phi_{|V|}(X) \hookrightarrow \mathbb{P}(V)$ is closed immersion. So π can be restricted to $\phi_{|V|}(X)$ which we denote it as $\pi|_X$.

And

$$\begin{array}{ccc} \pi|_X : \phi_{|V|}(X) & \longrightarrow & \mathbb{P}(W) \\ & \searrow & \downarrow g \\ & & \mathrm{Spec}\ \mathbb{C} \end{array}$$

where $g \circ \pi|_X$ is proper since $\phi_{|V|}(X)$ is a projective variety, and g is separated, [Har77, II,ex4.8] $\Rightarrow \pi|_X$ is proper, and also be affine since it is a linear projection.

$\Rightarrow \pi|_X$ is finite via [Har77, II, ex4.6]

So $\phi_{|V|} : X \to \mathbb{P}(V)$ and $\phi_{|W|} : X \to \mathbb{P}(W)$ differ by a finite projection of $\phi_{|V|}(X)$.

1.1C Intersection Numbers via Cohomological Approach

We follow the [Deb01, sec 1.2]

All schemes considered here will be of finite type over an arbitrary field $\mathbf{k}$, but for variety we'll follow Hartshorne's setting which is always over algebraically closed field.

If $X \subseteq \mathbb{P}^N_{\mathbb{C}}$ as a subvariety of dimension n, [Har77, I, 7.5] says that function:

$$m \longmapsto \chi(X, \mathscr{O}_X(m))$$

is a polynomial of degree n with rational coefficients. Indeed, for $m \gg 0$, Serre proves that higher cohomology of $\mathscr{O}_X(m)$ vanishes, see [Har77, III, 5.2(b)], so $\chi(X, \mathscr{O}_X(m)) = \dim_{\mathbb{C}} H^0(X, \mathscr{O}_X(m))$. This polynomial is called *Hilbert polynomial* of X.

The *degree* of X in $\mathbb{P}^N$ is defined as $n!$ times the leading coefficient(of m^n) in the classical Hilbert polynomial $P_X(m) = \frac{\deg X}{n!}m^n + O(m^{n-1})$, see [Har77, p.52]. If H is a hyperplane section, $\deg H$ is denoted by (H^n).

To define $(D_1 \cdots D_n)$ for Cartier divisors $D_1, \ldots, D_n$ on a proper scheme of dimension n, we use the *Euler Characteristics number*: (Mainly follows [Kle66], also can see [KM98, 1.36], [Deb01, 1.5], [Laz04a, 1.1.24] and [Iit82, Chap8])

Theorem 1.1.19. (Asymptotic Riemann-Roch) *Let $D_1, \ldots, D_r$ be Cartier divisor on a proper scheme X and let $\mathscr{F}$ be a coherent sheaf on X. Then function:*

$$(m_1, \ldots, m_r) \longmapsto \chi(X, \mathscr{F}(m_1 D_1 + \cdots + m_r D_r))$$

takes the same value on $\mathbb{Z}^r$ as a (numerical) polynomial with rational coefficients of degree at most the dimension of the support of $\mathscr{F}$ ($\leq dim X$).

In particular, if X is an integral proper scheme of dimension n,

$$\chi(X, mD) := \chi(X, \mathscr{O}_X(mD)) = \frac{(D^n)}{n!}m^n + O(m^{n-1})$$
$$\text{and } \chi(X, \mathscr{F}(mD)) = rank(\mathscr{F}) \cdot \frac{(D^n)}{n!}m^n + O(m^{n-1}),$$

where (D^n) is the n-power self-intersection of D, $rank(\mathscr{F}) = \dim_{k(X)} \mathscr{F} \otimes k(X)$ is the length of stalk of $\mathscr{F}$ at the generic point of X (this is indeed finite by coherence).

Proof. Any coherent sheaf $\mathscr{F}$ on a scheme X has a finite filtration:

$$\mathscr{F} = \mathscr{F}_0 \supseteq \mathscr{F}_1 \supseteq \cdots \supseteq \mathscr{F}_m = 0$$

by coherent subsheaves such that $\mathscr{F}_i/\mathscr{F}_{i+1}$ is torsion-free on an integral subscheme of X, and $\chi(\mathscr{F})$ is determined by these $\chi(\mathscr{F}_i/\mathscr{F}_{i+1})$, so we may assume that X is integral and $\mathscr{F}$ is torsion-free.

Then there exists a $U \underset{open}{\subset} X$ over which $\mathscr{F}$ is free of rank r:

$$\mathscr{F}|_U = \mathscr{F} \underset{\mathscr{O}_X}{\otimes} \mathscr{O}_U \cong \mathscr{O}_U^{\oplus r}$$

$\mathscr{F}$ torsion-free and X integral

$$\Rightarrow \mathscr{F}_x \hookrightarrow \mathscr{F}_\eta \text{ is injective, } \forall x \in X.$$
$$\Rightarrow \mathscr{F}_x \hookrightarrow \mathscr{F}_\eta = (\mathscr{F}|_U)_\eta \cong (\mathscr{O}_U^r)_\eta = \mathscr{O}_{U,\eta}^r \cong \mathscr{K}_X^r = \mathscr{K}_{X,x}^r, \forall x \in X.$$
$$\Rightarrow \mathscr{F} \hookrightarrow \mathscr{K}_X^r \text{ is injective.}$$

Set $\mathscr{G} = \mathscr{F} \cap \mathscr{O}_X^r$ (intersecting in $\mathscr{K}_X^r$ via a given suitable embedding). In the following self-defining exact sequence:

$$0 \to \mathscr{G} \to \mathscr{F} \ \to \mathscr{G}_1 \to 0$$
$$0 \to \mathscr{G} \to \mathscr{O}_X^r \to \mathscr{G}_2 \to 0$$

Since on $U : \mathscr{G}|_U \cong \mathscr{F}|_U \cong \mathscr{O}_X^r|_u$ so cokernel $= 0$, hence the support of the sheaves $\mathscr{G}_1$ and $\mathscr{G}_2$ do not meet U, they have smaller dimension than X, so theorem holds for $\mathscr{G}_1$ and $\mathscr{G}_2$ by induction hypothesis where induction is on the dimension of X.

It follows from the two short exact sequences, $\chi(\mathscr{F})$ is determined by $\chi(\mathscr{G})$ and $\chi(\mathscr{G}_1)$ where $\chi(\mathscr{G})$ is determined by $\chi(\mathscr{O}_X^r) = r \cdot \chi(\mathscr{O}_X)$ and $\chi(\mathscr{G}_2)$. Already know about $\chi(\mathscr{G}_1)$ and $\chi(\mathscr{G}_2)$, so enough to prove $\chi(\mathscr{O}_X)$ satisfies the theorem.

We are reduced to the case $\mathscr{F} = \mathscr{O}_X$(and X integral).

Step 1: Assume first $D_1 = \cdots D_r =: D$ and choose embedding $\mathscr{O}_X(D)$ into $\mathscr{K}_X$. Set

$$\mathscr{I}_1 = \mathscr{O}_X(-D) \cap \mathscr{O}_X \qquad\qquad \mathscr{I}_2 = \mathscr{O}_X(D) \cap \mathscr{O}_X$$

and let Y_j be the subscheme of X defined by the ideal $\mathscr{I}_j$. Since X is integral, Y_j has smaller dimension than X. And notice that

$$\begin{aligned}\mathscr{I}_1(D) := \mathscr{I}_1 \otimes \mathscr{O}_X(D) &= \mathscr{O}_X(-D) \otimes \mathscr{O}_X(D) \cap \mathscr{O}_X \otimes \mathscr{O}_X(D) \\ &\cong \mathscr{O}_X \cap \mathscr{O}_X(D) \\ &= \mathscr{I}_2\end{aligned}$$

so there are exact sequences:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathscr{I}_1(mD) & \longrightarrow & \mathscr{O}_X(mD) & \longrightarrow & \mathscr{O}_{Y_1}(mD) & \longrightarrow & 0 \\ & & \| & & & & & & \\ 0 & \longrightarrow & \mathscr{I}_2((m-1)D) & \longrightarrow & \mathscr{O}_X((m-1)D) & \longrightarrow & \mathscr{O}_{Y_2}((m-1)D) & \longrightarrow & 0 \end{array}$$

Using $\chi(\mathscr{I}(mD)) = \chi(\mathscr{I}_2((m-1)D))$ as a bridge, we get that:

$$\chi(X, mD) - \chi(X, (m-1)D) = \chi(Y_1, mD) - \chi(Y_2, (m-1)D).$$

By induction hypothesis, the right hand side satisfies the theorem, that is, it's a polynomial function in m with rational coefficients, of degree $d < \dim X$. Denote $\chi(X, mD)$ by f.

But [Har77, I, 7.3(b)] says that if a function $f : \mathbb{Z} \longrightarrow \mathbb{Z}$ such that

$$m \longmapsto f(m) - f(m-1)$$

is polynomial of degree d with rational coefficients, then the function f is a numerical polynomial of degree $d+1$ for $m \gg 0$; therefore, $\chi(X, mD)$ is a rational coefficient polynomial function in m of degree $\leq d+1 \leq \dim X$.

Step 2: Now treat the general case, still assume $\mathscr{F} = \mathscr{O}_X$.

Lemma 1.1.20. *Let $d \in \mathbb{Z}_{>0}$, and let $f : \mathbb{Z}^r \longrightarrow \mathbb{Z}$ be a map such that for each $(n_1, \ldots, n_{(i-1)}, n_{(i+1)}, \ldots, n_r) \in \mathbb{Z}^{r-1}$, the map:*

$$m \longmapsto f(n_1, \ldots, n_{(i-1)}, m, n_{(i+1)}, \ldots, n_r)$$

is polynomial of degree at most d, which f takes the same values as a polynomial in r variables with rational coeffs.

Proof. Using induction on r:

- $r = 1$: [Har77, I, 7.3(a)]
- Assume $r - 1 > 0$
- Now for r: there exist functions $f_0, \cdots, f_d : \mathbb{Z}^{r-1} \to \mathbb{Z}$ such that

$$f(m_1, \ldots, m_r) = \sum_{j=0}^{d} f_j(m_1, \ldots, m_{r-1}) \cdot m_r^j.$$

  Now pick distinct integers $c_0, \ldots, c_d$. For each $i \in \{0, \ldots, d\}$, by induction hypothesis, there exist polynomials P_i with rational coefficients such that

$$f(m_1, \ldots, m_{r-1}, c_r) = \sum_{j=0}^{d} f_j(m_1, \ldots, m_{r-1}) \cdot c_i^j = P_i(m_1, \ldots, m_{r-1})$$

  i.e.

$$\begin{pmatrix} c_0^0 = 1 & c_0^1 & \cdots & c_0^d \\ c_1^0 = 1 & c_1^1 & \cdots & c_1^d \\ \vdots & \vdots & \ddots & \vdots \\ c_d^0 = 1 & c_d^1 & \cdots & c_d^d \end{pmatrix} \cdot \begin{pmatrix} f_0(m_1, \ldots, m_{r-1}) \\ f_1(m_1, \ldots, m_{r-1}) \\ \vdots \\ f_d(m_1, \ldots, m_{r-1}) \end{pmatrix} = \begin{pmatrix} P_0(m_1, \ldots, m_{r-1}) \\ P_1(m_1, \ldots, m_{r-1}) \\ \vdots \\ P_d(m_1, \ldots, m_{r-1}) \end{pmatrix}$$

  where the matrix (c_i^j) is Vandermonde matrix, since all c_i's are distinct, so $\det \neq 0 \Rightarrow (c_i^j)$ is invertible. Clearly its inverse has rational coefficients, then each f_j is a linear combination of $P_0, \ldots, P_d$ with rational coefficients, so is f.

□

Using induction on r, the number of Cartier divisors:

- $r = 1$: OK, proved in the lemma above.
- Assume $r-1 > 0$ holds, that is $(m_1, \ldots, m_{r-1}) \mapsto \chi(X, m_1D_1 + \cdots + m_{r-1}D_{r-1})$ satisfies the theorem.
- Now for r, using the lemma, so there exists a polynomial $P \in \mathbb{Q}[T_1, \ldots, T_r]$ such that
$$\chi(X, m_1D_1 + \cdots + m_rD_r) = P(m_1, \ldots, m_r)$$
for all integers $m_1, \ldots, m_r$. Let d be its total degree, we want to show $d \leq \dim X$, so let $n_1, \ldots, n_r$ be integers such that the degree of the polynomial
$$Q(T) = P(n_1T, \ldots, n_rT)$$
is still d. And denoted $n_1D_1 + \cdots + n_rD_r$ by D', so $Q(T)$ sends m to $\chi(X, mD')$, then follows from the case $r = 1$ we get that $d \leq \dim X$.

□

Definition 1.1.21. Let $D_1, \ldots, D_r$ be Cartier divisors on a proper scheme X with $r \geq \dim X$.

The *intersection number*
$$(D_1 \cdots D_r)$$
is the coefficient of $m_1 \cdots m_r$ in the polynomial
$$\chi(X, m_1D_1 + \cdots + m_rD_r)$$

Our next result (Proposition 1.1.26) will show that this number is an integer, and by Theorem 1.19 we proved that it vanishes for $r > \dim X$.

Remark 1.1.22. When $D_1 = \cdots = D_n =: D$, the $\chi(X, (m_1 + \cdots + m_n)D)$ is a polynomial with rational coefficients as theorem mentioned:
$$a_n \cdot (m_1 + \cdots + m_n)^n + O((m_1 + \cdots m_n)^{n-1})$$
in $m_1 + \cdots + m_n$. So the coefficient of $m_1 \cdots m_n$ is $a_n \cdot n! = (D^n)$, the n-power self-intersection of D, can see that $\frac{(D^n)}{n!} = a_n$, and write $m := m_1 + \cdots + m_n$ to get the asymptotic Riemann-Roch formula.

Moreover, for X is projective and H is a hyperplane section on X (as the pullback of the hyperplane in $\mathbb{P}^N$ to X), then H is ample on X so higher cohomologies vanish, in this case, we see that $\deg H$ is just the n-power self-intersection of H.

Example 1.1.23. If $D_1, \ldots, D_r$ are effective and meet properly in a finite number of points, then:

$$(D_1 \cdots D_r) = \#(D_1 \cap \cdots \cap D_r) < \infty$$

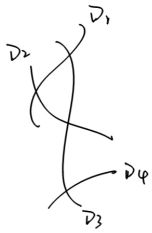


Figure 1.3.: Transversally



Figure 1.4.: Non-transversally

Now if $Y \subseteq X$ of dim $Y \leq s$, we set

$$(D_1 \cdots D_s \cdot Y) = (D_1|_Y \cdots D_s|_Y),$$

also if $\mathscr{F}$ is a coherent sheaf on X with dim Supp$\mathscr{F} \leq r$, we can define

$$(D_1 \cdots D_r \cdot \mathscr{F})$$

as the coefficient of $m_1 \cdots m_r$ in the polynomial

$$\chi(X, \mathscr{F}(m_1 D_1 + \cdots + m_r D_r)).$$

Example 1.1.24.

$$\begin{aligned} \text{Riemann-Roch Theorem} \Rightarrow \chi(C, mD) &= \deg(mD) + 1 - g \\ &= m \cdot \deg(D) + \chi(C, \mathscr{O}_C) \end{aligned}$$

where D is a Cartier divisor, C is a curve, see [Har77, IV, 1.3].

So

$$\begin{aligned} (D \cdot C) &= \deg(D) \text{ which is the coefficient of } m \\ &= \deg(\mathscr{O}_X(D)|_C) = \deg(\mathscr{O}_C(D)), \end{aligned}$$

notice here C is a non-singular curve, so for the general case, we set $\tau : \hat{C} \longrightarrow C$ is normalization, then $\hat{C}$ is smooth, hence

$$(D \cdot C) := (D \cdot \hat{C}) = \deg(\tau^*(\mathscr{O}_X(D)|_C)) = \deg(\tau^* \mathscr{O}_C(D)).$$

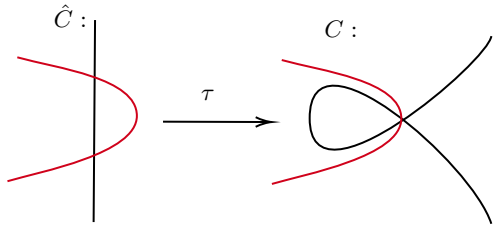


Figure 1.5.: "Sort of moving lemma"

Proposition 1.1.25. *Let $D_1, \ldots, D_n$ be Cartier divisors on a proper scheme X of dim n.*

(a) The map

$$(D_1, \ldots, D_n) \longmapsto (D_1 \cdots D_n)$$

is multilinear, symmetric and takes integer value.

(b) If D_n is effective with associated subscheme Y,

$$(D_1 \cdots D_n) = (D_1 \cdots D_{n-1} \cdot Y)$$

Proof. for(a):

- Symmetric: OK by $m_1 D_1 + m_2 D_2 = m_2 D_2 + m_1 D_1$.
- Taking integer values:

  Fact:

  If $P(T_1, \ldots, T_r)$ is any polynomial of total degree $\leq r$ (notice that P is taking values in $\mathbb{Z}$), then the coefficient of $\prod_{i=1}^{r} T_i$ in P is

  $$\sum_{I \subseteq \{1,\ldots,r\}} \varepsilon_I \cdot P(-m^I) = \sum_{I \subseteq \{1,\ldots,r\}} \varepsilon_I \cdot P(-m_1^I, -m_2^I, \ldots, -m_r^I)$$

  where $\varepsilon_I = (-1)^{\#I}$, and $m_i^I = 1$ if $i \in I$, 0 for otherwise.

  It is easy to see $P(-m_1^I, -m_2^I, \ldots, -m_r^I) = \chi(X, -\sum_{i \in I} D_i)$, assume $r \geq n = \dim X$, the above shows $(D_1 \cdots D_r) = \sum_{I \subseteq \{1,\ldots,r\}} \varepsilon_I \cdot \chi(X, -\sum_{i \in I} D_i)$ is an integer.

- Multilinear:

  Notice by Asymptotic Riemann-Roch, the degree of this polynomial is less than $\dim X = n$, so it only has non-zero coefficient at $r = n$, and vanishes in the case

of $r > n$. So for **any** Cartier divisor $D_1, D_1', D_2, \ldots, D_n$, the sum

$$\begin{aligned}
&\sum_{I\subseteq\{1,\ldots,n+1\}} \varepsilon_I\big(\chi(X,-\sum_{i\in I}D_i)\big) \text{ regard } D_1' \text{ as } D_{n+1}\\
=&\sum_{I\subseteq\{2,\ldots,n\}} \varepsilon_I\cdot\chi(X,-\sum_{i\in I}D_i)+\varepsilon_I\cdot(-1)\cdot\chi(X,-D_1-\sum_{i\in I}D_i)\\
&+\varepsilon_I\cdot(-1)\cdot\chi(X,-D_1'-\sum_{i\in I}D_i)+\varepsilon_I\cdot(-1)^2\chi(X,-D_1-D_1'-\sum_{i\in I}D_i)\\
=&\sum_{I\subseteq\{2,\ldots,n\}} \varepsilon_I\big(\chi(X,-\sum_{i\in I}D_i)-\chi(X,-D_1-\sum_{i\in I}D_i)\\
&-\chi(X,-D_1'-\sum_{i\in I}D_i)+\chi(X,-D_1-D_1'-\sum_{i\in I}D_i)\big)
\end{aligned}$$

$=0$ since it is $n+1$ Cartier divisors intersecting on a proper scheme of dimension n.

On the other hand:

$$\begin{aligned}
\big((D_1+D_1')\cdot D_1\cdots D_n\big) &= \sum_{I\subseteq\{1,\ldots,n\}} \varepsilon_I\cdot\chi(X,-\sum_{i\in I}D_i), \text{ regard } D_1+D_1' \text{ as } D_1.\\
\text{(fixed index 1)} &= \sum_{I\subseteq\{2,\ldots,n\}} \varepsilon_I\cdot\chi(X,-\sum_{i\in I}D_i)+\varepsilon_I\cdot(-1)\cdot\chi(X,-D_1-D_1'-\sum_{i\in I}D_i)\\
&= \sum_{I\subseteq\{2,\ldots,n\}} \varepsilon_I\big(\chi(X,-\sum_{i\in I}D_i)-\chi(X,-D_1-D_1'-\sum_{i\in I}D_i)\big)
\end{aligned}$$

and

$$\begin{aligned}
(D_1\cdot D_2\cdots D_n) &= \sum_{I\subseteq\{1,\ldots,n\}} \varepsilon_I\cdot\chi(X,-\sum_{i\in I}D_i)\\
\text{(fixed index 1)} &= \sum_{I\subseteq\{2,\ldots,n\}} \varepsilon_I\big(\chi(X,-\sum_{i\in I}D_i)-\chi(X,-D_1-\sum_{i\in I}D_i)\big)\\
(D_1'\cdot D_2\cdots D_n) &= \sum_{I\subseteq\{2,\ldots,n\}} \varepsilon_I\cdot\big(\chi(X,-\sum_{i\in I}D_i)-\chi(X,-D_1'-\sum_{i\in I}D_i)\big)
\end{aligned}$$

Hence $(D_1\cdot D_2\cdots D_n)+(D_1'\cdot D_2\cdots D_n)-((D_1+D_1')\cdot D_2\cdots D_n)=0$, holds by multilinearity.

for (b):

$$(D_1\cdots D_{n-1}\cdot D_n) = \sum_{I\subseteq\{1,\ldots,n-1\}} \varepsilon_I\big(\chi(X,-\sum_{i\in I}D_i)-\chi(X,-D_n-\sum_{i\in I}D_i)\big) \text{ (fixed index n)}$$

and the famous exact sequence $0 \to \mathscr{O}_X(-D_n) \to \mathscr{O}_X \to \mathscr{O}_Y \to 0$ tensoring with the locally free coherent sheaf $\mathscr{O}_X(-\sum_{i\in I} D_i)$ we get

$$0 \longrightarrow \mathscr{O}_X(-D_n - \sum_{i\in I} D_i) \longrightarrow \mathscr{O}_X(-\sum_{i\in I} D_i) \longrightarrow \mathscr{O}_Y(-\sum_{i\in I} D_i) \longrightarrow 0$$

so

$$\begin{aligned}\chi(X, -\sum_{i\in I} D_i) - \chi(X, -D_n - \sum_{i\in I} D_i) =& \chi\big(X, i_*\mathscr{O}_Y(-\sum_{i\in I} D_i)\big) \\ =& \chi(Y, -\sum_{i\in I} D_i)\end{aligned}$$

where $i : Y \hookrightarrow X$ is a closed immersion.

We get:

$$(D_1 \cdots D_n) = \sum_{I\subseteq\{1,\ldots,n-1\}} \varepsilon_I \cdot \chi(Y, -\sum_{i\in I} D_i) = (D_1 \cdots D_{n-1} \cdot Y).$$

□

Projection formula:

Let $\pi : X \longrightarrow Y$ be a proper morphism between varieties, $C \subseteq X$ curve. We define the 1-cycle $\pi_* C$ as follows:

- π contracts C to a point in Y: $\pi_* C = 0$;
- $\pi(C)$ is a curve on Y: $\pi_* C = d\pi(C)$, where d is the degree of the morphism $C \longrightarrow \pi(C)$.

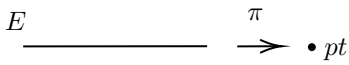


Figure 1.6.: 1st case

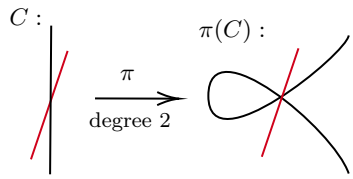


Figure 1.7.: 2nd case

Corollary 1.1.26. (Projection formula for a simple case) *If D is a Cartier divisor on Y, and $\pi : X \longrightarrow Y$ as above.*

Then

$$(\pi^* D \cdot C) = (D \cdot \pi_* C).$$

Proof. We give a sketch of the proof. As a consequence of the Riemann-Roch theorem, we have that

$$\begin{aligned}(\pi^* D \cdot C) &= \deg(\mathscr{O}_C(\pi^* D)) \\ &= \deg(\pi|_C^* \mathscr{O}_{\pi(C)}(D)) \\ &= \deg(\pi|_C) \cdot \deg(\mathscr{O}_{\pi(C)}(D)) \\ &= \deg(\mathscr{O}_{\pi_* C}(D))\end{aligned}$$

□

Example 1.1.27. (Expansion of II, 6.9 in [Har77]) Let $\pi : Y \longrightarrow X$ be a finite dominant morphism between varieties, then π induces $\pi^* : K(X) \longrightarrow K(Y)$ is a finite field extension, so $\deg \pi$ is the degree of this field extension.

The next result comes from [Kle66, P.299, 6], the proof will be given in (D.6.4).

Proposition 1.1.28. *Let $\pi : Y \longrightarrow X$ be a morphism between proper varieties. Let $D_1, \ldots, D_r$ be Cartier divisors on X with $r \geq \dim Y$.*

Then:

$$(\pi^* D_1 \cdots \pi^* D_r) = \deg(\pi)(D_1 \cdots D_r).$$

In particular, if π is birational, then

$$(\pi^* D_1 \cdots \pi^* D_r) = (D_1 \cdots D_r).$$

Remark 1.1.29. In Lazarsfeld's book, $(D_1 \cdots D_r)$ is denoted by $\int_X D_1 \cdots D_r$ and the projection formula is

$$\int_Y \pi^* D_1 \cdots \pi^* D_r = \deg(\pi) \cdot \int_X D_1 \cdots D_r$$

And for line bundles $(L_1 \cdots L_r)$ is denoted by $\int_X c_1(L_1) \cdots c_1(L_r)$, where c_1 is the *first Chern class* sending line bundles into codimension-1 cycles on a variety, see [EH16, 1.30] or [Ful98].

Moreover if we're on schemes, the *first Chern class* can be regarded an operator pairing with a line bundle L of the form $c_1(L) \cap _$, sending k-cycles to be $k-1$-cycles, thus of course by applying multiple operators, $c_1(L_1) \cap \cdots c_1(L_n) \cap _$ can sending dimension n scheme X to be numerical result $\int_X c_1(L_1) \cdots c_1(L_n)$.

The notation we use $\int$ comes from the complex geometry, where the first Chern class c_1 sending a line bundle L into a cohomological class, and via Poincare duality on compact complex manifold we can get an intersecting pair. This we'll call the *intersection number via topological approach*, for details see [GH78].

Numerical Equivalence:

X is a proper scheme.

Definition 1.1.30. Two Cartier divisors D_1, $D_2 \in \mathrm{Div}(X)$ are *numerically equivalent* written $D_1 \equiv_{num} D_2$, if

$$(D_1 \cdot C) = (D_2 \cdot C) \text{ for every irreducible curve } C \subseteq X.$$

A Cartier divisor D is *numerically trivial* if $D \equiv_{num} 0$, these Cartier divisors form a subgroup of $\mathrm{Div}(X)$, denoted by $\mathrm{Num}(X)$.

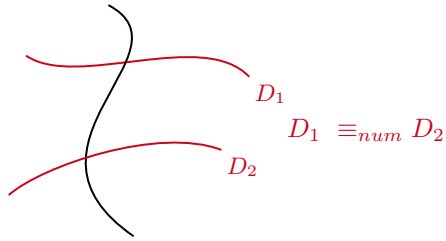


Figure 1.8.: Numerically equivalent

Definition 1.1.31. (Néron-Severi group) The *Néron-Severi group* of X is the group

$$N^1(X) := \mathrm{Div}(X)/\mathrm{Num}(X)$$

of numerically equivalent classes of Cartier divisors on X. Or we denote it by $\mathrm{NS}(X)$.

Basic Fact: see [Har77, V, ex1.7] or [GH78, P.461], we have

$$N^1(X) \text{ is free of finite rank.}$$

So rank $N^1(X) =: \rho(X)$, is called the *Picard number* of X.

Remark 1.1.32. Also the Picard number is usually defined by

$$\dim_{\mathbb{R}} N^1(X)_{\mathbb{R}} = \dim_{\mathbb{R}} N^1(X) \underset{\mathbb{Z}}{\otimes} \mathbb{R}$$

we will discuss this in the $\mathbb{R}$-Cartier divisor section.

1.2 The Classical Theory

Question:

Given a Cartier divisor on a projective variety X, what should it mean for D to be positive?

Idea:

Asking that D be a hyperplane section under a suitable projective embedding,

$$\text{i.e. } D = H \cap X, \text{ where } H^0\big(X, \mathscr{O}_X(D)\big) \text{ induces embedding } X \hookrightarrow \mathbb{P}^N$$

one says that D is very ample.

But this is not easy to find such embedding.

Good Idea:

More conveniently we can find some positive multiple of D to be very ample, in this case we say D is *ample*.

Definition 1.2.1. (Ampleness and very ampleness on complete scheme) Let X be a **complete scheme**, and L a line bundle on X,

(i) L is *very ample* if there exist a **closed** embedding $X \subseteq \mathbb{P}^N$ of X such that

$$L = \mathscr{O}_X(1) := \mathscr{O}_{\mathbb{P}^N}(1)|_X$$

(ii) L is *ample* if $L^{\otimes m}$ is very ample for some $m > 0$.

A Cartier divisor D on X is *ample* or *very ample* if the corresponding line bundle $\mathscr{O}_X(D)$ is so.

Example 1.2.2. (Ample line bundle on Curve) X: irreducible curve. L: line bundle on X.

[Har77, IV, 3.3] says that L ample $\iff \deg(L) > 0$.

Example 1.2.3. (Intersection products) If $D_1, \ldots, D_n$ be ample Cartier divisor on a dim n proj. variety X, then $(D_1 \cdots D_n) > 0$.

Since D_i being ample implies that mD_i are very ample $\forall i$, and intersection number is multilinear, so enough to assume D_i are very ample, which are all effective.

We consider the intersection $D_1 \cap \cdots \cap D_{n-1}$, each of them has codimension 1 due to Hauptidealsatz, so the result is on curve case, hence $(D_1 \cdots D_n) = \sum_C \deg \mathscr{O}_C(D_n)$ where C in irreducible curve run-out in the $D_1 \cap \cdots \cap D_{n-1}$, and by D_n is ample on C which we are going to prove later.

1.2A Cohomological Properties

As in [Har77] we already know the following holds in projective case; now we begin with the general case:

Theorem 1.2.4. *Let L be a line bundle on a complete scheme X.*

The following are equivalent:

(i) *L is ample.*

(ii) *[SP25, Tag 0B5T] Given any coherent sheaf $\mathscr{F}$ on X, there exists $m_1 = m_1(\mathscr{F}) \in \mathbb{Z}_{>0}$ such that*

$$H^i(X, \mathscr{F} \otimes L^{\otimes m}) = 0,\ \forall i > 0 \text{ and } m \geq m_1.$$

(iii) *[SP25, Tag 01Q3] Given any coherent sheaf $\mathscr{F}$ on X, there exists $m_2 = m_2(\mathscr{F}) \in \mathbb{Z}_{>0}$ such that*

$$\mathscr{F} \underset{\mathscr{O}_X}{\otimes} L^{\otimes m} \text{ is generated by its global sections}$$

for all $m \geq m_2$.

(iv) *There is a $m_3 \geq m_3(L) \in \mathbb{Z}_{>0}$ such that $L^{\otimes m}$ is very ample for every $m \geq m_3$.*

Example 1.2.5. (Sum of Cartier divisors) Let D and E be (Cartier) Cartier divisors on a projective scheme X. If D is ample, then there exists an $m_1 = m_1(D)$ such that mD is very ample for $m \geq m_1$, and $\exists m_2 = m_2(E)$ such that $mD + E$ is free for $m \geq m_2$.

Using [Har77, II, ex7.5(d)], we see that $mD + E$ is very ample for $m \geq m_1 + m_2$ which implies $mD + E$ is ample by Serre.

Example 1.2.6. (Ample line bundle on a product) If L and M are ample line bundles on proj. schemes X and Y respectively, then $\mathrm{pr}_1^* L \otimes \mathrm{pr}_2^* M$ is ample on $X \times Y$.

Proof. Picking N sufficiently large such that

$$L^{\otimes N} \text{ is very ample on } X,$$
$$M^{\otimes N} \text{ is very ample on } Y.$$

by [Har77, II,ex7.5(d)].

So

$$L^{\otimes N} \cong i_1^* \mathscr{O}_{\mathbb{P}^r}(1),\ i_1 : X \hookrightarrow \mathbb{P}^r,$$
$$M^{\otimes N} \cong i_2^* \mathscr{O}_{\mathbb{P}^s}(1),\ i_2 : Y \hookrightarrow \mathbb{P}^s.$$

then

$$\begin{aligned} \mathrm{pr}_1^* L^{\otimes N} \otimes \mathrm{pr}_2^* M^{\otimes N} &\cong (i_1 \circ \mathrm{pr}_1)^* \mathscr{O}_{\mathbb{P}^r}(1) \otimes (i_2 \circ \mathrm{pr}_2)^* \mathscr{O}_{\mathbb{P}^s}(1) \\ &\cong \mathscr{O}_{X\times Y}(1) \text{ by [Har77, II, ex5.12]} \\ &= i^* \mathscr{O}_{\mathbb{P}^{rs+r+s}}(1) \end{aligned}$$

where $i : X \times Y \hookrightarrow \mathbb{P}^r \times \mathbb{P}^s \hookrightarrow \mathbb{P}^{rs+r+s}$ is Segre embedding,

$$\Longrightarrow \mathrm{pr}_1^* L \otimes \mathrm{pr}_2^* M \cong (\mathrm{pr}_1^* L \otimes \mathrm{pr}_2^* M)^{\otimes N} \text{ is ample on } X \times Y,$$
$$[\text{Har77}, \text{II}, 7.6] \Longrightarrow \mathrm{pr}_1^* L \otimes \mathrm{pr}_2^* M \text{ is ample on } X \times Y.$$

□

Proposition 1.2.7. (Finite Pullbacks, I) *Let $f : Y \longrightarrow X$ be a finite morphism of complete schemes, and L be an ample line bundle on X.*

Then f^*L is an ample line bundle on Y.

Moreover, if $Y \subseteq X$ is a subscheme of X, then $L|_Y \cong L \otimes \mathscr{O}_Y$ is ample on Y.

Remark 1.2.8. Can see [Har77, II, ex5.7(a)+(d)] "only if" part.

Proof. Let $\mathscr{F}$ be a coherent sheaf on Y.

Then $f_*(\mathscr{F} \otimes f^*L^{\otimes m}) = f_*\mathscr{F} \otimes L^{\otimes m}$ by [Har77, II, ex5.1(d)] the "projection formula", and $f_*\mathscr{F}$ is indeed coherent by [Har77, II, ex5.5(c)].

And since f is a finite map, it is an affine map, so using [Har77, III, ex8.2], for $m \gg 0$:

$$\begin{aligned} H^i(Y, \mathscr{F} \otimes f^*L^{\otimes m}) &\cong H^i\big(X, f_*(\mathscr{F} \otimes f^*L^{\otimes m})\big) \\ &\cong H^i(X, f_*\mathscr{F} \otimes L^{\otimes m}) \\ &= 0 \text{ because } L \text{ is ample on } X, \end{aligned}$$

so f^*L is ample on Y.

In particular when $f : Y \hookrightarrow X$ is a closed embedding, $f^*L = L \otimes \mathscr{O}_Y = L|_Y$. □

Corollary 1.2.9. (Globally generated line bundle) *Suppose that L is globally generated, and let*

$$\phi = \phi_{|L|} : X \longrightarrow \mathbb{P} = \mathbb{P}H^0(X, L)$$

be the resulting map to the projective space defined by the complete linear system $|L|$.

Then the following are equivalent:

(i) L *is ample.*

(ii) ϕ *is a finite map.*

(iii) $(\mathscr{O}_X(C) \cdot L) = deg(L|_C) > 0$ *for every irreducible curve* $C \subseteq X$ *(up to normalization and projection formula).*

Proof. "(ii)⇒(i)":

ϕ is finite, and $\mathscr{O}_{\mathbb{P}}(1)$ is ample on $\mathbb{P}$, so $L \cong \phi^* \mathscr{O}_{\mathbb{P}}(1)$ is ample by previous proposition.

"(i)⇒(ii)":

Suppose ϕ is not finite map, then there is a subvariety $Z \subseteq X$ of dim > 0 such that is contracted by ϕ to a point, since by [Har77, III, ex11.2] we have under projective morphism ϕ, quasi-finite $\iff$ finite.

Then

$$\begin{aligned} L \cong \phi^* \mathscr{O}_{\mathbb{P}}(1), \text{ then } L|_Z \cong & \big(\phi^* \mathscr{O}_{\mathbb{P}}(1)\big)|_Z \\ = & (\phi|_Z)^* \mathscr{O}_{\mathbb{P}}(1)|_{\{p\}} \\ \cong & (\phi|_Z)^* \mathscr{O}_{\mathbb{P},p} \\ \cong & \mathscr{O}_Z \end{aligned}$$

is a trivial line bundle, and $L|_Z$ not ample, and thus L is not ample by previous proposition.

"(i)⇒(iii)":

$$\begin{aligned} L \text{ ample on } X \Longrightarrow & L|_C \text{ ample on } C \\ \Longrightarrow & \deg(L|_C) > 0 \text{ by [Har77, IV, 3.3]} \end{aligned}$$

"(iii)⇒(ii)":

Also as above where ϕ is not finite, then for an irreducible curve $C \subseteq Z$ irreducible curve, we have

$$\begin{aligned} \deg(L|_C) = & (\mathscr{O}_X(C) \cdot L) \\ \cong & (\mathscr{O}_Z(C) \cdot L|_Z) \\ \cong & \big(\mathscr{O}_Z(C) \cdot \mathscr{O}_Z\big) \\ = & 0. \end{aligned}$$

□

Proposition 1.2.10. *Let X be a complete scheme, and L is line bundle on X.*

(i): L is ample on $X \iff L_{red}$ is ample on X_{red}.

(ii): L is ample on $X \iff L$ is ample on the irreducible components of X.

Proof. (i):

"$\Rightarrow$" $A \overset{onto}{\to} A/\sqrt{(0)} \rightsquigarrow X_{\mathrm{red}} \overset{i}{\to} X$ is a finite map, so $L_{\mathrm{red}} = i^*L$ is ample on X_{red}.

"$\Leftarrow$" Fix a coherent sheaf $\mathscr{F}$ on X, and let $\mathscr{N}$ be the nilradical ideal sheaf of $\mathscr{O}_X$ such that $\mathscr{N}^r = 0$ for some r.

Consider the filtration:

$$\mathscr{F} \supseteq \mathscr{N} \cdot \mathscr{F} \supseteq \mathscr{N}^2 \cdot \mathscr{F} \supseteq \cdots \supseteq \mathscr{N}^r \cdot \mathscr{F} = 0$$

where the quotients $\mathscr{N}^i\mathscr{F}/\mathscr{N}^{i+1}\mathscr{F} \cong \mathscr{N}^i\mathscr{F} \underset{\mathscr{O}_X}{\otimes} \mathscr{O}_X/\mathscr{N} \cong \mathscr{N}^i\mathscr{F} \otimes \mathscr{O}_{X_{\mathrm{red}}}$ are coherent $\mathscr{O}_{X_{\mathrm{red}}}$-modules.

For $i : X_{\mathrm{red}} \hookrightarrow X$ closed immersion,

$$\begin{aligned}
H^j\big(X, (\mathscr{N}^i\mathscr{F}/\mathscr{N}^{i+1}\mathscr{F}) \otimes L^{\otimes m}\big) \cong & H^j\big(X, (\mathscr{N}^i\mathscr{F} \underset{\mathscr{O}_X}{\otimes} \mathscr{O}_{X_{\mathrm{red}}}) \otimes L^{\otimes m}\big) \\
\cong & H^j\big(X, (\mathscr{N}^i\mathscr{F} \underset{\mathscr{O}_X}{\otimes} \mathscr{O}_{X_{\mathrm{red}}}) \underset{\mathscr{O}_{X_{\mathrm{red}}}}{\otimes} L^{\otimes m} \underset{\mathscr{O}_X}{\otimes} \mathscr{O}_{X_{\mathrm{red}}}\big) \\
\cong & H^j\big(X, i_*(\mathscr{N}^i\mathscr{F} \underset{\mathscr{O}_X}{\otimes} \mathscr{O}_{X_{\mathrm{red}}} \underset{\mathscr{O}_{X_{\mathrm{red}}}}{\otimes} L_{\mathrm{red}}^{\otimes m})\big) \\
[\text{Har77}, \text{III}, 2.10] \cong & H^j\big(X_{\mathrm{red}}, (\mathscr{N}^i\mathscr{F} \underset{\mathscr{O}_X}{\otimes} \mathscr{O}_{X_{\mathrm{red}}}) \underset{\mathscr{O}_{X_{\mathrm{red}}}}{\otimes} L_{\mathrm{red}}^{\otimes m}\big) \\
= & 0 \text{ because } L_{\mathrm{red}} \text{ is ample on } X_{\mathrm{red}}, \text{ for } j > 0 \text{ and } m \gg 0.
\end{aligned}$$

Now twisting the exact sequences

$$0 \longrightarrow \mathscr{N}^{i+1}\mathscr{F} \longrightarrow \mathscr{N}^i\mathscr{F} \longrightarrow \mathscr{N}^i\mathscr{F}/\mathscr{N}^{i+1}\mathscr{F} \longrightarrow 0$$

by $L^{\otimes m}$ and taking cohomology, induces the long exact sequence:

$$\cdots \to H^j(X, \mathscr{N}^{i+1}\mathscr{F} \otimes L^{\otimes m}) \to H^j(X, \mathscr{N}^i\mathscr{F} \otimes L^{\otimes m}) \to H^j\big(X, (\mathscr{N}^i\mathscr{F}/\mathscr{N}^{i+1}\mathscr{F}) \otimes L^{\otimes m}\big) \to$$

where for i large enough, $\mathscr{N}^i = 0$ so cohomologies vanish for all j, so we can use decreasing induction on i: the left term is 0 by the induction hypothesis, right term is 0 by the above computation, so the middle term is vanishes. Hence

$$H^j(X, \mathscr{N}^i\mathscr{F} \otimes L^{\otimes m}) = 0 \text{ for } j > 0 \text{ and } m \gg 0$$

which holds for every i, and the case $i = 0$ exactly gives the ampleness of L.

(ii): We have already proved that ampleness is the same on the reduced scheme structure, so can just assume X is reduced.

"⇒": OK, using previous proposition.

"⇐": let $X = X_1 \cup \cdots \cup X_r$ for X_i irreducible components (We're working with algebraic schemes over $\mathbb{C}$!), and assume that $L|_{X_i}$ is ample for $\forall i$.

Fix a coherent sheaf $\mathscr{F}$ on X, let $\mathscr{I}$ be ideal sheaf of X_1 in $\mathscr{O}_X$, and consider the exact sequence:

$$0 \longrightarrow \mathscr{I} \cdot \mathscr{F} \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}/\mathscr{I} \cdot \mathscr{F} \longrightarrow 0$$

when the first term is supported on $X_2 \cup \cdots \cup X_r$ since it vanishes on X_1, and last term is supported on X_1 since $\mathscr{F}/\mathscr{I}\mathscr{F} \cong \mathscr{F} \underset{\mathscr{O}_X}{\otimes} \mathscr{O}_X/\mathscr{F}$ such that $\mathrm{Supp}(\mathscr{F} \otimes \mathscr{O}_X/\mathscr{I}) \subseteq \mathrm{Supp}(\mathscr{O}_X/\mathscr{I}) = X_1$.

So by induction on the number of irreducible components, we get that

$$H^j(X, \mathscr{I}\mathscr{F} \otimes L^{\otimes m}) = H^j\big(X, (\mathscr{F}/\mathscr{I}\mathscr{F}) \otimes L^{\otimes m}\big) = 0$$

for $j > 0$ and $m \gg 0$, then the middle term $H^j(X, \mathscr{F} \otimes L^{\otimes m}) = 0$ for $j > 0$ and $m \gg 0$, which implies that L is ample on X. □

The following result can find in [EGA III, 4.7.1] or in [KM98, 1.41].

Theorem 1.2.11. (Ampleness in families) *Let $f : X \longrightarrow T$ be a proper morphism of schemes, and L a line bundle on X. Given $t \in T$, write*

$$X_t := X \times \mathrm{Spec}\ k(t) = f^{-1}(t)$$
$$L_t := L|_{X_t} = L \underset{\mathscr{O}_X}{\otimes} \mathscr{O}_{X_t}$$

Assume that L_0 is ample on X_0 for some point $0 \in T$. Then there is an open neighborhood U of 0 in T such that L_t is ample on X_t for all $t \in U$.

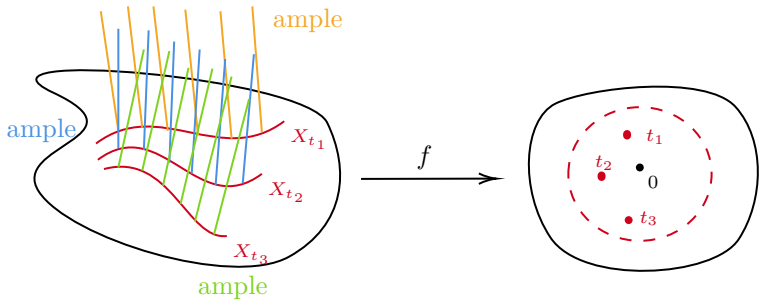


Figure 1.9.: Ampleness stable under suitable continuous deformation.

Proof. The statement is local on T, so we can assume $T = \mathrm{Spec}A$ is affine.

Step 1:

For an arbitrary fixed coherent sheaf $\mathscr{F}$ on X, we claim that there is a positive integer $m(\mathscr{F})$ such that

$$R^i f_*(\mathscr{F} \otimes L^{\otimes m}) = 0 \text{ in a neighborhood } U_m \subseteq T \text{ of } 0 \tag{1.1}$$

for all $i \geq 1$ and $m \geq m(\mathscr{F})$, where $R^i f_*(\mathscr{F} \otimes L^{\otimes m}) \cong H^i(X, \widetilde{\mathscr{F} \otimes L^{\otimes m}})$ by [Har77, III, 8.5].

In fact, by Grothendieck vanishing theorem, this is true for sufficiently large i, so we prove it by decreasing induction on i. Assuming that the equation is known for given $i \geq 2$, need to show that it holds for $i - 1$.

Consider the maximal ideal $m_0 \subseteq A$ of 0 in T, and choose generators $u_1, \ldots, u_p \in m_0$. This gives rise to a presentation:

$$A^{\oplus p} \xrightarrow{u} A \longrightarrow A/m_0 = k(0) \longrightarrow 0$$

of A/m_0, where $u(a_1, \ldots, a_p) = \sum a_i u_i$, which induces

$$\mathscr{O}_T^{\oplus p} \xrightarrow{u} \mathscr{O}_T \longrightarrow \mathscr{O}_T/m_0\mathscr{O}_T \longrightarrow 0$$

pulling back to f:

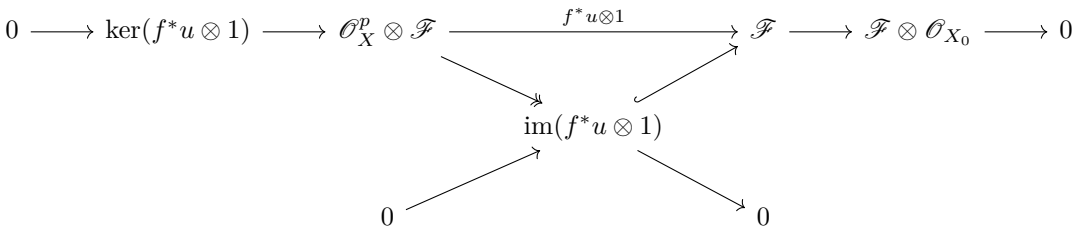
$$f^*\mathscr{O}_T^{\oplus p} \longrightarrow f^*\mathscr{O}_T \longrightarrow f^*(\mathscr{O}_T/m_0\mathscr{O}_T) \longrightarrow 0$$

which is just

$$\mathscr{O}_X^{\oplus p} \longrightarrow \mathscr{O}_X \longrightarrow \mathscr{O}_{X_0} \longrightarrow 0$$

since $\mathscr{O}_T/m_0\mathscr{O}_T$ is a skyscraper sheaf supported on $0 \in T$, so after pullback it must be supported on $f^{-1}(0) = X_0$.

Tensoring with $\mathscr{F}$, running an epi-mono factorization:



using the induction hypothesis on the kernel sheaf:

$$R^i f_*\big(\ker(f^* \otimes 1) \otimes L^{\otimes m}\big) = 0 \text{ near } 0 \text{ for } m \gg 0.$$

Furthermore, since L_0 is ample, by Serre, the index $i-1$ higher direct image of $\mathscr{F} \underset{\mathscr{O}_X}{\otimes} \mathscr{O}_{X_0} \underset{\mathscr{O}_X}{\otimes} L^{\otimes m} \cong \mathscr{F} \underset{\mathscr{O}_X}{\otimes} \mathscr{O}_{X_0} \underset{\mathscr{O}_{X_0}}{\otimes} L_0^{\otimes m}$ vanish when m sufficiently large, which is just the cohomology group on X_0, follows from [Har77, III, 2.10 and 5.3].

Then we get a morphism after tensoring $L^{\otimes m}$ and chasing the above diagram:

$$R^{i-1}f_*(\mathscr{O}_X^p \underset{\mathscr{O}_X}{\otimes} \mathscr{F} \otimes L^{\otimes m}) \cong R^{i-1}f_*(\mathscr{F} \otimes L^{\otimes m}) \underset{\mathscr{O}_T}{\otimes} \mathscr{O}_T^p \xrightarrow{1\otimes u} R^{i-1}f_*(\mathscr{F} \otimes L^{\otimes m})$$

where the equality follows from [Har77, III, ex8.3], the "projection formula", and this is surjective in a neighborhood U'_m of 0 for $m \gg 0$.

On the other hand, by our construction $1 \otimes u$ factors through the inclusion

$$\begin{array}{ccc} R^{i-1}f_*(\mathscr{F} \underset{\mathscr{O}_X}{\otimes} L^{\otimes m}) \underset{\mathscr{O}_T}{\otimes} \mathscr{O}_T^p & \xrightarrow{1\otimes u} & R^{i-1}f_*(\mathscr{F} \otimes L^{\otimes m}) \\ & \searrow \quad \nearrow & \\ & m_0 \cdot R^{i-1}f_*(\mathscr{F} \otimes L^{\otimes m}) & \end{array}$$

since using projection formula again we have:

$$R^{i-1}f_*(\mathscr{F} \underset{\mathscr{O}_X}{\otimes} \mathscr{O}_{X_0} \otimes L^{\otimes m}) \cong R^{i-1}f_*(\mathscr{F} \otimes L^{\otimes m}) \underset{\mathscr{O}_T}{\otimes} \mathscr{O}_T/m_0\mathscr{O}_T.$$

which is the term on the right maps into.

In other words, for m is sufficiently large, then

$$R^{i-1}f_*(\mathscr{F}L^{\otimes m}) = m_0 \cdot R^{i-1}f_*(\mathscr{F} \otimes L^{\otimes m})$$

in a neighborhood of 0. But due to Nakayama's lemma with the coherence is at [Har77, III, 8.8.1], implies that

$$R^{i-1}f_*(\mathscr{F} \otimes L^{\otimes m}) = 0$$

near 0, as we verified the claim (1.1).

Step 2:

Next we claim that the canonical mapping

$$\rho_m : f^*f_*L^{\otimes m} \longrightarrow L^{\otimes m}$$

which is induced by the adjoint pair, is surjective along X_t for all t in a neighborhood U''_m of 0 provided that $m \gg 0$.

To see this, apply the (1.1) to the ideal sheaf $\mathscr{I}_{X_0/X}$ of X_0 in $\mathscr{O}_X$, can find that

$$R^if_*(\mathscr{I}_{X_0/X} \otimes L^{\otimes m}) = 0 \text{ in a neighborhood } U_m \subseteq T \text{ of } 0$$

for all $i \geq 1$, $m \gg 0$, so that by

$$0 \longrightarrow \mathscr{I}_{X_0/X} \longrightarrow \mathscr{O}_X \longrightarrow \mathscr{O}_X/\mathscr{I}_{X_0/X} \cong \mathscr{O}_{X_0} \longrightarrow 0$$

which inducs

$$H^0(\widetilde{X, L^{\otimes m}}) \longrightarrow H^0(X, \widetilde{L^{\otimes m}} \underset{\mathscr{O}_X}{\otimes} \mathscr{O}_{X_0})$$

is surjective in a neighborhood $U_m \subseteq T$ of 0, but the right term is just $H^0(\widetilde{X_0, L_0^{\otimes m}})$ support on a neighborhood of 0, so

$$f_*(L^{\otimes m}) \longrightarrow f_*(L^{\otimes m} \otimes \mathscr{O}_{X_0}) = H^0(\widetilde{X, L_0^{\otimes m}}) \tag{1.2}$$

is surjective when $m \gg 0$. But by [Har77, II, 7.5], all sufficiently large powers of the ample line bundle L_0 are globally generated.

Composting (1.2) with the evaluation map

$$\begin{aligned} H^0(X_0, L_0^{\otimes m}) \underset{\mathbb{C}}{\otimes} \mathscr{O}_{X_0} &\longrightarrow L_0^{\otimes m} \\ \sum v_i \otimes f_i &\longmapsto \sum f_i \cdot v_i \end{aligned}$$

which is also surjective, we get

$$f_*(L^{\otimes m}) \longrightarrow L_0^{\otimes m}$$

is surjective along X_0, i.e.

$$f_* L^{\otimes m} \underset{\mathscr{O}_T}{\otimes} \mathscr{O}_{X_0} \cong f^* f_* L^{\otimes m}|_{X_0} \longrightarrow L^{\otimes m}|_{X_0} = L_0^{\otimes m}$$

says that ρ_m is surjective along X_0, for $m \gg 0$.

This implies $\mathrm{Coker}\rho_m$ is 0 along X_0, so $(\mathrm{Coker}\rho_m)_x = 0$ for $\forall x \in X_0$, implies $\mathrm{Coker}\rho_m = 0$ along X_t for t near 0 by coherence, as we claimed.

Step 3:

Shrinking T, we just suppose that ρ_m is globally surjective for some fixed $m \gg 0$ (Why it is fixed? Because all neighborhoods we discussed are about some sufficiently large m, we can not shrinking T and moving m simultaneously), then $f_*(L^{\otimes m})$ is also globally generated since T is affine, [Har77, III, 8.5]. So we choose finitely many sections $\{s_i\}_{i=0}^r \subseteq H^0(T, f_* L^{\otimes m})$ generating $f_*(L^{\otimes m})$:

$$\mathscr{O}_T^{r+1} \longrightarrow f_*(L^{\otimes m})$$

which is surjective, pulling back on X to get an onto map

$$f^* \mathscr{O}_T^{r+1} \longrightarrow f^* f_*(L^{\otimes m})$$

then composing with ρ_m, we get

$$f^* \mathscr{O}_T^{r+1} \longrightarrow L^{\otimes m}$$

is surjective of sheaves on X. This defines a mapping

$$\phi : X \longrightarrow \mathbb{P}(\mathscr{O}_T^{r+1}) = \mathbb{P}^r \times T$$

over T, where $L^{\otimes m} \cong \phi^* \mathscr{O}_{\mathbb{P}(\mathscr{O}_T^{r+1})}(1)$, by [Har77, II, 7.12].

Since L_0 is ample on X_0, so ϕ restrict on X_0 is

$$\phi_0 := \phi|_{X_0} : X_0 \longrightarrow \mathbb{P}^r \times \{0\} \cong \mathbb{P}^r$$

then $L_0^{\otimes m} \cong (\phi_0)^* \mathscr{O}_{\mathbb{P}(\mathscr{O}_T^{r+1})}(1) \cong (\phi_0)^* \mathscr{O}_{\mathbb{P}^r}(1)$ says that ϕ_0 is fibre finite, since if $C \subseteq X_0$ such that $\phi_0(C)$ is a point, then

$$L_0^{\otimes m}|_C \cong \phi_0^* \mathscr{O}_{\mathbb{P}^r}(1)|_C = (\phi_0|_C)^* \mathscr{O}_{\mathbb{P}^r}(1)|_{\{pt\}} \cong \mathscr{O}_C$$

is just $\deg(L_0^{\otimes m}|_C) = m \cdot \deg(L_0|_C) = 0$ so $L_0|_C$ is not ample, thanks to Proposition 1.2.7, neither is L_0, which is a contradiction. Hence ϕ_0 is finite as in [GW20, 12.89].

Hence $\phi_t := \phi|_{X_t} : X_t \longrightarrow \mathbb{P}^r$ is finite for t in a neighborhood of 0 ([SP25, Tag 02OH]), and $L_t^{\otimes m} = \phi_t^* \mathscr{O}_{\mathbb{P}^r}(1)$ implies that $L_t^{\otimes m}$ is ample by Proposition 1.2.7, so is L_t. □

Example 1.2.12. (Asymptotic Riemann-Roch on Ample case) Let D be an ample Cartier divisor on an irreducible projective variety X of dimension n. Then

$$h^0\big(X, \mathscr{O}_X(mD)\big) = \frac{(D^n)}{n!} \cdot m^n + O(m^{n-1})$$

More generally, if $\mathscr{F}$ is any coherent sheaf on X, then

$$h^0\big(X, \mathscr{F} \otimes \mathscr{O}_X(mD)\big) = \operatorname{rank}(\mathscr{F}) \cdot \frac{(D^n)}{n!} \cdot m^n + O(m^{n-1})$$

This immediately follows from Serre's vanishing for ampleness.

Example 1.2.13. (Upper bounds on h^0) If E is any Cartier divisor on an irreducible projective variety X of dimension n, then there is a constant $C > 0$ such that

$$h^0\big(X, \mathscr{O}_X(mE)\big) \leq C \cdot m^n$$

for all m sufficiently large.

Proof. Fix an ample Cartier divisor D on X. Then for sufficiently large a, $aD - E$ is an effective Cartier divisor, so $\mathscr{O}_X(aD - E)$ has global section ([SP25, Tag 01X0]). Picking a non-zero section $s \in H^0(X, \mathscr{O}_X(aD - E))$, it defines

$$\begin{aligned} \mathscr{O}_X(mE) &\hookrightarrow \mathscr{O}_X(maD) \\ t &\mapsto s^m \cdot t \end{aligned}$$

which is injective since $s \neq 0$ and X is integral.

$H^0(X, -)$ is left exact:

$$H^0\big(X, \mathscr{O}_X(mE)\big) \hookrightarrow H^0\big(X, \mathscr{O}_X(maD)\big)$$

so is injective, hence

$$h^0\big(X, \mathscr{O}_X(mE)\big) \leq h^0\big(X, \mathscr{O}_X(maD)\big) = \frac{(aD)^n}{n!} \cdot m^n + O(m^{n-1}) \leq C \cdot m^n$$

□

Example 1.2.14. (Resolutions of a sheaf) Let X be a proj. variety, and D an ample Cartier divisor on X. Then any coherent $\mathscr{F}$ on X admits a resolution:

$$\cdots \to \oplus \mathscr{O}_X(-p_1 D) \to \oplus \mathscr{O}_X(-p_0 D) \to \mathscr{F} \to 0$$

for suitable integers $0 \ll p_0 \ll p_1 \ll \cdots$ (might not terminate).

Proof. (Comparing with [Har77, II, 5.18])

Choose $p_0 \gg 0$ such that $\mathscr{F} \otimes \mathscr{O}_X(p_0 D)$ is globally generated, so has surjective map:

$$\oplus \mathscr{O}_X \to \mathscr{F} \otimes \mathscr{O}_X(p_0 D) \to 0$$

Twisting by $\mathscr{O}_X(-p_0 D)$, then gives rise to an exact sequence

$$\text{Kernel} \to \oplus \mathscr{O}_X(-p_0 D) \to \mathscr{F} \to 0$$

then use the same process on the kernel: kernel$\otimes \mathscr{O}_X(p_1 D)$ is globally generated... □

1.2B Numerical Properties

The ampleness of Cartier divisors can be characterized numerically:

Theorem 1.2.15. *(Nakai-Moishezon-Kleiman criterion) Let L be a line bundle on a projective scheme X.*

Then L is ample if and only if

$$\int_V c_1(L)^{\dim V} = (L^{\dim V} \cdot V) > 0$$

for every positive-dimensional irreducible subvariety $V \subseteq X$

In particular, D is ample if and only if $(D^{\dim V} \cdot V) > 0$.

Before giving a proof, we use this consequence to derive two important corollaries. First is of that the ampleness of a Cartier divisor depends only on its numerical equivalence class:

Corollary 1.2.16. (Numerical nature of ampleness) *If $D_1, D_2 \in Div(X)$ are numerically equivalent Cartier divisors on a projective variety or scheme X, then*

$$D_1 \text{ is ample if and only if } D_2 \text{ is.}$$

We need a lemma to prove this corollary.

Lemma 1.2.17. *Let X be a complete variety or scheme, and let*

$$D_1, \ldots, D_k, D_1', \ldots, D_k' \in Div(X)$$

be Cartier divisors on X. If $D_i \equiv_{num} D_i'$ for each i, then

$$(D_1 \cdot D_2 \cdots D_k \cdot V) = (D_1' \cdot D_2' \cdots D_k' \cdot V)$$

for every irreducible subscheme $V \subseteq X$ of dimension k

Proof. We assert first that $E \equiv_{num} 0$ is a numerically trivial Cartier divisor on X, then as $(D_2 \cdots D_k \cdot V)$ represented (in Chow ring) by a one-cycle γ on X where $1 = \dim V - (k-1)$.

So $(E \cdot D_2 \cdots D_k \cdot V) = (E \cdot \gamma) = 0$. By multilinear, we get that

$$(D_1 \cdot D_2 \cdots D_k \cdot V) = (D_1' \cdot D_2 \cdots D_k \cdot V)$$

as $D_1 \equiv_{num} D_1'$, hence the lemma follows by induction on k. □

Now we begin to prove the Corollary 2.16.

Proof. Due to the lemma, we get that

$$(D_1^{\dim V} \cdot V) = (D_2^{\dim V} \cdot V)$$

as strictly larger than 0 simultaneously. □

Definition 1.2.18. (Ample classes) A numerical class $\delta \in N^1(X)$ is *ample* if it is the class of an ample Cartier divisor.

Example 1.2.19. (Varieties with Picard number 1) If X is a projective variety having Picard number $\rho(X) = 1$, i.e. $N^1(X) \cong \mathbb{Z}$, then any non-zero effective Cartier divisor on X is ample.

Here we discuss a corollary in Lazarsfled's book, that has a little problem about the construction of a finite preimage. Here I think the condition with f^*L is ample is too strong automatically ensure that the map f is finite, so when Lazarsfled writes down the step about "cutting down the general hyperlanes" he might thought of the case when f is just surjective. To show respect, we still write down the full details here.

Corollary 1.2.20. (Finite Pullbacks, II) *Let $f : Y \to X$ be a (finite) surjective morphism of projective schemes, and L is a line bundle on X. Then:*

$$f^*L \text{ ample on } Y \Longrightarrow L \text{ is ample on } X.$$

Proof. Let $V \subseteq X$ be an irreducible variety. Since f is surjective, there is an irreducible variety $W \subseteq Y$ mapping finitely onto V.

We can construct W as follows:

- f is surjective so it must have a generic point in Y that is mapped to the generic point of V, we pick this one, denote it as η.
- Taking closure $\overline{\{\eta\}}$ gives the irreducible component $W \subseteq f^{-1}(V)$.
- But this map $f|_W : W \to V$ might not be finite, i.e. has infinite fibre. So need cutting down the general hyperplanes to ensure quasi-finite.

  So we get a new map $f|_W : W \to V$ for which we using the same notation.
- Deligne's result ([GW20, 12.80]) about proper and quasi-finite is equivalent to finite is what we want for $f : W \to V$.

Using the projection formula:

$$\int_W c_1(f^*L)^{\dim W} = \deg f \cdot \int_V c_1(L)^{\dim V} > 0$$

where $\deg f = [k(W) : k(V)]$. □

Now we return to proving the **NMK criterion**:

Proof. "⇒":

Suppose first that L is ample. Then $L^{\otimes m}$ is very ample for some $m \gg 0$:

$$\int_V c_1(L^{\otimes m})^{\dim V} = m^{\dim V} \cdot \int_V c_1(L)^{\dim V} > 0$$

since the left term is the $\deg V$ corresponding to projective embedding of X via $L^{\otimes m}$, as you can see [KM98, 1.35(3)].

"⇐":

Assuming that intersection number is strictly positive. Since ampleness is stable under finite pullbacks - so as on reduced scheme structure - we can assume X is irreducible and reduced, i.e. a projective variety. By and large, we can associate L to a Cartier divisor D so we write $L = \mathscr{O}_X(D)$, as [Har77, II, 6.15].

Now we use the induction on $\dim X = n$.

- $n = 1$: X is an irreducible curve, then
$$\int_X c_1(L) = \deg\big(\mathscr{O}_X(D)|_C\big) > 0$$
implies L is ample by [Har77, IV, 3.3].
- Assuming it holds for dimension $\leq n_1$.
- Now for dim $X = n$:

  Step 1:
  Asserting first that
$$H^0\big(X, \mathscr{O}_X(mD)\big) \neq 0 \text{ for } m \gg 0.$$

In fact, asymptotic Riemann-Roch gives that
$$\chi(X, mD) = \frac{(D^n)}{n!} \cdot m^n + O(m^{n-1})$$
and $(D^n) = \int_X c_1(L)^n > 0$ by our condition.

Now picking H as a hyperplane section of X which is ample. So can set: $A = mH + D$, $B = mH$ for $m \gg 0$, both are very ample by example 1.2.5. By cancelling out the poles on A using $\mathrm{div}(f)$, $f \in k(X)^*$ we can write $D \sim_{\mathrm{lin}} A - B$ as difference of very ample effective Cartier divisors. Or just see Lemma 4.1.13.

We derive two exact sequences:
$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathscr{O}_X(mD - B) & \xrightarrow{\cdot A} & \mathscr{O}_X\big((m+1)D\big) & \longrightarrow & \mathscr{O}_A\big((m+1)\big) & \longrightarrow & 0 \\
 & & \| & & & & & & \\
0 & \longrightarrow & \mathscr{O}_X(mD - B) & \xrightarrow{\cdot B} & \mathscr{O}_X(mD) & \longrightarrow & \mathscr{O}_B(mD) & \longrightarrow & 0
\end{array}$$
where the associated schemes of A and B are have strictly smaller dimension than X since they're both effective, so by induction hypothesis, $\mathscr{O}_A(D)$ and $\mathscr{O}_B(D)$ are ample.

Consequently the higher cohomology of each of the two schemes on the two right terms vanishes when $m \gg 0$. Using the left term as a bridge, we have:
$$H^i(X, mD) = H^i(X, mD - B) = H^i\big(X, (m+1)D\big)$$
for $i \geq 2$. In other words, for $i \geq 2$, the dimension $h^i(X, mD)$ are eventually constant as m growing.

Therefore
$$\chi(X, mD) = h^0(X, mD) - h^1(X, mD) + C$$
for some non-zero constant C and $m \gg 0$. So if $H^0 = 0$, then the coefficient of m^n is negative, contradicting $(D^n) > 0$, we proved what we asserted.

Since D is ample if and only if mD is, then without loss of generality, we replace D by mD, which is effective by our assertion, so we henceforth suppose that D is effective.

Step 2:

Next show that $\mathscr{O}_X(mD)$ is globally generated for sufficiently large m.

Since D is assumed to be effective, we do not care about the points away from the associated subscheme, i.e. Supp$D$, so the issue is to show that no point of the associated subscheme - we also denote it by D - is a base point of the linear series $|mD|$.

To this end, consider the exact sequence:

$$0 \longrightarrow \mathscr{O}_X\big((m-1)D\big) \xrightarrow{\cdot D} \mathscr{O}_X(mD) \longrightarrow \mathscr{O}_D(mD) \longrightarrow 0$$

Since D has strictly smaller dimension than X, $\mathscr{O}_D(D)$ is ample by induction hypothesis. So $\mathscr{O}_D(mD)$ is globally generated by very ampleness and $H^1(X, mD) = 0$ for $m \gg 0$ by Serre's vanishing. Then we pull the long exact sequence to get the natural homomorphism

$$H^1\big(X, (m-1)D\big) \longrightarrow H^1(X, mD)$$

is surjective for $m \gg 0$.

By coherence, $\dim_{\mathbb{C}} H^1(X, mD)$ is finite, and is a number sequence varying as m growing, must form a non-increasing sequence - as surjectivity might "cut down" some basis - which is eventually stationary. Now since a surjection between vector spaces of the same dimension must be an isomorphism, the restriction mapping

$$H^0(X, mD) \longrightarrow H^0\big(X, \mathscr{O}_D(mD)\big) = H^0(D, mD)$$

is surjective for $m \gg 0$.

But since $\mathscr{O}_D(mD)$ is globally generated, by surjectivity sections in left can't be vanish simultaneously, i.e. no point of Supp$D$ is a base point of $|mD|$, as we get that $|mD|$ is a complete linear series.

Step 3:

Using Corollary 2.9 and assumption that

$$\int_C c_1\big(\mathscr{O}_X(mD)\big) = (mD \cdot C) > 0$$

for every irreducible curve $C \subseteq X$, get that $\mathscr{O}_X(mD)$ is ample, so is $\mathscr{O}_X(D)$. □

Remark 1.2.21. (Nakai's criterion on proper scheme) The statement of theorem remains true for any complete scheme X, without assuming that X is projective, of

which we use to express a Cartier divisor D as a difference of two very ample Cartier divisors.

See [Deb01, 1.5] or [KM98, 1.37] the **proper** case specific to Cartier divisor:

Theorem 1.2.22. (Nakai-Moishezon Criterion) *A Cartier divisor D on a proper scheme X is ample if and only if for every (closed) integral subscheme $Y \subseteq X$, one has*

$$(D^{\dim Y} \cdot Y) = (D|_Y^{\dim Y}) > 0$$

Proof. "⇒":

If D is ample, then mD is very ample for some $m > 0$, hence defines an embedding $f : X \hookrightarrow \mathbb{P}^N$ such that $f^*\mathscr{O}_{\mathbb{P}^N}(1) \cong \mathscr{O}_X(mD)$.

Then the self-intersection of this very ample Cartier divisor is the degree of this subscheme under the embedding:

$$\big((mD)^{\dim Y} \cdot Y\big) = \big((mD|_Y)^{\dim Y}\big) = \deg\big(f(Y)\big) > 0$$

that is

$$m^{\dim Y} \cdot (D|_Y^{\dim Y}) > 0 \Longrightarrow (D|_Y^{\dim Y}) > 0.$$

"⇐":

Let D be a Cartier divisor such that $(D|_Y^{\dim Y}) > 0$, where $Y \subseteq X$ is the arbitrary integral subscheme. We will use the induction on $\dim X$.

Since ampleness is stable under finite pullbacks, it is enough assuming that X is an integral proper scheme.

Choosing an embedding $\mathscr{O}_X(D) \hookrightarrow \mathscr{K}_X$, and set

$$\mathscr{I}_1 = \mathscr{O}_X(-D) \cap \mathscr{O}_X \quad \text{and} \quad \mathscr{I}_2 = \mathscr{O}_X(D) \cap \mathscr{O}_X$$

and let Y_j be the subscheme of X defined by $\mathscr{I}_j$, which have strictly smaller dimension than X, so by induction hypothesis, D is ample on Y_1 and Y_2.

Consider the exact sequence:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathscr{I}_1(mD) & \longrightarrow & \mathscr{O}_X(mD) & \longrightarrow & \mathscr{O}_{Y_1}(mD) & \longrightarrow & 0 \\ & & \| & & & & & & \\ 0 & \longrightarrow & \mathscr{I}_2\big((m-1)D\big) & \longrightarrow & \mathscr{O}_X\big((m-1)D\big) & \longrightarrow & \mathscr{O}_{Y_2}\big((m-1)D\big) & \longrightarrow & 0 \end{array}$$

Notice that by Serre's vanishing, $H^i(Y_j, mD) = 0$ for all $i > 0$ and all m sufficiently large, it follows that for $i \geq 2$:

$$h^i(X, mD) = h^i\big(X, \mathscr{I}_1(mD)\big) = h^i\Big(X, \mathscr{I}_2\big((m-1)D\big)\Big) = h^i\big(X, (m-1)D\big)$$

for m sufficiently large. Since we assume that $(D^{\dim X}) > 0$, by asymptotic Riemann-Roch, $\chi(X, mD)$ goes to infinity as m growing. It follows that $h^0(X, mD) - h^1(X, mD)$ hence also $h^0(X, mD)$ goes to infinity as m grows (the other terms are eventually stationary has no influence), so $\mathscr{O}_X(mD)$ has global sections which implies that mD is effective, [SP25, Tag 01X0].

D is ample if and only if mD is, for some $m > 0$, so we may replacing D by mD therefore assmuing D is effective.

The exact sequence

$$0 \longrightarrow \mathscr{O}_X\big((m-1)D\big) \longrightarrow \mathscr{O}_X(mD) \longrightarrow \mathscr{O}_D(mD) \longrightarrow 0$$

where the associated subscheme D has strictly smaller dimension than X therefore $\mathscr{O}_D(mD)$ is ample so $H^1(D, mD) = 0$ for sufficiently large m. This yields a surjection

$$\rho_m : H^1\big(X, (m-1)D\big) \longrightarrow H^1(X, mD)$$

The dimension $h^1(X, mD)$ forms a non-increasing sequence which is eventually stationary, in which case ρ_m becoms an isomorphism, so the restriction

$$H^0(X, mD) \longrightarrow H^0(D, mD)$$

is surjective.

D is ample on D hence $\mathscr{O}_D(mD)$ is globally generated i.e. $|mD|$ is base point free on D, so define a proper morphism $\phi : X \longrightarrow \mathbb{P}^r$ which must fibre finite because if $C \subseteq X$ a curve such that $\phi(C)$ is a point then

$$\mathscr{O}_X(mD)|_C \cong \phi^*\mathscr{O}(1)|_C = (\phi|_C)^*\,\mathscr{O}(1)|_{\{pt\}} \cong \mathscr{O}_C$$

which is just $(D \cdot C) = 0$ contradicting with our assumption. Now ϕ is finite and $\mathscr{O}_X(mD) = \phi^*\mathscr{O}(1)$ enough to see mD is ample on X, so the D. □

Example 1.2.23. (Further characterization of ampleness) Follows from [Kle66].

X is a projective scheme and D is Cartier divisor on X. Then D is ample if and only if for every irreducible subvariety $V \subseteq X$ of positive dimension, $\exists m = m(V) \in \mathbb{Z}_{>0}$ together a non-zero section $s_V \in H^0(V, \mathscr{O}_V(mD))$ such that s_V vanishes at some point of V.

Proof. One direction is trivial. Now prove the converse:

It is enough to assume X is integral. As $\mathscr{O}_V(mD)$ has a global section, mD is effective on V see [SP25, Tag 01X0]. Taking first $V = X$ and replacing D by mD, we can assume D is effective.

By induction on dimension, $\mathscr{O}_D(D)$ is ample where D is the associated subscheme. So $\mathscr{O}_X(mD)$ is base point free for same reason for m sufficiently large, that is, $|mD|$ is complete.

As the morphism $\phi : X \longrightarrow \mathbb{P}H^0(X, mD)$ used in Proposition 2.9, by induction hypothesis, D is ample on curve $C \subseteq X$ so $\mathscr{O}_X(D) \cdot C > 0$ implies that $\mathscr{O}_X(D)$ is ample. □

Example 1.2.24. (Growth of cohomology) Let X is a projective scheme of dimension n and D a Cartier divisor on X. If $\mathscr{F}$ is any coherent sheaf on X then

$$h^i\big(X, \mathscr{F}(mD)\big) = O(m^n) \text{ for every } i.$$

Proof. Write $D = A - B$ as the difference of very ample Cartier divisors having the property that neither A nor B contains any of the subvarieties of X defined by the associated primes of $\mathscr{F}$.

Consider two sequences:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathscr{F}(mD - B) & \xrightarrow{\cdot A} & \mathscr{F}((m+1)D) & \longrightarrow & \mathscr{F} \otimes \mathscr{O}_A((m+1)D) & \longrightarrow & 0 \\ & & \| & & & & & & \\ 0 & \longrightarrow & \mathscr{F}(mD - B) & \xrightarrow{\cdot B} & \mathscr{F}(mD) & \longrightarrow & \mathscr{F} \otimes \mathscr{O}_B(mD) & \longrightarrow & 0 \end{array}$$

are exact by what we setting that $\cdot A$ and $\cdot B$ are indeed injective.

So by induction on dimension n:

- $n = 0$: higher cohomologies are trivial so $h^i = O(1)$.
- Suppose it holds for all dimension $< n$.
- Now for n:

$$\begin{aligned} -O(m^{n-1}) =& - h^{i-1}\Big(A, \mathscr{F}\big((m+1)D\big)|_A\Big) \\ \leq& h^i\Big(X, \mathscr{F}\big((m+1)D\big)\Big) - h^i\big(X, \mathscr{F}(mD - B)\big) \\ \leq& h^i\Big(A, \mathscr{F}\big((m+1)D\big)|_A\Big) = O(m^{n-1}) \\ -O(m^{n-1}) =& - h^{i-1}\big(B, \mathscr{F}(mD)|_B\big) \\ \leq& h^i\big(X, \mathscr{F}(mD)\big) - h^i\big(X, \mathscr{F}(mD - B)\big) \\ \leq& h^i\big(B, \mathscr{F}(mD)|_B\big) = O(m^{n-1}), \end{aligned}$$

so taking sum together, we get:

$$\Big|h^i\Big(X, \mathscr{F}\big((m+1)D\big)\Big) - h^i\big(X, \mathscr{F}(mD)\big)\Big| = O(m^{n-1})$$

that is $h^i\big(X, \mathscr{F}(mD)\big) = O(m^n)$.

□

1.3 $\mathbb{Q}$-Cartier divisors and $\mathbb{R}$-Cartier divisors

Goal:

Give a "small pertubation" on Cartier divisor classes to see what happens.

1.3A Definition of $\mathbb{Q}$-Cartier divisors.

A $\mathbb{Q}$-Cartier divisor is simply a $\mathbb{Q}$-linear combination of integral Cartier divisors.

Definition 1.3.1. ($\mathbb{Q}$-Cartier divisors) Let X be a variety or scheme.

A $\mathbb{Q}$-*Cartier divisor* on X is an element of the $\mathbb{Q}$-vector space

$$\mathrm{Div}_{\mathbb{Q}}(X) := \mathrm{Div}(X) \underset{\mathbb{Z}}{\otimes} \mathbb{Q}$$

We represent a $\mathbb{Q}$-Cartier divisor $D \in \mathrm{Div}_{\mathbb{Q}}(X)$ as a finite sum

$$D = \sum c_i \cdot A_i$$

where $c_i \in \mathbb{Q}$ and $A_i \in \mathrm{Div}(X)$.

By clearing denominators, can write $D = c \cdot A$ for $c \in \mathbb{Q}$, $A \in \mathrm{Div}(X)$ which we call an *integral Cartier divisor*, and if $c \neq 0$ then $cA = 0$ if and only if A is a torsion element of $\mathrm{Div}(X)$.

The $\mathbb{Q}$-Cartier divisor D is *effective* if it is of the form $D = \sum c_i A_i$ with $c_i \geq 0$ and A_i are effective integral Cartier divisors.

Definition 1.3.2. (Supports) Let $D \in \mathrm{Div}_{\mathbb{Q}}(X)$ be a $\mathbb{Q}$-Cartier divisor. A codimension 1 subset $E \subseteq X$ *supports* D, or is a *support* of D, if D is of the form $\sum c_i A_i$ in which $\bigcup_i \mathrm{Supp} A_i \subseteq E$.

Notice that the expression $D = \sum c_i A_i$ may not be unique, hence E is canonically determined. But this is always fine anyway.

All the usual operations and properties of Cartier divisors can be extended naturally to $\mathbb{Q}$-Cartier divisors.

Definition 1.3.3. (Equivalences and operations on $\mathbb{Q}$-Cartier divisors) Assume that X is complete.

(i) Given a subvariety or subscheme $V \subseteq X$ of dimension k, has a $\mathbb{Q}$-valued intersection product:

$$\begin{aligned} \mathrm{Div}_{\mathbb{Q}}(X) \times \cdots \times \mathrm{Div}_{\mathbb{Q}}(X) &\longrightarrow \mathbb{Q} \\ (D_1, \ldots, D_k) &\longmapsto (D_1 \cdots D_k \cdot V) = \int_V D_1 \cdots D_k \end{aligned}$$

is defined via extension of scalars from the analogous product on $\mathrm{Div}(X)$.

(ii) Two $\mathbb{Q}$-Cartier divisors D_1, $D_2 \in \mathrm{Div}_{\mathbb{Q}}(X)$ are *numerically equivalent*, written

$$D_1 \equiv_{num} D_2 \text{ or } D_1 \equiv_{num,\mathbb{Q}} D_2$$

if

$$(D_1 \cdot C) = (D_2 \cdot C)$$

for every curve $C \subseteq X$. We denote its equivalence classes, which form a finite dimensional vector space, by $N^1_{\mathbb{Q}}(X)$.

(iii) Two $\mathbb{Q}$-Cartier divisors D_1, $D_2 \in \mathrm{Div}_{\mathbb{Q}}(X)$ are *linearly equivalent*, written

$$D_1 \sim_{\mathrm{lin}} D_2 \text{ or } D_1 \sim_{\mathbb{Q}} D_2$$

if there is an integer r such that rD_1 and rD_2 are integral and linearly equivalent in the usual sense, i.e. $r(D_1 - D_2) \in$ image of $\mathrm{Div}(X) \longrightarrow \mathrm{Div}_{\mathbb{Q}}(X)$.

(iv) If $f : Y \longrightarrow X$ is a morphism such that the image of every associated subvariety of Y meets $\mathrm{Supp}D$ for $D \in \mathrm{Div}_{\mathbb{Q}}(X)$ properly, so we can define $f^*D \in \mathrm{Div}_{\mathbb{Q}}(Y)$ by extension of scalar from the corresponding pullback on integral Cartier divisors. See Remark 1.1.6.

(v) If $f : Y \longrightarrow X$ is arbitrary morphism of complete schemes, extension by scalars gives rise to a functorial induced homomorphism

$$f^* : N^1(X)_{\mathbb{Q}} \longrightarrow N^1(Y)_{\mathbb{Q}}$$

compatible with the Cartier divisor-level pullback.

Remark 1.3.4. More concretely, these operations and equivalences are determined from the writing Cartier divisors $D = \sum c_i A_i$ - or, after clearing denominators, $D = cA$ - and expanding by linearity.

So for instance $D \equiv_{num,\mathbb{Q}} 0$ if and only if $\sum c_i (A_i \cdot C) = 0$ for every curve $C \subseteq X$.

Note also that there is an isomorphism

$$N^1(X)_{\mathbb{Q}} \cong N^1(X) \underset{\mathbb{Z}}{\otimes} \mathbb{Q}$$

i.e. $\mathrm{Div}_{\mathbb{Q}}(X)/\mathrm{Num}_{\mathbb{Q}}(X) \cong (\mathrm{Div}(X)/\mathrm{Num}(X)) \underset{\mathbb{Z}}{\otimes} \mathbb{Q}$, which follows from that a numerically trivial $\mathbb{Q}$-Cartier divisor is a $\mathbb{Q}$-linear combination of numerically trivial integral Cartier divisors.

Remark 1.3.5. It can happen that two integral Cartier divisors in the distinct linear equivalence classes might be linearly equivalent under the sense of (iii). So be careful when dealing with linear equivalence. But the numerical equivalence is always fine.

Continue to assume X is complete which is an indispensable condition in Nakai's criterion.

Definition 1.3.6. (Ampleness for $\mathbb{Q}$-Cartier divisors) A $\mathbb{Q}$-Cartier divisor

$$D \in \mathrm{Div}_{\mathbb{Q}}(X)$$

is *ample* if the following equivalent conditions hold:

(i) D is of the form $D = \sum c_i A_i$ where $c_i \in \mathbb{Q}_{>0}$ and A_i is an ample Cartier divisor.

(ii) There is a $r \in \mathbb{Z}_{>0}$ such that $r \cdot D$ is integral and ample.

(iii) D satisfies the statement of Nakai's criterion, i.e.

$$(D^{\dim V} \cdot V) > 0$$

for every irreducible subvariety $V \subseteq X$ of positive dimension.

Via Nakai's criterion, ampleness is presented by numerical equivalence, and we can speak of ample classes in $N^1(X)_{\mathbb{Q}}$.

Now we prove that ampleness is an open condition under small perturbations of a Cartier divisor:

Proposition 1.3.7. *Let X be a projective variety. H an ample $\mathbb{Q}$-Cartier divisor on X, and E an arbitrary $\mathbb{Q}$-Cartier divisor. Then $H + \varepsilon E$ is ample for all sufficiently small rational numbers $0 \leq |\varepsilon| \ll 1$.*

More generally, given finitely many $\mathbb{Q}$-Cartier divisors $E_1, \ldots, E_r$ on X,

$$H + \varepsilon_1 E_1 + \cdots + \varepsilon_r E_r$$

is ample for all sufficiently small rational numbers $0 \leq |\varepsilon_i| \ll 1$.

Proof. By clearing denominators, we may assume H and each E_i are integral, so as in Proposition 2.10, we can reduce to the integral scheme case; this is why we assume X is a projective variety.

By taking $m \gg 0$, as in Example 2.5, we can arrange for each of the $2r$ Cartier divisors $mH \pm E_1, \ldots, mH \pm E_r$ to be ample.

Now provided that $|\varepsilon_i| \ll 1$, any Cartier divisor of the form

$$H + \varepsilon_1 E_1 + \cdots + \varepsilon_r E_r$$

can be written as a positive $\mathbb{Q}$-linear combination of H and some of the $\mathbb{Q}$-Cartier divisors $H \pm \frac{1}{m}E_i = \frac{1}{m}(mH \pm E_i)$ which is ample, for example:

$$H + \varepsilon_1 E_1 + \cdots + \varepsilon_r E_r = \frac{1}{r+1}H + \sum_{i=1}^{r} \frac{1}{r+1}(H \pm \frac{1}{m}E_i)$$

But a positive linear combination of ample $\mathbb{Q}$-Cartier divisors is ample, by definition. □

Remark 1.3.8. (Weil $\mathbb{Q}$-divisors) As in definition 1.1.3, we write $\mathrm{WDiv}(X)$ be the additive group of Weil divisors on an irreducible variety X.

The group of Weil $\mathbb{Q}$-divisors is defined to be

$$\mathrm{WDiv}_{\mathbb{Q}}(X) := \mathrm{WDiv}(X) \underset{\mathbb{Z}}{\otimes} \mathbb{Q}$$

So a Weil $\mathbb{Q}$-divisor is just a $\mathbb{Q}$-linear combination of codimension 1 subvarieties.

As before, there is a cycle class map:

$$\begin{aligned} [\quad] : \mathrm{Div}_{\mathbb{Q}}(X) &\longrightarrow \mathrm{WDiv}_{\mathbb{Q}}(X) \\ D &\longmapsto [D] = \sum \mathrm{ord}_V(D) \cdot [V] \end{aligned}$$

where $\mathrm{ord}_V(D) = v_V(D) \in \mathbb{Q}$.

If we assuming that X is normal, then cycle map is injective, in this case can identify $\mathrm{Div}_{\mathbb{Q}}(X)$ as a subgroup of $\mathrm{WDiv}_{\mathbb{Q}}(X)$.

A Weil $\mathbb{Q}$-divisor $E \in \mathrm{WDiv}_{\mathbb{Q}(X)}$ is said to be $\mathbb{Q}$-*Cartier* if it lies in $\mathrm{Div}_{\mathbb{Q}}(X)$. So if X is normal, all of operations and equivalences we defined above can make sense for $\mathbb{Q}$-Cartier Weil $\mathbb{Q}$-divisors.

Example 1.3.9. (Fails when X is not normal) Consider the quadric cone $X \subseteq \mathbb{P}^3$ with vertex O, which is not normal. Let $E \subseteq X$ be a line through O. It is a Weil divisor, but not Cartier, since it is not locally principal.

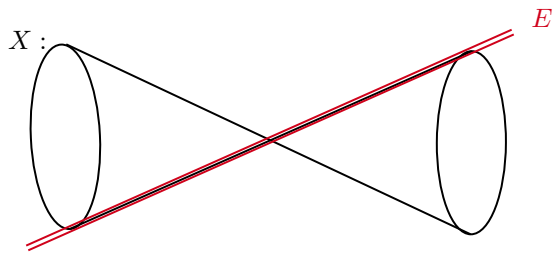


Figure 1.10.: Quadric Cone in $\mathbb{P}^3$

But $2 \cdot E$ is, see [Har77, II, e.g. 6.11.3].

So if A is the Cartier divisor obtained by intersecting X with the hyperplane in $\mathbb{P}^3$ tangent to X along E, we get $A = 2 \cdot E$, which implies E is $\mathbb{Q}$-Cartier, and

$$\begin{aligned} E \cdot E &= (\frac{1}{2}A \cdot \frac{1}{2}A) \\ &= \frac{1}{4}(A \cdot A) \\ &= \frac{1}{4} \cdot 2 \text{ since } A \text{ is defined by linear form} \\ &= \frac{1}{2} \end{aligned}$$

1.3B $\mathbb{R}$-Cartier divisors and Their ampleness

Definition 1.3.10. ($\mathbb{R}$-Cartier divisors) Let X be an algebraic variety or scheme.

An $\mathbb{R}$-Cartier divisor on X is an element of the $\mathbb{R}$-vector space

$$\mathrm{Div}_{\mathbb{R}}(X) = \mathrm{Div}(X) \underset{\mathbb{Z}}{\otimes} \mathbb{R}.$$

We represent a $\mathbb{R}$-Cartier divisor $D \in \mathrm{Div}_{\mathbb{R}}(X)$ as a finite sum

$$D = \sum c_i \cdot A_i$$

where $c_i \in \mathbb{R}$, $A_i \in \mathrm{Div}(X)$. Write $D_1 \sim_{\mathbb{R}} D_2$ if $D_1 - D_2 \in \sum \mathbb{R}(\text{principal divisor})$.

- Suppose X is complete, then the intersection theory is $\mathbb{R}$-valued depends on numerical equivalence.
- It is *numerically trivial* if and only if $\sum c_i(A_i \cdot C) = 0$ for every curve $C \subseteq X$.
- It is *effective* if $D = \sum c_i A_i$ with $c_i \geq 0$ and A_i effective.
- *Pullbacks* and *supports* are defined as before.

Example 1.3.11. One has the isomorphism

$$N^1(X)_{\mathbb{R}}{}^{1} = N^1(X) \underset{\mathbb{Z}}{\otimes} \mathbb{R}$$

which follows from the topological fact that a numerically trivial $\mathbb{R}$-Cartier divisor is an $\mathbb{R}$-linear combination of numerically trivial integral Cartier divisors.

[1] Sometimes we will denote it by $N^1(X)$ if there is no confuse of notation.

Definition 1.3.12. (Ampleness for $\mathbb{R}$-Cartier divisors) Assume that X is complete. An $\mathbb{R}$-Cartier divisor is *ample* if it can be expressed as a finite sum

$$D = \sum c_i \cdot A_i$$

where $c_i \in \mathbb{R}_{>0}$ and A_i is an ample Cartier divisor.

Remark 1.3.13. (Nakai inequalities for $\mathbb{R}$-Cartier divisors) If D is an ample $\mathbb{R}$-Cartier divisor then certainly

$$(D^{\dim V} \cdot V) > 0$$

for every irreducible $V \subseteq X$ of positive dimension. The converse is not that clear because clearing the denominators is unavailable, but it is indeed true, can see the theorem of Campana and Peternell in [Laz04a, 2.3.18], or the original paper [CP90].

Similar as before, ampleness depends only on numerical equivalence classes:

Proposition 1.3.14. (Ample class for $\mathbb{R}$-Cartier divisors) *The ampleness of an $\mathbb{R}$-Cartier divisor depends only on its numerical equivalence class.*

Proof. To show an $\mathbb{R}$-Cartier divisor D being ample implies its class is ample in $N^1(X)_{\mathbb{R}}$, it sufficient to show that for $B \equiv_{num} 0$, then $D + B$ is again ample.

We first admit a fact that B is an $\mathbb{R}$-linear combination of numerically trivial integral Cartier divisors.

So can reduce the case for showing if A and B are integral Cartier divisors, with A ample and $B \equiv_{num} 0$, then $A + rB$ is ample for $\forall r \in \mathbb{R}$.

- For r is rational: we already know this by Nakai-Moishezon.
- For r in general: we can fix two rational numbers

  $$r_1 < r < r_2,\ r_i \in \mathbb{Q}$$

  together with a real number $t \in [0, 1]$, such that

  $$r = tr_1 + (1 - t)r_2.$$

  Then

  $$A + rB = t(A + r_1B) + (1 - t)(A + r_2B)$$

  in which $A + r_iB$ is ample by rational case, thus exhibiting the ampleness of $A + rB$.

□

Next we show that the ampleness is open under the usual topology in $\mathbb{R}$-vector space $N^1(X)_{\mathbb{R}}$.

Example 1.3.15. (Openness of ampleness for $\mathbb{R}$-Cartier divisors) Let X be a projective variety and H an ample $\mathbb{R}$-Cartier divisor on X. Given finitely many $\mathbb{R}$-Cartier divisors $E_1, \ldots, E_r$, the $\mathbb{R}$-Cartier divisor

$$H + \varepsilon_1 E_1 + \cdots + \varepsilon_r E_r$$

is ample for all sufficiently small real numbers $0 \leq |\varepsilon_i| \ll 1$.

Proof. The case of each H and E_i are rational has been proven. So we turn to the general case.

To begin with, since each E_i is a finite $\mathbb{R}$-linear combination of integral Cartier divisors, so without loss of generality, we can assume all the E_i are integral.

Now write $H = \sum c_j A_j$ with $c_j > 0$ and A_j ample and integral. We fix a rational number $0 < c < c_1$, then

$$H + \sum \varepsilon_i E_i = c(A_1 + \sum \varepsilon_i E_i) + (c_1 - c)A_1 + \sum_{j \geq 2} c_j A_j$$

Since we just need all ε_i to be sufficiently small to exhibit the openness, we can assume they are all rational. The first term on the right is governed by the case of $\mathbb{Q}$-Cartier divisors, and remaining summands are indeed ample. □

Corollary 1.3.16. *If X is projective, then the finite-dimensional vector space $N^1(X)_{\mathbb{R}}$ is spanned by the classes of ample Cartier divisors on X.*

Proof. By the nature of openness, the ample Cartier divisors form a non-empty open subset in this finite-dimensional vector space, which therefore contains a basis. □

1.4 Nef line bundle and Cartier divisors

We have seen that if X is a projective variety, then a class $\delta \in N^1(X)_{\mathbb{Q}}$ is ample if and only if it satisfies the Nakai inequalities:

$$(\delta^{\dim V} \cdot V) > 0$$

for all irreducible subvarieties $V \subseteq X$ of positive dimension.

So we want this inequality to be weaker:

$$(\delta^{\dim V} \cdot V) \geq 0$$

for all $V \subseteq X$, this is related to the nef condition and Kleiman's theorem, it suffices to test when V is just a curve.

1.4A Definitions and Formal Properties

Definition 1.4.1. (Nef line bundles and Cartier divisors) Let X be a complete variety or scheme.

A line bundle L on X is *numerically effective*, or *nef*, if

$$\int_C c_1(L) \geq 0$$

for every irreducible curve $C \subseteq X$.

Similarly, a Cartier divisor D on X (arbitrary coefficients) is *nef* if

$$(D \cdot C) \geq 0$$

for every irreducible curve $C \subseteq X$.

Since the intersection with a curve depends only on numerical equivalence classes, the definition can be extended to nef classes in $N^1(X)$ with arbitrary coefficients. Note that any ample class is nef.

Remark 1.4.2. (Chow's lemma) In dealing with nefness, we often use Chow's lemma to reduce statements about complete varieties or schemes to the projective case:

Suppose X is a complete variety or scheme, then there exists a projective variety or scheme X', together with a surjective morphism

$$f : X' \longrightarrow X$$

such that $f^{-1}(U) \longrightarrow U$ is an isomorphism for $U \subseteq X$ open dense.

This lemma we will often use this lemma since over a proper scheme, the existence of the ample Cartier divisor is a question, but if we run this lemma to reduce the case over projective scheme, then it is easy to find an ample Cartier divisor.

We admit these formal properties of nefness, which follow from [Kle66]:

Example 1.4.3. (Formal properties of nefness) Let X be a complete variety or scheme, L a line bundle on X.

(i) Let $f : Y \longrightarrow X$ be a proper morphism. If L is nef, then f^*L is a nef line bundle on Y.

In particular, as a closed immersion is proper, restrictions of a nef line bundle to subschemes remain nef.

(ii) Moreover if f is surjective and f^*L is nef on Y, then L itself is nef.

(iii) L is nef if and only if L_{red} is nef on X_{red}.

(iv) L is nef if and only if its restriction on each irreducible component of X is nef.

Example 1.4.4. Let X be a complete variety or scheme, and L a globally generated line bundle on X. Then L is nef.

Proof. As [Har77, III, 8.8.1] says that cohomologies on proper scheme has coherence, i.e. $H^0(X, L)$ still a finite $\mathbb{C}$-module, so due to globally generated:

$$\phi_{|L|} : X \longrightarrow \mathbb{P}H^0(X, L)$$

is a morphism such that $L \cong \phi_{|L|}^* \mathscr{O}(1)$.

So by projection formula:

$$(L \cdot C) = (\phi_{|L|}^* \mathscr{O}(1) \cdot C) = (\mathscr{O}(1) \cdot \phi_{|L|^*} C) \geq 0$$

implies L is nef. □

Example 1.4.5. (Cartier divisor with nef normal bundle) Let X be a complete variety, and $D \subseteq X$ an effective Cartier divisor on X. If the normal bundle $\mathscr{N}_{D/X} = \mathscr{O}_D(D)$ to D in X is nef, then D itself is a nef Cartier divisor.

Proof.

$$\int_C c_1\big(\mathscr{O}_D(D)\big) = \deg\big(\mathscr{O}_D(D)|_C\big) = \deg \mathscr{O}_C(D) = (D \cdot C) \geq 0$$

for every irreducible curve $C \subseteq D$ after normalization. For irreducible curves C not contained in D, we automatically have $(D \cdot C) \geq 0$, so D is nef. □

Notice that in particular for X is a surface and $C \subseteq X$ an irreducible curve with $(C^2) \geq 0$, then C is nef.

1.4B Kleiman's Theorem

We will discuss the fundamental result due to [Kle66].

Theorem 1.4.6. (Kleiman's theorem) *Let X be a complete variety or scheme. If D is a nef $\mathbb{R}$-Cartier divisor on X, then*

$$(D^k \cdot V) \geq 0$$

for every irreducible subvariety $V \subseteq X$ of dimension k. Similarly

$$\int_V c_1(L)^{\dim V} \geq 0$$

for every nef line bundle L on X.

Before giving a proof, we discuss some applications below.

Corollary 1.4.7. *Let X be a projective variety or scheme, and D a nef $\mathbb{R}$-Cartier divisor on X. If H is any ample $\mathbb{R}$-Cartier divisor on X, then*

$$D + \varepsilon \cdot H$$

is ample for every $\varepsilon > 0$.

Conversely, if D and H are any two Cartier divisors such that $D + \varepsilon H$ is ample for all sufficiently small $\varepsilon > 0$, then D is nef.

Proof. "⇐": If $D + \varepsilon H$ is ample for $\varepsilon > 0$, then

$$(D \cdot C) + \varepsilon(H \cdot C) = \big((D + \varepsilon H) \cdot C\big) > 0$$

by NMK criterion, for every irreducible curve $C \subseteq X$. Letting $\varepsilon \to 0$, it follows that $(D \cdot C) \geq 0$ and hence D is nef.

"⇒":

Assume that D is nef and H is ample. Replacing εH by H it suffices to show that $D + H$ is ample.

We can approach this problem in an approximate way:

- For $D + H$ is (numerically equivalent to) a rational Cartier divisor, if it satisfies Nakai inequalities, it will be ample.

- With these ample Cartier divisors as the center, the open nature of ampleness in the finite-dimensional vector space $N^1(X)_{\mathbb{R}}$ ensures that the required $\mathbb{R}$-Cartier divisor lies nearby after a small perturbation.

  Rational Case:

  Fix an irreducible subvariety $V \subseteq X$ of positive dimension k. Then

$$\big((D+H)^k \cdot V\big) = \sum_{s=0}^{k} \tbinom{k}{s} (H^s \cdot D^{k-s} \cdot V)$$

Since H is a positive $\mathbb{R}$-linear combination of integral ample Cartier divisors, the intersection $(H^s \cdot V)$ is represented (in the Chow ring) by an effective real $(k-s)$-cycle.

Since D is nef, we can apply Kleiman's theorem to each components of this cycle, we get

$$\big(D^{k-s} \cdot (H^s \cdot V)\big) = (H^s \cdot D^{k-s} \cdot V) \geq 0$$

Thus each of the terms in that binomial expansion, is non-negative, and the last term $(H^k \cdot V)$ is strictly positive, since it is related to a degree of V under a suitable closed immersion via mH.

Therefore $\big((D+H)^k \cdot V\big) > 0$ for every such V, so the ampleness of $D+H$ follows from the Nakai inequalities.

General Case:

Choose ample Cartier divisors $H_1, \ldots, H_r$ whose classes span $N^1(X)_{\mathbb{R}}$. By the open nature of ampleness (Example 3.15), the $\mathbb{R}$-Cartier divisor

$$H(\varepsilon_1, \ldots, \varepsilon_r) := H - \varepsilon_1 H_1 - \cdots - \varepsilon_r H_r$$

remain ample for all $0 \leq \varepsilon_i \ll 1$.

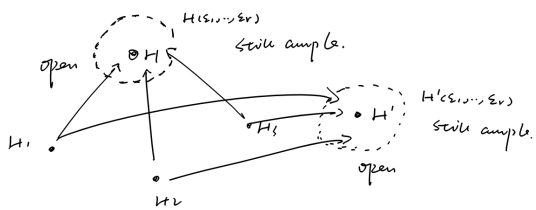


Figure 1.11.: Give a small perturbation.

But the classes of these Cartier divisors fill up as an open subset of $N^1(X)_{\mathbb{R}}$, consequently there exist $0 < \varepsilon_i \ll 1$ such that $D' = D + H(\varepsilon_1, \ldots, \varepsilon_r)$ represents a

rational class in $N^1(X)_{\mathbb{R}}$. But the case we already treated shows that D' is ample. So

$$D + H = D' + \varepsilon_1 H_1 + \cdots + \varepsilon_r H_r$$

remains ample as in Example 3.15. □

Corollary 1.4.8. (Test ampleness ny intersecting with curves) *Let X be a projective variety or scheme, and H an ample $\mathbb{R}$-Cartier divisor on X. Fix an $\mathbb{R}$-Cartier divisor D on X.*

Then D is ample if and only if there exists a $\varepsilon > 0$ such that

$$\frac{(D \cdot C)}{(H \cdot C)} \geq \varepsilon$$

for every irreducible curve $C \subseteq X$.

Proof. "⇐": $\frac{(D \cdot C)}{(H \cdot C)} \geq \varepsilon$ is just $(D \cdot C) - \varepsilon \cdot (H \cdot C) \geq 0$ by $(H \cdot C) > 0$, which is equivalent to $D - \varepsilon \cdot H$ being nef. So from the above corollary, $D = (D - \varepsilon H) + \varepsilon H$ is ample.

"⇒":

If D is ample, then so is $D - \varepsilon H$ for $0 \leq \varepsilon \ll 1$, which is just the statement of the Nakai inequalities. □

Example 1.4.9. (Intersection of an ample Cartier divisor with a curve is bounded) If H_1 and H_2 are ample Cartier divisors on a projective variety X, then there is $M, m \in \mathbb{Q}_{>0}$ such that

$$m \cdot (H_1 \cdot C) \leq (H_2 \cdot C) \leq M \cdot (H_1 \cdot C)$$

for every irreducible curve $C \subseteq X$.

Proof. Using the open nature of ampleness (Example 1.3.15), we can choose $M, m \in \mathbb{Q}_{>0}$ such that

$$M \cdot H_1 - H_2 \text{ and } m \cdot H_1 - H_2$$

are both ample, with $0 \leq |\frac{1}{M}|, |\frac{1}{m}| \ll 1$.

So by testing ampleness on curves, we have

$$\frac{\big((M \cdot H_1 - H_2) \cdot C\big)}{(H_2 \cdot C)} \geq \varepsilon_1 \text{ and } \frac{(H_2 \cdot C)}{\big((m \cdot H_1 - H_2) \cdot C\big)} \geq \varepsilon_2$$

for $\varepsilon_1, \varepsilon_2 > 0$, which is equivalent to

$$\frac{1}{\varepsilon_2 + 1} \cdot m \cdot (H_1 \cdot C) \leq (H_2 \cdot C) \leq \frac{1}{\varepsilon_1 + 1} \cdot M \cdot (H_1 \cdot C)$$

□

Theorem 1.4.10. (Seshadri's criterion) *Let X be a projective variety and D a Cartier divisor on X. Then D is ample if and only if there exists $\varepsilon > 0$ such that*

$$\frac{(D \cdot C)}{\mathrm{mult}_x C} \geq \varepsilon$$

for every point $x \in X$ and every irreducible curve $C \subseteq X$ passing through x.

Proof. "⇒":

Note that if E_x is an effective Cartier divisor passing through x and meeting an irreducible curve C properly, then the local intersection number $i(E_x, C; x)$ of E_x and C at x is bounded below by $\mathrm{mult}_x C$:

$$(E_x \cdot C) \geq \mathrm{mult}_x C$$

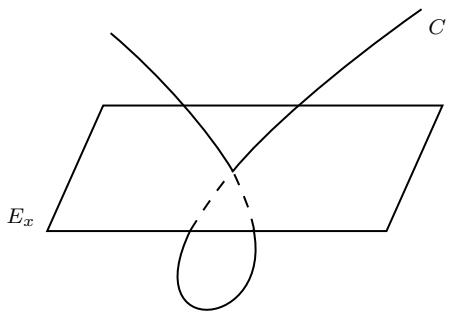


Figure 1.12.: Intersection number bounded below

But if D is ample, so that mD is very ample for $m \gg 0$, corresponding to a closed immersion into a projective space, so we can find a hyperplane section whose pullback on X is E_x, such that

$$E_x \sim_{\mathrm{lin}} mD$$

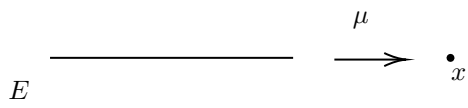
satisfies the inequality, so $(D \cdot C) \geq \frac{1}{m}\mathrm{mult}_x C$ for all x and such C.

"⇐":

Conversely, suppose that inequality holds for some $\varepsilon > 0$. Using induction on $n = \dim X$, we can assume that $\mathscr{O}_V(D)$ is ample on every irreducible proper subvariety $V \subseteq X$, that is $(D^{\dim V} \cdot V) > 0$ by the Nakai-Moishezon criterion. So it suffices to show that $(D^n) > 0$.

Now we fix any smooth point $x \in X$, and consider the blowing up

$$\mu : X' = \mathrm{Bl}_x(X) \longrightarrow X$$



of X at x, with exceptional Cartier divisor $E = \mu^{-1}(x)$.

Step 1:

We claim that the $\mathbb{R}$-Cartier divisor

$$\mu^* D - \varepsilon \cdot E$$

is nef on X.

So if this holds, Kleiman's theorem implies that

$$\begin{aligned}((\mu^* D - \varepsilon \cdot E)^n)_{X'} &= \sum_{s=0}^{n} \binom{n}{s} (-1)^s (\mu^* D^{n-s} \cdot (\varepsilon E)^s)_X \\ &= (D^n)_X - \varepsilon^n \\ &\geq 0\end{aligned}$$

where $(\mu^* D^{n-s} \cdot (\varepsilon E)^s)_{X'} = (\varepsilon^s \cdot D^{n-s} \cdot \mu_* E^s)_X = (\varepsilon^s \cdot D^{n-s} \cdot (\mathrm{pt})^s)_X = 0$, and the subscript below is indicate the variety on which intersection number is computed.

So $(D^n) \geq \varepsilon^n > 0$ as required.

Step 2:

For the nefness of $(\mu^* D - \varepsilon \cdot E)$, fix an irreducible curve $C' \subseteq X'$ not contained in E and set $C = \mu(C')$, so that C' is the strict transformation of C.

Then

$$(C' \cdot E) = \mathrm{mult}_x C$$

as in [Ful98, P.79], we will remark on this below.

On the other hand,

$$(C' \cdot \mu^* D)_{X'} = (\mu_* C' \cdot D)_X = (C \cdot D)_X$$

by projection formula, so the hypothesized inequality implies that

$$\begin{aligned}&(\mu^* D \cdot C')_{X'} - (\varepsilon E \cdot C')_{X'} \\ &= (D \cdot C)_X - (\varepsilon \cdot \mu_* E \cdot C)_X \\ &= (D \cdot C)_X - \varepsilon \cdot \mathrm{mult}_x C \\ &\geq 0\end{aligned}$$

Since $\mathscr{O}_E(E)$ is a negative normal bundle so for $C' \subseteq E$, $\mathscr{O}_E(E)|_{C'} = \mathscr{O}_{C'}(E)$ is also a negative line bundle, the equality holds. Therefore $(\mu^* D - \varepsilon \cdot E)$ is a nef Cartier divisor. □

Remark 1.4.11. $((C' \cdot E) = \mathrm{mult}_x C)$ As in [Ful98, p.79], Y is the fixed $x \in V$, and X is the variety V. $\mathbb{P}(C_X Y)$ is the exceptional Cartier divisor E. So $p_*(\widetilde{X^n})$ corresponding to $E^{\dim V}$, which the intersection number is computed on V.

So $e_X Y[X] = (-1)^{n-1} p_*(\widetilde{X^n})$ is just $\mathrm{mult}_x V = (-1)^{(1+\dim V)} \cdot (E^{\dim V})$.



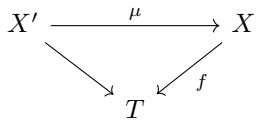
Figure 1.13.: strict transform

Proposition 1.4.12. (Nefness in families) *Let $f : X \longrightarrow T$ be a surjective proper morphism of varieties, and let L be a line bundle on X. For $t \in T$ put*

$$X_t := f^{-1}(t),\ L_t := L|_{X_t}$$

If L_0 is nef for some given $0 \in T$, then there is a countable union $B \subseteq T$ of proper subvarieties of T, not containing 0, such that L_t is nef for all $t \in T - B$.

Proof. By Chow's lemma we get a birational map μ,



and by projection formula intersection numbers do not change sign. Hence can assume f is projective.

Since we're proving the local property, we may shrink T, write $L = \mathscr{O}_X(D)$ where D is a Cartier divisor on X where $\mathrm{Supp}\, D$ does not contain any of the fibres X_t.

Now on a projective scheme X, can fix an ample Cartier divisor A on X such that $A_t = A|_{X_t}$ is ample for all t. According to Corollary 4.7, D_t being nef is equivalent to $D_t + \frac{1}{m} A_t$ being ample for all $m \in \mathbb{Z}_{>0}$. By assumption this holds when $t = 0$, and it follows from ampleness in families that the locus on T where $D_t + \frac{1}{m} A_t$ fails to be ample, is contained in a proper algebraic subset $B_m \subseteq T$ not containing 0, so taking $B = \bigcup B_m$ is fine. □

Now we turn to prove Kleiman's theorem.

Proof. One can assume that X is irreducible and reduced, as by Chow's lemma - as intersection number doesn't change the sign - can assume in addition that X is projective.

We use the induction on $n = \dim X$.

- $n = 1$: just the definition of the nef Cartier divisor.
- Suppose holds for $n - 1$:
$$(D^k \cdot V) \geq 0$$
for all irreducible $V \subseteq X$ of dimension $\leq n - 1$.
- Now for n: the issue is to show $(D^n \cdot X) = (D^n) \geq 0$.

We begin with that D is a $\mathbb{Q}$-Cartier divisor, then to the general case by small perturbations.

Rational Case: As X is a projective scheme, can fix an ample Cartier divisor H on X, and consider for $t \in \mathbb{R}$ the self-intersection number:

$$P(t) := (D + tH)^n \in \mathbb{R}$$

Expanding the right-hand side, we can view $P(t)$ as a polynomial in t with real coefficients, and we are required to show that $P(0) \geq 0$.

To prove it by contradiction, we assume that $P(0) < 0$.

Step 1:

Note that for $k < n$, then

$$(D^k \cdot H^{n-k}) \geq 0$$

In fact, H being ample means mH is effective for $m \gg 0$, so $(mH)^{n-k} = m^{n-k} \cdot H^{n-k}$ represents an effective k-cycle Z, and therefore H^{n-k} represents $\frac{1}{m^{n-k}} \cdot Z$ which is a k-cycle with rational coefficients. So the inequality follows from applying induction hypothesis to the components of this k-cycle.

In particular, for $k < n$ the coefficient of t^{n-k} in $P(t)$ is non-negative. Since by assumption $P(0) < 0$, we know that $P(t)$ has a single real root $t_0 > 0$ by Calculus.

Step 2:

Claim next that for $\forall t \in \mathbb{Q}, t > t_0$, the $\mathbb{Q}$-Cartier divisor $D + tH$ is ample.

Using Nakai inequalities, this is equivalent to checking that

$$\big((D + tH)^k \cdot V\big) > 0$$

for every irreducible $V \subseteq X$ of dimension k.

If $V = X$, it just follows the fact that $t > t_0$ implies $P(t) > P(t_0) = 0$ as the coefficient of t^{n-k} is non-negative for $k < n$.

If $V \subsetneq X$, we expand out to get

$$((D+tH)^k \cdot V) = \sum_{s=0}^{k} \binom{k}{s} (D^s \cdot (tH)^{k-s} \cdot V).$$

Similarly as above:

- when $s < k$ we have $(D|_V)^s \cdot ((tH)|_V)^{k-s} \geq 0$,
- and when $s = k$, $(D|_V)^k \geq 0$ is just the induction hypothesis.

This implies that all coefficients are non-negative, while the leading coefficient $(H^k \cdot V)$ is strictly positive, so the claim is established.

Step 3:

Now write $P(t) = Q(t) + R(t)$, where

$$Q(t) = (D \cdot (D+tH)^{n-1}),$$
$$R(t) = (tH \cdot (D+tH)^{n-1}).$$

If $t > t_0$, $(D+tH)$ is ample, and hence $(D \cdot (D+tH)^{n-1})$ is the intersection of a nef $\mathbb{Q}$-Cartier divisor with an effective rational 1-cycle. Therefore $Q(t) \geq 0$ for all rational $t > t_0$, and consequently $Q(t_0) \geq 0$ by continuity.

On the other hand, thanks to $(D^k \cdot H^{n-k}) \geq 0$, all the coefficients in

$$R(t) = t \cdot \sum_{s=0}^{n-1} \binom{n-1}{s} \left(D^s \cdot (tH)^{n-s}\right)$$

are non-negative, and the highest one (H^n) is strictly positive as it is ample (Example 1.2.3). Therefore $R(t_0) > 0$ by $t_0 > 0$.

But then $P(t_0) = Q(t_0) + R(t_0) > 0$, contradiction.

General Case: Check that when D is arbitrary nef $\mathbb{R}$-Cartier divisor.

We choose ample Cartier divisors $H_1, \ldots, H_r$ whose classes span $N^1(X)_{\mathbb{R}}$. Then $\varepsilon_1 H_1 + \cdots + \varepsilon_r H_r$ is ample for all $\varepsilon_i > 0$.

In particular,

$$D(\varepsilon_1, \ldots, \varepsilon_r) = D + \varepsilon_1 H_1 + \cdots + \varepsilon_r H_r$$

is indeed nef since H_i are ample and by the Nakai inequalities. But the classes of these Cartier divisors fill up an open subset in $N^1(X)_{\mathbb{R}}$, so we can find arbitrarily small $0 < \varepsilon_i \ll 1$ such that $D(\varepsilon_1, \ldots, \varepsilon_r)$ is (numerically equivalent to) a rational Cartier divisor. Hence

$$\left(D(\varepsilon_1, \ldots, \varepsilon_r)^k \cdot V\right) \geq 0$$

for all irreducible $V \subseteq X$ of dimension k as in rational case. Letting the $\varepsilon_i \to 0$, we get $(D^k \cdot V) \geq 0$. □

Example 1.4.13. (Intersection products of nef classes) Let

$$\delta_1, \ldots, \delta_n \in N^1(X)_{\mathbb{R}}$$

be nef Cartier divisors on a complete variety or scheme X. Then

$$\int_X \delta_1 \cdots \delta_n \geq 0$$

Proof. By Chow's lemma and projection formula, we can assume that X is projective to ensure the ample Cartier divisors H_i indeed exist. Then using Corollary 4.7 to perturb the δ_i's to be $\delta_i + \varepsilon_i H_i$ slightly, which become ample classes, using Example 2.3 we have

$$\int_X (\delta_1 + \varepsilon_1 H_1) \cdots (\delta_n + \varepsilon_n H_n) > 0$$

Letting the ε_i's $\to 0$, get the required inequality. □

Remark 1.4.14. (Mumford's counter-example) Let X be a projective variety and L a line bundle on X.

By analogy with Kleiman's theorem, if

$$\int_C c_1(L) > 0$$

on every irreducible curve $C \subseteq X$, does it imply that

$$\int_V c_1(L)^{\dim V} > 0$$

for every irreducible $V \subseteq X$? The answer is NO! This is the famous counter-example due to Mumford and we'll discuss it later.

Definition 1.4.15. (Minimal surface) A smooth projective surface X is a *minimal surface*, if it contains no (-1)-curves, i.e. a curve $E \cong \mathbb{P}^1$ and $(E^2) = -1$.

Example 1.4.16. (Minimal Surfaces) Let X be a smooth projective surface with $\kappa(X) \geq 0$, i.e. there is $m > 0$ such that $|mK_X| \neq \emptyset$.

Then X is minimal if and only if the canonical Cartier divisor K_X is nef.

Proof. "⇒":

Suppose that K_X is not nef. Since we are working on a smooth variety, Cartier divisors and Weil divisors are the same. We fix $D \in |mK_X| \neq \emptyset$, and write $D = \sum a_i C_i$ with $a_i > 0$ and C_i irreducible. If there is an irreducible curve $C \subseteq X$ with $(K_X \cdot C) < 0$. As $D \sim_{\text{lin}} mK_X$, we have

$$(K_X \cdot C) = \frac{1}{m} \cdot (D \cdot C)$$

so C must appear as one of the C_i, since otherwise, the intersection number must be non-negative with different curves.

Say $C = C_1$, then

$$(a_1 C \cdot C) \leq (D \cdot C) = m \cdot (K_X \cdot C) < 0$$

which is $C^2 < 0$. From [Mat02, 1-1-4], we know C is a (-1)-curve.

"$\Leftarrow$": Also follows from [Mat02, 1-1-4]. □

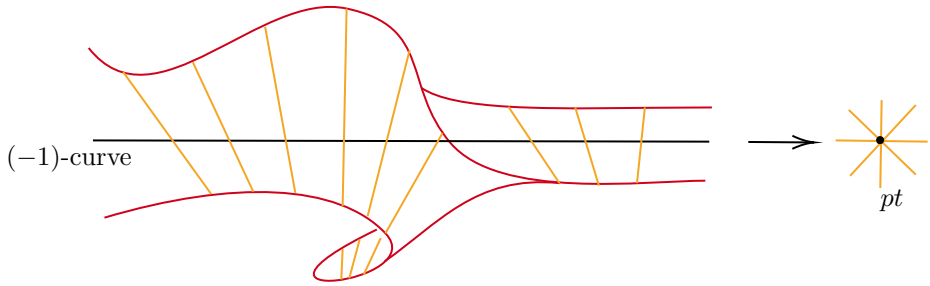


Figure 1.14.: Contraction of (-1)-curve.

Remark 1.4.17. Why we must choose this special Cartier divisor K_X? I think the reason is, it always appears in Riemann-Roch formula.

1.4C Cones

Thanks to Grothendieck's student Kleiman and his work [Kle66], turned algebraic geometry problems into Linear algebra. Now we consider the *cone* in a finite dimensional real vector space V, which is the subset $K \subseteq V$ stable under multiplication by positive scalars and 0 is in it. And the *convex cone* is the cone is also stable under addition.

Definition 1.4.18. (Ample and nef cones) The *ample cone*

$$\mathrm{Amp}(X) \subseteq N^1(X)_{\mathbb{R}}$$

of X is the convex cone spanned by all ample $\mathbb{R}$-Cartier divisor classes on X.

The *nef cone*

$$\mathrm{Nef}(X) \subseteq N^1(X)_{\mathbb{R}}$$

is the convex cone of all nef $\mathbb{R}$-Cartier divisor classes.

Concretely, $\mathrm{Amp}(X)$ is spanned by the classes of all ample integral (or rational) Cartier divisors with positive real coefficients: $\sum a_i \cdot \delta_i$ for $a_i \in \mathbb{R}_{>0}$ and $\delta_i \in N^1(X)$ or $N^1(X)_{\mathbb{Q}}$.

Remark 1.4.19. As soon as $\rho(X) = \dim N^1(X)_{\mathbb{R}} \geq 3$ the structure of these cones can be quite complicated. For example, they may or may not be a polyhedron.

Remark 1.4.20. (Visualization) It is sometimes convenient to represent a cone by drawing its intersection with a hyperplane not passing through the origin.

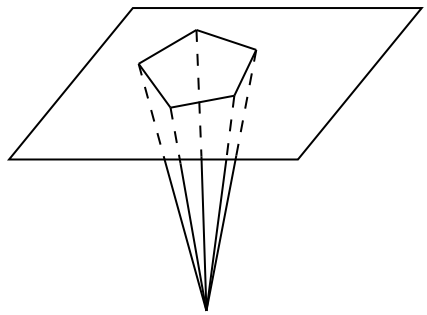


Figure 1.15.: Representing cones

We view $N^1(X)_{\mathbb{R}}$ as a finite-dimensional vector space under its standard Euclidean topology, to discuss the conceptions like closure and interior points.

Theorem 1.4.21. ([Kle66]) *Let X be any projective variety or scheme.*

(i) The nef cone is the closure of the ample cone:

$$\mathrm{Nef}(X) = \overline{\mathrm{Amp}}(X).$$

(ii) The ample cone is the interior of the nef cone:

$$\mathrm{Amp}(X) = \mathrm{int}\big(\mathrm{Nef}(X)\big).$$

Proof. As taking limits preserves signs, it is evident that the nef cone is closed, and it follows from Example 3.15 that $\mathrm{Amp}(X)$ is open. This gives the inclusions

$$\overline{\mathrm{Amp}}(X) \subseteq \mathrm{Nef}(X) \quad \text{and} \quad \mathrm{Amp}(X) \subseteq \mathrm{int}\big(\mathrm{Nef}(X)\big).$$

For the converse inclusions

$$\mathrm{Nef}(X) \subseteq \overline{\mathrm{Amp}}(X) \quad \text{and} \quad \mathrm{int}\big(\mathrm{Nef}(X)\big) \subseteq \mathrm{Amp}(X).$$

To establish the first inclusion, we let H be an ample Cartier divisor on X. Picking any nef $\mathbb{R}$-Cartier divisor D on the left, Corollary 4.7 shows that $D + \varepsilon H$ is ample for all $\varepsilon > 0$. Letting $\varepsilon \to 0$, we see that D is a limit of ample Cartier divisors.

For the second, observe that as a cone, if the class of D lies in int $\big(\mathrm{Nef}(X)\big)$, then $D - \varepsilon H$ remains nef for $0 < \varepsilon \ll 1$. Consequently

$$D = (D - \varepsilon H) + \varepsilon H$$

is ample again by Corollary 4.7. □

Remark 1.4.22. (Non-projective complete varieties) [Kle66, P.325-326] show that the above theorem (ii) possibly holds on a non-projective complete variety X.

The $\mathbb{R}$-Cartier divisors form a vector space, and the intersection product provides a linear functional, yielding a dual pairing with the vector space of curves.

Definition 1.4.23. (Numerical equivalence classes of curves) Le X be a complete variety. We denote by $Z_1(X)_{\mathbb{R}}$ the $\mathbb{R}$-vector space of *real one-cycles* on X, consisting of all finite $\mathbb{R}$-linear combinations of irreducible curves on X:

$$\gamma = \sum a_i \cdot C_i$$

for $\gamma \in Z_1(X)_{\mathbb{R}}$ where $a_i \in \mathbb{R}$ and $C_i \subseteq X$ is an irreducible curve.

Two one-cycles $\gamma_1, \gamma_2 \in Z_1(X)_{\mathbb{R}}$ are *numerically equivalent* if

$$(D \cdot \gamma_1) = (D \cdot \gamma_2)$$

for every $D \in \mathrm{Div}_{\mathbb{R}}(X)$.

The corresponding vector space of *numerical equivalence classes* of one-cycles is written as $N_1(X)_{\mathbb{R}}$. Thus by construction we have a perfect pairing

$$\begin{aligned} N^1(X)_{\mathbb{R}} \times N_1(X)_{\mathbb{R}} &\longrightarrow \mathbb{R} \\ (\delta, \gamma) &\longmapsto (\delta \cdot \gamma) \in \mathbb{R} \end{aligned}$$

In particular, $N_1(X)_{\mathbb{R}}$ is of finite dimensional real vector space on which we put the standard Euclidean topology. For numerical equivalence of integral and rational one-cycles are defined similarly.

For effective curves, the coefficients are all non-negative; they span a convex cone.

Definition 1.4.24. (Cone of curves) Let X be a complete variety. The *cone of curves*

$$\mathrm{NE}(X) \subseteq N_1(X)_{\mathbb{R}}$$

is the convex cone spanned by the classes of all effective one-cycles on X.

Concretely,

$$\mathrm{NE}(X) = \Big\{ \sum a_i [C_i] \;\Big|\; C_i \subseteq X \text{ an irreducible curve, } a_i \geq 0 \Big\}.$$

Its closure

$$\overline{\mathrm{NE}}(X) \subseteq N_1(X)_{\mathbb{R}}$$

is the *closed cone of curves* on X, which has the name *Kleiman-Mori cone*.

Remark 1.4.25. $\mathrm{NE}(X)$ itself might not be closed.

Now we need a basic fact about convexity, which follows from [BV04, P.53], where we recall that the *dual of a cone* $K \subseteq V$ is defined to be the cone in V^* given by

$$K^* = \{\phi \in V^* \mid \phi(x) \geq 0 \text{ for } \forall x \in K\}$$

Lemma 1.4.26. *For a cone K which is both convex and closed, then $K^{**} = K$.*

Then we prove that $\overline{\mathrm{NE}}(X)$ and $\mathrm{Nef}(X)$ are dual:

Proposition 1.4.27. *In the situation of Definition 1.4.23, $\overline{\mathrm{NE}}(X)$ is the closed cone dual to $\mathrm{Nef}(X)$, i.e.*

$$\overline{\mathrm{NE}}(X) = \big\{\gamma \in N_1(X)_{\mathbb{R}} \;\big|\; (\delta \cdot \gamma) \geq 0 \text{ for all } \delta \in \mathrm{Nef}(X)\big\}.$$

Proof. Taking

$$V = N_1(X)_{\mathbb{R}} \;,\; K = \overline{\mathrm{NE}}(X).$$

Then $\mathrm{Nef}(X) = \overline{\mathrm{NE}}(X)^*$ by definition. Consequently we have

$$\overline{\mathrm{NE}}(X) = \mathrm{Nef}(X)^*$$

which follows from the lemma. □

Now we continue assuming that X is complete and fix a Cartier divisor $D \in \mathrm{Div}_{\mathbb{R}}(X)$, not numerically trivial. We denote by

$$\begin{aligned} \phi_D : N_1(X)_{\mathbb{R}} &\longrightarrow \mathbb{R} \\ \gamma &\longmapsto (D \cdot \gamma) \end{aligned}$$

the linear functional determined by intersection with D, and we set

$$\begin{aligned} D^{\perp} &= \{\gamma \in N_1(X)_{\mathbb{R}} \mid (D \cdot \gamma) = 0\} \\ D_{>0} &= \{\gamma \in N_1(X)_{\mathbb{R}} \mid (D \cdot \gamma) > 0\} \end{aligned}$$

Thus $D^{\perp} = \ker \phi_D$ is a hyperplane and $D_{>0}$ is an open half-space in $N_1(X)_{\mathbb{R}}$.

We can define $D_{\geq 0}$, $D_{\leq 0}$, $D_{<0} \subseteq N_1(X)_{\mathbb{R}}$ similarly.

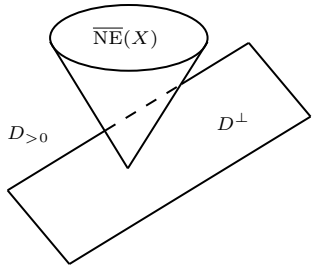


Figure 1.16.: Test for ampleness via the cone of curves

Theorem 1.4.28. (Kleiman's criterion: Ampleness via cones) *Let X be a projective variety or scheme, and let D be an $\mathbb{R}$-Cartier divisor on X. Then D is ample if and only if*

$$\overline{\mathrm{NE}}(X) - \{0\} \subseteq D_{>0}.$$

Moreover, for any ample Cartier divisor H and any integer k, the set

$$\{\gamma \in \overline{\mathrm{NE}}(X) \mid (H \cdot \gamma) \leq k\}$$

is compact, hence contains only finitely many classes of effective one-cycles $Z = \sum a_i Z_i$ with integral coefficients such that $(Z \cdot H) \leq k$.

Proof. We will follow the proof in [Deb01].

To the first statement:

"⇒": Assume D is ample, by Nakai inequalities, clearly we have $(D \cdot \gamma) \geq 0$ for all non-zero $\gamma \in \overline{\mathrm{NE}}(X)$. Now assume that $(D \cdot \gamma) = 0$ and $\gamma \neq 0$. Since this pairing is perfect which is non-degenerate, there is an $\mathbb{R}$-Cartier divisor E such that $(E \cdot \gamma) < 0$, hence $\big((D + tE) \cdot \gamma\big) < 0$ for all $t > 0$. In particular, $D + tE$ can not be ample, which contradicts Example 3.15.

"⇐": Assume that D is positive on $\overline{\mathrm{NE}}(X) - \{0\}$. We choose a norm $\| \cdot \|$ on $N_1(X)_{\mathbb{R}}$. The set

$$S := \{\gamma \in \overline{\mathrm{NE}}(X) \mid \|\gamma\| = 1\}$$

is compact as an half-sphere in a finite-dimensional real vector space. The linear functional $\phi_D : \gamma \longmapsto (D \cdot \gamma)$ is positive on S, so it is bounded below by $a \in \mathbb{Q}_{>0}$ due to compactness:

$$\phi_D(\gamma) \geq a \text{ for } \forall \gamma \in S.$$

For an ample Cartier divisor H on X, ϕ_H is bounded above by $b \in \mathbb{Q}_{>0}$, also due to compactness:

$$\phi_H(\gamma) \leq b \text{ for } \forall \gamma \in S.$$

It follows that $\big((D - \frac{a}{b}H) \cdot \gamma\big) = (D \cdot \gamma - \frac{a}{b}H \cdot \gamma) \geq 0$ on S, hence on $\overline{\mathrm{NE}}(X)$, which is equivalent to saying that $(D - \frac{a}{b}H)$ is nef. Then by Corollary 4.7

$$D = (D - \frac{a}{b}H) + \frac{a}{b}H$$

is ample.

Now we prove the second statement, following the [KM98, 1.19]:

We fix a norm $\|\cdot\|$ on $N_1(X)_{\mathbb{R}}$ and assume that

$$W_k := \{\gamma \in \overline{\mathrm{NE}}(X) \mid (H \cdot \gamma) \leq k\}$$

is not compact. Since under the Euclidean topology of the finite dimensional real vector space, compactness is equivalent to being closed and bounded, we can we find a sequence $\{\gamma_i\} \subseteq W_k$ such that

$$\lim_{i\to\infty} \|\gamma_i\| \to \infty.$$

As $\{\gamma_i/\|\gamma_i\|\}$ is a bounded sequence, we can find a converge subsequence $\{\gamma'_j/\|\gamma'_j\|\}$ which converges to $\gamma \in \overline{\mathrm{NE}}(X) - \{0\}$, with $\|\gamma\| = \lim_{j\to\infty} \|\frac{\gamma'_j}{\|\gamma'_j\|}\| = 1$. But

$$(H \cdot \gamma) = \lim_{j\to\infty} \frac{(H \cdot \gamma'_j)}{\|\gamma'_j\|} = 0,$$

since for $\gamma'_j \in W_k$, we have $(H \cdot \gamma'_j) \leq k$ while $\|\gamma'_j\| \to \infty$. Thus contradicts the fact that H is ample.

Finally, one-cycles with integral coefficients correspond to a discrete set in $N_1(X)_{\mathbb{R}}$, and so it has only finitely many points in any compact set. □

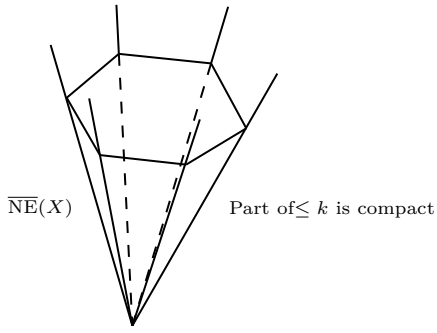


Figure 1.17.: $\{\gamma \in \overline{\mathrm{NE}}(X) \mid (H \cdot \gamma) \leq k\}$ is compact

We now give an example in [KM98, 1.19].

Example 1.4.29. Let X be a projective variety or scheme, then the closed cone of curves $\overline{\mathrm{NE}}(X) \subseteq N_1(X)_{\mathbb{R}}$ does not contain any infinite straight line, since if $\gamma \in N_1(X)_{\mathbb{R}}$ is a class such that both γ, $-\gamma \in \overline{\mathrm{NE}}(X)$, then γ must be 0.

Definition 1.4.30. (Extremals) Let $K \subseteq V$ be a convex cone in a finite dimensional real vector space.

An *extremal* $F \subseteq K$ is a subcone having the property that if $v + w \in F$ for some vectors v, $w \in K$, then necessarily v, $w \in F$. F is also called an *extremal face* of K.

A 1-dimensional extremal subcone is called an *extremal ray*, denoted by $\mathbf{r}$ or R.

Example 1.4.31. (Curves on a surface) If X is a **smooth projective surface**, then by [Har77, II, 6.11], a one-cycle is the same thing as a Cartier divisor. Hence

$$N^1(X)_{\mathbb{R}} = N_1(X)_{\mathbb{R}}.$$

and in particular the various cone we have defined before all live in a same finite dimensional real vector space.

(i) One has the inclusion
$$\mathrm{Nef}(X) \subseteq \overline{\mathrm{NE}}(X),$$
with equality holding if and only if $(C^2) \geq 0$ for every irreducible curve $C \subseteq X$.

(ii) If $C \subseteq X$ is an irreducible curve with $(C^2) \leq 0$, then $\overline{\mathrm{NE}}(X)$ is spanned by $[C]$ and the subcone
$$\overline{\mathrm{NE}}(X)_{C \geq 0} := C_{\geq 0} \cap \overline{\mathrm{NE}}(X).$$

(iii) In the situation of (ii), $[C]$ lies on the boundary of $\overline{\mathrm{NE}}(X)$. If in addition $(C^2) < 0$ then $[C]$ spans an extremal ray in that cone.

Proof. For (i), the fact that the intersection number of different curves is non-negative means that they must have positive coefficients. But notice that the only case where the intersection number is negative arises in self-intersections.

For (ii), this is obvious, as effective curves different from C must lie in $\overline{\mathrm{NE}}(X)_{C \geq 0}$.

Now for (iii), in situation (ii) we have

$$\overline{\mathrm{NE}}(X) = \overline{\mathrm{NE}}(X)_{C \geq 0} + \mathbb{R}_{\geq 0}[C].$$

If $(C^2) = 0$, then the linear function $D \mapsto (D \cdot C)$ is non-negative on $\overline{\mathrm{NE}}(X)$ and zero on C, so if $[C]$ were in the interior, a sufficiently small negative perturbation would yield that $(D \cdot C)$ is still non-negative, which is absurd. If $(C^2) < 0$, clearly $[C] \notin \overline{\mathrm{NE}}(X)_{C \geq 0}$, thus $[C]$ generates an extremal ray $\mathbb{R}_{\geq 0}[C]$. □

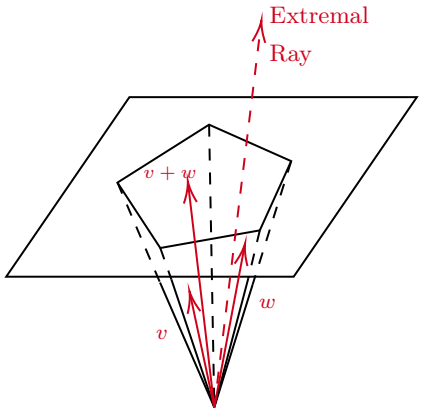


Figure 1.18.: $v + w$ not in extremal ray $\mathbf{r}$.

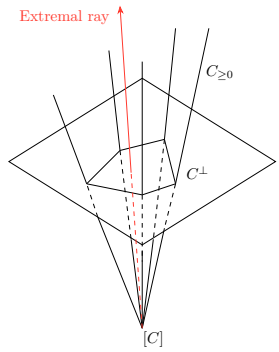


Figure 1.19.: $[C]$ spans the extremal ray.

1.4D Applications of Cones on Examples

To begin with, we recall the simplest blow down case, following the [Har77] and [Mat02], not necessarily restrict on $\mathbb{C}$:

Theorem 1.4.32. (Castelnuovo's Contraction theorem) *Let X be a smooth projective surface over an algebraically closed field $\mathbf{k}$ with E is a (-1)-curve in X. Then there is a proper birational map*

$$\mu : X \longrightarrow X_0$$

such that $\mu(E) = p$ where X_0 is a smooth surface, and μ is the blow-up of X_0 at p.

Proof. We want an L such that $\phi_{|L|}$ is a morphism with $\phi_{|L|} = \mu$. In particular, we need $(L \cdot E) = 0$ because its pushforward intersects with a point, and $(L \cdot C) \neq 0$ for any other curve C, since we only want to contract the curve E.

Pick a very ample Cartier divisor A on X which determines an embedding into projective space. Set $k = (A \cdot E) > 0$. Since $(E^2) = -1$, setting $L = A + kE$ satisfies $(L \cdot E) = 0$.

We also have $(L \cdot C) > 0$ for any curve $C \neq E$, because $(A \cdot C) > 0$ after pushing out and intersecting with a hyperplane, and $(E \cdot C) \geq 0$ for any different curves.

Let $L_n := A + nE$, $n = 0, 1, \ldots, k + 1$.

Step 1:

We claim that $H^1(X, \mathscr{O}_X(L_n)) = 0$ for all $0 \leq n \leq k + 1$. In fact, by induction:

- $n = 0$: $L_n = A$ is very ample, so A is ample on X. Then $H^i(X, \mathscr{O}_X(nA)) = 0$ for sufficiently large n due to Serre's vanishing. By replacing A with nA and k with nk, which does not change anything, we can assume that $H^1(X, \mathscr{O}_X(A)) = 0$.
- Suppose holds for $n - 1$.
- Now for $n > 0$:

  For the effective Cartier divisor E, we have the short exact sequence

$$0 \longrightarrow \mathscr{O}_X(A + (n-1)E) \longrightarrow \mathscr{O}_X(A + nE) \longrightarrow \mathscr{O}_E(A + nE) \longrightarrow 0$$

which is induced by twisting

$$0 \longrightarrow \mathscr{O}_X(-E) \longrightarrow \mathscr{O}_X \longrightarrow \mathscr{O}_E \longrightarrow 0$$

with the locally free sheaf $\mathscr{O}_X(L_n)$.

Now we pull the long exact sequence

$$\cdots \longrightarrow H^1(X, \mathscr{O}_X(L_{n-1})) \longrightarrow H^1(X, \mathscr{O}_X(L_n)) \longrightarrow H^1(E, \mathscr{O}_E(A + nE)) \longrightarrow \cdots$$

where the left term is 0 by induction hypothesis. And notice that $\deg \mathscr{O}_E(A + nE) = (E \cdot (A + nE)) = k - n$ so $\mathscr{O}_E(A + nE) \cong \mathscr{O}_{\mathbb{P}^1}(k - n)$ vanishing on H^1 when $k - n > -2$, see the details in [Har77, III, 5.1].

Therefore $H^1(X, \mathscr{O}_X(L_n)) = 0$.

Step 2:

We claim that $|A + nE|$ is base point free, for all $0 \leq n \leq k$. Since $|A| + nE \subseteq |A + nE|$, so $\mathrm{Bs}(|A + nE|) \subseteq E$.

To show $\mathrm{Bs}(|A + nE|) = \emptyset$, we only need to restrict on E.

Now

$$H^0(X, \mathscr{O}_X(A + nE)) \longrightarrow H^0(E, \mathscr{O}_E(A + nE))$$

is surjective because we have shown that $H^1(X, \mathscr{O}_X(L_{n-1})) = 0$, and $\mathscr{O}_E(A + nE) \cong \mathscr{O}_{\mathbb{P}^1}(k - n)$ by degree reasons, so it has global sections since $k - n \geq 0$.

Picking arbitrary non-zero global sections in $H^0(E, \mathscr{O}_E(A + nE))$, they can be lifted to $H^0(X, \mathscr{O}_X(A + nE))$ under the surjective map. THis implies they do not vanish on the whole of E, so E contains no base points, hence $|A + nE|$ is base point free.

Step 3:

We verify that the morphism

$$\phi = \phi_{|A+kE|} : X \longrightarrow X_1 \subseteq \mathbb{P}^N$$

is an isomorphism away from E, and $\phi(E) = p_1$ is a point.

- Since A is very ample, so the sections corresponding to linear series $|A|$ which separate points and separate tangent vectors, i.e. a closed immersion away from E, so ϕ is indeed an isomorphism away from E.
- By our construction, $(L \cdot E) = 0$, so using projection formula

$$0 = (L \cdot E) = (H \cdot \phi_* E)$$

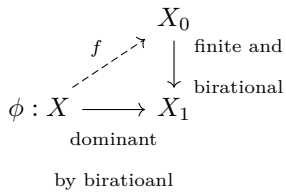
where the hyperplane section $H \subseteq \mathbb{P}^N$, whose pullback to X is L. So E must be mapped to a point.

Step 4:

Since a smooth variety must be normal, we can normalize first:

$$v : X_0 \longrightarrow X_1$$

So we have the diagram



in which ϕ factor through f, so f is also an isomorphism away from E, and $f(E) =: p \mapsto p_1$.

f is (proper) birational between normal varieties, by Zariski Main Theorem [Har77, III, 11.4], know that $f_* \mathscr{O}_X = \mathscr{O}_{X_0}$.

Step 5:

We check that X_0 is smooth at p.

The theorem on formal functions [Har77, III, 11.1] implies

$$\widehat{\mathscr{O}}_{X_0,p} = \widehat{(f_* \mathscr{O}_X)}_p = \varprojlim H^0(X, \mathscr{O}_{E_n})$$

where

$$\begin{aligned} E_n = f^{-1}\mathrm{Spec}(\mathscr{O}_{X_0,p}/m_p^n) &= V(m_p^n \mathscr{O}_X) \\ &= V(I_E^n) \end{aligned}$$

so the above is equal to $\varprojlim H^0(X, \mathscr{O}_{V(m_p^n \mathscr{O}_{\mathscr{X}})}) = \varprojlim H^0(X, \mathscr{O}_{V(I_E^n)})$.

Now we use the induction on n:

- n=1:

$H^0(X, \mathscr{O}_E) = H^0(E, \mathscr{O}_E) = \mathbb{C}$ since $E \cong \mathbb{P}^1$. As $(E^2) = -1$, we know that $\mathscr{O}_E(E) \cong \mathscr{N}_{E/X} \cong \mathscr{O}_{\mathbb{P}^1}(-1)$, then for $I_E^1 = I$, we have $I/I^2 \cong \mathscr{O}_{\mathbb{P}^1}(1)$. In addition,

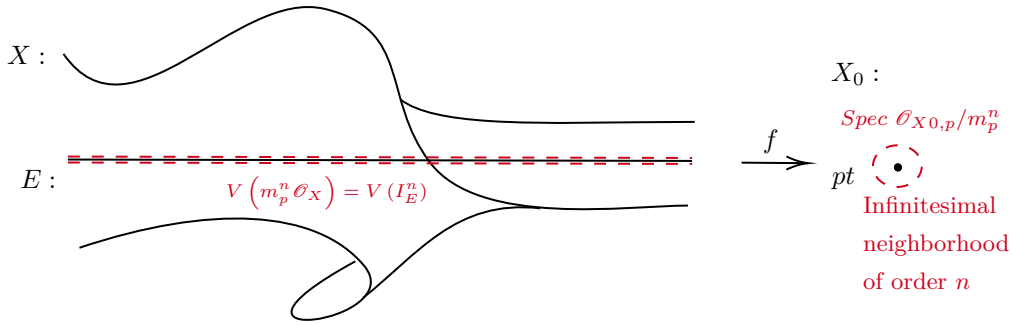


Figure 1.20.: Local information near the point p

E is a Cartier divisor, corresponding to a locally principal subscheme, which implies that I is locally generated by one element. So we have a diagram and a short exact sequence:

$$\begin{array}{ccc} f^n & \in & I^n/I^{n+1} \cong \mathscr{O}_{\mathbb{P}^1}(n) \\ \uparrow & & \uparrow \\ f & \in & I/I^2 \cong \mathscr{O}_{\mathbb{P}^1}(1) \end{array} \qquad \text{and} \quad 0 \longrightarrow I^n/I^{n+1} \longrightarrow \mathscr{O}_{E_{n+1}} \longrightarrow \mathscr{O}_{E_n} \longrightarrow 0$$

Taking global sections gives:

$$0 \longrightarrow H^0(E, I^n/I^{n+1}) \longrightarrow H^0(E, \mathscr{O}_{E_{n+1}}) \longrightarrow H^0(E, \mathscr{O}_{E_n}) \longrightarrow 0$$

where H^1 of I^n/I^{n+1} vanishes by $I^n/I^{n+1} \cong \mathscr{O}_{\mathbb{P}^1}(n)$ and $n > -2$.

Now

$$H^0(E, \mathscr{O}_E) = H^0(\mathbb{P}^1, \mathscr{O}_{\mathbb{P}^1}) = \mathbf{k}$$
$$H^0(E, I/I^2) = H^0(\mathbb{P}^1, \mathscr{O}_{\mathbb{P}^1}(1)) \cong (x, y)$$

so we have a short exact sequence

$$0 \longrightarrow (x, y) \longrightarrow H^0(\mathbb{P}^1, \mathscr{O}_{E_2}) \longrightarrow \mathbf{k} \longrightarrow 0$$

and notice that by Serre's result, $H^0(\mathbb{P}^1, \mathscr{O}_{E_2})$ is a finite dimensional $\mathbf{k}$-vector space, so must contains $\mathbf{k}$ and x, y terms. Thereforch $H^0(\mathbb{P}^1, \mathscr{O}_{E_2}) \cong \mathbf{k}[x, y]/(x, y)^2$.

- Now assume the statement holds for n, $H^0(E, \mathscr{O}_{E_n}) \cong \mathbf{k}[x, y]/(x, y)^n$.
- Then for $n + 1$, we have:

$$H^0(\mathbb{P}^1, \mathscr{O}_{\mathbb{P}^1}(n)) = (x^n, x^{n-1}y, \ldots, xy^{n-1}, y^n)$$
$$H^0(E, \mathscr{O}_{E_n}) = H^0(\mathbb{P}^1, \mathscr{O}_{E_n}) = \mathbf{k}[x, y]/(x, y)^n$$

and we have the short exact sequence

$$0 \longrightarrow (x^n, x^{n-1}y, \ldots, y^n) \longrightarrow H^0(\mathscr{P}^1, \mathscr{O}_{E_{n+1}}) \longrightarrow \frac{\mathbf{k}[x,y]}{(x,y)^n} \longrightarrow 0$$

for the same reason, it contains degree n forms and $\mathbf{k}$, which implies that

$$H^0(\mathbb{P}^1, \mathscr{O}_{E_{n+1}}) \cong \mathbf{k}[x,y]/(x,y)^{n+1}.$$

Then

$$\begin{aligned}
\widehat{\mathscr{O}}_{X_0,p} &= \varprojlim H^0(X, \mathscr{O}_{E_n}) \\
&= \varprojlim H^0(E.\mathscr{O}_{E_n}) \\
&= \varprojlim \mathbf{k}[x,y]/(x,y)^n \\
&\cong \mathbf{k}[[x,y]]
\end{aligned}$$

which is a regular local ring. So X_0 is indeed smooth at p.

Step 6:

Taking blow-up of $X_0 : \mathrm{Bl}_p X_0 \longrightarrow X_0$ with respect to the ideal sheaf m_p, and for $f : X \longrightarrow X_0$, the sheaf $f^{-1}m_p \cdot \mathscr{O}_X$ is corresponds to the (-1)-curve E. Since E is a Cartier divisor, this implies that $f^{-1}m_p \cdot \mathscr{O}_X$ is a line bundle. By the universal property of blow-ups, we have the diagram:

$$\begin{array}{ccc} & & \mathrm{Bl}_p X_0 \\ & \overset{u}{\nearrow} & \downarrow \\ \mu := f : X & \longrightarrow & X_0 \end{array}$$

where u is birational by construction, and the number of contracted points is 1, i.e. just p. So using [Bea78, II, 11] we see that u is an isomorphism. □

Remark 1.4.33. As in [KM98] the statement is for "smooth proper surface", this is true, because by Zariski's result, any non-singular complete surface is projective.

Corollary 1.4.34. *For any smooth projective surface, there is a finite sequence of blow-ups:*

$$X = X_0 \longrightarrow X_1 \longrightarrow \cdots \longrightarrow X_m$$

such that

1) $X_i = Bl_p X_{i+1}$,

2) X_i *is smooth surface,*

3) X_m *is minimal surface.*

Proof. If X_i has a (-1)-curve, then by Castelnuovo's theorem there is a blow-down $X_i \to X_{i+1}$ in which $\rho(X_i) - 1 = \rho(X_{i+1})$. Since each step blows down a (-1)-curve, the dimension of N_1 must drop by 1. This process must terminate because $N_1(X)_{\mathbb{R}}$ is finite-dimensional. □

Definition 1.4.35. (Relative cone of curves) Let X and Y both be projective varieties, the *relative cone* of a morphism

$$\pi : X \longrightarrow Y$$

as the convex subcone $\mathrm{NE}(\pi)$ of $\mathrm{NE}(X)$ generated by the classes of curves contracted by π.

Note that as Y is projective, we can define numerical equivalence on it, then an irreducible curve C on X is contracted by π if and only if $\pi_*[C] = 0$, that is, $\mathrm{NE}(\pi)$ is the intersection of $\mathrm{NE}(X)$ with the vector space $\ker(\pi_*)$. Therefore it is closed in $\mathrm{NE}(X)$.

Before our exploration, we need a well-known rigidity result, can see [Deb01, 1.15] and [KM98, 1.6], and we give the different details below:

Lemma 1.4.36. (Rigidity lemma I) *Let X, Y and Y' be varieties, let a surjective map $\pi : X \longrightarrow Y$ and $\pi' : X \longrightarrow Y'$ be proper morphisms. Assume that $\pi_*\mathscr{O}_X \cong \mathscr{O}_Y$ i.e. every fibre of π is connected.*

(a) If π' contracts one fibre $\pi^{-1}(y_0)$ of π, that is, $\pi'(\pi^{-1}(y_0))$ is a point, then there is an open neighborhood Y_0 of y_0 in Y and a factorization

$$\pi'|_{\pi^{-1}(Y_0)} : \pi^{-1}(Y_0) \xrightarrow{\pi} Y_0 \longrightarrow Y'$$

(b) If π' contracts each fibre of π, it factors through π, that is of the form $\pi'(\pi^{-1}(y))$ which is a point.

Proof. Here we use the different situation from [Deb01], for which requiring π to be proper is sufficient; there is no need for π' to be proper.

(a): First notice that the morphism

$$g := \pi \times \pi' : X \longrightarrow Y \times Y'$$

can be factored into

$$X \xrightarrow{\Gamma_{\pi'}} X \times Y' \xrightarrow{\pi \times id} Y \times Y',$$

where the $\Gamma_{\pi'}$ is a closed immersion as we are in the variety case, and $\pi \times id$ is proper since π is, hence g is proper. We denote the image of g to be Z which is closed and proper in $Y \times Y'$, and let $p : Z \longrightarrow Y$ and $p' : Z \longrightarrow Y'$ be the two projections:

$$\begin{array}{ccccc} & & & & Y' \\ & \overset{\pi'}{\nearrow} & & & \uparrow{\scriptstyle p'} \\ g : X & \longrightarrow & Z = \mathrm{Im}(g) & \hookrightarrow & Y \times Y' \\ & \underset{\pi}{\searrow} & & & \downarrow{\scriptstyle p} \\ & & & & Y. \end{array}$$

As $\pi^{-1}(y_0) = g^{-1}(p^{-1}(y_0))$ is contracted by π', so it is contracted by g, since p is just a projection and doesn't contract anything.

It follows that $p^{-1}(y_0) = g(g^{-1}(p^{-1}(y_0)))$ is a point, and p is proper by [Har77, II, ex4.8], which is fibre-finite at point y_0, so by [SP25, Tag 02OH] we see that there is an open neighborhood Y_0 of y_0 in Y such that

$$p|_{p^{-1}(Y_0)} : p^{-1}(Y_0) \longrightarrow Y_0$$

is a finite map.

Set $X_0 = \pi^{-1}(Y_0)$ (this is fine due to surjectivity) and $Z_0 = p^{-1}(Y_0)$

$$\begin{array}{ccc} X_0 & \xrightarrow{g|_{X_0}} & Z_0 \\ & \underset{\pi|_{\pi^{-1}(Y_0)}}{\searrow} & \downarrow{\scriptstyle p|_{p^{-1}(Y_0)}} \\ & & Y_0 \end{array}$$

where $g|_{X_0}$ is surjective between varieties, hence the map $g^{\#} : \mathscr{O}_{Z_0} \longrightarrow g_*\mathscr{O}_{X_0}$ is injective [Har77, II, ex2.18(b)], that is $\mathscr{O}_{Z_0} \subseteq g_*\mathscr{O}_{X_0}$. For the same reason, we have $\mathscr{O}_{Y_0} \subseteq p_*\mathscr{O}_{Z_0}$. Then

$$\mathscr{O}_{Y_0} \subseteq p_*\mathscr{O}_{Z_0} \subseteq p_*g_*\mathscr{O}_{X_0} = \pi_*\mathscr{O}_{X_0} = \mathscr{O}_{Y_0}$$

we get that $p_*\mathscr{O}_{Z_0} = \mathscr{O}_{Y_0}$. Since p is finite over Y_0, and a finite map is always affine, it follows that $p_0 : Z_0 \longrightarrow Y_0$ is an isomorphism and

$$\pi'|_{\pi^{-1}(Y_0)} = \pi'|_{X_0} = p' \circ p_0^{-1} \circ \pi|_{X_0}.$$

(b): If π' contracts **each** fibre of π, then as the proof in (a), the morphism p above is finite over whole of Y, can take $Y_0 = Y$ hence π' factors through π. □

Lemma 1.4.37. (Rigidity lemma II) *Let Y be an irreducible variety and let $f : Y \longrightarrow Z$ be proper and surjective morphism. Assume that every fibre of f is connected*

and of dimension n. Let $g : Y \longrightarrow X$ be a morphism such that g contracts one fibre of f, that is $g(f^{-1}(z_0))$ is a point for some $z_0 \in Z$. Then $g(f^{-1}(z))$ is a point for every $z \in Z$.

Proof. First consider the morphism $f \times g : Y \longrightarrow Z \times X$ can be factored as

$$Y \xrightarrow{\Gamma_g} Y \times X \xrightarrow{f\times id} Z \times X,$$

where Γ_g is a closed immersion because we working with separated algebraic schemes, and $f \times id$ also clearly proper, so $f \times g$ is proper. Now we set $W = \mathrm{im}(f \times g) \subseteq Z \times X$ is a closed subvariety by properness, and consider $h : Y \longrightarrow W$, it is the morphism cancel out by

$$f \times g : Y \xrightarrow{h} W \underset{\text{separated}}{\longrightarrow} Z \times X$$

hence also be proper by [Har77, II, 4.8(e)].

Moreover we need to show that the morphism $p : W \longrightarrow Z$ is proper. We use the valuative criterion. Let R be a valuation ring and K its quotient field,

$$\begin{array}{ccc} \mathrm{Spec}\ K & \longrightarrow & W \\ \downarrow & & \downarrow \\ \mathrm{Spec}\ R & \longrightarrow & Z. \end{array}$$

Since $h : Y \longrightarrow W$ is surjective, we can lift the point $\mathrm{Spec}\ K \to W$ on Y, hence we get

$$\begin{array}{ccc} \mathrm{Spec}\ K & \longrightarrow & Y \\ \downarrow & & \downarrow f \\ \mathrm{Spec}\ R & \longrightarrow & Z. \end{array}$$

But as f is proper, there is a unique lifting $\mathrm{Spec}\ R \to Y$ making the diagram commutative. We descend this morphism via h to get our desired unique lifting $\mathrm{Spec}\ R \to W$.

So we obtain proper morphisms

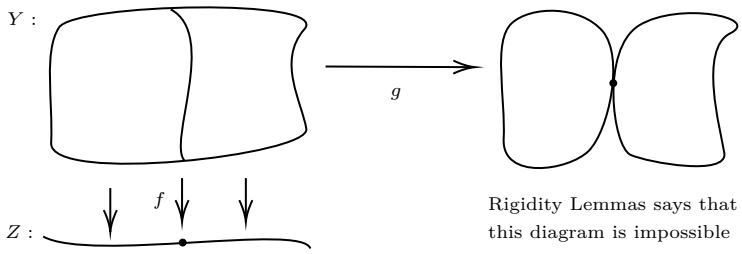
$$f : Y \xrightarrow{h} W \xrightarrow{p} Z,$$

with $p^{-1}(z) = h(f^{-1}(z))$ and $\dim p^{-1}(z_0)$, so

$$p^{-1}(z_0) = h(f^{-1}(z_0) = (f \times g)(f^{-1}(z_0)) = (z_0, \mathrm{pt})$$

and $\dim\ p^{-1}(z_0) = 0$. By upper semi-continuity [Har77, II, ex3.22(d)] we see that E_1 is closed, so there is an open set $U \subseteq Z$ containing z_0 such that $\dim p^{-1}(z) = 0$ for every $z \in U$. Thus h has fibre dimension n over $p^{-1}(U)$ as $h^{-1}(p^{-1}(z)) = f^{-1}(z)$, hence h has fibre dimension at least n everywhere again by upper semi-continuity.

Now for any $w \in W$, $h^{-1}(w) \subseteq f^{-1}(p(w))$, $\dim h^{-1}(w) \geq n$ and $\dim f^{-1}(p(w)) = n$. Therefore $\dim h^{-1}(w) = n$ and $h^{-1}(w)$ is a union of irreducible components of $f^{-1}(p(w))$. But notice that the number of irreducible components of $f^{-1}(p(w))$ is fnite, so $h(f^{-1}(p(w))) = p^{-1}(p(w))$ consists of finitely many points. It must be a single point since $f^{-1}(p(w))$ is connected. Thus, $p^{-1}(p(w)) = (p(w), g(f^{-1}(p(w))))$ is a single point due to the surjectivity of p. □



Proposition 1.4.38. *Let X, Y and Y' be projective varieties, let $\pi : X \longrightarrow Y$ be a morphism.*

(a) The subcone $\mathrm{NE}(\pi)$ of $\mathrm{NE}(X)$ is extremal.

(b) Assume that $\pi_* \mathscr{O}_X \cong \mathscr{O}_Y$, and let $\pi' : X \longrightarrow Y'$ be another morphism.

- *If $\mathrm{NE}(\pi)$ is contained in NE(π'), there is a unique morphism $f : Y \longrightarrow Y'$ such that $\pi' = f \circ \pi$.*
- *The morphism π is uniquely determined by NE(π) up to isomorphism.*

Proof. "(a)": Let $a = \sum a_i[C_i]$ and $a' = \sum a'_j[C'_j]$ be elements of $\mathrm{NE}(X)$, where a_i and $a'_j \in \mathbb{R}_{>0}$. If $a + a' \in \mathrm{NE}(\pi)$, there is a decomposition

$$a + a' = \sum a_i[C_i] + \sum a'_j[C'_j] = \sum a''_k[C''_k]$$

where the C''_k are irreducible curves contracted by morphism π and $a''_k \in \mathbb{R}_{>0}$. Now applying π_*, we get $\sum a_i \pi_*[C_i] + \sum a'_j \pi_*[C'_j] = 0$ in $N_1(Y)_{\mathbb{R}}$. But Y is projective, there is no non-trivial numerical equivalence on it (see remark), the C_i and C'_j all must be contracted by π hence a and a' are all in $\mathrm{NE}(\pi)$ which shows it is extremal.

"(b)": Assume now $\pi_* \mathscr{O}_X \cong \mathscr{O}_Y$ and $\mathrm{NE}(\pi) \subseteq \mathrm{NE}(\pi')$. This just says that every irreducible curve contracted by π must be contracted by π', hence every (connected) fibre of π must contracted by π'. As in Rigidity lemma I (b), π' factor through π, so there is $f : Y \longrightarrow Y'$ such that $\pi' = f \circ \pi$.

The uniqueness of f follows from the universal property of product. Assume $f' : Y \longrightarrow Y'$ is another morphism satisfying $\pi' = f' \circ \pi$. Using the same notation in the proof of lemma, the composition

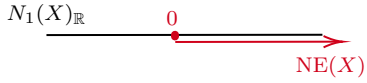
$$Z \xrightarrow{p} Y \xrightarrow{f'} Y'$$

must be the second projection $p' : Z \longrightarrow Y'$, hence $f' \circ p = p'$, which implies $f' = p' \circ p^{-1} = f$. □

Remark 1.4.39. As Y is projective so can be embedded into a projective space $\mathbb{P}$, then pulling back the hyperplane section on Y such that to be an ample Cartier divisor (by Serre's conclusion).

For a curve C mapped by π, there are only 2 possibilities: its image is either a 1-cycle or a point. If the image is a 1-cycle, then by the Nakai-Moishezon criterion, an ample Cartier divisor would have a non-zero intersection with this 1-cycle, which contradicts $\pi_*[C] = 0$.

Example 1.4.40. (One-dimensional $N_1(X)_{\mathbb{R}}$) On a projective variety X for which the vector space $N_1(X)_{\mathbb{R}}$ has dimension 1 (such as projective space by [Har77, II, 6.4]), there is the simplest case about extremals in $\mathrm{NE}(X) = \overline{\mathrm{NE}}(X)$: $\{0\}$ and $\mathrm{NE}(X)$ itself.



These two subcones of $\mathrm{NE}(X)$, corresponding to the morphisms $\pi : X \longrightarrow pt$ and $\pi : X \longrightarrow X$ the identity map.

Example 1.4.41. (Product of projective spaces) Let X be a product of $\mathbb{P}$ and $\mathbb{P}'$ two projective spaces. The vector space $N_1(X)_{\mathbb{R}}$ has dimension 2, spanned by two class of lines which just the dual of the hyperplanes. We denoted the class of line in $\mathbb{P}$ by l, and respectively by l' for $\mathbb{P}'$.

The cone of curves of X is

$$\mathrm{NE}(X) = \mathbb{R}_{\geq 0} l + \mathbb{R}_{\geq 0} l'$$

which has only two proper extremal rays, the rays generated by l and by l'.

They correspond to the projections $X \longrightarrow \mathbb{P}$ and $X \longrightarrow \mathbb{P}'$ respectively.

1.4E Fujita's Vanishing Theorem

We now discuss a theorem of Fujita showing that Serre-type vanishings can be made to operate uniformly with respect to twists by nef Cartier divisors.

Theorem 1.4.42. (Fujita's vanishing theorem) *Let X be a complex projective scheme, and let H be an ample Cartier divisor on X. Given any coherent sheaf $\mathscr{F}$ on X, there exists an integer $m(\mathscr{F}, H)$ such that*

$$H^i\big(X, \mathscr{F} \otimes \mathscr{O}_X(mH + D)\big) = 0$$

for all $i > 0$ and $m \geq m(\mathscr{F}, H)$, and any nef Cartier divisor D on X.

The essential point here is that the integer $m(\mathscr{F}, H)$ is independent of the nef Cartier divisor D, that's why we call it "Serre-type vanishing" because it can't be a simple corollary of Serre's vanishing, which that number must dependent on D.

Proof. By replacing X with $\operatorname{Supp}\mathscr{F}$, we may assume that $X = \operatorname{Supp}\mathscr{F}$. We prove the theorem by using induction on $\dim X$:

Step 1: For $\dim X = 0$ vanishing obviously holds due to Grothendieck vanishing. So from now on we assume that the vanishing holds in lower-dimensional cases. Now we give some reductions:

Step 2: We reduce to the case when X is reduced. Assuming that theorem holds for reduced schemes. Let $\mathscr{N}$ be the nilradical of $\mathscr{O}_X$, so that $\mathscr{N}^r = 0$ for some $r > 0$ as we're dealing with algebraic schemes. Consider the filtration

$$\mathscr{F} \supseteq \mathscr{N} \cdot \mathscr{F} \supseteq \mathscr{N}^2 \cdot \mathscr{F} \supseteq \cdots \supseteq \mathscr{N}^r \cdot \mathscr{F} = 0.$$

The quotients $\mathscr{N}^i\mathscr{F}/\mathscr{N}^{i+1}\mathscr{F} \cong (\mathscr{O}_X/\mathscr{N}) \otimes \mathscr{F}$ are coherent $\mathscr{O}_{X_{red}}$-modules, and therefore by assumption

$$H^j\big(X, (\mathscr{N}^i\mathscr{F}/\mathscr{N}^{i+1}\mathscr{F}) \otimes \mathscr{O}_X(mH + D)\big) = 0$$

for $j > 0$ and $m \geq m(\mathscr{N}^i\mathscr{F}/\mathscr{N}^{i+1}\mathscr{F}, H)$, while H_{red} is ample on X_{red}. Twisting the exact sequence

$$0 \longrightarrow \mathscr{N}^{i+1}\mathscr{F} \longrightarrow \mathscr{N}^i\mathscr{F} \longrightarrow \mathscr{N}^i\mathscr{F}/\mathscr{N}^{i+1}\mathscr{F} \longrightarrow 0$$

by $\mathscr{O}_X(mH + D)$ and taking cohomology,

$$\to H^j\big(X, \mathscr{N}^{i+1}\mathscr{F}(mH+D)\big) \to H^j\big(X, \mathscr{N}^i\mathscr{F}(mH+D)\big) \to H^j\big(X, \mathscr{N}^i\mathscr{F}/\mathscr{N}^{i+1}\mathscr{F}(mH+D)\big) \to$$

where as i large enough, $\mathscr{N}^i = 0$ so cohomologies vanished for all j.Thus, we can use decreasing induction on i. Then the left term is 0 by induction hypothesis, and the right term vanished by our preceding computation; therefore

$$H^j\big(X, \mathscr{N}^i\mathscr{F}(mH + D)\big) = 0$$

for all $j > 0$ and $m \geq m(\mathscr{N}^i\mathscr{F}, H)$. When $i = 0$ this gives the desired vanishing.

From now on, we assume that X is reduced.

Step 3: We reduce to the case when X is irreducible. Assume that the theorem holds for integral schemes. Let $X = X_1 \cup \cdots \cup X_k$ be its irreducible decomposition, and let $\mathscr{I}$ be the ideal sheaf of X_1 in X. We consider the exact sequence

$$0 \longrightarrow \mathscr{I} \cdot \mathscr{F} \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}/\mathscr{I} \cdot \mathscr{F} \longrightarrow 0,$$

where $\operatorname{Supp} \mathscr{I} \cdot \mathscr{F} = X_2 \cup \cdots \cup X_k$ and $\operatorname{Supp} \mathscr{F}/\mathscr{I} \cdot \mathscr{F} = X_1$. So taking cohomology and using induction on the number of irreducible components, we may assume that

$$H^j\big(X, \mathscr{I}\mathscr{F}(mH + D)\big) = 0$$

for $j > 0$ and $m \geq m(\mathscr{I} \cdot \mathscr{F}, H|_{X_2 \cup \cdots \cup X_k})$ and

$$H^j\big(X, (\mathscr{F}/\mathscr{I}\mathscr{F})(mH + D)\big) = 0$$

for $j > 0$ and $m \geq m(\mathscr{F}/\mathscr{I}\mathscr{F}, H|_{X_1})$. It then follows from the above exact sequence that

$$H^j\big(X, \mathscr{F}(mH + D)\big) = 0$$

for $j > 0$ and $m \geq m(\mathscr{F}, H) = \max\big\{m(\mathscr{I}\mathscr{F}, H|_{X_2 \cup \cdots \cup X_k}), m(\mathscr{F}/\mathscr{I}\mathscr{F}, H|_{X_1})\big\}$, as we desired.

So from now on, we assume that X is an integral scheme.

Step 4: We reduce to the case where H is very ample. Let l be a positive integer such that lH is very ample. We assume that the theorem holds for lH. Then apply this vanishing on coherent sheaf $\mathscr{F}(nH)$ for $0 \leq n \leq l-1$ with lH, we obtain numbers $m(\mathscr{F}(nH), lH)$ for each n. We put

$$m(\mathscr{F}, H) = l \cdot \big(\max m(\mathscr{F}(nH), lH) + 1\big),$$

thus for $m \geq m(\mathscr{F}, H)$, we can take part the $\mathscr{F}(mH + D)$ to be $\mathscr{F}(nH) \otimes \mathscr{O}_X(q(lH) + D)$ for $0 \leq n \leq l-1$ such that

$$H^i\big(X, \mathscr{F}(nH) \otimes \mathscr{O}_X(q(lH) + D)\big) = 0$$

for $i > 0$, i.e. $m(\mathscr{F}, H)$ satisfies the desired property.

So we can assume that H is very ample from now on.

Step 5: We reduce to the case when $\mathscr{F} = \mathscr{O}_X$. Assume that the theorem holds for $\mathscr{F} = \mathscr{O}_X$. So for arbitrary coherent sheaf $\mathscr{F}$ on X, we claim that there is an injective homorphism

$$\alpha : \mathscr{O}_X \longrightarrow \mathscr{F} \otimes \mathscr{O}_X(aH)$$

for some positive integer a. We consider the exact sequence

$$0 \longrightarrow \mathscr{O}_X \xrightarrow{\alpha} \mathscr{F}(aH) \longrightarrow \mathscr{G} \longrightarrow 0,$$

where $\mathscr{G}$ is the cokernel. The rank $\operatorname{rk}\mathscr{G} = \dim_{\mathbb{C}(X)} \mathscr{G}_\eta$ is strictly cut down by 1. So we can use the induction on $\operatorname{rk}\mathscr{F}$, which for rank 0 case is implicated by $\dim_{\mathbb{C}(X)} \mathscr{F}_\eta = 0$, then $\dim \operatorname{Supp}\mathscr{F} < \dim X$ as the generic point η not in its support. Twisting by $\mathscr{O}_X\big((m-a)H + D\big)$ and taking cohomology, we get

$$\cdots \to H^i\big(X, (m-a)H+D\big) \to H^i\big(X, \mathscr{F}(mH+D)\big) \to H^i\big(X, \mathscr{G}((m-a)H+D)\big) \to \cdots$$

where the left term vanishes by assumption, and the right term vanishes by the induction hypothesis, so the middle term also vanishes. Thus we can find $m(\mathscr{F}, H)$ satisfying

$$m(\mathscr{F}, H) \geq \max\big\{m(\mathscr{O}_X, H) + a, m(\operatorname{coker}\alpha, H) + a\big\}.$$

Now we prove our claim. In fact, since H is very ample, the coherent sheaf $\mathscr{F}(aH)$ is globally generated, so we can take a non-zero global section $s \in H^0(X, \mathscr{F}(aH))$ which induces the morphism on generic point $\eta \in \operatorname{Supp}\mathscr{F} = X$:

$$\alpha_\eta : \mathscr{O}_{X,\eta} = \mathbb{C}(X) \xrightarrow{\cdot s_\eta} \mathscr{F}(aH)_\eta,$$

which must be injective. So we conclude that $(\ker\alpha)_\eta = 0$ for the coherent (ideal) sheaf $\ker\alpha$. Locally see, on any affine open $U = \operatorname{Spec} A$, the above says that for $I = \ker\alpha|_U \subseteq A$, we have

$$I_\eta = I \underset{A}{\otimes} K(A) = 0.$$

But as $K(A)$ is flat over A, $I \underset{A}{\otimes} K(A) \cong IK(A)$ is just the image of injections

$$I \hookrightarrow A \hookrightarrow K(A),$$

so we see $I = IK(A) = 0$. As $U \subseteq X$ is arbitrarily chosen, we know $\ker\alpha = 0$.

From now on we assume $\mathscr{F} = \mathscr{O}_X$.

Step 6: Let $f : Y \longrightarrow X$ be a resolution of singularities, and consider the torsion-free coherent sheaf $f_*\omega_Y$. Moreover, $\operatorname{rk} f_*\omega_Y = 1$ since Y is smooth and f is birational. So we consider $(f_*\omega_Y)^\vee \otimes \mathscr{O}_X(bH)$ which is globally generated by H very ample, for a suitable integer $b > 0$. Take a non-zero global section s in $H^0\big(X, \mathscr{O}_X(bH) \otimes (f_*\omega_Y)^\vee\big)$, then at the generic point $\eta \in X$ the homomorphism

$$(f_*\omega_Y)_\eta \cong \mathscr{O}_{X,\eta} \xrightarrow{\cdot s_\eta} \mathscr{O}_{X,\eta}$$

is injective, as f is birational. But $\ker(f_*\omega_Y \to \mathscr{O}_X(bH))$ as a subsheaf of the torsion-free sheaf $f_*\omega_Y$ is also torsion-free, so the kernel itself must be zero. Therefore this non-zero global section s induces an injective homomorphism

$$f_*\omega_Y \longrightarrow \mathscr{O}_X(bH).$$

Then we complete it with a cokernel to get the following exact sequence

$$0 \longrightarrow f_*\omega_Y \longrightarrow \mathscr{O}_X(bH) \longrightarrow \mathscr{C} \longrightarrow 0,$$

where $\dim \operatorname{Supp} \mathscr{C} < \dim X$ since it has rank zero. On the other hand,

$$H^i\big(X, f_*\omega_Y \otimes \mathscr{O}_X(mH + D)\big) = 0$$

for every $m > 0$ and $i > 0$ by Kollár's vanishing theorem [Laz04a, 4.3.8], for which $mH + D$ **is ample** by Nakai-Moishezon. Therefore

$$H^i\big(X, \mathscr{O}_X((b+m)H + D)\big) = 0$$

for every $i > 0$ and $m \geq m(\mathscr{C}, H)$. Thus the theorem holds. □

Remark 1.4.43. This vanishing also holds over a positive characteristic field; see [Fuj11].

Theorem 1.4.44. (Higher cohomology of nef Cartier divisors, II) *Let X be a projective variety or scheme of dimension n, and D be a nef Cartier divisor on X. Then for any coherent sheaf $\mathscr{F}$ on X,*

$$h^i\big(X, \mathscr{F}(mD)\big) = O(m^{n-i}).$$

for sufficiently large m.

Proof. First fix an arbitrary coherent sheaf $\mathscr{F}$ on X. We use induction on dimension. For $n = 0$ it is clear, so we assume that the statement holds for all schemes of dimension $\leq n - 1$. Thanks to Fujita's vanishing, there exists a very ample Cartier divisor H having the property that $H^i(X, \mathscr{F}(mD + H)) = 0$ for all $i > 0$ and every $m \geq 0$. (Notice here m is the coefficient of this nef Cartier divisor, so the growing thing in Fujita's vanishing is the twisting power of the ample Cartier divisor!) We pick a general Cartier divisor $H' \in |H|$ which doesn't contain any of the subvarieties of X defined by the associated primes of $\mathscr{F}$ and replace H, i.e. we ensure that $\mathscr{F}(mD) \xrightarrow{\cdot H} \mathscr{F}(mD+H)$ must be injective (see remark). We have the exact sequence

$$0 \longrightarrow \mathscr{F}(mD) \xrightarrow{\cdot H} \mathscr{F}(mD + H) \longrightarrow \mathscr{F}(mD + H) \otimes \mathscr{O}_H \longrightarrow 0.$$

Therefore when $i \geq 1$ and $m \geq m(\mathscr{F}, H)$, we have

$$h^i\big(X, \mathscr{F}(mD)\big) \leq h^{i-1}\big(H, \mathscr{F}(mD+H) \otimes \mathscr{O}_H\big) = O(m^{(n-1)-(i-1)}),$$

in particular for $i = 1$ it is lesser than, and for $i \geq 2$ it is exactly equal, as desired. □

Remark 1.4.45. For

$$\mathscr{F}(mD) \xrightarrow{\cdot H'} \mathscr{F}(mD+H),$$

if a subvariety defined by an associated prime $\mathfrak{p}$ is contained in H, i.e. locally we see $V(\mathfrak{p}) \subseteq H|_U$ for which $U \subseteq X$ such that $\mathscr{F}|_U \cong \widetilde{M}$ where M is a finitely generated $\mathscr{O}_U$-module. Say H is locally cut out by single h, then $h \in \mathfrak{p}$. But the homomorphism

$$\mathscr{F}(mD)|_U \cong \widetilde{M} \otimes \mathscr{O}_U \cdot \frac{1}{f^m} \xrightarrow{\cdot H|_U} \mathscr{F}(mD+H)|_U \cong \widetilde{M} \otimes \mathscr{O}_U \cdot \frac{1}{f^m} \otimes \mathscr{O}_U \cdot \frac{1}{h}$$

which concretely is

$$x \otimes \frac{1}{f^m} \longmapsto h\left(x \otimes \frac{1}{f^m} \otimes \frac{1}{h}\right) = x \otimes \frac{1}{f^m},$$

we expect it should be injective. So if $h \in \mathfrak{p}$ then $\cdot h$ will kill the element x.

Corollary 1.4.46. (Asymptotic Riemann-Roch for nef Cartier divisors) *Let X be a projective variety or scheme of dimension n, and let D be a nef Cartier divisor on X. Then*

$$h^0\big(X, \mathscr{O}_X(mD)\big) = \frac{(D^n)}{n!} \cdot m^n + O(m^{n-1}).$$

More generally,

$$h^0\big(X, \mathscr{F} \otimes \mathscr{O}_X(mD)\big) = \mathrm{rk}(\mathscr{F}) \cdot \frac{(D^n)}{n!} \cdot m^n + O(m^{n-1})$$

for any coherent sheaf $\mathscr{F}$ on X.

1.5 Ampleness in Relative Version

The ampleness can be a relative concept, and relative is a distinctive philosophy from Grothendieck.

We consider a proper mapping $f : X \longrightarrow T$ of schemes, and a coherent sheaf $\mathscr{F}$ on X. As in [Har77, III, 8.8.1] then $f_*\mathscr{F}$ is a coherent sheaf on T, so one can form the T-scheme

$$\pi : \mathbb{P}(f_*\mathscr{F}) := \underline{\mathrm{Proj}}_T \mathrm{Sym}(f_*\mathscr{F}) \longrightarrow T$$

whose fibre over a given point $t \in T$ is the projective space of one-dimensional quotients of the fibre $f_*\mathscr{F} \otimes \mathbb{C}(t)$. This is the analogue in the relative setting of the projective space of sections of a sheaf on a fixed complete scheme, see Example 4.4. Moreover if $f_*\mathscr{F}$ is free of rank $n+1$ over an open set $U \subseteq T$, then $\pi^{-1}(U) \cong \mathbb{P}^n_U$. And there is a natural mapping $f^*f_*\mathscr{F} \longrightarrow \mathscr{F}$ whose surjectivity is the analogue of the global generation of $\mathscr{F}$ in the absolute situation, which indeed, when the base scheme is a single point $\mathrm{Spec}\,\mathbb{C}$, gives $f^*f_*\mathscr{F} = f^*\widetilde{H^0(X,\mathscr{F})} = \widetilde{H^0(X,\mathscr{F})} \underset{\mathbb{C}}{\otimes} \mathscr{O}_X$, corresponding to the classical evaluation map.

1.5A Relative Ampleness

Definition 1.5.1. (Ampleness for a map) Let $f : X \longrightarrow T$ be a proper morphism of schemes or varieties, and let L be a line bundle on X.

(i) L is *very ample relative to* f, or *f-very ample*, if the canonical map

$$\rho : f^*f_*L \longrightarrow L$$

is surjective (while now we call it is f-free) and defines an embedding

$$\begin{array}{ccc} X & \xhookrightarrow{\ j\ } & \mathbb{P}(f_*L) \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle \pi} & \\ & T & \end{array}$$

of schemes over T.

(ii) L is *ample relative to* f, or *f-ample*, if $L^{\otimes m}$ is f-very ample for some $m > 0$.

A Cartier divisor D on X is *very ample for* f if the corresponding line bundle is, and f-ampleness for Cartier $\mathbb{Q}$-Cartier divisors is defined by clearing denominators.

Remark 1.5.2. (Relatively ampleness is local on the base) Equivalently, f-very ampleness is depend on an open covering $\{U_\alpha\}$ of T, that is for all α, get an embedding j_α

$$\begin{array}{ccc} X_\alpha := f^{-1}(U_\alpha) & \xrightarrow{\quad j_\alpha \quad} & P_\alpha := \mathbb{P}^{n_\alpha}_{U_\alpha} \\ & {\scriptstyle f}\searrow \quad \swarrow & \\ & U_\alpha & \end{array}$$

over U_α such that $L|_{X_\alpha} \cong j_\alpha^* \mathscr{O}_{P_\alpha}(1)$ as in the absolute sense. See [GW20, 13.56 and 13.63]. Moreover, since our schemes are all assumed as separated algebraic schemes, the existence of f-very ampleness line bundle gives that the proper morphism f is projective. One can see [SP25, Tag 01VV and 01W7], where need schemes are qcqs.

Example 1.5.3. If E is a vector bundle on a scheme T, then the Serre line bundle $\mathscr{O}_{\mathbb{P}(E)}(1)$ on $X = \mathbb{P}(E)$ is π-ample for the natural morphism $\pi : \mathbb{P}(E) \longrightarrow T$.

The cohomological properties of relative ampleness are the following:

Theorem 1.5.4. *Let $f : X \longrightarrow T$ be a proper morphism of schemes, and L a line bundle on X. Then the following are equivalent:*

(i) L is ample for f.

(ii) Given any coherent sheaf $\mathscr{F}$ on X, there exists a $m_1 = m_1(\mathscr{F}) \in \mathbb{Z}_{>0}$ such that

$$R^i f_*(\mathscr{F} \otimes L^{\otimes m}) = 0 \text{ for all } i > 0 \ , \ m \geq m_1(\mathscr{F})$$

(iii) Given any coherent sheaf $\mathscr{F}$ on X, there is a $m_2 = m_2(\mathscr{F})$ such that the canonical mapping

$$f^* f_*(\mathscr{F} \otimes L^{\otimes m}) \longrightarrow \mathscr{F} \otimes L^{\otimes m}$$

is surjective whenever $m \geq m_3$.

(iv) There is a $m_3 \in \mathbb{Z}_{>0}$ such that $L^{\otimes m}$ is f-very ample for every $m \geq m_3$.

Example 1.5.5. Suppose given a diagram

$$\begin{array}{ccc} Y & \xrightarrow{\quad \mu \quad} & X \\ & {\scriptstyle g}\searrow \quad \swarrow{\scriptstyle f} & \\ & T & \end{array}$$

of schemes over T, with μ being finite. If L is an f-ample line bundle on X, then $\mu^* L$ is ample relative to g.

Proof. Since L is f-ample, then there is an open covering $\{U_i\}$ of T such that $L|_{f^{-1}U_i}$ is an ample line bundle in the absolute sense. But as the restriction

$$\mu|_{g^{-1}U_i} : \mu^{-1} \circ f^{-1}(U_i) = g^{-1}(U_i) \longrightarrow f^{-1}(U_i)$$

is finite and surjective. So by finite pullbacks property we see $\mu^* L|_{g^{-1}U_i}$ is absolutely ample. We see $\mu^* L$ is g-ample. □

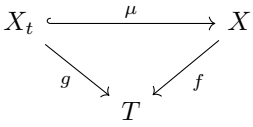
The following useful result allows one in practice to reduce to the absolute setting:

Theorem 1.5.6. (Fibre-wise ampleness) *Let $f : X \longrightarrow T$ be a proper morphism of schemes, let L be a line bundle on X, and for a closed point $t \in T$ we set*

$$X_t = f^{-1}(t) \ , \ L_t = L|_{X_t}.$$

Then L is ample for f if and only if L_t is ample on X_t for every $t \in T$.

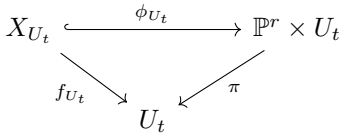
Proof. There is a commutative diagram



where μ is a closed immersion, hence finite. By the previous example, $\mu^* L = L|_{X_t}$ is g-ample. But g sends X_t to a single closed point, so this relative ampleness is nothing but the absolute meaning. Conversely, suppose that L_t is ample for every $t \in T$. By the last step of the proof of ampleness in families, every point $t \in T$ has a neighborhood U_t with the property that

$$L^{\otimes m_t}|_{f^{-1}U_t} = \phi^* \big(\mathscr{O}_{\mathbb{P}^r \times U_t}(1) \big)$$

for a finite morphism $X_{U_t} := f^{-1}(U_t) \xrightarrow{\phi_{U_t}} \mathbb{P}^r \times U_t$ of schemes over U_t. It follows from the previous example, that for



with $\mathscr{O}_{\mathbb{P}^r \times U_t}(1)$ is π-ample, so $L|_{f^{-1}U_t}$ is f_{U_t}-ample on each U_t of $t \in T$. Recalling that f-ampleness is local on T, we see L is indeed ample relative to f. □

Corollary 1.5.7. (Nakai-Moishezon criterion in relative version) *In the setting of the previous theorem, a $\mathbb{Q}$-Cartier divisor D on X is ample with respect to f if and only if $(D^{\dim V} \cdot V) > 0$ for every irreducible subvariety $V \subseteq X$ of positive dimension that* ***maps to a closed point*** *in T.*

The next result gives a connection between relative and absolute ampleness.

Proposition 1.5.8. *Consider a morphism*

$$f : X \longrightarrow T$$

of projective schemes. Let L be a line bundle on X, and let A be an (absolute) ample line bundle on T. Then L is f-ample if and only if $L \otimes f^*(A^{\otimes m})$ is an (absolute) ample line bundle on X for all $m \gg 0$.

Proof. "⇒": Assume that L is f-ample. Up to replacing L by its higher power, we can suppose that $f^*f_*L \longrightarrow L$ is surjective. Since A is ample, $f_*L \otimes A^{\otimes p}$ is globally generated for p sufficiently large. Therefore its pullback $f^*f_*L \otimes f^*A^{\otimes p}$ is likewise globally generated since, recalling that the pullback f^* is right exact (on evaluation map). So we choose generators gives rise to a morphism

$$\phi : X \longrightarrow \mathbb{P} \times T$$

of schemes over T, where $\mathbb{P} = \mathbb{P}H^0(T, f_*L \otimes A^{\otimes p})$ and ϕ is induced from the universal property of the fibre product. And as $H^0(T, f_*L \otimes A^{\otimes p}) \cong H^0(X, L \otimes f^*A^{\otimes p})$ thanks to the projection formula, we have $L \otimes f^*A^{\otimes p} \cong \phi^* \operatorname{pr}_1^* \mathscr{O}_{\mathbb{P}}(1)$. Moreover, ϕ is finite since it is evidently so on each fibre of f. Therefore

$$L \otimes f^*A^{\otimes p+1} = \phi^*(\mathscr{O}_{\mathbb{P}}(1) \boxtimes A)$$

is the pullback of an ample line bundle under a finite map, and consequently is ample.

"⇐ ": Easily to deduce from the Theorem 5.6, since we have

$$\big(L \otimes f^*(A^{\otimes m})\big)_t \cong \big(L_t \otimes f^*(A_t^{\otimes m})\big) \cong L_t,$$

where f pulls back a trivial line bundle. This direction just holds for some fixed m. □

1.5B Intersections in Relative Version

In Corollary 5.7 we proved the relative ampleness can be characterized by take the intersection with fibre varieties. This gives us a motivation to talk with the intersection situation in relative case. Let $f : X \longrightarrow T$ be a proper morphism of schemes.

Definition 1.5.9. (Relative subvariety) A closed subvariety Z of X is called a *relative subvariety* over T if $f(Z)$ is a closed point of T.

Definition 1.5.10. (Relative numerically equivalent) Two Cartier divisors $D_1, D_2 \in \mathrm{Div}(X)$ are *relatively numerically equivalent*, written by $D_1 \equiv_T D_2$ or $D_1 \equiv_f D_2$ if

$$(D_1 \cdot C) = (D_2 \cdot C) \text{ for every relative curve } C \subseteq X.$$

A Cartier divisor $D \in \mathrm{Div}(X)$ is *relatively numerically trivial* if $D \equiv_T 0$, these Cartier divisors from a subgroup of $\mathrm{Div}(X)$, denoted by $\mathrm{Num}(X/T)$.

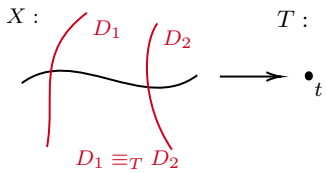


Figure 1.21.: Relatively numerically equivalent.

Definition 1.5.11. (Relative Néron-Severi group) The *relative Néron-Severi group* of proper morphism $f : X \longrightarrow T$ is the group

$$N^1(X/T) := \mathrm{Div}(X)/\mathrm{Num}(X/T)$$

of relatively numerically equivalent classes of Cartier divisors on X. Or usually denote it by $NS(X/T)$ in some reference.

Since numerically trivial is on all curves, must contains the case of relative curves, so we have $\mathrm{Num}(X) \subseteq \mathrm{Num}(X/T)$. Thus, we have a surjective map

$$N^1(X) = \mathrm{Div}(X)/\mathrm{Num}(X) \longrightarrow N^1(X/T) = \mathrm{Div}(X)/\mathrm{Num}(X) \cong \frac{\mathrm{Div}(X)/\mathrm{Num}(X)}{\mathrm{Num}(X/T)/\mathrm{Num}(X)}.$$

We can see that $\rho(X/T) := \mathrm{rk}\, N^1(X/T)$ is also finite.

Definition 1.5.12. (For $\mathbb{R}$-Cartier divisors) Two $\mathbb{R}$-Cartier divisors $D_1, D_2 \in \mathrm{Div}_{\mathbb{R}}(X)$ are said to be *relatively numerically equivalent*, which also denote by $D_1 \equiv_{T,\mathbb{R}} D_2$ or just $D_1 \equiv_T D_2$, if

$$(D_1 \cdot C) = (D_2 \cdot C) \text{ for every relative curve.}$$

Where the intersection is act on integral Cartier divisors and take out the real coefficients.

Definition 1.5.13. (Relatively numerical equivalence class of curves) Let $f : X \longrightarrow T$ be a proper morphism from an algebraic variety. We denote by $Z_1(X/T)_{\mathbb{R}}$ the

$\mathbb{R}$-vector space of *real relative one-cycles* on X, consisting of all finite $\mathbb{R}$-linear combinations of relatives irreducible curves on X:

$$\gamma = \sum a_i \cdot C_i$$

for $\gamma \in Z_1(X/T)$ where $a_i \in \mathbb{R}$ and $C_i \subseteq X$ is an irreducible curve sends to a closed point in T.

Two relative one-cycles $\gamma_1, \gamma_2 \in Z_1(X/T)_{\mathbb{R}}$ are *numerically equivalent*, denoted by $\gamma_1 \equiv_T \gamma_2$, if

$$(D \cdot \gamma_1) = (D \cdot \gamma_2)$$

for every $D \in \mathrm{Div}(X)$.

The corresponding real vector space of *numerical equivalent classes* of relative one-cycles is written as $N_1(X/T)_{\mathbb{R}}$. Thus by construction we have a perfect pairing

$$\begin{aligned} N^1(X/T)_{\mathbb{R}} \times N_1(X/T)_{\mathbb{R}} &\longrightarrow \mathbb{R} \\ (\delta, \gamma) &\longmapsto (\delta \cdot \gamma) \in \mathbb{R}. \end{aligned}$$

In particular, $N_1(X/T)_{\mathbb{R}}$ is of finite dimensional real vector space on which the standard Euclidean topology. For numerical equivalence of relative integral and rational one-cycles are defined similarly.

For another proper morphism $g : Y \longrightarrow X$ over T from another algebraic variety, and for a relative subvariety $Z \subseteq Y$ over T. We define its *push-out* g_*Z as follows:

- If g contracts Z, i.e. $\dim g(Z) < \dim Z$: $g_*Z = 0$;
- If $\dim g(Z) = \dim Z$: $g_*Z = dZ$, where d is the degree of the morphism $Z \longrightarrow g(Z)$.

Similarly we have the *projection formula*.

Proposition 1.5.14. *Let $f : X \longrightarrow T$ and $g : Y \longrightarrow X$ be two proper morphisms of schemes. Consider the pullback g^*L of a line bundle $L \in \mathrm{Pic}(X)$.*

(1) If $L \equiv_T 0$, then $g^*L \equiv_T 0$. Therefore g induces a natural linear map $g^* : N^1(X/T) \longrightarrow N^1(Y/T)$.

(2) Where g is surjective and conversely if $g^*L \equiv_T 0$, then $L \equiv_T 0$. Hence the linear map g^* is injective.

This obviously holds when $T = \mathrm{Spec}\,\mathbb{C}$.

Proof. (1): For any relative curve C on Y, by the projection formula we have

$$(g^*L \cdot C) = (L \cdot g_*C) = d(L \cdot g(C)) = 0,$$

so $g^*L \equiv_T 0$.

(2): If g is surjective, then for any relative curve C on X, there exist a relative curve C' on Y such that $C = g(C')$. Hence

$$(L \cdot C) = \frac{1}{d}(L \cdot g_*C') = \frac{1}{d}(g^*L \cdot C') = 0,$$

which proves the assertion. □

Definition 1.5.15. Let $f : X \longrightarrow T$ be proper morphism of schemes. We say an $\mathbb{R}$-Cartier divisor D on X is f-ample if D is a finite $\mathbb{R}_{>0}$-linear combination of f-ample integral Cartier divisors on X. Similarly for $\mathbb{Q}$-Cartier divisor case.

This definition is compatible for T is just a point.

Proposition 1.5.16. (Openness for f-ampleness for $\mathbb{R}$-Cartier divisors) *Let $f : X \longrightarrow T$ be a proper morphism of projective varieties. And H be a f-ample $\mathbb{R}$-Cartier divisors on X. Given finitely many $\mathbb{R}$-Cartier divisors $E_1, \ldots, E_r$, the $\mathbb{R}$-Cartier divisor*

$$H + \varepsilon_1 E_1 + \cdots + \varepsilon_r E_r$$

is f-ample for all sufficiently small real numbers $0 \leq |\varepsilon_i| \ll 1$.

Proof. **Step 1:** First we prove it for $\mathbb{Q}$-Cartier divisor case. By clearing denominators, we may assume that H and E_i are integral. By taking $m \gg 0$, we can arrange for each of the $2r$ Cartier divisors $mH \pm E_1, \ldots mH \pm E_r$ to be ample. That is because from Proposition 5.8, we can turn it into absolute case. For A is an ample Cartier divisor on T, we have $H + nf^*A$ is ample for all sufficiently large n. So by Example 2.5 we see

$$m(H + nf^*A) + E$$

is ample, for sufficiently large m depending on n. Therefore we can use Proposition 5.8 again, applying on

$$m(H + nf^*A) + E = (mH + E) + mnf^*A,$$

which the ampleness gives that $mH + E$ is f-ample. Moreover as we are dealing with finitely many Cartier divisors $\{\pm E_i\}_1^r$, so take sufficiently large m can holds that for these $2r$ Cartier divisors.

Now provided that rational numbers $|\varepsilon_i| \ll 1$, any Cartier divisor of the form

$$H + \varepsilon_1 E_1 + \cdots + \varepsilon_r E_r$$

can be written as a positive $\mathbb{Q}$-linear combination of H and some of the $\mathbb{Q}$-Cartier divisors $H \pm \frac{1}{m} E_i = \frac{1}{m}(mH \pm E_i)$ which are f-ample. But a positive linear combination of f-ample $\mathbb{Q}$-Cartier divisors is f-ample, by definition.

Step 2: Now we turn to the general case. We write $H = \sum a_i A_i$ with $a_i \in \mathbb{R}_{>0}$ and A_i are f-ample and integral Cartier divisors. Without loss of generality we can assume that all the E_i are integral. We fix a rational number $0 < c < c_1$, then

$$H + \sum \varepsilon_i E_i = c(A_1 + \sum \varepsilon_i E_i) + (c_1 - c)A_1 + \sum_{i \geq 2} c_i A_i.$$

Since we just need all ε_i to be sufficiently small to exhibit the openness, we can assume they are all rational. The first term on the right hand side is governed by the case of $\mathbb{Q}$-Cartier divisors, and remaining summands are indeed f-ample. □

1.5C Nefness in Relative Version

Taking the analogue of Theorem 5.6 we say a few words about the relative version of nefness:

Definition 1.5.17. (Nefness in relative version) Given a proper morphism $f : X \longrightarrow T$ as above, a line bundle L on X is *nef relative to* f if the restrictions $L_t = L|_{X_t}$ of L to each fibre are nef, or equivalently if $\int_C c_1(L) \geq 0$ for every curve $C \subseteq X$ **mapping to a point** under T.

Remark 1.5.18. Everything in this section is defined over a field of arbitrary characteristic.

Theorem 1.5.19. (Kleiman's theorem in relative version) *Let $f : X \longrightarrow T$ be a proper morphism of schemes or varieties. If D is a f-nef $\mathbb{R}$-Cartier divisor on X, then*

$$(D^k \cdot V) \geq 0$$

for every relative subvariety $V \subseteq X$ of dimension k. Similarly,

$$\int_V c_1(L)^{\dim V} \geq 0$$

for every f-nef line bundle on X.

Proof. For relative subvariety V such that $f(V) = t \in T$, obviously we have

$$(D^k \cdot V)_X = (D_t^k \cdot V)_{X_t} \geq 0$$

for $D_t = D|_{X_t}$ is an $\mathbb{R}$-nef Cartier divisor on the complete scheme X_t, where the right hand side follows from the absolute version of Kleiman's theorem. □

Corollary 1.5.20. *Let $f : X \longrightarrow T$ be a projective morphism of varieties or schemes, and D is a f-nef $\mathbb{R}$-Cartier divisor on X. If H is any f-ample $\mathbb{R}$-Cartier divisor on X, then*

$$D + \varepsilon \cdot H$$

is f-ample for every $\varepsilon > 0$.

Conversely, if D and H are any two Cartier divisors such that $D+\varepsilon H$ is f-ample for all sufficiently small $\varepsilon > 0$, then D is f-nef.

Proof. "$\Leftarrow$": If $D+\varepsilon H$ is f-ample for $\varepsilon > 0$, then for every relative subvariety $V \subseteq X$, we have

$$\big((D + \varepsilon H)^{\dim V} \cdot V\big) > 0$$

by Nakai-Moishezon criterion in relative version. In particular,

$$\big((D \cdot C) + \varepsilon(H \cdot C)\big) = \big((D + \varepsilon H) \cdot C\big) > 0$$

for every irreducible relative curve C. Letting $\varepsilon \to 0$ it follows that $(D \cdot C) \geq 0$, and hence D is f-nef.

"$\Rightarrow$": Similarly, for relative subvariety V such that $f(V) = t \in T$, we have

$$\big((D + \varepsilon H)^{\dim V} \cdot V\big)_X = \big((D + \varepsilon H)^{\dim V} \cdot V\big)_{X_t} > 0$$

for every $\varepsilon > 0$, where $D + \varepsilon H$ is ample on the projective scheme X_t applying the Corollary 4.7 and NMK criterion. So by relative Nakai-Moishezon criterion, we see $D + \varepsilon H$ is f-ample for every $\varepsilon > 0$. □

The reader can similarly verify the remaining propositions in Section 4, since everything can be replaced by the corresponding relative subvarieties.

1.5D Kleiman's Criterion in Relative Version

Definition 1.5.21. (Relative ample and relative nef cone) For $f : X \longrightarrow T$ be a proper morphism of schemes. The *relative ample cone*

$$\mathrm{Amp}(X/T) \subseteq N^1(X/T)_{\mathbb{R}}$$

of X is the convex cone spanned by all f-ample $\mathbb{R}$-Cartier divisor classes on X.

The *relative nef cone*

$$\mathrm{Nef}(X/T) \subseteq N^1(X/T)_{\mathbb{R}}$$

is the convex cone of all f-nef $\mathbb{R}$-Cartier divisor classes.

Concretely, $\mathrm{Amp}(X/T)$ is spanned by the classes of all f-ample integral (or rational) Cartier divisors with positive real coefficients: $\sum a_i\delta_i$ for all $a_i \in \mathbb{R}_{>0}$ and $\delta_i \in N^1(X/T)$ or $N^1(X/T)_{\mathbb{Q}}$.

Theorem 1.5.22. *Let $f : X \longrightarrow T$ be a projective morphism of varieties or schemes.*

(1) The relative nef cone is the closure of the relative ample cone:

$$\mathrm{Nef}(X/T) = \overline{\mathrm{Amp}}(X/T).$$

(2) The relative ample cone is the interior of the relative nef cone:

$$\mathrm{Amp}(X/T) = \mathrm{int}\big(\mathrm{Nef}(X/T)\big).$$

Proof. As taking limits preserves signs, it is evident that the relative nef cone is closed, and follows from Proposition 5.16 that $\mathrm{Amp}(X/T)$ is open. This gives the inclusion

$$\overline{\mathrm{Amp}}(X/T) \subseteq \mathrm{Nef}(X/T) \quad \text{and} \quad \mathrm{Amp}(X/T) \subseteq \mathrm{int}\big(\mathrm{Nef}(X/T)\big).$$

For the converse inclusions

$$\mathrm{Nef}(X/T) \subseteq \overline{\mathrm{Amp}}(X/T) \quad \text{and} \quad \mathrm{int}\big(\mathrm{Nef}(X/T)\big) \subseteq \mathrm{Amp}(X/T).$$

To establish the first inclusion, we let H be a f-ample Cartier divisor on X. Picking any f-nef $\mathbb{R}$-Cartier divisor D on the left, Corollary 5.20 shows that $D+\varepsilon H$ is f-ample for all $\varepsilon > 0$. Letting $\varepsilon \to 0$, we see that D is a limit of f-ample Cartier divisors.

For the second, observe that as a cone, if the class of D lies in $\mathrm{int}\big(\mathrm{Nef}(X/T)\big)$, then $D - \varepsilon H$ remains f-nef for $0 < \varepsilon \ll 1$. Consequently

$$D = (D - \varepsilon H) + \varepsilon H$$

is f-ample again by Corollary 5.20. □

The above merely constitutes the book-keeping for the proof of Theorem 4.21.

Definition 1.5.23. (Relative cone of curves) Let $f : X \longrightarrow T$ be a proper morphism of varieties. The *relative cone of curves*

$$\mathrm{NE}(X/T) \subseteq N_1(X/T)_{\mathbb{R}}$$

is the convex cone spanned by the classes of all effective relative one-cycles on X.

Concretely,

$$\mathrm{NE}(X/T) = \Big\{ \sum a_i[C_i] \Big| \, C_i \subseteq X \text{ an irreducible curve contracted by } f, a_i \geq 0 \Big\}.$$

Its closure

$$\overline{\mathrm{NE}}(X/T) \subseteq N_1(X/T)_{\mathbb{R}}$$

is the *closed cone of relative curves* on X.

Remark 1.5.24. $\mathrm{NE}(X/T)$ also might not be closed.

Then we similarly can prove that $\overline{\mathrm{NE}}(X/T)$ and $\mathrm{Nef}(X/T)$ are dual:

Proposition 1.5.25. *$\overline{\mathrm{NE}}(X/T)$ is the closed cone dual to $\mathrm{Nef}(X/T)$, i.e.*

$$\overline{\mathrm{NE}}(X/T) = \big\{ \gamma \in N_1(X/T)_{\mathbb{R}} \;\big|\; (\delta \cdot \gamma) \geq 0 \text{ for all } \delta \in \mathrm{Nef}(X/T) \big\}.$$

Now we turn to the relative Kleiman's criterion:

Theorem 1.5.26. (Kleiman's criterion in relative version) *Let $f : X \longrightarrow T$ be a projective morphism of varieties or schemes, and let D be an $\mathbb{R}$-Cartier divisor on X. Then D is f-ample if and only if*

$$\overline{\mathrm{NE}}(X/T) - \{0\} \subseteq D_{>0}.$$

Moreover, for any f-ample Cartier divisor H and any integer k, the set

$$\big\{ \gamma \in \overline{\mathrm{NE}}(X/T) \big| (H \cdot \gamma) \leq k \big\}$$

is compact, hence contains finitely many classes of effective relative one-cycles $Z = \sum a_i Z_i$ with integral coefficients such that $(Z \cdot H) \leq k$.

Proof. "$\Rightarrow$ ": Assume that D is f-ample, by Nakai inequalities in relative version (easy to verify!), clearly we have $(D \cdot \gamma) \geq 0$ for all non-zero $\gamma \in \overline{\mathrm{NE}}(X/T)$. Now assume that $(D \cdot \gamma) = 0$ and $\gamma \neq 0$. Since this pairing is perfect which is non-degenerate, there is an $\mathbb{R}$-Cartier divisor E such that $(E \cdot \gamma) < 0$, hence $\big((D + tE) \cdot \gamma\big) < 0$ for all $t > 0$. In particular, $D + tE$ can not be f-ample, otherwise will contradicts Proposition 5.16.

"⇐": Assume that D is positive on $\overline{\mathrm{NE}}(X/T) - \{0\}$. We choose a norm $\|\cdot\|$ on $N_1(X/T)_{\mathbb{R}}$. The set

$$S := \big\{\gamma \in \overline{\mathrm{NE}}(X/T)\big|\|\gamma\| = 1\big\}$$

is compact as an half-sphere in a finite-dimensional real vector space. The linear functional $\phi_D : \gamma \mapsto (D \cdot \gamma)$ is positive on S, so it is bounded below by $a \in \mathbb{Q}_{>0}$ due to compactness:

$$\phi_D(\gamma) \geq a \text{ for } \forall \gamma \in S.$$

For a f-ample Cartier divisor H on X, ϕ_H os bounded above by $b \in \mathbb{Q}_{>0}$, also due to compactness:

$$\phi_H(\gamma) \leq b \text{ for } \forall \gamma \in S.$$

It follows that $\big((D - \frac{a}{b}H) \cdot \gamma\big) = (D \cdot \gamma - \frac{a}{b} \cdot \gamma) \geq 0$ on S, hence on $\overline{\mathrm{NE}}(X/T)$, which is equivalent to saying that $D - \frac{a}{b}H$ is nef. Then by Corollary 5.20

$$D = (D - \frac{a}{b}H) + \frac{a}{b}H$$

is f-ample.

Now we prove the second statement.

We fix a norm $\|\cdot\|$ on $N_1(X/T)_{\mathbb{R}}$ and assume that

$$W_k := \big\{\gamma \in \overline{\mathrm{NE}}(X/T)\big|(H \cdot \gamma) \leq k\big\}$$

is not compact. Since under the Euclidean topology of the finite dimensional real vector space, compactness is equivalent to being closed and bounded, we can find a sequence $\{\gamma_i\} \subseteq W_k$ such that

$$\lim_{i\to\infty} \|\gamma_i\| \to \infty.$$

As $\gamma_i/\|\gamma_i\|$ is a bounded sequence, we can find a converge subsequence $\{\gamma_j'\|\gamma_j'\|\}$ which converges to $\gamma \in \overline{\mathrm{NE}}(X/T) - \{0\}$, with $\|\gamma\| = \lim_{j\to\infty} \|\frac{\gamma_j/}{\|\gamma_j'\|}\| = 1$. But

$$(H \cdot \gamma) = \lim_{j\to\infty} \frac{(H \cdot \gamma_j')}{\|\gamma_j'\|} = 0.$$

Since for $\gamma_j' \in W_k$, we have $(H \cdot \gamma_j') \leq k$ while $\|\gamma_j'\| \to \infty$. Thus contradicts the fact that H is f-ample.

Finally, relative one-cycles with integral coefficients correspond to a discrete set in $N_1(X/T)_{\mathbb{R}}$, and so it has only finitely many points in any compact set. □

In Proposition 5.14 we discuss the pullback map between the finite vector space N^1, and now we turn to its dual space N_1 and the pushforwards:

Proposition 1.5.27. *Let $f : X \longrightarrow Y$ be a morphism of projective varieties. Then the following hold.*

(1) f induces an $\mathbb{R}$-linear map $f_* : N_1(X) \longrightarrow N_1(Y)$.

(2) $f_*\mathrm{NE}(X) \subseteq \mathrm{NE}(Y)$ and $f_*\overline{\mathrm{NE}}(X) \subseteq \overline{\mathrm{NE}}(Y)$.

(3) $N_1(X/Y) \subsetneq \ker f_*$, and the equality does not hold in general.

(4) $\overline{\mathrm{NE}}(X/Y) = \overline{\mathrm{NE}}(X) \cap N_1(X/Y)$.

Proof. (1): It is due to the definition of the pushforward. Moreover, it is well-defined by the projection formula, that is f_* is compatible with the numerical equivalence relation.

(2): Since $f_*(\sum a_i[C_i]) = \sum a_i f_*[C_i]$ by f_* is $\mathbb{R}$-linear, and this map is obviously compatible with taking limit. Thus we tale closure also will be fine.

(3): Trivial, by the definition. The counterexample can see [KM98, 1.46].

(4): One direction "$\subseteq$" is obviously. Now assume $\overline{\mathrm{NE}}(X/Y) \subsetneq \overline{\mathrm{NE}}(X) \cap N_1(X/Y)$. We derive a contradiction. There exists a linear function $(H \cdot -)$ on $N_1(X/Y)$ such that $H_{>0} \supseteq \overline{\mathrm{NE}}(X/Y)\backslash\{0\}$ but $(H \cdot \xi) < 0$ for some $\xi \in \overline{\mathrm{NE}}(X) \cap N_1(X/Y)$, by the separating hyperplane theorem, see [BV04, 2.5.1]. Perturbing the coefficients of H a little, we can assume that H is a $\mathbb{Q}$-Cartier divisor on X. The relative Kleiman's criterion of ampleness implies that H is f-ample. Hence by Proposition 5.8, $H + f^*M$ is ample on X for some ample Cartier divisor M on Y. Then

$$0 < \big((H + f^*M) \cdot \xi\big) = (H \cdot \xi) < 0,$$

gets a contradiction. Thus (4) proved.

□

Chapter 2

Rational Curves & Bend and Break

2.1 Parametrizing Morphisms

2.1A Rational Curves on Exceptional locus

> The more rational curves X contains, the more complicated the birational geometry of X.
>
> *János Kollár*

Rational curves play an important role in classifying algebraic varieties. For example, for smooth projective surfaces, Corollary 1.4.34 says that there is a **minimal surface** X_m with no (-1)-curves. But there are two cases:

- either X is not covered by rational curves, in which case the minimal surface X_m is unique and K_{X_m} is nef.
- or else X is covered by rational curves, the minimal surface X_m is not unique and K_{X_m} is not nef.

[Bea78, II, 14(b)] gives a simple example for the second case:

Starting with $\mathbb{P}^2$ blown up at 2 points to get $\hat{Q}$, then blow downs the strict transform of the line joining them to get a smooth quadric surface $\mathbb{P}^1 \times \mathbb{P}^1$. The surfaces $\mathbb{P}^2$ and $\mathbb{P}^1 \times \mathbb{P}^1$ are two distinct birational minimal surfaces.

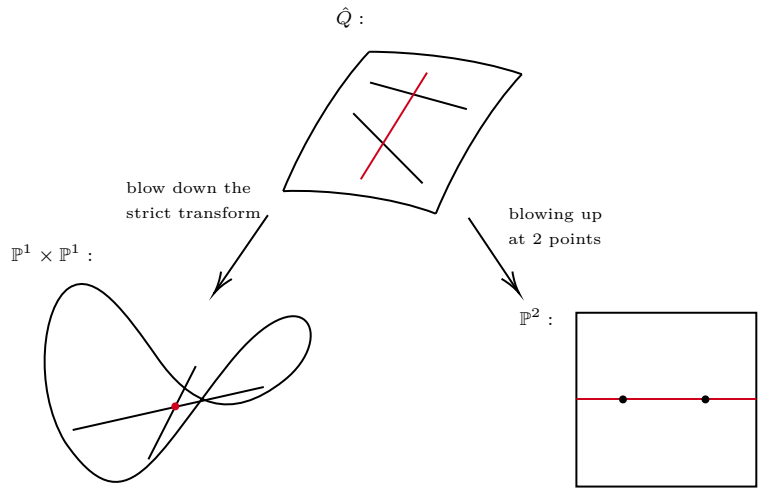


Figure 2.1.: Two distinct minimal surfaces.

Notice that even [Bea78] is over $\mathbb{C}$, all the cited results can hold over $\mathbf{k}$ by [Har77].

Next we follow [Deb01, 1.10], and in this chapter all varieties will over arbitrary algebraically closed field.

Definition 2.1.1. Let $\pi : Y \longrightarrow X$ be a birational morphism (i.e. a morphism which is a birational map) of varieties.

The *exceptional locus* $\mathrm{Ex}(\pi)$ of π is the locus of points of Y where π is not a local isomorphism, i.e. π^{-1} is not a morphism at there, and we denote the $\mathrm{Ex}(\pi)$ by E.

Notice that E is a closed subset, since any point p not in E can find a neighborhood U such that π is a local isomorphism on U.

If X is normal and π is proper, the fibres of π are connected by Zariski's connectedness theorem, see [GW23, 24.58].

If $y \in E$ and $\pi^{-1}(\pi(y))$ has at least 2 points, then by connectedness, it has positive dimension everywhere. And $\pi^{-1}(\pi(y)) \subseteq E$ also by connectedness, since if there is a point $p \in \pi^{-1}(\pi(y))$ which is not in E, then near p, π is a local isomorphism making p isolated.

This implies $E = \pi^{-1}(\pi(E))$.

Now by upper semi-continuity of the fibre dimension, see [Har77, II, ex3.22(d)], [GW20, 14.112] or [Sha13, p.75], the set

$$E_1 = \{x \in \mathrm{im}(\pi) = X \mid \dim \pi^{-1}(x) \geq 1\}$$

is closed in Y, which equals to $\pi(E)$ because π is proper.

Suppose that $\pi(E)$ has a codimension 1 irreducible component W, i.e. $\dim W = n-1$ for which $n = \dim X = \dim Y$, and for points $x \in W \subseteq E_1$, it has $\dim \pi^{-1}(x) = d \geq 1$. Then consider the closed subset $\pi^{-1}(W) \subseteq Y$, it has dimension

$$\begin{aligned} n \geq \dim \pi^{-1}(W) &\geq \dim W + d \\ &\geq n. \end{aligned}$$

But as Y is irreducible, so $\pi^{-1}(W) = Y$ by [Har77, I, ex1.10(d)], that is $\pi(Y) \subseteq W$ contradicting the fact that π is surjective.

So $\pi(E)$ must have codimension at least two, and the largest open subset on which $\pi^{-1} : X \dashrightarrow Y$ is defined is $X - \pi(E)$.

Remark 2.1.2. (Proper and Birational is Surjective) In the above statements, we often used that π is surjective. In fact, since $\pi : Y \longrightarrow X$ is proper, $\pi(Y) \subseteq X$ is closed by [Har77, II, ex4.4]. Moreover since π is birational, it induces an isomorphism between dense open subsets $U \subseteq Y$ and $V \subseteq X$, so $V \subseteq \pi(Y) \subseteq X$ implies that $\pi(Y) = X$.

The name of the next important result comes from [EGA IV$_4$, 21.12.12].

Proposition 2.1.3. (Van der Waerden) *With Defn2.1.1 situation, furthermore if X is normal and $\mathbb{Q}$-factorial, that is every $\mathbb{Q}$-divisor is $\mathbb{Q}$-Cartier (e.g. smooth), then every irreducible component of E has codimension 1 in Y.*

Proof. We let $y \in E$ and $x = \pi(y)$, identifying the function fields $k(X)$ and $k(Y)$ by the isomorphism π^*, so that $\mathscr{O}_{X,x}$ is a proper subring of $\mathscr{O}_{Y,y}$.

Let t be an element of m_y but not in $\mathscr{O}_{X,x}$, and write $\mathrm{div}(t) = D' - D''$ as a difference of two effective divisors (on X) without common components.

As X is $\mathbb{Q}$-factorial, there exists an $m \in \mathbb{Z}_{>0}$ such that mD' and mD'' are Cartier divisors, hence locally define elements u and v of $\mathscr{O}_{X,x}$ such that

$$t^m = \frac{u}{v}.$$

We claim that they're both in m_x. In fact, $v \in m_x$ because t^m is not in $\mathscr{O}_{X,x}$, i.e. not defined at point $x \in X$, otherwise t would be in $\mathscr{O}_{X,x}$ by X is normal. so $\mathscr{O}_{X,x}$ is integrally closed. And $u = t^m \cdot v \in m_x$ since it is in $m_y \cap \mathscr{O}_{X,x} = m_x$.

Then the equation $u = v = 0$ defines a subscheme Z containing x, which has codimension 2 in some neighborhood of x, since it is the intersection of two codimension 1 subschemes mD' and mD'' with no common components. So via a pullback, the subscheme $\pi^{-1}(Z)$ passing through point y, is defined by

$$t^m \cdot v = v = 0,$$

hence by the sole equation $v = 0$, which implies that $\pi^{-1}(Z)$ is codimension 1 in Y due to the Hauptidealsatz.

We have now proved that for every point of E, there is a codimension 1 component of E passing through it, which is what we desired.

□

Ramification divisor of a morphism between smooth varieties:

Now we assume that X and Y are smooth. For this morphism $\pi : Y \to X$ with locally free differential sheaves Ω_X and Ω_Y, π induces the morphism:

$$\pi^*\Omega_X \longrightarrow \Omega_Y.$$

If X and Y have the same dimension n, their highest exterior powers induce a morphism:

$$\pi^*\omega_X \longrightarrow \omega_Y$$

of line bundles, then by [Har77, II, ex5.1(b)] this morphism defines a section of

$$\pi^*\omega_X^{-1} \underset{\mathscr{O}_Y}{\otimes} \omega_Y \cong \mathscr{O}_Y(K_Y - \pi^*K_X).$$

Moreover, if this section is not identically zero (Over $\mathbb{C}$ we just need π is surjective), then it vanishes on an effective divisor $\mathrm{Ram}(\pi)$ called the *ramification divisor* of π, which is the hypersurface defined by the zero locus of the Jacobian determinant. In particular,

$$K_Y \equiv \pi^*K_X + \mathrm{Ram}(\pi).$$

and if π is birational morphism, the Support of $\mathrm{Ram}(\pi) = \mathrm{Ex}(\pi)$.

Next we show that exceptional locus can be covered by rational curves, which follows from [Deb01, 1.43] and [KM98, 1.3]:

Proposition 2.1.4. (Abhyankar lemma) *Let X and Y be varieties with X smooth, and let*

$$\pi : Y \longrightarrow X$$

be a proper birational morphism which is not an isomorphism. Through a general point of each component of $Ex(\pi)$, there exists a rational curve contracted by π.

Proof. We set $E = \mathrm{Ex}(\pi)$. Upon replacing Y with its normalization, we may assume that Y is nonsingular in codimension 1. As the previous result Proposition 1.3, each component of E **has codimension 1**, i.e. it is a divisor.

Hence we can shrink Y to assume that E and Y are smooth and irreducible.

Set $U_0 = X - \mathrm{Sing}\pi(E)$, so that the closure of the image of $E \cap \pi^{-1}(U_0)$ in U_0 is smooth, and of codimensio at least 2 by the discussion under the Def 2.1.1.

Now let $\varepsilon_1 : X_1 \longrightarrow U_0$ be the blow up along $\overline{\pi(E \cap \pi^{-1}(U_0)}$ in U_0. By [Har77, II, 7.14] the universal property of blowing up, there is a factorization

$$\begin{array}{ccccc} V_1 := \pi^{-1}(U_0) & \xrightarrow{\exists \pi_1} & X_1 & \hookleftarrow & E_1 := \mathrm{Ex}(\varepsilon_1) \\ & \searrow^{\pi|_{V_1}} & \downarrow{\scriptstyle \varepsilon_1} & & \downarrow \\ & & U_0 & \hookleftarrow & \overline{\pi(E \cap \pi^{-1}(U_0))} \end{array}$$

such that π_1 is a birational morphism since $k(V_1) \cong k(U_0) \cong k(X_1)$, and the complement of $V_1 = \pi^{-1}(U_0)$ is just $\pi^{-1}(\mathrm{Sing}\pi(E))$, which has codimension at least 2 in Y by

$$\begin{aligned} \mathrm{codim}_Y \pi^{-1}(\mathrm{Sing}\pi(E)) &\geq \mathrm{codim}_Y E + \mathrm{codim}_E \pi^{-1}(\mathrm{Sing}\pi(E)) \\ &\geq 1 + \mathrm{codim}_{\pi(E)} \mathrm{Sing}\pi(E) \\ &\geq 1 + 1 = 2 \end{aligned}$$

and $\overline{\pi_1(E \cap V_1)}$ is contained in the support of the $\mathrm{Ex}(\varepsilon_1)$ because the diagram commutes.

If the codimension of $\overline{\pi_1(E \cap V_1)}$ in X_1 is at least 2 and divisor $E \cap V_1$ is contained in $\mathrm{Ex}(\pi_1)$, we can replace X_1 by an open (dense, by [Har77, II, 7.16] X_1 is also a variety) subset $U_1 = X_1 - \mathrm{Sing}\overline{\pi_1(E \cap V_1)}$ to ensure the closure in U_1 of the image of $E \cap \pi^{-1}(U_1)$ is smooth, and replace V_1 by $V_2 = \pi_1^{-1}(U_1)$, which has complement $\pi_1^{-1}(\mathrm{Sing}\overline{\pi_1(E \cap V_1)})$ in Y such that

$$\begin{aligned}\mathrm{codim}_Y \pi_1^{-1}(\mathrm{Sing}\overline{\pi_1(E \cap V_1)}) &\geq \mathrm{codim}_Y E + \mathrm{codim}_E \pi_1^{-1}(\mathrm{Sing}\overline{\pi_1(E \cap V_1)}) \\ &\geq 1 + \mathrm{codim}_{\overline{\pi_1(E)}} \mathrm{Sing}\overline{\pi_1(E)} \\ &\geq 2.\end{aligned}$$

Similarly to the above processes, we can repeat the construction, and after i steps get a factorization:

$$\pi : V_i \xrightarrow{\pi_i} X_i \xrightarrow{\varepsilon_i} U_{i-1} \subseteq X_{i-1} \xrightarrow{\varepsilon_{i-1}} \cdots \xrightarrow{\varepsilon_2} U_1 \subseteq X_1 \xrightarrow{\varepsilon_1} U_0 \subseteq X$$

as long as the codimension of $\overline{\pi_{i-1}(E \cap V_{i-1})}$ in X_{i-1} is at least 2, and where V_i is the complement of a closed subset in Y, of codimension is at least 2. Notice that each ε_i is a blowing up along the nonsingular - by our construction - subvariety $\overline{\pi_{i-1}(E \cap V_{i-1})} = \overline{\pi_{i-1}(E \cap \pi_{i-1}^{-1}(U_{i-2}))}$, so we can use [Har77, II, ex8.5] to get that

$$\begin{aligned}K_{X_i} &= \varepsilon_i^* K_{U_{i-1}} + c_i E_i \\ &= (\varepsilon_1 \circ \cdots \circ \varepsilon_i)^* K_{U_0} + c_i E_i + c_{i-1} E_{i,i-1} + \cdots c_1 E_{i,1}\end{aligned}$$

where $E_j \subseteq X_j$ be the $\mathrm{Ex}(\varepsilon_j)$, where $E_{i,j}$ in the inverse image of E_j in X_i and

$$c_i = \mathrm{codim}_{X_{i-1}} \overline{\pi_{i-1}(E \cap V_{i-1})} - 1 > 0.$$

Since π_i is a birational morphism in each step (via the identification of function fields), and since V_i is an open subset of a smooth variety Y, it is still smooth; thus by the previous discussion we have

$$K_{V_i} \sim_{\mathrm{lin}} \pi_i^* K_{X_i} + \mathrm{Ex}(\pi_i).$$

This implies that $K_{V_i} \geq \pi_i^* K_{X_i}$, so $\pi_i^* \mathscr{O}_{X_i}(K_{X_i})$ is a subsheaf of $\mathscr{O}_{V_i}(K_{V_i})$. Moreover, for each j, since $\pi_j(E \cap V_j) = (\pi_i \circ \varepsilon_i \circ \cdots \varepsilon_{j+1})(E \cap V_j)$ is contained in the support of $E_j = \mathrm{Ex}(\varepsilon_j)$, the divisor $\pi_j^* E_j - E|_{V_j}$ is effective, so is the $\pi_i^* E_{i,j} - E|_{V_i}$.

It follows that

$$\begin{aligned}\pi_i^* K_{X_i} &= \pi_i^*((\varepsilon_1 \circ \cdots \varepsilon_i)^* K_{U_0}) + c_i \pi_i^* E_i + c_{i-1}\pi_i^* E_{i,i-1} + \cdots c_1 \pi_i^* E_{i,1} \\ &= \pi^* K_{U_0} + c_i \pi_i^* E_i + c_{i-1}\pi_i^* E_{i,i-1} + \cdots + c_1 \pi_i^* E_{i,1} \\ &\geq \pi^* K_{U_0} + (c_i + c_{i-1} + \cdots + c_1) E|_{V_i}\end{aligned}$$

and hence

$$K_{V_i} \geq \pi^* K_{U_0} + (c_i + c_{i-1} + \cdots + c_1) E|_{V_i},$$

therefore $\mathscr{O}_Y(\pi^* K_{U_0} + (c_i + c_{i-1} + \cdots + c_1)E)|_{V_i}$ is a subsheaf of $\mathscr{O}_{V_i}(K_{V_i}) = \mathscr{O}_Y(K_Y)|_{V_i}$. Since Y is normal and the complement of V_i in Y has codimension at least 2, then picking affine opens on Y such that they are both trivialize as $\mathscr{O}_U$, where $U \cap V_i$ has complement at least 2 in U,

$$\text{Algebraic Hartogs: } H^0(U, \mathscr{O}_U) \longrightarrow H^0(U \cap V_i, \mathscr{O}_U) \text{ is bijective}$$

which is just saying that the injection

$$\mathscr{O}_Y(\pi^* K_{U_0} + (c_i + c_{i-1} + \cdots + c_1)E)|_{V_i} \hookrightarrow \mathscr{O}_Y(K_Y)|_{V_i}$$

can be extended the whole on Y:

$$\mathscr{O}_Y(\pi^* K_{U_0} + (c_i + c_{i-1} + \cdots + c_1)E) \hookrightarrow \mathscr{O}_Y(K_Y)$$

is still injective.

Since there are no infinite ascending chain

$$\cdots \geq \pi^* K_{U_0} + \sum_{i=1}^{k+1} c_i E_i \geq \pi^* K_{U_0} + \sum_{i=1}^{k} c_i E_i \geq \cdots$$

$$\cdots \hookrightarrow \mathscr{O}_Y(\pi^* K_{U_0} + \sum_{i=1}^{k} c_i E_i) \hookrightarrow \mathscr{O}_Y(\pi^* K_{U_0} + \sum_{i=1}^{k+1} c_i E_i) \hookrightarrow \cdots$$

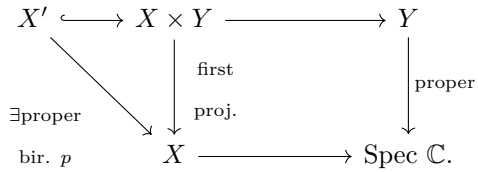
of subsheaves of a coherent sheaf $\mathscr{O}_Y(K_Y)$ on a noetherian scheme Y, the process must terminate at some step such as $\overline{\pi(E \cap V_i)}$ is a divisors in X_i for some i, where $E \cap V_i$ is not contained in $\mathrm{Ex}(\pi_i)$. The $\pi_i|_{E \cap V_i}$ is a birational morphism between $E \cap V_i$ and $E_i = \mathrm{Ex}(\varepsilon_i)$, as note that $E \cap V_i$ is irreducible by E is and V_i open dense. E_i is irreducible by it is the exceptional locus of blow up $\varepsilon_i : X_i \longrightarrow U_{i-1}$ along the irreducible centre $\overline{\pi_{i-1}(E \cap V_{i-1})}$.

Mention that E_i is ruled, since it is locally isomorphic to $\overline{\pi_{i-1}(E \cap V_{i-1})} \times \mathbb{P}^{r-1}$ where $r = \mathrm{codim}_{U_{i-1}} \overline{\pi_{i-1}(E \cap V_{i-1})}$, by [Har77, II, 8.24]. More precisely, through every point of E_i there is a rational curve contracted by ε_i, and after pullback via the birational morphism $\pi_i|_{E \cap V_i}$ over E_i, it is a rational curve which can be contracted by π, which is exactly what we desired. □

The application of this proposition is in [KM98, 1.5]

Corollary 2.1.5. *Let Y and X be varieties. Assume that X is smooth and that Y is proper contains no rational curves. Any rational map $f : X \dashrightarrow Y$ is defined everywhere.*

Proof. Let $X' \subseteq X \times Y$ be the closure of the graph of f. The first projection induces a proper birational morphism $p : X' \longrightarrow X$:



Now we assume that $f : X \dashrightarrow Y$ is not defined everywhere on X, that is, $\mathrm{Ex}(p)$ is non-empty. So by Proposition 1.4. there exists a rational curve on $\mathrm{Ex}(p)$ which can be contracted by p.

Since Y contains no rational curves, it must also be contracted by the second

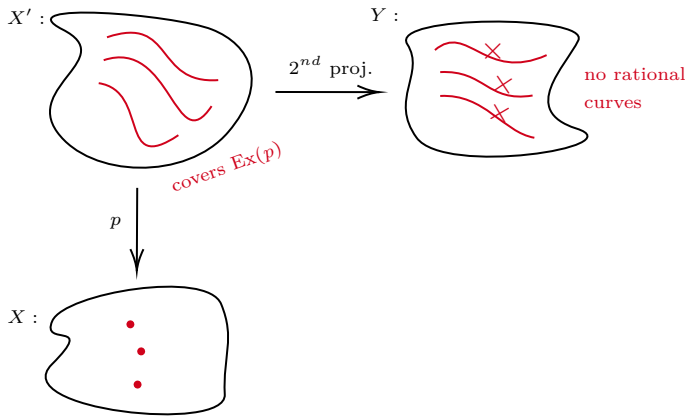


Figure 2.2.: 2nd projection also contracting by Y has no rational curves.

projection, which is absurd since it is contained in $X \times Y$ equipped with 2 projections but does not project rational curves!

Hence $\mathrm{Ex}(p)$ must be non-empty and f is defined everywhere. □

Remark 2.1.6. According to [KM98, 1.3], the content of that proposition cannot be proved entirely; it requires the consequence of [KM98, 2.45], which says that one obtains a DVR after finite blow-ups. Alternatively, one can just see the appendix of [Kol96].

Proposition 2.1.7. (Negativity Lemma in Simplest Form) *Let X and Y be smooth projective varieties and let $\pi : Y \longrightarrow X$ be a birational morphism which is not an isomorphism. Then there exists a rational curve C on Y contracted by π such that $(K_Y \cdot C) < 0$.*

Proof. Let $E = \mathrm{Ex}(\pi)$. As X is smooth, the image $\pi(E)$ of the exceptional locus E of π has codimension at least 2 in the previous discussion.

By Bertini theorem, a general hyperplane section of X is smooth, and connected by [Har77, III, 7.9], also a general hyperplane section of X passing through a fixed point $x \in \pi(E)$ has the **same property**.

It follows that by taking $(\dim X - 2)$ general hyperplane sections will cut out a smooth surface S in X that meets $\pi(E)$ in a finite set containing x.

Moreover taking one more hyperplane section to cut, we can get a smooth curve C_0 that meets $\pi(E)$ only at x and a smooth curve C that does not meet $\pi(E)$. This implies they C_0 and C are linearly equivalent, since the two cutting hyperplane sections are and also for their restrictions.

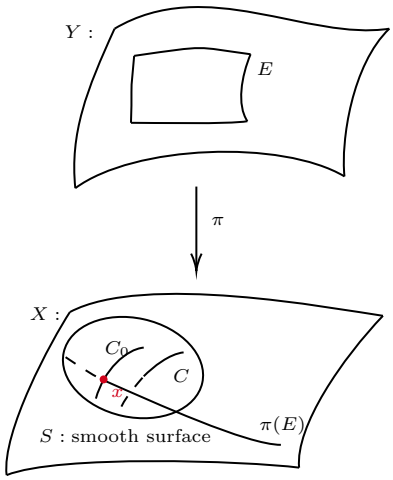


Figure 2.3.: Construction of two smooth curves C_0 and C.

By our construction, C and C_0 are the intersections of general hyperplane sections, so intersecting them with any divisor will give the same numerical result; that is,

$$(K_X \cdot C) = (K_X \cdot C_0).$$

Recall from the previous discussion about the ramification divisor

$$K_Y \sim_{\mathrm{lin}} \pi^* K_X + \mathrm{Ram}(\pi)$$

where π is a birational morphism, so the support of $\mathrm{Ram}(\pi)$ is exactly E. Since the curve $C' = \pi^{-1}(C)$ does not meet E, we have

$$\begin{aligned}(K_Y \cdot C') &= (\pi^* K_X + \mathrm{Ram}(\pi)) \cdot C' \\ &= (\pi^* K_X \cdot C') \\ &= (K_X \cdot \pi_* C') \\ &= (K_X \cdot C).\end{aligned}$$

On the other hand, since the strict transform

$$C_0' = \overline{\pi^{-1}(C_0 - \pi(E))}$$

of C_0 does meet E:

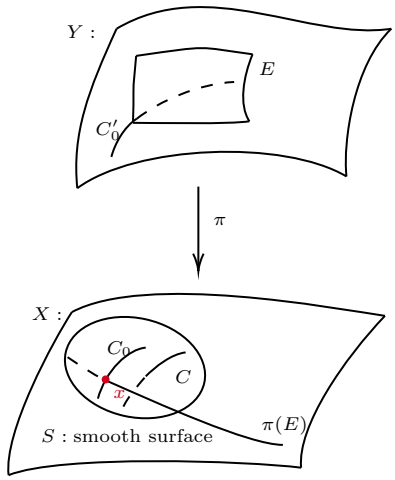


Figure 2.4.: Strict transform C_0' does meet E.

So we have the inequality:

$$\begin{aligned}(K_Y \cdot C_0') = (\pi^* K_X + \mathrm{Ram}(\pi)) \cdot C_0' > (\pi^* K_X \cdot C_0') &= (K_X \cdot \pi_* C_0') \\ &= (K_X \cdot C_0) \\ &= (K_X \cdot C) \\ &= (K_Y \cdot C')\end{aligned}$$

The rational map $\pi^{-1} : S \dashrightarrow Y$ might not be a morphism, and image is dimension 2 by π is surjective, so we reduce to the case of between birational surfaces. Now by

[Bea78, II.11], this rational map can be factorized into a finite composition of blow-ups up to isomorphism, where the points blowing-up lie in $S \cap \pi(E)$, which is a finite point set. Then we get a map

$$g : \widetilde{S} \xrightarrow{\varepsilon} S \overset{\pi^{-1}}{\dashrightarrow} Y$$

whose image is the strict transform of S. The curve $C'' := \varepsilon^* C$ is irreducible (as it has no intersection with $\pi(E)$) and we set $g_* C'' =: C' \subseteq Y$.

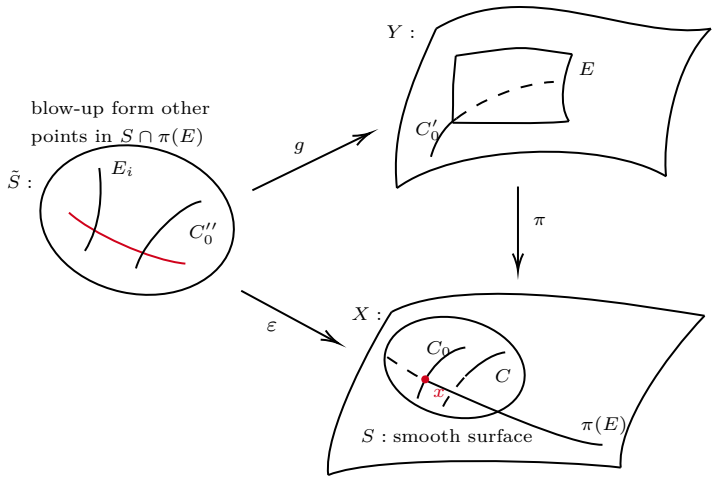


Figure 2.5.: Blowing-up at finite points.

For C_0, as it intersects with $\pi(E)$, by [Bea78, Lem II.2] we have

$$\varepsilon^* C_0 = C_0'' + \sum_i^{\text{finite}} m_i E_i$$

where the m_i are non-negative integers, the multiplicity of C_0 at the points we are blowing-up. And the E_i are exceptional divisors for ε, hence in particular are rational (-1)-curves, and $g_* C_0'' = C_0'$.

Since C and C_0 are linearly equivalent on S, we have

$$C'' = \varepsilon^* C \sim_{\text{lin}} \varepsilon^* C_0 = C_0'' + \sum_i m_i E_i$$

on $\widetilde{S}$, hence, by applying g_* to get

$$C' \sim_{\text{lin}} C_0' + \sum_i m_i (g_* E_i).$$

Taking the intersection with K_Y, we get

$$(K_Y \cdot C') = (K_Y \cdot C'_0) + \sum_i m_i (K_Y \cdot g_* E_i).$$

It follows from the inequality $(K_Y \cdot C'_0) > (K_Y \cdot C')$ that there must be some i such that

$$(K_Y \cdot g_* E_i) < 0.$$

In particular this says $g(E_i)$ is not a point, hence is a rational curve on Y. Moreover, $\pi(g(E_i)) = \varepsilon(E_i)$ is a point and hence $g(E_i)$ is contracted by π. □

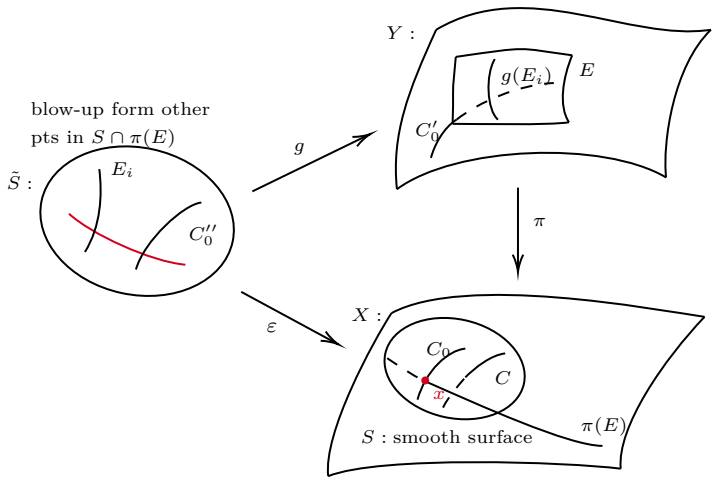


Figure 2.6.: Construction of a rational curve $g(E_i)$ in the E.

Notice that in the proof of the simplest negativity lemma, a key aspect is the assertion that there is indeed a smooth general hyperplane section passing through the closed point $x \in \pi(E)$. Hence, we can procedurally cut a smooth surface and, once more, a smooth curve passing through it. While a beginner taking a glimpse at this might think it is a trivial fact, my advisor had a doubt about it, which gives a complete proof of that claim here.

Lemma 2.1.8. *Let X be a smooth projective variety of dimension at least two, and $x \in X$ a closed point. Then a general hyperplane section passing through x is smooth and irreducible.*

Proof. Pick L to be a very ample divisor on X, and let

$$\pi : \widetilde{X} := \mathrm{Bl}_x X \longrightarrow X$$

be the blow-up, as in [Har77, II, ex7.14(b)] we see that for all $m \gg 0$,

$$-E + m\pi^* L$$

is very ample on $\widetilde{X}$, with $E \cong \mathbb{P}^{n-1}$ for $n = \dim X$.

So by Bertini's theorem, a general member

$$H' \in |m\pi^* L - E|$$

is smooth and irreducible, see [Har77, III, 10.9].

Set $H = \pi(H')$, then H is smooth outside the point $x \in X$. And since we're proving the local property - after linear translation - we can view it on an affine open $X = U$ with $\widetilde{X} = \mathrm{Bl}_0 U$, where H is given by the local equation f and $H'|_E$ is given by the homogeneous local equation F.

Since

$$(m\pi^* L \cdot E) = (mL \cdot \pi_* E) = (mL \cdot \{x\}) = 0$$

then $\mathscr{O}_{\widetilde{X}}(m\pi^* L|_E) \cong \mathscr{O}_E(\pi^* L|_E) \cong \mathscr{O}_E$ is a trivial line bundle, that is

$$\mathscr{O}_{\widetilde{X}}(H'|_E) \cong \mathscr{O}_E(-E|_E) \cong \mathscr{O}_E(1)$$

hence degree of F is 1.

Now $f = F + \{\text{higher degree terms}\}$ implies that f is non-singular at 0, so H is non-singular at x. □

2.1B Parametrizing Rational Curves

Let $\mathbf{k}$ be an algebraically closed field not necessarily char 0, we aim to construct a space $\mathrm{Mor}(\mathbb{P}^1_{\mathbf{k}}, X)$ where $X \subseteq \mathbb{P}^N_{\mathbf{k}}$ is a projective variety over $\mathbf{k}$, which parametrizes all morphisms $f : \mathbb{P}^1_{\mathbf{k}} \longrightarrow X$. That is to say, the morphism f "can be deformed".

To begin with, study the case of $X = \mathbb{P}^N_{\mathbf{k}}$ with $N \geq 1$.

A morphism $f : \mathbb{P}^1_{\mathbf{k}} \longrightarrow \mathbb{P}^N_{\mathbf{k}}$ is given by a set of homogeneous polynomials $F_0, \ldots, F_N$ of the same degree d

$$\begin{aligned} f : \mathbb{P}^1_{\mathbf{k}} &\longrightarrow \mathbb{P}^N_{\mathbf{k}} \\ [u : v] &\longmapsto [F_0(u,v) : F_1(u,v) : \cdots : F_N(u,v)] \end{aligned}$$

and we assume that $F_0, F_1, \ldots, F_N$ have no nontrivial common zeros, i.e. have no nonconstant common factor in $\mathbf{k}[U, V]$, and two morphisms f and f' are the same if and only if each F_i and F'_i differ by a constant factor.

Since each degree d homogeneous polynomial $F_i = a_0U^d + a_1U^{d-1}V + \cdots + a_dV^d$ has $d+1$ coefficients, we see that f is uniquely determined by a point

$$[f] \in \mathbb{P}(\mathbf{k}^{(d+1)(N+1)}) \cong \mathbb{P}_{\mathbf{k}}^{(d+1)(N+1)-1}$$

where

$$(a_I^0, a_I^1, \ldots, a_I^N) \in \mathbf{k}^{(d+1)(N+1)} \ , \ I = \{0, 1, \ldots, d\}.$$

Moreover,

$$\mathrm{Mor}_d(\mathbb{P}_{\mathbf{k}}^1, \mathbb{P}_{\mathbf{k}}^N) \subseteq \mathbb{P}_{\mathbf{k}}^{(d+1)(N+1)-1}$$

consisting of all ordered tuples $(F_0, \ldots, F_N)$ of homogeneous polynomials of degree d, which have no nontrivial common zero in $\mathbb{C}$.

Lemma 2.1.9. *The subset*

$$\mathrm{Mor}_d(\mathbb{P}_{\mathbf{k}}^1, \mathbb{P}_{\mathbf{k}}^N) \subseteq \mathbb{P}_{\mathbf{k}}^{(d+1)(N+1)-1}$$

is Zariski open, that is, a quasi-projective variety.

Proof. The F_i have no non-constant common factor in $\mathbf{k}[U, V]$ if and only if they have no non-trivial common zero in some algebraic closure $\bar{\mathbf{k}}$, by the Nullstellensatz, if and only if $\sqrt{I = (F_0, \ldots, F_N)} = (U, V)$ in $\bar{\mathbf{k}}[U, V]$, that is

$$(U, V)^m \subseteq I$$

for some $m > 0$, the only possible zero is at $(0, 0)$. But this is equivalent to saying that the degree m part I_m is equal to $\bar{\mathbf{k}}[U, V]_m$.

Hence for some integer $m \geq d$, the linear map

$$\begin{aligned} \phi_m : (\bar{\mathbf{k}}[U, V]_{m-d})^{N+1} &\longrightarrow \bar{\mathbf{k}}[U, V]_m \\ (G_0, \ldots, G_N) &\longmapsto \sum_{i=0}^{N} G_i F_i \end{aligned}$$

is surjective, that is, the matrix corresponding to ϕ_m has the rank $\dim_{\bar{\mathbf{k}}} \bar{\mathbf{k}}[U, V]_m = \binom{1+m}{m} = m + 1$.

Therefore, $F_0, \ldots, F_N$ do have a non-constant common factor if and only if, for all m, all $(m+1)$-minors of the matrix corresponding to ϕ_m vanish, which is a Zariski closed condition. Then we take the union of these Zariski closed subset for all m, the complement just consists of points of the form

$$[F_0(u, v) : F_1(u, v) : \cdots : F_N(u, v)]$$

which are well-defined. □

Finally, morphisms from $\mathbb{P}^1_{\mathbf{k}}$ to $\mathbb{P}^N_{\mathbf{k}}$ are parametrized by the locally noetherian disjoint union

$$\mathrm{Mor}(\mathbb{P}^1_{\mathbf{k}}, \mathbb{P}^N_{\mathbf{k}}) = \bigsqcup_{d\geq 0} \mathrm{Mor}_d(\mathbb{P}^1_{\mathbf{k}}, \mathbb{P}^N_{\mathbf{k}})$$

of quasi-projective varieties.

Remark 2.1.10. These morphisms of degree d can fit together into a universal morphism

$$\begin{aligned} f^{\mathrm{univ}} : \mathbb{P}^1_{\mathbf{k}} \underset{\mathbf{k}}{\times} \mathrm{Mor}_d(\mathbb{P}^1_{\mathbf{k}}, \mathbb{P}^N_{\mathbf{k}}) &\longrightarrow \mathbb{P}^N_{\mathbf{k}} \\ ([u:v], f) &\longmapsto [F_0(u,v) : \cdots : F_N(u,v)] \end{aligned}$$

Now let X be a subscheme of $\mathbb{P}^N_{\mathbf{k}}$ cut out by homogeneous equations $G_1, \ldots, G_m$. The morphism of degree d from $\mathbb{P}^1_{\mathbf{k}}$ to X can be written as

$$\begin{aligned} &f : \mathbb{P}^1_{\mathbf{k}} \longrightarrow X \\ &[u:v] \longmapsto [F_0(u,v), \ldots, F_N(u,v)] \ \text{ satisfying } \ G_j(F_0, \ldots, F_N) = 0 \ , \ j = 1, \ldots, m. \end{aligned}$$

Hence again, same as in the lemma, morphisms from $\mathbb{P}^1$ to X are parametrized by the disjoint union

$$\mathrm{Mor}(\mathbb{P}^1_{\mathbf{k}}, X) = \bigsqcup_{d\geq 0} \mathrm{Mor}_d(\mathbb{P}^1_{\mathbf{k}}, X)$$

of quasi-projective varieties.

2.1C Parametrizing Morphisms

We admit the following theorem generalizing the preceding construction:

Theorem 2.1.11. (Grothendieck's Hilbert scheme theorem) *Let X and Y be varieties over $\mathbf{k}$. If X is quasi-projective and Y is projective, then there is a locally noetherian scheme*

$$\mathrm{Mor}(Y, X)$$

over $\mathbf{k}$ which we call the deformation space, parametrizing all morphisms $Y \to X$ in the following sense:

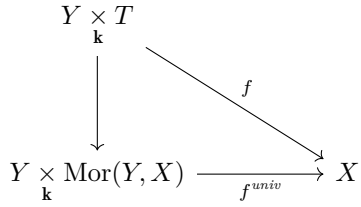
(a) There is a universal evaluation morphism

$$\begin{aligned} f^{univ} : Y \underset{\mathbf{k}}{\times} \mathrm{Mor}(Y, X) &\longrightarrow X \\ (*, f) &\longmapsto f(*) \end{aligned}$$

(b) For any $\mathbf{k}$*-scheme* T *and any morphism of* $\mathbf{k}$*-schemes*

$$f : Y \underset{\mathbf{k}}{\times} T \longrightarrow X$$

there is a unique $\mathbf{k}$*-morphism* $\varphi : T \longrightarrow Mor(Y, X)$ *making the following diagram commute*



such that

$$f(y, t) = f^{\mathrm{univ}}(y, \varphi(t))$$

Moreover, even though $\mathrm{Mor}(Y, X)$ *is in general not of finite type over* $\mathbf{k}$*, the subscheme*

$$\mathrm{Mor}_P(Y, X) \subseteq \mathrm{Mor}(Y, X)$$

which parametrizes those morphisms $f : Y \longrightarrow X$ *whose Hilbert polynomial* $\chi(Y, f^*\mathscr{O}_X(m))$ *coincides with a given polynomial* $P(m)$*, is quasi-projective over* $\mathbf{k}$ *and hence of finite type over* $\mathbf{k}$*.*

We have:

$$\mathrm{Mor}(Y, X) = \bigsqcup_P \mathrm{Mor}_P(Y, X)$$

where the union runs through all possible Hilbert polynomials.

Remark 2.1.12. Notice that Y being projective is an essential condition: the space $\mathrm{Mor}(\mathbb{A}^1_{\mathbf{k}}, \mathbb{A}^N_{\mathbf{k}})$ is **not** a disjoint union of quasi-projective varieties.

In the language of moduli theory, the above Grothendieck's theorem says that the functor

$$\begin{aligned} \mathscr{M}or(Y, X) : \mathrm{Sch}_{\mathbf{k}} &\longrightarrow \mathrm{Sets} \\ T &\longrightarrow \mathrm{Hom}_{\mathrm{Sch}_{\mathbf{k}}}(Y \underset{\mathbf{k}}{\times} T, X) \end{aligned}$$

is representable by a locally noetherian $\mathbf{k}$-scheme $\mathrm{Mor}(Y, X)$, that is, the above functor is canonically isomorphic to $\mathrm{Hom}_{\mathrm{Sch}_{\mathbf{k}}}(-, \mathrm{Mor}(Y, X))$, sending

$$\begin{aligned} T &\longrightarrow \mathrm{Hom}_{\mathrm{Sch}_k}(T, \mathrm{Mor}(Y, X)) \\ * &\longmapsto \varphi : t \mapsto \varphi(t) \end{aligned}$$

where we think of φ as a family of morphisms $f_t : Y \times \{t\} \longrightarrow X$, and given f (or φ), we call the morphism

$$\begin{aligned} ev : Y \underset{\mathbf{k}}{\times} T &\longrightarrow X \\ (y, t) &\longmapsto f_t(y) = \varphi(t)(y) \end{aligned}$$

the *evaluation map*.

This means that the above moduli problem has a fine moduli space.

Example 2.1.13. If $Y = \operatorname{Spec} \mathbf{k}$, then the $\operatorname{Mor}(Y, X) = \operatorname{Hom}_{\mathrm{Sch}_{\mathbf{k}}}(\operatorname{Spec} \mathbf{k}, X) \cong X$. They are therefore tautologically parametrized by X itself, with the universal morphism being:

$$\begin{aligned} f^{\mathrm{univ}} : \operatorname{Spec} \mathbf{k} \underset{\mathbf{k}}{\times} X &\xrightarrow{\cong} X \\ (*, x) &\longmapsto x \end{aligned}$$

More generally, if Y is a reduced scheme of dimension 0, then the scheme

$$\operatorname{Mor}(Y, X) \cong X^{|Y|} = \underbrace{X \times X \times \cdots \times X}_{|Y| \text{ times}}$$

Example 2.1.14. (Dual number) If $Y = \operatorname{Spec} \mathbf{k}[\varepsilon]/(\varepsilon^2)$, then a map $Y \longrightarrow X$ corresponding to the data of a $\mathbf{k}$-valued point $x \in X$ and a tangent vector in $T_{X,x}$.

When X is smooth, $\operatorname{Mor}(\operatorname{Spec} \mathbf{k}[\varepsilon]/(\varepsilon^2), X)$ is the total space of the tangent bundle T_X, see [GW20, (6.4)].

Example 2.1.15. Using the universal property to determine the Zariski tangent space $T_{\operatorname{Mor}(Y,X),[f]}$ to $\operatorname{Mor}(Y, X)$ at a $\mathbf{k}$-point $[f]$.

As mentioned above, this vector space parametrized by morphisms

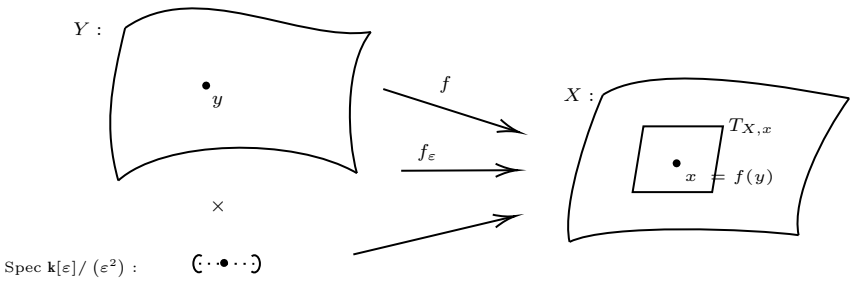
$$\begin{aligned} \varphi : T = \operatorname{Spec} \mathbf{k}[\varepsilon]/(\varepsilon^2) &\longrightarrow \operatorname{Mor}(Y, X) \\ * &\longmapsto [f] \end{aligned}$$

whose image is $[f]$, hence by extensions of f to morphisms

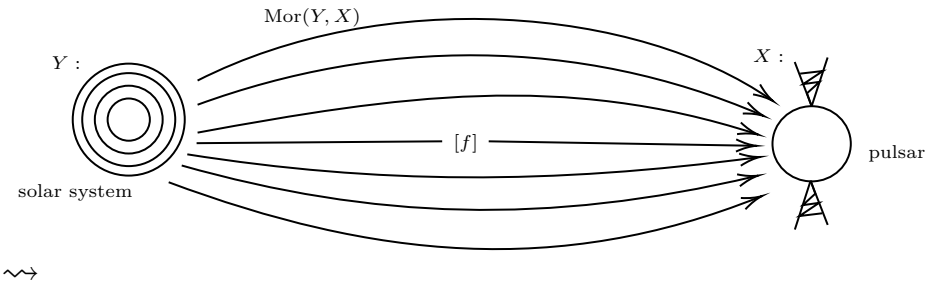
$$f_\varepsilon : Y \times \operatorname{Spec} \mathbf{k}[\varepsilon]/(\varepsilon^2) \longrightarrow X$$

which should be thought of first-order infinitesimal deformation of Y, making the following diagram commute

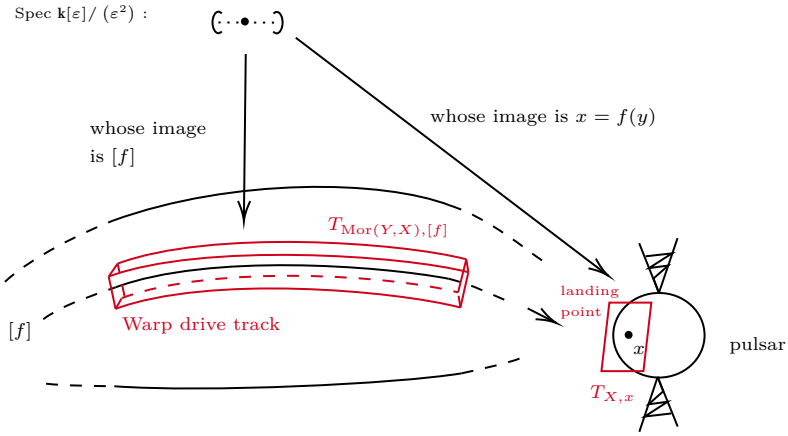
$$\begin{array}{ccc} Y \times \operatorname{Spec} \mathbf{k}[\varepsilon]/(\varepsilon^2) & & \\ \downarrow{\scriptstyle \mathrm{id} \times \varphi} & \searrow^{f_\varepsilon} & \\ Y \times \operatorname{Mor}(Y, X) & \xrightarrow[f^{\mathrm{univ}}]{} & X \end{array}$$



we can view that



$\rightsquigarrow$



Proposition 2.1.16. (Tangent space of $\mathrm{Mor}(Y,X)$) *Let X and Y be varieties, with X quasi-projective and Y projective, and let $f: Y \longrightarrow X$ be a morphism. One has*

$$T_{\mathrm{Mor}(Y,X),[f]} \cong H^0\big(Y, \mathscr{H}om(f^*\Omega_X, \mathscr{O}_Y)\big)$$

Proof. Assume first that Y and X are affine, write $Y = \mathrm{Spec}\ B$ and $X = \mathrm{Spec}\ A$, where A and B are finitely generated $\mathbf{k}$-algebras.

Let $f^\# : A \longrightarrow B$ be the corresponding map, then as mentioned above, we are looking for morphisms corresponding to first-order deformations:

$$\begin{aligned} f_\varepsilon^\# : A &\longrightarrow B \underset{\mathbf{k}}{\otimes} \mathbf{k}[\varepsilon]/(\varepsilon^2) \cong B[\varepsilon]/(\varepsilon^2) \\ a &\longmapsto f^\#(a) + \varepsilon \cdot g(a) \end{aligned}$$

for some $g : A \longrightarrow B$, and this determines $f_\varepsilon^\#$ uniquely. Then the equality

$$f_\varepsilon^\#(aa') = f_\varepsilon^\#(a) f_\varepsilon^\#(a')$$

is equivalent to

$$\begin{aligned} f^\#(a)f^\#(a') + \varepsilon \cdot g(aa') &= f^\#(aa') + \varepsilon \cdot g(aa') \\ &= (f^\#(a) + \varepsilon \cdot g(a)) \cdot (f^\#(a') + \varepsilon \cdot g(a')) \\ &= f^\#(a) \cdot f^\#(a') + \varepsilon\big(f^\#(a) \cdot g(a') + f^\#(a') \cdot g(a)\big), \end{aligned}$$

that is,

$$g(aa') = f^\#(a)g(a') + f^\#(a')g(a).$$

Notice that this is just the Leibniz rule, saying $g : A \longrightarrow B$ is a $\mathbf{k}$-linear derivation of the A-module B, via the structure map $f^\#$. Hence, by the universal property of differentials, g factors uniquely through Ω_A,

$$g : A \longrightarrow \Omega_A \longrightarrow B.$$

Altogether, we see that

$$\begin{array}{ccccc} \mathrm{Spec}\ \mathbf{k}[\varepsilon]/(\varepsilon^2) & \longrightarrow & \mathrm{Spec}\ B \times \mathrm{Spec}\ \mathbf{k}[\varepsilon]/(\varepsilon^2) & & \\ \Big\downarrow{\scriptstyle T_{\mathrm{Mor}(\mathrm{Spec}\ B, \mathrm{Spec}\ A),[f]} \overset{can.}{\longleftrightarrow}} & & \Big\downarrow{\scriptstyle id \times \varphi} & \searrow{\scriptstyle f_\varepsilon \overset{can.}{\longleftrightarrow} f_\varepsilon^\# : A \longrightarrow B[\varepsilon]/(\varepsilon^2)} & \\ \mathrm{Mor}(\mathrm{Spec}\ B, \mathrm{Spec}\ A) & \longrightarrow & \mathrm{Spec}\ B \times \mathrm{Mor}(\mathrm{Spec}\ B, \mathrm{Spec}\ A) & \longrightarrow & \mathrm{Spec}\ A \end{array}$$

and $f_\varepsilon^\#$ is uniquely determined by derivation g, canonical corresponding to elements of $\mathrm{Hom}_A(\Omega_A, B)$. So there is a canonical isomorphism

$$T_{\mathrm{Mor}(\mathrm{Spec}\ B, \mathrm{Spec}\ A),[f]} \cong \mathrm{Hom}_A(\Omega_A, B) \cong \mathrm{Hom}_B(\Omega_A \underset{A}{\otimes} B, B).$$

Now in general, cover the variety X by affine opens $U_i = \operatorname{Spec} A_i$, and Y by affine opens $V_i = \operatorname{Spec} B_i$ with $f(V_i) \subseteq U_i$. By Grothendieck's theorem, the tangent space $T_{\operatorname{Mor}(Y,X),[f]}$ is canonically isomorphic to the extensions of f to morphisms

$$f_\varepsilon : Y \times \operatorname{Spec} \mathbf{k}[\varepsilon]/(\varepsilon^2) \longrightarrow X.$$

Such a morphism is uniquely determined by its restrictions

$$f_{\varepsilon,i} : V_i \times \operatorname{Spec} \mathbf{k}[\varepsilon]/(\varepsilon^2) \longrightarrow U_i$$

which corresponds to a section of

$$\operatorname{Hom}_{B_i}(\Omega_{A_i} \underset{A_i}{\otimes} B_i, B_i).$$

And these sections are compatible on overlaps, so they can be glued together to get a global section of $\mathscr{H}om(f^*\Omega_X, \mathscr{O}_Y)$, hence get a canonical isomorphism

$$T_{\operatorname{Mor}(Y,X),[f]} \cong H^0(Y, \mathscr{H}om(f^*\Omega_X, \mathscr{O}_Y)).$$

□

Corollary 2.1.17. (Very important result) *When X is smooth along the image $f(Y)$, that is the image $f(Y)$ has no intersection with* $\operatorname{Sing}(X)$. *Then*

$$T_{\operatorname{Mor}(Y,X),[f]} \cong H^0(Y, f^*T_X).$$

Proof. Since the question is local on $\operatorname{Mor}(Y,X)$, we may shrink X and assume that X is smooth. Then Ω_X is a locally free sheaf of finite rank, hence

$$\begin{aligned}\mathscr{H}om(f^*\Omega_X, \mathscr{O}_Y) &\cong (f^*\Omega_X)^\vee \otimes \mathscr{O}_Y \\ &\cong f^*T_X,\end{aligned}$$

by [Har77, II, ex5.1(b)]. □

Example 2.1.18. When X is a smooth projective variety, then we consider the automorphism on X which is just the parameter space $\operatorname{Aut}(X)$. Then tangent space at one point $f \in \operatorname{Aut}(X)$ is given by $H^0(X, f^*T_X)$.

Moreover, if X has no global tangent fields, then the tangent space is trivial, saying that X has a discrete automorphism group.

Now we discuss the local structure of the parameter space, for which we need the following lemmas:

Lemma 2.1.19. *Let $(R, \mathfrak{m})$ a finitely generated local $\mathbf{k}$-algebra where $\mathbf{k}$ is algebraically closed, and let $I \subseteq \mathfrak{m}$ be an ideal such that $\mathfrak{m}I = 0$.*

Let $f : Y \longrightarrow X$ be a morphism of varieties with the same situation as above such that X is smooth along $f(Y)$, and let

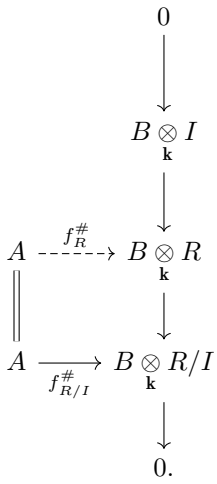
$$f_{R/I} : Y \times \text{Spec } R/I \longrightarrow X$$

be an extension of f. Then the obstruction to extending $f_{R/I}$ to a morphism

$$f_R : Y \times \text{Spec } R \longrightarrow X$$

lies in $H^1(Y, f^*T_X) \underset{\mathbf{k}}{\otimes} I$.

Proof. Assume first that Y and X are affine, write $Y = \text{Spec } B$, $X = \text{Spec } A$. We're looking for R-algebra liftings $f_R^\#$ fitting into the diagram



Notice that B is always flat as $\mathbf{k}$-module.

Since X is smooth along $f(Y)$, we can shrink X to assume Spec A is nonsingular, and $I^2 = 0$ so can use the infinitesimal lifting property (B.2.8) to get the lifting $f_R^\#$, and two liftings differ by a $\mathbf{k}$-derivation from A to $B \underset{\mathbf{k}}{\otimes} I$. Since $I^2 = 0$, a $B \underset{\mathbf{k}}{\otimes} I$ has a natural structure of $B \underset{\mathbf{k}}{\otimes} R/I$-module, and hence also of an A-module. So this $\mathbf{k}$-linear derivation can be regarded as an element of $\text{Hom}_A(\Omega_A, B \underset{\mathbf{k}}{\otimes} I)$. This can simplifies to

$$\begin{aligned}
\text{Hom}_A(\Omega_A, B \underset{\mathbf{k}}{\otimes} I) &\cong \text{Hom}_B(\Omega_A \underset{A}{\otimes} B, B \underset{\mathbf{k}}{\otimes} I) \\
\text{(see remark for details)} \quad &\cong \text{Hom}_B(\Omega_A \underset{A}{\otimes} B, B) \underset{\mathbf{k}}{\otimes} I \\
&\cong H^0(Y, \mathscr{H}om(f^*\Omega_X, \mathscr{O}_Y)) \underset{\mathbf{k}}{\otimes} I \\
X \text{ smooth along } f(Y) \quad &\cong H^0(Y, f^*T_X) \underset{\mathbf{k}}{\otimes} I.
\end{aligned}$$

Now assume X is covered by affine opens U_i, Y is covered by affine opens V_i with $f(V_i) \subseteq U_i$. Then a global extension of f will be given by a collection of local extensions that agree on overlaps. But there is an obstruction to doing that: on each intersection $V_i \cap V_j$, two extensions differ by an element of $H^0(V_i \cap V_j, f^*T_X) \underset{\mathbf{k}}{\otimes} I$. These elements define a 1-cocycle, hence an element in $H^1(Y, f^*T_X) \underset{\mathbf{k}}{\otimes} I$, whose vanishing is equivalent to the existence of a global extension. □

Remark 2.1.20. Write $I = (f_1, \ldots, f_n)$, as $mI = 0$, I is an $R/m = \mathbf{k}$-module as $\mathbf{k}$ is algebraically closed, and is isomorphic to $\mathbf{k}^n$. Then

$$\begin{aligned}
\mathrm{Hom}_B(\Omega_A \underset{A}{\otimes} B, B \underset{\mathbf{k}}{\otimes} I) &\cong \mathrm{Hom}_B(\Omega_A \underset{\mathbf{k}}{\otimes} B, B \underset{\mathbf{k}}{\otimes} \mathbf{k}^n) \\
&\cong \mathrm{Hom}_B(\Omega_A \underset{A}{\otimes} B, B^n) \\
&\cong \mathrm{Hom}_B(\Omega_A \underset{A}{\otimes} B, B)^n \\
&\cong \mathrm{Hom}_B(\Omega_A \underset{A}{\otimes} B, B) \underset{\mathbf{k}}{\otimes} \mathbf{k}^n \\
&\cong \mathrm{Hom}_B(\Omega_A \underset{A}{\otimes} B, B) \underset{\mathbf{k}}{\otimes} I.
\end{aligned}$$

Remark 2.1.21. (Important case) An important case of the above lemma is the case when $R = \mathbf{k}[\varepsilon]/(\varepsilon^m)$ for some $m \geq 2$, $I = (\varepsilon^{m-1})$ and $m = (\varepsilon)$. In this case the lemma says that the obstruction to extending a deformation of f from

$$Y \times \mathrm{Spec}\ \mathbf{k}[\varepsilon]/(\varepsilon^{m-1}) \longrightarrow X$$

to

$$Y \times \mathrm{Spec}\ \mathbf{k}[\varepsilon]/(\varepsilon^{m}) \longrightarrow X$$

lies in $H^1(Y, f^*T_X)$.

Lemma 2.1.22. *Let $(R, \mathfrak{m})$ be a noetherian local ring, and let $I \subseteq \mathfrak{m}^2$ be an ideal. If the canonical projection $\pi : R \longrightarrow R/I$ has a section, i.e.a ring homomorphism $R/I \xrightarrow{s} R$ with $\pi \circ s = id$, then $I = 0$.*

Proof. If $a, b \in R$, then $\pi(s(\pi(a))) = \pi(a)$ and $\pi(s(\pi(b))) = \pi(b)$ as $\pi \circ s = id$. But this implies that

$$s(\pi(a)) = a + a' \text{ and } s(\pi(b)) = b + b'$$

with some $a', b' \in I$. Furthermore we assume that $a, b \in m$, then

$$\begin{aligned}
s(\pi(ab)) &= s(\pi(a)) \cdot s(\pi(b)) \\
&= (a + a') \cdot (b + b') \\
&= ab + a'b + ab' + a'b' \in ab + \mathfrak{m}I + I^2 = ab + \mathfrak{m}I.
\end{aligned}$$

Since $I \subseteq \mathfrak{m}^2$, the above implies for any $x \in I \subseteq \mathfrak{m}^2$,

$$0 = s(\pi(x)) \in x + \mathfrak{m}I.$$

Hence we must have $x \in \mathfrak{m}I$, i.e. $I \subseteq \mathfrak{m}I$, then we use Nakayama's lemma to get $I = 0$. □

Recall that all *abstract varieties* are assumed to be over an algebraically closed field.

Theorem 2.1.23. (Local structure of $\mathrm{Mor}(Y, X)$) *Let X and Y be projective varieties, and $f : Y \longrightarrow X$ be a morphism such that X is smooth along $f(Y)$. Then locally around $[f]$, the deformation space $\mathrm{Mor}(Y, X)$ can be defined by $h^1(Y, f^*T_X)$ equations in a non-singular variety of dimension $h^0(Y, f^*T_X)$.*

Proof. By Grothendieck's theorem, the scheme $\mathrm{Mor}(Y, X)$ is locally noetherian, so locally around the $\mathbf{k}$-valued point $[f]$, there are finitely many polynomials $P_1, \ldots, P_m$ in a polynomial ring of n variables. The rank r of the corresponding Jacobian matrix at $[f]$

$$\left(\frac{\partial P_i}{\partial x_j}\right)_{\substack{i=1,\ldots,m \\ j=1,\ldots,n}}$$

is the codimension of the Zariski tangent space $T_{\mathrm{Mor}(Y,X),[f]}$ in $\mathbb{A}^n$, that is, due to Corollary 1.17, $r \geq n - h^0(Y, f^*T_X) =: r'$, where $h^0(Y, f^*T_X)$ is the dimension of linear space $T_{\mathrm{Mor}(Y,X),[f]}$.

Then we pick r' polynomials among these $P_1, \ldots, P_m$, that have a full rank Jacobian matrix at point $[f]$. Hence $\mathrm{Mor}(Y, X)$ is locally cut out from the variety

$$V := \{P_{i_1} = \cdots = P_{i_{r'}} = 0\} \subseteq \mathbb{A}^n$$

which is smooth at $[f]$ and has dimension $h^0(Y, f^*T_X)$. Notice that by our construction $T_{\mathrm{Mor}(Y,X),[f]} \subseteq T_{V,[f]}$ with same dimension, so they are equal. Up to shrinking V, we may assume that V is smooth of dimension $h^0(Y, f^*T_X)$. And we know that $\mathrm{Mor}(Y, X)$ locally around point $[f]$, is a subscheme of V.

Next we need to show that this subscheme is cut out in V by at most $h^1(Y, f^*T_X)$ equations. But this is enough to show that in the regular local ring

$$\mathscr{O}_{V,[f]} =: R,$$

the ideal I that locally defines $\mathrm{Mor}(Y, X)$, can be generated by $h^1(Y, f^*T_X)$ elements.

Let $\mathfrak{m} \subseteq R$ be the unique maximal ideal, since the Zariski tangent space are the same, we have

$$\mathfrak{m}/\mathfrak{m}^2 = (\mathfrak{m} + I)/(\mathfrak{m}^2 + I) = \mathfrak{m}/(\mathfrak{m}^2 + I)$$

and hence

$$I \subseteq \mathfrak{m}^2.$$

Moreover, by Nakayama's lemma, it is enough to show the $R/\mathfrak{m} = \mathbf{k}$-vector space

$$I/\mathfrak{m}I$$

has dimension at most $h^1(Y, f^*T_X) =: h^1$.

Now consider the canonical morphism

$$T = \text{Spec } R/I \longrightarrow \text{Mor}(Y, X)$$

which corresponds to a morphism

$$f_{R/I} : Y \times \text{Spec } R/I \longrightarrow X$$

extending $f : Y \longrightarrow X$. Since $I \subseteq \mathfrak{m}^2$, the obstruction to extending $f_{R/I}$ to a morphism

$$f_{R/\mathfrak{m}I} : Y \times \text{Spec } R/\mathfrak{m}I \longrightarrow X$$

lies in $H^1(Y, f^*T_X) \underset{\mathbb{C}}{\otimes} (I/\mathfrak{m}I)$ by lemma 1.19, so we can write this obstruction as

$$\sum_{i=1}^{h^1} a_i \otimes \bar{b_i}$$

where $a_1, \ldots, a_{h^1}$ form a basis of $H^1(Y, f^*T_X)$, and $b_1, \ldots, b_{h^1}$ are elements in I. But this element can vanish in

$$H^1(Y, f^*T_X) \underset{\mathbb{C}}{\otimes} (I/(\mathfrak{m}I + (b_1, \ldots, b_{h^1}))),$$

hence by lemma 1.19 again can lift to a morphism

$$Y \times \text{Spec } R/(\mathfrak{m}I + (b_1, \ldots, b_{h^1}) \longrightarrow X$$

which induces to

$$\begin{array}{ccc} \text{Spec } R/(\mathfrak{m}I + (b_1, \ldots, b_{h^1}) & \longrightarrow & \text{Mor}(Y, X) \\ \uparrow & \nearrow & \\ \text{Spec } R/I. & & \end{array}$$

Since Spec $R/I \longrightarrow \text{Mor}(Y, X)$ is a local isomorphism, the above diagram can be modified into

$$\begin{array}{ccc} \text{Spec } R/(\mathfrak{m}I + (b_1, \ldots, b_{h^1}) & \longrightarrow & \text{Spec } R/I \\ \uparrow & \overset{id}{\nearrow} & \\ \text{Spec } R/I, & & \end{array}$$

that is, the identity $R/I \longrightarrow R/I$ can be factored as

$$R/I \longrightarrow R/(\mathfrak{m}I + (b_1, \ldots, b_{h^1}) \xrightarrow{\pi} R/I$$

where π is the canonical projection. So by lemma 1.22

$$I = \mathfrak{m}I + (b_1, \ldots, b_{h^1})$$

hence $I/\mathfrak{m}I$ can be generated by $b_1, \ldots, b_{h^1}$ as we want. □

Corollary 2.1.24. *In the above situation, every irreducible component of* $\mathrm{Mor}(Y, X)$ *passing through the point* $[f]$ *has dimension at least* $h^0(Y, f^*T_X) - h^1(Y, f^*T_X)$.

So far, we finish the classical conclusion in [Kol96, I, 2.16] restricted to over $\mathbb{C}$, (indeed this holds on arbitrary algebraically closed field).

2.1D Morphisms With Fixed Points

Fix a subscheme $B \subseteq Y$ and a morphism $g : B \longrightarrow X$

$$\begin{array}{ccc} Y & \xrightarrow{f} & X \\ \uparrow & \nearrow_{g} & \\ B. & & \end{array}$$

To study those morphisms $Y \longrightarrow X$ that agree with g on B, we need to consider the restriction morphism

$$\rho : \mathrm{Mor}(Y, X) \longrightarrow \mathrm{Mor}(B, X)$$
$$f \longmapsto f|_B$$

and we denote the fibre of this morphism over the point $[g]$ by

$$\mathrm{Mor}(Y, X; g).$$

If $f : Y \longrightarrow X$ is such that X is smooth along $f(Y)$, then the tangent map of ρ is given by

$$d\rho : H^0(Y, f^*T_X) \longrightarrow H^0(B, g^*T_X).$$

From the short exact sequence

$$0 \longrightarrow \mathscr{I}_B \longrightarrow \mathscr{O}_Y \longrightarrow \mathscr{O}_B \longrightarrow 0$$

where $\mathscr{I}_B$ is the ideal sheaf of the subscheme B. Tensoring with a locally free sheaf f^*T_X

$$0 \longrightarrow f^*T_X \otimes \mathscr{I}_B \longrightarrow f^*T_X \longrightarrow f^*T_X|_B \longrightarrow 0$$

and notice that $f^*T_X|_B \cong g^*T_X$, so we take the cohomology

$$0 \to H^0(Y, f^*T_X \otimes \mathscr{I}_B) \to H^0(Y, f^*T_X) \xrightarrow{d\rho} H^0(B, g^*T_X) \to H^1(Y, f^*T_X \otimes \mathscr{I}_B) \to \cdots$$

Hence we see the tangent space of $\mathrm{Mor}(Y, X; g)$ at the point $[f]$ is the kernel of $d\rho$, which is nothing but

$$H^0(Y, f^*T_X \otimes \mathscr{I}_B).$$

More generally, we have the theorem

Theorem 2.1.25. (Local structure of $\mathrm{Mor}(Y, X; g)$) *Let X and Y be projective varieties, let $B \subseteq Y$ be a closed subscheme equipped with a morphism $g : B \longrightarrow X$. And further $f : Y \longrightarrow X$ be a morphism with $f|_B = g$ such that X is smooth along $f(Y)$. Then locally around the* **k***-point $[f]$, $\mathrm{Mor}(Y, X; g)$ can be defined by $h^1(Y, f^*T_X \otimes \mathscr{I}_B)$ equations in a non-singular variety of dimension $h^0(Y, f^*T_X \otimes \mathscr{I}_B)$.*

2.1E Morphism From a Curve

If $Y = C$ is a smooth projective curve and B is a finite subscheme, then for any $f : C \longrightarrow X$ such that X is smooth along $f(C)$, by Corollary 1.24 one has

$$\begin{aligned}\dim_{[f]} \mathrm{Mor}(C, X) &\geq h^0(C, f^*T_X) - h^1(C, f^*T_X) \\ &= \chi(C, f^*T_X).\end{aligned}$$

But by Hirzebruch-Riemann-Roch formula, we have

$$\chi(C, f^*T_X) = \deg(\mathrm{ch}(f^*T_X) \cdot \mathrm{td}(T_C))_1$$

where $\mathrm{ch}(f^*T_X) = \dim X + c_1(f^*T_X)$ and $\mathrm{td}(T_C) = 1 + \frac{1}{2}c_1(T_C)$, hence

$$\begin{aligned}\deg\left(\mathrm{ch}(f^*T_X) \cdot \mathrm{td}(T_C)\right)_1 &= \deg\left((\dim X + c_1(f^*T_X)) \cdot (1 + \frac{1}{2}c_1(T_C))\right)_1 \\ &= \deg\left(\dim X + c_1(f^*T_X) + \frac{\dim X}{2}c_1(T_C) + \frac{1}{2}c_1(f^*T_X)c_1(T_C)\right)_1 \\ &= \deg\left(\dim X + c_1(f^*T_X) + \frac{\dim X}{2}c_1(T_C)\right) \\ &= \deg(c_1(f^*T_X)) + \frac{\dim X}{2}\deg(c_1(T_C)) \\ (*) &= \deg_C(f^*\omega_X^{-1}) + \frac{\dim X}{2}\deg(-K_C) \\ &= (f^*(-K_X) \cdot C) + \frac{\dim X}{2}(2 - 2g(C)) \\ &= -(K_X \cdot f_*C) + \dim X(1 - 1g(C)),\end{aligned}$$

where the equation $(*)$ follows from $c_1(f^*T_X) = c_1(\det(f^*T_X))$, can see [Har77, A C.5]. Thus, we get

$$\dim_{[f]} \operatorname{Mor}(C, X) \geq -(K_X \cdot f_*C) + (1 - g(C)) \cdot \dim X.$$

which is an important formula in the proof of the Bend and Break lemma, see [KM98, P.12 line -7].

For the fixed points version, due to the exact sequence

$$0 \longrightarrow f^*T_X \otimes \mathscr{I}_B \longrightarrow f^*T_X \longrightarrow f^*T_X|_B \longrightarrow 0$$

we get

$$\chi(C, f^*T_X \otimes \mathscr{I}_B) = \chi(C, f^*T_X) - \chi(C, f^*T_X|_B).$$

But as B is zero-dimensional, we have

$$\begin{aligned} \chi(C, f^*T_X|_B) &= \chi(B, f^*T_X) \\ &= h^0(B, f^*T_X) \\ &= \operatorname{length}(B) \cdot \dim X, \end{aligned}$$

hence by Theorem 1.25, we have

$$\begin{aligned} \dim_{[f]} \operatorname{Mor}(C, X; f|_B) &\geq \chi(C, f^*T_X \otimes \mathscr{I}_B) \\ &= \chi(C, f^*T_X) - \operatorname{length}(B) \cdot \dim X \\ &= -(K_X \cdot f_*C) + (1 - g(C)) \cdot \dim X - \operatorname{length}(B) \cdot \dim X \\ &= -(K_X \cdot f_*C) + (1 - g(C) - \operatorname{length}(B)) \cdot \dim X. \end{aligned}$$

In particular, when B is a single point denoted by $\{0\}$, then

$$\dim_{[f]} \operatorname{Mor}(C, X; f|_{\{0\}}) \geq -(K_X \cdot f_*C) - g(C) \cdot \dim X$$

is nothing but just the formula in [KM98, P.13 line 3].

2.2 Bend-and-Break Lemmas

Olivier Debarre in his book [Deb01, Chap3] says that:

The whole story began in 1979, with Mori's spectacular proof of a conjecture of Hartshorne characterizing projective spaces as the only smooth projective varieties with ample tangent bundle. The techniques that Mori introduced to solve this conjecture have turned out to have more far-reaching applications than Hartshorne's conjecture itself.

Recall that a *1-cycle* on X is a formal sum $\sum_{i=1}^{s} n_i C_i$, where the $n_i \in \mathbb{Z}$ and C_i are integral curves on X. It is called *rational* if the C_i are rational curves. If C is a curve with irreducible components $C_1, \ldots, C_r$, and $f : C \longrightarrow X$ a morphism, we will write f_*C for the effective 1-cycle $\sum_{i=1}^{r} (\deg\ f|_{C_i}) f(C_i)$ and coefficient is zero if be contracted.

The following is the original bend-and-break lemma, which is Mori's result from 1979. It says that a curve deforming non-trivially keeping a point fixed, must break into an effective 1-cycle with a rational component passing through that fixed point.

Now we give a proof of the bend and break techniques, in [KM98] the Abhyankar lemma is used to produce the rational curves, but I think as we're dealing with the source $C \times \bar{T}$ is a smooth projective surface, blowing up to resolve the indeterminacy is enough to produce rational curves, there is no need to use the Abhyankar lemma which applies to arbitrary dimensions.

Lemma 2.2.1. (Bend and Break I) *Let X be a projective variety, C a smooth projective curve, $c_0 \in C$ a point and $f : C \longrightarrow X$ a non-constant morphism. Assume that there is a smooth, connected non-proper pointed curve $t_0 \in T$ and a morphism*

$$F : C \times T \longrightarrow X$$

such that

(1) $F|_{C \times \{t_0\}} = f$;

(2) $F(\{c_0\} \times T) = f(c_0)$ is a point in X, and

(3) for general $t \in T$, $F|_{C \times \{t\}}$ is different from f.

Then there is a (possibly constant) morphism $f' : C \longrightarrow X$, and a non-zero effective rational 1-cycle $Z = \sum a_i Z_i$ on X such that

(1) $f_*C \sim_{alg} f'_*C + Z$ (see [Ful98, 10.3]),

(2) $f(c_0) \in \bigcup_i Z_i$.

In particular, X contains a rational curve passing through the point $f(c_0)$.

Remark 2.2.2. According to formula in 2.1E, when X is smooth along $f(C)$, a sufficient condition for the hypotheses of Lemma 2.1 is

$$-(K_X \cdot f_*C) - g(C) \cdot \dim X \geq 1.$$

Proof. Let $\bar{T}$ be the smooth compactification of T, then we get a rational map induced from F:

$$F' : C \times \bar{T} \dashrightarrow X.$$

Now we claim that F' is undefined somewhere along $\{c_0\} \times \bar{T}$. Suppose not, then for the proper morphism

$$\pi : U \times \bar{T} \longrightarrow U$$

where every fibre is connected and $U \ni c_0$ is an open neighborhood of C, we have, by our assumption, F' will contracts one fibre $\pi^{-1}(c_0)$ for $c_0 \in U$. So by the Rigidity lemma we see F' will contract every fibre of π over U. That is for all $c \in U$, we have

$$F'(c,t) = F'(c,t_0) = f(c), \text{ for } t \in T \text{ is a general point,}$$

moreover this U is just C minus finite points, so this equality can be extended to the whole of C by [Har77, I, 6.8], contradicting our assumption (3).

Thus there is at least one point $t \in \bar{T}$ where F' is undefined. Using [Bea78, II, 7] to resolve these indeterminacies we get a morphism $\varepsilon : S \longrightarrow C \times \bar{T}$ which is a composite of a finite number of blow-ups and a commutative diagram

$$\begin{array}{ccc} & S & \\ {\scriptstyle\varepsilon}\swarrow & & \searrow{\scriptstyle F_S} \\ C \times \bar{T} & \xrightarrow[F']{} & X \end{array}$$

where ε is not an isomorphism locally over $\{c_0\} \times \{t\}$ for $t \in \bar{T}$ given above.

Hence the pullback by ε over $C \times \{t\}$ which is passes through the undefined point, it will be given by

$$\varepsilon^*(C \times \{t\}) = C' + \sum a_i E_i =: C' + E$$

where $a_i > 0$, $C' \cong C$ because it is the strict transform of $C \times \{t\}$ and C itself is smooth, and E is the cycle of rational curves E_i which is ε-exceptional (recall that we use the same construction in Proposition 1.7 for E_i's). Let $f' : C \longrightarrow X$ be the restriction of F_S to C' and set

$$Z = (F_S)_*(E) = \sum_{\dim F_S(E_i)=1} a_i d_i [Z_i], \quad Z_i := F_S(E_i)_{\text{red}}, \quad d_i = [k(E_i) : k(Z_i)].$$

As f can be identified with F_S restricted to the fibre of $S \to \bar{T}$ at t_0, then we have an algebraically equivalent relation

$$f_*C \sim_{alg} f'_*C + Z.$$

Since for each i such that E_i is not contracted by F_S, E_i is a rational curve and the map $E_i \to Z_i$ induces the embeddings $k(Z_i) \hookrightarrow k(E_i) \cong k(t)$, hence by Lüroth's theorem, we see Z_i are all rational curves, that is the (1).

Moreover notice that there must be at least one Z_i passing through $f(c_0)$, otherwise all of these E_i will be contracted to the point $f(c_0)$, consequently the F' would be defined locally around $\{c_0\} \times \{t\}$, which proves (2). □

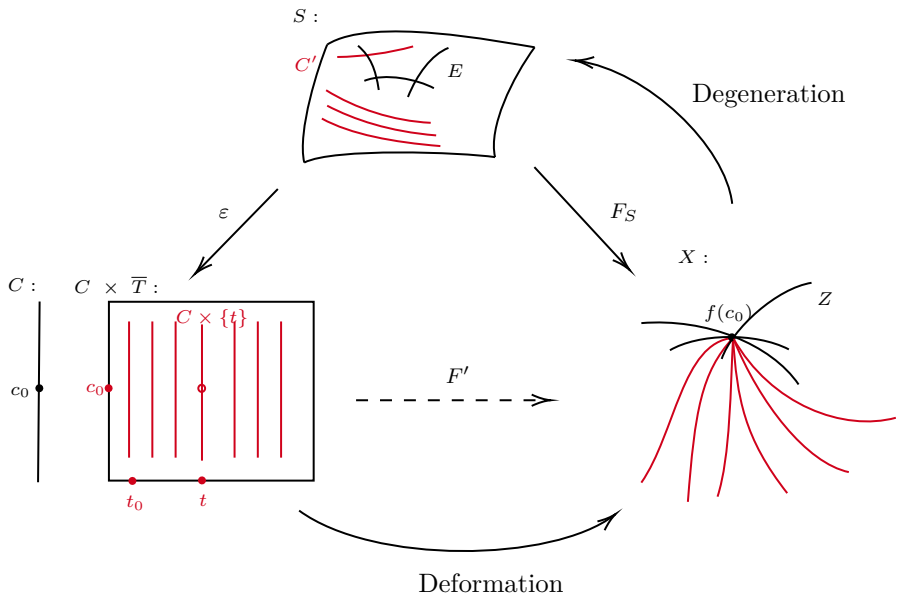


Figure 2.7.: Bend and Break I

Remark 2.2.3. (Thanks to *YuHong Zhou*) The assumptions imply that T is necessarily non-proper. Thus the wording "possibly non-proper" in [KM98] is harmless; the proper case is excluded by rigidity. In the case when T is indeed proper, there is no extra compactification, as every smooth proper curve must be projective. See [Har77, II, 4.10.2]. So the conditions (2) and (3) imply that this curve must be non-proper.

Suppose we have a proper curve T with assumption (2) that $F(\{c_0\} \times T) = f(c_0)$ and $\bar{T} = T$, meaning F' is defined everywhere and it is nothing but F. Then for the proper morphism

$$\pi : C \times T \longrightarrow C$$

we see that F contracts the fibre $\pi^{-1}(c_0)$. By the Rigidity lemma, F will contract every fibre $\pi^{-1}(c) = \{c\} \times T$ for each $c \in C$ to a point. That is, for any $c \in C$, we have

$$F(c,t) = F(c,t_0) = f(c) \text{ for any } t \in T,$$

contradicting our assumption (3).

Lemma 2.2.4. (Bend and Break II) *Let X be a projective variety and $f : \mathbb{P}^1 \longrightarrow X$ be a non-constant morphism. Assume that there is a smooth connected (possibly non-proper) pointed curve $t_0 \in T$ and a morphism $F : \mathbb{P}^1 \times T \longrightarrow X$ such that*

(1) $F|_{\mathbb{P}^1 \times \{t_0\}} = f$;

(2) $F(\{0\} \times T) = f(0)$, $F(\{\infty\} \times T) = f(\infty)$ and

(3) $F(\mathbb{P}^1 \times T)$ is a surface.

Then the 1-cycle $f_*\mathbb{P}^1$ is algebraically equivalent to a 1-cycle Z that passes through $f(0)$ and $f(\infty)$, such that Z is either

(1) a reducible 1-cycle of rational curves, or

(2) a multiple rational curve (i.e. of the form $a\Gamma$ for some $a > 1$).

Remark 2.2.5. Since the automorphism group of $\mathbb{P}^1$ fixing two points is $\mathbb{G}_m = \mathbf{k}^\times$ is one-dimensional, so the assumptions of the lemma are satisfied if $\deg f = 1$[1] and

$$\dim_{[f]} \operatorname{Mor}(\mathbb{P}^1, X; f|_{\{0,\infty\}}) \geq 2,$$

since 2 morphisms appearing in $\operatorname{Mor}(\mathbb{P}^1, X; f|_{\{0,\infty\}})$ are differ from an element of $\mathbb{G}_m$

$$\begin{array}{ccc} & & X \\ & \overset{f'}{\nearrow} & \uparrow f \\ \mathbb{P}^1 & \xrightarrow[0,\infty]{\text{fixed}} & \mathbb{P}^1 \end{array}$$

and we must ensure that the images of $\mathbb{P}^1$ lie in X must be deformed by the condition (3), informally speaking we say "moved".

If X is smooth along $f(\mathbb{P}^1)$, then according to the formula 2.1E, condition is enough to be

$$-(K_X \cdot f_*\mathbb{P}^1) - \dim X \geq 2.$$

[1]For $\deg f = d > 1$, the pushforward cycle $f_*\mathbb{P}^1 = d\Gamma$ is already gives the conclusion. Note that f is a finite morphism since it is fibre-finite and proper.

Proof. Let $\bar{T}$ be a smooth compactification of T, and let $\pi : P \longrightarrow \bar{T}$ be a $\mathbb{P}^1$-bundle over $\bar{T}$ containing $T \times \mathbb{P}^1$ as an open subset such that the diagram commutes

$$\begin{array}{ccc} P & \xrightarrow{\pi} & \bar{T} \\ \uparrow & & \uparrow \text{open} \\ T \times \mathbb{P}^1 & \longrightarrow & T \end{array} \quad \text{im.}$$

where P hence a smooth surface.

Let $F' : P \dashrightarrow X$ be the rational map induced from F:

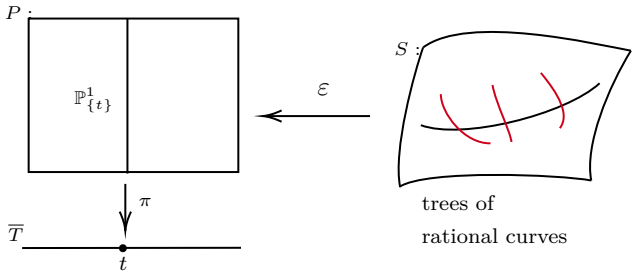
$$\begin{array}{ccc} P & \overset{F'}{\dashrightarrow} & X \\ \text{open}\uparrow & \nearrow_{F} & \\ T \times \mathbb{P}^1. & & \end{array}$$

Similarly, we get a morphism $\varepsilon : S \longrightarrow P$ which is a composite of a finite number of blow-ups and a commutative diagram

$$\begin{array}{ccccc} & & S & & \\ & \swarrow^{\varepsilon} & & \searrow^{F_S} & \\ P & & \xrightarrow[F']{} & & X \end{array}$$

such that F_S is a morphism. We may assume that, the $\mathbb{P}^1$-bundle is chosen in such a way that the number r of blow-ups of ε is minimal with the property F_S is a morphism.

Then the fibres of the morphism $p : S \xrightarrow{\varepsilon} P \xrightarrow{\pi} \bar{T}$ are trees of rational curves and so their pushout on X are algebraically equivalent 1-cycles whose components are



rational curves due to Lüroth's theorem.

Note that $p = \pi \circ \varepsilon : S \longrightarrow \bar{T}$ admits two sections from $\bar{T}_0$ and $\bar{T}_\infty$ which are respectively given by the closure of $\{0\} \times T$ and $\{\infty\} \times T$ in P, that is two strict transforms of smooth curves lying in S that are contracted via F' to be $f(0)$ and $f(\infty)$ respectively. Hence we get that

$$(F_S)_* p^{-1}(t)$$

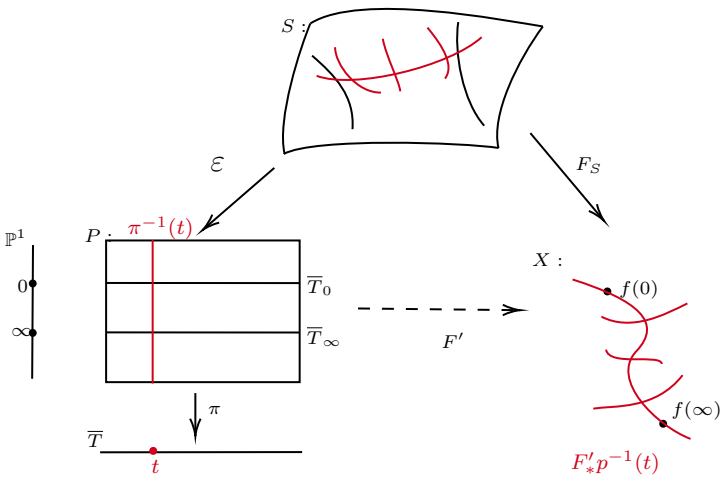


Figure 2.8.: 1-cycle passing through $f(0)$ and $f(\infty)$.

is a 1-cycle on X passing through $f(0)$ and $f(\infty)$, algebraically equivalent to $f_*\mathbb{P}^1$.

To prove the lemma, we need to show that for some $t \in \bar{T}$, the $(F_S)_* p^{-1}(t)$ is not integral.

Assume that for all $t \in \bar{T}$, the $(F_S)_* p^{-1}(t)$ is integral. But this implies that for all $t \in \bar{T}$, there is only one component R of $p^{-1}(t)$ that is not contracted by F_S. Moreover, the coefficient of $(F_S)_* p^{-1}(t) = (F_S)_*(R)$ on the left is 1, and coefficient on the right is the multiplicity of R along the divisor $p^{-1}(t)$ also must be 1. This implies R must be a reduced component of $p^{-1}(t)$.

We aim to arrive at a contradiction of this assumption, using the induction on the number r of blow-ups in ε to resolve F'.

- $r = 0$, then $F' : P \longrightarrow X$ itself is a morphism, contracts two different sections

$$\bar{T}_0, \bar{T}_\infty \subseteq P$$

of the $\mathbb{P}^1$-bundle $P \longrightarrow \bar{T}$ given as compactification of $\{0\}\times T$ and $\{\infty\}\times T$ respectively. This implies if we fix an ample divisor H on X, the morphism induce the pullback map $F'^* : N^1(X) \longrightarrow N^1(P)$ gives

$$(\bar{T}_0 \cdot F'^* H) = (\bar{T}_\infty \cdot F'^* H) = 0$$

by projection formula, and moreover by ampleness we have $(F'^* H)^2 > 0$. But by Hodge Index theorem [Har77, V, 1.9], we have

$$(\bar{T}_0^2) < 0, \ (\bar{T}_\infty^2) < 0,$$

and so $F'^{*}H$, $\bar{T}_0$ and $\bar{T}_\infty$ are linearly independent element in $N^1(P)$. But on the other hand, P is $\mathbb{P}^1$-bundle over $\bar{T}$ which must has $\rho(P) = 2$, see [Har77, V, 2.3], hence arrive a contradiction.

- Consider the sequence of blow-ups

$$S = S_r \longrightarrow S_{r-1} \longrightarrow \cdots \longrightarrow S_1 \longrightarrow S_0 = P$$

and suppose satisfies the condition that 1-cycle is integral for all $t \in \bar{T}$, but the necessary condition is for $\varepsilon_1 : S_r \longrightarrow S_1$ the fibre is must be reduced, and let this be our induction hypothesis.

- Now for the whole sequence, and consider the first blow-up

$$S_1 \xrightarrow{\varphi} P$$

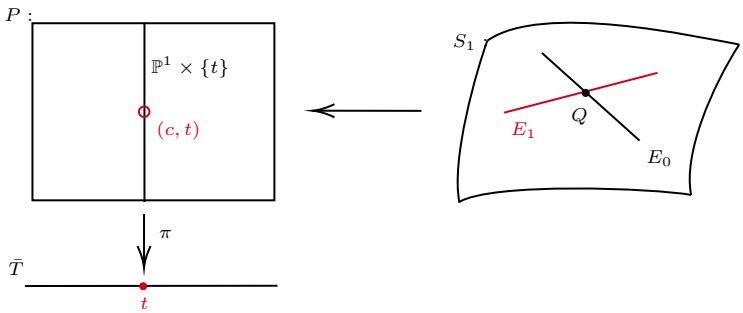
in a single point (c, t). This implies the fibre $p_1 : S_1 \longrightarrow \bar{T}$ above point t has two components

$$p_1^{-1}(t) = E_0 \cup E_1$$

where the E_i are (-1)-curves, since for total transformation $\varphi^*\mathbb{P}^1 = E_0 + E_1$ we have

$$(E_0)^2 = (\varphi^*\mathbb{P}^1 - E_1)^2 = (\varphi^*\mathbb{P}^1)^2 - 2(\varphi^*\mathbb{P}^1 \cdot E_1) + (E_1)^2 = (\mathbb{P}^1)^2 - 1 = -1,$$

meeting in a single point $Q \mapsto (c, t)$. Here E_0 denotes the strict transform of $\mathbb{P}^1 \times \{t\} \subseteq P$, and E_1 is the exceptional divisor.



Consider the natural map $\varepsilon_1 : S = S_r \longrightarrow S_1$, as our choice of P the number of blow-ups r is minimal, implies that there is at least one component of $\varepsilon_1^{-1}(E_1)$ is not contracted by F_S, otherwise if F_S contracts all of $\varepsilon_1^{-1}(E_1)$ the rational map $S_1 \dashrightarrow X$ will sends the whole of E_1 to a single point, F' will be defined locally around (c, t) contradicting the minimality of the number r. Moreover since F_S contracts all but one component R of $p^{-1}(t) = \varepsilon_1^{-1}(E_0 \cup E_1) = \varepsilon_1^{-1}(E_0) \cup \varepsilon_1^{-1}(E_1)$, we see that F_S contracts

all of $\varepsilon_1^{-1}(E_0)$, so the rational map $S_1 \dashrightarrow X$ is defined along $E_0 - \{Q\}$, and so ε_1 is a local isomorphism along $E_0 - \{Q\}$.

We claim that $S_1 \dashrightarrow X$ is also defined at point Q. If not, then $\varepsilon_1 : S \longrightarrow S_1$ also blows up Q, then the unique component R of $p^{-1}(t) = \varepsilon_1^{-1}(E_0 \cup E_1)$ which is not contracted by F_S will lie in $\varepsilon_1^{-1}(Q)$. But since Q is a point of multiplicity 2 in $E_0 \cup E_1$, any component of $\varepsilon_1^{-1}(Q)$ has multiplicity at least 2, which contradicts the induction hypothesis. So, $S_1 \dashrightarrow X$ is defined locally at the (-1)-curve E_0, it follows that the morphism F_S can descends on a blow-down of this (-1)-curve E_0 on P, contradicting the minimality of the number $r \geq 1$.

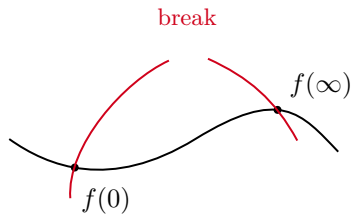
Hence we have shown that $F_{S*}p^{-1}(t)$ must not be integral. □

Remark 2.2.6. The two fixed points need not be 0 and ∞; any two distinct points of $\mathbb{P}^1$ can be used. In fact, we can choose an automorphism

$$\begin{pmatrix} q_1' & q_1 \\ q_2' & q_2 \end{pmatrix} \in \mathrm{PGL}(1, \mathbf{k})$$

such that sending $0 = [0 : 1]$, $\infty = [1 : 0]$ to $q = [q_1 : q_2]$, $q' = [q_1' : q_2']$ where $q \neq q'$.

So after change of coordinates, we even can arrive at conclusion that there must exist undefined points lying in $\bar{T}_0$ and $\bar{T}_\infty$ such that $f_*\mathbb{P}^1$ can break into a 1-cycle with one component passing through $f(0)$ and another component passing through $f(\infty)$:



As we will see later when proving the Mori's 1982 result, we just fix 2 points q and q' which can be arbitrary.

Fix $t \in \bar{T}$ such that the non-trivial effective cycle $Z_t := (F_S)_*p^{-1}(t)$ non-integral. Suppose that every ε-exceptional component of $p^{-1}(t)$ is contracted by F_S. Then only the strict transform R_t of $\pi^{-1}(t)$ contributes to Z_t. Since R_t occurs with multiplicity 1 in the fibre and as Z_t is non-integral, we have $d > 1$, i.e. $F_S|_{R_t}$ is a finite morphism of degree greater than one onto its image.

More generally, there is a stronger version of bend-and-break lemma, which does not need to restrict the number of fixed points, see [Deb01, 3.5].

2.3 Application: Finding Rational Curves

2.3A Rational Curves On Fano Varieties

We are ready to formulate and prove the first main result proved in 1982 by Mori, about the existence of rational curves on *Fano variety*, the smooth variety with an ample $-K_X$ divisor.

Example 2.3.1. (Simplest Fano variety) By the adjunction formula, a smooth projective hypersurface $X_d \subseteq \mathbb{P}^{n+1}$ of degree d satisfies

$$\begin{aligned} K_{X_d} &= (K_{\mathbb{P}^{n+1}} + dH)|_{X_d} \\ &= (d-n-2)H|_{X_d}. \end{aligned}$$

Hence, X_d is Fano if $d < n+2$.

Example 2.3.2. (Construct Fano variety) If Y is a Fano variety with nef divisors $D_1, \dots, D_r$ such that

$$-K_Y - D_1 - \cdots - D_r$$

is ample, then $X = \mathbb{P}\left(\bigoplus_{i=1}^{r} \mathscr{O}_Y(D_i)\right)$ is a Fano variety.

In fact, for canonical projection $\pi : X \to Y$, follows from [Har77, III, ex8.4(b)]we have

$$0 \to \Omega_{X/Y} \to \pi^* \left(\bigoplus_{i=1}^{r} \mathscr{O}_Y(D_i)\right) \otimes \mathscr{O}_X(-1) \to \mathscr{O}_X \to 0$$

is an exact sequence, so taking the highest power wedge to get

$$\begin{aligned} \det(\pi^* \left(\bigoplus_{i=1}^{r} \mathscr{O}_Y(D_i)\right) \otimes \mathscr{O}_X(-1)) &\cong \pi^* \det \left(\bigoplus_{i=1}^{r} \mathscr{O}_Y(D_i)\right) \otimes \mathscr{O}_X(-r) \\ &\cong \det \Omega_{X/Y}. \end{aligned}$$

Moreover, we see π is a flat morphism as locally it is of the form $R \to R[X_1, \dots, X_{r-1}]$ is flat ring map, hence by [SP25, Tag 01V8] we see that π is a smooth morphism; thus

$$0 \to \pi^*\Omega_Y \to \Omega_X \to \Omega_{X/Y} \to 0$$

is exact by [SP25, Tag 02K4].

Combining the above results, we get

$$-K_X \sim_{\text{lin}} -\pi^* K_Y - K_{X/Y} \sim_{\text{lin}} rD + \pi^* M$$

where D is the divisor associated with $\mathscr{O}_X(1)$ and M denotes the $-K_Y - D_1 - \cdots - D_r$. Then since each D_i is nef on Y, we know that $M + D_i$ is ample by Corollary 1.4.7. Consider the tensor product

$$\left(\bigoplus_{i=1}^{r} \mathscr{O}_Y(D_i)\right) \otimes \mathscr{O}_Y(M) \cong \bigoplus_{i=1}^{r} \mathscr{O}_Y(M + D_i)$$

which is an ample vector bundle (we can just admit it. In fact can see [Laz04b, 6.1.13]). As Y is a smooth variety, by [Har77, II, ex7.9] we have

$$\varphi : X' := \mathbb{P}\left(\bigoplus_{i=1}^{r} \mathscr{O}_Y(M + D_i)\right) \xrightarrow{\cong} \mathbb{P}\left(\bigoplus_{i=1}^{r} \mathscr{O}_Y(D_i)\right).$$

But the left hand side admits an ample line bundle by definition of ample vector bundle, and furthermore

$$\mathscr{O}_{X'}(1) \cong \varphi^*\big(\mathscr{O}_X(1) \otimes \pi^* \mathscr{O}_Y(M)\big),$$

because $\mathrm{Sym}^d(\mathscr{E} \otimes \mathscr{O}_Y(M)) = \mathrm{Sym}^d \mathscr{E} \otimes \mathscr{O}_Y(M)^{\otimes d}$ and using [Har77, II, 7.9]. So we see $D + \pi^* M$ is indeed ample. Hence $rD + \pi^* M$ is also ample by D is nef on X and Corollary 1.4.7 again, which implies that $-K_X$ is ample.

Lemma 2.3.3. *Let R be a finitely generated $\mathbb{Z}$-algebra.*

Then

(1) for any maximal ideal $\mathfrak{m} \subseteq R$, $R/\mathfrak{m}$ is a finite field.

(2) the closed points of Spec R is Zariski dense.

Proof. (1): Notice that $\mathbb{Z} \cap \mathfrak{m}$ must be a prime ideal of $\mathbb{Z}$. If this prime ideal is 0, i.e. $\mathbb{Z} \cap \mathfrak{m} = (0)$, then the field $(R/\mathfrak{m})_{\mathfrak{m}}$ is a finitely generated $\mathbb{Z}_{(0)} = \mathbb{Q}$-algebra, hence

$$\mathbb{Q} \hookrightarrow k(\mathfrak{m}) := (R/\mathfrak{m})_{\mathfrak{m}} \cong R/\mathfrak{m}$$

is a finite field extension.

Now let $e_1, \ldots, e_n$ be a $\mathbb{Q}$-basis of $R/\mathfrak{m}$ and let $x_1, \ldots, x_n \in R/\mathfrak{m}$ be elements that generate $R/\mathfrak{m}$ as a $\mathbb{Z}$-algebra. Then there is an integer $N \gg 0$ such that $Nx_j \in \bigoplus^n \mathbb{Z} e_i$, which just clears denominators. Hence

$$(\bigoplus \mathbb{Z} x_i =) \bigoplus \mathbb{Q} e_i = R/\mathfrak{m} \subseteq \bigoplus \mathbb{Z}[\frac{1}{N}] \cdot e_i$$

which is absurd.

Hence, $\mathbb{Z} \cap \mathfrak{m} = (p)$ so $R/\mathfrak{m}$ is a field extension of the finite field $\mathbb{Z}/(p)_{(p)} =: \mathbb{F}_p$. Since R/m is finitely generated over $\mathbb{Z}$, it is also finitely generated over $\mathbb{Z}/(p)$; this implies that $\mathbb{F}_p \hookrightarrow R/\mathfrak{m}$ is a finite extension, hence a finite field.

(2): We need to show that for any $a \neq 0 \in R$, the open subset

$$D(a) = \text{Spec } R_a \subseteq \text{Spec } R$$

contains a closed point.

Let $\mathfrak{n}$ be a maximal ideal in R_a, and since R_a is still an integral domain finitely generated over $\mathbb{Z}$, we know $R_a/\mathfrak{n}$ is a finite field by (1).

Consider the natural map $\phi : R \to R_a$, and a prime ideal

$$\mathfrak{m} := \phi^{-1}(\mathfrak{n}) \in \text{Spec } R.$$

Then $R/\mathfrak{m}$ is a subring of $R_a/\mathfrak{n}$ implies that $R/\mathfrak{m}$ is a finite domain, hence a field. So $\mathfrak{m} \subseteq R$ is a maximal ideal does not contain a, i.e. $\mathfrak{m} \in D(a)$ as we want. □

Lemma 2.3.4. *If a homogeneous system of algebraic equations with $\mathbb{Z}$-coefficients has a non-trivial solution in $\bar{\mathbb{F}}_p$ for infinitely many p (in the general case below, having a non-trivial solution over a Zariski dense subset of Spec R).*

Then it has a non-trivial solution in arbitrary algebraically closed field of characteristic 0.

Proof. A chosen homogeneous system

$$F_1, \ldots, F_r \in \mathbb{Z}[y_0, \ldots, y_N] \subseteq R[y_0, \ldots, y_N]$$

defines a closed subscheme $Z \subseteq \mathbb{P}^N_R$. And consider the structure morphism

$$\pi : \mathbb{P}^N_R \longrightarrow \text{Spec } R$$

which is proper, so $\pi(Z) \subseteq \text{Spec } R$ is closed. As the condition states, this homogeneous system has a non-trivial solution over a Zariski dense subset of closed points, that is the fibre

$$Z_s = Z \underset{R}{\times} \text{Spec } k(s) \overset{\pi|_Z}{\longmapsto} \{s\}$$

is non-empty, as s runs through the closed points in Spec R.

So $\pi(Z)$ contains this Zariski dense subset, therefore $\pi(Z) = \text{Spec } R$ also contains the generic point η of Spec R. It follows that the generic fibre

$$Z_\eta = Z \underset{R}{\times} \text{Spec } k(\eta)$$

is non-empty, where $k(\eta)$ has characteristic 0 since it contains $\mathbb{Q}$.

Now for arbitrary algebraically closed field K of char 0, we have $k(\eta) \hookrightarrow K$, so after base extension

$$\begin{array}{ccc} Z_K & \longrightarrow & Z_{k(\eta)} \\ \downarrow & & \downarrow \\ \text{Spec } K & \longrightarrow & \text{Spec } k(\eta) \end{array}$$

shows that Z_K is non-empty, that is this homogeneous system has a non-trivial solution in K. □

Remark 2.3.5. (Functor point view of schemes) In the above statement, saying it has a non-trivial solution corresponds to the fibre being non-empty; this is because for projective space $\mathbb{P}^N_R$, for any field extension $R \to k(\eta) \hookrightarrow K$, the *$K$-valued points* of $\mathbb{P}^N_R$ is defined by

$$\mathbb{P}^N_R(K) := \text{Hom}_R(\text{Spec } K, \mathbb{P}^N_R)$$

the morphisms Spec $K \to \mathbb{P}^N_R$ making the following diagram commute

$$\begin{array}{ccc} \text{Spec } K & \longrightarrow & \mathbb{P}^N_R \\ & \searrow & \downarrow \\ & & \text{Spec } R. \end{array}$$

And also one has

$$\mathbb{P}^N_R(K) \cong \left\{ [x_0 : x_1 : \cdots : x_N] \in K^{N+1} \mid x_i \text{ are not all zero simultaneously } \right\},$$

which comes from the fact that the morphisms

$$\begin{aligned} R[T_1, \ldots, T_N]/(f_1, \ldots, f_r) &\longrightarrow A \\ T_i &\longmapsto a_i \end{aligned}$$

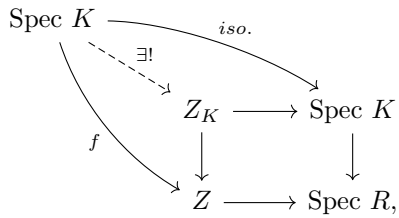
are in one-to-one correspondence with the elements $a = (a_1, \ldots, a_N) \in A^N$ satisfying $f_1(a) = \cdots = f_r(a) = 0$, where A is a R-algebra.

So for $Z = V_+(F_1, \ldots, F_r)$, we have

$$Z(K) = \{ [x_0 : \cdots : x_N] \mid F_1(x) = \cdots = F_r(x) = 0 \}$$

that is the non-trivial solutions in K; we wish this set to be non-empty. But the morphism Spec $K \to Z$ exists if and only if the fibre $Z_K = Z \underset{R}{\times} \text{Spec } K$ is non-empty:

" $\Rightarrow$ ": If there is a R-morphism $f : \operatorname{Spec} K \to Z$, then



so Z_K is non-empty.

" $\Leftarrow$ ": If $Z_K \neq \emptyset$, then as finite type scheme, must contain a closed point x, so has $K \hookrightarrow k(x)$ which is a finite field extension, But as K is algebraically closed, we get a morphism

$$\operatorname{Spec} K \cong \operatorname{Spec} k(x) \longrightarrow Z_K \longrightarrow Z.$$

Theorem 2.3.6. ([Mor82]) *For every point of a smooth projective Fano variety X of positive dimension, there is a rational curve D passing through it such that*

$$0 < -(D \cdot K_X) \leq \dim X + 1.$$

Proof. This is done in several steps.

Step 1: To find this rational curve, we intend to use the bend-and-break technique. Thus we have to find a morphism $f : C \longrightarrow X$ which we will be able to deform, then we bend these deformations together over a fixed point, and break them into some rational curves.

Pick any integral projective curve C' passing through x, and let $f : C \longrightarrow X$ be its normalization. Then C is a smooth projective curve. Pick c_0 such that $f(c_0) = x$, which is the target point we are bending.

Step 2: According to the formula in 2.1E,

$$\dim_{[f]} \operatorname{Mor}(C, X; f|_{\{c_0\}}) \geq -(K_X \cdot f_* C) - g(C) \cdot \dim X.$$

Thus whenever the quantity $-(K_X \cdot f_* C) - g(C) \dim X$ is positive, there must be a non-trivial one-dimensional subvariety of deformations of the map $f : C \to X$ keeping the image of c_0 fixed.

So using Bend and Break I, therefore, we obtain a rational curve in X through x.

Step 3: Now we show how to get the positive quantity

$$-(K_X \cdot f_* C) - g(C) \cdot \dim X > 0.$$

To do this, we need to get $-(K_X \cdot f_* C)$ to be big enough. We consider three cases:

(1) $g(C) = 0$.

Then automatically $-(K_X \cdot f_* C) > 0$ by $-K_X$ is ample and Nakai-Moishezon criterion. Actually by Lüroth theorem, the image of C is already a rational curve through x.

(2) $g(C) = 1$.

For the same reason we have $-(K_X \cdot f_* C) > 0$, but as we want it to be big enough, we apply some endomorphism on C fixing the point c_0, which is a finite map.

In particular, via the injective ring homomorphism [Har77, IV, 4.10]

$$\mathbb{Z} \longrightarrow \mathrm{End}(C, c_0)$$

we can pick some endomorphism n_C to compose with f. Then

$$-(K_X \cdot (f \circ n_C)_* C) - \dim X = -n^2 (K_X \cdot f_* C) - \dim X,$$

where the equality follows from $\deg n_C = n^2$ by [Har77, IV, 4.17], thus we have that $n_{C*} C = n^2 C$; at this time some multiple of C (which is connected) moves, and one point of some sheet over the image stays fixed.

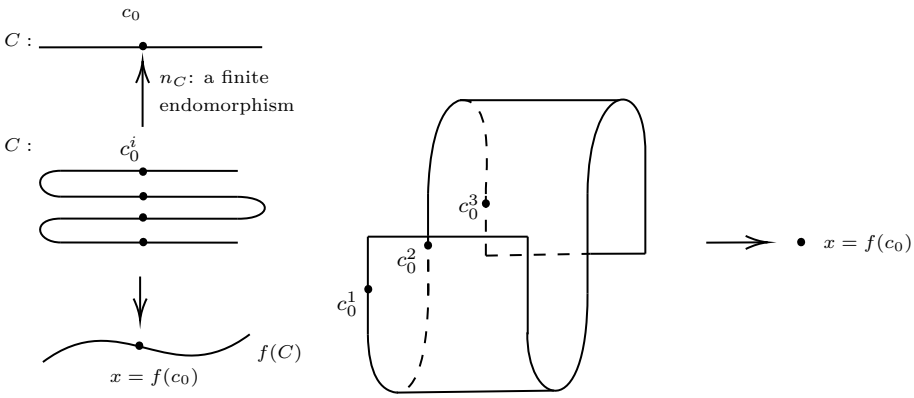


Figure 2.9.: Endomorphism n_c rise the degree of smooth curve C.

(3) $g(C) \geq 2$.

By Hurwitz[2] formula for a endomorphism of degree d, we have

$$2g(C) - 2 = d \cdot (2g(C) - 2) + \deg R_{\mathrm{ram}}.$$

[2]Since our ground field has characteristic zero, the endomorphism is automatically separable.

If $d > 1$, the right-hand side would be strictly larger than the left, since $\deg R_{\text{ram}} \geq 0$, so any curve of genus ≥ 2 has no endomorphism of degree greater than 1.

Thus we can try to consider another smooth projective curve $h : C" \longrightarrow C$ and deform $f \circ h$. But the problem here is that even though

$$((f \circ h)_* C" \cdot K_X) = \deg h \cdot (f_* C \cdot K_X),$$

the genus also changes. For example, if $C"$ is a connected m-sheeted unbranched cover of C, then by the Hurwitz formula again:

$$2g(C") - 2 = \deg h \cdot (2g(C) - 2) = m \cdot (2g(C) - 2),$$

hence

$$-(K_X \cdot (f \circ h)_* C") - g(C") \cdot \dim X = -m((K_X \cdot f_* C) + g(C) \cdot \dim X) + (m-1) \dim X.$$

This does not necessarily get positive even by making m large enough.

Step 4: The remaining difficulty is the case $g(C) > 1$, because C does not admit an endomorphism of high degree. However, over fields of positive characteristic, a curve C can admit an endomorphism in high degree.

The *Frobenius morphism* is **such** an endomorphism.

We see how to pass our original situation to one over a field of characteristic $p > 0$.

Step 5: Take a smooth projective curve C and a smooth projective variety X.

First suppose that both C and X are defined by equations with integral coefficients:

$$h_1(x_0, \ldots, x_n), \ldots, h_r(x_0, \ldots, x_n) \text{ define } X,$$
$$c_1(y_0, \ldots, y_m), \ldots, c_s(y_0, \ldots, y_m) \text{ define } C.$$

Let $\mathbb{F}_p$ be the field with p elements and $\bar{\mathbb{F}}_p$ be its algebraic closure. Then the equations above define varieties C_p and X_p in the projective space $\bar{\mathbb{F}}_p \mathbb{P}^m$ and $\bar{\mathbb{F}}_p \mathbb{P}^n$ respectively.

These varieties are non-singular, and $\dim C_p = 1$, for all but finitely many p, the mapping

$$(y_0, \ldots, y_m) \longrightarrow (y_0^p, \ldots, y_m^p)$$

gives an endomorphism F_p of C_p, called the *Frobenius* endomorphism, it is a finite morphism of degree p, see [Har77, IV, 2.4.3].

Now, by the "generic flatness" theorem ([FGA05, 5.12] or [GW20, 10.84]) for the morphism $C \to \operatorname{Spec} \mathbb{Z}$, there is an open dense subset $U \subseteq \operatorname{Spec} \mathbb{Z}$ containing all but finitely many primes p such that $C_U \to U$ is flat, hence C_p is flat over $\operatorname{Spec} \mathbb{Z}_p$; also

$$T_X|_{C_p} \text{ and } (f^* \omega_X)|_{C_p} \cong f_p^* \omega_{X_p}$$

are flat over $\mathscr{O}_{\mathbb{Z},p}$, where $f_p : C_p \to X_p$ is the base change of f.

But these quantities, $((f_p)_* C_p \cdot K_{X_p}) = (C_p \cdot (f_p)^* K_{X_p})$, $g(C_p)$ and $\chi(f^* T_X|_{C_p})$ are all related to the Euler characteristic number, which is a locally constant function over U, since $C_U \to U$ is a flat family, see [FGA05, 5.10(2)]. Moreover notice that $C_U \to U$ is proper by base change

$$\begin{array}{ccc} C_U & \longrightarrow & C \\ \downarrow & & \downarrow \text{proper} \\ U & \longrightarrow & \operatorname{Spec} \mathbb{Z}. \end{array}$$

As U is open dense hence connected, the quantities $((f_p)_* C_p \cdot K_{X_p})$, $g(C_p)$ and $\chi(f^* T_X|_{C_p}) = \chi(f_p^* T_{X_p})$ are constant for almost all primes p. Therefore the "base-pointed" deformation space of the composite morphism

$$C_p \xrightarrow{F_p^m} C_p \xrightarrow{f_p} X_p$$

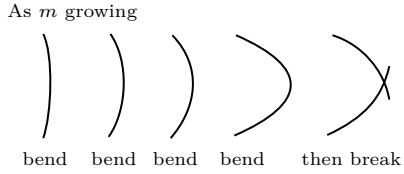
has dimension $\dim_{[f_p \circ F_p^m]} \operatorname{Mor}(C_p, X_p; f_p \circ F_p^m|_{\{c_{0,p}\}})$ bounded below by

$$-(K_{X_p} \cdot (f_p \circ F_p^m)_* C_p) - g(C_p) \cdot \dim X = -p^m (K_{X_p} \cdot (f_p)_* C_p) - g(C_p) \cdot \dim X.$$

As ampleness is stable under finite pullbacks, Proposition 1.2.7, we see that $-K_{X_p}$ still an ample divisor, hence by Nakai-Moishezon again the quantity $-(K_{X_p} \cdot (f_p)_* C_p)$ is positive and independent of p. So the above expression

$$-p^m (K_{X_p} \cdot (f_p)_* C_p) - g(C_p) \cdot \dim X$$

is positive for $m \gg 1$.



Then as in *Step 2*, we produce a rational curve A_p on X_p for almost all p.

Step 6: Suppose now that we are in the general case, the coefficients of the $h_j(x_0, \ldots, x_n)$ (defining X in $\mathbb{P}^n$), the $c_j(y_0, \ldots, y_m)$ (defining C in $\mathbb{P}^m$), the $b_j(x_0, \ldots, y_m)$ (defining the graph $\Gamma(f)$ in $\mathbb{P}^n \times \mathbb{P}^m$) and the coordinates of $c_0 \in C \subseteq \mathbb{P}^m$ are not necessarily integers.

In any case, these coefficients generate a finitely generated $\mathbb{Z}$-algebra R. Now let $\mathfrak{m}$ be any maximal ideal corresponding the closed point $P \in \operatorname{Spec} R$, by lemma 3.3, $R/\mathfrak{m}$

is a finite field which is isomorphic to $\mathbb{F}_{p^k}$, the finite field with p^k elements for some prime p. In this case, our *Frobenius* morphism is given by raising the homogeneous coordinates of $\bar{\mathbb{F}}_{p^k}\mathbb{P}^m$ to the p^k-th power.

The rest of the argument is similar to the above. Using the generic flatness again, we find an open dense subset $U \subseteq \operatorname{Spec} R$ such that $C_U \to U$ is flat, and "base-pointed" deformation space again has dimension bounded below by a positive number raised up by degree p^k morphism F_{p^k}. This gives us a rational curve A_P, for all closed points P in some Zariski open dense subset $U \subseteq \operatorname{Spec} R$.

Step 7: Now X is embedded by linear series $|-mK_X|$ for some positive integer m. In this step, we wish to obtain that inequality, so we wish to replace A_p with another rational curve B_p such that

$$-(K_{X_p} \cdot B_p) \leq \dim X + 1.$$

To do this, notice that if $-(K_{X_p} \cdot A_p) > \dim X + 1$, then according to 2.1E to the morphism

$$\varphi : \bar{\mathbb{F}}_p\mathbb{P}^1 \longrightarrow X_p \ , \ A_p = \varphi(\bar{\mathbb{F}}_p\mathbb{P}^1)$$

with two points q, $q' \in \bar{\mathbb{F}}_p\mathbb{P}^1$ fixed, we have

$$\begin{aligned}\dim_{[\varphi]} \operatorname{Mor}(\bar{\mathbb{F}}_p\mathbb{P}^1, X_p; \varphi|_{\{q,q'\}}) &\geq -(K_{X_p} \cdot \varphi_*\bar{\mathbb{F}}_p\mathbb{P}^1) + (1 - g(\bar{\mathbb{F}}_p\mathbb{P}^1) - 2) \cdot \dim X \\ &= -(K_{X_p} \cdot A_p) - \dim X \\ &> 1,\end{aligned}$$

so this deformation space has dimension at least two. But notice that the automorphism group of $\bar{\mathbb{F}}_p\mathbb{P}^1$ fixing two points is $\mathbb{G}_m = \bar{\mathbb{F}}_p^{\times}$, is 1-dimensional, thus two morphisms in $\operatorname{Mor}(\bar{\mathbb{F}}_p\mathbb{P}^1, X_p; \varphi|_{\{q,q'\}})$ differ by an automorphism of $(\bar{\mathbb{F}}_p\mathbb{P}^1; q, q')$:

$$\begin{array}{ccc} & & A_p \\ & \overset{\varphi'}{\nearrow} & \uparrow{\scriptstyle \varphi} \\ \bar{\mathbb{F}}_p\mathbb{P}^1 & \xrightarrow[\substack{\text{fixed} \\ q,q'}]{} & \bar{\mathbb{F}}_p\mathbb{P}^1. \end{array}$$

Hence image of $\bar{\mathbb{F}}_p\mathbb{P}^1 \subseteq X_p$ must move. By bend-and-break II, we see

$$A_p = \varphi_*\bar{\mathbb{F}}_p\mathbb{P}^1 \sim_{alg} \sum a_i C_i$$

decompose into a sum of rational curves of lower degree, with

$$-(K_{X_p} \cdot C_i) < -(K_{X_p} \cdot A_p).$$

Continue this process of deformation and decomposition, we must be able to find a rational curve of lower degree as long as $-(K_{X_p} \cdot A_p) > \dim X + 1$. This process must terminate in finite steps, to get our desired rational curve B_p, with

$$0 < -(K_{X_p} \cdot B_p) \leq \dim X + 1.$$

The general case is similar, deform the morphism $\varphi : \bar{\mathbb{F}}_{p^k}\mathbb{P}^1 \to X_P$ for all closed points $P \in U$, counting the lower bound of dimension and continuing this process to arrive at a rational curve B_P with

$$0 < -(K_{X_P} \cdot B_P) \leq \dim X + 1.$$

Step 8: In this step, we must conclude the existence of a rational curve on the Fano variety X of characteristic 0, from the existence of the bounded degree rational curve B_p for almost all p. (The general case is for B_P for all closed points $P \in U$).

In our situation, X is embedded into $\mathbb{P}^n$ via the very ample linear series $|-mK_X|$, which is embedded into $\mathbb{P}H^0(X, -mK_X)$. But recall by generic flatness, the dimension of $H^0(X, -mK_X)$ is independent of the choice of almost all primes p (in the general case, $\dim_{k(P)} H^0(X_P, -mK_{X_P})$ is independent with the choice of closed points $P \in U$), so we have embedding $X_p \hookrightarrow \mathbb{P}H^0(X_p, -mK_{X_p}) = \bar{\mathbb{F}}_p\mathbb{P}^n$ for this fixed m, since closed immersion is stable under base change.

But the degree of B_p under the embedding is equal to $-(mK_{X_p} \cdot B_p)$ bounded above by $m(\dim X + 1)$. So for reliability we just choose the $m(\dim X + 1)$-uple embedding for almost all p, giving the map $\bar{\mathbb{F}}_p\mathbb{P}^1 \longrightarrow X_p \subseteq \bar{\mathbb{F}}_p\mathbb{P}^n$ via the homogeneous coordinates

$$[g_{p,0}(t_0, t_1) : g_{p,1}(t_0, t_1) : \cdots : g_{p,n}(t_0, t_1)],$$

where $[t_0 : t_1]$ is the coordinate of $\bar{\mathbb{F}}_p\mathbb{P}^1$, and satisfying

$$h_i(g_{p,0}, g_{p,1}, \ldots, g_{p,n}) = 0$$

identically in (t_0, t_1) for all i. But this condition can be expressed as a homogeneous system of equations in the coefficients of the $g_{p,k}$, for almost all p this system has a solution, hence by lemma 3.4 has a solution in any algebraically closed field of characteristic 0. This solution defines a rational curve lying in X (recall Lüroth theorem, as it is image of $\mathbb{P}^1$) over char 0, passing through the point x.

Step 9: This rational curve over char 0, which we denote by D, by our choice of $g_{p,i}$'s it has degree at most $m(\dim X + 1)$, then return to the char 0 these equations $g_0, \ldots, g_n$ are still bounded above by degree $m(\dim X + 1)$, that is

$$-m(K_X \cdot D) \leq m(\dim X + 1),$$

hence

$$-(K_X \cdot D) \leq \dim X + 1,$$

and is strictly bounded below by 0 since $-K_X$ is ample, using the Nakai-Moishezon criterion.

For the general case is again similar, yielding a solution which is a rational curve over char 0, and counting the degree which is bounded by the choice of $g_{P,i}$'s, for all closed points $P \in U$. □

2.3B Finding Rational Curves when K_X is not Nef

Now let us weaken our hypothesis on the smooth projective variety X, no longer requiring $-K_X$ to be ample, just only assuming that it is nef, which is

$$\big(-K_X \cdot f(C)\big) > 0$$

for our fixed morphism $f : C \to X$, rather than assuming the positivity of $-K_X$.

We also fix an ample divisor H on X.

Theorem 2.3.7. ([Mor82]) *Let X be a smooth projective variety and H an ample divisor on X. Assume that there is an irreducible curve $C' \subseteq X$ such that $-(C' \cdot K_X) > 0$.*

Then there is a rational curve $E \subseteq X$ such that

$$\dim X + 1 \geq -(E \cdot K_X) > 0 \text{ and } \frac{-(E \cdot K_X)}{(E \cdot H)} \geq \frac{-(C' \cdot K_X)}{C' \cdot H}$$

Proof. For finding rational curves, we can use the similar technique as previously, turning the question over char p fields to ensure that the dimension of deformation space is large enough.

Step 1: Let $f : C \to X$ be the normalization of C', which is a birational morphism of degree 1 onto C such that $f_*C = f(C)$. If

$$\begin{aligned}\dim_{[f]} \operatorname{Mor}(C, X; f|_{\{c_0\}}) &\geq -(f(C) \cdot K_X) - g(C) \cdot \dim X \\ &> 0,\end{aligned}$$

then C can deform with one point c_0 fixed. As before, using bend-and-break I, this family must break into $f'(C) +$ (sum of rational curves).

Step 2: (Repeating bend and break): In order to achieve above lower bound of deformation space, we pass to finite characteristic and compose $f_p : C_p \to X_p$ with the m-th power of Frobenius morphism $F_p : C_p \to C_p$, **bend** the curve into case

$$\dim_{[f_p \circ F_p^m]} \operatorname{Mor}(C_p, X_p; f_p \circ F_p^m|_{\{c_{0,p}\}}) \geq -p^m(f_p(C_p) \cdot K_{X_p}) - g(C_p) \cdot \dim X$$

which is positive for $m \gg 0$. So we are able to **break** $f_p(F_{p,*}^m C_p) = p^m \cdot f_p(C_p)$ into

$$p^m \cdot f_p(C_p) \sim_{alg} f_{p,m}^1(C_p) + Z_{p,m}^1$$

where $Z_{p,m}^1$ is a sum of rational curves.

Now we set $M = -(C' \cdot K_X)/(C' \cdot H)$. The ratio $-(f_p(C_p) \cdot K_{X_p})/(f_p(C_p) \cdot H_{X_p})$ equals to M for almost all p (in general case is for all closed points in an open dense, and we'll not repeatedly recall such a fact) and it does not change if we replace f_p with $f_p \circ F_p^m$. If

$$\dim_{[f_{p,m}^1]} \mathrm{Mor}(C_p, X_p; f_{p,m}^1|_{\{c_{0,p}\}}) \geq -(f_{p,m}^1(C_p) \cdot K_{X_p}) - g(C_p) \cdot \dim X > 0,$$

we can deform $f_{p,m}^1(C_p)$ as before, without composing again with the Frobenius morphism. And we recall that algebraic equivalence can imply numerical equivalence[3], hence each time the intersection number of H_p with $f_{p,m}^j(C_p)$ must goes down, so the process must stop after finite steps until the families can no longer be deformed. Thus we reach an algebraic equivalence

$$p^m \cdot f_p(C_p) \sim_{alg} f_{p,m}^s(C_p) + Z_{p,m}^s$$

such that

$$-(f_{p,m}^s(C_p) \cdot K_{X_p}) \leq g(C_p) \cdot \dim X.$$

Let

$$a = -(f_{p,m}^s(C_p) \cdot K_{X_p}); \quad b = -(Z_{p,m}^s \cdot K_{X_p});$$
$$c = (f_{p,m}^s(C_p) \cdot H_p); \qquad d = (Z_{p,m}^s \cdot H_p).$$

For large m, $(c+d) = p^m(f_p(C_p) \cdot H_p)$ is large, and $(a+b)/(c+d) = M$ so $(a+b)$ must get large too. But a is bounded, so b must get large.

Step 3: We claim the following.

For any $\varepsilon > 0$, if $m \gg 0$ then there exists an irreducible component E_p of rational cycle $Z_{p,m}^s$ such that

$$\frac{-(E_p \cdot K_{X_p})}{(E_p \cdot H_p)} > M - \varepsilon.$$

To see this, we need a lemma

[3]Since algebraically equivalent is about the different fibres of the flat morphism $X \times T \supseteq \mathscr{V} \to T$, the intersection number is a quantity related to Euler characteristic which is a locally constant function over T, thus constant by T is a non-singular variety (in particular, connected). See [Ful98, 10.3.2].

Lemma 2.3.8. *If c, $d > 0$, then $\frac{a+b}{c+d} \leq \max\left\{\frac{a}{c}, \frac{b}{d}\right\}$.*

First, if $a/c < M$, then $b/d \geq M$ by lemma. But as $Z^s_{p,m}$ is a sum of rational curves, we write these irreducible components are $E_{p,i}$, and set

$$b_i = -(E_{p,i} \cdot K_{X_p}) \text{ and } d_i = (E_{p,i} \cdot H_p).$$

Hence by lemma again we have

$$\max\left\{\frac{b_i}{d_i}\right\} \geq \frac{\sum b_i}{\sum d_i} = \frac{b}{d} \geq M > M - \varepsilon$$

for such E_p exists as claimed. So for large m, if c gets large then we eventually get $a/c < M$ and we are done as above. If c stays bounded, then a, c stay bounded and b, d get large with

$$\left|M - \frac{b}{d}\right| = \left|\frac{a - Mc}{d}\right| < \varepsilon \text{ as } d \to \infty,$$

so $b/d + \varepsilon > (a+b)/(c+d) = M$. Hence for $m \gg 0$, one has

$$\max\left\{\frac{b_i}{d_i}\right\} \geq \frac{b}{d} = \frac{-(Z^s_{p,m} \cdot K_{X_p})}{(Z^s_{p,m} \cdot H_p)} > M - \varepsilon$$

by lemma again, the required E_p exists.

Step 4: Suppose now that

$$-(E_p \cdot K_{X_p}) > \dim X + 1$$

for the above rational curve E_p. Then as we count the lower bound of deformation space (where genus is 0 here), we can move the rational curve $E_p \subseteq X_p$ and the moving curve must break into a sum of two or more (not necessarily distinct) rational curves.

We use lemma again to conclude that the inequality

$$-\frac{(- \cdot K_{X_p})}{(- \cdot H_p)} > M - \varepsilon \tag{2.1}$$

must hold for at least one of the components E'_p of the broken E_p. If

$$-(E'_p \cdot K_{X_p}) > \dim X + 1,$$

then E'_p can move again and as above break, we find E''_p for which (2.1) holds. This process cannot continue infinitely, since at each step $(- \cdot H_p)$ must go down, but by Nakai-Moishezon criterion this number is strictly positive. So eventually we arrive at a rational curve (which we again call E_p) such that

$$\dim X + 1 \geq -(E_p \cdot K_{X_p}) > 0 \text{ and } \frac{-(E_p \cdot K_{X_p})}{(E_p \cdot H_p)} > M - \varepsilon.$$

This implies that $(E_p \cdot H_p) < (\dim X + 1)/(M - \varepsilon)$, so the possible values of $(E_p \cdot H_p)$ is a finite number of positive integers, hence also for value $-(E_p \cdot K_{X_p})/(E_p \cdot H_p)$ has finitely many possible values strictly larger than $M - \varepsilon$, for any $\varepsilon > 0$, so can let $\varepsilon \to 0$ to get our desired inequality.

Furthermore, $(E_p \cdot H_p)$ is bounded for almost all p. We can reason as in Theorem 3.6 *Step 8* to conclude the existence of a rational curve E over characteristic 0, which satisfies the requirement of we want. □

From this theorem we see the strength of "bend and break" technique, it transfers the question from $\mathbb{C}$ to a finite characteristic base field, solving the problem that even experts in complex geometry might be unable to achieve.

2.4 Mori's Cone Theorem

Recall that in Chapter 1 we introduce some concepts like *cone*, *extremal ray* and *nef cone* over a proper variety X is the subcone of the $\mathbb{R}$-vector space $N_1(X)$ with positive coefficients, related to the perfect pairing

$$N^1(X)_{\mathbb{R}} \times N_1(X)_{\mathbb{R}} \longrightarrow \mathbb{R},$$

has of the form

$$\begin{aligned}
\mathrm{NE}_{\mathbb{Q}}(X) &= \Big\{ \sum a_i[C_i] \;\Big|\; C_i \subseteq X \text{ an irreducible curve, } 0 \le a_i \in \mathbb{Q} \Big\}; \\
\mathrm{NE}(X) &= \Big\{ \sum a_i[C_i] \;\Big|\; C_i \subseteq X \text{ an irreducible curve, } 0 \le a_i \in \mathbb{R} \Big\}, \text{ and} \\
\overline{\mathrm{NE}}(X) &= \text{ the closure of } \mathrm{NE}(X) \text{ in } N_1(X)_{\mathbb{R}}.
\end{aligned}$$

And recall that for any Cartier divisor D,

$$\begin{aligned}
D_{\ge 0} &:= \{\gamma \in N_1(X)_{\mathbb{R}} \mid (D \cdot \gamma) \ge 0\}; \\
D^{\perp} &:= \{\gamma \in N_1(X)_{\mathbb{R}} \mid (D \cdot \gamma) = 0\}.
\end{aligned}$$

We also use the notation

$$\overline{\mathrm{NE}}(X)_{D \ge 0} := \overline{\mathrm{NE}}(X) \cap D_{\ge 0},$$

all of these are defined similarly for > 0, ≤ 0 and < 0.

Now we are ready to state the first result of Mori for smooth varieties of arbitrary dimension.

2.4A Cone Theorem on the Nonsingular Case

Theorem 2.4.1. (Cone Theorem) *Let X be a non-singular projective variety and K_X fails to be nef.*

(1) There are countably many rational curves $C_i \subseteq X$ such that $0 < -(C_i \cdot K_X) \le \dim X + 1$, and

$$\overline{\mathrm{NE}}(X) = \overline{\mathrm{NE}}(X)_{K_X \ge 0} + \sum_i \mathbb{R}_{\ge 0}[C_i].$$

(2) For any $\varepsilon > 0$ and an ample divisor H, we have

$$\overline{\mathrm{NE}}(X) = \overline{\mathrm{NE}}(X)_{(K_X + \varepsilon H) \ge 0} + \sum_{\text{finite}} \mathbb{R}_{\ge 0}[C_i].$$

Proof. From Grothendieck's theorem 1.11 we know that the scheme $\mathrm{Mor}(\mathbb{P}^1_{\mathbf{k}}, X)$ is the disjoint union of countably many schemes of finite type over $\mathbf{k}$, hence itself has countably many irreducible components. Moreover we pick a component $B \subseteq \mathrm{Mor}(\mathbb{P}^1_{\mathbf{k}}, X)$ and consider the flat family

$$\pi : \mathbb{P}^1_{\mathbf{k}} \times B \longrightarrow B,$$

and the universal evaluation map

$$f^{\mathrm{univ}} : \mathbb{P}^1 \times \mathrm{Mor}(\mathbb{P}^1_{\mathbf{k}}, X) \longrightarrow X.$$

Picking a divisor class $D \in N^1(X)$, as $\{\mathbb{P}^1_b\}_b$ is a flat family, the function

$$b \mapsto \chi(\mathbb{P}^1_b, f_b^* \mathscr{O}_X(D))$$

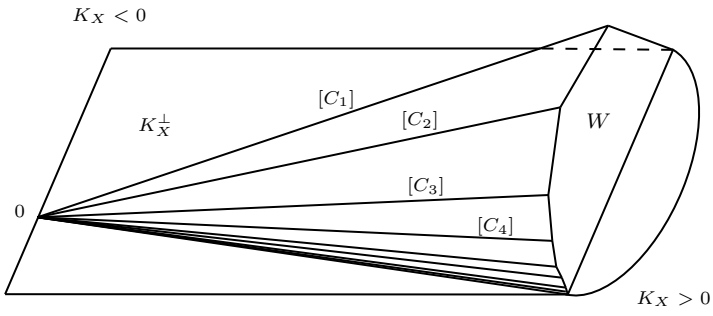
is locally constant. But as B is irreducible it is automatically connected, this function is constant. This implies that the intersection number $((f_b)_* \mathbb{P}^1_{\mathbf{k}} \cdot D)$ is a constant, thus these rational curves on X parametrized by the same irreducible component are numerically equivalent.

In each numerical equivalence class $\gamma \in N_1(X)$ with

$$0 < -(\gamma \cdot K_X) \leq \dim X + 1,$$

then we pick (if possible) a rational curve C_i. This leads to a countable index set $I \ni i$ and we define

$$W := \text{the closure of} \left(\overline{\mathrm{NE}}(X)_{K_X \geq 0} + \sum_{i \in I} \mathbb{R}_{\geq 0}[C_i] \right)$$



Step 1: We here aim to prove $W = \overline{\mathrm{NE}}(X)$.

Assume the contrary, that the natural inclusion $W \subseteq \overline{\mathrm{NE}}(X)$ is strict.

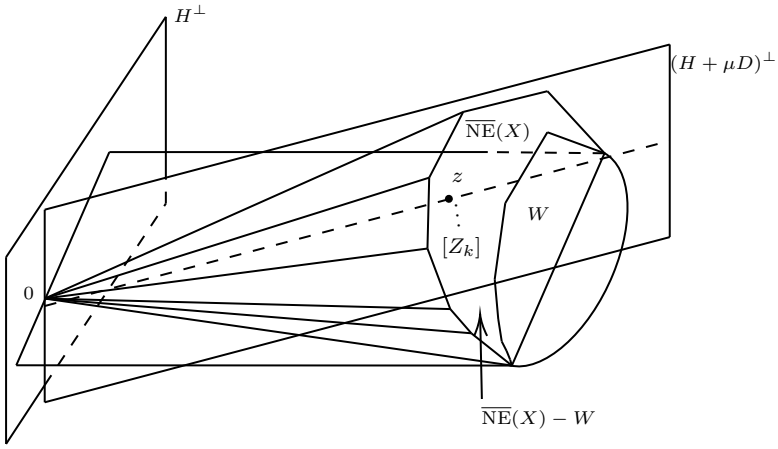
Then there is a divisor D on X whose associated linear function $(D \cdot -)$ on $N_1(X)_{\mathbb{R}}$ is positive in $W - \{0\}$ but negative somewhere in $\overline{\mathrm{NE}}(X)$, see [BV04, 2.5.1]. Now let

H be an ample divisor on X and let $\mu > 0$ be the largest (real) number such that $H + \mu D$ is nef, that is the largest number such that H can suppress the tendency of D to become negative. We choose a nonzero element $z \in \overline{\mathrm{NE}}(X)$ with $((H+\mu D)\cdot z) = 0$, then $(D\cdot z) < 0$ because $(H\cdot z) > 0$ via Kleiman's criterion 1.4.28, so $z \notin W$. Moreover $(K_X \cdot z) < 0$ as $\overline{\mathrm{NE}}(X)_{K_X \geq 0} \subseteq W$.

By our choice, there is a sequence of effective 1-cycles $\{Z_k = \sum_j a_{kj}[Z_{kj}]\}$ with

$$Z_k \to z \text{ as } k \to \infty.$$

The situation is illustrated by the following picture.



Now for any rational number $\mu' < \mu$, the $\mathbb{Q}$-Cartier divisor $H + \mu' D$ is ample on X by it is the sum of an ample and a nef. By lemma 3.8, we have

$$\max_j \frac{-(Z_{kj}\cdot K_X)}{(Z_{kj}\cdot (H+\mu' D))} \geq \frac{-(Z_k \cdot K_X)}{(Z_k \cdot (H+\mu' D))}.$$

We rearrange the indices so that the maximum is achieved for Z_{k0}. Next we apply the Theorem 3.7 to Z_{k0} and the ample divisor $H + \mu' D$. We obtain that there is a rational curve E_k such that

$$0 < -(E_k \cdot K_X) \leq \dim X + 1 \tag{2.2}$$

and

$$\frac{-(E_k \cdot K_X)}{(E_k \cdot (H+\mu' D))} \geq \frac{-(Z_{k0} \cdot K_X)}{(Z_{k0} \cdot (H+\mu' D))} \geq \frac{-(Z_k \cdot K_X)}{(Z_k \cdot (H+\mu' D))}.$$

By the formula (2.2), E_k must be numerically equivalent to one of the rational curves $\{C_i\}$ and so $(E_k \cdot D) > 0$ as our choice of $(D \cdot -)$ is positive on $W - \{0\}$.

Hence for all rational numbers $0 < \mu' < \mu$, we have

$$\frac{-(E_k \cdot K_X)}{(E_k \cdot H)} \geq \frac{-(E_k \cdot K_X)}{(E_k \cdot (H + \mu' D))} \geq \frac{-(Z_k \cdot K_X)}{(Z_k \cdot (H + \mu' D))}.$$

So for $k \to \infty$, we will find that

$$\liminf_{k\to\infty} \frac{-(E_k \cdot K_X)}{(E_k \cdot H)} \geq \frac{-(z \cdot K_X)}{(z \cdot (H + \mu' D))}.$$

But notice that as $k \to \infty$,

$$-(Z_k \cdot K_X) \to -(z \cdot K_X) > 0 \text{ and } (Z_k \cdot (H + \mu D)) \to (z \cdot (H + \mu D)) = 0,$$

thus for μ' sufficiently close to μ, the fraction

$$\frac{-(z \cdot K_X)}{(z \cdot (H + \mu' D))}$$

will become arbitrarily large, and so the limit

$$\liminf_{k\to\infty} \frac{-(E_k \cdot K_X)}{(E_k \cdot H)}$$

also will be arbitrarily large. On the other hand, there is a positive constant $c \gg 0$ such that $K_X + cH$ is ample on X, in particular

$$(E_k \cdot (K_X + cH)) > 0$$

and so

$$c > \frac{-(E_k \cdot K_X)}{(E_k \cdot H)}$$

for all k, which contradicts the above observation: for μ' sufficiently close to μ, the right-hand side tends to arbitrarily large as $k \to \infty$. Hence we conclude *Step 1.*

Step 2: Prove that the part (2) of the theorem holds.

Fix now H and $\varepsilon > 0$. If $(C_i \cdot (K_X + \varepsilon H)) < 0$,then

$$(C_i \cdot H) \leq \frac{-(C_i \cdot K_X)}{\varepsilon} \leq \frac{\dim X + 1}{\varepsilon}.$$

By the 2nd statement of Theorem 1.4.28, there are only finitely many such curve C_i. Set

$$V_{H,\varepsilon} := \overline{\mathrm{NE}(X)}_{(K_X+\varepsilon H)\geq 0} + \sum_{(C_i \cdot (K_X+\varepsilon H))<0} \mathbb{R}_{\geq 0}[C_i].$$

$V_{H,\varepsilon}$ is a closed cone, and obviously contained in $\overline{\mathrm{NE}}(X)$. But also it contains W, and we have proved that $W = \overline{\mathrm{NE}}(X)$, so the part (2) of the theorem holds.

Step 3: Prove that the part (1) of the theorem holds.

The only missing ingredient is to show that

$$\overline{\mathrm{NE}}(X)_{K_X\geq 0}+\sum_i \mathbb{R}_{\geq 0}[C_i]$$

is a closed cone. This is clear as if the index is finite, but in the general case, one can see [Kol96, P.161] and we skip the proof here. □

Example 2.4.2. (Fano varieties) Let X be a Fano variety. Then $\overline{\mathrm{NE}}(X)\subseteq N_1(X)_{\mathbb{R}}$ is a finite rational polytope, spanned by the classes of rational curves. (Apply statement (2) of the cone theorem with $H=-K_X$, $\varepsilon=1$.)

Now we introduce a key step, the observation that *extremal rays correspond to morphisms of a variety.* This can connect the Cone Theorem, which reveals the structure of the cone of curves, with the structure of more traditional geometric properties of a variety.

Definition 2.4.3. Let X be a projective variety and $F\subseteq\overline{\mathrm{NE}}(X)$ an extremal face. A morphism $\mathrm{cont}_F: X\longrightarrow Z$ is called the *contraction* of F if the following holds:

(1) $\mathrm{cont}_F(C)$ is a point for an irreducible curve $C\subseteq X$ if and only if $[C]\in F$,

(2) $(\mathrm{cont}_F)_*\mathscr{O}_X=\mathscr{O}_Z$, that is cont_F has connected fibres.

Remark 2.4.4. By our Proposition 1.4.38, the contraction of F is unique up to isomorphism if it exists. Moreover any morphism between two projective varieties $g: X\longrightarrow Z$ with $g_*\mathscr{O}_X=\mathscr{O}_Z$ is the contraction of some extremal face, denoted by $\mathrm{NE}(g)$ as previous. Thus we have $\mathrm{cont}_{\mathrm{NE}(g)}=g$. Now we give a lemma will be used in Chapter 5.

Lemma 2.4.5. *Let X be a normal projective variety and let $\mathbf{r}\subseteq\overline{\mathrm{NE}}(X)$ be an extremal ray. Assume that X is $\mathbb{Q}$-factorial. Assume that the contraction of $\mathbf{r}$ between projective varieties exists*

$$f:=\mathrm{cont}_{\mathbf{r}}: X\longrightarrow Z.$$

Then one of the following holds:

- *f is birational and $\mathrm{Ex}(f)$ is an irreducible divisor (we say that f is a **divisorial contraction**);*
- *f is birational and $\mathrm{Ex}(f)$ is of codimension at least two (we say that f is a **small contraction**);*

- $\dim Z < \dim X$ *(we say that f is* ***of fibre type****).*

Proof. The only assertion that has to be proven, is the claim that $\mathrm{Ex}(f)$ is irreducible in the case where it contains a divisor.

Assume that $\mathrm{Ex}(f)$ contains a divisor $D \subseteq \mathrm{Ex}(f)$, and let $H \subseteq Z$ be a general hyperplane section that contains $f(D)$. Since $f(D) \subseteq Z$ is irreducible of codimension at least two, as we have proved under Defn 1.1, the hyperplane section H exists and irreducible.

Let $H' \subseteq X$ be the strict transform of H, then H' meets the fibre $f^{-1}(z)$ over any point $z \in f(D)$ non-trivially. On the other hand, H' cannot completely contains the fibre $f^{-1}(z)$ for all $z \in f(D)$, that is because it would imply $D \subseteq f^{-1}(f(D)) \subseteq H'$, but D is a divisor and by f birational and $H' \sim_{bir} H$, H' has same dimension as D which different from D. Thus we have seen that for $z \in f(D)$ general, the fibre $f^{-1}(z)$ meets H' but is not contained in H'.

By definition of contraction morphism, the fibre $f^{-1}(z)$ is connected. And as X is projective, we always can find a curve $C \subseteq f^{-1}(z)$ that meets H' non-trivially but is not contained completely in H'. Note that C is contracted via f to the point z and so $\mathbf{r} = \mathbb{R}_{\geq 0}[C]$. Since H is a hyperplane section of Z, it is Cartier and so we may consider the pullback

$$f^*H = H' + \sum a_i E_i$$

where $E_i \subseteq \mathrm{Ex}(f)$ are some divisors on X that are contracted by f and $a_i > 0$ because if the local equation of H is given by $g = 0$, then the pullback is given by $g \circ f = 0$ locally, the coefficients a_i is just the order of vanishing of $g \circ f$ in $\mathscr{O}_{X,\eta_i}$, where η_i are the generic points of the E_i.

By assumption, each E_i is $\mathbb{Q}$-Cartier. Moreover, $(f^*H \cdot C) = (H \cdot f_*C) = 0$, hence so

$$-(H' \cdot C) = (\sum a_i E_i \cdot C).$$

Since $a_i > 0$ and $(H' \cdot C) > 0$, we conclude that there must be a divisor $E_i =: E' \subseteq \mathrm{Ex}(f)$ with $(E' \cdot C) < 0$.

Since f has connected fibres, through any point $x \in \mathrm{Ex}(f)$ there is a curve $C' \subseteq X$ such that $f_*C' = 0$. But f is a contraction of an extremal ray, the C' is numerically proportional to C, i.e.

$$[C'] = \lambda[C] \text{ for some } \lambda \in \mathbb{R}_{>0},$$

so $(E' \cdot C') = \lambda(E' \cdot C) < 0$, this implies that $C' \subseteq E'$ otherwise the intersection number will be non-negative. Hence $x \in E'$, that is $E' = \mathrm{Ex}(f)$ as we want. □

In general not every extremal face can be contracted, as we see by the following example.

Example 2.4.6. Let $g : X \longrightarrow \mathbb{P}^2$ be the blow up of $\mathbb{P}^2$ at 12 points $p_1, \ldots, p_{12}$ on a smooth cubic plane curve D. Let $C \subseteq X$ be the strict tramsform of that plane cubic. Then by [Har77, II, ex8.5(b)] we have

$$\begin{aligned}(K_X \cdot C) &= ((g^* K_{\mathbb{P}^2} + \sum_{i=1}^{12} E_i) \cdot C) \\ &= \int_D c_1(\mathscr{O}_{\mathbb{P}^2}(-3)) + \sum_{i=1}^{12} (E_i \cdot C) \\ &= -3 \times 3 + 12 = 3,\end{aligned}$$

and as $g^* D = C + \sum_{i=1}^{12} E_i$, one has

$$\begin{aligned}(C^2) &= (g^* D - \sum_{i=1}^{12} E_i)^2 \\ &= 9 - 12 = -3.\end{aligned}$$

Since $(C^2) < 0$, as Example 1.4.31, the curve $[C]$ spans an extremal ray $\mathbf{r}$ of $\overline{\mathrm{NE}}(X)$. On the other hand, we claim that $\mathbf{r}$ can in general not be contracted.

Indeed, assume that the contraction of $\mathbf{r}$ exists: $\mathrm{cont}_{\mathbf{r}} : X \longrightarrow Z$. Then X must carry a non-trivial line bundle

$$L = g^* \mathscr{O}_{\mathbb{P}^2}(d) + \sum_{i=1}^{12} a_i E_i$$

whose restriction on C is trivial. So we would have the line bundle

$$\mathscr{O}_{\mathbb{P}^2}(d)|_D + \sum_{i=1}^{12} a_i p_i$$

is trivial on D, which is impossible for a general choice of the p_i, unless $d = a_1 = \cdots = a_{12} = 0$. Hence $\mathrm{cont}_{\mathbf{r}}$ does not exist, as we claimed.

2.4B Application: Minimal Models of Surfaces

To begin with, we need two lemmas:

Lemma 2.4.7. *If D is a Cartier divisor on an integral proper surface X with $(D^2) > 0$, then two linear series have either $|nD| \neq \emptyset$ or $|-nD| \neq \emptyset$ for $n \gg 0$.*

Proof. By Riemann-Roch Theorem on surface, we pick out the *superabundance* - i.e. h^1 - terms to get

$$h^0(nD) + h^0(K_X - nD) \geq \frac{n^2}{2}(D^2) - \frac{n}{2}(D \cdot K_X) + \chi(\mathscr{O}_X),$$
$$h^0(-nD) + h^0(K_X + nD) \geq \frac{n^2}{2}(D^2) + \frac{n}{2(D \cdot K_X)} + \chi(\mathscr{O}_X).$$

Now we letting n be sufficiently large, and notice that the right-hand-side of each inequality gets big. But it cannot be true that both $h^0(K_X - nD)$ and $h^0(K_X + nD)$ get big, since we have the natural inclusion as we're on integral scheme:

$$H^0(X, K_X - nD) \otimes H^0(X, K_X + nD) \hookrightarrow H^0(X, 2K_X)$$
$$f \otimes g \mapsto f \cdot g$$

their sum to a fixed linear series $|2K_X|$, which has the fixed dimension. Thus it is $h^0(nD)$ or $h^0(-nD)$ will gets big quadratically with n, that is either nD or $-nD$ will be effective. □

Corollary 2.4.8. *Let X be an integral projective surface with an ample Cartier divisor H. The set*

$$Q := \{z \in N_1(X)_{\mathbb{R}} \mid (z^2) > 0\}$$

has two connected components

$$Q^+ := \{z \in Q \mid (z \cdot H) > 0\} \text{ and } Q^- := \{z \in Q \mid (z \cdot H) < 0\}.$$

Furthermore, $Q^+ \subseteq \overline{\mathrm{NE}}(X)$.

Proof. We have the intersection form on $N_1(X)_{\mathbb{R}}$

$$(-,-) : N_1(X)_{\mathbb{R}} \times N_1(X)_{\mathbb{R}} \longrightarrow \mathbb{R}$$
$$(z_1, z_2) \mapsto (z_1 \cdot z_2)$$

Now we denote $\rho = \rho(X)$ and let $\{D_i\}_{i=1}^{\rho}$ be a basis of $N_1(X)_{\mathbb{R}}$. Now by the Hodge Index Theorem, we structure of $N_1(X)_{\mathbb{R}}$ can be decomposed into

$$N_1(X)_{\mathbb{R}} \cong \mathbb{R}\{H\} \oplus H^{\perp} \text{ where } H^{\perp} = \{z \in N_1(X)_{\mathbb{R}} \mid (z \cdot H) = 0\},$$

and $\mathbb{R}\{H\}$ is the dimension 1 subspace of positive definite, $H^{\perp}$ is the dimension $\rho - 1$ subspace of negative definite.

More precisely, we can write the intersection form $(-,-)$ to be a real symmetric matrix $M := (M_{ij})_{i,j=1,\cdots,\rho}$, with $M_{ij} = (D_i \cdot D_j)$. Thus for any Cartier divisor $z_1 = \sum_{i=1}^{\rho} x_i D_i$ and $z_2 = \sum_{i=1}^{\rho} y_i D_i$, we have

$$(z_1 \cdot z_2) = \mathbf{x}^T M \mathbf{y}.$$

From linear algebra. every real symmetric matrix can be orthogonal diagonalizable, and does not change the index of inertia by Sylvester, see [Lan00, P.577 Cor4.2]. So after orthogonal diagonalization, M is of the form

$$\begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & -1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & -1 \end{pmatrix}$$

with 1 positive sign and $\rho - 1$ negative signs, which says that under suitable orthogonal basis $\{e_i\}_{i=1}^{\rho}$ the self-intersection of $D = \sum x_i e_i$ can be written as $x_1^2 - \sum_{i\geq 2} x_i^2$. Moreover we can choose such basis such that

$$[H] = (\sqrt{(H \cdot H)}, 0, \ldots, 0)$$

up to numerically equivalent.

Now with $z = \sum x_i e_i \in Q$, we have $x_1^2 - \sum_{i\geq 2} x_i^2 > 0$ which just nothing but

$$|x_1| > \sqrt{x_2^2 + \cdots + x_\rho^2}.$$

This gives the two connected components

$$Q^+ = (x_1 > \sqrt{\sum_{i\geq 2} x_i^2}) \text{ and } Q^- = (x_1 < -\sqrt{\sum_{i\geq 2} x_i^2}),$$

where in the component Q^+, one has the intersection number $(z \cdot [H]) = x_1 \cdot \sqrt{(H^2)} > \sqrt{\sum_{i\geq 2} x_i^2} \cdot 0 = 0$, and in the Q^- one has $(z \cdot [H]) = x_1 \cdot \sqrt{(H^2)} < -\sqrt{\sum_{i\geq 2} x_i^2} \cdot 0 = 0$.

For the second statement, picking any $[D] \in Q^+ \subseteq Q$ we have that either nD or $-nD$ is effective for $n \gg 0$ by previous lemma. An effective curve has positive intersection with H, so if we have $(-nD \cdot H) > 0$ then $(D \cdot H) < 0$ contradicts with our choice, hence we must has nD is effective which is nothing but $[D] = \frac{1}{n}[nD] \in \overline{\mathrm{NE}}(X)$. □

Now we give the main result in this section (even in this chapter!), classifying surfaces from the point of view of extremal ray theory, when K_X fails to be nef.

Theorem 2.4.9. (MMP for Surfaces) *Let X be a smooth projective surface and $R \subseteq \overline{\mathrm{NE}}(X)$ an extremal ray such that $(R \cdot K_X) < 0$, i.e. R is generated by a K_X-negative curve. Then the contraction morphism $\mathrm{cont}_R : X \longrightarrow Z$ exists and is one of the following types:*

(1) Z is a smooth surface and X is obtained from Z by blowing up a closed point; $\rho(Z) = \rho(X) - 1$.

(2) Z is a smooth curve and X is a minimal ruled surface over Z; $\rho(X) = 2$.

(3) Z is a point, $\rho(X) = 1$ and $-K_X$ is ample. (In fact, $X \cong \mathbb{P}^2$ but this is harder to prove.)

Proof. **Step 1:** Let $C \subseteq X$ be an irreducible rational curve such that $[C] \in R$ with the intersection number

$$0 < -(K_X \cdot C) \leq \dim X + 1 = 3,$$

which we can always do this by Theorem 3.7. Hence $(K_X \cdot C) \in \{-1, -2, -3\}$. Recall that the *arithmetic genus* is defined by

$$p_a(C) := 1 - \chi(C, \mathscr{O}_C) = h^1(C, \mathscr{O}_C),$$

since here C is integral and proper. As a consequence of the Riemann-Roch formula for surfaces, there is the following formula (see [Har77, V, ex1.3(a)]):

$$2p_a(C) - 2 = (C \cdot (C + K_X)) = (C^2) + (K_X \cdot C).$$

Since $p_a(C) \geq 0$ (notice here curve C may singular!), this implies $(C^2) \geq -2 - (K_X \cdot C)$. Moreover since $(K_X \cdot C) \in \{-1, -2, -3\}$, we find that $(C^2) \geq -1$, and the **equality holds** if and only if $(K_X \cdot C) = -1$ and $p_a(C) = 0$. In this case, by [Har77, IV, ex1.8(b)] we see that $C \cong \mathbb{P}^1$, that is C is a (-1)-curve therefore we reduce to case (1) by Castelnuovo's contraction theorem.

Step 2: Now it remains to deal with the cases $(C^2) > 0$ and $(C^2) = 0$. We deal with the case when $(\mathbf{C^2}) > \mathbf{0}$. By previous corollary 2.4.8, we see the class $[C]$ lies in Q^+ as H is ample and C is an effective curve, so $[C]$ is an interior point of $\overline{\mathrm{NE}}(X)$. Since $[C]$ also spans an extremal ray by our assumption, we have $N_1(X)_{\mathbb{R}} \cong \mathbb{R}[C] = \mathbb{R}$, so cont_R will contract the whole of X. By assumption $(K_X \cdot C) < 0$, thus $-K_X$ is positive on $\overline{\mathrm{NE}}(X) - \{0\}$ and so by Kleiman's ampleness criterion we see $-K_X$ is ample (Theorem 1.4.28). For $X \cong \mathbb{P}^2$ can see [Mat02, 1-4-4].

Step 3: Thirdly we consider the case $(\mathbf{C^2}) = \mathbf{0}$. Again by $2p_a(C) - 2 = (K_X \cdot C)$ and the three possibilities for the last term, we see that $p_a(C) = 0$ and hence $C \cong \mathbb{P}^1$. We prove that the linear series $|mC|$ gives our desired contraction morphism for $m \gg 1$. Since C is effective, and we use the Serre duality [Har77, III, 7.7] to get

$$H^2(X, \mathscr{O}_X(mC)) \cong H^0(X, \mathscr{O}_X(K_X - mC))^\vee = 0$$

for $m \gg 1$. Thus again by Riemann-Roch formula on surface we have

$$h^0(X, \mathscr{O}_X(mC)) \geq \chi(X, \mathscr{O}_X(mC)) = \frac{-(K_X \cdot C)}{2} + \chi(X, \mathscr{O}_X)$$

which grows linearly because $-(K_X \cdot C)$ is positive. In particular, for large enough m we find that $h^0(X, \mathscr{O}_X(mC)) \geq 2$ which implies that there is a curve $D \subseteq X$ with $D \sim_{\text{lin}} mC$ and $D \neq mC$, for $m \gg 1$. Notice that for any such effective divisor $D \in |mC|$, we can write D as $D = F + M$ where $[D] = [F] + [M] = m[C]$, F is the fixed component and M is the moving part. By the definition of the extremal ray, we see the numerical class of F is still in $R = \mathbb{R}_{\geq 0}[C]$, i.e. any fixed component is a multiple of C which we say $k[C]$. So the moving part is

$$[M] = [mC] - k[C] = (m - k)[C]$$

and we denote $m' := m - k$. Thus, there is an $m' > 0$ such that $|m'C|$ has no fixed component for $m \gg 1$.

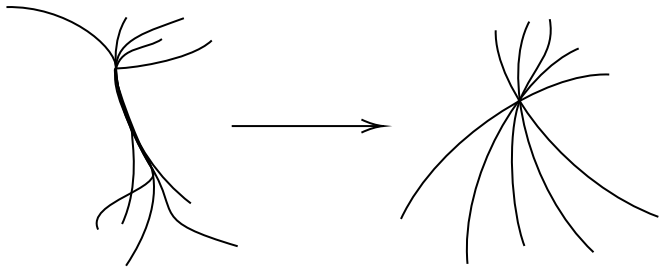


Figure 2.10.: As m growing, the moving part separates from the fixed component.

Next we need to verify that it also base point free, this is trivial since $(C^2) = 0$, so we have $(D \cdot C) = m'(C^2) = 0$ for any $D \in |m'C|$. This implies that either D is disjoint from C or D is the C itself, so the general member of $|m'C|$ is disjoint from C, which implies $|m'C|$ is base point free. We let $\text{cont}_R : X \longrightarrow Z$ be the Stein factorization of the corresponding morphism of $|m'C|$, i.e.

$$\phi_{|m'C|} : X \xrightarrow{\text{cont}_R} Z \longrightarrow \mathbb{P}^N_{\mathbf{k}},$$

where Z is the $\underline{\mathrm{Spec}}_{\mathbb{P}^N_{\mathbf{k}}}(\phi_{|m'C|})_*\mathscr{O}_X$.

Let $\sum a_i C_i$ be a fibre of cont_R at *any point of* Z, with $a_i \in \mathbb{Z}_{>0}$ for each i. Then one has $\sum a_i[C_i] = [C] \in R$ by the definition of contraction morphism. Moreover since R is an extremal ray, this implies that $[C_i] \in R$ for every i. Thus $(C_i^2) = 0$ and $(K_X \cdot C_i) < 0$. By a similar analysis we know

$$0 < -(K_X \cdot C_i) \leq 3$$

and hence $2p_a(C_i) - 2 = (K_X \cdot C_i) \in \{-1, -2, -3\}$, which implies $C_i \cong \mathbb{P}^1$ and $(K_X \cdot C_i) = -2$. Thus

$$-2 = (K_X \cdot C) = (K_X \cdot \sum a_i C_i) = -2 \sum a_i$$

which shows that $\sum a_i = 1$, that is only one term of coefficients is nonzero (specifically equal to 1) and others are 0, so $\sum a_i C_i$ is an irreducible and reduced curve which is isomorphic to $\mathbb{P}^1$. Thus X is a minimal ruled surface over Z. And by [Har77, V, 2.8] we know it is possible to write X to be the projectivilization of some locally free sheaf. Thus, the Picard number $\rho(X)$ grows 1 as stated in [Har77, II, ex7.9(a)]. □

Summary:

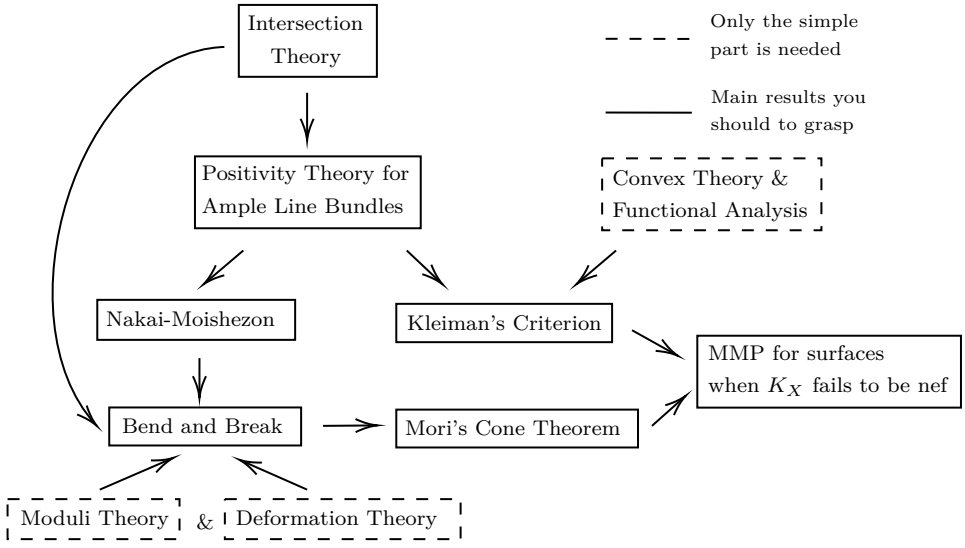


Figure 2.11.: Knowledge Framework

Chapter 3

Linear Series

In this chapter all schemes will be considered over $\mathbb{C}$.

3.1 Asymptotic Theory

3.1A Basic Definitions

This definition may looks a little bit scary.

Caucher Birkar on asymptotic definition of Kodaira dimension

The asymptotic behavior of the linear series $|L^{\otimes m}|$ as $m \to \infty$, due to Iitaka and Ueno, is that up to birational equivalence they define an essentially unique rational mapping $\varphi : X \dashrightarrow Y$, and L close to trivial (in suitable sense) along the fibres of φ. Throughout this section X always means projective variety.

Definition 3.1.1. (Semigroup and exponent of a line bundle) Let L be a line bundle on a projective variety X. The *semigroup* of L consists of those non-negative powers of L that have a non-zero section:

$$\mathbb{N}(L) = \mathbb{N}(X, L) = \left\{ m \geq 0 \;\middle|\; H^0(X, L^{\otimes m}) \neq 0 \right\}.$$

And we clearly have $\mathbb{N}(L) = (0)$ if for all $m > 0$, the $H^0(X, L^{\otimes m}) = 0$.

Assuming that $\mathbb{N}(L) \neq (0)$, then we let $e = e(L)$ to be the greatest common divisor of $\{m \in \mathbb{N}(L) \mid m > 0\}$, call the *exponent* of L. So for all sufficiently large elements $m \in \mathbb{N}(X, L)$ it is a multiple of this e. We often use the word "*sufficiently divisible*" to imply that m is sufficiently large in $\mathbb{N}(X, L)$.

The semigroup $\mathbb{N}(X, D)$ and exponent $e = e(D)$ for Cartier divisor D are defined via the line bundle $L = \mathscr{O}_X(D)$.

Example 3.1.2. Let T be a projective variety of dimension $d \geq 1$ that carries a non-trivial torsion line bundle L_η having order e in $\mathrm{Pic}(T)$, that is $L_\eta^{\otimes e} \cong \mathscr{O}_T$. Let Y be any projective variety of dimension k, and B a very ampe line bundle on Y. Set

$$X = Y \times T \text{ and } L = \mathrm{pr}_1^*(B) \otimes pr_2^*(L_\eta).$$

Then $e(L) = e$ and $\mathbb{N}(L) = \mathbb{N}e$ because by Künneth formula we have the decomposition of global section: for $L^{\otimes m} \cong \mathrm{pr}_1^*(B^{\otimes m}) \otimes \mathrm{pr}_2^* L_\eta^{\otimes m}$, we have

$$H^0(X, L^{\otimes m}) \cong H^0(Y, B^{\otimes m}) \otimes H^0(T, L_\eta^{\otimes m}),$$

which one can see in [SP25, Tag 0BED]. And we use a fact here that over a projective variety T, $H^0(T, \mathscr{L}) = 0$ if $\mathscr{L}$ is a non-trivial torsion line bundle, that is because we can pick an arbitrary global section $s \in H^0(T, \mathscr{L})$, then it's power s^e gives a global section in $\mathscr{L}^{\otimes e} \cong \mathscr{O}_T$ which is a constant, hence s itself is a constant, which contradicts the fact that $\mathscr{L}$ is non-trivial.

Now for given $m \in \mathbb{N}(L)$, we consider the rational map

$$\phi_m = \phi_{|L^{\otimes m}|} : X \dashrightarrow \mathbb{P}H^0(X, L^{\otimes m})$$

associated to the complete linear series $|L^{\otimes m}|$. We denote by

$$Y_m := \phi_m(X) \subseteq \mathbb{P}H^0(X, L^{\otimes m})$$

the closure of its image, i.e. the image of the closure of the graph of ϕ_m. Our purpose is to understand the asymptotic behavior of these rational maps ϕ_m under the condition as $m \to \infty$.

Definition 3.1.3. (Iitaka dimension) Assume that X is normal (the reason will explain later). Then the *Iitaka dimension* of L is defined to be

$$\kappa(L) = \kappa(X, L) = \max_{m \in \mathbb{N}(L)} \{\dim \phi_m(X)\},$$

provided that $\mathbb{N}(L)$ is non-trivial. Moreover we define that $\kappa(L) = -\infty$, if $\mathbb{N}(L)$ is trivial.

If X is non-normal, we pass to its normalization $\nu : X' \to X$ and set

$$\kappa(X, L) = \kappa(X', \nu^* L).$$

The definition for Cartier divisor D is similar.

Thus we get that either $\kappa(X, L) = -\infty$, or else

$$0 \leq \kappa(X, L) \leq \dim X.$$

Example 3.1.4. For the pair (X, L) in Example 1.2, we consider

$$H^0(X, L^{\otimes m}) \cong H^0(Y, B^{\otimes m}) \otimes H^0(T, L_\eta^{\otimes m})$$

wuth m sufficiently large and is a multiple of the exponent e. Then we have

$$H^0(X, L^{\otimes m}) \cong H^0(Y, B^{\otimes m}) \otimes \mathbb{C},$$

so the image $\phi_m(X)$ is just the image of Y embedding into a projective space via very ample line bundle $B^{\otimes m}$, hence $\kappa(X, L) = \dim Y = k$.

Example 3.1.5. (Kodaira dimension) Let X be a smooth projective variety, and K_X a canonical divisor on X. Then we define $\kappa(X) = \kappa(X, K_X)$ to be the *Kodaira dimension* of X. The Kodaira dimension of a singular variety is defined to be the Kodaira dimension of any smooth model.

Example 3.1.6. (Iitaka dimension of restrictions) Iitaka dimension can be very unpredictable under restriction to subvarieties. We consider $X = \mathrm{Bl}_p(\mathbb{P}^2)$ to be the blowing up of $\mathbb{P}^2$ at a point, with the exceptional curve E and H be the pullback of a hyperplane Cartier divisor. Then $\mathscr{O}_X(H)$ and $\mathscr{O}_X(H+E)$ both have the maximal Iitaka dimension 2, but their restrictions on subvariety E are very different, being 0 and $-\infty$ respectively. The reason for this is that because the line bundle $\mathscr{O}_X(H)|_E \cong \mathscr{O}_{\mathbb{P}^1}$ since

$$(H \cdot E) = (L \cdot \{p\}) = 0 \text{ for the hyperplane } L,$$

and similarly $\mathscr{O}_X(H+E)|_E \cong \mathscr{O}_{\mathbb{P}^1}(-1)$. So in these cases the Iitaka dimension **drops** upon restriction.

On the other hand, we consider another example let $X = \mathbb{P}^1 \times \mathbb{P}^1$ and take the line bundle $L = \mathrm{pr}_1^* \mathscr{O}_{\mathbb{P}^1}(-1) \otimes \mathrm{pr}_2^* \mathscr{O}_{\mathbb{P}^1}(1)$, we can see that

$$H^0(X, L^{\otimes m}) \cong H^0(\mathbb{P}^1, \mathscr{O}_{\mathbb{P}^1}(-1)) \otimes H^0(\mathbb{P}^1, \mathscr{O}_{\mathbb{P}^1}(1)).$$

Hence the Iitaka dimension $\kappa(X, L) = -\infty$. But if we restrict it to $Y = \{\text{point}\} \times \mathbb{P}^1$, we have $\kappa(Y, L|_Y) = 1$, so that here the Iitaka dimension has **increased** upon restriction.

Example 3.1.7. (Non-normal varieties) Linear series will have different behavior than their pullback via the normalization $\nu : X' \to X$. For example we consider the X be the nodal plane cubic curve, from [Har77, II, ex6.7] we have the degree zero Picard group $\mathrm{Pic}^0 X \cong \mathbb{G}_m$ and we picking a line bundle $L \in \mathrm{Pic}^0 X$ which is non-torsion i.e. corresponding an element $\lambda \in \mathbb{C}^\times$ which is not the roots of unity.

Now for normalization $\nu : \mathbb{P}^1 \to X$ gluing 2 points of $\mathbb{P}^1$ together. As L is degree zero, we have $\nu^* L \cong \mathscr{O}_{\mathbb{P}^1}$, so if we pick an element $s \in H^0(X, L)$, then its pullback $s' = \nu^* s$ is the global section of $\mathscr{O}_{\mathbb{P}^1}$ over $\mathbb{P}^1$, which is a constant that we denote it by $c \in \mathbb{C}$. As we want the section s' to descend to X, it must satisfy the relation

$$s'(p) = \lambda \cdot s'(q)$$

for two points p and $q \in \mathbb{P}^1$ gluing together, and this relation is comes from the transition map

$$s|_{U_i} = \lambda \cdot s|_{U_j}$$

where λ is the transition function on $U_i \cap U_j$. But this implies that we must have the relation on $L^{\otimes}$ is

$$s'(p) = \lambda^m s'(q) \text{ is nothing but } c = \lambda^m c,$$

which says that $(1-\lambda^m)c = 0$. Hence if $c \neq 0$ we must have that λ is a root of unity, so we must have $H^0(X, L^{\otimes m}) = 0$, but $H^0(\mathbb{P}^1, \nu^* L)$ is not. If we don't pass to the normalization in the definition, we are quite hard to obtain a birational invariant theory.

Example 3.1.8. Assume that X is normal. If $\kappa(X, L) = \kappa$, then $\dim \phi_m(X) = \kappa$ for all sufficiently large $m \in \mathbb{N}(X, L)$.

Proof. By replacing L by $L^{\otimes e}$ can suppose that L has exponent $e(L) = 1$, so there exists a $p_0 \in \mathbb{Z}_{>0}$ s.t. $H^0(X, L^{\otimes p})$ for all $p \geq p_0$. Fix some k s.t. $\dim \phi_k(X) = \kappa$. Multiplying by a non-zero section in $H^0(X, L^{\otimes p})$ determines for every $p \geq p_0$ an embedding (as we're dealing with integral schemes)

$$H^0(X, L^{\otimes k}) \subseteq H^0(X, L^{\otimes k+p}).$$

This gives rise to a factorization $\phi_k = \nu_p \circ \phi_{k+p}$, where

$$\nu_p : \mathbb{P}H^0(X, L^{\otimes k}) H^0(X, L^{\otimes k}) \dashrightarrow \mathbb{P}H^0(X, L^{\otimes k+p})$$

is a rational map induced from the linear projection associated with above embedding. This obviously cuts down the dimension, so we have $\dim Y_{k+p} \geq \dim Y_k = \kappa$, and the reverse direction is just the definition. □

Now we need deal with the case for morphisms $f : X \longrightarrow Y$ having property that $H^0(Y, L) = H^0(X, f^* L)$ for every line bundle L on Y:

Lemma 3.1.9. *(Coincide for Pullbacks) Let $f : X \longrightarrow Y$ be the projective surjective morphism between varieties, satisfying $f_* \mathscr{O}_X = \mathscr{O}_Y$, and let L be a line bundle on Y. Then*

$$H^0(X, f^* L^{\otimes m}) = H^0(Y, L^{\otimes m}) \text{ for all } m \geq 0.$$

In particular, $\kappa(Y, L) = \kappa(X, f^* L)$.

Proof. Since f is surjective, we have $H^0(X f^* L^{\otimes m}) = H^0(Y, f_*(f^* L^{\otimes m}))$. But by projection formula and $f_* \mathscr{O}_X = \mathscr{O}_Y$ we get

$$f_*(f^* L^{\otimes m}) = f_*(\mathscr{O}_Y \otimes f^* L^{\otimes m}) \cong f_* \mathscr{O}_X \otimes L^{\otimes m} = L^{\otimes m}.$$

□

Thus we make this assumption to be a definition to realize our original idea, also to connect with Stein factorization.

Definition 3.1.10. (Algebraic fibre space) An *algebraic fibre space* is a surjective projective morphism $f : X \longrightarrow Y$ of varieties such that $f_*\mathscr{O}_X = \mathscr{O}_Y$.

Now for arbitrary surjective projective morphism $g : V \longrightarrow W$ of varieties, we have the Stein factorization [Har77, III, 11.5] expressing g as a composition

$$V \xrightarrow{g'} W' \xrightarrow{h} W,$$

where $W' = \underline{\mathrm{Spec}}_W(g_*\mathscr{O}_V)$, and g' is an algebraica fibre space and h is finite. So g itself is a algebraic fibre space if and only if the finite part of its Stein factorization is trivial. In particular by theorem of the formal function, we see all fibres of a fibre space f are connected.

Conversely, if Y is normal then any projective surjective morphism $f : X \longrightarrow Y$ with connected fibres is a fibre space, that is because we factor f via the Stein factorization:

$$X \xrightarrow{g} Y' \xrightarrow{h} Y$$

where $Y' = \underline{\mathrm{Spec}}_Y(f_*\mathscr{O}_X)$, g is an algebraic fibre space and h is a finite map. Now for any point $y \in Y$ we have the disjoint union

$$f^{-1}(y) = \coprod_{y' \in h^{-1}(y)} g^{-1}(y'),$$

but as g is fibre connected, every $g^{-1}(y')$ is a connected component which remains only one here since f also fibre connected. So for each $y \in Y$, the set $h^{-1}(y)$ consists of exactly one point, i.e. h is a set-theoretic bijection. Hence the degree of h must be 1, that is we have

$$[k(Y') : k(Y)] = 1 \Longrightarrow k(Y') \cong k(Y),$$

so h is a birational map. Now as Y is normal, can easily get that $\mathscr{O}_{Y'} = \mathscr{O}_Y$ via the proof of [Har77, III, 11.4].

By the same token, if $f : X \longrightarrow Y$ is an algebraic fibre space with X normal, and moreover we equipped with a projective birational morphism $\mu : X' \longrightarrow X$ such that also an algebraic fibre space by [Har77, III, 11.4], then the composition $f \circ \mu : X' \longrightarrow Y$ is again a fibre space.

Example 3.1.11. (Fibre spaces and function fields) Let $f : X \longrightarrow Y$ be a projective surjective morphism of normal varieties, and

$$\mathbb{C}(Y) \subseteq \mathbb{C}(X)$$

to be the corresponding finitely generated extension of function fields. Then f is an algebraic fibre space if and only if $\mathbb{C}(Y)$ is algebraically closed in $\mathbb{C}(X)$.

Proof. " $\Rightarrow$ ": Suppose not, then there is a field K s.t.

$$\mathbb{C}(Y) \subsetneq K \subsetneq \mathbb{C}(X)$$

with $[K : \mathbb{C}(Y)] \geq 2$. So we can choose a normal variety Y' with rational maps

$$X \overset{u}{\dashrightarrow} Y' \overset{v}{\dashrightarrow} Y$$

which is the factorization of f, such that Y' is generically finite of degree ≥ 2 over Y. Then we can use [Har77, II, 7.17.3] to eliminate the points of indeterminacy in 2 steps:

Step 1: Embedding Y into a projective space, then we get an elimination:

$$\begin{array}{ccccc} & & \tilde{Y'} & & \\ & & \downarrow{\scriptstyle \pi_1} & \searrow{\scriptstyle v'} & \\ X & \dashrightarrow & Y' & \dashrightarrow & Y. \end{array}$$

So we get a rational map $X \dashrightarrow \tilde{Y'}$.

Step 2: Embedding $\tilde{Y'}$ into another projective space, then do again a elimination:

$$\begin{array}{ccc} \tilde{X} & & \\ \downarrow{\scriptstyle \pi_2} & \searrow{\scriptstyle u'} & \\ X & \dashrightarrow & \tilde{Y'}. \end{array}$$

So after replacing Y' and X by suitable birational modifications, we can assume that u and v are morphisms. Moreover,

$$\begin{array}{ccccc} \tilde{X} & & & & \\ \downarrow & \searrow & & & \\ X & \dashrightarrow & \tilde{Y'} & & \\ & \searrow & \downarrow & \searrow & \\ & & Y' & \dashrightarrow & Y \end{array}$$

by our condition X is normal and $\tilde{X} \longrightarrow X$ is projective surjective morphism with connected fibres, so $\tilde{X} \longrightarrow X$ is an algebraic fibre space. And by commutative diagram

$$\begin{array}{ccc} \tilde{X} & & \\ \downarrow & \searrow & \\ X & \xrightarrow[f]{} & Y \end{array}$$

we know that $\tilde{X} \longrightarrow Y$ is also an algebraic fibre space, so this modification does not affect the hypothesis. As f is an algebraic fibre space we know that all fibres of f are connected, but it passes throught a degree ≥ 2 generically finite map making generic fibre is a finite set with cardinality ≥ 2, so we get a contradiction.

" $\Leftarrow$ ": If $\mathbb{C}(Y) \subsetneq \mathbb{C}(X)$ is algebraically closed in the latter, then we run a Stein factorization

$$X \longrightarrow Y' \longrightarrow Y$$

such that $Y' \longrightarrow Y$ is a finite map which must be trivial since $\mathbb{C}(Y') \cong \mathbb{C}(Y)$ and Y normal (check that birational finite morphism between varieties with target normal must be isomorphism!), consequently f is a fibre space. □

The above example allows one to attach a meaning to algebraic fibre spaces in the **birational category**: asking that a dominant rational mapping $X \dashrightarrow Y$ of normal varieties induce an algebraically closed extension of function fields.

Example 3.1.12. (Injectivity of Picard groups) Let X and Y be projective varieties, and $f : X \longrightarrow Y$ an algebraic fibre space. Then the induced pullback

$$f^* : \mathrm{Pic}(Y) \longrightarrow \mathrm{Pic}(X)$$

is injective.

Proof. We pick $B = \mathscr{O}_Y(D)$ a line bundle on Y s.t. maps to zero, i.e. $f^*B \cong \mathscr{O}_X$. Then

$$H^0(Y, B) = H^0(X, f^*B) \neq 0$$

thanks to Lemma 1.9, so there is an effective Cartier divisor $E_1 \sim_{\mathrm{lin}} D$. And similarly

$$H^0(Y, B^\vee) = H^0(X, f^*B^\vee) \neq 0,$$

that is E_2 is also effective with $E_2 \sim_{\mathrm{lin}} -D$, therefore $E_1 + E_2 \sim_{\mathrm{lin}} 0$ means that they are both linearly equivalent to 0, and so is D. □

Example 3.1.13. (Normality of fibre spaces) Let $f : X \longrightarrow Y$ be an algebraic fibre space. If X is normal, then Y will also be normal.

Proof. By universal property of normalization $\nu : Y' \longrightarrow Y$, we have

$$\begin{array}{ccc} X & & \\ {\scriptstyle g}\downarrow & \searrow^{f} & \\ Y' & \xrightarrow[\nu]{} & Y \end{array}$$

with $\mathscr{O}_Y \subseteq \nu_* \mathscr{O}_{Y'}$ and $\mathscr{O}_{Y'} \subseteq g_* \mathscr{O}_X$ as g and ν are dominant morphisms.

So $f_* \mathscr{O}_X = \mathscr{O}_Y$ implies that

$$\mathscr{O}_Y = (\nu \circ g)_* \mathscr{O}_X = \nu_*(g_* \mathscr{O}_X) \supseteq \nu_* \mathscr{O}_{Y'} \supseteq \mathscr{O}_Y,$$

that is $\nu_* \mathscr{O}_{Y'} \cong \mathscr{O}_Y$. Moreover as finite map is affine, by [Har77, II, ex5.17(d)] we have

$$Y' = \underline{\mathrm{Spec}}_Y(\nu_* \mathscr{O}_{Y'}) \cong \underline{\mathrm{Spec}}_Y(\mathscr{O}_Y) = Y.$$

□

Example 3.1.14. (Exceptional twists of birational pullbacks) Consider a proper birational morphism

$$\mu : X' \longrightarrow X$$

of varieties, with X' normal. Let L be a line bundle on X, and E a μ-exceptional effective Cartier divisor on X' (i.e. an effective Cartier divisor on X' with the property that each of the irreducible components of the corresponding Weil divisor maps under μ to a variety of codimension ≥ 2 in X). Then the natural inclusion

$$H^0(X', \mu^* L) \longrightarrow H^0(X', \mu^* L \otimes \mathscr{O}_{X'}(E))$$

is an isomorphism.

Proof. Take a normalization $X^\nu \xrightarrow{\nu} X$ then we get a new map $\mu' : X' \longrightarrow X^\nu$ by the universal property of the normalization, with satisfies $\mu = \nu \circ \mu'$. But then we have for $L^\nu := \nu^* L$ with $L \in \operatorname{Pic} X$, the $\mu^* L = (\nu \circ \mu')^* L = (\mu')^* L$ says the coincidence on X. Moreover as ν is a finite map, it is fibre finite so it does not contract anything, thus the μ-exceptioanl Cartier divisor and the μ'-exceptional Cartier divisor are exactly the same.

Now we can assume X is normal without loss of generality. By the Zariski's Main theorem, we have $\mu_* \mathscr{O}_{X'} = \mathscr{O}_X$, so we have

$$\mu_*(\mu^* L) = \mu_*(\mu^* L \otimes \mathscr{O}_{X'}) \cong L \otimes \mu_* \mathscr{O}_{X'} \cong L,$$

thanks to the projection formula. Thus we have

$$H^0(X', \mu^* L) \cong H^0(X, \mu_*(\mu^* L)) \cong H^0(X, L)$$

and we denote the image of E to be Z. Let $U := X \backslash Z$ and $U' := \mu^{-1}(U)$, we have that the line bundle $\mathscr{O}_{X'}(E)$ restricted to U' is the trivial line bundle, as its support is empty, so it corresponds to the 0 Cartier divisor.

Thus if we pick a section $s \in H^0(X', \mu^* L \otimes \mathscr{O}_{X'}(E))$ and restrict it on U', it will be a section in $H^0(U', \mu^* L|_{U'})$ which is isomorphic to $H^0(U, L|_U)$. Now as L is a line bundle, we can choose an open covering $\{V_i\}$ such that $L|_{V_i} \cong \mathscr{O}_{V_i}$ for each i. Then we restrict section s on V_i again, to get a regular section in $H^0(V_i \cap U, \mathscr{O}_{V_i})$. But by the normality we have the algebraic Hartogs, i.e. the map

$$H^0(V_i \cap U, \mathscr{O}_{V_i}) = H^0(V_i \backslash Z, \mathscr{O}_{V_i}) \longrightarrow H^0(V_i, \mathscr{O}_{V_i})$$

is bijection, thus the section s automatically glues to be a section in $H^0(X, L)$. This proves that this natural inclusion is surjective, hence isomorphism as they're both finite dimensional vector spaces. □

Definition 3.1.15. (Section ring associated to a line bundle) Given a line bundle L on a projective variety X, the *graded ring* or *section ring* associated to L is the graded $\mathbb{C}$-algebra

$$R(L) = R(X, L) = \bigoplus_{m \geq 0} H^0(X, L^{\otimes m}).$$

The graded ring $R(D) = R(X, D)$ associated to a Cartier divisor D is defined similarly.

Example 3.1.16. If $X = \mathbb{P}^n$ and $L = \mathscr{O}_{\mathbb{P}^n}(1)$, then $R(L) = \mathbb{C}[T_0, \ldots, T_n]$ is the homogeneous coordinate ring of $\mathbb{P}^n$, as we have $H^0(X, L^{\otimes m}) = H^0(X, \mathscr{O}_X(m))$.

In above example we see the section ring $R(L)$ is a finitely generated $\mathbb{C}$-algebra, but not always so. This important property greatly influences the geometric properties of L.

Definition 3.1.17. (Finitely generated line bundles and Cartier divisors) A line bundle L on a projective variety X is *finitely generated* if its section ring $R(X, L)$ is a finitely generated $\mathbb{C}$-algebra. A Cartier divisor D is *finitely generated* if $\mathscr{O}_X(D)$ is so.

Passing to additive notation, we will discuss next the stable base locus of a Cartier divisor.

Suppose then that X still a projective variety with D is a Cartier divisor on X subject to $\kappa(X, D) \geq 0$. Given an integer $m \geq 1$ we consider the *set-theoretic* base locus $\mathrm{Bs}(|mD|) := \bigcap_{D' \in |mD|} \mathrm{Supp}(D')$ of $|mD|$, with the convention that $\mathrm{Bs}(|mD|) = X$ if $|mD| = \emptyset$. Here we use the same notation as in Chapter 1, but need to remember that the *set-theoretic* base locus is the *base scheme* equipped with reduced structure.

Definition 3.1.18. (Stable base locus) The *stable base locus* of D is the algebraic set

$$\mathbf{B}(D) = \bigcap_{m \geq 1} \mathrm{Bs}(|mD|).$$

Notice here we don't wish to view the righthand side as an intersection of schemes, that's the reason why we're **set-theoretically** speaking.

Proposition 3.1.19. *The stable base locus $\boldsymbol{B}(D)$ is the unique minimal element of the family of algebraic sets*

$$\{\mathrm{Bs}(|mD|)\}_{m\geq 1}.$$

Moreover, the base loci $\mathrm{Bs}(|mD|)$ actually stabilize to $\boldsymbol{B}(D)$ for sufficiently large and divisible m, i.e. there exists an $m_0 \in \mathbb{Z}_{>0}$ such that

$$\boldsymbol{B}(D) = \mathrm{Bs}(|km_0D|) \text{ for all } k \gg 0.$$

Proof. Given any natural numbers $m, l \geq 1$, there is a set-theoretic inclusion

$$\mathrm{Bs}(|lmD|) \subseteq \mathrm{Bs}(|mD|)$$

thanks to the reverse inclusion $\mathfrak{b}(|mD|)^l \subseteq \mathfrak{b}(|lmD|)$ on the base ideals, see Example 1.1.15. This implies the uniqueness of the minimal element appearing in the family $\{\mathrm{Bs}(|mD|)\}_{m\geq 1}$, which by definition must be coincide with $\mathbf{B}(D)$. The existence of the minimal element is trivial as X is a noetherian space, descending chain condition on Zariski-closed subsets $\{\mathrm{Bs}(|mD|)\}_{m\geq 1}$ of X must hold.

For the second statement, notice that if $\mathrm{Bs}(|pD|)$ and $\mathrm{Bs}(|qD|)$ are each minimal, then they are both coincide with $\mathrm{Bs}(|pqD|)$. □

Remark 3.1.20. (Realizing the stable base locus) It is not possible to take $m_0 = 1$ in the last statement: there is a $m_0 \in \mathbb{Z}_{>0}$ such that $\mathbf{B}(D) = \mathrm{Bs}(|km_0D|)$ for all $k \gg 0$. To begin with, if $e(D) = e > 1$, then by definition for m which is not divisible by e, we have $H^0(X, mD) = 0$, i.e. $|mD| = \mathbb{P}H^0(X, mD) = \emptyset$, so $\mathrm{Bs}(|mD|) = X$ unless we have $e|m$. Therefore $\mathbf{B}(D) = \mathrm{Bs}(|mD|)$ only if $e|m$.

However, even when $e(D) = 1$, it can happen that $\mathbf{B}(D) \neq \mathrm{Bs}(|mD|)$ for arbitrary large values of m.

Example 3.1.21. (Multiples) For any Cartier divisor D with $\kappa(X, D) \geq 0$, one has

$$\mathbf{B}(pD) = \mathbf{B}(D) \text{ for all } p \geq 1.$$

Proof. By second statement of Proposition 1.19, we have

$$\exists m_0 \in \mathbb{Z}_{>0} \text{ s.t. } \mathbf{B}(D) = \mathrm{Bs}(|km_0D|) \text{ for } k \gg 0,$$
$$\exists m_0' \in \mathbb{Z}_{>0} \text{ s.t. } \mathbf{B}(pD) = \mathrm{Bs}(|k'm_0'D|) \text{ for } k' \gg 0.$$

This shows that for suitable large k, $k' \gg 0$, we have $\mathbf{B}(D) = \mathbf{B}(pD)$. □

Remark 3.1.22. ($\mathbb{Q}$-Cartier divisors) The previous example leads to a natural definition of $\mathbf{B}(D)$ for an arbitrary $\mathbb{Q}$-Cartier divisor D: choose p such that pD is integral, then we define $\mathbf{B}(D)$ to be $\mathbf{B}(pD)$.

Example 3.1.23. (Scheme structures on base locus) Recall that $\mathrm{Bs}(|mD|)$ carries a natural scheme structure determined by the base-ideal $\mathfrak{b}(|mD|) \subseteq \mathscr{O}_X$. However, if they are considered as schemes, these base loci may be distinct even when their underlying sets coincide.

For a simple example, take X to be $\mathrm{Bl}_p\mathbb{P}^2$ and denote by E and H respectively the exceptional Cartier divisor over point p and the pullback of a line, and we set $D = H + E$. Then we have $|mD| = |mH| + mE$, that is because at the first from short exact sequence

$$0 \longrightarrow \mathscr{O}_X(mD - E) \xrightarrow{\times E} \mathscr{O}_X(mD) \longrightarrow \mathscr{O}_E(mD|_E) \longrightarrow 0$$

but where $\deg_E(mD) = (mD \cdot E) = -m$ so after taking global sections, we will see

$$H^0(X, mD - E) \cong H^0(X, mD)$$
$$s \mapsto s \cdot s_E$$

Thus $|mD - E| = |mD| + E$. Inductively, $|mD| = |mH| + mE$, where the $|mH|$ is free part and the mE is fixed component. Then next we recall that base-ideal $\mathfrak{b}(|mD|)$ is defined via the evaluation map

$$H^0(X, mD) \otimes \mathscr{H}om_{\mathscr{O}_X}(\mathscr{O}_X(mD), \mathscr{O}_X) \longrightarrow \mathscr{O}_X,$$

but as $|mD| = |mH| + mE$ we have $\mathfrak{b}(|mD|) = \mathscr{O}_X(-mE) \cdot \mathfrak{b}(|mH|) = \mathscr{O}_X(-mE) \cdot \mathscr{O}_X$ which is just $\mathscr{O}_X(-mE)$, i.e. the scheme-theoretic base locus is the mth-order neighborhood mE of E.

3.1B Semiample Line Bundles

As we're investigating the asymptotic behavior of the mappings ϕ_k determined by $|L^{\otimes k}|$ for large $k \in \mathbb{N}(X, L)$. This subsection focuses on the important case that some power of L is free, so we're restrict on complete scheme that we need the coherence of H^0 to realize the projectivization.

Definition 3.1.24. A line bundle on a complete scheme is *semi-ample* if $L^{\otimes m}$ is globally generated, i.e. $|L^{\otimes m}|$ is free, for some $m > 0$.

A Cartier divisor D is *semi-ample* if $\mathscr{O}_X(D)$ is so.

Now we fix a semi-ample line bundle L, and we denote by $\mathrm{M}(X,L) \subseteq \mathbb{N}(X,L)$ the sub-semigroup:

$$\mathrm{M}(X,L) := \{m \in \mathbb{N} \mid L^{\otimes m} \text{ is free}\},$$

and we write $f = f(L)$ for the "exponent" of $\mathrm{M}(X,L)$, i.e. the largest number such that every element of $\mathrm{M}(X,L)$ is a multiple of f. (In particular, $|L^{\otimes kf}|$ is free for $k \gg 0$).

Given $m \in \mathrm{M}(X,L)$, write as above $Y_m = \phi_m(X)$ for image of the morphism

$$\phi_m = \phi_{|L^{\otimes m}|} : X \longrightarrow Y_m \subseteq \mathbb{P} := \mathbb{P}H^0(X, L^{\otimes m})$$

determined by the complete linear series $|L^{\otimes m}|$. We usually abuse of notation to view ϕ_m as a morphism from X to Y_m rather than to $\mathbb{P}$.

Theorem 3.1.25. (Semi-ample fibrations) *Let X be a normal projective variety, and let L be a semi-ample line bundle on X. Then there is an algebraic fibre space*

$$\phi : X \longrightarrow Y$$

having the property that for any sufficiently large $m \in M(X,L)$, we have

$$Y_m \cong Y \text{ and } \phi_m = \phi \text{ up to differ by an isomorphism.}$$

Furthermore, there is an ample line bundle A on Y s.t. $\phi^* A = L^{\otimes f}$, where $f = f(L)$ is the exponent of $M(X,L)$.

In other words, for sufficiently large $m \in \mathrm{M}(X,L)$, the morphisms ϕ_m will stabilize to define a fibre space structure on X. When $\phi^* A = L^{\otimes f}$, we see this fibre space structure characterized by the fact that $L^{\otimes f}$ is trivial on fibres:

$$L^{\otimes f}|_{X_y} = \phi^* A|_{X_y} \cong (\phi|_{X_y})^*(A \otimes k(y)) \cong \mathscr{O}_{X_y}.$$

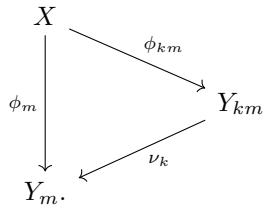
And observe that Lemma 1.9 implies that

$$H^0(X, L^{\otimes kf}) = H^0(Y, A^{\otimes k})$$

for all $k \geq 0$, while it follows from Example 1.13 that Y is normal. Notice that as we have a natural multiplication map

$$\begin{aligned} \mu_k : \mathrm{Sym}^k H^0(X, L^{\otimes m}) &\longrightarrow H^0(X, L^{\otimes km}) \\ s_1 \cdot s_2 \cdots s_k &\longmapsto s_1 \otimes s_2 \otimes \cdots \otimes s_k \end{aligned}$$

while the $\operatorname{im}\mu_k$ is a vector subspace of $H^0(X, L^{\otimes km})$, thus $\operatorname{Sym}^k H^0(X, L^{\otimes m})$ determines a free subseries of $|L^{\otimes km}|$, which corresponding to the kth Veronse re-embedding of Y_m sending into $\mathbb{P}H^0(X, L^{\otimes km})$



As in Example 1.1.18, for $W = H^0(X, L^{\otimes m}) \subseteq V = H^0(X, L^{\otimes km})$, we see ϕ_{km} and ϕ_m differ by a finite projection ν_k on Y_{km}: $\phi_m = \nu_k \circ \phi_{km}$.

Note also that Y_m carries a very ample line bundle A_m, that is $\mathscr{O}_{\mathbb{P}}(1)|_{Y_m}$ for $\mathbb{P} = \mathbb{P}H^0(X, L^{\otimes m})$ such that

$$\phi_m^* A_m = \phi_m^*(\mathscr{O}_{\mathbb{P}}(1)|_{Y_m}) \cong L^{\otimes m}$$

so we have $H^0(X, L^{\otimes m}) \cong H^0(Y_m, \mathscr{O}_{\mathbb{P}(1)|_{Y_m}}) = H^0(Y_m, A_m)$.

By above discussion, we can consider the following lemma:

Lemma 3.1.26. *In the situation of above theorem, **fix** any $m \in M(X, L)$. Then for all sufficiently large $k \in \mathbb{Z}$, the composition:*

$$X \xrightarrow{\phi_{km}} Y_{km} \xrightarrow{\nu_k} Y_m$$

gives a Stein factorization of ϕ_m, i.e. ϕ_{km} is an algebraic fibre space. In particular, Y_{km} and ϕ_{km} are independent of k for $k \gg 0$.

Proof. Let $X \xrightarrow{\psi} V \xrightarrow{\mu} Y_m$ be the Stein factorization of ϕ_m, so that ψ is a fibre space, and V is normal as X is, and μ is finite.

Let A_m to be the ample line bundle on Y_m that pulls back to $L^{\otimes m}$ on X. Since μ is finite, $B := \mu^* A_m$ is an ample line bundle on V. Therefore $B^{\otimes k}$ is very ample on V for all $k \gg 0$. On the other hand,

$$\psi^* B^{\otimes k} = (\mu \circ \psi)^* A_m^{\otimes k} = L^{\otimes km}$$

and

$$H^0(X, L^{\otimes km}) \cong H^0(V, B^{\otimes k})$$

which is thanks to the Lemma 1.9. But notice that as $B^{\otimes k}$ is very ample, the projective variety V actually embeds in $\mathbb{P}H^0(V, B^{\otimes k}) \cong \mathbb{P}H^0(X, L^{\otimes km})$, that is the morphism

$$\phi_{km} = \phi_{|L^{\otimes km}|} : X \longrightarrow Y_{km} \subseteq \mathbb{P}H^0(X, L^{\otimes km}) \cong \mathbb{P}H^0(V, B^{\otimes k}).$$

As ψ^* is an isomorphism sending the basis $\{t_i\}_{i=0}^N$ of $H^0(V, B^{\otimes k})$ to the basis $\{\psi^* t_i\}_{i=0}^N$ of $H^0(X, L^{\otimes km})$, we have

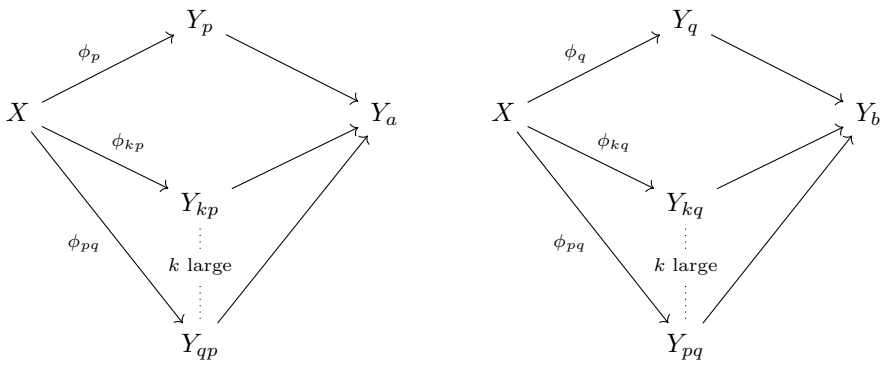
$$\begin{aligned}\phi_{km} : x &\mapsto [\psi^* t_0(x) : \psi^* t_1(x) : \cdots : \psi^* t_N(x)] \\ &= [t_0(\psi(x)) : t_1(\psi(x)) : \cdots : t_N(\psi(x))] \\ &= \psi_{|B^{\otimes k}|}(\psi(x))\end{aligned}$$

where $\psi_{|B^{\otimes k}|}$ is just a closed immersion as $B^{\otimes k}$ is very ample, therefore the image of ϕ_{km} coincide with V up to an isomorphism, i.e. $\phi_{km} = \psi$ differ only by an isomorphism for $k \gg 0$. □

Now we can turn to prove the semi-ample fibrations:

Proof. In order to lighten notation we assume that $f = f(L) = 1$, that is, by replacing L by $L^{\otimes f}$, so that every sufficiently large power of L is free.

Now we pick two sufficiently large relatively prime integers a, b which, of course, lie in $\mathrm{M}(X, L)$. So by Lemma 1.26 for this fixed a, there is an integer $K_a > 0$ such that for arbitrary $k \geq K_a$, we have $\phi_{ka} = \phi_{K_a a}$ up to differ by an isomorphism, which stabilizes to be the fibre space part of the Stein factorization of $X \longrightarrow Y_a$. Similarly for fixed $b \in \mathrm{M}(X, L)$, we also have $K_b > 0$ such that $\phi_{kb} = \phi_{K_b b}$ for arbitrary $k \geq K_b$. Thus we can choose prime numbers $l_a \geq K_a$ and $l_b \geq K_b$, satisfying that a, b, l_a and l_b are relatively prime in each pairs. Then we let $p := l_a a$, $q := l_b b$ which still relatively prime, and they satisfy $\phi_{kp} = \phi_p$, $\phi_{kq} = \phi_q$ up to differ by an isomorphism for all $k \geq 1$. Moreover as Stein factorization is unique up to isomorphism, we know that



then $Y_p \cong Y_{pq} \cong Y_q$, and $\phi_p = \phi_{pq} = \phi_q$ up to differ by an isomorphism. So we denote them by Y and ϕ: write $\phi : X \longrightarrow Y$ for the resulting fibre space.

We next produce a line bundle A on Y that pullbacks to $L = L^{\otimes f}$. In fact, Y carries very ample line bundles A_p and A_q, respectively is $\mathscr{O}_{\mathbb{P}}(1)|_Y$ and $\mathscr{O}_{\mathbb{P}'}(1)|_Y$ for

$\mathbb{P} = \mathbb{P}H^0(X, L^{\otimes p})$ and $\mathbb{P}' = \mathbb{P}H^0(X, L^{\otimes q})$, such that $\phi^* A_p = L^{\otimes p}$, $\phi^* A_q = L^{\otimes q}$. But since p and q are relatively prime, we can write $1 = rp + sq$ for suitable $r, s \in \mathbb{Z}$, and it suffices to take

$$A := A_p^{\otimes r} \otimes A_q^{\otimes s}.$$

Now as $\phi^* : \mathrm{Pic}(Y) \longrightarrow \mathrm{Pic}(X)$ is injective by ϕ is an algebraic fibre space (Example 1.12). So by $\phi^* A = L^{\otimes rp} \otimes L^{\otimes sq} = L$ and by $\phi^* A_p = L^{\otimes p} = \phi^* A^{\otimes p}$, $\phi^* A_q = L^{\otimes q} = \phi^* A^{\otimes q}$, we have $A_p \cong A^{\otimes p}$ and $A_q \cong A^{\otimes q}$, in particular A is ample.

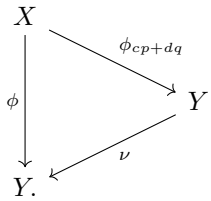
It remains to show that $Y_m \cong Y$ and $\phi_m = \phi$ for all $m \gg 0$. Fix $c, d \in \mathbb{Z}_{\geq 1}$, then the product

$$\mathrm{Sym}^c H^0(Y, A^{\otimes p}) \otimes \mathrm{Sym}^d H^0(Y, A^{\otimes q}) \longrightarrow H^0(Y, A^{\otimes (cp+dq)})$$

making the image determine a free linear subseries of

$$H^0(Y, A^{\otimes (cp+dq)}) = H^0(X, \phi^* A^{\otimes (cp+dq)}) = H^0(X, L^{\otimes (cp+dq)}).$$

As in the discussion preceding Lemma 1.26, it follows that ϕ factors as the composition of ϕ_{cp+dq} with a finite morphism:



Since ϕ is a fibre space, by Lemma 1.26 we have the finite part of Stein factorization of ϕ is trivial, i.e. $\phi = \phi_{cp+dq}$ differ by an isomorphism ν. But any sufficiently large integer m is of the form $cp + dq$, and the theorem follows. □

Semi-ample fibrations leads to some important finite generation properties of semi-ample line bundles.

Example 3.1.27. (Surjectivity of multiplication maps) Let X be a normal projective variety with a line bundle L generated by global sections. Then there exists $m_0 = m_0(L) \in \mathbb{Z}$ such that the mappings

$$H^0(X, L^{\otimes a}) \otimes H^0(X, L^{\otimes b}) \longrightarrow H^0(X, L^{\otimes (a+b)})$$

determined by multiplication are surjective whenever a, $b \geq m_0$.

More generally, for any coherent sheaf $\mathscr{F}$ on X,

$$H^0(X, \mathscr{F} \otimes L^{\otimes a}) \otimes H^0(X, L^{\otimes b}) \longrightarrow H^0(X, \mathscr{F} \otimes L^{\otimes (a+b)})$$

is onto for a, $b \gg 0$.

Proof. We first consider a projective morphism induced by base point free series $|L|$:

$$\psi = \psi_{|L|} : X \longrightarrow \psi(X) \subseteq \mathbb{P}^N,$$

then take the Stein factorization of ψ:

$$X \xrightarrow{\phi} Y \xrightarrow{\nu} \psi(X) = Y_1 \subseteq \mathbb{P}^N.$$

Using the semi-ample fibrations theorem, with $f = f(L) = 1$, we have that $\phi_m = \phi$ and $\phi^* A = L$ for some ample line bundle on Y when $m \gg 0$, in particular $m \geq m_0(L)$.

So $\phi^* L^{\otimes m} = \phi_*(\phi^* A^{\otimes m}) \cong A^{\otimes m} \otimes \phi_* \mathscr{O}_X$, and since ϕ is an algebraic fibre space, we have $\phi_* \mathscr{O}_X = \mathscr{O}_Y$, that is

$$H^0(X, \mathscr{F} \otimes L^{\otimes m}) = H^0(Y, \phi_*(\mathscr{F} \otimes L^{\otimes m})) = H^0(Y, \phi_* \mathscr{F} \otimes A^{\otimes m})$$

for any coherent sheaf $\mathscr{F}$, therefore we reduce to the case to proving that

$$H^0(Y, \phi_* \mathscr{F} \otimes A^{\otimes a}) \otimes H^0(Y, A^{\otimes b}) \longrightarrow H^0(Y, \phi_* \mathscr{F} \otimes A^{\otimes (a+b)})$$

with $\phi_* \mathscr{F}$ is coherent by ϕ projective, is surjective when a, b sufficiently large. In particular for $\mathscr{F} = \mathscr{O}_X$.

This is nothing but the Castelnuovo-Mumford regularity, that is as $\mathscr{F}$ is m-regular with respect to A on Y:

$$H^i(Y, \mathscr{F} \otimes A^{\otimes (m-i)}) = 0, \forall i > 0$$

for any coherent sheaf $\mathscr{F}$ and $m = m(\mathscr{F})$. Then by [Laz04a, 1.8.5] we see the multiplication map

$$H^0(Y, \phi_* \mathscr{F} \otimes A^{\otimes a}) \otimes H^0(Y, A^{\otimes b}) \longrightarrow H^0(Y, \phi_* \mathscr{F} \otimes A^{\otimes (a+b)})$$

is surjective for a, $b \gg 0$.

Now we verify the above regularity clain for any coherent sheaf $\mathscr{F}$ on Y. Assume $\dim Y = d$, then obviously

$$H^i(Y, \mathscr{F} \otimes A^{\otimes (m-i)}) = 0$$

for $i > d$, by Grothendieck vanishing theorem. Thus we can choose m to be large enough which depends on the number in the Serre vanishing with respect to $\mathscr{F}$ in [Har77, III, 5.3], i.e. by choose $m = m_0 = n_0 + d$, such that $m - i$ is always large than n_0, where m_0 is dependent on $\mathscr{F}$ and $\mathscr{A}$. □

Theorem 3.1.28. (Finite generation of semi-ample line bundles) *Let L be a semi-ample line bundle on a normal projective variety X. Then L is finitely generated, i.e. the section ring $R(X, L)$ of L is finitely generated as a $\mathbb{C}$-algebra.*

Proof. We suppose $L^{\otimes k}$ is free, then by Example 1.27 the multiplication map

$$H^0(X, L^{\otimes ka}) \otimes H^0(X, L^{\otimes kb}) \longrightarrow H^0(X, L^{\otimes k(a+b)})$$

is surjective for any a, $b \geq m_0(L^{\otimes k})$, that is any element of the Veronese subalgebra $R(X, L)^{(k)} = \bigoplus_{m \geq 0} H^0(X, L^{\otimes km})$, can be written as the sum of lower degree elements' multiplication, i.e. $R(X, L)^{(k)}$ is finitely generated as $\mathbb{C}$-algebra.

Now we can apply the Example 1.27 again on L, $L^{\otimes 2}$, $\ldots$, $L^{\otimes k-1}$ for second statement, can easily get that $R(X, L)$ is finitely generated as $\mathbb{C}$-algebra. □

Furthermore, Zariski proved a surprising fact about linear series having finite base loci in 1962 at [Zar62, II, 6.2], and Fujita generalized it in 1980 at [Fuj80]:

Theorem 3.1.29. (Zariski-Fujita) *Let L be a line bundle on a projective variety X, with the property that the base locus $\mathrm{Bs}(|L|)$ is a finite set. Then L is semi-ample, i.e. $L^{\otimes m}$ is free for some $m > 0$.*

3.1C Iitaka Fibration

We now give the asymptotic analysis of the rational mappings:

$$\phi_k = \phi_{|L^{\otimes k}|} : X \dashrightarrow Y_k \subseteq \mathbb{P} = \mathbb{P}H^0(X, L^{\otimes k})$$

in the general setting. We always view the map ϕ_k as a rational mapping to Y_k.

Theorem 3.1.30. (Iitaka fibrations) *Let X be a normal projective variety, and L a line bundle on X such that $\kappa(X, L) > 0$. Then for all sufficiently large $k \in \mathbb{N}(X, L)$, the rational mappings $\phi_k : X \dashrightarrow Y_k$ are birationally equivalent to a fixed algebraic fibre space*

$$\phi_\infty : X_\infty \longrightarrow Y_\infty$$

of normal varieties, and the restriction of L to a very general fibre of ϕ_∞ has Iitaka dimension is 0.

More specifically, there exists for large $k \in \mathbb{N}(X, L)$ a commutative diagram

$$\begin{array}{ccc} X & \xleftarrow{u_\infty} & X_\infty \\ {\scriptstyle \phi_k}\downarrow & & \downarrow{\scriptstyle \phi_\infty} \\ Y_k & \xleftarrow{\nu_k} & Y_\infty \end{array}$$

of rational maps and morphisms, where ν_k are birational and u_∞ is a morphism. One has $\dim Y_\infty = \kappa(X, L)$. Moreover, if we set $L_\infty = u_\infty^* L$, and take $F \subseteq X_\infty$ to be a very general fibre of ϕ_∞, then

$$\kappa(F, L_\infty|_F) = 0.$$

Definition 3.1.31. (Iitaka fibration associated to a line bundle) In the same situation of above theorem, the morphism $\phi_\infty : X_\infty \longrightarrow Y_\infty$ is the *Iitaka fibration* associated to L. It is unique up to the birational equivalence (whereas a Semi-ample fibration is unique up to isomorphism).

The *Iitaka fibration* of a Cartier divisor D is defined by passing to $\mathscr{O}_X(D)$.

Definition 3.1.32. (Iitaka fibration of a variety) The *Iitaka fibration of a variety* X is by definition the Iitaka fibration associated to the canonical bundle on any non-singular model of X.

Now we turn to prove the Iitaka fibration theorem:

Proof. Fix any $m \in \mathbb{N}(X, L)$ such that $\dim Y_m = \kappa(X, L)$.

Step 1: We first prove that if $k \gg 0$, then all the mappings $\phi_{km} : X \dashrightarrow Y_{km}$ are birationally equivalent to a fixed algebraic fibre space $\psi_{(m)} : X_{(m)} \longrightarrow Y_{(m)}$ of normal varieties.

In fact, we consider the blow-up with respect to the base ideal $\mathfrak{b} := \mathfrak{b}(|L^{\otimes m}|)$ for each divisible m, that this is the $\tilde{X} = \mathrm{Proj}(\bigoplus_{d\geq 0} \mathfrak{b}^d)$ with a birational morphism $\varepsilon : \tilde{X} \longrightarrow X$, and an ideal sheaf $\tilde{\mathfrak{b}} = \varepsilon^{-1}\mathfrak{b} \cdot \mathscr{O}_{\tilde{X}} \cong \mathscr{O}_{\tilde{X}}(-E)$ where E is the ε-exceptional Cartier divisor. Moreover we equip it once more with a normalization $X_{(m)} \xrightarrow{\nu} \tilde{X}$ to get a birational morphism $u_m : X_{(m)} \longrightarrow X$. Since the base ideal is coming from the evaluation map

$$H^0(X, L^{\otimes m}) \underset{\mathbb{C}}{\otimes} (L^{\otimes m})^\vee \longrightarrow \mathscr{O}_X$$

so we have a surjective morphism

$$H^0(X, L^{\otimes m}) \underset{\mathbb{C}}{\otimes} \mathscr{O}_X \longrightarrow \mathfrak{b} \underset{\mathscr{O}_X}{\otimes} L^{\otimes m}.$$

Then pulls it back via u_m:

$$H^0(X_{(m)}, u_m^* L^{\otimes m}) \otimes \mathscr{O}_{X_{(m)}} \longrightarrow u_m^* \mathfrak{b} \otimes u_m^* L^{\otimes m}$$

is surjective by $H^0(X_{(m)}, u_m^* L^{\otimes m}) \cong H^0(X, L^{\otimes m})$, use X normal and u_m birational. And futher more

$$H^0(X_{(m)}, u_m^* L^{\otimes m}) \otimes \mathscr{O}_{X_{(m)}} \longrightarrow u_m^{-1}\mathfrak{b} \cdot \mathscr{O}_{X_{(m)}} \otimes u_m^* L^{\otimes m}$$

is also surjective since the $u_m^{-1}\mathfrak{b} \cdot \mathscr{O}_{X_{(m)}}$ is defined by the image of $u_m^*\mathfrak{b} \to \mathscr{O}_{X_{(m)}}$. Moreover the commutative diagram

$$\begin{array}{ccc} X_{(m)} & & \\ {\scriptstyle\nu}\downarrow & \searrow^{u_m} & \\ \tilde{X} & \xrightarrow[\varepsilon]{} & X \end{array}$$

implies that

$$\begin{aligned} u_m^{-1}\mathfrak{b} \cdot \mathscr{O}_{X_{(m)}} =& (\varepsilon \circ \nu)^{-1}\mathfrak{b} \cdot \mathscr{O}_{X_{(m)}} \\ =& \nu^{-1}(\varepsilon^{-1}\mathfrak{b} \cdot \mathscr{O}_{\tilde{X}}) \cdot \mathscr{O}_{X_{(m)}} \\ =& \nu^{-1}\mathscr{O}_{\tilde{X}}(-E) \cdot \mathscr{O}_{X_{(m)}} \\ =& \nu^*\mathscr{O}_{\tilde{X}}(-E) \end{aligned}$$

where the last equality holds because $\mathscr{O}_{\tilde{X}}(-E)$ is locally free, so $\otimes$ is exact ([Har77, 7.12.2]), and result is denoted by $\mathscr{O}_{X_{(m)}}(-F_m)$.

Thus we now define $M_m := \mathscr{O}_{X_{(m)}}(-F_m) \otimes u_m^*L^{\otimes m}$, which is a globally generated line bundle since the surjective evaluation map

$$H^0(X_{(m)}, u_m^*L^{\otimes m}) \underset{\mathbb{C}}{\otimes} \mathscr{O}_{X_{(m)}} \longrightarrow M_m$$

gives the surjective map $\mathscr{O}_{X_{(m)}}^N \longrightarrow M_m$.

Therefore we get $u_m : X_{(m)} \longrightarrow X$ a resolution of indeterminacies of ϕ_m, i.e. a birational morphism with

$$u_m^*|L^{\otimes m}| = |M_m| + F_m,$$

where F_m is an effective fixed Cartier divisor of $|u_m^*L^{\otimes m}|$, and

$$\psi_m : X_{(m)} \longrightarrow Y_m \subseteq \mathbb{P}H^0(X_{(m)}, M_m) = \mathbb{P}H^0(X, L^{\otimes m})$$

is the morphism defined by $|M_m|$, where the last equality holds because we can regard $u_m^*L^{\otimes m}$ as corresponding to the Cartier divisor mD, so the right H^0 consists of f with $\mathrm{div}(f)+mD \geq 0$. But as F_m is the fixed component automatically have $\mathrm{div}(f)+mD \geq F_m$, thus lies in the left.

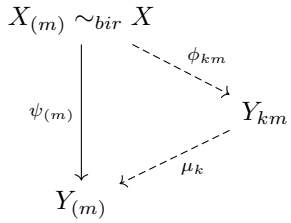
The situation is summarized by the top triangle of the following commutative diagram:

$$\begin{array}{ccc} X & \xleftarrow{u_m} & X_{(m)} \\ {\scriptstyle\phi_m}\vdots & \swarrow_{\psi_m} & \downarrow{\scriptstyle\psi_{km}} \\ Y_m & \xleftarrow[\lambda_k]{} & Y'_{km}. \end{array}$$

Consider now the morphism $\psi_{km} : X_{(m)} \longrightarrow Y'_{km}$ defined by $|M_m^{\otimes k}|$. Then as indicated in the diagram, the ψ_m factors as the composition

$$X_{(m)} \xrightarrow{\psi_{km}} Y'_{km} \xrightarrow{\lambda_k} Y_m$$

with λ_k is finite, by the discussion under the Semi-ample fibration. Moreover follows from the Semi-ample fibration theorem, that for $k \gg 0$ the ψ_{km} will stabilizes to a fixed algebraic fibre space $\psi_{(m)} : X_{(m)} \longrightarrow Y_{(m)}$. On the other hand, $|M_m^{\otimes k}|$ defines a linear subseries of $|u_m^* L^{\otimes km}|$, thus also defines a linear subseries of $|L^{\otimes km}|$. So we can view $\psi_{(m)}$ as a rational mapping from X to $Y_{(m)}$ by $X_{(m)} \sim_{bir} X$, and then we get a factorization when k sufficiently large:



where μ_k is generically finite. That is because as λ_k is a finite map so we have $\dim Y'_{km} = \dim Y_m = \kappa(X, L)$, and by ψ_{km} will stabilizes to $\psi_{(m)}$, we have that $\dim Y_{(m)} = \dim Y'_{km}$. But $\dim Y_{(m)} \leq \dim Y_{km} \leq \dim Y_m = \dim Y_{(m)}$, implies that

$$\dim Y_{(m)} = \dim Y_{km},$$

so μ_k is indeed a generically finite map. Moreover by $\mathbb{C}(Y_{(m)})$ is algebraically closed in $\mathbb{C}(X_{(m)}) = \mathbb{C}(X)$ by $\psi_{(m)}$ is an algebraic fibre space (Example 1.11), so we see the degree of finite field extension part of $\mathbb{C}(Y_{(m)}) \subseteq \mathbb{C}(Y_{km}) \subseteq \mathbb{C}(X)$ is one, that is μ_k is a birational map. So $\phi_{km} : X \dashrightarrow Y_{km}$ are birationally equivalent to the algebraic fibre space $\psi_{(m)} : X_{(m)} \longrightarrow Y_{(m)}$, as asserted.

Step 2: The fibre spaces $X_{(m)} \longrightarrow Y_{(m)}$ just constructed are all birationally isomorphic, and the next step is to choose a common model.

Specifically, replacing L by $L^{\otimes e}$ we may assume first of all that L has exponent $e(L) = 1$. Fix them two large enough relatively prime integers p and q such that

$$\dim Y_p = \dim Y_q = \kappa(X, L)$$

by Example 1.8. With notation as in *Step 1* there exists, by construction, an integer $m \gg 0$ such that $Y_{(p)}$ and $Y_{(q)}$ are the images of $X_{(p)}$ and $X_{(q)}$ under the morphisms defined by the complete linear series $|M_p^{\otimes(p^{m-1})}|$ and $|M_q^{\otimes(q^{m-1})}|$ respectively (by our choice of m, their indices are sufficiently large to ensure the morphisms stabilized).

Now we need construct a common resolution X_∞ of $u_p : X_{(p)} \longrightarrow X$ and $u_q : X_{(q)} \longrightarrow X$ of ϕ_p and ϕ_q, satisfying that there is a commutative diagram

$$\begin{array}{ccc} X_\infty & \xrightarrow{v_p} & X_{(p)} \\ {\scriptstyle v_q}\downarrow & & \downarrow{\scriptstyle u_p} \\ X_{(q)} & \xrightarrow[u_q]{} & X. \end{array}$$

To begin with, we consider the fibre product $X_{(p)} \underset{X}{\times} X_{(q)} := Z$, there is a Cartesian diagram

$$\begin{array}{ccc} Z & \xrightarrow{pr_1} & X_{(p)} \\ {\scriptstyle pr_2}\downarrow & & \downarrow{\scriptstyle u_p} \\ X_{(q)} & \xrightarrow[u_q]{} & X. \end{array}$$

As u_p and u_q are birational morphism like the construction in *Step 1*, so we have that $Z_U := u_p^{-1}(U) \underset{U}{\times} u_q^{-1}(U) \cong U \underset{U}{\times} U = U$ is irreducible. Then taking closure of Z_U in Z we get an irreducible component of Z, denote it by W which satisfying that $pr_1|_W$ and $pr_2|_W$ are both birational morphisms. After take a normalization to W, we get the result which is denoted by X_∞. Thus, we can fix a normal variety X_∞ together with birational morphisms

$$v_p : X_\infty \longrightarrow X_{(p)} \text{ and } v_q : X_\infty \longrightarrow X(q)$$

such that $u_p \circ v_p = u_q \circ v_q =: u_\infty : X_\infty \longrightarrow X$.

Now consider on X_∞ the globally generated line bundle

$$M_{p,q} := v_p^* M_p^{\otimes(p^{m-1})} \otimes v_q^* M_q^{\otimes(q^{m-1})},$$

where in exactly $M_p^{\otimes(p^{m-1})}$ and $M_q^{\otimes(q^{m-1})}$ coming from, by our construction, the natural ample line bundles on $Y_{(p)}$ and $Y_{(q)}$ by restricting the hyperplane sections on them, i.e.

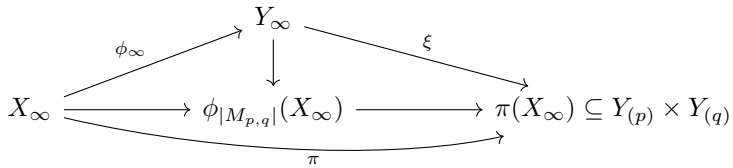
$$M_p^{\otimes(p^{m-1})} = \psi_{(p)}^* \mathscr{O}_{Y_{(p)}}(1) \text{ and } M_q^{\otimes(q^{m-1})} = \psi_{(q)}^* \mathscr{O}_{Y_{(q)}}(1).$$

Thus by [Har77, II, ex5.11], the $\mathscr{O}_{Y_{(p)} \times Y_{(q)}}(1) = pr_{(p)}^* \mathscr{O}_{Y_{(p)}}(1) \otimes pr_{(q)}^* \mathscr{O}_{Y_{(q)}}(1)$ and satisfying that $\pi^* \mathscr{O}_{Y_{(p)} \times Y_{(q)}}(1) = M_{p,q}$ (abuse of notation), where

$$\pi = (\psi_{(p)} \circ v_p, \psi_{(q)} \circ v_q) : X_\infty \longrightarrow Y_{(p)} \times Y_{(q)}.$$

Denote by Y_∞ the normalization of the image of X_∞ under the morphism defined by the complete linear series $|M_{p,q}|$, and by $\phi_\infty : X_\infty \longrightarrow Y_\infty$ the corresponding

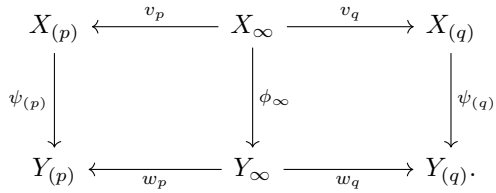
morphism. So we have the commutative diagram



Then Y_∞ maps finitely to the Segre embedding of the product $Y_{(p)} \times Y_{(q)}$, the reason is simply that the morphism $\phi_{|M_{p,q}|}(X_\infty) \longrightarrow \pi(X_\infty)$ pulls back the $\mathscr{O}(1)$ to $\mathscr{O}(1)$, so it is fibre finite. And so we have morphisms

$$w_p : Y_\infty \longrightarrow Y_{(p)} \text{ and } w_q : Y_\infty \longrightarrow Y_{(q)}$$

sitting in the commutative diagram:



Now we have $\dim Y_\infty = \kappa(X, L)$, since the morphisms v_p and $\psi_{(p)}$ (resp. v_q and $\psi_{(q)}$) are surjective, so w_p (resp. w_q) also is surjective by diagram commutes, hence

$$\dim Y_\infty \geq \dim Y_{(p)} = \dim Y_{(q)} = \kappa(X, L)$$

where the last equality holds by similar analysis in *Step 1*. On the other hand, $|M_p^{\otimes(p^{m-1})}|$ (resp. $|M_q^{\otimes(q^{m-1})}|$) defines a linear subseries of $|u_p^* L^{\otimes p^m}|$ (resp. $|u_q^* L^{\otimes q^m}|$), and hence also of $|L^{\otimes p^m}|$ (resp. $|L^{\otimes q^m}|$). Then by our construction, the linear series $|M_{p,q}|$ defines a linear subseries of $|u_\infty^* L^{\otimes(p^m+q^m)}|$ hence also of $|L^{\otimes(p^m+q^m)}|$, by u_∞ is an algebraic fibre space ([Har77, II, 11.4]). Therefore

$$\dim Y_\infty \leq \dim Y_{p^m+q^m} \leq \kappa(X, L).$$

So we see w_p and w_q are generically finite.

Moreover, since they factor the algebraic fibre spaces $\psi_{(p)} \circ v_p$ and $\psi_{(q)} \circ v_q$, where v_p and v_q are fibre spaces by our construction $X_{(p)}$ and $X_{(q)}$ are normal and they're birational. Thus we see the finite field extension part of $\mathbb{C}(Y_{(p)}) \subseteq \mathbb{C}(X_\infty)$ (resp. $\mathbb{C}(Y_{(q)}) \subseteq \mathbb{C}(X_\infty)$) is trivial, so w_p (resp. w_q) is birational. By same token, Example 1.11, we see ϕ_∞ is an algebraic fibre space since the composition $w_p \circ \phi_\infty$ is. Note that by construction Y_∞ carries an ample line bundle $A_{p,q}$ because ξ is finite and pulls back

the hyperplane section on $Y_{(p)} \times Y_{(q)}$, satisfying $\phi_\infty^* A_{p,q} = M_{p,q}$ because the diagram commutes.

Now fix positive integers c, $d \geq 1$. Then we have

$$\begin{aligned} H^0(X_\infty, M_{p,q}) \subseteq & H^0(X_\infty, v_p^* M_p^{\otimes(cp^{m-1})} \otimes v_q^* M_q^{\otimes(dq^{m-1})}) \\ \subseteq & H^0(X_\infty, u_\infty^* L^{\otimes(cp^m+dq^m)}), \end{aligned}$$

the first inclusion being defined via multiplication by a fixed section of $u_p^* M_p^{\otimes(c-1)p^{m-1}} \otimes u_q^* M_q^{\otimes(d-1)q^{m-1}}$, which is injective. This gives rise in the usual way to a factorization

$$\begin{array}{ccc} X_\infty & \xrightarrow{\quad v_q \quad} & X \\ \Big\downarrow{\scriptstyle \phi_\infty} & & \Big\vdots{\scriptstyle \phi_{cp^m+dq^m}} \\ Y_\infty & \xleftarrow[\mu_{cp^m+dq^m}]{} & Y_{cp^m+dq^m} \end{array}$$

where $\mu_{cp^m+dq^m}$ is generically finite, since we have

$$\kappa(X, L) = \dim Y_\infty \leq \dim Y_{cp^m+dq^m} \leq \kappa(X, L).$$

But, since ϕ_∞ is an algebraic fibre space, so by Example 1.11 implies that the finite field extension part is trivial, i.e. $\mu_{cp^m+dq^m}$ must be birational. And since every $k \gg 0$ is of the form $cp^m + dq^m$, taking $\nu_k = \mu_k^{-1}$ gives the diagram appearing in the statement of this theorem:

$$\begin{array}{ccc} X & \xleftarrow{\quad u_\infty \quad} & X_\infty \\ \Big\vdots{\scriptstyle \phi_k} & & \Big\downarrow{\scriptstyle \phi_\infty} \\ Y_k & \xleftarrow[\nu_k]{} & Y_\infty. \end{array}$$

Step 3: It remains to show that a very general fibre F of ϕ_∞ has L-dimension zero.

Set $L_\infty = u_\infty^* L$. Evidently $\kappa(F, L_\infty|_F) \geq 0$, since by u_∞ is an algebraic fibre space we have $|L_\infty| = |L|$ so the map $\phi_{|L_\infty^{\otimes k}|} = \phi_{|L^{\otimes k}|}$ restricted to F has a non-empty image. The issue is to establish the reverse inequality.

To this end, we fix a very general point $y \in Y_\infty$ (that is, outside of countably many Zariski closed subsets), and let

$$F = F_y = \phi_\infty^{-1}(y) \subseteq X_\infty.$$

We can assume that ν_k is defined and regular at y for every $k \gg 0$, and that $u_\infty(F) \subseteq X \overset{\phi_k}{\dashrightarrow} Y_k$ is not contained in the locus of indeterminacy of any of the ϕ_k. (Here we

are imposing countably many open conditions on y). Then $\phi_k \circ u_\infty : F \longrightarrow \{\nu_k(y)\}$ maps F to a point. But this means that for the rational map $\phi_k \circ u_\infty : X_\infty \dashrightarrow Y_k$ is defined by global sections of $H^0(X_\infty, L_\infty^{\otimes k})$:

$$\begin{aligned} X_\infty \dashrightarrow Y_k \subseteq \mathbb{P}H^0(X_\infty, L_\infty^{\otimes k}) = \mathbb{P}H^0(X, L^{\otimes k}) \\ x \mapsto [s_0(x) : s_1(x) : \cdots : s_N(x)] \end{aligned}$$

for which we have $F \mapsto [c_0 : c_1 : \cdots : c_N] = y$, i.e. $s_i|_F = c_i \cdot s_F$ for some nowhere vanishing section in $H^0(F, L_\infty^{\otimes k}|_F)$. Thus the restriction map

$$\rho_k : H^0(X_\infty, L_\infty^{\otimes k}) \longrightarrow H^0(F, L_\infty^{\otimes k}|_F)$$

has rank one for every $k \gg 0$.

Fix next a very ample line bundle B on Y_∞, which we can simply choose the hyperplane section of $\mathbb{P}H^0(X_\infty, M_{p,q})$ pulls back on Y_∞. We claim that there is a large positive integer $m_0 > 0$ such that

$$H^0(X_\infty, L_\infty^{\otimes m} \otimes \phi_\infty^*(B^\vee)) \neq 0.$$

This will follow that we get, for a fixed k and any $r > 0$, a diagram

$$\begin{array}{ccc} H^0(X_\infty, L_\infty^{\otimes k} \otimes \phi_\infty^* B^{\otimes r}) & \hookrightarrow & H^0(X_\infty, L_\infty^{\otimes (k+rm_0)}) \\ \downarrow{\scriptstyle \beta_{k,r}} & & \downarrow{\scriptstyle \rho_{k+rm_0}} \\ H^0(F, (L_\infty^{\otimes k} \otimes \phi_\infty^* B^{\otimes r})|_F) & \hookrightarrow & H^0(F, L_\infty^{\otimes (k+rm_0)}|_F) \end{array}$$

where the horizontal map is given by multiplication by sections of $H^0(X_\infty, L_\infty^{\otimes rm_0} \otimes \phi_\infty^*(B^\vee)^{\otimes r})$, and the vertical maps arise via restriction to F. Now for very general fibre $F = F_y$, the homomorphism $\beta_{k,r}$ on the left is identified with the map

$$\begin{aligned} H^0(X_\infty^{\otimes k} \otimes \phi_\infty^* B^{\otimes r}) =& H^0(Y_\infty, \phi_{\infty,*}(L_\infty^{\otimes k} \otimes \phi_\infty^* B^{\otimes r})) \\ =& H^0(Y_\infty, \phi_{\infty,*}(L_\infty^{\otimes k}) \otimes B^{\otimes r}) \\ \longrightarrow& H^0(\{y\}, \phi_{\infty,*}(L_\infty^{\otimes k}) \otimes B^{\otimes r}|_{\{y\}}) \\ =& ((\phi_{\infty,*}(L_\infty^{\otimes k}) \otimes B^{\otimes r}) \otimes \mathscr{O}_{\{y\}})_y \\ =& (\phi_{\infty,*}(L_\infty^{\otimes k}) \otimes B^{\otimes r})_y \underset{\mathscr{O}_{Y_\infty,y}}{\otimes} \mathbb{C}(y) \end{aligned}$$

obtained by evaluating the section of the indicated direct image sheaf at the point $y \in Y$, where $\mathbb{C}(y) = \mathscr{O}_{Y_\infty,y}/m_y$ is the residue field of Y_∞ at y. But since B is

ample, for fixed k the sheaf $\phi_{\infty,*}(L_\infty^{\otimes k}) \otimes B^{\otimes r}$ is globally generated for $r \gg 0$, i.e. the evaluation map

$$H^0(Y_\infty, \phi_{\infty,*}(L_\infty^{\otimes k}) \otimes B^{\otimes r}) \underset{\mathbb{C}}{\otimes} (\phi_{\infty,*}(L_\infty^{\otimes k}) \otimes B^{\otimes r})^\vee \longrightarrow \mathscr{O}_{Y_\infty}$$

is surjective, so is the

$$H^0(Y_\infty, \phi_{\infty,*}(L_\infty^{\otimes k}) \otimes B^{\otimes r}) \underset{\mathbb{C}}{\otimes} \mathscr{O}_{Y_\infty, y} \longrightarrow (\phi_{\infty,*}(L_\infty^{\otimes k}) \otimes B^{\otimes r})_y.$$

Then we tensor it with $\underset{\mathscr{O}_{Y_\infty,y}}{\otimes} \mathbb{C}(y)$:

$$H^0(Y_\infty, \phi_{\infty,*}(L_\infty^{\otimes k}) \otimes B^{\otimes r}) \underset{\mathbb{C}}{\otimes} \mathbb{C}(y) \longrightarrow (\phi_{\infty,*}(L_\infty^{\otimes k}) \otimes B^{\otimes r})_y \otimes \mathbb{C}(y)$$

is also surjective, but it is easy to see that $\beta_{k,r}$ is surjective for fixed k and $r \gg 0$, since for closed point y, $\mathbb{C}(y) \cong \mathbb{C}$.

On the other hand, we have seen that ρ_{k+rm_0} has rank one, so by the commutativity of the diagram we can see $\beta_{k,r}$ also has rank one, and therefore

$$\dim H^0(F, (L_\infty^{\otimes k} \otimes \phi_\infty^* B^{\otimes r})|_F) = 1.$$

But since $\phi_\infty^* B$ is trivial along F, this implies that

$$h^0(F, L_\infty^{\otimes k}|_F) = 1$$

for fixed k, hence for every $k \gg 0$. But notice that this dimension also holds for every k since if not, we can take the higher tensor powers to make h^0 sufficiently large for some $k \gg 0$. So $\kappa(F, L_\infty|_F) \leq 0$ as required.

It remains only to verify our claim, the non-vanishing of $h^0(X_\infty, L_\infty^{\otimes m_0} \otimes \phi_\infty^*(B^\vee))$. Recall that by commutative diagram

$$\begin{array}{ccccc} & & Y_\infty & & \\ & \overset{\phi_\infty}{\nearrow} & \downarrow & \overset{\xi}{\searrow} & \\ X_\infty & \longrightarrow & \phi_{|M_{p,q}|}(X_\infty) & \longrightarrow & \pi(X_\infty), \end{array}$$

that Y_∞ carries an ample and globally generated line bundle $A = A_{p,q}$ pulling back to $M_{p,q}$ on X_∞. Then $A^{\otimes m_1} \otimes B^\vee$ has a non-zero global section for some $m_1 \gg 0$, as the corresponding Cartier divisor is effective. So we pull it back via ϕ_∞ to get

$$\phi_\infty^*(A^{\otimes m_1}) \otimes \phi_\infty^*(B^\vee) = M_{p,q}^{\otimes m_1} \otimes \phi_\infty^*(B^\vee).$$

But by our construction, recall that, we have injections

$$M_p = \mathscr{O}_{X_{(p)}}(-F_p) \otimes u_p^* L^{\otimes p} \hookrightarrow u_p^* L^{\otimes p}$$
$$M_q = \mathscr{O}_{X_{(q)}}(-F_q) \otimes u_q^* L^{\otimes q} \hookrightarrow u_q^* L^{\otimes q}$$

so

$$M_{p,q} = v_p^* M_p^{\otimes(p^{m-1})} \otimes v_q^* M_q^{\otimes(q^{m-1})} \hookrightarrow u_\infty^* L^{\otimes(p^m+q^m)} = L_\infty^{\otimes(p^m+q^m)}$$

is also injective, that is, we can regard $M_{p,q}^{\otimes m_1}$ as a subsheaf of $L_\infty^{\otimes(p^m+q^m)m_1}$. Then we tensor with a locally free sheaf $\phi_\infty^*(B^\vee)$ and take global section; since each step is left exact, we have

$$H^0(X_\infty, L_\infty^{\otimes(p^m+q^m)m_1} \otimes \phi_\infty^*(B^\vee)) \neq 0,$$

thus the claim declared as we desired. □

Remark 3.1.33. At the beginning of the proof, we use a method to remove the base locus of a linear system, which is a useful technique. See [Kaw24, 1.2.5]. We give a general version below.

Example 3.1.34. Let $|V|$ be a linear series of Cartier divisors on a normal variety. Then there exists a birational projective morphism $f : Y \longrightarrow X$ from a normal variety Y such that the pullback has the form

$$f^*|V| = |V_1| + F,$$

where F is the fixed Cartier divisor of $|f^*V|$ and the linear system $|V_1|$ is free.

Proof. Let $V \subseteq H^0(X, D)$ be the finite-dimensional subspace corresponding to the linear series. The image of the evaluation map

$$V \underset{\mathbb{C}}{\otimes} \mathscr{O}_X \longrightarrow \mathscr{O}_X(D)$$

can be written as $\mathfrak{b} \otimes \mathscr{O}_X(D) = \mathfrak{b} \cdot \mathscr{O}_X(D)$, where $\mathfrak{b}$ is an ideal sheaf on X. Take $f = \nu \circ \varepsilon$ to be the normalization of the blow-up along $\mathfrak{b}$. Then the inverse image ideal sheaf

$$\begin{aligned} f^{-1}\mathfrak{b} \cdot \mathscr{O}_Y =& \nu^{-1}(\varepsilon^{-1}\mathfrak{b} \cdot \mathscr{O}_{\tilde{X}}) \cdot \mathscr{O}_Y \\ =& \nu^{-1}(\mathscr{O}_{\tilde{X}}(-E)) \cdot \mathscr{O}_Y \\ =& \nu^* \mathscr{O}_{\tilde{X}}(-E) \end{aligned}$$

is an invertible sheaf on Y, for $\varepsilon : \mathrm{Bl}_{\mathfrak{b}} X =: \tilde{X} \longrightarrow X$ the blow-up and $\nu : Y \longrightarrow \tilde{X}$ is normalization. Notice that here for a general variety, the blow-up along arbitrary ideal sheaf is in general proper. But here, since we have surjective map

$$V \underset{\mathbb{C}}{\otimes} \mathscr{O}_X \cong \mathscr{O}_X^{\oplus(n+1)} \twoheadrightarrow \mathfrak{b} \cdot \mathscr{O}_X(D),$$

it induces a surjective map

$$\mathrm{Sym}(\mathscr{O}_X^{\oplus(n+1)}) \twoheadrightarrow \bigoplus_{d\geq 0}\mathfrak{b}^d\cdot\mathscr{O}_X(dD).$$

So we get an injection

$$\mathrm{Bl}_{\mathfrak{b}}X = \underline{\mathrm{Proj}}_X\left(\bigoplus_{d\geq 0}\mathfrak{b}^d\right) \cong \underline{\mathrm{Proj}}_X\left(\bigoplus_{d\geq 0}\mathfrak{b}^d\cdot\mathscr{O}_X(dD)\right) \hookrightarrow \mathbb{P}^n_X,$$

where the isomorphism is from [Har77, II, ex7.11(b)]. Therefore ε must be projective as $\mathrm{Bl}_{\mathfrak{b}}X$ admits an ample line bundle (see [Har77, II, 7.10(b)]).

Now $f^{-1}\mathfrak{b}\cdot\mathscr{O}_Y\otimes f^*\mathscr{O}_X(D) = f^{-1}\mathfrak{b}\cdot\mathscr{O}_Y(f^*D)$ is the image of the evaluation map

$$V\underset{\mathbb{C}}{\otimes}\mathscr{O}_Y \longrightarrow \mathscr{O}_Y(f^*D).$$

But as $f^{-1}\mathfrak{b}\cdot\mathscr{O}_Y \cong \nu^*\mathscr{O}_{\tilde{X}}(-E)$, the $f^{-1}\mathfrak{b}\cdot\mathscr{O}_Y(f^*D)$ can be written in the form $\mathscr{O}_Y(f^*D-F)$ for some effective Cartier divisor F. That is, the evaluation map

$$V\underset{\mathbb{C}}{\otimes}\mathscr{O}_Y \cong \mathscr{O}_Y^{\oplus(n+1)} \longrightarrow \mathscr{O}_Y(f^*D-F)$$

being surjective gives the freeness of $f^*|V|-F =: |V_1|$, and F is the fixed part of $|f^*V|$. □

Corollary 3.1.35. *Let L be a line bundle on a normal projective variety X, and set $\kappa=\kappa(X,L)$. Then there are constants a, $A>0$ such that*

$$a\cdot m^\kappa \leq h^0(X,L^{\otimes m}) \leq A\cdot m^\kappa$$

for all sufficiently large $m\in\mathbb{N}(X,L)$.

Proof. Since Iitaka dimension is a birational invariant by Lemma 1.9, so the assertion is invariant under replacing X with a birational modification. We may assume that $\phi: X\longrightarrow Y$ is the Iitaka fibration associated to L, i.e. we take $X=X_\infty$, $Y=Y_\infty$ and $L=L_\infty$. By resolving singularities, we may suppose also that X is non-singular. It follows from the proof in *Step 3* of the theorem that there is a large positive integer m_0 together with an ample line bundle on Y, such that

$$H^0(X,L^{\otimes m_0}\otimes\phi^*(B^\vee))\neq 0.$$

But this means we have an injection

$$H^0(Y,B^{\otimes l}) = H^0(X,\phi^*B^{\otimes l}) \hookrightarrow H^0(X,L^{\otimes(l\cdot m_0)}),$$

given by the multiplication by a section in $H^0(X, L^{\otimes(l\cdot m_0)} \otimes \phi^*(B^\vee)^{\otimes l})$. So we have

$$h^0(X, L^{\otimes(l\cdot m_0)}) \geq h^0(Y, B^{\otimes l}) = \frac{l^\kappa}{\kappa!}\int_Y c_1(B)^\kappa + O(l^\kappa),$$

where the right is given by asymptotic Riemann-Roch. This gives a lower bound on $h^0(X, L^{\otimes m})$ of the approriate shape for all **sufficiently divisible** $m \in \mathbb{N}(X, L)$. But for all **sufficiently large** $m \in \mathbb{N}(X, L)$, we have to verify the similar argument by replacing l with m. Picking a sufficiently large $n \in \mathbb{N}(X, L)$, satisfying that for any integer r in $\{0, e, 2e, \ldots, m_0 - e\}$, $n + r \in \mathbb{N}(X, L)$ with $e = e(L)$. Since all sufficiently large elements in $\mathbb{N}(X, L)$ are multiple of e, $n + r$ must be a multiple of e. (As $H^0(X, L^{\otimes(l\cdot m_0)}) \neq 0$, so $l \cdot m_0 \in \mathbb{N}(X, L)$ for every l goes big, i.e. a multiple of e. So m_0 is also a multiple of e).

Now, for sufficiently large $m \in \mathbb{N}(X, L)$, as it is a mutliple of e we have the division $m - n = l \cdot m_0 + r$ where l is the quotient, $0 \leq r \leq m_0$. Since $e|m - n$ and $e|m_0$, we have $e|r$ by our choice of $n + r$. Then we pick a section $s \in H^0(X, L^{\otimes(n+r)}) \neq 0$, and we get an injection:

$$H^0(X, L^{\otimes(l\cdot m_0)}) \xhookrightarrow{\cdot s} H^0(X, L^{\otimes m}).$$

Therefore $h^0(X, L^{\otimes m}) \geq h^0(X, L^{\otimes(l\cdot m_0)}) \geq C \cdot l^\kappa$ for some constant $C > 0$. But by our construction, $l = \frac{m-n-r}{m_0} = \frac{m}{m_0} - \frac{n+r}{m_0}$, so

$$C \cdot l^\kappa = C \cdot (\frac{m}{m_0} - \frac{n+r}{m_0})^\kappa \geq a \cdot m^\kappa$$

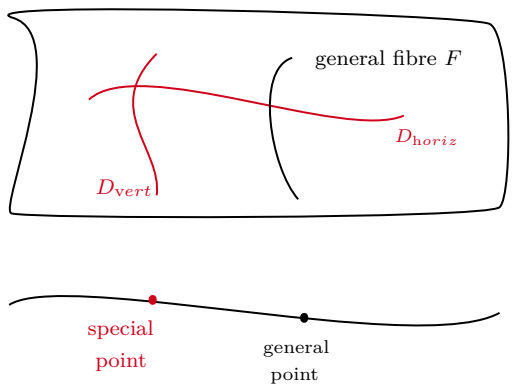
for some constant $a > 0$.

For the reverse inequality, as we're inverstigating the asymptotic behavior when $m \to \infty$, so we will suppose for simplicity that $H^0(X, L) \neq 0$ so that $\mathbb{N}(X, L) = \mathbb{N}$. Then take D to be an effective Cartier divisor on X with $\mathscr{O}_X(D) = L$. As X is normal, we can view the Cartier divisor D to be a Weil divisor, and then decompose it as a sum

$$D = D_{\text{vert}} + D_{\text{horiz}},$$

where D_{vert} consists of those components of D that map via ϕ to a proper subvariety of Y, while every component of D_{horiz} maps onto Y. Thus every fibre F meets D_{horiz} but the general fibre is disjoint from D_{vert}. We assert that

$$H^0(X, \mathscr{O}_X(mD_{\text{vert}})) = H^0(X, \mathscr{O}_X(mD)) \text{ for all } m \gg 0 \tag{3.1}$$



Grant this for the moment. We can find an ample Cartier divisor H on Y such that $D_{\text{vert}} \leq \phi^* H$, and then we have

$$\begin{aligned} h^0(X, \mathscr{O}_X(mD)) =& h^0(X, \mathscr{O}_X(mD_{\text{vert}})) \\ \leq& h^0(X, \phi^* \mathscr{O}_Y(mH)) \\ =& h^0(Y, \mathscr{O}_Y(mH)). \end{aligned}$$

The required upper bound follows from the asymptotic Riemann-Roch as before, i.e. $\frac{m^\kappa}{\kappa!}(H^\kappa) + O(m^\kappa) \leq A \cdot m^\kappa$.

Therefore, it only remains to show (3.1) and gives a construction of H. As for (3.1), we automatically have $H^0(X, \mathscr{O}_X(mD_{\text{vert}})) \subseteq H^0(X, \mathscr{O}_X(mD))$ since

$$\operatorname{div}(f) + mD \geq \operatorname{div}(f) + mD_{\text{vert}} \geq 0.$$

So it is equivalent to show that every section s of $\mathscr{O}_X(mD)$ contains mD_{horiz} as a base-Cartier divisor, since for $(s)_0 = \operatorname{div}(f) + mD \geq 0$ the $\operatorname{div}(f) + mD_{\text{vert}} \geq 0$ is nothing but $((s)_0 - mD) + mD_{\text{vert}} = (s)_0 - mD_{\text{vert}} \geq 0$.

Now we set $D_{\text{vert}} = \sum_{i=1}^{k} a_i \Gamma_i$ where $\phi(\Gamma_i)$ is the proper subvariety Z_i of Y, and let Z be the union of them. Then we take the general fibre F over $Y \backslash Z$, for which $F \cap \operatorname{Supp}(D_{\text{vert}}) = \emptyset$, so $D|_F = D_{\text{horiz}}|_F$. If the condition fails, we can choose two global sections s_1 and s_2 of $\mathscr{O}_X(mD)$ such that $(s_1)_0 = mD$ and $(s_2)_0 =: E$ does not contain mD_{horiz} as base-Cartier divisor. So there is a coefficient of E strictly smaller than mD_{horiz} on some prime Cartier divisor component. Therefore

$$(s_1)_0|_F = mD|_F = mD_{\text{horiz}}|_F \neq E|_F = (s_2)_0|_F,$$

that is $h^0(F, \mathscr{O}_F(mD)) \geq 2$ for a general fibre F, and this contradicts the fact $\kappa(F, L|_F) = 0$ for very general fibre F.

It remains to give the construction of the ample Cartier divisor H on Y. First we fix a very ample Cartier divisor A on Y, which linearly equivalent to a hyperplane setion. Set, again, $Z_i = \phi(\Gamma_i)$ for $D_{\text{vert}} = \sum a_i\Gamma_i$. We define the ideal sheaf of Z_i to be $\mathscr{I}_{Z_i}$, then $\mathscr{I}_{Z_i} \otimes \mathscr{O}_Y(m_iA)$ is globally generated for some $m_i > 0$ and for each i. But this means we have sections $s_i \in H^0(Y, \mathscr{I}_{Z_i}(m_iA))$ with letting $H_i := (s_i)_0$, and $Z_i \subseteq \text{Supp}(H_i)$ as each s_i comes from $\mathscr{I}_{Z_i}$. Then by $\Gamma_i \subseteq \phi^{-1}(Z_i) \subseteq \text{Supp}(\phi^*H_i)$, we can write

$$\phi^*H_i = b_i\Gamma_i + E_i \text{ for } b_i \geq 1 \text{ and } E_i \geq 0.$$

Notice since $H^0(Y, \mathscr{I}_{Z_i}(m_iA)) \hookrightarrow H^0(Y, \mathscr{O}_Y(m_iA))$ where each $H_i \sim_{\text{lin}} m_iA$, so we can choose sufficiently large integers c_i for each i, subject to $c_ib_i \geq a_i$. Then define $H := \sum c_iH_i$ which is an ample Cartier divisor. One can easily check that $\phi^*H \geq D_{\text{vert}}$ by

$$\begin{aligned}\phi^*H - D_{\text{vert}} =& \sum_i c_ib_i\Gamma_i + c_iE_i - \sum_i a_i\Gamma_i \\ =& \sum_i (c_ib_i - a_i)\Gamma_i + \sum_i c_iE_i \\ \geq& 0.\end{aligned}$$

So we complete the proof. □

Remark 3.1.36. It seems to be unknown whether or not

$$\liminf_{m\to\infty} \frac{h^0(X, L^{\otimes m})}{m^\kappa} = \limsup_{m\to\infty} \frac{h^0(X, L^{\otimes m})}{m^\kappa}$$

for $m \in \mathbb{N}(X, L)$. However when $\kappa(X, L) = \dim X$ - i.e. when L is big - then in fact $h^0(X, L^{\otimes m})/m^{\dim X}$ has a limit. See [Laz04b, 11.4.7].

Remark 3.1.37. By above corollary, we often the *Iitaka dimension* $\kappa(X, L)$ of L to be the largest number in $\mathbb{N} \cup \{-\infty\}$ such that

$$\limsup_{m\to\infty} \frac{h^0(X, L^{\otimes m})}{m^{\kappa(X,L)}} > 0,$$

where $m \in \mathbb{N}(X, L)$ (actually we usually omit it since for m not in, h^0 vanished not effects the limit). This definition has arised in Birkar's lecture note of birational geometry, and the quote at the beginning of this chapter is what he said in that class.

Proof. We use the notation κ_{class} and κ_{def} to distinguish between the classical definition and the one defined above.

Case 1: When $h^0(X, L^{\otimes m}) = 0$ for all $m > 0$, then there no integers in $\mathbb{N}$ such that

$$\limsup_{m\to\infty} \frac{h^0(X, L^{\otimes m})}{m^{\kappa(X,L)}} > 0,$$

so the only possibility is to choose $-\infty$.

Case 2: We first prove that $\kappa_{def} \geq \kappa_{class}$. By previous corollary we have

$$\frac{h^0(X, L^{\otimes m})}{m^{\kappa_{class}}} \geq a > 0$$

for all $m \in \mathbb{N}(X, L)$. So clearly we have $\kappa_{def} \geq \kappa_{class}$ by

$$\limsup_{m\to\infty} \frac{h^0(X, L^{\otimes m})}{m^{\kappa_{class}}} > 0.$$

since the κ_{def} is the largest number satisfying this condition.

Now for the reverse direction, we suppose that there is $\kappa > \kappa_{class}$ such that it is the largest number in $\mathbb{N} \cup \{-\infty\}$ making

$$\limsup_{m\to\infty} \frac{h^0(X, L^{\otimes m})}{m^{\kappa}} > 0.$$

Then

$$\frac{h^0(X, L^{\otimes m})}{m^{\kappa}} \leq \frac{A \cdot m^{\kappa_{class}}}{m^{\kappa}} = A \cdot m^{\kappa_{class}-\kappa} \to 0$$

as $m \to \infty$. We get the contradiction. □

3.2 Big Line Bundles and Cartier divisors

3.2A Basic Properties of Big Cartier divisors

Definition 3.2.1. (Big) A line bundle L on a projective variety X is *big* if $\kappa(X, L) = \dim X$. A Cartier divisor D on X is *big* if $\mathscr{O}_X(D)$ is so.

Example 3.2.2. The pullback of an ample line bundle under a generically finite morphism $f : X \longrightarrow Y$ is big.

Proof. As f is generically finite, so f must be dominant since if not, the generic point of Y must have an empty preimage. Since the Iitaka dimension is computed by passing to the normalization of X, so can assume X is normal.

By Corollary 1.35, for ample line bundle A on Y, it is enough to show that

$$\limsup_{m\to\infty} \frac{h^0(X, f^*A^{\otimes m})}{m^n} > 0$$

for m sufficiently divisible, and we write $\dim X = \dim Y = n$. Need to prove that there is a constant $C > 0$ such that

$$h^0(X, f^*A^{\otimes m}) \geq C \cdot m^n.$$

As f is dominant, $\mathscr{O}_Y \hookrightarrow f_*\mathscr{O}_X$ is injective, so is for

$$A^{\otimes m} \hookrightarrow A^{\otimes m} \otimes f_*\mathscr{O}_X = f_*(f^*A^{\otimes m}).$$

Then

$$h^0(Y, A^{\otimes m}) \leq h^0(Y, f_*(f^*A^{\otimes m})) = h^0(X, f^*A^{\otimes m}).$$

But as A is ample, so by asymptotic Riemann-Roch,

$$h^0(Y, A^{\otimes m}) = \frac{m^n}{n!} \int_Y c_1(B)^n + O(m^n),$$

so $h^0(X, f^*A^{\otimes m}) \geq C \cdot m^n$ for some constant $C > 0$, so f^*A is big. □

Example 3.2.3. When X is normal, then L is big if and only if for the rational mapping $\phi_m : X \dashrightarrow \mathbb{P}H^0(X, L^{\otimes m})$, X is birational to its image $\phi_m(X) =: Y_m$ for some $m > 0$.

Proof. The sufficient direction is trivial. We prove the necessary direction. By Iitaka fibration theorem, for all sufficiently divisible $m \gg 0$, there is an algebraic fibre space

$$\phi_\infty : X_\infty \longrightarrow Y_\infty$$

birationally equivalent to $\phi_m : X \dashrightarrow Y_m$. Moreover the finite field extension part of the $\mathbb{C}(Y_\infty) \subseteq \mathbb{C}(X_\infty)$ is trivial, so it is with respect to ϕ. Hence $X \sim_{bir} Y_m$ for some $m > 0$. □

In Example 2.2 we compute the Iitaka dimension by passing to the normal case, but this is unnecessary if we extend our method in the proof for general variety, and general line bundle under this big setting. The following Lemma is usually used as the definition of a *big line bundle*, see [KM98, 2.59].

Lemma 3.2.4. *Assume that X is a projective variety of dimension n, which is not necessarily normal. A line bundle L on X is big if and only if there is a constant $C > 0$ such that*

$$h^0(X, L^{\otimes m}) \geq C \cdot m^n$$

for all sufficiently divisible m.

Proof. Let $\nu : X' \longrightarrow X$ be the normalization of X, and put $L' = \nu^* L$. We will show that L' has maximal Iitaka dimension on X' if the lower bound of above holds. Using the projection formula, one has

$$h^0(X', L'^{\otimes m}) = h^0(X, \nu_* L'^{\otimes m}) = h^0(X, L^{\otimes m} \otimes \nu_* \mathscr{O}_{X'})$$

for all m. On the other hand, as ν is dominant there is an exact sequence

$$0 \longrightarrow \mathscr{O}_X \longrightarrow \nu_* \mathscr{O}_{X'} \longrightarrow \eta \longrightarrow 0$$

of sheaves on X, where η is supported on a subscheme of dimension $\leq n-1$. Therefore

$$\begin{aligned} h^0(X, L^{\otimes m}) \leq & h^0(X, L^{\otimes m} \otimes \nu_* \mathscr{O}_{X'}) \\ \leq & h^0(X, L^{\otimes m}) + h^0(X, \eta \otimes L^{\otimes m}). \end{aligned}$$

But $h^0(X, \eta \otimes L^{\otimes m}) = O(m^{n-1})$ since η is supported on a proper subscheme (Example 1.2.24). So the $h^0(X, L^{\otimes m})$ grows like m^n if and only if $h^0(X', L'^{\otimes m})$ does. But again by Example 1.2.24, both they are both equal to $O(m^n)$. Therefore L is big on X if and only if $\kappa(X, L) \underset{\text{def}}{=} \kappa(X', L') = n$, if and only if n is the largest number such that

$$\limsup_{m \to \infty} \frac{h^0(X, L^{\otimes m})}{m^n} = \limsup_{m \to \infty} \frac{h^0(X', L'^{\otimes m})}{m^n} > 0,$$

if and only if there is a constant $C > 0$ such that

$$h^0(X, L^{\otimes m}) \geq C \cdot m^n,$$

where each holds for sufficiently divisible m. □

Example 3.2.5. (Blow-up of projective space) The blow-up of projective space at a finite set of points provides some interesting examples. Let $S \subseteq \mathbb{P}^n$ be a finite set, equipped with a reduced scheme structure. And let

$$\mu : X = \mathrm{Bl}_S(\mathbb{P}^n) \longrightarrow \mathbb{P}^n$$

be the blowing up of $\mathbb{P}^n$ along S. Denote $E := E_S$ and $H := H_S$ respectively the exceptional Cartier divisor and the pullback of a hyperplane class, and put $D := dH - rE$. One has

$$\begin{aligned}\mu_*\mathscr{O}_X(mD) &= \mu_*\mathscr{O}_X(mdH - mrE)\\ &= \mu_*(\mathscr{O}_X(mdH) \otimes \mathscr{I}_E^{\otimes mr})\\ &= \mathscr{O}_{\mathbb{P}^n}(md) \otimes \mathscr{I}^{\otimes mr},\end{aligned}$$

where $\mathscr{I}$ is the ideal sheaf of S in $\mathbb{P}^n$, and therefore $H^0(X, \mathscr{O}_X(mD)) = H^0(\mathbb{P}^n, \mu_*\mathscr{O}_X(mD))$ is identified with the space of hypersurface of degree md that vanish to order $\geq mr$ at each point of S.

Assume $\#S = s$. Then for D is big which we need

$$h^0(X, mD) \geq C \cdot m^n$$

for some constant $C > 0$, for sufficiently divisible m. Instantly we have

$$\begin{aligned}h^0(\mathbb{P}^n, mdH) &= \binom{n+md}{md}\\ &= \frac{d^n}{n!}m^n + O(m^{n-1}).\end{aligned}$$

But as global sections in $H^0(X, \mathscr{O}_X(mD))$ has order of vanishing $\geq mr$ at each point of S, that is under the affine coordinates we locally chosen, their polynomial forms vanishing in all degrees of strictly less than mr in Taylor expansions. So can calculate it has the number

$$\binom{mr-1+n}{mr-1} = \frac{r^n}{n!}m^n + O(m^{n-1}),$$

which is calculated at each point of S. Therefore we have

$$\begin{aligned}h^0(X, \mathscr{O}_X(mD)) &= h^0(\mathbb{P}^n, \mathscr{O}_{\mathbb{P}^n}(mdH) \otimes \mathscr{I}^{\otimes mr})\\ &\geq \frac{d^n - s \cdot r^n}{n!}m^n + O(m^{n-1})\end{aligned}$$

implies that D is big provided that $d^n - s \cdot r^n > 0$. The our desired constant C.

The first essential result is due to Kodaira:

Proposition 3.2.6. (Kodaira's lemma) *Let D be a big Cartier divisor and F be an arbitrary effective Cartier divisor on X. Then*

$$H^0(X, \mathscr{O}_X(mD - F)) \neq 0$$

for all sufficiently divisible m.

Proof. Suppose that $\dim X = n$, and consider the exact sequence

$$0 \longrightarrow \mathscr{O}_X(mD - F) \longrightarrow \mathscr{O}_X(mD) \longrightarrow \mathscr{O}_F(mD) \longrightarrow 0.$$

Since D is a big Cartier divisor, there is a constant $C > 0$ such that $h^0(X, mD) \geq C \cdot m^n$ for sufficiently divisible m. On the other hand, F is a codimension 1 subscheme, so $h^0(F, mD) = O(m^{n-1})$ (Example 1.2.24). Therefore

$$h^0(X, \mathscr{O}_X(mD)) > h^0(F, \mathscr{O}_F(mD))$$

for sufficiently divisible m, this ensures that $h^0(X, mD - F) \neq 0$. □

Kodaira's lemma has several important consequences. First, it leads to a useful characterization of bigness:

Corollary 3.2.7. (Characterization of big Cartier divisors) *Let D be a Cartier divisor on a projective variety X. Then the following are equivalent:*

(i) D is big.

(ii) For any ample integral Cartier divisor A on X, there exists a positive integer $m > 0$ and an effective Cartier divisor N on X such that $mD \sim_{\text{lin}} A + N$.

*(iii) Same as in (ii) for **some** ample Cartier divisor A.*

(iv) There exists an ample Cartier divisor A, a positive integer $m > 0$, and an effective Cartier divisor N such that $mD \equiv_{num} A + N$.

Proof. "(i) ⇒ (ii)": Assuming that D is big, take $r \gg 0$ so that $rA \sim_{lin} H_r$ and $(r+1)A \sim_{lin} H_{r+1}$ are both effective. The reason we can do this is by $\mathscr{O}_X \otimes \mathscr{O}_X(rA)$ and $\mathscr{O}_X(A) \otimes \mathscr{O}_X(rA)$ are globally generated for suitable $r \gg 0$. Apply Kodaira's lemma with $F = H_{r+1}$ to find a positive integer m and an effective Cartier divisor N' with

$$mD \sim_{lin} H_{r+1} + N' \sim_{lin} A + (H_r + N').$$

Taking $N = H_r + N'$ gives (ii).

"(ii) ⇒ (iii) ⇒ (iv)" is trivial.

" (iv) $\Rightarrow$ (i)": If $mD \equiv_{num} A + N$, then $mD - N$ is numerically equivalent to an ample Cartier divisor, so it is also ample by Nakai-Moishezon. So after possibly passing to an even larger multiple we can assume $mD - N' \sim_{lin} H$ where H is very ample and N' is effective. But then

$$\kappa(X, D) \geq \kappa(X, H) = \dim X,$$

so D is big. □

From statement (iv) we can directly obtain the following conclusion:

Corollary 3.2.8. (Numerical nature of bigness) *The bigness of a Cartier divisor D depends only on its numerical equivalence class.*

Example 3.2.9. (Cohomological characterization of big Cartier divisors) Let D be a Cartier divisor on projective variety X. Then D is big if and only if the following holds:

> For any coherent sheaf $\mathscr{F}$ on X, there exists a positive integer $m = m(\mathscr{F})$ such that $\mathscr{F} \otimes \mathscr{O}_X(mD)$ is generically globally generated.

Proof. "$\Rightarrow$": D is big so for any ample Cartier divisor A, we have $mD \sim_{lin} A + N$ for some $m > 0$ and some Cartier divisor $N \geq 0$, i.e. $\mathscr{O}_X(mD) \cong \mathscr{O}_X(A) \otimes \mathscr{O}_X(N)$. So for any coherent sheaf $\mathscr{F}$, we have that $\mathscr{F} \otimes \mathscr{O}_X(mD) \cong \mathscr{F} \otimes \mathscr{O}_X(N') \otimes \mathscr{O}_X(m'A)$ is globally generated by after passing to an even larger multiple of A, for some $m \gg 0$.

"$\Leftarrow$": Trivial. Since D is an ample Cartier divisor by definition of ampleness. So D is big by choosing $m = 1$ and $N = 0$ in (iii). □

Concerning the exponent of a big Cartier divisor, one finds:

Corollary 3.2.10. (Exponent of big Cartier divisors) *If D is a big Cartier divisor on a projective variety X, then $e(D) = 1$, i.e. all sufficiently large multiple of D is effective (up to linear equivalence).*

Proof. Given such big D, and we can choose a very ample Cartier divisor H on X such that $H - D \sim_{lin} H_1$ is effective (by choosing an ample Cartier divisor and relate to globally generated). So by Corollary 2.7 there is an integer $m \in \mathbb{N}(X, D)$ such that $mD \sim_{lin} H + N$ for some effective Cartier divisor N. Then

$$(m-1)D \sim_{lin} (H - D) + N \sim_{lin} H_1 + N$$

is also effective, i.e. has global sections. So the two consecutive integers m and $m - 1$ both lie in the semi-group $\mathbb{N}(X, D)$, hence $e(D) = 1$. □

Finally, due to Kodaira's lemma we can get that the restriction of a big line bundle to a "sufficiently general" subvariety remains big:

Corollary 3.2.11. (Restrictions of big Cartier divisors) *Let L be a big line bundle on a projective variety X. There is a proper Zariski-closed subset $V \subseteq X$ having the property that, if $Y \subseteq X$ is any subvariety of X not contained in V then the restriction $L_Y := L|_Y$ is a big line bundle on Y.*

In particular, if H is a general member of a very ample linear series, then L_H is big.

Proof. Say $L = \mathscr{O}_X(D)$, and using Corollary 2.7 we write $mD \sim_{lin} H + N$ where N is effective and H is very ample (by passing to large multiple). If $Y \nsubseteq \operatorname{Supp} N$, then the restriction mD_Y of mD to Y is again the sum of a very ample and an effective, hence D_Y is big by Corollary 2.7 again. □

Example 3.2.12. (Restriction fails to preserve bigness) Let $X = \mathrm{Bl}_p\mathbb{P}^2$ be the blowing up at a point $p \in \mathbb{P}^2$. And let E and H denote respectively the exceptional Cartier divisor and the pullback of a line. Then $\mathscr{O}_X(H+E)$ is big, but $\mathscr{O}_X(H+E)|_E \cong \mathscr{O}_{\mathbb{P}^2}(-1)$ which is not big.

Note that an integral Cartier divisor A is big if and only if every (or equivalently some) positive multiple of A is big. This leads to a natural notion of bigness for a $\mathbb{Q}$-Cartier divisor.

Definition 3.2.13. (Big $\mathbb{Q}$-Cartier divisors) A $\mathbb{Q}$-Cartier divisor D is *big* if there is a positive integer $m > 0$ such that mD is integral and big.

As before, by passing to integral Cartier divisors, bigness is a numerical property of $\mathbb{Q}$-Cartier divisors. We now give a basic result of Yum-Tong Siu giving a numerical condition, for a difference of nef-Cartier divisors to be big.

Theorem 3.2.14. (Numerical criterion for bigness) *Let X be a projective variety of dimension n, and let D and E be nef $\mathbb{Q}$-Cartier divisors on X. Assume that*

$$(D^n) > n \cdot (D^{n-1} \cdot E).$$

Then $D - E$ is big.

Proof. We fix an ample Cartier divisor H on X, and notice that the polynomial

$$f(t) = (D + tH)^n - n \cdot \big((D + tH)^{n-1} \cdot (E + tH)\big)$$

is continuous, so it remains positive under a very small perturbation. Thus we can replace D and E respectively by $D + \varepsilon H$ and $E + \varepsilon H$ for $0 < \varepsilon \ll 1$. Therefore we may suppose that both D and E are ample, by Corollary 1.4.7. Then since condition is preserved under multiplying D and E by a large integer, we can assume also that D and E are integral and very ample.

Choose a sequence $E_1, E_2, E_3, \ldots, E_m, \cdots \in |E|$ of general members Cartier divisors linearly equivalent to E, and fix an integer $m \geq 1$. Then

$$\mathscr{O}_X\big(m(D-E)\big) \cong \mathscr{O}_X(mD - \sum_{i=1}^{m} E_i),$$

so $H^0(X, m(D-E)) \cong H^0(X, mD - \sum_{i=1}^{m} E_i)$ is identified with the group of sections of $\mathscr{O}_X(mD)$ vanishing on each of codimension 1 subschemes $E_1, \ldots, E_m$. This in turn may be analyzed via the exact sequence

$$0 \longrightarrow H^0\big(X, \mathscr{O}_X(mD - \sum_{i=1}^{m} E_i)\big) \longrightarrow H^0\big(X, \mathscr{O}_X(mD)\big) \longrightarrow \bigoplus_{i=1}^{m} H^0\big(E_i, \mathscr{O}_{E_i}(mD)\big)$$
$$s \mapsto (s|_{E_1}, s|_{E_2}, \ldots, s|_{E_m}).$$

For large m, the dimensions $h^0(E_i, \mathscr{O}_{E_i}(mD))$ are independent of i, and then the asymptotic Riemann-Roch on X and each E_i yields

$$\begin{aligned} h^0\Big(X, \mathscr{O}_X\big(m(D-E)\big)\Big) &\geq h^0(X, mD) - \sum_{i=1}^{m} h^0\big(E_i, \mathscr{O}_{E_i}(mD)\big) \\ &= \frac{(D^n)}{n!} m^n - \sum_{i=1}^{m} \frac{(D^{n-1} \cdot E)}{(n-1)!} m^{n-1} + O(m^{n-1}) \\ &= \frac{(D^n)}{n!} m^n - n\frac{(D^{n-1} \cdot E)}{n!} m^n + O(m^{n-1}). \end{aligned}$$

In particular, hypothesis says that $h^0\big(X, m(D-E)\big)$ grows like a positive multiple of m^n, so $m(D-E)$ is big, hence also for $D - E$. □

Now we can obtain an important characterization of bigness for nef Cartier divisor using Siu's result.

Theorem 3.2.15. (Bigness of nef Cartier divisors) *Let D be a nef Cartier divisor on a projective variety of dimension n. Then D is big if and only if its top self-intersection is strictly positive, i.e. $(D^n) > 0$.*

Proof. "⇐": Suppose that $(D^n) > 0$. Then the hypothesis of Theorem 2.14 is satisfied with $E = 0$. Hence D is big.

"⇒ ": Suppose that D is nef and big. Then $mD \sim_{lin} H + N$ for some very ample H and effective N, and suitable $m > 0$. But as D is nef, we have

$$(D^{n-1} \cdot N) \geq 0$$

by Kleiman's theorem, and therefore

$$\begin{aligned} m \cdot (D^n) &= \big((H+N) \cdot D^{n-1}\big) \\ &\geq (H \cdot D^{n-1}). \end{aligned}$$

But by suitable choice of very ample Cartier divisor H, and by Corollary 2.11 we may assume moreover that the restriction $D|_H$ is a big and a nef Cartier divisor on H. So we can use induction on the dimension:

- $n = 1$: for D is a big Cartier divisor on a curve, i.e. $h^0(X, mD) \geq C \cdot m$ for some $C > 0$. Then by Riemann-Roch and Serre duality we have

  $$\begin{aligned} h^0(X, mD) &= h^0(X, mD) - h^0(X, K_X - mD) \\ &= \deg(D) \cdot m + 1 - g, \end{aligned}$$

  for m goes large. So we must have $\deg(D) > 0$.

- Suppose the assertion holds for $n-1$, i.e. $(D^{n-1} \cdot H) > 0$.

Now for n the inequality $(D^n) > 0$ follows immediately. □

Remark 3.2.16. (Alternative approach to Theorem 2.15) See Theorem 1.4.42, by Fujita's vanishing theorem we can deduce a beautiful asymptotic result for nef Cartier divisor D:

$$h^0(X, mD) = \frac{(D^n)}{n!} m^n + O(m^{n-1}).$$

So clearly it is equivalent to $(D^n) > 0$ by Lemma 2.3.

Example 3.2.17. (Big line bundles with negative numerics) We should keep in mind that if we drop the nefness condition, the intersection number associated with a big line bundle can be very negative.

For example, let $X = \mathrm{Bl}_p \mathbb{P}^2$ with E and A respectively be exceptional Cartier divisor and a very ample Cartier divisor on X. Then

$$D_l := A + lE$$

is big for all $l \geq 0$, but $(D_l \cdot D_l) \ll 0$ for very large l.

Example 3.2.18. (Alternative characterization of big and nef Cartier divisors) Let D be a Cartier divisor on a projective variety X. Then D is nef and big if and only if there is an effective Cartier divisor N such that $D - \frac{1}{k}N$ is ample for all $k \gg 0$.

Proof. "$\Leftarrow$": Bigness is by Corollary 2.7, and nefness is follows from Nakai-Moishezon.

"$\Rightarrow$": We assume that D is big and nef. Then there is an $m \in \mathbb{Z}_{>0}$, an effective Cartier divisor N, and an ample Cartier divisor A with $mD \equiv_{num} A + N$. Thus for $k > m$,

$$kD \equiv_{num} \big((k-m)D + A\big) + N,$$

and the term in parentheses, being the sum of a nef and an ample Cartier divisor, is ample. □

3.2B Pseudo-effective and Big Cones

In this subsection we will use the language of numerical geometry to rephrase the corollary of Kodaira's lemma, that is we will establish the big cone. As before, X is projective variety of dimension n.

Definition 3.2.19. (Big $\mathbb{R}$-Cartier divisors) An $\mathbb{R}$-Cartier divisor $D \in \mathrm{Div}_{\mathbb{R}}(X)$ is *big* if it can be written in the form

$$D = \sum a_i \cdot D_i$$

where each D_i is a big integral Cartier divisor and a_i is a positive real number.

It is immediate that this definition is stable under addition and positive scalar multiplication.

Proposition 3.2.20. (Formal properties of big $\mathbb{R}$-Cartier divisors) *Let D and D' be $\mathbb{R}$-Cartier divisors on X. Then*

(1) If $D \equiv_{num} D'$, then D is big if and only if D' is.

(2) D is big if and only if $D \equiv_{num} A + N$ where A is an ample and N is an effective $\mathbb{R}$-Cartier divisor.

Proof. (1) is easy to get as we can pass to integral Cartier divisors.

(2) If D is big then immediately know that each $\mathbb{R}$-linear components have that form, so is D. Conversely, we **reduce to** showing that if B and N are integral Cartier divisors, with B is big and N effective, then $B + sN$ is big for any real number $s > 0$.

If $s \in \mathbb{Q}$ this follows again from Corollary 2.7. In general, we choose positive rational numbers $s_1 < s < s_2$ and $t \in [0,1]$ such that $s = ts_1 + (1-t)s_2$. Then

$$B + sN = t(B + s_1N) + (1-t)(B + s_2N)$$

exhibits $B + sN$ as a positive linear combination of big $\mathbb{Q}$-Cartier divisors, so is big. Now as A is big by ampleness, all things are clear now. □

Example 3.2.21. (Big and nef $\mathbb{R}$-Cartier divisors) Let D be a nef and big $\mathbb{R}$-Cartier divisor. Then there is an effective $\mathbb{R}$-Cartier divisor N such that $D - \frac{1}{k}N$ is an ample $\mathbb{R}$-Cartier divisor for every sufficiently large $k \in \mathbb{N}$.

Proof. By Example 2.18 we see each $\mathbb{R}$-linear components has such a form which must be ample, so for $D - \frac{1}{k}N$ as it can be written as their $\mathbb{R}$-linear combinations. □

Corollary 3.2.22. (Openness of big Cartier divisors) *Let $D \in \mathrm{Div}_{\mathbb{R}}(X)$ be a big $\mathbb{R}$-Cartier divisor, and let $E_1, \ldots, E_t \in \mathrm{Div}_{\mathbb{R}}(X)$ be arbitrary $\mathbb{R}$-Cartier divisors. Then*

$$D + \varepsilon_1 E_1 + \cdots + \varepsilon_t E_t$$

remains big for all sufficiently small real numbers $0 \leq |\varepsilon_i| \ll 1$.

Proof. It follows from the statement (2) of above proposition and the open nature of ampleness, while after a small perturbation $D + \sum \varepsilon_i E_i$ will still be numerically equivalent to the sum of an ample Cartier divisor and an $\mathbb{R}$-effective Cartier divisor. □

Note that statement (1) of Proposition 2.20 allows us to talk about the big $\mathbb{R}$-class in $N^1(X)_{\mathbb{R}}$.

Definition 3.2.23. (Big and pseudo-effective cones) The *big cone*

$$\mathrm{Big}(X) \subseteq N^1(X)_{\mathbb{R}}$$

is the convex cone of all big $\mathbb{R}$-Cartier divisor classes on X.

The *pseudo-effective cone*

$$\overline{\mathrm{Eff}}(X) \subseteq N^1(X)_{\mathbb{R}}$$

is the closure of the convex cone spanned by the classes of all effective $\mathbb{R}$-Cartier divisors.

A Cartier divisor $D \in \mathrm{Div}_{\mathbb{R}}(X)$ is *pseudo-effective* if its class lies in the $\overline{\mathrm{Eff}}(X)$.

Directly we have:

Theorem 3.2.24. *The big cone is the interior of the pseudo-effective cone and the pseudo-effective cone is the closure of the big cone:*

$$\mathrm{Big}(X) = \mathrm{int}(\overline{\mathrm{Eff}}(X)) \ , \ \overline{\mathrm{Eff}}(X) = \overline{\mathrm{Big}(X)}.$$

Proof. The pseudo-effective cone is closed by definition, and the big cone is open by Corollary 2.22, so $\mathrm{Big}(X) \subseteq \overline{\mathrm{Eff}}(X)$ thanks to Proposition 2.20(2) and the ample $A \equiv_{num} \frac{1}{m}E \in \mathrm{Eff}(X)$ for some $m > 0$. Thus also $\overline{\mathrm{Big}(X)} \subseteq \overline{\mathrm{Eff}}(X)$.

Now it remains to establish the inclusions

$$\overline{\mathrm{Eff}}(X) \subseteq \overline{\mathrm{Big}(X)} \ , \ \mathrm{int}(\overline{\mathrm{Eff}}(X)) \subseteq \mathrm{Big}(X).$$

We focus on the first of these. Given $\eta \in \overline{\mathrm{Eff}}(X)$, we can write η as the limit $\eta = \lim_{k\to\infty} \eta_k$ of the numerical classes of effective Cartier divisors. Fixing an ample class $\alpha \in N^1(X)_{\mathbb{R}}$ we have

$$\eta = \lim_{k\to\infty} (\eta_k + \frac{1}{k}\alpha),$$

where each of the classes $\eta_k + \frac{1}{k}\alpha$ is big thanks to Corollary 2.20(2), so η is a limit of big classes, finishing the first inclusion. And the second then goes without saying. □

Remark 3.2.25. (Pseudo-effective Cartier divisors on a surface) On a smooth projective surface X, where we can identify curves with Cartier divisors, the pseudo-effective cone coincides with the Kleiman-Mori cone $\overline{\mathrm{NE}}(X) \subseteq N_1(X)_{\mathbb{R}}$. So in particular, it follows from Proposition 1.4.27 that a divisor D on a smooth projective surface X is pseudo-effective if and only if it lies in its dual $\mathrm{Nef}(X)$, i.e. $(D \cdot H) \geq 0$ for every nef divisor H on X.

3.2C Volume of a Big Cartier divisor

There is an interesting invariant of a big Cartier divisor D that measures the asymptotic growth of the linear series $|mD|$ when $m \to \infty$.

Definition 3.2.26. (Volume of a line bundle) Let X be a projective variety of dimension n, and let L be a line bundle on X. The *volume* of L is defined to be the non-negative real number

$$\mathrm{vol}(L) = \mathrm{vol}_X(L) := \limsup_{m\to\infty} \frac{h^0(X, L^{\otimes m})}{m^n/n!}.$$

The volume $\mathrm{vol}(D) = \mathrm{vol}_X(D)$ of a Cartier divisor D is defined by passing to $\mathscr{O}_X(D)$.

Note that $\mathrm{vol}(L) > 0$ if and only if L is big. Moreover if L is **nef**, then by Fujita's vanishing theorem and asymptotic Riemann-Roch, we have that

$$h^0(X, L^{\otimes m}) = \frac{\int_X c_1(L)^n}{n!} \cdot m^n + O(m^{n-1}),$$

so we have $\mathrm{vol}(L) = \int_X c_1(L)^n$.

Example 3.2.27. (Volume on a blow-up of projective space) Again we blow-up $\mathbb{P}^n$ at $\#S = s$ points, and put $D = dH - rE$, each of them corresponding to the pullback of a hyperplane and exceptional Cartier divisor. Then

$$h^0(X, mD) \geq \frac{d^n - s \cdot r^n}{n!} \cdot m^n + O(m^{n-1})$$

by our preceding computation, so $\mathrm{vol}(D) \geq d^n - s \cdot r^n$.

Proposition 3.2.28. (Properties of the volume) *Let D be a big Cartier divisor on a projective variety X of dimension n. Then*

(i) For a fixed natural number $a > 0$,

$$\mathrm{vol}(aD) = a^n \,\mathrm{vol}(D).$$

(ii) For any Cartier divisor N on X, and any $\varepsilon > 0$. Then there exists an integer $p_0 = p_0(N, \varepsilon)$ such that

$$\frac{1}{p^n} \cdot |\,\mathrm{vol}(pD - N) - \mathrm{vol}(pD)| < \varepsilon$$

for every $p > p_0$.

We start with two lemmas:

Lemma 3.2.29. *Let L be a big line bundle and A a very ample Cartier divisor on X. If E, $E' \in |A|$ are very general Cartier divisors, then*

$$\mathrm{vol}_E(L|_E) = \mathrm{vol}_{E'}(L|_{E'}).$$

Proof. First we need to consider the parameter space $\mathscr{D} \subseteq X \times |A|$, equipped with two projections p_1 and p_2, as follows.

For vector space $H^0(X, \mathscr{O}_X(A))$, we choose a basis $s_0, \ldots, s_n$ which correspond to homogeneous coordinates $y_0, \ldots, y_n$ of $|A| = \mathbb{P}H^0(X, \mathscr{O}_X(A)) := \mathbb{P}$, form a basis of $H^0(\mathbb{P}, \mathscr{O}_{\mathbb{P}}(1))$. Then we have a canonical global section $s \in H^0(X \times \mathbb{P}, \mathscr{L})$, where $\mathscr{L} = \mathscr{O}_X(A) \boxtimes \mathscr{O}_{\mathbb{P}}(1)$, and $s = \sum_{i=0}^{n} s_i \boxtimes y_i$. We define $\mathscr{D}$ to be the Cartier divisor of zeros

$(s)_0$ in $X \times \mathbb{P}$, which naturally is an effective Cartier divisor on $X \times \mathbb{P}$. The restriction of $\mathscr{D}$ on closed point $t = [c_0 : c_1 : \cdots : c_n] \in \mathbb{P}$, the section s on $X \times \{t\} \cong X$ will be restricted to be

$$s|_t = \sum_{i=0}^{n} c_i s_i \in H^0(X, \mathscr{O}_X(A)),$$

its Cartier divisor of zeros corresponds to an effective Cartier divisor $E_t \in |A|$. So by our construction, for each closed point $t \in \mathbb{P}$, $\mathscr{D}_t$ will be some effective Cartier divisor in $|A|$.

Next we will show that $\mathscr{D} \longrightarrow |A|$ is flat. First we introduce a lemma on the local criterion of flatness:

> On a noetherian scheme (Y/S) and an exact sequence of coherent sheaves
>
> $$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{G} \longrightarrow \mathscr{H} \longrightarrow 0$$
>
> with $\mathscr{G}$ is flat over S. Then $\mathscr{H}$ is flat over S if and only if for each closed point $t \in S$, the restriction map $\mathscr{F}_t \longrightarrow \mathscr{G}_t$ is injective.

This is easily deduced from the algebraic version of local criterion of flatness, see [SP25, Tag 00MK], for which we have

$$0 = \mathrm{Tor}^1(k, \mathscr{G}) \longrightarrow \mathrm{Tor}^1(k, \mathscr{H}) \longrightarrow \mathscr{F}_t \longrightarrow \mathscr{G}_t \longrightarrow \mathscr{H}_t \longrightarrow 0.$$

Now back to our analysis, we have an exact sequence

$$0 \longrightarrow \mathscr{I}_{\mathscr{D}} \cong \mathscr{L}^{-1} \longrightarrow \mathscr{O}_{X \times \mathbb{P}} \longrightarrow \mathscr{O}_{\mathscr{D}} \longrightarrow 0,$$

with $\mathscr{O}_{X \times \mathbb{P}}$ is flat over $\mathscr{O}_{\mathbb{P}}$, by flatness is stable under base change and $|A| \to \operatorname{Spec} \mathbb{C}$ is. But at each closed point $t \in \mathbb{P}$, we have

$$\mathscr{L}_t^{-1} \longrightarrow \mathscr{O}_{X \times \mathbb{P}, X \times \{t\}}$$

is nothing but

$$\mathscr{O}_X(-E_t) \xrightarrow{\cdot s|_t} \mathscr{O}_X$$

which clearly is injective. So $\mathscr{O}_{\mathscr{D}}$ is flat over $\mathscr{O}_{\mathbb{P}}$, i.e. the morphism $\mathscr{D} \to |A|$ is flat.

Finally, we use the upper semi-continuity theorem [Har77, III, 12.8], with that $\mathscr{F} = (p_1^* L^{\otimes m})|_{\mathscr{D}}$ is flat over $|A|$ for each $m > 0$ (since it is locally free so flat over $\mathscr{D}$, then composed with the flatness $\mathscr{D} \to |A|$). Then we have the function

$$t \mapsto h^0(X, \mathscr{F}_t) = h^0(E, L^{\otimes m}|_{E_t})$$

is upper semi-continuous, i.e. the $\{E \in |A| \mid h^0(E, L^{\otimes m}|_E) < k\}$ is an open subset of $|A|$. We take $E^{\min} \in |A|$ such that

$$h^0(E^{\min}, L^{\otimes m}|_{E^{\min}}) = \min_{E \in |A|} h^0(E, L^{\otimes m}|_E).$$

Then there exists a Zariski-open dense subset $U_m \subseteq |A|$ such that

$$h^0(E, L^{\otimes m}|_E) = h^0(E', L^{\otimes m}|_{E'})$$

for every E, $E' \in U_m$, for arbitrary fixed $m > 0$. □

Lemma 3.2.30. *Let D be any Cartier divisor on X, and $a \in \mathbb{N}$ a fixed positive integer. Then*

$$\limsup_m \frac{h^0(X, mD)}{m^n/n!} = \limsup_k \frac{h^0(X, akD)}{(ak)^n/n!}.$$

Proof. If $\kappa(X, D) < n = \dim X$ then both sides vanish, so we suppose that D is big, that is $e(D) = 1$ while we don't label the semi-group under limits.

Given $r \in \mathbb{N}$, put

$$v_r := \limsup_k \frac{h^0(X, (ak + r)D)}{(ak + r)^n/n!}$$

so that we reduce to show $\mathrm{vol}(D) = v_0$. For any fixed natural number $r_0 \in \mathbb{N}$, the Euclidean division gives that there is a unique $k \in \mathbb{Z}$ and unique remainder $r \in \{r_0 + 1, \ldots, r_0 + a\}$ such that

$$m = ak + r,$$

while $m \to \infty$ equivalent to $k \to \infty$. So our numerical sequence $\{x_m\}_{m \geq 1}$ is partitioned into subsequences $\{x_{ak+r}\}_{k \geq 1}$ for $r \in \{r_0 + 1, \ldots, r_0 + a\}$, so

$$\limsup_m x_m = \max_{r \in \{r_0+1, \ldots, r_0+a\}} \{\limsup_k x_{ak+r}\},$$

that is $\mathrm{vol}(D) = \max\{v_{r_0+1} \ldots, v_{r_0+a}\}$. So it is enough to produce any $r_0 > 0$ for which we can show that

$$v_0 = v_r \text{ for all } r \in \{r_0 + 1, \ldots, r_0 + a\},$$

where the right term must arise the $\mathrm{vol}(D)$.

To this end, fix $r_0 \gg 0$ such that $h^0(X, rD) \neq 0$ for any $r \geq r_0$, by our assumption D is big. Then choose $q \gg 0$ such that $qa - (r_0 + a) \geq r_0$. For $r \in \{r_0 + 1, \ldots, r_0 + a\}$ we can then fix Cartier divisors

$$D_r \in |rD| \ , \ D'_r \in |(qa - r)D| \text{ for which } qa - r \geq r_0.$$

These in turn give rise to inclusions

$$\mathscr{O}_X(kaD) \overset{\cdot D_r}{\hookrightarrow} \mathscr{O}_X((ka+r)D) \overset{\cdot D'_r}{\hookrightarrow} \mathscr{O}_X((k+q)aD)$$

for every $k \in \mathbb{N}$ and $r \in \{r_0+1, \ldots, r_0+a\}$. Hence

$$h^0(X, kaD) \leq h^0(X, (ka+r)D) \leq h^0(X, (k+q)aD).$$

But

$$\limsup_k \frac{h^0(X,(ka+r)D)}{(ka)^n/n!} = \limsup_k \frac{h^0(X,(ka+r)D)}{(ka+r)^n/n!} = v_r$$

by multiplying on the left by $\frac{(ka)^n}{(ka+r)^n}$. And similarly

$$\limsup_k \frac{h^0(X,(k+q)aD)}{(ka)^n/n!} = \limsup_k \frac{h^0(X,(k+q)aD)}{((k+q)a)^n/n!}$$

by multiplying on the left $\frac{(ka)^n}{((k+q)a)^n}$, while the right hand side is equal to v_0. Moreover we have

$$\begin{aligned} v_0 = \limsup_k \frac{h^0(X,kaD)}{(ka)^n/n!} &\leq \limsup_k \frac{h^0(X,(ka+r)D)}{(ka)^n/n!} = v_r \\ &\leq \limsup_k \frac{h^0(X,(k+q)aD)}{(ka)^n/n!} = v_0 \end{aligned}$$

which implies $v_0 = v_r$, for $r \in \{r_0+1, \ldots, r_0+a\}$, as we desired. □

Now we turn to prove the proposition 2.28:

Proof. The first statement follows immediately from Lemma 2.30, which yields

$$\begin{aligned} \operatorname{vol}(aD) &= \limsup_k \frac{h^0(X,akD)}{k^n/n!} \\ &= a^n \limsup_k \frac{h^0(X,akD)}{(ak)^n/n!} \\ &= a^n \operatorname{vol}(D). \end{aligned}$$

Now for (ii) we start with some reductions. By Example 1.2.5, we set $A = mH+N$, $B = mH$ then write $A \sim_{lin} A-B$ as the difference of two very ample Cartier divisors. Since D is big, we can write $rD - B \sim_{lin} B_1$ for some effective Cartier divisor B_1 and suitable integer $r > 0$, so that

$$pD - N \sim_{lin} (p+r)D - (A+B_1).$$

Moreover by (i),

$$\lim_{p\to\infty} \frac{\operatorname{vol}((p+r)D)}{p^n} = \lim_{p\to\infty} \frac{\operatorname{vol}(pD)}{p^n} = \operatorname{vol}(D),$$

while the statement (ii) is equivalent to the limit

$$\lim_{p\to\infty}\frac{\mathrm{vol}(pD-N)}{p^n}=\lim_{p\to\infty}\frac{\mathrm{vol}(D)}{p^n}=\mathrm{vol}(D).$$

But we have

$$\begin{aligned}\lim_{p\to\infty}\frac{\mathrm{vol}(pD-N)}{p^n}&=\lim_{p\to\infty}\frac{\mathrm{vol}((p+r)D-(A+B_1))}{p^n}\\&=\lim_{p\to\infty}\left[\frac{(p+r)^n}{p^n}\cdot\frac{\mathrm{vol}((p+r)D-(A+B_1))}{(p+r)^n}\right]\\&=\lim_{q\to\infty}\frac{\mathrm{vol}(qD-(A+B_1))}{q^n},\end{aligned}$$

where $A+B_1$ is effective by our choice. Thus we can assume without loss of generality that N is effective. If N' is a second effective Cartier divisor, then evidently

$$\mathrm{vol}(pD-(N+N'))\le\mathrm{vol}(pD-N)\le\mathrm{vol}(pD)$$

by the inclusions

$$\mathscr{O}_X(pD-(N+N'))\overset{\cdot N'}{\hookrightarrow}\mathscr{O}_X(pD-N)\overset{\cdot N}{\hookrightarrow}\mathscr{O}_X(pD).$$

So it is enough to prove the statement for $N+N'$ in place of N. In particular, by taking N' to be sufficiently large multiple of an ample Cartier divisor, we have $N+N'$ is very ample by Example 1.2.5. So we are reduced to the case in which N is very ample.

Assuming that N is very ample (and in particular effective). Choose a sequence $E_1,E_2,\ldots,E_m\in|N|$ of very general Cartier divisors linearly equivalent to N. Then

$$\mathscr{O}_X(m(pD-N))\cong\mathscr{O}_X(mpD-\sum_{i=1}^{m}E_i),$$

so $H^0(X,\mathscr{O}_X(m(pD-N)))\cong H^0(X,\mathscr{O}_X(mpD-\sum\limits_{i=1}^{m}E_i))$ is identified with the group of sections of $\mathscr{O}_X(mpD)$ vanishing on each of the Cartier divisors $E_1,\ldots,E_m$. In particularwe can choose them to be the one very general Cartier divisor $E\in|N|$ which do not affect h^0 by Lemma 2.29. This turn may be analyzed via the exact sequence

$$0\longrightarrow H^0\big(X,\mathscr{O}_X(mpD-mE)\big)\longrightarrow H^0\big(X,\mathscr{O}_X(mpD)\big)\longrightarrow\bigoplus^{m}H^0\big(E,\mathscr{O}_E(mpD|_E)\big).$$

And then we have

$$h^0\big(X,m(pD-N)\big)\ge h^0(X,mpD)-m\cdot h^0(E,mpD|_E).$$

This implies that

$$\mathrm{vol}_X(pD - N) \geq \mathrm{vol}_X(pD) - n \cdot \mathrm{vol}_E(pD|_E),$$

but by (i) we have $\mathrm{vol}_E(pD|_E) = p^{n-1} \cdot \mathrm{vol}_E(D|_E)$, so statement (ii) follows from

$$\frac{\mathrm{vol}_X(pD)}{p^n} \geq \frac{\mathrm{vol}_X(pD - N)}{p^n} \geq \frac{\mathrm{vol}_X(pD)}{p^n} - \frac{n}{p} \cdot \mathrm{vol}_E(D|_E).$$

□

Remark 3.2.31. One can view the second statement as hinting at the continuity properties of the volume.

Remark 3.2.32. (Volume of a $\mathbb{Q}$-Cartier divisor) When D is a $\mathbb{Q}$-Cartier divisor, we can define the volume $\mathrm{vol}(D)$ of D as taking the limsup over those m for which mD is integral. However it is perhaps quicker to choose some $a \in \mathbb{N}$ for which aD is integral, and then define $\mathrm{vol}(D) = \frac{1}{a^n}\,\mathrm{vol}(aD)$ while the statement (i) says that this is independent of the choice of a.

Example 3.2.33. (Volume of difference of nef Cartier divisors, I) If D and E are nef $\mathbb{Q}$-Cartier divisors, then as in the proof of Theorem 2.14, by after dividing the same large enough number k we have

$$\mathrm{vol}(D - E) \geq (D^n) - n(D^{n-1} \cdot E).$$

Using Fujita's vanishing theorem, we next deduce from Proposition 2.28 that the volume of a line bundle depends only upon on its numerical equivalence class.

Proposition 3.2.34. (Numerical nature of volume) *If D, D' are numerically equivalent $\mathbb{Q}$-Cartier divisors on X, then*

$$\mathrm{vol}(D) = \mathrm{vol}(D').$$

In particular, it makes sense to talk of the volume $\mathrm{vol}(\xi)$ *of a class $\xi \in N^1(X)_{\mathbb{Q}}$.*

Lemma 3.2.35. *There exists a fixed Cartier divisor N having the property that*

$$H^0\big(X, \mathscr{O}_X(N + P)\big) \neq 0$$

for every numerically trivial Cartier divisor P.

In particular, if P_0 is any numerically trivial Cartier divisor, then for any $m \in \mathbb{Z}$, $N \pm mP_0$ is linearly equivalent to an effective Cartier divisor.

Proof. Fix a very ample Cartier divisor B on X. By Fujita's vanishing theorem, we have if F is a sufficiently large multiple of B, then $H^i(X, \mathscr{O}_X(F+A)) = 0$ for $i > 0$ and every nef Cartier divisor A. In particular, this applies to $A = P + (n-i)B$ where as always $0 < i \leq n$ for $n = \dim X$.

It follows by Catelnuovo-Mumford regularity that $F+P+nB$ is globally generated for every numerically trivial P. In particular, it suffices to take $N = F+nB$. Moreover for every $m \in \mathbb{Z}$, mP_0 is also numerically trivial so we can apply the above on $A = mP_0 + (n-i)B$. □

Now we return to prove the Proposition 2.34:

Proof. As $\mathrm{vol}(D) > 0$ if and only if D is big, we can assume that D and D' are big, and the issue is to show that $\mathrm{vol}(D) = \mathrm{vol}(D+P)$ for any numerically trivial Cartier divisor P. To this end, fix any integer $p > 0$, and take $A = -pD + (n-i)B$ as in preceding lemma with $N = F + nB$, then we have

$$H^0\big(X, \mathscr{O}_X(N - pP)\big) \neq 0.$$

Choosing a non-zero global section s in $\mathscr{O}_X(N-pP)$ gives rise for every $m > 0$ to an inclusion

$$\mathscr{O}_X\Big(m\big(p(D+P) - N\big)\Big) \overset{\cdot s^{\otimes m}}{\hookrightarrow} \mathscr{O}_X\big(m(pD)\big)$$

which comes from twisting with

$$\mathscr{O}_X(pP - N) \hookrightarrow \mathscr{O}_X.$$

Therefore $\mathrm{vol}\big(p(D+P) - N\big) \leq \mathrm{vol}(pD) = p^n \,\mathrm{vol}(D)$. On the other hand,

$$\frac{1}{p^n} \cdot \mathrm{vol}\big(p(D+P) - N\big) \to \mathrm{vol}(D+P)$$

as $p \to \infty$ thanks to Proposition 2.28 (ii). So we conclude that $\mathrm{vol}(D+P) \leq \mathrm{vol}(D)$. By replacing P by $-P$ and D by $D+P$, we have $\mathrm{vol}(D) \leq \mathrm{vol}(D+P)$. □

The volume of a Cartier divisor is also a birational invariant.

Proposition 3.2.36. (Birational invariance of volume, I) *Consider a birational morphism*

$$\nu : X' \longrightarrow X$$

of projective varieties of dimension n. Given an integral or $\mathbb{Q}$-Cartier divisor D on X, put $D' = \nu^* D$. Then

$$\mathrm{vol}_{X'}(D') = \mathrm{vol}_X(D).$$

Proof. It suffices to treat the case of integral D, and then by the projection formula we have

$$\begin{aligned} H^0\big(X', \mathscr{O}_{X'}(mD')\big) &= H^0\Big(X, \nu_*\big(\mathscr{O}_{X'}(mD')\big)\Big) \\ &= H^0\big(X, (\nu_*\mathscr{O}_{X'}) \otimes \mathscr{O}_X(mD)\big). \end{aligned}$$

Since ν is birational there is an exact sequence

$$0 \longrightarrow \mathscr{O}_X \longrightarrow \nu_*\mathscr{O}_{X'} \longrightarrow \eta \longrightarrow 0$$

where η is supported on a scheme of dimension $\leq n-1$ as the generic point is not in the support. By what we said above, this leads to the inequalities

$$\begin{aligned} h^0\big(X, \mathscr{O}_X(mD)\big) &\leq h^0\big(X', \mathscr{O}_{X'}(mD')\big) \\ &= h^0\big(X, (\nu_*\mathscr{O}_{X'}) \otimes \mathscr{O}_X(mD)\big) \\ &\leq h^0\big(X, \mathscr{O}_X(mD)\big) + h^0\big(X, \eta \otimes \mathscr{O}_X(mD)\big) \\ \text{(Example 1.2.24)} \quad &= h^0\big(X, \mathscr{O}_X(mD)\big) + O(m^{n-1}), \end{aligned}$$

and therefore $\mathrm{vol}_{X'}(D') = \mathrm{vol}_X(D)$ by taking the limit superior. □

Finally, we can study the continuity properties of the volume.

Theorem 3.2.37. (Continuity of volume) *Let X be a projective variety of dimension n, and fix any norm $\| \ \|$ on $N^1(X)_{\mathbb{R}}$ inducing the usual topology on that finite-dimensional vector space. Then there is a positive constant $C > 0$ such that*

$$|\mathrm{vol}(\xi) - \mathrm{vol}(\xi')| \leq C \cdot \frac{\|\xi - \xi'\|}{\big(\max(\|\xi\|, \|\xi'\|)\big)^{n-1}}$$

for any two classes ξ, $\xi' \in N^1(X)_{\mathbb{Q}}$.

Proof. We can assume classes ξ and $\xi'$are big in $N^1(X)_{\mathbb{Q}}$. So both sides of the inequality are homogeneous of degree n in (ξ, ξ'), we are free to prove the inequality after replacing the given classes by a common large multiple. Having said this, fix very ample Cartier divisors $A_1, \ldots, A_r$ on X whose classes from a basis of $N^1(X)_{\mathbb{Q}}$. It is enough to prove that holds for Cartier divisors

$$\begin{aligned} D &= a_1 \cdot A_1 + \cdots + a_r \cdot A_r \\ D' &= a_1' \cdot A_1 + \cdots + a_r' \cdot A_r \end{aligned}$$

after passing to the large multiple, with $a_i, a_i' \in \mathbb{Z}$. As we are over a finite dimensional vector space, every norm is equivalent so it will be convenient to work with norm given by $\|D\| = \max\{|a_i|\}$.

First we assume to begin with that $a_i - a_i' = b_i \geq 0$, so write

$$D' = D - B \text{ with } B = \sum b_i \cdot A_i \geq 0 \text{ effective.}$$

Let $E_j \in |A_j|$ be a very general Cartier divisor, such that in particular E_j meets each A_i properly. We claim that

$$\mathrm{vol}(D - B) \geq \mathrm{vol}(D) - n \cdot \sum_{j=1}^{r} b_j \cdot \mathrm{vol}_{E_j}(D_j),$$

where $D_j = D|_{E_j}$. We prove this by using induction, but as we **want**

$$\mathrm{vol}(D - \sum_{j=1}^{r-1} b_j A_j =: \tilde{D}) \geq \mathrm{vol}(D) - n \cdot \sum_{j=1}^{r-1} b_j \cdot \mathrm{vol}_{E_j}(D_j)$$

implies that

$$\begin{aligned}
\mathrm{vol}(\tilde{D} - b_r A_r) &\geq \mathrm{vol}(\tilde{D}) - n \cdot b_r \cdot \mathrm{vol}_{E_r}(\tilde{D}_r) \\
&\geq \mathrm{vol}(D) - n \cdot \sum_{j=1}^{r-1} b_j \cdot \mathrm{vol}_{E_j}(D_j) - n \cot b_r \cdot \mathrm{vol}_{E_r}(\tilde{D}_r) \\
&\geq \mathrm{vol}(D) - n \cdot \sum_{j=1}^{r} b_j \cdot \mathrm{vol}_{E_j}(D_j)
\end{aligned}$$

as $\mathrm{vol}_{E_r}(\tilde{D}_r) \leq \mathrm{vol}_{E_r}(D_r)$, for $D' = D - \sum\limits_{j=1}^{r-1} b_j A_j$. Then it suffices to prove this in the special case for $B = b_1 A_1$, we use Siu's method while $b_1 > 0$. Choosing mb_1 very general Cartier divisors $F_\alpha \in |A_1|$, for given $m \gg 0$. And consider the exact sequence

$$0 \longrightarrow \mathscr{O}_X(mD - mb_1 A_1) \longrightarrow \mathscr{O}_X(mD) \longrightarrow \bigoplus_{\alpha=1}^{mb_1} \mathscr{O}_{F_\alpha}(mD).$$

Using lemma 2.29, each of the terms on the right have the same volume as $\mathrm{vol}_{E_1}(D_1)$. Moreover, by the asymptotic Riemann-Roch and the definition of volume says that, the dimension of H^0 of them is approximated by $\mathrm{vol}_{E_1}(D_1) \cdot \frac{m^{n-1}}{(n-1)!}$ when m tends to infinity:

$$\begin{aligned}
\mathrm{vol}_{F_\alpha}(D) = \mathrm{vol}_{E_1}(D_1) &\Longleftrightarrow \limsup_m h^0(F_\alpha, mD) \sim \mathrm{vol}_{E_1}(D_1) \cdot \frac{m^{n-1}}{(n-1)!}, \\
&\Longleftrightarrow mb_1 \cdot \limsup_m h^0(F_\alpha, mD) \sim n \cdot b_1 \cdot \mathrm{vol}_{E_1}(D_1) \frac{m^n}{n!}.
\end{aligned}$$

So $\mathrm{vol}(D - b_1 A_1) \geq \mathrm{vol}(D) - n \cdot b_1 \cdot \mathrm{vol}_{E_1}(D_1)$, as the assertion follows.

Now let $D^+ = \sum |a_i| \cdot A_i$ be the Cartier divisor obtained by making all coefficient of D non-negative, and let $D_j^+ = D^+|_{E_j}$. Then $D_j^+ - D_j$ is effective, so we have the injection $\mathscr{O}_{E_j}(D_j - D_j^+) \hookrightarrow \mathscr{O}_{E_j}$, and consequently

$$\mathrm{vol}_{E_j}(D_j) \leq \mathrm{vol}_{E_j}(D_j^+).$$

On the other hand, D_j^+ is nef via Nakai-Moishezon, so by asymptotic Riemann-Roch we have

$$\begin{aligned}
\mathrm{vol}_{E_j}(D_j^+) &= ({D_j^+}^{n-1} \cdot E_j) \\
&= \Big((\sum |a_i| \cdot A_i)^{n-1} \cdot A_j \Big) \\
&\leq C_j \cdot (\max\{|a_i\})^{n-1} \\
&= C_j \cdot (\|D\|)^{n-1}
\end{aligned}$$

for suitable constant $C_j = C_j(\{A_i\}_{i=1}^r) > 0$. So taking $C = 2rn \cdot \max\{C_j\}$ and noting that $\sum b_j \leq r \cdot \max\{b_j\} = r \cdot ||B||$, then we have

$$\begin{aligned}
\mathrm{vol}(D) - \mathrm{vol}(D - B) &\leq n \cdot \sum b_j \cdot \mathrm{vol}_{E_j}(D_j) \\
&\leq nr \cdot \|B\| \cdot \max\{C_j\} \cdot (\|D\|)^{n-1} \\
&= \frac{C}{2} \cdot (\|D\|)^{n-1} \cdot \|B\|.
\end{aligned}$$

Now consider the general case

$$D' = D + E - F$$

where E and F are effective integral linear combinations of the A_i's, with no non-zero "coordinates" in common. Then previous inequality gives

$$\begin{aligned}
\mathrm{vol}(D) - \mathrm{vol}(D - F) &\leq \frac{C}{2} \cdot (\|D\|)^{n-1} \cdot \|F\|, \\
\mathrm{vol}(D + E - F) - \mathrm{vol}((D + E - F) - E) &\leq \frac{C}{2} \cdot (\|D + E - F\|)^{n-1} \cdot \|E\|.
\end{aligned}$$

Observing that $\max\{\|E\|, \|F\|\} = \|E - F\|$ by our choice, so the triangle inequality gives

$$|\mathrm{vol}(D) - \mathrm{vol}(D + E - F)| \leq C \cdot (\max\{\|D\|, \|D + E - F\|\})^{n-1} \cdot \|E - F\|,$$

as desired. □

Corollary 3.2.38. (Volume of the real classes) *The function $\xi \mapsto \mathrm{vol}(\xi)$ on $N^1(X)_{\mathbb{Q}}$ can be extended uniquely to a continuous function*

$$\mathrm{vol} : N^1(X)_{\mathbb{R}} \longrightarrow \mathbb{R}.$$

Proof. By theorem 2.37, if we choose a sequence $\{\xi_i\}$ in $N^1(X)_{\mathbb{Q}}$ which converges to a given real class $\xi \in N^1(X)_{\mathbb{R}}$, then $\lim_{i\to\infty} \mathrm{vol}(\xi_i)$ exists and is independent of the choice of $\{\xi_i\}$, which defines the $\mathrm{vol}(\xi)$ taht we extend. □

Example 3.2.39. (Volume on blow-up of projective space) Let $X = \mathrm{Bl}_P\mathbb{P}^n$ be the blowing up of $\mathbb{P}^n$ at a point P, and we denote $e, h \in N^1(X)_{\mathbb{R}} \cong \mathbb{R}^2$ respectively the classes of exceptional Cartier divisor E and the pullback H of a hyperplane. The nef cone of X is generated by h and $h - e$. In the sector of the plane $x \geq 0, y \geq 0$ they

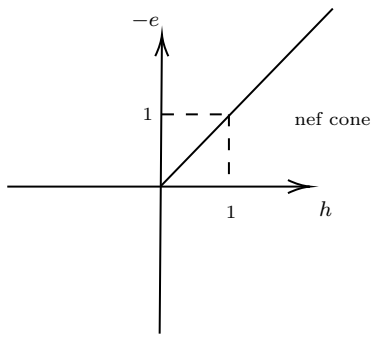


Figure 3.1.: Plane of $N^1(X)_{\mathbb{R}}$.

span, the volume is given by

$$\mathrm{vol}(x \cdot h - y \cdot e) = ((x \cdot h - y \cdot e)^n) = x^n - y^n,$$

as $(h \cdot e) = 0$ by projection formula.

On the other hand, when $k, l \geq 0$, the linear series $|kH + lE|$ contains lE as a fixed component. But we have

$$\begin{aligned} H^0(X, \mathscr{O}_X(kH + lE)) &= H^0(\mathbb{P}^n, \mathscr{O}_{\mathbb{P}^n}(k) \otimes \pi_*\mathscr{O}_X(lE)) \\ &= H^0(\mathbb{P}^n, \mathscr{O}_{\mathbb{P}^n}(k)) \end{aligned}$$

as $\pi_*\mathscr{O}_X(lE) \cong \mathscr{O}_{\mathbb{P}^n}$ by $X\backslash E \cong \mathbb{P}^n\backslash\{P\}$ and algebraic Hartogs, for here we insisting $n \geq 2$. So in the region $x \leq 0, -y \geq 0$ spanned by h and e, which corresponds to effective Cartier divisors that are not nef, the volume is given by

$$\mathrm{vol}(x \cdot h - y \cdot e) = x^n,$$

since, for simplicity, assuming x is an integer, $h^0(\mathbb{P}^n, mx) = \binom{mx+n}{n} = \frac{(mx)^n}{n!} + O(m^{n-1})$. And elsewhere the volume is zero. This shows the volume $\mathrm{vol}(x \cdot h - y \cdot e)$ as a function defined on $(x, y) \in \mathbb{R}^2$.

By reducing to the case of $\mathbb{Q}$-Cartier divisors hence integral, and using the continuity, we can easily to obtain the following results.

Example 3.2.40. (Volume of difference of nef Cartier divisors, II) Let $\xi, \eta \in \mathrm{Nef}(X)_{\mathbb{R}}$ be real nef classes on an n-dimensional projective variety. Then

$$\mathrm{vol}(\xi - \eta) \geq (\xi^n) - n \cdot (\xi^{n-1} \cdot \eta).$$

Example 3.2.41. (Volume increases in effective directions) If $\xi \in N^1(X)_{\mathbb{R}}$ is big and $e \in N^1(X)_{\mathbb{R}}$ is effective, then

$$\mathrm{vol}(\xi) \leq \mathrm{vol}(\xi + e).$$

Example 3.2.42. (Birational invariance of volume, III) Let

$$\nu : X' \longrightarrow X$$

be a birational morphism of projective varieties. Then

$$\mathrm{vol}_X(\xi) = \mathrm{vol}_{X'}(\nu^* \xi)$$

for any class $\xi \in N^1(X)_{\mathbb{R}}$.

3.3 Bigness in Relative Version

In this section we will consider about the bigness for a morphism. By way of preparation, consider the proper morphism $f : X \longrightarrow S$ from a normal algebraic variety X to a projective variety S, and we **always** assume that f is surjective with irreducible geometric generic fibre.

Definition 3.3.1. (Relative Iitaka dimension) The *relative Iitaka dimension* is defined by the Iitaka dimension on the geometric generic fibre, that is

$$\kappa(L) = \kappa(X/S, L) = \kappa(X_{\bar{\eta}}, L|_{X_{\bar{\eta}}})$$

for a line bundle L on X, where $X_{\bar{\eta}} = X \underset{S}{\times} \overline{k(\eta)}$ for the generic point $\eta \in S$.

Remark 3.3.2. If we drop the condition that f has irreducible geometric generic fibre, then at this time, for all points $\bar{x} \in X_{\bar{\eta}}$

$$\mathscr{O}_{X_{\bar{\eta}},\bar{x}} \cong \left(\mathscr{O}_{X_\eta,x} \underset{k(\eta)}{\otimes} \overline{k(\eta)}\right)_{q_{\bar{x}}} \cong \left((\mathscr{O}_{X,x}/m_\eta \mathscr{O}_{X,x}) \underset{k(\eta)}{\otimes} \overline{k(\eta)}\right)_{q_{\bar{x}}} \cong \left(\mathscr{O}_{X,x} \underset{k(\eta)}{\otimes} \overline{k(\eta)}\right)_{q_{\bar{x}}}$$

is reduced, by [SP25, Tag 030U]. Therefore $X_{\bar{\eta}}$ is a reduced algebraic scheme. Moreover, $X_{\bar{\eta}}$ is a normal algebraic scheme since $\mathscr{O}_{X,p} \underset{k(\eta)}{\otimes} \overline{k(\eta)}$ is a normal ring by [SP25, Tag 0C30]. But we may be careful X might be a disjoint union of normal algebraic varieties, one can see [SP25, Tag 0C3M] and [SP25, Tag 033M].

Definition 3.3.3. (Relative bigness) A line bundle L on a normal variety X is *big relative to* f or f*-big*, if $\kappa(X/S, L) = \dim X_{\bar{\eta}}$. That is, $L|_{X_{\bar{\eta}}}$ is a big line bundle on the normal variety $X_{\bar{\eta}}$.

A Cartier divisor D on X is f*-big* if $\mathscr{O}_X(D)$ is so.

Remark 3.3.4. (Equivalent definition of relative bigness) In [KMM87] the definition of relative bigness is only on the generic fibre X_η, this is more convenient. But in fact, they are equivalent since by [Har77, III, 9.3] we have

$$H^0\big(X_\eta, L^{\otimes m}|_{X_\eta}\big) \underset{k(\eta)}{\otimes} \overline{k(\eta)} \cong H^0\big(X_{\bar{\eta}}, L^{\otimes m}|_{X_{\bar{\eta}}}\big),$$

so $\kappa(X_\eta, L|_{X_\eta}) = \kappa(X_{\bar{\eta}}, L|_{X_{\bar{\eta}}})$.

We may alternately use these two equivalent definitions, for the convenience in the proofs.

Similarly, we can leave the "normal" assumption out and extend this definition to a general variety.

Lemma 3.3.5. *Assume that $f : X \longrightarrow S$ is a projective surjective[1] morphism of varieties with X_η is of dimension n, not necessarily normal. A line bundle L on X is f-big if and only if there is a constant $C > 0$ such that*

$$\operatorname{rk} f_* L^{\otimes m} \geq C \cdot m^n$$

for all sufficiently large m.

Proof. Since $\operatorname{rk} f_* L^{\otimes m} = \dim_{k(\eta)} (f_* L^{\otimes m})_\eta = h^0(X_\eta, L^{\otimes m}|_{X_\eta}) \geq C \cdot m^n$ by apply Lemma 2.4 on the projective variety X_η. □

Proposition 3.3.6. (Relative Kodaira's lemma) *Let D be a f-big Cartier divisor and F be an arbitrary effective Cartier divisor on X. Then*

$$f_* \mathscr{O}_X(mD - F) \neq 0$$

for all sufficiently large m.

Proof. By $f_* \mathscr{O}_X(mD-F)_\eta = H^0\big(X_\eta, \mathscr{O}_X(mD-F)|_{X_\eta}\big) \neq 0$ and Kodaira's lemma. □

Corollary 3.3.7. (Characterization of relative big Cartier divisors) *Let $f : X \longrightarrow S$ be a projective surjective morphism of varieties, and let D be a Cartier divisor on X. The following are equivalent*

(i) D is f-big.

(ii) For any f-ample Cartier divisor A on X, there is a positive integer $m > 0$ and an effective Cartier divisor N on X such that $mD \sim_S A + N$. (See Appendix A.)

*(iii) Same as in (ii)[2] for **some** f-ample Cartier divisor A.*

(iv) There exist an f-ample Cartier divisor A, a positive integer $m > 0$, and an effective Cartier divisor N such that $mD \equiv_S A + N$.

Proof. "(i)⇒(ii)": Assume that D is f-big, take $r \gg 0$ such that

$$rA \sim_S H_r \quad \text{and} \quad (r+1)A \sim_S H_{r+1}$$

are both effective. Apply relative Kodaira's lemma with $F = H_{r+1}$, we find that

$$H^0\big(X, \mathscr{O}_X(mD - H_{r+1} + f^*B)\big) = H^0\big(S, f_* \mathscr{O}_X(mD - H_{r+1}) \otimes \mathscr{O}_S(B)\big) \neq 0,$$

[1] When f is not surjective, we have $\operatorname{rk} f_* L^{\otimes m} = 0$.

[2] Since for ample divisor A on X and a Cartier divisor B on S, we have $A + f^*B$ is still ample, the result (iii) can also for $\sim_{lin}$.

thanks to the projection formula. Then we have

$$mD + f^*B \sim_{lin} H_{r+1} + N' \sim_{lin} A + (H_r + N').$$

Taking $N = H_{r+1} + N'$ gives (ii).

"(ii)⇒(iii)⇒(iv)" is trivial.

"(iv)⇒(i)": If $mD \equiv_S A + N$, then $mD - N$ is relatively numerically equivalent to a f-ample Cartier divisor, so it is also f-ample by Nakai-Moishezon. So after possibly passing to an even large multiple we can assume that $mD - N' \sim_S H$ where H is f-very ample and N' is effective. But then

$$\kappa(X/S, D) \geq \kappa(X/S, H) = \dim X_\eta,$$

so D is f-big. □

Corollary 3.3.8. *The relative bigness of a Cartier divisor D depends only on its relative numerical equivalence class.*

Remark 3.3.9. Since the rank is computed on the generic fibre, the remaining properties such as volume, $\mathbb{R}$-Cartier f-big Cartier divisors and pseudo-effective cones are easily deduced.

Remark 3.3.10. In Lemma 2.4, we assume that X is a projective variety because, in the proof of Lemma 1.2.24, we need to express a Cartier divisor D as the difference of two very ample Cartier divisors. If this growth result can be extended to a proper scheme, then the definitions of big Cartier divisors and relative big Cartier divisors can be applied to proper varieties and proper morphisms, respectively, as in [KM98, 3.22]. But be careful, the existence of f-ample divisor implies that f is projective, see Remark 1.5.2, so Corollary 3.7 no need to weaken the assumption.

Lemma 3.3.11. (Growth of cohomology for proper scheme) *Let X be a proper scheme of dimension n and D is a Cartier divisor on X. If $\mathscr{F}$ is any coherent sheaf on X, then*

$$h^i\big(X, \mathscr{F}(mD)\big) = O(m^n).$$

Proof. We prove this by using induction on the dimension.

Step 1: We introduce a projective scheme X' with morphism $f : X' \longrightarrow X$ which is an isomorphism over an open dense subset $U \subseteq X$, by Chow's lemma. Then for the natural map $\mathscr{F} \longrightarrow f_*f^*\mathscr{F}$ which is an isomorphism on this $U \subseteq X$. Now for the short exact sequence

$$0 \longrightarrow \eta \longrightarrow \mathscr{F} \longrightarrow f_*f^*\mathscr{F} \longrightarrow 0$$

where the kernel η is supported outside U, that is supported on a proper subscheme of X has dimension $< n$. So by induction hypothesis its growth of cohomology is like $O(m^{n-1})$. Taking long exact sequence

$$\cdots \to H^i\big(X, \eta(mD)\big) \to H^i\big(X, \mathscr{F}(mD)\big) \to H^i\big(X, f_*f^*\mathscr{F}(mD)\big) \to H^{i+1}\big(X, \eta(mD)\big) \to \cdots$$

we have

$$\begin{aligned} -O(m^{n-1}) &= -h^i\big(X, \eta(mD)\big) \\ &\le h^i\big(X, f_*f^*\mathscr{F}(mD)\big) - h^i\big(X, \mathscr{F}(mD)\big) \\ &\le h^{i+1}\big(X, \eta(mD)\big) \\ &= O(m^{n-1}), \end{aligned}$$

so we can get:

$$\Big|h^i\big(X, \mathscr{F}(mD)\big) - h^i\big(X, f_*f^*\mathscr{F}(mD)\big)\Big| = O(m^{n-1}).$$

Thus we only need to prove that $h^i\big(X, f_*f^*\mathscr{F}(mD)\big) = O(m^n)$.

Step 2: Thanks to the projection formula, we have

$$f_*f^*\mathscr{F} \otimes \mathscr{O}_X(mD) \cong f_*\big(f^*\mathscr{F} \otimes f^*\mathscr{O}_X(mD)\big) =: f_*\mathscr{G}.$$

Now consider the convergent spectral sequence (see (D.6.3))

$$E_2^{p,q} = H^p\big(X, R^qf_*\mathscr{G}\big) \underset{p}{\Rightarrow} R^{p+q}\big(H^0(X, f_*\mathscr{G})\big) = H^n(X', \mathscr{G}),$$

there is a induced filtration $\Phi^\bullet H^n(X', \mathscr{G})$ such that

$$E_\infty^{p,q} \cong \Phi^pH^n/\Phi^{p+1}H^n.$$

But as we are in finite dimensional vector spaces, there are short exact sequences

$$0 \to \Phi^pH^n(X', \mathscr{G}) \to \Phi^{p+1}H^n(X', \mathscr{G}) \to E_\infty^{p,q} \to 0$$

equipping with an additive function $\dim_{\mathbb{C}}$ for each p. So respect to the filtration

$$\{0\} = \Phi^sH^n \subseteq \Phi^{s-1}H^n \subseteq \cdots \subseteq \Phi^{t+1}H^n \subseteq \Phi^tH^n = H^n(X', \mathscr{G}),$$

we have

$$\begin{aligned} h^n(X', \mathscr{G}) &= \dim_{\mathbb{C}} \Phi^tH^n - \dim_{\mathbb{C}} \Phi^sH^n \\ &= \dim_{\mathbb{C}} \Phi^tH^n - \dim_{\mathbb{C}} \Phi^{t+1}H^n + \dim_{\mathbb{C}} \Phi^{t+1}H^n - \dim_{\mathbb{C}} \Phi^{t+2}H^n \\ &+ \cdots + \dim_{\mathbb{C}} \Phi^{s-1}H^n - \dim_{\mathbb{C}} \Phi^sH^n \\ &= \dim_{\mathbb{C}} E_\infty^{t,n-t} + \dim_{\mathbb{C}} E_\infty^{t+1,n-(t+1)} + \cdots + \dim_{\mathbb{C}} E_\infty^{s-1,n-(s-1)}. \end{aligned}$$

But as our spectral sequence is first quadrant, so $s = s(n) = n + 1$ and $t = t(n) = 0$; that is, by change the superscript we have

$$\begin{aligned} h^i(X', \mathscr{G}) &= \sum_{k=0}^{i} \dim_{\mathbb{C}} E_\infty^{k,i-k} \\ &= \sum_{p+q=i} \dim_{\mathbb{C}} E_\infty^{p,q} \\ &= \dim_{\mathbb{C}} E_\infty^{i,0} + \sum_{p+q=i, q>0} \dim_{\mathbb{C}} E_\infty^{p,q}. \end{aligned}$$

On the other hand, recall that f is an isomorphism over an open dense subset $U \subseteq X$, the higher direct image sheaf $R^q f_*(f^*\mathscr{F}) = 0$ on U, supported on a proper subscheme of X has dimension $< n$, so is $R^q f_*\mathscr{G}$. Thus by induction hypothesis, we have

$$\dim_{\mathbb{C}} E_2^{p,q} = h^p(X, R^q f_*\mathscr{G}) = O(m^{n-1})$$

for all $q > 0$. With the fact $\dim_{\mathbb{C}} E_\infty^{o,q} < \dim_{\mathbb{C}} E_2^{p,q}$, we have

$$h^i(X', \mathscr{G}) = \dim_{\mathbb{C}} E_\infty^{i,0} + O(m^{n-1}). \tag{3.2}$$

Moreover, recall that

$$E_{r+1}^{i,0} = \frac{\ker(d_r : E_r^{i,0} \to E_r^{i+r,-r+1})}{\operatorname{im}(d_r : E_r^{i-r,r-1} \to E_r^{i,0})} = \frac{E_r^{i,0}}{\operatorname{im} d_r}$$

since $-r + 1 < 0$ outside the first quadrant. Theorefore,

$$\begin{aligned} &\dim_{\mathbb{C}} E_{r+1}^{i,0} = \dim_{\mathbb{C}} E_r^{i,0} - \dim_{\mathbb{C}}(\operatorname{im} d_r), \\ &\dim_{\mathbb{C}}(\operatorname{im} d_r) \leq \dim_{\mathbb{C}} E_r^{i-r,r-1} \leq \dim_{\mathbb{C}} E_2^{i-r,r-1} = O(m^{n-1}) \text{ by } r - 1 > 0. \end{aligned}$$

And this spectral sequence will stable by (D.5.2), i.e. $E_\infty^{i,0} = E_m^{i,0}$ for some large enough m, so after finite steps above we finally can get

$$\dim_{\mathbb{C}} E_\infty^{i,0} = \dim_{\mathbb{C}} E_2^{i,0} - O(m^{n-1}),$$

and combining this result with (3.2) gives that

$$\begin{aligned} h^i(X', \mathscr{G}) &= \dim_{\mathbb{C}} E_2^{i,0} + O(m^{n-1}) \\ &= h^i(X, f_*\mathscr{G}) + O(m^{n-1}). \end{aligned}$$

But by X' is a projective scheme, applying the Lemma 1.2.24 on the left gives

$$h^i(X, f_*\mathscr{G}) = h^i\big(X, f_* f^*\mathscr{F}(mD)\big) = O(m^n),$$

is our desired approximation. □

Example 3.3.12. Let X be a proper variety and let N be a nef and big Cartier divisor on X. If T is a nef $\mathbb{R}$-Cartier divisor on X, then $N+T$ is also nef and big.

Proof. By Chow's lemma, there is a projective variety X' and a proper birational morphism $f : X' \longrightarrow X$. Due to Lemma 3.11, the Proposition 2.36 holds for proper birational morphisms and hence f^*N is big. It is also nef by Example 1.4.3. So on the projective variety X', we have the decomposition

$$f^*N = A_k + \frac{1}{k}E$$

where $k \gg 0$, A_k is an ample Cartier divisor and $E \geq 0$, by Example 2.21. Moreover, f^*T is nef on X', and we have

$$f^*(N+T) = f^*T + A_k + \frac{1}{k}E$$

where $f^*T + A_k$ is ample by Corollary 1.4.7. Therefore, can use Example 2.21 again to see that $f^*(N+T)$ is nef and big, so is for $N+T$. □

Furthermore, we have the analogue version of Proposition 1.5.8 to the f-bigness:

Proposition 3.3.13. *Consider a projective surjective morphism*

$$f : X \longrightarrow S$$

of varieties onto a proper variety S. Let D be a Cartier divisor on X, and let M be a (absolute) big Cartier divisor on S. Then D is f-big if and only if $D + mf^*M$ is a (absolute) big Cartier divisor on X for all $m \gg 0$.

Proof. "⇒": By Corollary 3.7(iii), there is a positive integer $r > 0$, an f-ample Cartier divisor A and an effective Cartier divisor N such that

$$rD \sim_{lin} A + N.$$

Since M is big on the proper variety S, there is a positive integer $q > 0$, an ample Cartier divisor H and an effective E such that

$$qM \sim_{lin} H + E.$$

Therefore, we have

$$\begin{aligned} rq(D + mf^*M) &\sim_{lin} qA' + qN + rmf^*(H+E) \\ &= \underbrace{qA' + rmf^*H}_{=:P_m} + \underbrace{qN + rmf^*E}_{=:G_m}. \end{aligned}$$

Since qA' is f-ample and H is ample on S, by Proposition 1.5.8 P_m is ample for all $m \gg 0$. Moreover, f is surjective so f^*E is effective, that is $G_m \geq 0$. Hence that

$$rq(D + mf^*M) \sim_{lin} (\text{ample}) + (\text{effective}),$$

which gives that $D + mf^*M$ is big for all $m \gg 0$, by Kodaira's lemma.

"$\Leftarrow$": Suppose that for some $m > 0$, $D + mf^*M$ is big. So we can write

$$D + mf^*M \sim_{lin} A + E$$

where A is ample on X and E effective. But, this is equivalent to

$$D \sim_S A + E$$

where A is f-ample. So by Corollary 3.7 we see that D is f-big. □

3.4 Examples

3.4A Zariski's Construction

3.4B Cutkosky's Construction

3.4C Okounkov's Construction

Chapter 4

Vanishing Theorems

In this chapter, we are still working with varieties over complex numbers, and unless otherwise stated, all varieties are assumed to be **non-singular**.

4.1 Preliminaries

> If the cohomology of a sheaf $\mathscr{F}$ comes from topological cohomology, then there is a Kodaira-type vanishing theorem.
>
> *János Kollár*

4.1A Simple Normal Crossings and Resolution of Singularities

Let X be a non-singular variety of dimension n.

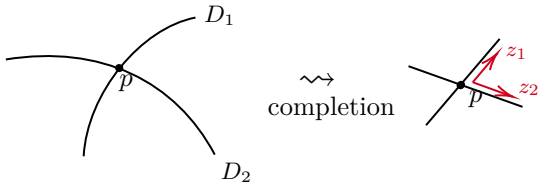
Definition 4.1.1. An effective divisor $D = \sum D_i$ on X has *simple normal crossings* if D is reduced with each component D_i is smooth, and D is defined in a neighborhood of any point by an equation in local analytic coordinates of the type

$$z_1 \cdots z_k = 0,$$

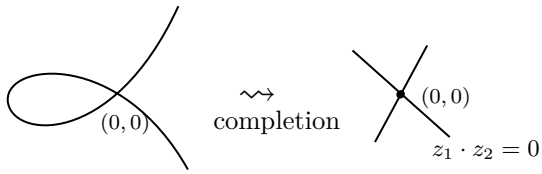
for some $k \leq n$.

A $\mathbb{Q}$-divisor $E = \sum a_i D_i$ has *simple normal crossing support* if the underlying reduced divisor $\sum D_i$ has simple normal crossings.

Remark 4.1.2. In the above case we say D is a simple normal crossing divisor, or snc divisor. Notice that by local analytic coordinates we mean taking the completion of the local ring at that point, while by our setting X is non-singular so $\widehat{\mathscr{O}}_{X,p} \cong \mathbb{C}[[z_1, \ldots, z_n]]$ by Cohen Structure theorem, see [Eis95, 10.16]. And the number $k \leq n$ is determined by intersection of components of D, for example if $k = 0$ we have $p \notin \operatorname{Supp} D$; and if $k \geq 1$, it indicates the number of components containing p.



Remark 4.1.3. In our definition we empharize that each component must be smooth, that's why we call it "simple" normal crossing. In fact, other reference associated with complex algebraic manifold will abandon this word, just say *normal crossing divisor*. For example in the affine plane $\mathbb{A}^2$ with coordinates x, y, the nodal curve $y^2 = x^2(x+1)$ is irreducible but has self-intersection at the origin, therefore it is a normal crossing divisor on the complex manifold, but not a simple normal crossing divisor.



In other words, intersections among the components of D should occur "as transversely as possible", i.e. the singularities of D should locally look no worse than a union of coordinate hyperplanes.

Example 4.1.4. (Hyperplane arrangements) Let $D = \sum H_i$ be the reduced divisor on $\mathbb{P}^n$ whose components are a collection of hyperplanes $H_i \subseteq \mathbb{P}^n$, corresponding to a finite set of point $S \subseteq (\mathbb{P}^n)^\vee$. Then D is snc divisor if and only if the points of S are in linearly general position, i.e. for $r < n$, no $r+2$ points lie on an r-dimensional linear subspace of $(\mathbb{P}^n)^\vee$.

Proof. First by the Rank-Nullity theorem in linear algebra, we see the hyperplanes of $\mathbb{P}^n$ corresponding to the points of its dual space $(\mathbb{P}^n)^\vee$. So we begin with

"$\Leftarrow$": Suppose S is not in linearly general position, that is there are $r+2$ points $(p_1, \ldots, p_{r+2})$ lying on an r-dimensional linear subspace of $(\mathbb{P}^n)^\vee$ for $r < n$. Translating this into linear algebra, it means there are $r+2$ linear forms

$$L_1, \ldots, L_{r+2} \in \mathrm{Hom}(\mathbb{C}^{n+1}, \mathbb{C})$$

contained in an $r+1$-dimensional subspace of right hand side, i.e. they are linearly dependent,

$$a_1 L_1 + \cdots + a_{r+2} L_{r+2} = 0$$

for some nonzero a_i's. Now consider $Y = \bigcap_{i=1}^{r+2} H_i$ in $\mathbb{P}^n$, the codimension of Y is equal to the dimension of a subspace spanned by these L_i's which is at most $r+1 \leq n$, so Y must be non-empty enabling us to choose a point $p \in Y$ where these $r+2$ linear forms are linearly dependent. Thus locally we find that these linear forms cannot form local analytic coordinates in $\widehat{\mathscr{O}}_{Y,p}$, as they cannot be a regular system of parameters.

"⇒": Suppose D is not a snc divisor, then there is a point $p \in \mathbb{P}^n$ such that the local equations of H_i cannot be extended to local analytic coordinates, i.e. the linear forms $L_1, \ldots, L_k$ are linearly dependent fail to be a regular sequence. Therefore, assuming $d = \dim \operatorname{span}\{L_1, \ldots, L_k\}$ with $d < k$, we have that the codimension of $Y = \bigcap_{i=1}^{k} H_i$ is at most $d \leq n$. So choose $r = d - 1 < n$, such that the the corresponding points $p_1, \ldots, p_k \in (\mathbb{P}^n)^\vee$ are in an r-dimensional linear subspace of $(\mathbb{P}^n)^\vee$. But then $k \geq d + 1 = r + 2$ by our choice. □

The normal crossing condition is very important because it singles out a class of divisors whose singularities we can understand completely.

Theorem 4.1.5. (Resolution of singularities) *Let (X, D) be a pair where X is a complex algebraic variety (possibly singular), and let $D \subseteq X$ be an effective Cartier divisor on X.*

(i) There is a projective birational morphism

$$\mu : X' \longrightarrow X$$

where X' is non-singular and μ has divisorial exceptional locus $\operatorname{Ex}(\mu)$, such that

$$\mu^* D + \operatorname{Ex}(\mu)$$

is a snc divisor.

(ii) Can construct X' via a sequence of blowing-up along smooth centers supported in the singular locus of D and X. In particular, we can assume that μ is an isomorphism over

$$X - \big(\operatorname{Sing}(X) \cup \operatorname{Sing}(D)\big).$$

We call X' a *log resolution* of D, or of the pair (X, D). Moreover log resolution exists for varieties over arbitrary field of characteristic zero by Hironaka and Lefschetz principle.

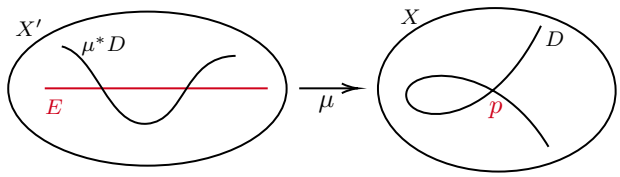


Figure 4.1.: Simple example of log resolution.

Corollary 4.1.6. *Assume X is smooth, and that $\mu : X' \longrightarrow X$ is a log resolution of the pair (X, D) constructed as above by a sequence of blowing-up along smooth centers.*

(i) For canonical divisors K_X and $K_{X'}$ of X and X' respectively, we have

$$\mu_* \mathscr{O}_{X'}(K_{X'}) = \mathscr{O}_X(K_X) \quad \text{and} \quad R^i \mu_* \mathscr{O}_{X'}(K_{X'}) = 0 \text{ for all } i > 0.$$

(ii) Assume that X is projective with an ample divisor H. Then for suitable integers $p \gg 0$ and $b_j > 0$,

$$\mu^*(pH) - \sum b_j E_j$$

is an ample divisor on X', the sum being taken over the μ-exceptional divisors.

Proof. By Hironaka's resolution of singularities, the log resolution $\mu : X' \longrightarrow X$ can be factored as

$$\mu : X' = X_m \xrightarrow{\pi_m} X_{m-1} \xrightarrow{\pi_{m-1}} \cdots \xrightarrow{\pi_2} X_1 \xrightarrow{\pi_1} X_0 = X,$$

where for each $1 \leq k \leq m$, the projective birational morphism $\pi_k : X_k \longrightarrow X_{k-1}$ is the blowing-up along the smooth center $Z_{k-1} \subseteq X_{k-1}$ of codimension $c_k \geq 2$. We denote its exceptional divisor $E_k \subseteq X_k$, and $\mu_k := \pi_1 \circ \cdots \circ \pi_k : X_k \longrightarrow X$ with $\mu_m = \mu$.

(i) First we consider the single blow-up case. By [Har77, II, ex8.5] we have

$$K_{X_1} = \pi^* K_X + (c_1 - 1)E_1,$$

so thanks to the projection formula, we get an isomorphism

$$\begin{aligned} R^i \pi_{1*} \mathscr{O}_{X_1}(K_{X_1}) &= R^i \pi_{1*} \big(\pi_1^* \mathscr{O}_X(K_X) \otimes \mathscr{O}_{X_1}((c_1 - 1)E_1) \big) \\ &\cong R^i \pi_{1*} \mathscr{O}_{X_1}\big((c_1 - 1)E_1\big) \otimes \mathscr{O}_X(K_X). \end{aligned}$$

We hence begin to compute $R^i \pi_{1*} \mathscr{O}_{X_1}\big((c_1 - 1)E_1\big)$.

- $i = 0$. For $\operatorname{codim}_X Z_0 \geq 2$, the algebraic Hartogs says $\pi_{1*} \mathscr{O}_{X_1}\big((c_1 - 1)E_1\big) \cong \mathscr{O}_X$.
- $i > 0$. Since we blowing-up along a smooth center, for any point $x \in Z_0$, its fibre is $\{x\} \times \mathbb{P}^{c_1 - 1} \cong \mathbb{P}^{c_1 - 1}$ with $\mathscr{O}_{X_1}(E_1)|_{\pi_1^{-1}(x)} \cong \mathscr{O}_{\mathbb{P}^{c_1 - 1}}(-1)$, see [Har77, II, 8.24]. Therefore

  $$\mathscr{O}_{X_1}((c_1 - 1)E_1)|_{\pi_1^{-1}(x)} \cong \mathscr{O}_{\mathbb{P}^{c_1 - 1}}(-(c_1 - 1)) \text{ for } c_1 \geq 2.$$

  By computation in [Har77, III, 5.2] we see $H^i(\mathbb{P}^{c_1 - 1}, \mathscr{O}_{\mathbb{P}^{c_1 - 1}}(-(c_1 - 1))) = 0$ for all $i > 0$. However, since π_1 is not flat, we cannot immediately deduce $(R^i \pi_{1*} \mathscr{O}_{X_1}((c_1 - 1)E_1))_x = 0$, we need to extend this result to the whole space.

Now consider the "thickened fibre" of X_1 over the point $x \in X$:

$$F_n := X_1 \underset{X}{\times} \operatorname{Spec} \mathscr{O}_{X,x}/m_x^n$$

while we denote $\mathscr{F} = \mathscr{O}_{X_1}((c_1-1)E_1)$ and $\mathscr{F}_n$ to be the pullback $v^*\mathscr{F}$ on F_n.

$$\begin{array}{ccc} F_n & \xrightarrow{\quad v \quad} & X_1 \\ \downarrow & & \downarrow{\scriptstyle \pi_1} \\ \operatorname{Spec} \mathscr{O}_{X,x}/m_x^n & \longrightarrow & X. \end{array}$$

Then by the theorem of formal functions (see [Har77, III, 11.1]), we have

$$\widehat{(R^i\pi_{1*}\mathscr{F})}_x \cong \varprojlim H^i(F_n, \mathscr{F}_n).$$

At the same time we have shown that $H^i(F_1, \mathscr{F}_1) = H^i(\mathbb{P}^{c_1-1}, \mathscr{O}_{\mathbb{P}^{c_1-1}}(-(c_1-1))) = 0$ for all $i > 0$, so we assume the vanishing holds for $H^i(F_n, \mathscr{F}_n)$ for any $i > 0$. Then via the surjection $\mathscr{O}_{X,x}/m_x^{n+1} \twoheadrightarrow \mathscr{O}_{X,x}/m_x^n$ we can regard $F_n \hookrightarrow F_{n+1}$ as a closed subscheme with ideal sheaf $\mathscr{I}_{F_n} = m_x^n/m_x^{n+1} \underset{k(x)}{\otimes} \mathscr{O}_{F_1}$. This gives the short exact sequence

$$0 \longrightarrow m_x^n/m_x^{n+1} \underset{k(x)}{\otimes} \mathscr{O}_{F_1} \longrightarrow \mathscr{O}_{F_{n+1}} \longrightarrow \mathscr{O}_{F_n} \longrightarrow 0.$$

Twisting by $\mathscr{F}$ then induces

$$0 \longrightarrow \mathscr{F}_1 \underset{k(x)}{\otimes} (m_x^n/m_x^{n+1}) \longrightarrow \mathscr{F}_{n+1} \longrightarrow \mathscr{F}_n \longrightarrow 0$$

where the left hand side is $\bigoplus_{\text{finite}} \mathscr{F}_1$, since m_x^n/m_x^{n+1} is a finite dimensional $k(x)$-vector space. Therefore by taking cohomology we have

$$\cdots \longrightarrow \bigoplus_{\text{finite}} H^i(F_1, \mathscr{F}_1) \longrightarrow H^i(F_{n+1}, \mathscr{F}_{n+1}) \longrightarrow H^i(F_n, \mathscr{F}_n) \longrightarrow \cdots$$

where the outside terms are vanish, so $H^i(F_{n+1}, \mathscr{F}_{n+1}) = 0$ for all $i > 0$. Thus $\widehat{(R^i\pi_{1*}\mathscr{F})}_x = 0$ for any $x \in Z_0$, i.e. $(R^i\pi_{1*}\mathscr{F})_x = 0$ by coherence and faithfully flatness, see [Eis95, 7.2]. Outside Z_0 the fibre is a single point, higher cohomologies just vanish.

Now we suppose that the statement holds for k-fold composition of blow-ups. And consider $\mu_{k+1} = \mu_k \circ \pi_{k+1}$. Applying the theorem of Grothendieck spectral sequence (D.6.3) to the object $\mathscr{O}_{X_{k+1}}(K_{X_{k+1}})$:

$$E_2^{p,q} = (R^p\mu_{k*})(R^q\pi_{k+1*})\mathscr{O}_{X_{k+1}}(K_{X_{k+1}}) \underset{p}{\Rightarrow} R^{p+q}\mu_{k+1*}\mathscr{O}_{X_{k+1}}(K_{X_{k+1}}).$$

But by our preceding discussion on the single blow-up π_{k+1}, we have for all $q > 0$ the coherent sheaves $R^q\pi_{k+1*}\mathscr{O}_{X_{k+1}}(K_{X_{k+1}}) = 0$, that is this spectral sequence collapses on the p-axis. So by (D.5.11) we see

$$E_2^{n,0} = R^n\mu_{k*}\big(\pi_{k+1*}\mathscr{O}_{X_{k+1}}(K_{X_{k+1}})\big) \cong R^n\mu_{k+1*}\mathscr{O}_{X_{k+1}}(K_{X_{k+1}}).$$

Then, substituting $\pi_{k+1*}\mathscr{O}_{X_{k+1}}(K_{X_{k+1}}) \cong \mathscr{O}_{X_k}(K_{X_k})$ into it to get

$$R^n\mu_{k*}\mathscr{O}_{X_k}(K_{X_k}) \cong R^n\mu_{k+1*}\mathscr{O}_{X_{k+1}}(K_{X_{k+1}}).$$

The left side term is isomorphic to $\mathscr{O}_{X_k}(K_{X_k})$ when $n = 0$ and vanishes when $n > 0$.

(ii) We also use induction on the number of single blow-ups. First we consider the single blow-up case $\pi_1 : X_1 \longrightarrow X$. As X is projective, we see $\mathscr{O}_{X_1}(-E_1) \cong \mathscr{O}_{X_1}(1)$ is very ample. So for ample divisor H on X we have that

$$\mathscr{O}_{X_1}(-E_1) \otimes \pi_1^*\mathscr{O}_X(p_1H)$$

is very ample on X_1 for suitable $p_1 > 0$, by [Har77, II, 7.10(b)]. Therefore the divisor $A_1 := \pi_1^*p_1H - E_1$ is very ample on X_1.

Now we suppose the statement is holds for k-fold composition of blow-ups, that is for $\mu_k : X_k \longrightarrow X$, there exists suitable integers $p_k > 0$ and $c_i^{(k)} > 0$ such that the divisor

$$A_k := \mu_k^*(p_kH) - \sum_{i=1}^{k} c_i^{(k)}E_i^{(k)}$$

is very ample on X_k, while $E_i^{(k)}$ denotes the strict transform of π_i-exceptional divisor E_{k+1}. As A_k is very ample on X_k, so use [Har77, II, 7.10] again we have

$$\mathscr{O}_{X_{k+1}}(-E_{k+1}) \otimes \pi_{k+1}^*\mathscr{O}_{X_k}(p'_{k+1}A_k)$$

is very ample for suitable $p'_{k+1} > 0$, which we denote its divisor by A_{k+1}. So

$$\begin{aligned}
A_{k+1} &= \pi_{k+1}^*(p'_{k+1}A_k) - E_{k+1} \\
&= \pi_{k+1}^*\left(p'_{k+1}\mu_k^*(p_kH) - \sum_{i=1}^{k} p'_{k+1}c_i^{(k)}E_i^{(k)}\right) - E_{k+1} \\
&= (\mu_k \circ \pi_{k+1})^*(p'_{k+1} \cdot p_kH) - \sum_{i=1}^{k} p_{k+1}c_i^{(k)}\pi_{k+1}^*E_i^{(k)} - E_{k+1} \\
&= \mu_{k+1}^*(p_{k+1}H) - \sum_{i=1}^{k} p'_{k+1}c_i^{(k)}(E_i^{(k+1)} + m_{k+1,i}E_{k+1}) - E_{k+1} \\
&= \mu_{k+1}^*(p_{k+1}H) - \sum_{i=1}^{k+1} c_i^{(k+1)}E_i^{(k+1)}.
\end{aligned}$$

□

Remark 4.1.7. (Tangent description of simple normal crossing) In the above definition we describe the *simple normal crossing* by using the local analytic coordinates, to demonstrate that they intersect transversally everywhere. But in some time we will discuss it via the **regular system of parameters**, i.e. the minimal generators of m_p corresponding to the basis of Zariski cotangent space m_p/m_p^2. We say with each component D_i is smooth, such that D is defined in a neighborhood of any point by an equation of the form $f_1 \cdots f_k$ for some $k \leq n$, with $m_p/m_p^2 = \mathsf{Span}_{k(p)}\{\bar{f}_1, \ldots, \bar{f}_n\}$.

It is easy to see that this definition implies the local analytic coordinates' definition, as we have the natural isomorphism

$$\mathrm{gr}_m R \cong \mathrm{gr}_{\hat{m}} \widehat{R}$$

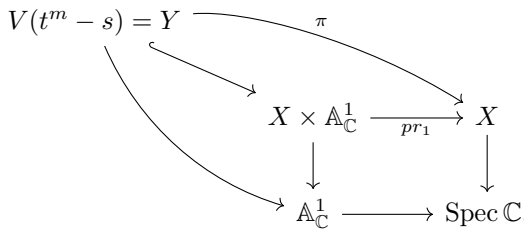
which sends the generators $\bar{f}_i$ to the formal variables z_i in $\widehat{\mathscr{O}}_{X,p} \cong \mathbb{C}[[z_1, \ldots, z_n]]$.

4.1B Covering Trick

We introduce an exquisite method to extract "roots" of divisors or line bundles.

Cyclic Coverings. "Local description of the m-fold cyclic covering branched along a given divisor on a variety."

Suppose that X is an affine variety with coordinate ring $H^0(\mathscr{O}_X, X) = \mathbb{C}[X]$, let $s \in \mathbb{C}[X]$ be a non-zero regular function. We wish to define a variety Y on which the m-th root $\sqrt[m]{s}$ of s makes sense. To begin with, we start with fibre product $X \times \mathbb{A}^1_{\mathbb{C}}$ with an affine line of coordinate t. Consider the subvariety $Y \subseteq X \times \mathbb{A}^1_{\mathbb{C}}$ defined by the equation $t^m - s = 0$:



The morphism $\pi : Y \longrightarrow X$ is a cyclic covering branched along the divisor $D = \mathrm{div}(s)$. We regard the coordinate t as a regular function in $\mathbb{C}[X \times \mathbb{A}^1] \cong \mathbb{C}[X][t]$, then we have

$$\mathbb{C}[Y] = \mathbb{C}[X][t]/(t^m - s).$$

So if we set $s' = t|_Y \in \mathbb{C}[Y]$, we have the equality

$$(s')^m = \pi^* s$$

of functions on Y, so we have indeed extracted the desired root of s.

This local construction can be globalized to:

Proposition 4.1.8. (Cyclic coverings) *Let X be a variety, and L be a line bundle on X. Suppose given a positive integer $m \geq 1$ associated to a non-zero section*

$$s \in H^0(X, L^{\otimes m})$$

defining an effective Cartier divisor $D = \mathrm{div}(s) \subseteq X$. Then there exists a finite flat covering map

$$\pi : Y \longrightarrow X,$$

where Y is a scheme having the property that, the pullback $L' := \pi^* L$ of L carries a section

$$s' \in H^0(Y, L') \quad \textit{with} \quad (s')^m = \pi^* s.$$

The Cartier divisor $D' = \mathrm{div}(s') \subseteq Y$ maps isomorphically to D. Moreover if X and D are non-singular, then so are Y and D'.

As above, we can regard of s' as giving the m-th root $\sqrt[m]{s}$ of s.

Proof. Thinking locally, we can take an affine open covering $\{U_i\}$ of X which locally trivializes L, and carry out the preceding construction over each U_i. But we need to ask whether this construction compatible on the intersection? That is, after locally we construct as above, can they be glued?

Consider the (geometric) line bundle $\mathbf{L} = \mathbb{V}(L^\vee) = \underline{\mathrm{Spec}}_X \mathrm{Sym}(L^\vee)$, with the bundle projection $p : \mathbf{L} \longrightarrow X$. By [Har77, II, ex5.17(c)], we have $p_* \mathscr{O}_{\mathbf{L}} = \mathrm{Sym}(L^\vee)$, so we pick a "tautological" section

$$\begin{aligned} T \in H^0(\mathbf{L}, p^* L) &= \mathrm{Hom}_{\mathrm{Mod}_{\mathscr{O}_{\mathbf{L}}}}(\mathscr{O}_{\mathbf{L}}, p^* L) \\ &\cong \mathrm{Hom}_{\mathrm{Mod}_{p_* \mathscr{O}_{\mathbf{L}}}}(p_* \mathscr{O}_{\mathbf{L}}, p_* p^* L) \text{ by [Har77, II, ex5.17(e)]} \\ &= \mathrm{Hom}_{\mathrm{Mod}_{p_* \mathscr{O}_{\mathbf{L}}}}(p_* \mathscr{O}_{\mathbf{L}}, L \otimes p_* \mathscr{O}_{\mathbf{L}}) \text{ by projection formula} \\ &= \mathrm{Hom}_{\mathrm{Mod}_{\mathrm{Sym}(L^\vee)}}\big(\mathrm{Sym}(L^\vee), L \otimes \mathrm{Sym}(L^\vee)\big). \end{aligned}$$

And we recall that in rank 1 case, the symmetric algebra is nothing but the tensor algebra, as the commutator $x \otimes y - y \otimes x$ is stalk-wise trivial everywhere, so globally trivial. Therefore we have the linear map

$$\mathrm{Sym}(L^\vee) \longrightarrow L \otimes \mathrm{Sym}(L^\vee)$$

is isomorphic to

$$\bigoplus_{n \geq 0} (L^\vee)^{\otimes n} \longrightarrow L \otimes \left(\bigoplus_{n \geq 0} (L^\vee)^{\otimes n} \right) = \bigoplus_{n \geq 0} \big(L \otimes (L^\vee)^{\otimes n}\big).$$

When $n = 0$ the term is L, and when $n \geq 1$ we have $L \otimes (L^\vee)^{\otimes n} \cong (L^\vee)^{\otimes(n-1)}$. So the right hand side is

$$L \oplus \mathscr{O}_X \oplus L^\vee \oplus (L^\vee)^{\otimes 2} \oplus \cdots = L \oplus \operatorname{Sym}(L^\vee).$$

Thus we see T is determined by the canonical inclusion $\operatorname{Sym}(L^\vee) \hookrightarrow L \oplus \operatorname{Sym}(L^\vee)$.

Now we take $Y \subseteq \mathbf{L}$ to be the divisor of the of the section

$$T^m - p^*s \in H^0(\mathbf{L}, p^*L^{\otimes m})$$

and $s' = T|_Y$. We verify this construction satisfy our assertions. For $\pi = p|_Y : Y \to X$, we have $s' = T|_Y \in H^0(Y, \pi^*L)$ and directly have $(s')^m = \pi^*s$. Locally, restricting to an affine open $U = \operatorname{Spec} A$ where $L|_U \cong \mathscr{O}_U = \widetilde{A}$, then s corresponding to an element $f \in A$. The scheme

$$\begin{aligned}
\mathbf{L}|_U &= \underline{\operatorname{Spec}}_U \operatorname{Sym}\left(H^0(U, L^\vee)\right) \\
&= \operatorname{Spec} \operatorname{Sym}(A\langle t \rangle) \\
&= \operatorname{Spec} A[t] \\
&= U \times \mathbb{A}^1,
\end{aligned}$$

where $t \in H^0(U, L^\vee)$ arising from the dual pairing $\langle t, e \rangle = 1 \in A$, for rank 1 free module $H^0(U, L) \cong A\langle e \rangle$. So locally $T|_U \in H^0(U \times \mathbb{A}^1, p^*L)$ is the coordinate t (we will describe it in detail in the remark), p^*s is just the element $f \in A[t]$. Thus Y is locally of the form $t^m - f = 0$, with the coordinate ring $A[t]/(t^m - f)$. Now the divisors $D = \operatorname{div}(s)$ and $D' = \operatorname{div}(s')$ are locally of the form

$$f = 0 \text{ in } A \quad \text{and} \quad t = 0 \text{ in } A[t]/(t^m - f),$$

thus they are isomorphic via $\frac{A[t]/(t^m - f)}{(t)} = A/(f)$. Furthermore, since as A-module we have $A[t]/(t^m - f) \cong \mathsf{Span}_A\{1, t, \ldots, t^{m-1}\} \cong \bigoplus^m A$ is finitely generated and free, immediately to get $Y \to X$ is a finite flat covering.

Now for the singularity statements, if X and D are non-singular, then D' is obviouly non-singular by $D \cong D'$, and Y is non-singular since locally for $F = t^m - f$,

$$dF = mt^{m-1}dt - df = 0$$

if and only if $df = 0$, as we are over $\mathbb{C}$. □

Remark 4.1.9. (Geometrical grasp) We locally restrict to $U = \operatorname{Spec} A$ where the line bundle L trivializes. So $\mathbf{L}|_U = \operatorname{Spec} A[t]$ where t is the dual basis of $H^0(U, L) =$

$A\langle e\rangle$. Then $H^0(\mathbf{L}|_U, p^*L) = H^0(\operatorname{Spec} A[t], p^*L|_U) = A\langle e\rangle \underset{A}{\otimes} A[t]$ by the definition of coherence. This is nothing but the $L \otimes \operatorname{Sym}(L^\vee)$ restrict on U after sheafification, so T comes from the canonical inclusion which is

$$\begin{aligned} \phi : A[t] &\hookrightarrow A\langle e\rangle \underset{A}{\otimes} A[t] \\ 1 &\mapsto e \otimes t \end{aligned}$$

And we have

$$\begin{aligned} p^* : L|_U = \widetilde{A\langle e\rangle} &\longrightarrow p^*L|_U = \widetilde{A\langle e\rangle \underset{A}{\otimes} A[t]} \\ e &\longmapsto p^*e = e \otimes 1 \end{aligned}$$

So T restricted to $\mathbf{L}|_U$ is the image of 1, i.e. $T = t \cdot p^*e$. Then for a closed point $a \in \mathbf{L}|_U \cong U \times \mathbb{A}^1$ where $x = p(a)$ in X, we have the fibre

$$\begin{aligned} \mathbf{L}_x &= \mathbf{L} \underset{X}{\times} \operatorname{Spec} k(x) \\ &= \mathbf{L}|_U \underset{U}{\times} \operatorname{Spec} \mathbb{C} \\ &= \operatorname{Spec}(A[t] \underset{A}{\otimes} \mathbb{C}) \\ &= \mathbb{A}^1_{\mathbb{C}}. \end{aligned}$$

At the same time

$$\begin{aligned} \mathbf{L}_x &= \mathbf{L}|_U \underset{U}{\times} \operatorname{Spec} \mathbb{C} \\ &= \operatorname{Spec}\left(\operatorname{Sym} H^0(U, L^\vee)\right) \underset{A}{\otimes} \mathbb{C} \\ &= \operatorname{Spec} \operatorname{Sym}\left(H^0(U, L^\vee) \underset{A}{\otimes} \mathbb{C}\right) \\ &= \operatorname{Spec} \operatorname{Sym}\left(H^0(U, L) \underset{A}{\otimes} \mathbb{C}\right)^\vee \\ &= \operatorname{Spec} \operatorname{Sym}\left(A\langle e\rangle \underset{A}{\otimes} \mathbb{C}\right)^\vee. \end{aligned}$$

From the functorial point of view, the closed points in $\mathbf{L}_x$ are of the form

$$\begin{aligned} \mathbf{L}_x(\mathbb{C}) &= \operatorname{Hom}_{\mathrm{Sch}}(\operatorname{Spec} \mathbb{C}, \mathbf{L}_x) \\ &= \operatorname{Hom}_{\mathrm{Ring}}\left(\operatorname{Sym}(A\langle e\rangle \underset{A}{\otimes} \mathbb{C})^\vee, \mathbb{C}\right) \\ &\cong \operatorname{Hom}_{\mathrm{Mod}}\left((A\langle e\rangle \underset{A}{\otimes} \mathbb{C})^\vee, \mathbb{C}\right) \\ &= A\langle e\rangle \underset{A}{\otimes} \mathbb{C} \\ &= \mathbb{C}\langle e\rangle. \end{aligned}$$

So by choosing a basis e in $H^0(U, L) = A\langle e\rangle$, we can write $a = \lambda e \in \mathbf{L}_x(\mathbb{C})$ for $\lambda \in \mathbb{C}$, then

$$T(a) = (t \cdot p^* e)(a) = t(a)(e \otimes 1) = \lambda e \otimes 1 = \lambda e = a.$$

That is why we call it the "tautological" section, playing a role as a coordinate function like in the affine case, sending everything to itself.

Remark 4.1.10. From the construction, we have

$$p_* \mathscr{O}_{\mathbf{L}} = \mathrm{Sym}(L^\vee) = \mathscr{O}_X \oplus L^{\otimes -1} \oplus L^{\otimes -2} \oplus \cdots \oplus L^{\otimes -(m-1)} \oplus L^{\otimes -m} \oplus \cdots$$

Locally on $U = \operatorname{Spec} A$, it is of the form

$$\mathrm{Sym}\, H^0(U, L^\vee) = \big(A \oplus A\langle t\rangle \oplus A\langle t^2\rangle \oplus \cdots \oplus A\langle t^{m-1}\rangle \oplus A\langle t^m\rangle \oplus \cdots\big) \cong A[t].$$

But pulling it back on Y, since we have $t^m = f \in A$ so it will be

$$A \oplus A\langle t\rangle \oplus A\langle t^2\rangle \oplus \cdots \oplus A\langle t^{m-1}\rangle \cong A[t]/(t^m - f),$$

i.e. global realizing to a canonical isomorphism

$$\pi_* \mathscr{O}_Y = (p|_Y)_* \mathscr{O}_{\mathbf{L}}|_Y = \mathscr{O}_X \oplus L^{\otimes -1} \oplus L^{\otimes -2} \oplus \cdots \oplus L^{\otimes (1-m)}.$$

Remark 4.1.11. This cyclic covering construction is all that is needed for the classical Kodaira vanishing theorem in the next section.

Proposition 4.1.12. (snc divisor behaves well) *As in the above situation, we suppose X and D are smooth. And $\sum D_i$ is a reduced effective divisor on X such that $D + \sum_2^r D_i$ has simple normal crossings, i.e. each component D_i is smooth and in a neighborhood of any point in $D \cup (\bigcup D_i)$ is defined by an equation of the form*

$$z_1 \cdots z_k = 0$$

in local analytic coordinates, for some $k \leq \dim X$.

Then by our construction of the cyclic covering, put $D_i' = \pi^* D_i$, we have that $mD' + \sum_2^r D_i'$ has simple normal crossing support on Y.

Proof. By Remark 1.7 we use the tangent description to verify that it is a snc divisor. Now we pick an arbitrary point p lies in the support. As $D + \sum D_i$ is snc divisor, locally around p there exists a regular system of parameters $\{f, f_2, \ldots, f_n\}$ defining the divisor D in the form $f = 0$ and D_i as $f_i = 0$ for $i = \mathbf{2}, \ldots, k$. By our construction of cyclic covering, we have

$$\mathscr{O}_{Y,q} = \mathscr{O}_{X,p}[t]/(t^m - f) \text{ for } p = \pi(q) \text{ in } X.$$

Now we need to verify that $\{t, f_2, \ldots, f_n\}$ is again a regular system of parameters of $\mathscr{O}_{Y,q}$. As $t^m = f$ in $\mathscr{O}_{Y,q}$, we have

$$m_q = (t, f, f_2, \ldots, f_n)\mathscr{O}_{Y,q} = (t, f_2, \ldots, f_n)\mathscr{O}_{Y,q},$$

that is they are minimal generators as Y is also smooth of dimension equal to $\dim X$.

Here we only need to check that $R = \mathscr{O}_{X,p}[t]/(t^m - f)$ is indeed a local ring, but this is trivial. Since we have the correspondence

$$\big\{\text{maximal ideals of } R\big\} \overset{1:1}{\longleftrightarrow} \big\{ \text{ maximal ideals of } \mathscr{O}_{X,p}[t] \text{ which contain } (t^m - f)\big\},$$

but the maximal ideal on the right must contain m_p, so it reduces to

$$\big\{\text{maximal ideals of } R/m_pR\big\} \overset{1:1}{\longleftrightarrow} \big\{ \text{ maximal ideals of } \mathbb{C}[t] \text{ which contains } (t^m)\big\}$$

as $R/m_pR \cong \mathbb{C}[t]/(t^m)$ for $f \in m_p$, we see R is a local ring with maximal ideal $m_pR + (t)R$ contracted to m_p in $\mathscr{O}_{X,p}$. □

Bloch–Gieseker and Kawamata coverings.

Lemma 4.1.13. *For an arbitrary quasi-projective variety X, an arbitrary line bundle L can be written in the form*

$$L = A \otimes B^\vee,$$

where A and B are very ample line bundles.

Proof. As X is quasi-projective, we can find an immersion

$$X \hookrightarrow Z \hookrightarrow \mathbb{P}^N$$

where X is the an open subvariety of Z, and Z is the closed subvariety of $\mathbb{P}^N$. So X admits a natural very ample line bundle H, which is the pullback of hyperplane bundle of $\mathbb{P}^N$.

Note that on Z which is a projective variety, by Serre's result in [Har77, II, 5.17] the very ampleness can implies ampleness. And on quasi-projective variety X, for very ample line bundle H is the pull back of $\mathscr{O}_Z(1)$. Now for every coherent sheaf $\mathscr{F}$ on X, its extension $\mathscr{F}'$ is a coherent sheaf on Z such that $\mathscr{F}'|_X \cong \mathscr{F}$, by [Har77, II, ex5.15]. So $\mathscr{F} \otimes H^{\otimes n}$ is generated by global sections by $\mathscr{F}' \otimes \mathscr{O}Z(1)^{\otimes n}$ is, we see H is an ample line bundle.

For arbitrary line bundle L and for H is ample, by definition of ampleness there is a suitable $n_1 > 0$, such that $L \otimes H^{\otimes n_1}$ is globally generated. Then by [Har77, II, ex7.5(d)] we see there is a suitable $n > 0$ such that $L \otimes H^{\otimes n}$ is very ample. So we just choose $A = L \otimes H^{\otimes n}$ and $B = H^{\otimes n}$ is fine. □

Theorem 4.1.14. (BG coverings) *Let X be a quasi-projective variety of dimension n, let L be a line bundle on X, and fix any positive integer m. Then there exists a variety Y, with a finite flat surjective morphism $f : Y \to X$ and a line bundle N on Y such that $f^*L = N^{\otimes m}$.*

If X is non-singular then we can take Y to be non-singular as well. In this case, given a simple normal crossing $D = \sum D_i$ on X, we can arrange that each f^*D_i is smooth and its pullback f^*D is a simple normal crossing divisor on Y.

Finally if $\dim X \geq 2$, then we can further ensure that the preimage $D_i' = f^*D_i$ of each irreducible components of D is irreducible.

Proof. **Step 1:** We first make some reductions. Suppose that the result is known for the pullback of the hyperplane bundle under a fibre-finite morphism $X \to \mathbb{P}^r$. By above lemma, we write an arbitrary line bundle L as the difference $L = A \otimes B^\vee$ of two such bundles, as an immersion is trivially fibre-finite. Dealing with them respectively, first we apply the results to A, find a variety Y_1 with a finite flat surjective morphism

$$f_1 : Y_1 \longrightarrow X$$

satisfying $f_1^*A \cong N_A^{\otimes m}$. Pulling back B via f_1, where B comes from a fibre-finite morphism $X \to \mathbb{P}^N$. Then apply the results to f_1^*B, since $Y_1 \to \mathbb{P}^N$ is also fibre-finite, we can find a variety Y with a finite flat surjective morphism

$$f_2 : Y \longrightarrow Y_1$$

satisfies $f_2^*f_1^*B \cong N_B^{\otimes m}$. Now for finite flat surjective morphism $f = f_1 \circ f_2$, we have

$$\begin{aligned} f^*L &= f_2^*(f_1^*A) \otimes f_2^*(f_1^*B^\vee) \\ &\cong f_2^*N_A^{\otimes m} \otimes N_B^{\otimes -m} \\ &= (f_2^*N_A \otimes N_B^\vee)^{\otimes m}, \end{aligned}$$

such that $N = f_2^*N_A \otimes N_B^\vee$ is what we want.

Now, we begin to prove the special result we need.

Step 2: We may assume that there is a finite-to-one morphism $\phi : X \to \mathbb{P}^r$ such that $L = \phi^*\mathscr{O}_{\mathbb{P}^r}(1)$. Next we fix a branched covering

$$\begin{aligned} \nu : \mathbb{P}^r &\longrightarrow \mathbb{P}^r \\ [x_0 : \cdots : x_r] &\longmapsto [x_0^m : \cdots : x_r^m] \end{aligned}$$

with $\nu^*\mathscr{O}_{\mathbb{P}^r}(1) = \mathscr{O}_{\mathbb{P}^r}(m)$. This is the "Fermat covering". Given an element in a group variety

$$g \in G := \mathrm{GL}(r+1, \mathbb{C}),$$

acting on $\mathbb{P}^r$ as an automorphism where $(\mathbb{P}^r, G)$ is a homogeneous space, and denote $\nu_g = g \circ \nu : \mathbb{P}^r \to \mathbb{P}^r$. Define

$$Y := Y_g = X \underset{\mathbb{P}^r}{\times} \mathbb{P}^r$$

to be the fibre product of X and $\mathbb{P}^r$ with respect to ϕ and ν_g (not to ϕ and $id_{\mathbb{P}^r}$!), so that we have the Cartesian diagram

$$\begin{array}{ccc} Y_g & \xrightarrow{\bar{\phi}} & \mathbb{P}^r \\ {\scriptstyle f=f_g}\downarrow & & \downarrow{\scriptstyle \nu_g} \\ X & \xrightarrow[\phi]{} & \mathbb{P}^r. \end{array}$$

The indicated morphism $f : Y_g \to X$ is finite and flat since ν_g is, and they're stable under base change. By construction, we have

$$\begin{aligned} f^*L &= f^*\phi^*\mathscr{O}_{\mathbb{P}^r}(1) \\ &= \bar{\phi}^*\nu_g^*\mathscr{O}_{\mathbb{P}^r}(1) \\ &= \bar{\phi}^*\mathscr{O}_{\mathbb{P}^r}(m) \\ &= (\bar{\phi}^*\mathscr{O}_{\mathbb{P}^r}(1))^{\otimes m}. \end{aligned}$$

We will show that if $g \in G$ is a general point, then $f_g : Y_g \to X$ has all stated properties.

Note first that Y_g is the inverse image of the graph $\Gamma_{g^{-1}} \subseteq \mathbb{P}^r \times \mathbb{P}^r$ of g^{-1} under the morphism

$$\phi \times \nu : X \times \mathbb{P}^r \longrightarrow \mathbb{P}^r \times \mathbb{P}^r,$$

so [Laz04a, 3.3.10] implies that for general g, we have Y_g is irreducible.

For reducedness, we can *certainly* assume that for general $g \in G$, Y_g is generically reduced, i.e. the local ring $\mathscr{O}_{Y_g,\eta}$ is reduced at the generic point $\eta \in Y_g$. In detail, the generic reducedness of Y_g is coming from the smoothness. Indeed, consider the variety $W = X \underset{\mathbb{P}^r}{\times} (G \times \mathbb{P}^r)$ with the acting morphism

$$\begin{aligned} m : G \times \mathbb{P}^r &\longrightarrow \mathbb{P}^r \\ (g, y) &\longmapsto g \cdot \nu(y), \end{aligned}$$

and a commutative diagram

$$\begin{array}{ccccc} W & \xrightarrow{} & G \times \mathbb{P}^r & \xrightarrow[\mathrm{pr}_1]{} & G \\ {\scriptstyle n}\downarrow & & {\scriptstyle m}\downarrow & & \\ X & \xrightarrow[\phi]{} & \mathbb{P}^r & & \end{array} \qquad (b : W \to G \times \mathbb{P}^r)$$

such that the morphism $b : W \to G$ has fibre $b^{-1}(g) = Y_g$. Then we pick the smooth locus $b_{\mathrm{reg}} : W_{\mathrm{reg}} \to G$ and applying the generic smoothness, see [Har77, III, 10.7], we have a smooth morphism

$$b_{\mathrm{reg}} : b_{\mathrm{reg}}^{-1}U \longrightarrow U$$

over an open dense subset $U \subseteq G$, that is for general $g \in G$, the fibre $b_{\mathrm{reg}}^{-1}(g) = Y_g \cap W_{\mathrm{reg}}$ is a non-singular variety, see [Har77, III, 10.0.3]. So the local ring at the generic point $\eta \in Y_g \cap W_{\mathrm{reg}} \subseteq Y_g$ is reduced since regular local ring is U.F.D.

Lemma 4.1.15. *If $f : V \longrightarrow W$ is a finite flat morphism of schemes with W integral and V generically reduced, then V must be reduced everywhere.*

Proof. Locally we regard it as $\operatorname{Spec} B \to \operatorname{Spec} A$ which A is domain such that $A \hookrightarrow B$ is a flat ring extension. By generically reduced, if we extend slightly over an open subset $D(\delta)$, there exist a non-zero element $\delta \in A$ such that B_δ is reduced.

Then for an arbitrary nilpotent element $\phi \in B$, ϕ must die in B_δ, i.e. $\delta^m \phi = 0$ for some $m > 0$. By the multiplication δ^m gives an injective map $A \xrightarrow{\cdot\delta^m} A$, and by flatness we twist it via $- \underset{A}{\otimes} B$ it must remain injective as a multiplication $B \xrightarrow{\cdot\delta^m} B$. So $\phi = 0$, says that B is reduced. □

The reducedness of Y is immediately, with a finite flat morphism $f_g : Y_g \to X$.

Given a reduced irreducible Cartier divisor $E \subseteq X$, the same argument applied to E in place of X, shows that we can arrange for f^*E to be reduced and irreducible provided that $\dim X \geq 2$, so that $\dim E \geq 1$.

Step 3: Now suppose that X is non-singular, we draw the above diagram again

$$\begin{array}{ccccc} W & \longrightarrow & G \times \mathbb{P}^r & \xrightarrow{\mathrm{pr}_1} & G \\ {\scriptstyle n}\downarrow & & \downarrow{\scriptstyle m} & & \\ X & \xrightarrow[\phi]{} & \mathbb{P}^r. & & \end{array} \qquad (b : W \to G)$$

As in [Har77, III, 10.8] we see the morphism m is smooth, so is n by after base change. As X is smooth over $\operatorname{Spec} \mathbb{C}$, we see W is a non-singular variety. Therefore use generic smoothness again, Y_g is a non-singular variety.

Similarly, for $E \subseteq X$ is a smooth divisor, then for general $g \in G$ its preimage in $Y = Y_g$ will likewise be non-singular.

Step 4: Finally we assume X is non-singular with a given snc divisor $D = \sum D_i$ on X. We have already seen that we can arrange that each $D_i' = f^*D_i$ to be reduced and irreducible if $\dim X \geq 2$. It remains to show that we can take f^*D to be a snc

divisor. To this end, note that each component D_i is smooth and any collection of the D_i meet transversely or not at all. By our argument in *Step 3*, the same argument applied to D_i in place of X, shows that the preimage

$$F_i = n^{-1}(D_i) = D_i \underset{\mathbb{P}^r}{\times} \mathbb{P}^r \subseteq W$$

is non-singular for general $g \in G$, for each i.

Now for meeting transversely, we use the tangent description. At the underlying space X, for each point $x \in \bigcap^k D_i$ it is locally cutting by $f_1 \cdots f_k = 0$ with $\{f_1, \ldots, f_k\}$ is a subset of regular system of parameters in $\mathscr{O}_{X,x}$, by simple normal crossings. Via the smooth morphism $n : W \to X$ *of relative dimension* d, with the regular local ring $\mathscr{O}_{W,y}$ such that $x = n(y)$, has the ring homomorphism

$$\begin{aligned} \mathscr{O}_{X,x} &\longrightarrow \mathscr{O}_{W,y} \\ f_i &\longmapsto u_i. \end{aligned}$$

Notice that it is flat and $\mathscr{O}_{W,y}/m_x\mathscr{O}_{W,y} = \mathscr{O}_{n^{-1}(x),y}$ is a regular local ring of dimension d. Now by [Mat86, 15.1] and [Mat86, 14.2] we see $\{u_1, \ldots, u_k\}$ is a subset of regular system of parameters of $\mathscr{O}_{W,y}$, which locally defines the point $y \in \bigcap^k F_i$. Therefore we see, any collection of $\{F_i\}$ meet transversely or not at all.

The following lemma shows that if $g \in G$ is general, then the same statement holds for the fibres of F_i over g. But these are the components of f^*D, and this implies that f^*D has simple normal crossings. □

Lemma 4.1.16. *Let $b : W \longrightarrow G$ be a morphism of non-singular varieties, and suppose given a finite collection of smooth varieties $V_i \subseteq W$ meeting transversely. Then for general $g \in G$, the fibres $(V_i)_g$ of the V_i over g are smooth and meet transversely (or not at all) in W_g.*

Proof. For the "simple" part, easily see that if we applying the generic smoothness on each V_i and their intersections, then for general fibre the components are indeed smooth. Now the "normal crossing" part follows immediately. □

By combining cyclic covering and BG covering, we can show that given an snc divisor $D = \sum D_i$, we can pass to a covering on which each of the components D_i becomes divisible by any pre-assigned integer $m_j > 0$.

Proposition 4.1.17. (Kawamata coverings) *Let X be a non-singular quasi-projective variety, let $D = \sum_{i=1}^{r} D_i$ be a snc divisor on X, and fix positive integers $m_1, \ldots, m_r > 0$. Then there exists a smooth variety Y and a finite flat covering $h : Y \longrightarrow X$ with the property that*

$$h^* D_i = m_i D_i'$$

for some smooth divisor D_i' on Y, where $D' = \sum D_i'$ has simple normal crossings.

Proof. By induction on the number of components of D, it suffices to produce a covering $h : Y \longrightarrow X$ where

$$h^* D_1 = m_1 D_1',\ h^* D_i = D_i' \text{ for } 2 \leq i \leq r,$$

and $\sum_1^r D_i'$ is snc divisor.

To this end, first take a Bloch-Gieseker covering $f : X_1 \longrightarrow X$ to arrange that

$$D_1^+ := f^* D_1 \sim_{lin} m_1 A_1^+$$

for some (possibly non-effective) divisor A_1^+ on X_1, while Theorem 1.14 ensures that $f^* D$ still has simple normal crossings, that is $f^* D_1$ is smooth divisor. Then we take an m_1-fold cyclic covering $\pi : Y \longrightarrow X_1$, branched along the effective divisor D_1^+, so that $\pi^* D_1^+ = m_1 D_1'$ for a smooth divisor D_1'. Now set $h = \pi \circ f : Y \longrightarrow X$, and for $i \geq 2$ we write $D_i' = h^* D_i$. It follows from Proposition 1.12 that $m_1 D_1' + \sum_2^r D_i'$ has simple normal crossing support, i.e. $\sum D_i'$ is snc divisor. □

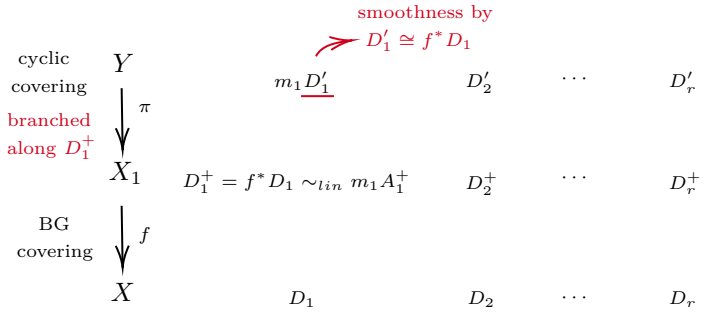


Figure 4.2.: Inductive construction.

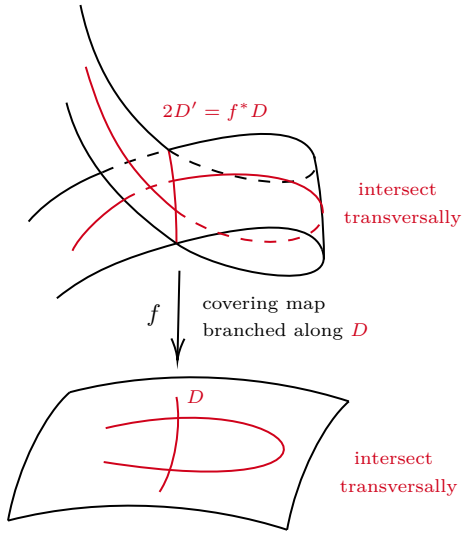


Figure 4.3.: Informal view of covering branched along the divisor D with upper space is singular, which I think it can not be drawn in 3-dimensional space.

Injectivity We now turn to use these covering constructions to establish vanishing theorems. Starting with a line bundle L on X for which one wants to prove a vanishing theorems, the general strategy is to pull L back to a covering $f : Y \longrightarrow X$ on which the problem simplifies in some way. The next lemma shows that it will be enough to get the vanishing on Y:

Lemma 4.1.18. *Let $f : Y \longrightarrow X$ be a finite surjective morphism of complex algebraic varieties, with X normal, and let E be a vector bundle on X. Then the natural homomorphism*

$$H^i(X, E) \longrightarrow H^i(Y, f^*E)$$

induced by f is injective. In particular, if

$$H^i(Y, f^*E) = 0 \text{ for some } i \geq 0,$$

then $H^i(X, E) = 0$.

Proof. First we have

$$\begin{aligned} H^i(Y, f^*E) &\cong H^i(X, f_*f^*E) \\ &\cong H^i(X, E \otimes f_*\mathscr{O}_Y) \end{aligned}$$

thanks to the projection formula and the finiteness of f, see [Har77, II, ex8.2]. On the

other hand, consider $\mathbb{C}(Y)$ as a finite $\mathbb{C}(X)$-vector space with linear transformations

$$\begin{aligned} \mathbb{C}(Y) &\longrightarrow \mathbb{C}(Y) \\ x &\longmapsto b \cdot x, \end{aligned}$$

where this linear transformation is represented by a matrix with entries in $\mathbb{C}(X)$, and we define the trace $\mathrm{Tr}(b)$ as the trace of this matrix. It induces a surjective linear map

$$\begin{aligned} \mathrm{Tr} : \mathbb{C}(Y) &\longrightarrow \mathbb{C}(X) \\ b &\longmapsto \mathrm{Tr}(b). \end{aligned}$$

Locally see, $f : \operatorname{Spec} B \longrightarrow \operatorname{Spec} A$ with B is a finite A-module, i.e. integral over A. So for any $b \in B$, we have that $\mathrm{Tr}(b) \in \mathbb{C}(X)$ is also integral over A. But since X is normal, we have $\mathrm{Tr}(b) \in A$, that is $\mathrm{Tr}(B) \subseteq A$. Gluing them together to get a trace map

$$\mathrm{Tr}_{Y/X} : f_* \mathscr{O}_Y \longrightarrow \mathscr{O}_X.$$

Consider the short exact sequence

$$0 \to A \hookrightarrow B \to \mathrm{coker} \to 0$$

where $\mathrm{Tr}_{Y/X}$ locally defines a left inverse (after suitable modification, as we are over char 0 field), making the above sequence split. Therefore, the inclusion $\mathscr{O}_X \hookrightarrow f_* \mathscr{O}_Y$ also splits, so is $E \hookrightarrow E \otimes f_* \mathscr{O}_Y$. Hence $H^i(X, E)$ embeds as a summand of

$$H^i(X, E \otimes f_* \mathscr{O}_Y) \cong H^i(Y, f^* E),$$

which states the injectivity. □

4.2 Kodaira and Nakano Vanishing Theorems

In this section we prove the classical vanishing theorems of Kodaira and Nakano. To prove these vanishing theorems, first we need to connect the singular cohomology of non-singular projective varieties with the cohomology of the constant sheaf. The main fundamental result is the *de Rham Theorem*, establishing a canonical isomorphism

$$H^i_{\text{sing}}(X, G) \cong H^i(X, G)$$

over a locally contractible topological space (e.g. smooth manifolds), where G is an arbitrary commutative ring. See [Voi02, 4.47] or [GH78, p.44]. On the other hand, we need an important result which is called *Lefschetz Hyperplane Theorem*, the reader can find it in [Laz04a, 3.1.17], [Voi03, 1.23] or [GH78, p.156], says that

Theorem 4.2.1. (Lefschetz hyperplane theorem) *Let X be a non-singular complex projective variety of dimension n, and let D be any effective ample divisor on X. Then the restriction*

$$r_i : H^i_{\text{sing}}(X, \mathbb{Z}) \longrightarrow H^i_{\text{sing}}(D, \mathbb{Z})$$

is an isomorphism for $i \leq n-2$ and is injective when $i = n-1$.

Thanks to de Rham theorem there is no issue to change the coefficients and turning into $\mathbb{C}$-sheaf cohomology, enable us to use the *Hodge decomposition* and the *Dolbeault isomorphism theorem*. Reader can find them in [GH78] or [Voi02].

Theorem 4.2.2. (Hodge decomposition) *For a compact Kähler manifold X, the cohomology of the constant complex sheaf satisfies*

$$\begin{cases} H^i(X, \mathbb{C}) \cong \bigoplus\limits_{p+q=i} H^{p,q}(X), \\ H^{p,q}(X) = \overline{H^{q,p}(X)}. \end{cases}$$

Where $H^{p,q}(X) \subseteq H^i(X, \mathbb{C})$ be the set of classes which are representable by a closed form α which is of type (p, q) at every point.

Moreover since $\Delta_d = 2\Delta_{\bar{\partial}}$, we have $H^{p,q}(X) \cong H^{p,q}_{\bar{\partial}}(X)$.

Notice that by construction of *Fubini-Study metric* every complex projective manifold is a compact Kähler manifold.

Theorem 4.2.3. (Dolbeault theorem) *For a complex manifold X, we have*

$$H^q(X, \Omega^p_X) \cong H^{p,q}_{\bar{\partial}}(X).$$

Now we trun to prove an application of Lefschetz hyperplane theorem:

4.2A Kodaira Vanishing Theorem

Lemma 4.2.4. *Let X be a smooth complex projective variety of dimension n, and let $D \subseteq X$ be a smooth effective ample divisor, Denote by*

$$r_{p,q} : H^q(X, \Omega_X^p) \longrightarrow H^q(D, \Omega_D^p)$$

the natural maps determined by restriction. Then

$$r_{p,q} \text{ is } \begin{cases} \text{bijective} & \text{for } p+q \leq n-2, \\ \text{injective} & \text{for } p+q = n-1. \end{cases}$$

Proof. By combining above three theorems. □

We shall now turn to the central topic.

Theorem 4.2.5. (Kodaira vanishing) *Let X be a smooth complex projective variety of dimension n, and let A be an ample divisor on X. Then*

$$H^i\big(X, \mathscr{O}_X(K_X + A)\big) = 0 \text{ for } i > 0.$$

Or equivalently, due to Serre duality,

$$H^j\big(X, \mathscr{O}_X(-A)\big) = 0 \text{ for } j < n.$$

Proof. Since A is ample, by Bertini's theorem there exists a smooth divisor $D \in |mA|$ for $m \gg 0$. Let $\pi : Y \longrightarrow X$ be the m-fold cyclic covering branched along D constructed in Proposition 1.8, and set $A' = \pi^* A$. By construction there is a smooth ample divisor D' such that $mD' = \pi^* D \in |mA'|$, where the ampleness comes from the finiteness of π. So $D' \in |A'|$, and Injectivity lemma at 1.18 says here is an injective map

$$H^j\big(X, \mathscr{O}_X(-A)\big) \hookrightarrow H^j\big(Y, \pi^* \mathscr{O}_X(-A)\big) = H^j\big(Y, \mathscr{O}_Y(-D')\big).$$

So it suffices to prove that $H^j(Y, \mathscr{O}_Y(-D')) = 0$ for $j < n$ for a smooth ample effective divisor $D' \subseteq Y$. Thus we are reduced to establishing the vanishing under the additional hypothesis that the given ample divisor is effective and non-singular.

Now suppose that $D \subseteq Y$ is a smooth ample divisor. The case of $p = 0$ and $q = j$ of Lemma 2.4 asserts that the restriction map

$$H^j(X, \mathscr{O}_X) \longrightarrow H^j(D, \mathscr{O}_D)$$

is an isomorphism when $j \leq n-2$ and is injective when $j = n-1$. But then the short exact sequence

$$0 \longrightarrow \mathscr{O}_X(-D) \longrightarrow \mathscr{O}_X \longrightarrow \mathscr{O}_D \longrightarrow 0$$

shows that $H^j(X, \mathscr{O}_X(-D)) = 0$ when $j \leq n-1$, as required. □

4.2B Differential Forms with Logarithmic Poles

Observe that the proof above just uses a very small part of the information provided by Lemma 2.4. The remaining instances lead to Nakano's generalization of Kodaira vanishing. But before give the proof, we need investigate the differential forms with logarithmic poles, see [EV92].

For a simple normal crossing divisor $D = \sum_1^r D_i$ on a non-singular variety X of dimension n, we write $\tau : U = X - D \longrightarrow X$ and

$$\Omega_X^p(*D) = \varinjlim_{\nu} \Omega_X^p(\nu \cdot D) = \tau_* \Omega_U^p,$$

that is $\Omega_X^p(*D)$ consists of differential forms which allow arbitrary degree poles along D. Of course $(\Omega_X^\bullet(*D), d)$ is a cochain complex.

Definition 4.2.6. In above situation, we write $\Omega_X^p(\log D)$ to be the subsheaf of $\Omega_X^p(*D)$ of *differential forms with logarithmic poles* on D, i.e. for each open subset $V \subseteq X$, it sends

$$H^0(V, \Omega_X^p(\log D)) = \{\alpha \in H^0(V, \Omega_X^p(*D)) \mid \alpha \text{ and } d\alpha \text{ has simple poles along } D\}.$$

Proposition 4.2.7. (Properties of differential log-forms) *We have the following properties of* $\Omega_X^p(\log D)$*:*

(a)

$$(\Omega_X^\bullet(\log D), d) \hookrightarrow (\Omega_X^\bullet(*D), d)$$

is a subcomplex.

(b)

$$\Omega_X^p(\log D) = \bigwedge^p \Omega_X^1(\log D)$$

(c) $\Omega_X^p(\log D)$ *is locally free. More precisely, for each point* $x \in X$ *such that* $x \in D_i$ *for* $i = 1, \ldots, s$ *and* $x \notin D_i$ *for* $i = s+1, \ldots, r$*, via the tangent description we choose parameters in* m_p *of* $f_1, \ldots, f_n$*, such that* D_i *is defined by* $f_i = 0$ *for* $i = 1, \ldots, s$*. Let us write*

$$\delta_i = \begin{cases} \frac{df_i}{f_i} & \text{if } i \leq s \\ df_i & \text{if } i > s \end{cases}$$

and for $I = \{i_1, \ldots, i_p\} \subseteq \{1, \ldots, n\}$ *with* $i_1 < i_2 < \cdots < i_p$

$$\delta_I = \delta_{i_1} \wedge \delta_{i_2} \wedge \cdots \wedge \delta_{i_p}.$$

Then $\{\delta_I \big| \#I = p\}$ *is a free system of generators for* $\Omega_X^p(\log D)$*.*

Proof. For these three properties:

(a) is trivial.

(b) is coming from the explicit form of the generators given in (c).

(c) As $d(df_i/f_i) = d(f_i^{-1}) \wedge df_i + f_i^{-1} \wedge d(df_i) = 0$, each δ_i is a closed form, so d won't give rise a new pole along D. Therefore δ_I is indeed a local section of $\Omega_X^p(\log D)$. Now we write the linear combination of δ_I by regular sections

$$\omega = \sum_I h_I \delta_I$$

which generates an $\mathscr{O}_X$-module Ω. By Leibniz rule,

$$d\omega = \sum_I dh_I \wedge \delta_I$$

still has simple poles along D, so the module Ω is contained in $\Omega_X^p(\log D)$. Easily to see that Ω is locally free, as each $\frac{1}{f_i}$ has no possibilities to become zero. Now in order to show $\Omega = \Omega_X^p(\log D)$, by symmetry, it is enough to consider the case $s = 1$. Each local section $\alpha \in \Omega_X^p(*D)$ can be written as

$$\alpha = \alpha_1 + \alpha_2 \wedge \frac{df_i}{f_i},$$

where α_1 is a local section in $\Omega_X^p(*D)$ and α_2 is in $\Omega_X^{p-1}(*D)$, each of them are in the subsheaf generated over $\mathscr{O}_X(*D)$ by wedge products of $df_2, \ldots, df_n$. So for $\alpha \in \Omega_X^p(\log D)$ which says that α and $d\alpha$ only have simple poles along D, implies that

$$f_1 \cdot \alpha = f_1 \cdot \alpha_1 + \alpha_2 \wedge df_1 \in \Omega_X^p \text{ and } f_1 d\alpha = f_1 d\alpha_1 + d\alpha_2 \wedge df_1 \in \Omega_X^{p+1}$$

are regular forms, Hence α_2 and $f_1\alpha_1$ are without poles. Since

$$\begin{aligned} d(f_1\alpha_1) &= df_1 \wedge \alpha_1 + f_1 d\alpha_1 \\ &= df_1 \wedge \alpha_1 + f_1 d\alpha - d\alpha_2 \wedge df_1, \end{aligned}$$

implies that the form $df_1 \wedge \alpha_1$ also has no poles, which implies $\alpha_1 \in \Omega_X^p$. That is, $\alpha \in \Omega$ as required. □

Notice in the analysis above, each D_i need not be a prime divisor, allowing it to have several connected components, since locally at the point it will always be cut out by a single function. In fact, in an snc divisor $\sum a_i D_i$ with different connected components sharing the same coefficent a_i, will be denoted together by D_i.

Using the notation from (c), we can define

$$\mathrm{Res}_D : \Omega_X^1(\log D) \longrightarrow \bigoplus_{i=1}^{s} \mathscr{O}_{D_i},$$

by locally sending

$$\mathrm{Res}_D : \sum_{i=1}^{n} a_i\delta_i \longmapsto \sum_{i=1}^{s} a_i|_{D_i}$$

where $\{a_i\}_{i=1}^s$ are the coefficients of $\left\{\frac{df_i}{f_i}\right\}_{i=1}^s$ in $\mathscr{O}_X$.

For $p \geq 1$ we have correspondingly a map

$$\mathrm{res}_{D_1} : \Omega_X^p(\log D) \longrightarrow \Omega_{D_1}^{p-1}\big(\log(D-D_1)|_{D_1}\big)$$

is locally given by: for φ is a local section of $\Omega_X^p(\log D)$, we can write

$$\varphi = \varphi_1 + \varphi_2 \wedge \frac{df_1}{f_1}$$

where φ_1 lies in the span of the δ_I with $1 \notin I$, and $\varphi_2 = \sum\limits_{1\in I} a_I\delta_{I\setminus\{1\}}$. Then

$$\mathrm{res}_{D_1} : \varphi \longmapsto \varphi_2|_{D_1} = \sum a_I\delta_{I\setminus\{1\}}|_{D_1}.$$

Of course, res_{D_i} will denotes the corresponding map for the i-th component of D.

Finally, the natural restriction of differential forms gives

$$\rho_{D_1} : \Omega_X^p\big(\log(D-D_1)\big) \longrightarrow \Omega_{D_1}^p\big(\log(D-D_1)|_{D_1}\big).$$

Since the sheaf on the left hand side is locally generated by

$$\{f_1\cdot\delta_I \big| 1\in I\} \text{ and } \{\delta_I\big|1\notin I\},$$

we can describe ρ_{D_1} locally by

$$\rho_{D_1} : \sum_{1\in I} f_1a_I\delta_I + \sum_{1\notin I} a_I\delta_I \longmapsto \sum_{1\notin I} a_I\delta_I|_{D_1}.$$

Similarly, ρ_{D_i} will correspond to the i-th component.

Thus, for these three maps we have the following properties.

Proposition 4.2.8. *We have the following three exact sequences:*

(a) $0 \longrightarrow \Omega_X^1 \longrightarrow \Omega_X^1(\log D) \overset{\mathrm{Res}_D}{\longrightarrow} \bigoplus\limits_{i=1}^{r} \mathscr{O}_{D_i} \longrightarrow 0$

(b) $0 \longrightarrow \Omega_X^p\big(\log(D-D_1)\big) \longrightarrow \Omega_X^p(\log D) \overset{\mathrm{res}_{D_1}}{\longrightarrow} \Omega_{D_1}^{p-1}\big(\log(D-D_1)|_{D_1}\big) \longrightarrow 0$

(c) $0 \longrightarrow \Omega_X^p(\log D)(-D_1) \longrightarrow \Omega_X^p\big(\log(D-D_1)\big) \overset{\rho_{D_1}}{\longrightarrow} \Omega_{D_1}^p\big(\log(D-D_1)|_{D_1}\big) \longrightarrow 0$

Proof. For (a), the surjectivity of Res_D is obvious, while notice that locally the prescript "r" is determined by the situation of intersection around the point $x \in X$. And the exactness is from the residue map Res_D which is just taking the coefficients with simple poles.

For (b), also obviously.

For (c), should recall that the local sections in $\mathscr{O}_X(-D_1)$ will vanish along D_1, so at this time we have $a_I\delta_I|_{D_1} = 0 \cdot \delta_I|_{D_1} = 0$. □

Now we deal with the simplest case:

Corollary 4.2.9. *Assume that D is a smooth divisor, i.e. $r = 1$ in above situation.*

(a) We have the following exact sequences

$$0 \longrightarrow \Omega_X^p \longrightarrow \Omega_X^p(\log D) \xrightarrow{\mathrm{res}_D} \Omega_D^{p-1} \longrightarrow 0$$
$$0 \longrightarrow \Omega_X^p(\log D)(-D) \longrightarrow \Omega_X^p \xrightarrow{\rho_D} \Omega_D^p \longrightarrow 0$$

(b) Let $\pi : Y \longrightarrow X$ be the m-fold cyclic covering branched along D, constructed as in Proposition 1.8, and let $D' \subseteq Y$ be the divisor mapping isomorphically to D, so that $\pi^*D = mD'$. Then

$$\pi^*\big(\Omega_X^p(\log D)\big) = \Omega_Y^p(\log D').$$

Proof. (a) is directly follows from proposition 2.8.

(b) is trivial, since locally see $t^m = f$, we have

$$\pi^*\left(\frac{df}{f}\right) = \frac{dt^m}{t^m} = \frac{mt^{m-1}dt}{t^m} = m \cdot \frac{dt}{t},$$

then use the structure theorem in Proposition 2.7(c). □

4.2C Nakano Vanishing Theorem

Theorem 4.2.10. (Nakano vanishing) *Let X be a smooth complex projective variety of dimension n, and let A be an ample divisor on X. Then*

$$H^q\big(X, \Omega_X^p \otimes \mathscr{O}_X(A)\big) = 0 \text{ for } p + q > n.$$

Or equivalently, by Serre duality,

$$H^s\big(X, \Omega_X^r \otimes \mathscr{O}_X(-A)\big) = 0 \text{ for } r + s < n,$$

where $s = n - q$ and $r = n - p$.

When $p = n$ this is the classical Kodaira vanishing theorem. As before, the strategy is to use cyclic coverings to reduce to the case in which A is represented by a smooth effective divisor D.

Proof. Also, we will establish the second vanishing after Serre duality. Since A is ample, by Bertini's theorem there exists a smooth divisor $D \in |mA|$. We claim first that it suffices to prove the vanishing

$$H^s\big(X, \Omega^r_X(\log D) \otimes \mathscr{O}_X(-A)\big) = 0 \text{ for } r + s < n. \tag{4.1}$$

In fact, by using induction on $\dim X$: for $\dim X = 1$ case, $r + s < 1$ has only one possibility is

$$H^0\big(X, \mathscr{O}_X(-A)\big) = 0,$$

but this is obviously since $\deg(-A) < 0$ by [Har77, IV, 3.3], and then by [Har77, IV, 1.2]. Now we suppose Nakano vanishing is holds for $\dim X - 1$ case, especially for D. That is,

$$H^s\big(D, \Omega^{r-1}_D(-A)\big) = 0 \text{ for each } (r-1) + s < n - 1,$$

notice that of course for prescript $s-1$ is also fine. Then for each fixed pair (r, s) with $r + s < n$, we have the exact sequence

$$0 \longrightarrow \Omega^r_X(-A) \longrightarrow \Omega^r_X(\log D)(-A) \longrightarrow \Omega^{r-1}_D(-A) \longrightarrow 0.$$

Which induces a long exact sequence

$$\cdots \to H^{s-1}\big(D, \Omega^{r-1}_D(-A)\big) \to H^s\big(X, \Omega^r_X(-A)\big) \to H^s\big(X, \Omega^r_X(\log D)(-A)\big) \to \cdots$$

Thus if the left hand side vanishes, then the injectivity of the nearby map gives the desired vanishing of $H^s\big(X, \Omega^r_X(-A)\big)$. Reader should be careful that when $r = 0$, the prescript is the wedge degree (so is zero sheaf), not the dual!

Now for vanishing (4.1), we consider the m-fold cyclic covering $\pi : Y \longrightarrow X$ branched along D. According to injectivity lemma 1.18, it suffices to prove that

$$H^s\Big(Y, \pi^*\big(\Omega^r_X(\log D) \otimes \mathscr{O}_X(-A\,)\big)\Big) = 0 \text{ for } r + s < n.$$

Now $\pi^*\big(\Omega^r_X(\log D)\big) = \Omega^r_Y(\log D')$ thanks to Corollary 2.9(c), and

$$mD' = \pi^* D \in |m\pi^* A|$$

gives that $\pi^* A$ is represented by the smooth prime divisor $D' \subseteq Y$ (primeness via D' is mapping isomorphically to D). So the question is reduced to the vanishing

$$H^s\big(Y, \Omega^r_Y(\log D') \otimes \mathscr{O}_Y(-D')\big) = 0 \text{ for } r + s < n.$$

It is immediately by Corollary 2.9 and Lemma 2.4, as $r + s \leq n - 1$. □

4.3 Vanishing for Big and Nef Line Bundles

This section is devoted to the basic vanishing theorem, due to Kawamata and Viehweg, for big and nef divisors.

4.3A Statement and Proof of the Theorem

We can weaken the ampleness hypothesis in the Kodaira vanishing, which also holds for nef and big line bundles. This result was independently discovered by Kawamata [Kaw82] and Viehweg [Vie82].

Theorem 4.3.1. (Kodaira vanishing for big and nef divisors) *Let X be a smooth complex projective variety of dimension n, and D is a nef and big divisor on X. Then*

$$H^i\big(X, \mathscr{O}_X(K_X + D)\big) = 0 \text{ for } i > 0.$$

Or equivalently, by Serre duality,

$$H^j\big(X, \mathscr{O}_X(-D)\big) = 0 \text{ for } j < n.$$

Remark 4.3.2. Recall from Theorem 3.2.15 that a nef divisor D is big if and only if its top self-intersection is strictly positive, i.e. $(D^n) > 0$. However, we do not use this fact in the proof.

Remark 4.3.3. The paper [Kaw82, Theorem 2] and [Vie82, Proposition 2.3] of Kawamata and Viehweg also contain the fundamental vanishing theorem for $\mathbb{Q}$-divisors, that we will discuss in the next section.

Example 4.3.4. The analogue of Nakano vanishing can fail for big and nef divisors. For instance, let $X = \mathrm{Bl}_p\mathbb{P}^3$ be the blowing-up of $\mathbb{P}^3$ at a point, and let $H \cong \mathbb{P}^2$ be the pullback of a general hyperplane of to X. Notice that H is nef and big, but

$$H^1(X, \Omega^1_X \otimes \mathscr{O}_X(-H)) \neq 0.$$

In fact, following from [Voi02, 7.31] we have an isomorphism

$$H^k\big(\mathbb{P}^3, \mathbb{C}\big) \oplus H^{k-2}\big(\{p\}, \mathbb{C}\big) \oplus H^{k-4}\big(\{p\}, \mathbb{C}\big) \cong H^k\big(X, \mathbb{C}\big).$$

Then aim to compute the dimension of $H^1(X, \Omega^1_X)$, we use the Hodge decomposition and pick $k = 2$:

$$\begin{aligned} H^2\big(X, \mathbb{C}\big) &\cong H^2\big(\mathbb{P}^3, \mathbb{C}\big) \oplus H^0\big(\{p\}, \mathbb{C}\big) \\ &\cong H^{0,2}(X) \oplus H^1(X, \Omega^1_X) \oplus H^{2,0}(X), \end{aligned}$$

where $H^2\big(\mathbb{P}^3,\mathbb{C}\big)\cong\mathbb{C}$ by [Voi02, 7.14], and trivially $H^0\big(\{p\},\mathbb{C}\big)\cong\mathbb{C}$.

Thus we see $h^2(X)=2$. Now we then compute $h^{0,2}(X)$ and $h^{2,0}(X)$. Recall that the plurigenus $h^{p,0}$ is a birational invariant, see [Har77, ex8.8]. So by symmetry of Hodge structure, we just need to compute $h^{2,0}(\mathbb{P}^3)$, but on $\mathbb{P}^n$ the Hodge structure is trivial, only has (p,p)-type, see [Voi02, 7.14] or [Har77, III, ex7.3]. Therefore

$$\dim H^1(X,\Omega^1_X)=2.$$

Also for $H\cong\mathbb{P}^2$, we have $\dim H^1(H,\Omega^1_H)=1$ again by [Voi02, 7.14]. Using the short exact sequences in Corollary 2.9, we can therefore get

$$\cdots\to H^1\big(X,\Omega^1_X(\log H)(-H)\big)\to H^1(X,\Omega^1_X)\xrightarrow{\phi}H^1(H,\Omega^1_H)\to\cdots$$

where $\dim\ker\phi\geq 2-1=1$, which ensures that the term on the left is non-zero. Then, after twisting by $-H$, we obtain

$$\cdots\to H^1\big(X,\Omega^1_X(-H)\big)\to H^1\big(X,\Omega^1_X(\log H)(-H)\big)\to H^1\big(H,\mathscr{O}_H(-H)\big)\to\cdots$$

where the right hand side is $H^1\big(\mathbb{P}^2,\mathscr{O}_{\mathbb{P}^2}(-1)\big)=0$. So the surjectivity gives that

$$H^1\big(X,\Omega^1_X\otimes\mathscr{O}_X(-H)\big)\neq 0.$$

Moreover it is isomorphic to $H^1\big(X,\Omega^1_X(\log H)(-H)\big)$, since $H^0\big(\mathbb{P}^2,\mathscr{O}_{\mathbb{P}^2}(-1)\big)=0$.

The point here is that while the plurigenera are birational invariant of a smooth variety, the spaces $H^{p,q}(X)$ for $p,q\geq 1$ are not. So we cannot expect Nakano-type statements to hold under essentially birational hypotheses of positivity.

We turn to the proof of Theorem 3.1. Following Kawamata' s original article [Kaw82], the plan is to reduce Theorem 3.1 to a statement showing that one can weaken positivity assumptions in the presence of strong geometric hypotheses, which in Kawamata's book it is called the *log version of the Kodaira vanishing*:

Lemma 4.3.5. (Norimatsu's lemma) *Let X be a smooth projective variety, with an ample divisor A on X, and E is an snc divisor on X. Then*

$$H^i\big(X,\mathscr{O}_X(K_X+A+E)\big)=0 \text{ for } i>0.$$

Or equivalently, by Serre duality

$$H^j\big(X,\mathscr{O}_X(-A-E)\big)=0 \text{ for } j<\dim X.$$

Proof. We write $E = \sum_{i=1}^{r} E_i$. We argue this by induction on the number of components of E. The case $r = 0$ being Kodaira vanishing. Assuming the result is known for snc divisor with $r-1$ components, then consider the natural exact sequence

$$0 \to \mathscr{O}_X\left(-A - \sum_{i=1}^{r} E_i\right) \to \mathscr{O}_X\left(-A - \sum_{i=1}^{r-1} E_i\right) \to \mathscr{O}_{E_r}\left(-A\sum_{i=1}^{r-1} E_i\right) \to 0$$

The induction hypothesis, applied on E_r and X, gives the vanishings

$$H^j\left(E_r, \mathscr{O}_{E_r}(-A - \sum_{i=1}^{r-1} E_i)\right) = 0 \text{ for } j < (n-1),$$
$$H^j\left(X, \mathscr{O}_X(-A - \sum_{i=1}^{r-1} E_i)\right) = 0 \text{ for } j < n.$$

This gives the vanishing for the term on the left. □

Now for Theorem 3.1:

Proof. Since D is big, by Kodaira's lemma (Corollary 3.2.7), for $m \gg 0$ we can write

$$mD \sim_{lin} H + N \tag{4.2}$$

where H is an ample divisor and N an effective divisor on X.

We now proceed in two steps:

Step 1: We first reduce to showing that it suffices to prove the vanishing under the additional assumption that the divisor N in (4.2) has snc support.

To this end, we will show to begin with that after passing to a birational modification $\mu : X^* \longrightarrow X$, we can arrange for $D^* = \mu^* D$ - which of course remains nef and big thanks to generalized projection formula - to admit an expression of the form

$$m'D^* \sim_{lin} H^* + N^*,$$

where H^* is ample and N^* is effective with snc support. Then we will argue that knowing the vanishing on X^* implies the statement on X.

Turning to the details, we start with an arbitrary singular divisor N in (4.2) and construct an embedded resolution

$$\mu : X^* \longrightarrow X$$

of N, via a sequence of blowings-up along smooth centers by using Hironaka's resolution of singularities. Thus $\mu^* N$ has simple normal crossing support, and moreover by Corollary 1.6, we have

$$\mu_* \mathscr{O}_{X^*}(K_{X^*}) = \mathscr{O}_X(K_X) \text{ and } R^i\mu_* \mathscr{O}_{X^*}(K_{X^*}) = 0 \text{ for } i > 0. \tag{4.3}$$

The plan of course is to pull back (4.2) to X^*, but we have to deal with the fact that μ^*H is no longer ample. Now for divisor N, we write

$$\mu^*N = \sum a_j F_j$$

where the divisors F_j appearing here include all μ-exceptional divisors, and each $a_j \geq 0$. Thanks to Corollary 1.6(ii), for $p \gg 0$ there exist $b_j \geq 0$ such that

$$\mu^*(pH) - \sum b_j F_j$$

is an ample divisor on X^*. Then we have

$$\mu^*\big(pmD\big) \sim_{lin} \Big(\mu^*(pH) - \sum b_j F_j\Big) + \Big(\sum (pa_j + b_j)F_j\Big)$$

which is the sum of an ample and an effective divisor with snc support. Thus we have constructed the promised decomposition in (4.3).

Now suppose we know that the vanishing holds for μ^*D on X^*, i.e. assume that

$$H^i\big(X^*, \mathscr{O}_{X^*}(K_{X^*} + \mu^*D)\big) = 0 \text{ for } j > 0.$$

Then it follows from the first equation in (4.3) we have

$$\begin{aligned} H^i\big(X^*, \mathscr{O}_{X^*}(K_{X^*} + \mu^*D)\big) &= H^i\big(X, \mu_*\mathscr{O}_{X^*}(K_{X^*} + \mu^*D)\big) \\ &\cong H^i\big(X, \mathscr{O}_X(K_X + D)\big) = 0 \text{ for } i > 0, \end{aligned}$$

thanks to the projection formula. So we get the desired vanishing on X. This completes *Step 1*, that is we can assume in (4.2) that N has snc support.

Step 2: We now prove the vanishing with assumption that

$$mD \sim_{lin} H + N$$

where H is ample and N has snc support.

Write $N = \sum_1^r e_i E_i$ with each integers $e_i > 0$, and set

$$e^* = e_1 \cdot \cdots \cdot e_r \ , \ e_i^* = \frac{e^*}{e_i}.$$

We use the Kawamata covering to construct a covering $h : Y \longrightarrow X$ of X by a smooth variety Y with respect to the fixed positive integers $me_1^*, \ldots, me_r^*$, to get

$$h^*E_i = me_i^*E_i' \text{ and set } E' := \sum E_i' \text{ an snc divisor on } Y.$$

Put $D' = h^*D$ and $H' = h^*H$, so that we have

$$mD' \sim_{lin} H' + me^*E'. \tag{4.4}$$

Now D' is nef because D is and by the projection formula, therefore $H' + m(e^* - 1)D'$ - being the sum of an ample (h is finite) and a nef divisor - is ample. So it follows from (4.4) that

$$me^*(D' - E') \sim_{lin} H' + m(e^* - 1)D'$$

is ample. Hence $D' - E'$ is ample, i.e. $D' \sim_{lin} A' + E'$ for some ample divisor A' on Y, and therefore

$$H^j\big(Y, \mathscr{O}_Y(-D')\big) = H^j\big(Y, \mathscr{O}_Y(-A' - E')\big) = 0 \text{ for } j < n$$

thanks to Norimatsu's lemma. Then by the Injectivity lemma, the injection

$$H^j\big(X, \mathscr{O}_X(-D)\big) \hookrightarrow H^j\big(Y, \mathscr{O}_Y(-D')\big) = 0$$

gives our desired vanishing. □

Example 4.3.6. (An extension of Theorem 3.1) Let D be a nef divisor on a smooth projective n-fold X. Assume that $(D^{n-k} \cdot H^k) > 0$ for some **natural number** k and ample divisor H. Then

$$H^i\big(X, \mathscr{O}_X(K_X + D)\big) = 0 \text{ for } i > k.$$

This statement is again due to Kawamata and Viehweg, and contains Theorem 3.1 as the case $k = 0$.

Proof. After replacing H by its high multiple and applying Bertini theorem, we can suppose that H is a very ample smooth divisor. We use the induction on the $\dim X$.

For $n = 1$, we only have the case $k = 0$, since for $k > 0$ it automatically vanishes by Grothendieck vanshing. But for $k = 0$ it is $\deg D > 0$, i.e. D is ample then use Kodaira vanishing theorem. Now suppose the statement holds for smooth $(n-1)$-fold Y, that is $\big((D|_Y)^{(n-1)=k'} \cdot (H|_Y)^k\big)$ for some natural number k with the vanishing. Then for this k, we have

$$\begin{aligned}(D^{n-1} \cdot H^k)_X &= (D^{(n-1)-(k-1)} \cdot H^{k-1} \cdot H)_X \\ &= \big((D|_H)^{(n-1)-k'} \cdot (H|_H)^{k'}\big)_H \\ &> 0.\end{aligned}$$

So we consider the exact sequence, by the adjunction formula:

$$0 \longrightarrow \mathscr{O}_X(K_X+D) \longrightarrow \mathscr{O}_X(K_X+D+H) \longrightarrow \mathscr{O}_X(K_X+D+H)|_H \cong \mathscr{O}_H(K_H+D) \longrightarrow 0$$

Then the induction hypothesis gives vanishing for the term on the right, whereas all the higher cohomology of the term in the middle vanishes thanks to Kodaira vanishing. □

We give two theorems without proof.

Theorem 4.3.7. (Kollár's injectivity theorem) *Let X be a smooth projective variety and L be a semi-ample divisor on X. Given $k \geq 1$, fix any divisor $D \in |kL|$. Then the homomorphisms*

$$H^i\big(X, \mathscr{O}_X(K_X + mL)\big) \xrightarrow{\cdot D} H^i\big(X, \mathscr{O}_X(K_X + (m+k)L)\big)$$

determined in the natural way by D are injective for all $i \geq 0$ and $m > 0$.

Observe that this implies Kodaira vanishing, since if L is ample and when $m = 1$, then

$$H^i\big(X, \mathscr{O}_X(K_X + kL)\big) = 0$$

for $i > 0$ and $k \gg 0$ by Serre vanishing.

The second theorem shows that the direct images $R^j f_* \mathscr{O}_X(K_X)$ have exceptionally good cohomological properties: in particular, they satisfy the same sort of vanishings as canonical bundles themselves.

Theorem 4.3.8. (Kollár vanishing theorem) *Let $f : X \longrightarrow Y$ be a surjective morphism of projective varieties, with X smooth. Then:*

(i) $R^j f_* \mathscr{O}_X(K_X)$ *is torsion-free for all* j.

(ii) $R^j f_* \mathscr{O}_X(K_X) = 0$ *for* $j > \dim X - \dim Y$.

(iii) For any ample divisor A on Y and every $j \in \mathbb{N}$,

$$H^i\big(Y, R^j f_* \mathscr{O}_X(K_X) \otimes \mathscr{O}_Y(A)\big) = 0 \text{ for } i > 0.$$

Which we have used it in proof of the Fujita's vanishing theorem.

4.3B Some Applications

Theorem 4.3.9. (Geauert-Riemenschneider vanishing) *Let $f : X \longrightarrow Y$ be a generically finite and surjective projective morphism of varieties, with X is smooth. Then*

$$R^i f_* \mathscr{O}_X(K_X) = 0$$

for every $i > 0$.

Proof. For origin proof can see [Laz04a, 4.3.9], but we will see this is just a special case of generalized Kawamata-Viehweg vanishing theorem. □

4.4 Kawamata-Viehweg vanishing theorem

We now come to the statement and proof of the basic vanishing theorem for $\mathbb{Q}$-divisors. Before this, we need some operations on $\mathbb{Q}$-divisors.

Definition 4.4.1. (Rounds) Let $D = \sum a_i D_i$ be a $\mathbb{Q}$-divisor on X. The *round up* $\lceil D \rceil$ and *round down* $\lfloor D \rfloor$ of D are the integral divisors

$$\lceil D \rceil = \sum \lceil a_i \rceil D_i,$$
$$\lfloor D \rfloor = \sum \lfloor a_i \rfloor D_i,$$

where as usual for $x \in \mathbb{Q}$ one denotes by $\lceil x \rceil$ the least integer $\geq x$, and by $\lfloor x \rfloor$ the greatest integer $\leq x$. The *fractional part* $\{D\}$ of D is an effective $\mathbb{Q}$-divisor

$$\{D\} = D - \lfloor D \rfloor.$$

Remark 4.4.2. Note that rounds are defined strictly at the cycle level. In particular, Definition 4.1 would **not** make sense if we worked with the Cartier divisors, since this operation need be compatible over the intersections $\{U_i \cap U_j\}$. This explains why we deal here with divisors. However, if X is **not** smooth, we then need extra hypotheses in order to guarantee that the usual functorial operations are defined.

Warning 4.4.3. (Warning on rounds) It is critical to note that *pullback* and *numerical equivalence* do **not** commute with roundings. For example, let $X = \mathbb{A}^2$ with coordinates x, y, and let $A \subseteq X$ be the parabola $V(y - x^2)$. Consider the $Y \subseteq X$ be the x-axis and P the origin, so that Y is tangent to A at point P. Then for the $\mathbb{Q}$-divisor $D = \frac{1}{2}A$ on X, we have $\lfloor D \rfloor = 0$ hence $\lfloor D \rfloor|_Y = 0$. But

$$\lfloor D|_Y \rfloor = \lfloor \frac{1}{2} \cdot 2P \rfloor = P.$$

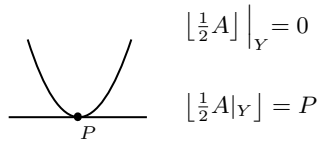


Figure 4.4.: Rounding does not commute with restriction.

Similarly, let $L, C \subseteq \mathbb{P}^2$ denote respectively a line and an irreducible conic. Then $L \equiv_{num} \frac{1}{2}C$, but

$$\lfloor \frac{1}{2}C \rfloor = 0 \text{ whereas } \lfloor L \rfloor = L.$$

4.4A Statement and Proof of the Theorem

Theorem 4.4.4. (Kawamata-Viehweg vanishing) *Let X be a non-singular projective variety of dimension n, and let N be an integral divisor on X. Assume that*

$$N \equiv_{num} B + \Delta$$

where B is a nef and big $\mathbb{Q}$-divisor, and $\Delta = \sum a_i \Delta_i$ is a $\mathbb{Q}$-divisor with simple normal crossing support and fractional coefficients:

$$0 \leq a_i < 1 \text{ for all } i.$$

Then

$$H^i\big(X, \mathscr{O}_X(K_X + N)\big) = 0 \text{ for } i > 0.$$

Or equivalently, by Serre duality,

$$H^j\big(X, \mathscr{O}_X(-N)\big) = 0 \text{ for } j < n.$$

Remark 4.4.5. In other words, the hypothesis on Δ is that it is an effective divisor whose support $\sum \Delta_i$ has simple normal crossings, with $\lfloor \Delta \rfloor = 0$. A $\mathbb{Q}$-divisor satisfying these conditions is sometimes called a *boundary divisor*.

We turn to prove the Kawamata-Viehweg vanishing theorem.

Proof. In the proof we allow the divisor Δ_i to have several connected components, as our discussion under the Proposition 2.7, with each being non-singular and reduced. Since in the $n = 1$ case the pullback may break up the irreducibility. We argue this theorem by induction on the number of fractional terms appearing in the Δ.

For trivial case, i.e. for $N \equiv_{num} B$, then the vanishing statement reduces to vanishing for big and nef N via Theorem 3.1. Now may assume that $a_1 \frac{c}{b} \neq 0$ for positive integers $0 < c < d$, we focus on the vanishing of $H^j(X, \mathscr{O}_X(-N))$ for $j < n$. Recall from the Injectivity lemma 1.18, that is suffices to exhibit a finite surjective map

$$f : X' \longrightarrow X,$$

with X non-singular, such that

$$H^j\big(X', f^*\mathscr{O}_X(-N)\big) = 0 \text{ for } j < n. \tag{4.5}$$

The vanishing above follows from a slight elaboration of Kawamata covering. Specifically, start by taking a Bloch-Gieseker covering to produce a non-singular branched covering

$$f : X' \longrightarrow X$$

such that $\Delta'_i := f^*\Delta_i$ is smooth for all i, and $\sum \Delta'_i$ has snc support after our arrangement. And for a given integer $d > 0$, we have

$$\Delta'_1 \sim_{lin} dA' \tag{4.6}$$

for some (possibly non-effective) divisor A' on X'. For ease of notation, we set $N' = f^*N$ and $B' = f^*B$.

Now Δ'_1 is divisible by d in $\mathrm{Pic}(X')$ thanks to (4.6), and so we next choose the d-fold cyclic covering

$$\pi : X'' \longrightarrow X'$$

branched along Δ'_1. Writing $\delta''_i = \pi^*\Delta'_i$, $A'' = \pi^*A'$, etc., we then have

$$\begin{aligned} d\Delta''_1 = \pi^*\Delta'_1 &\sim_{lin} dA'', \\ \text{and } N'' = \pi^*N' &\sim_{lin} B'' + cA'' + \sum_{i\geq 2} a_i\Delta''_i. \end{aligned}$$

Then $N'' - cA'' \equiv_{num} B'' - \sum\limits_{i\geq 2} a_i\Delta''_i$ and we have reduced the number of fractional divisors in play. So by the induction hypothesis we have that

$$H^j\big(X'', \mathscr{O}_{X''}(-N'' + cA'')\big) = 0$$

for $j < n$, and hence also that

$$H^j\big(X', \pi_*\mathscr{O}_{X''}(-N'' + cA'')\big) = 0. \tag{4.7}$$

But as π is a d-fold cyclic covering, by Remark 1.10 we get

$$\begin{aligned} \pi_*\mathscr{O}_{X''}(-N'' + cA'') &= (\pi_*\mathscr{O}_{X''})(-N' + cA') \\ &= \Big(\mathscr{O}_{X'} \oplus \mathscr{O}_{X'}(-A') \oplus \cdots \oplus \mathscr{O}_{X'}(-(d-1)A')\Big)(-N' + cA') \\ &= \mathscr{O}_{X'}(-N' + cA') \oplus \cdots \oplus \mathscr{O}_{X'}(-N' + (c-d+1)A'), \end{aligned}$$

thanks to the projection formula. Since $c \leq d-1$, the $\mathscr{O}_{X'}(-N') = f^*\mathscr{O}_X(-N)$ occurs as one of the summands on the right, and so (4.5) follows from (4.7). □

Corollary 4.4.6. *As in the theorem, assume that X is a smooth projective variety. Let B be a nef and big $\mathbb{Q}$-divisor whose fractional part $\{B\}$ has snc support. Then*

$$H^i\big(X, \mathscr{O}_X(K_X + \lceil B \rceil)\big) = 0 \text{ for } i > 0.$$

Proof. By a simple check, we have $\lceil B \rceil = -\lfloor -B \rfloor$, so

$$\begin{aligned} B - \lceil B \rceil &= B - (-\lfloor -B \rfloor) \\ &= B + \lfloor -B \rfloor \\ &= -(-B - \lfloor -B \rfloor) \\ &= -\{-B\}. \end{aligned}$$

This implies that $\lceil B \rceil \equiv_{num} B + \{-B\}$ with $\{-B\}$ having snc support same as $\{B\}$. Then applying Kawamata-Viehweg vanishing theorem on $\lceil B \rceil$. □

Corollary 4.4.7. *In the setting of Corollary above, let L be an integral divisor and D is a $\mathbb{Q}$-divisor on X. Assume that $L - D$ is nef and big, and that D has snc support. Then*

$$H^i\big(X, \mathscr{O}_X(K_X + L - \lfloor D \rfloor)\big) = 0 \text{ for } i > 0.$$

Proof. By $L - \lfloor D \rfloor \equiv_{num} L - D + \{D\}$ and Kawamata-Viehweg vanishing. □

4.4B Generalizations of the Vanishings in Relative Version

According [Kaw24] or [KMM87], we generalize the Kodaira-Viehweg vanishing theorem to different directions in order to apply it to higher dimensional algebraic geometry.

Recall that a $\mathbb{R}$-Cartier divisor B on X is said to be *f-nef* if its restriction to every fibre of f is nef, and D is *f-big* if its restriction to a generic fibre of f is big.

We first prove the generalized Kodaira vanishing theorem, before that we need one lemma:

Lemma 4.4.8. *Let X be a smooth variety and let D be an $\mathbb{R}$-divisor on X such that the fractional part $\{D\}$ has snc support. Let $f : Y \longrightarrow X$ be a proper birational morphism from a smooth variety Y. Then we have*

$$f_* \mathscr{O}_Y(K_Y + \lceil f^* D \rceil) \cong \mathscr{O}_X(K_X + \lceil D \rceil).$$

Proof. As X is smooth, so $\mathrm{Ex}(f) \subseteq Y$ is pure of codimension 1 by Proposition 2.1.3. By assumption $\{D\}$ has snc support, so we put $\Delta := \lceil D \rceil - \lfloor D \rfloor$ which is a reduced divisor on X having snc support. For pair (X, Δ) we can write

$$K_Y + f_*^{-1}\Delta = f^*(K_X + \Delta) + \sum_{E_i : f\text{-exceptional}} a(E_i, X, \Delta) E_i,$$

for which one can see Chapter 5.2. As the integral divisor Δ is snc, $\mathrm{discrep}(X,\Delta) = -1$, i.e. $a(E_i, X, \Delta) \geq -1$ for every i see Lemma 5.2.20. Then by $f^*\Delta = f^*\lceil D\rceil - f^*\lfloor D\rfloor$,

$$K_X + f_*^{-1}\Delta + f^*\lfloor D\rfloor = f^*(K_X + \lceil D\rceil) + \sum_{E_i : f\text{-exceptional}} a(E_i, X, \Delta)E_i.$$

Now we plus the $\lceil f^*D\rceil$ on both side to get

$$K_Y + \lceil f^*D\rceil = f^*(K_X + \lceil D\rceil) + \left(\lceil f^*D\rceil - (f_*^{-1}\Delta + f^*\lfloor D\rfloor) + \sum_{E_i : f\text{-exceptional}} a(E_i, X, \Delta)E_i \right).$$

Then we compute the coefficient of the last term on each exceptional divisor E_i:

$$\begin{aligned}
&\mathrm{mult}_{E_i}\left(\lceil f^*D\rceil - (f_*^{-1}\Delta + f^*\lfloor D\rfloor) + \sum_{E_i : f\text{-exceptional}} a(E_i, X, \Delta)E_i \right) \\
=&\,\mathrm{mult}_{E_i}\Big(\lceil f^*D\rceil - (f_*^{-1}\Delta + f^*\lfloor D\rfloor)\Big) + a(E_i, X, \Delta) \\
=&\,\mathrm{mult}_{E_i}\Big(\lceil f^*(\lfloor D\rfloor + \{D\})\rceil - (f_*^{-1}\Delta + f^*\lfloor D\rfloor)\Big) + a(E_i, X, \Delta) \\
=&\,\mathrm{mult}_{E_i}\big(f^*\lfloor D\rfloor + \lceil f^*\{D\}\rceil - f_*^{-1}\Delta - f^*\lfloor D\rfloor\big) + a(E_i, X, \Delta) \\
=&\,\mathrm{mult}_{E_i}(\lceil f^*\{D\}\rceil) - \mathrm{mult}_{E_i}(f_*^{-1}\Delta) + a(E_i, X, \Delta) \\
\geq&\, 1 - 0 + (-1) = 0,
\end{aligned}$$

for every f-exceptional divisor E_i with $a(E_i, X, \Delta) = -1$. Thus we can write

$$K_Y + \lceil f^*D\rceil = f^*(K_X + \lceil D\rceil) + F$$

where F is an effective f-exceptional divisor on Y. Therefore, we have

$$\begin{aligned}
f_*\mathscr{O}_Y(K_Y + \lceil f^*D\rceil) &= f_*\mathscr{O}_Y\big(f^*(K_X + \lceil D\rceil) + F\big) \\
&\cong \mathscr{O}_X(K_X + \lceil D\rceil) \otimes f_*\mathscr{O}_Y(F) \\
&\cong \mathscr{O}_X(K_X + \lceil D\rceil) \otimes \mathscr{O}_X,
\end{aligned}$$

by the projection formula and algebraic Hartogs as $f(F)$ has codimension ≥ 2. □

Theorem 4.4.9. (Generalized Kodaira vanishing in the projective case) *Let X be a smooth projective variety and let $f : X \longrightarrow S$ be a surjective morphism onto a projective variety S. Let D be a f-ample $\mathbb{Q}$-divisor on X whose fractional part $\{D\}$ has snc support. Then*

$$R^i f_*\mathscr{O}_X(K_X + \lceil D\rceil) = 0 \text{ for } i > 0.$$

Proof. **Step 1:** First we prove that we can reduce to the vanishing of

$$R^i f_* \mathscr{O}_X(K_X + \lceil D + f^*A \rceil) = 0 \text{ for i>0},$$

where A is an integral divisor on S. In fact, thanks to projection formula we have

$$\begin{aligned} R^i f_* \mathscr{O}_X(K_X + \lceil D + f^*A \rceil) &= R^i f_* \mathscr{O}_X(K_X + \lceil D \rceil + f^*A) \\ &\cong R^i f_* \mathscr{O}_X(K_X + \lceil D \rceil) \otimes \mathscr{O}_S(A). \end{aligned}$$

Hence if the left-hand side vanishes, then so does the $R^i f_* \mathscr{O}_X(K_X + \lceil D \rceil)$.

Step 2: Consider an ample integral divisor A on S. We clear the denominator that rD is an integral f-ample divisor on S, for some r large enough. Then by Proposition 1.5.8, we have that $rD + f^*(rA)$ is ample on X, so we see $D + f^*A$ is an ample $\mathbb{Q}$-divisor on X. Now we aim to prove the vanishing for $\lceil D + f^*A \rceil$.

Replacing D by $D + f^*A$, whose fractional part still has snc support, we will take an ample divisor H on S and a positive integer m. Consider the following spectral sequence

$$E_2^{p,q} = H^p\big(S, R^q f_* \mathscr{O}_X(K_X + \lceil D \rceil + m f^* H)\big) \underset{p}{\Rightarrow} R^{p+q}\Big(H^0\big(S, f_* \mathscr{O}_X(K_X + \lceil D \rceil + m f^* H)\big)\Big),$$

whose convergence follows from the Grothendieck spectral sequence (D.6.3). Thanks to the projection formula, this is nothing but

$$E_2^{p,q} = H^p\big(S, R^q f_* \mathscr{O}_X(K_X + \lceil D \rceil) \otimes \mathscr{O}_S(mH)\big) \underset{p}{\Rightarrow} H^{p+q}\big(X, \mathscr{O}_X(K_X + \lceil D \rceil + m f^* H)\big).$$

But for sufficiently large m, we have $E_2^{p,q} = 0$ for $p > 0$ by Serre's vanishing theorem. Therefore, this spectral sequence collapses on the q-axis, by (D.5.11) we can get

$$E_2^{0,n} = H^0\big(S, R^n f_* \mathscr{O}_X(K_X + \lceil D \rceil) \otimes \mathscr{O}_S(mH)\big) \cong H^n\big(X, \mathscr{O}_X(K_X + \lceil D \rceil + m f^* H)\big).$$

Moreover by similar analysis as above applying on $D + f^*(mH)$, we see it is an ample $\mathbb{Q}$-divisor on X, so it is automatically nef and big. Then by Corollary 4.6 can see

$$H^n\big(X, \mathscr{O}_X(K_X + \lceil D \rceil + m f^* H)\big) = H^n\big(X, \mathscr{O}_X(K_X + \lceil D + m f^* H \rceil)\big) = 0$$

for every $n > 0$. Now since H is ample on S, for $m \gg 0$ thanks to the projection formula, the

$$R^n f_* \mathscr{O}_X(K_X + \lceil D \rceil) \otimes \mathscr{O}_S(mH)$$

is generated by global sections, but we have $H^0 = 0$. Therefore

$$R^n f_* \mathscr{O}_X(K_X + \lceil D \rceil) = 0 \text{ for } n > 0.$$

□

Theorem 4.4.10. (Generalized Kodaira vanishing in the general case) *Let X be a smooth variety and let $f : X \longrightarrow S$ be a projective surjective morphism onto a variety S. Let D be a f-ample $\mathbb{Q}$-divisor on X whose fractional part $\{D\}$ has snc support. Then*

$$R^i f_* \mathscr{O}_X(K_X + \lceil D \rceil) = 0 \text{ for } i > 0.$$

This statement also holds when f is proper, one can see Chapter 1, Remark 1.5.2.

Proof. The idea is turning the problem into the projective case.

Let $f : X \longrightarrow S$ be a projective surjective morphism from a smooth variety X to an affine variety S, since the assertion is local on S. By taking compactifications, we have projective varieties $\bar{X}$ and $\bar{S}$ with a projective morphism $\bar{f} : \bar{X} \longrightarrow \bar{S}$, such that we have the commutative diagram

$$\begin{array}{ccc} X & \xhookrightarrow{\text{open}} & \bar{X} \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle \bar{f}} \\ S & \xhookrightarrow{\text{open}} & \bar{S} \end{array}$$

where $\bar{f}|_X$ is the composition of f with the open immersion $S \hookrightarrow \bar{S}$. By taking resolution of singularities of $\bar{X}$, we may assume that $\bar{X}$ is smooth.

Step 1: We **claim** that in the above setting, there exist a birational morphism $\bar{\mu} : \bar{Y} \longrightarrow \bar{X}$ from a smooth projective variety $\bar{Y}$ and a $\mathbb{Q}$-divisor $\bar{C}$ on $\bar{Y}$ such that

(1) $\bar{C}$ is relatively nef and relatively big over $\bar{S}$,

(2) $\{\bar{C}\} + \mathrm{Ex}(\bar{\mu})$ has snc support on $\bar{Y}$, and

(3) $\bar{C}|_Y = \mu^* D$, where $Y = \bar{\mu}^{-1}(X)$ and $\mu = \bar{\mu}|_Y : Y \longrightarrow X$ is the induced birational morphism.

Then we have an isomorphism

$$\mathscr{O}_X(K_X + \lceil D \rceil) \cong \mu_* \mathscr{O}_Y(K_Y + \lceil \mu^* D \rceil). \tag{4.8}$$

Proof. Assuming the existence of such a birational morphism and divisor satisfying (ii) and (iii), the isomorphism follows immediately from Lemma 4.8. Thus, it suffices to construct $\bar{Y}$ and $\bar{C}$. Consider the closure $\bar{D}$ of D in $\bar{X}$, which is a $\mathbb{Q}$-divisor. We begin by considering the natural homomorphism

$$\varphi_m : \bar{f}^* \bar{f}_* \mathscr{O}_{\bar{X}}(m\bar{D}) \longrightarrow \mathscr{O}_{\bar{X}}(m\bar{D})$$

for a sufficiently large fixed integer $m > 0$ such that mD is a f-very ample integral divisor. Then φ_m is surjective on X since D is f-ample. We denote the image of φ_m to be $\mathcal{I} \cdot \mathscr{O}_{\bar{X}}(m\bar{D})$ where $\mathcal{I}$ is an ideal sheaf of $\mathscr{O}_{\bar{X}}$ satisfying $\mathcal{I}|_X \cong \mathscr{O}_X$. We take further blow-ups along the subscheme of $\bar{X}$ defined by $\mathcal{I}$, whose support have empty intersection with X. Analogous to Example 3.1.34, we factor out the fixed part, we may assume that the image of φ_m can be expressed as $\mathscr{O}_{\bar{X}}(m\bar{D} - E)$ for an effective divisor E on $\bar{X}$ such that $E \cap X = \emptyset$. Therefore, the surjective map gives a morphism

$$\pi : \bar{X} \longrightarrow \mathbb{P}_{\bar{S}}(\bar{f}_* \mathscr{O}_{\bar{X}}(m\bar{D})),$$

see [Har77, II,7.12]. So we can find a projective variety $\bar{P} = \mathrm{im}(\pi)_{\mathrm{red}}$ over $\bar{S}$ which admits a relatively ample line bundle $\mathscr{O}_{\bar{P}}(1)$ over $\bar{S}$ such that

$$\pi^* H \sim_{lin} m\bar{D} - E$$

where H corresponds to $\mathscr{O}_{\bar{P}}(1)$. Since mD is f-very ample and $E \cap X = \emptyset$, we have

$$(\pi|_X)^* \big(H|_{\pi(X)}\big) \sim_{lin} mD.$$

In particular, the morphism

$$\pi|_X : X = \bar{X} \underset{\bar{S}}{\times} S \longrightarrow P := \bar{P} \underset{\bar{S}}{\times} S$$

is proper by properness is stable under base change. And if $\pi|_X$ contracts some curve, i.e. $\pi|_X(C)$ is a point, then we have

$$\big(H \cdot (\pi|_X)_* C\big) = (mD \cdot C) = 0$$

contradicts the fact mD is ample. (Notice that as S is affine, so relative very ample implies ampleness, see [GW20, 13.59]). Thus the morphism $\pi|_X$ is fibre-finite, hence is a finite morphism. In particular, π is a generically finite morphism. We set

$$D' = \bar{D} - \frac{1}{m} E \sim_{\mathbb{Q}} \frac{1}{m} \pi^* H,$$

such that D' is an $\bar{f}$-nef and $\bar{f}$-big $\mathbb{Q}$-divisor on $\bar{X}$, and $D'|_X = D$. We can take a birational morphism $\bar{\mu} : \bar{Y} \longrightarrow \bar{X}$ from another smooth projective variety $\bar{Y}$, thanks to Hironaka, such that $\bar{\mu}^*\{D'\} + \mathrm{Ex}(\bar{\mu})$ has snc support on $\bar{Y}$. We set $\bar{C} := \bar{\mu}^* D'$. □

Step 2: Having verified our claim, we will use the projective case Theorem 4.9 to continue, and notice that in that proof we do not have to assume that S is affine by shrinking S.

By applying the Theorem 4.9 for projective birational morphism $\bar{\mu} : \bar{Y} \longrightarrow \bar{X}$ which is obviously surjective, we have that

$$R^i\bar{\mu}_*\mathscr{O}_{\bar{Y}}(K_{\bar{Y}} + \lceil \bar{C} \rceil) = 0$$

for every $i > 0$ since $\bar{X}$ is projective. By applying Theorem 4.9 again on surjective morphism $\bar{f} \circ \bar{\mu} : \bar{Y} \longrightarrow \bar{S}$, then we have

$$R^i(\bar{f} \circ \bar{\mu})_*\mathscr{O}_{\bar{Y}}(K_{\bar{Y}} + \lceil \bar{C} \rceil) = 0$$

for every $i > 0$. Again, by Grothendieck spectral sequence

$$E_2^{p,q} = (R^p\bar{f}_*)(R^q\bar{\mu}_*)\mathscr{O}_{\bar{Y}}(K_{\bar{Y}} + \lceil \bar{C} \rceil) \underset{p}{\Rightarrow} R^{p+q}(\bar{f} \circ \bar{\mu})_*\mathscr{O}_{\bar{Y}}(K_{\bar{Y}} + \lceil \bar{C} \rceil),$$

and $E_2^{p,q}$ is collapses on p-axis, so via (D.5.11) we have

$$E_2^{n,0} = R^n\bar{f}_*\big(\bar{\mu}_*\mathscr{O}_{\bar{Y}}(K_{\bar{Y}} + \lceil \bar{C} \rceil)\big) \cong R^n(\bar{f} \circ \bar{\mu})_*\mathscr{O}_{\bar{Y}}(K_{\bar{Y}} + \lceil \bar{C} \rceil) = 0,$$

for every $n > 0$. By considering the restriction on S with the isomorphism (4.8), we therefore have

$$\begin{aligned} R^i\bar{f}_*\big(\bar{\mu}_*\mathscr{O}_{\bar{Y}}(K_{\bar{Y}} + \lceil \bar{C} \rceil)\big)\big|_S &\cong R^i f_*\big(\mu_*\mathscr{O}_Y(K_Y + \lceil \mu^* D \rceil)\big) \\ &\cong R^i f_*\mathscr{O}_X(K_X + \lceil D \rceil) \\ &= 0 \end{aligned}$$

for every $i > 0$. This is our desired vanishing. □

The generalized Kodaira vanishing theorem can be extended on $\mathbb{R}$-divisors, by we perturb the coefficients.

Theorem 4.4.11. (Generalized Kodaira vanishing in the general case for $\mathbb{R}$-divisors) *Let X be a smooth variety and let $f : X \longrightarrow S$ be a projective surjective morphism onto a variety S. Let D be a f-ample $\mathbb{R}$-divisor on X whose fractional part $\{D\}$ has snc support. Then*

$$R^i f_*\mathscr{O}_X(K_X + \lceil D \rceil) = 0 \text{ for } i > 0.$$

Proof. From Proposition 1.5.16 we see $\mathrm{Amp}(X/S)$ is open, so we can perturb the coefficients of $\mathbb{R}$-divisor D such that to be a $\mathbb{Q}$-divisor, with the same round up and same snc support of fractional part as in condition. Then the theorem can directly deduces from the $\mathbb{Q}$-version, Theorem 4.10. □

For the following theorem can see [KMM87, Theorem 1-2-3].

Theorem 4.4.12. (Generalized Kawamata-Viehweg vanshing) *Let X be a non-singular variety and let $f : X \longrightarrow S$ be a proper surjective morphism onto a variety S. Let D be a f-big and f-nef $\mathbb{Q}$-divisor on X whose fractional part $\{D\}$ has snc support. Then*

$$R^i f_* \mathscr{O}_X(K_X + \lceil D \rceil) = 0 \text{ for } i > 0.$$

Before providing a proof of this theorem, we require the following lemma from [KMM87, 0-3-6], which states that, by combining Kodaira's lemma with Hironaka's resolution of singularities, one can control the boundary divisor over a nef and big $\mathbb{Q}$-divisor such that its coefficients are sufficiently small.

Lemma 4.4.13. *Let $f : X \longrightarrow S$ be a proper surjective morphism from a normal variety onto an algebraic variety, and let D be a f-nef and f-big $\mathbb{Q}$-Cartier divisor on X. Then there exists a projective birational morphism $\mu : Y \longrightarrow X$ from a smooth variety Y which is projective over S and a family of divisors $\{F_j\}$ on Y such that*

$$\mu^* D + \sum_j F_j$$

has snc support, and such that $\mu^* D - \sum_j \delta_j F_j$ is $(f \circ \mu)$-ample for some $\delta_j \in \mathbb{Q}$ with $0 < \delta_j \ll 1$.

Proof. Our strategy is to apply the relative Kodaira's lemma to find a relative ample divisor. But as our morphism is only assumed to be proper, we need Chow's lemma to modify our source space.

Step 1: Applying Chow's lemma, there is a variety X' which is projective over S and a birational morphism $\phi : X' \longrightarrow X$. Furthermore, by [Har77, II, ex4.8] we have that ϕ is projective. Moreover, we can take a normalization $X^\nu \xrightarrow{\nu} X'$ and replacing X^ν by X' and $\phi \circ \nu$ by ϕ which ensures that ϕ still is a projective birational morphism, since a finite morphism is projective. We denote $f' : X' \longrightarrow S$.

Step 2: By our assumeption, D is a f-big $\mathbb{Q}$-divisor on X. Following from the relative version of Proposition 3.2.36, we see its pullback $D' = \phi^* D$ is f'-big. Passing to a large multiple, by Corollary 3.3.7 we have

$$mD' \sim_S A' + N'$$

for a f'-ample divisor A' and an effective divisor N'. That is

$$D' \sim_{S,\mathbb{Q}} \frac{1}{m} A' + \frac{1}{m} N'$$

where $\frac{1}{m}A'$ is a f'-ample $\mathbb{Q}$-divisor and $\frac{1}{m}N' =: E$ is effective.

Moreover, since D' is f'-nef by the projection formula, for any rational number $0 < \varepsilon < 1$ we can write that

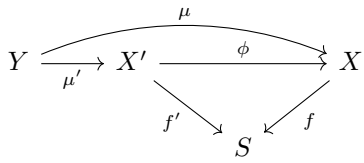
$$D' = (1-\varepsilon)D' + \varepsilon D' \sim_{S,\mathbb{Q}} (1-\varepsilon)D' + \frac{\varepsilon}{m}A' + \varepsilon E.$$

Set $A_\varepsilon = (1-\varepsilon)D' + \frac{\varepsilon}{m}A'$, by Corollary 1.5.20 it is a f'-ample $\mathbb{Q}$-divisor. Thus we can get

$$D' \sim_{S,\mathbb{Q}} A_\varepsilon + \varepsilon E,$$

and by taking ε sufficiently small, we can make the coefficient of E to be arbitrary small.

Step 3: Consider the pair $(X', D' + E)$, applying the Hironaka's resolution of singularities to get a log resolution $\mu' : Y \longrightarrow X'$



such that

$$\mu'^*(D' + E) + \mathrm{Ex}(\mu')$$

has snc support, where μ' is a projective birational morphism gives that Y is projective over S. Therefore, we have

$$\mu'^* D' \sim_{S,\mathbb{Q}} \mu'^* A_\varepsilon + \mu'^*(\varepsilon E).$$

Moreover as our construction of Y is the sequence of blowing-ups, there is a μ'-ample divisor $-F$ on Y which is supported in $\mathrm{Ex}(\mu')$, by [Har77, II, 7.13(a)]. By introducing a rational number $\xi > 0$, we write it to be

$$\mu'^* D' - \left(\mu'^*(\varepsilon E) + \xi F\right) \sim_{S,\mathbb{Q}} \mu'^* A_\varepsilon - \xi F,$$

where $\mu'^* A_\varepsilon - \xi F$ is a $(f' \circ \mu')$-ample $\mathbb{Q}$-divisor, which is follows from [EGA II, 4.6.13(ii)] or [GW20, 13.65] and we can choose $\xi = \frac{1}{n}$ for sufficiently small.

Step 4: Finally, we denote $\mu = \phi \circ \mu' : Y \longrightarrow X$ such that $\mu'^* D = \mu^* D$. We set

$$\sum_j \delta_j F_j := \mu'^*(\varepsilon E) + \xi F$$

where $\{F_j\}$ is the union of irreducible components of $\operatorname{Supp}\mu'^*(\varepsilon E)\cup\operatorname{Supp}\xi F$. But by our construction in *Step 3*, we have that

$$\operatorname{Supp}\mu^*D\cup\operatorname{Supp}(\bigcup_j F_j)\subseteq\operatorname{Supp}\mu'^*D'\cup\operatorname{Supp}\mu'^*E\cup\operatorname{Supp}\operatorname{Ex}(\mu')$$

where the right hand side has snc support, implies that $\mu^*D+\sum_j F_j$ also has snc support. And since our choice of ε and ξ are sufficiently small rational numbers, there δ_j are in $\mathbb{Q}$ and satisfies that $0<\delta_j\ll 1$. □

We now turn to prove the generalized Kawamata-Viehweg vanishing:

Proof. We prove this vanishing by using the generalized Kodaira vanishing. By Lemma 4.13, we can construct a projective birational morphism $\mu: Y\longrightarrow X$ such that

$$\mu^*D+\sum_j F_j+\operatorname{Ex}(\mu)$$

is a snc divisor on Y and $\mu^*D-\sum_j\delta_jF_j$ is $(f\circ\mu)$-ample for some $\delta_j\in\mathbb{Q}$ with $0<\delta_j\ll 1$. Then by applying the Theorem 4.10 to μ, we obtain

$$0=R^i\mu_*\mathscr{O}_Y(K_Y+\lceil\mu^*D-\sum_j\delta_jF_j\rceil)=R^i\mu_*\mathscr{O}_Y(K_Y+\lceil\mu^*D\rceil)\tag{4.9}$$

for every $i>0$. We can also see that

$$\mu_*\mathscr{O}_Y(K_Y+\lceil\mu^*D\rceil)\cong\mathscr{O}_X(K_X+\lceil D\rceil)$$

by Lemma 4.8. Follows from the Grothendieck spectral sequence, we have

$$E_2^{p,q}=(R^pf_*)(R^q\mu_*)\mathscr{O}_Y(K_Y+\lceil f^*D-\sum_j\delta_jF_j\rceil)\underset{p}{\Rightarrow}R^{p+q}(f\circ\mu)_*\mathscr{O}_Y(K_Y+\lceil f^*D-\sum_j\delta_jF_j\rceil),$$

and by (4.9) we see this spectral sequence collapses on the p-axis, so due to (D.5.11) we know

$$\begin{aligned}0&=R^i(f\circ\mu)_*\mathscr{O}_Y(K_Y+\lceil f^*D-\sum\delta_jF_j\rceil)\\&\cong E_2^{i,0}\\&=R^if_*\big(\mu_*\mathscr{O}_Y(K_Y+\lceil f^*D-\sum\delta_jF_j\rceil)\big)\\&=R^if_*\big(\mu_*\mathscr{O}_Y(K_Y+\lceil f^*D\rceil)\big)\\&\cong R^if_*\mathscr{O}_X(K_X+\lceil D\rceil)\end{aligned}$$

where the first equality comes from the Theorem 4.10 applied on $f\circ\mu: Y\longrightarrow S$. □

Theorem 4.4.14. (Generalized Kawamata-Viehweg vanshing for $\mathbb{R}$-divisors) *Let X be a non-singular variety and let $f : X \longrightarrow S$ be a proper surjective morphism onto a variety S. Let D be a f-big and f-nef $\mathbb{R}$-divisor on X whose fractional part $\{D\}$ has snc support. Then*

$$R^i f_* \mathscr{O}_X(K_X + \lceil D \rceil) = 0 \text{ for } i > 0.$$

Proof. The only crucial part is how to change the assumption of D in Lemma 4.13 to be a f-nef and f-big $\mathbb{R}$-divisor, to get such

$$\mu^* D - \sum_j \delta_j F_j$$

is a $(f \circ \mu)$-ample $\mathbb{R}$-divisor for some $\delta_j \in \mathbb{R}$ with $0 < \delta_j \ll 1$. Actually this is easy, since by relative Kodaira's lemma we can write

$$D' \sim_{S,\mathbb{R}} A' + E$$

where A' is f'-ample $\mathbb{R}$-divisor and $E \in \mathrm{Div}(X)_{\mathbb{R}}$ is effective. Then we write

$$D' = (1-\varepsilon)D' + \varepsilon D' \sim_{S,\mathbb{R}} (1-\varepsilon)D' + \varepsilon A' + \varepsilon E$$

and set $A_\varepsilon := (1-\varepsilon)D' + \varepsilon A'$ is a f'-ample $\mathbb{R}$-divisor. Taking resolution of singularities and introducing a real number $\xi > 0$ such that

$$\mu'^* D' - \left(\mu'^*(\varepsilon E) + \xi F\right) \sim_{S,\mathbb{R}} \mu'^* A_\varepsilon - \xi F,$$

for some μ'-ample divisor $-F$. The **main part** is here. We expand the $\mathbb{R}$-divisor A_ε into a linear combination

$$A_\varepsilon = \sum_i r_i H_i$$

where $r_i \in \mathbb{R}$ and H_i are f'-ample integral divisors. Applying [EGA II, 4.6.13(ii)] or [GW20, 13.65] on **each of these** H_i with $-F$, we find that each

$$n_i \mu'^* H_i - F$$

is a $(f' \circ \mu')$-ample integral divisor for some suitable. As they are finitely many terms we choose $n \gg 0$ satisfying on each n_i. Then we have

$$\sum_i r_i \left(\mu'^* H_i - \frac{1}{n} F\right) = \mu'^* A_\varepsilon - \frac{\sum r_i}{n} F$$

is a $(f' \circ \mu')$-ample $\mathbb{R}$-divisor, while we choose $\xi = \frac{\sum r_i}{n}$ can be arbitrarily small. Therefore, we write $\sum_j \delta_j F_j = \mu'^*(\varepsilon E) + \xi F$ is done.

The remain is just apply Theorem 4.11 on $(f \circ \mu)$-ample divisor $\mu^* D - \sum \delta_j F_j$ whose fractional part has snc support, like in the proof for $\mathbb{Q}$-divisor version. □

Now we turn to prove the Grauert-Riemenschneider vanishing theorem:

Proof. Since f is generically finite, that is the generic fibre X_η is of dimension 0, we have $K_X - K_X$ is f-nef and f-big. □

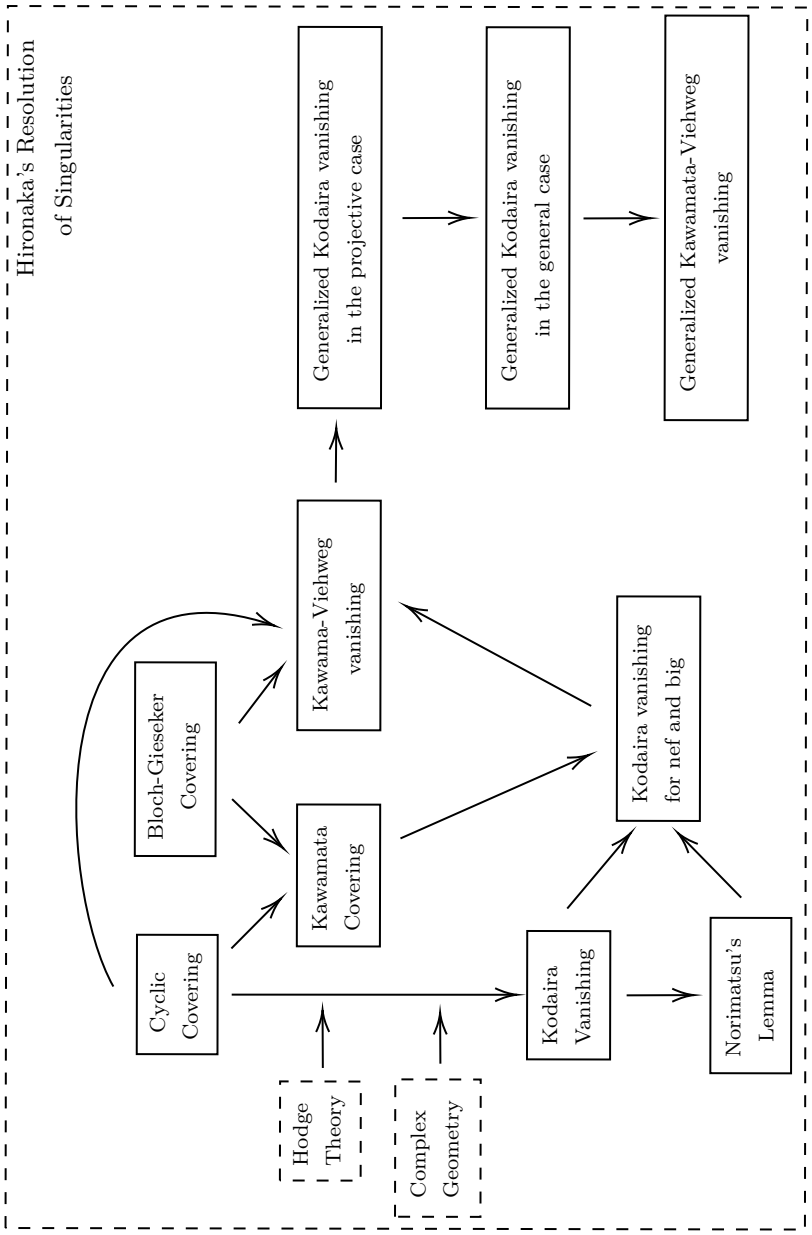


Figure 4.5.: Knowledge Framework

Chapter 5

Introduction to the Minimal Model Program

5.1 Introduction to the Mori's program

Recall that in chapter 2, we proved the result for contracting rational curves on smooth projective surface:

Theorem 5.1.1. (MMP for Surfaces) *Let X be a smooth projective surface and $R \subseteq \overline{\mathrm{NE}}(X)$ an extremal ray such that $(R \cdot K_X) < 0$, i.e. R is generated by a K_X-negative curve. Then the contraction morphism $\mathrm{cont}_R : X \longrightarrow Z$ exists and is one of the following types:*

(1) Z is a smooth surface and X is obtained from Z by blowing up a closed point; $\rho(Z) = \rho(X) - 1$.

(2) Z is a smooth curve and X is a minimal ruled surface over Z; $\rho(X) = 2$.

(3) Z is a point, $\rho(X) = 1$ and $-K_X$ is ample. (In fact, $X \cong \mathbb{P}^2$ but this is harder to prove.)

The cases 1.1(2) and 1.1(3) provide a structure theorem for X where we call these two contractions as *Mori fibre space*, and 1.1(1) introduces a new surface Z. We can apply theorem 1.1 to Z and continue if possible. At each step the Picard number drops by one, and so this process terminates. This gives the following:

Theorem 5.1.2. *Let X be a smooth projective surface. There is a sequence of contractions $X \to X_1 \to \cdots \to X_n = X'$ such that X' satisfies exactly one of the following conditions:*

(1) $K_{X'}$ is nef;

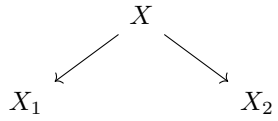
(2) X' is a minimal ruled surface over a curve C;

(3) $X' \cong \mathbb{P}^2$.

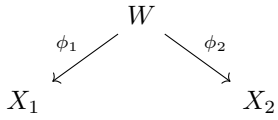
Definition 5.1.3. If $K_{X'}$ is nef as above, then X' is called a *minimal model* of X.

Remark 5.1.4. When X has non-negative Kodaira dimension, $K_{X'}$ being nef is equivalent to X' being a minimal **surface**. Without this assumption, K_X being nef **caon only** imply that it is a minimal surface with no (-1)-curves, but the reverse direction fails, see Figure 2.1 two minimal surfaces but are not minimal model. Therefore, the conception of *minimal surface* and *minimal model* are different. Even though [Har77] uses the term "minimal model", this is purely historical, since the modern *Minimal Model Program* had not yet been born in 1977.

It turns out that in this case the morphism $X \to X'$ is unique, thus X' is determined by the surface X. Indeed, if it is not unique, there are two minimal models X_1 and X_2 such that K_{X_1} and K_{X_2} are nef. Via diagram



we can take a rational map $\phi : X_1 \dashrightarrow X_2$ and then eliminate the indeterminacy to find a smooth surface W which is their common resolution,



with neither of each ϕ_i being an isomorphism. By Castelnuovo's contraction theorem there is a (-1)-curve E contracted by ϕ_1. Write $C := (\phi_2)_* E$ on X_2. Then by

$$K_W = \phi_2^* K_{X_2} + \sum a_i E_i$$

we have

$$(K_W \cdot E) = (K_{X_2} \cdot C) + \sum a_i (E_i \cdot E)$$

where the left term is -1, and the right term is ≥ 0 since K_{X_2} nef. Get contradiction. Similarly on X_1, we see ϕ_1 and ϕ_2 both must be isomorphism.

Now, run the minimal model program on a given smooth projective surface X:

Step 0. We start with a smooth projective surface X.

Step 1. If K_X is nef, then we go to *Step 5*. If K_X is not nef, then by the Cone Theorem we can choose a K_X-negative extremal ray $R \subseteq \overline{\mathrm{NE}}(X)$.

Step 2. By 1.1, the contraction morphism $\mathrm{cont}_R : X \longrightarrow Z$ exists. There are two possibilities.

Step 3. If $\dim Z = \dim X$, then go back to *Step 0* with Z replacing X.

Step 4. If $\dim Z < 2 = \dim X$, then 1.1(2) – (3) gives a structure theorem for X.

Step 5. If K_X is nef, then we stop. These surfaces should be investigated by other methods.

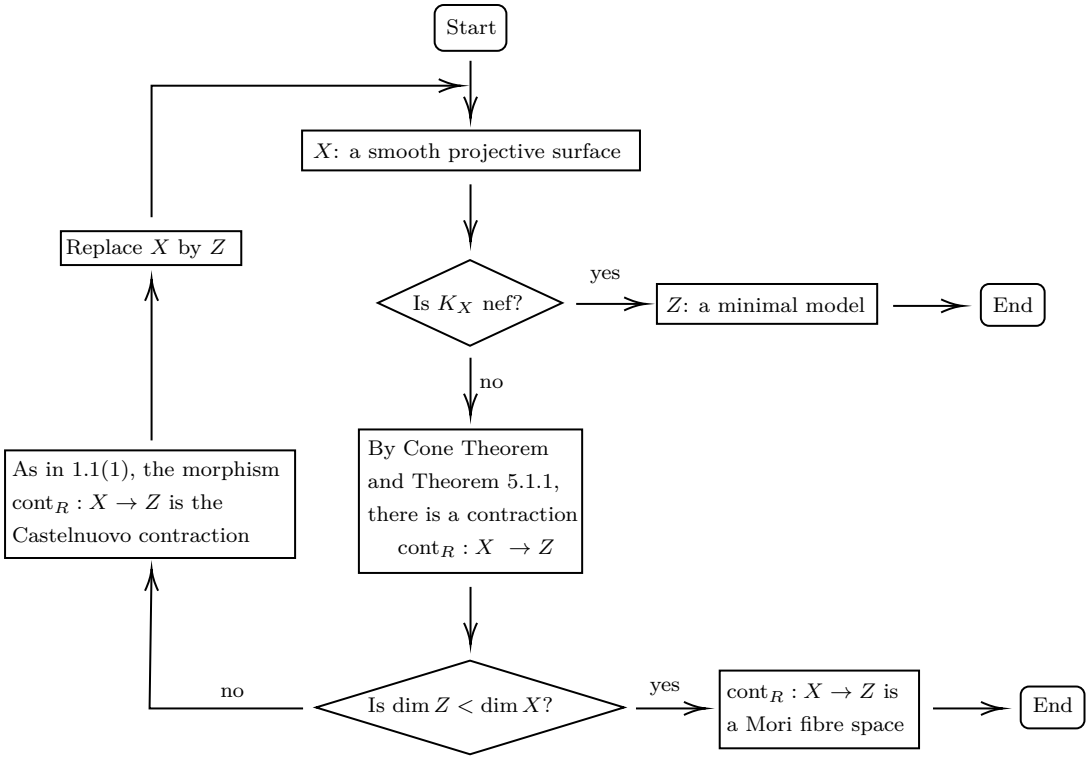


Figure 5.1.: Minimal Model Program for Smooth Surface

The aim of the Minimal Model Program is to provide a similar step by step approach to higher dimensional varieties.

The next step is to develop an analogue of Theorem 1.1 for higher dimensional varieties. There are already numerous cases for smooth 3-folds over $\mathbb{C}$, see [Mor82, Chapter 3].

Theorem 5.1.5. ([Mor82]) *Let X be a non-singular projective threefold over $\mathbb{C}$ and $\mathrm{cont}_R : X \longrightarrow Y$ be the contraction of a K_X-negative extremal ray $R \subseteq \overline{\mathrm{NE}}(X)$. The following is a list of all possibilities for cont_R:*

E: (Exceptional) $\dim Y = 3$, cont_R *is birational and there are five types of local behaviour near the contracted surface:*

E1: cont_R *is the blow-up of a smooth curve in the smooth threefold Y.*

E2: cont_R *is the blow-up of a smooth point of the smooth threefold Y.*

E3: cont_R *is the blow-up of an ordinary double point of Y. (Locally analytically, an ordinary double point is given by the equation $x^2 + y^2 + z^2 + w^2 = 0$.)*

E4: cont_R *is the blow-up of a point of* Y *which is locally analytically given by the equation* $x^2 + y^2 + z^2 + w^3 = 0$.

E5: cont_R *contracts a smooth* $\mathbb{P}^2_{\mathbb{C}}$ *with normal bundle* $\mathscr{O}(-2)$ *to a point of multiplicity* 4 *on* Y *which is locally analytically the quotient of* $\mathbb{C}^3$ *by the involution* $(x, y, z) \mapsto (-x, -y, -z)$.

C: (Conic bundle) $\dim Y = 2$ *and* cont_R *is a fibration whose fibres are plane conics. (General fibres are, of course, smooth.)*

D: (Del Pezzo fibration) $\dim Y = 1$ *and general fibres of* cont_R *are Del Pezzo surfaces.*

F: (Fano variety) $\dim Y = 0$, $-K_X$ *is ample and hence* X *is a Fano variety.*

Note that because of the singularities occurring in cases E3, E4 and E5, we cannot apply Theorem 1.5 to the resulting variety Y. Thus we need to go back and see if one can prove a version of 1.5 when we allow X to have certain singularities. It turns out that the singularities in E3 and E4 do not cause much trouble. But unfortunately, the singularity in case E5 is not a hypersurface singularity, and it is much harder to give a direct geometric analysis of varieties with such singularities.

So it seems that a more conceptual approach is required. As with many questions, there are two opposing points of view:

- Minimalist: We should try to identify the *smallest* class of singularities which we encounter starting with smooth varieties and applying 1.5 and its generalizations.
- Maximalist: We should try to identify the *largest* class of singularities where an analogue of 1.1 is possible.

For the beginning of the section we try to follow the maximalist approach. This is most suitable for general discussion. Unfortunately, essentially none of the results hold without rather strong restrictions on the singularities. Thus, when it comes to proofs, we have to get close to the minimalist approach.

In *Step 1* we first ask if K_X is nef. For this question to make any sense, the *intersection numbers* of K_X with curves have to be defined. These numbers **make sense** if K_X is Cartier, or more generally, when K_X is $\mathbb{Q}$-Cartier. It seems that, starting with dimension three, nothing can be done without this assumption.

Next we have to establish the Cone Theorem for X. For smooth projective varieties this was done at Chapter 2. The main use of smoothness is in the *bend and break* method to get the lower bound of the deformation space, which frequently fails

when X is singular, recall the Corollary 2.1.17. There are, however, a large class of singularities where this lower bound formula holds, and this leads to the Cone Theorem for many singular varieties. It turns out that the Cone Theorem holds for any normal threefold, but it fails in higher dimensions.

In *Step 2* we need to contract the extremal ray $R \subseteq \overline{\mathrm{NE}}(X)$. It seems that this is a much more delicate result, which holds only for rather special singularities.

For the moment we recall a result proved in Chapter 2, which classifies the resulting extremal contractions into three types.

Lemma 5.1.6. *Let X be a normal projective variety and let $R \subseteq \overline{\mathrm{NE}}(X)$ be an extremal ray. Assume that X is $\mathbb{Q}$-factorial. Assume that the contraction of* $\mathbf{r}$ *between projective varieties exists*

$$f := \mathrm{cont}_R : X \longrightarrow Z.$$

Then one of the following holds:

- *f is birational and $\mathrm{Ex}(f)$ is an irreducible divisor (we say that f is **divisorial contraction**);*
- *f is birational and $\mathrm{Ex}(f)$ is of codimension at least two (we say that f is a **small contraction**);*
- $\dim Z < \dim X$ *(we say the f is **of fibre type**).*

The three cases of Lemma 1.6 do behave very differently from the point of view of our program:

Case 1. **Fibre type contractions** correspond to *Step 4*. If $(R \cdot K_X) < 0$, then the general fibre F of f is an algebraic variety where $-K_F = -K_X|_F$ is ample, by Nakai-Moishezon. We call it *Fano fibre space*. Thus, at least in principle, we reduce the problem of understanding X to the study of the lower dimensional variety Y and the fibres of f. Moreover the fibres are of a very special kind - they are analogues of $\mathbb{P}^1_{\mathbb{C}}$ and of Del Pezzo surfaces.

Case 2. **Divisorial contractions** should correspond to *Step 3*. For this to be true, we need to establish that K_Y is $\mathbb{Q}$-Cartier, but preferably that Y is $\mathbb{Q}$-factorial. If this holds, then Y can be considered to be "simpler" than X, as Picard number goes down by 1.

Case 3. **Small contractions**. This is a new situation that never happens for surfaces for dimensional reasons. And it did not happen for smooth three-folds X. We

claim that in this case, Y has "very bad" singularities where no multiple of K_Y is Cartier.

Indeed, assume that mK_Y and mK_X are both Cartier for some suitable $m > 0$. Then mK_X and $f^*(mK_Y)$ are two Cartier divisors on X which are linearly equivalent outside the codimension two set $\mathrm{Ex}(f)$, since f is an isomorphism outside there. But by algebraic Hartogs, they are linearly equivalent everywhere on X. This is, however, impossible since $(R \cdot K_X) < 0$ but $(R \cdot f^*(mK_Y)) = 0$, as R be contracted to a point.

So we were led out of the class of varieties that we can control. In order to continue at this point, we have to introduce a new operation called a *flip*.

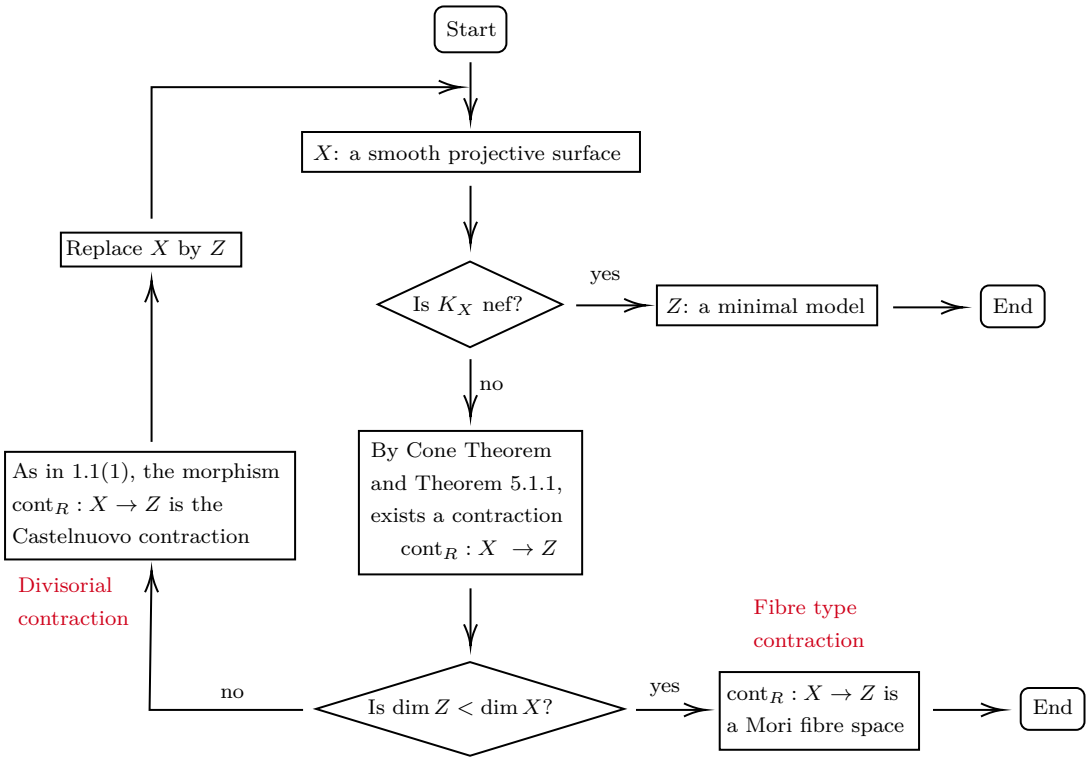


Figure 5.2.: Two possibilities correspond to different contractions.

This flip operation will produce a new canonical divisor K_{X^+} such that making $(R \cdot K_{X^+})$ to be positive, which we can take a substantial step toward the goal of minimal model (K_X is nef).

Example 5.1.7. (Example of flips) In the example, the flip removes the curve $C \cong \mathbb{P}^1$ from the singular variety X_n and replaces it with $D \cong \mathbb{P}^1$ to achieve the "improved" variety X_n^+ which in this case is non-singular.

We start with an auxiliary construction. Let us consider $Y = V(xy - uv) \subseteq \mathbb{A}^4$, which has an isolated singularity at the origin.

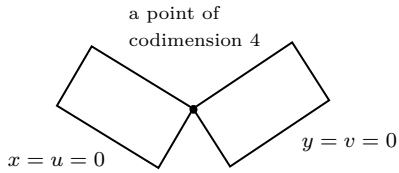


Figure 5.3.: Informal diagram.

To investigate $\tilde{X} = \mathrm{Bl}_0 Y$, we blow up $\mathbb{A}^4$ at the origin to get

$$\mathrm{Bl}_0 \mathbb{A}^4 = \left\{ \big((x,y,u,v),[X:Y:U:V]\big) \in \mathbb{A}^4 \times \mathbb{P}^3 \;\middle|\; \begin{bmatrix} x & y & u & v \\ X & Y & U & V \end{bmatrix} \text{ has rank } 1 \right\} \xrightarrow{\pi} \mathbb{A}^4$$

that is we have equations

$$\begin{cases} xY = yX, xU = uX, xV = vX, \\ yU = uY, yV = vY, uV = vU. \end{cases}$$

So at the origin ($x = y = u = v = 0$), the preimage is $\{0\} \times \mathbb{P}^3 \cong \mathbb{P}^3$, while the relation of X, Y, U, V is free. Now on $\mathbb{P}^3$ which is covered by 4 affine pieces

- On $X \neq 0$, can set $X = 1$. Then for $\tilde{X}$ we have equations

  $$xU = u, xY = y, xV = v, xy - uv = 0.$$

  Substituting into the last equation, we get $x^2(Y - UV) = 0$. Here, $x^2 = 0$ defines the exceptional divisor $E \cong \mathbb{P}^3$ of $\mathrm{Bl}_0\mathbb{A}^4$, and $Y - UV = 0$ defines the strict transform $\tilde{X}$ in this chart. The exceptional set Q of $\tilde{X}$ is the intersection $E \cap \tilde{X}$.

- On $Y \neq 0$, can set $Y = 1$. Then for $\tilde{X}$ we have equations

  $$x = yX, u = yU, v = yV, xy - uv = 0.$$

  Substituting, we get $y^2(X - UV) = 0$ where $y^2 = 0$ defines E and $X - UV$ defines $\tilde{X}$.

Similarly we can get for $U \neq 0$ and $V \neq 0$. Therefore the strict transform $\tilde{X}$ is defined by

$$\left\{ \big((x,y,u,v),[X:Y:U:V]\big) \in \mathbb{A}^4 \times \mathbb{P}^3 \;\middle|\; XY - UV = 0 \text{ and } \begin{bmatrix} x & y & u & v \\ X & Y & U & V \end{bmatrix} \text{ has rank } 1 \right\}.$$

So $\tilde{X}$ restrict on fibre $\pi^{-1}(0)$ is just $V(XY - UV) \subseteq \mathbb{P}^3$, which is just the exceptional set Q of the blowing-up $\mathrm{Bl}_0 Y \to Y$. Hence Q is a quadric surface which isomorphic to the Segre embedding $\mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^3$, so Q has two families of lines

$$\begin{cases} X = cV \\ Y = c^{-1}U \end{cases} \quad \text{and} \quad \begin{cases} X = cU \\ Y = c^{-1}V \end{cases}$$

since by $\det \begin{bmatrix} X & V \\ U & Y \end{bmatrix} = 0$, we have $\begin{bmatrix} X \\ U \end{bmatrix} = c \begin{bmatrix} V \\ Y \end{bmatrix}$ or $[X\ V] = c[U\ Y]$, for constant c. These two families can be blown down to smooth three-folds X (resp. X^+), that is X (resp. X^+) can be obtained by blowing up the ideals (x, v) (resp. (x, u)). Write in details:

- Blowing up $(xy - uv = 0)$ along the plane $(x = v = 0)$:

$$\begin{aligned} X = \mathrm{Bl}_{(x,v)} Y &= \underline{\mathrm{Proj}}_Y \bigoplus_{d \geq 0} (x,v)^d \\ &\cong \underline{\mathrm{Proj}}_Y (A[S,T]/\ker\varphi) \\ &= \left\{ \big((x,y,u,v),[S:T]\big) \in \mathbb{A}^4 \times \mathbb{P}^1 \;\middle|\; xT = vS, yS = uT \right\} \\ &\subseteq \mathbb{A}^4 \times \mathbb{P}^1_{[S:T]}, \end{aligned}$$

  where $A = \mathbb{C}[x,y,u,v]/(xy - uv)$ and a surjection

$$\begin{aligned} \varphi : A[S,T] &\longrightarrow \bigoplus_{d \geq 0} (x,v)^d \\ S &\longmapsto x \\ T &\longmapsto v \end{aligned}$$

  with $\ker\varphi = (xT - vS, yS - uT)$. Here we omit the relation $xy - uv = 0$, since by $xT = vS$ we have

$$xyT = vyS = vuT,$$

  that is $T(xy - uv) = 0$. Similarly $S(xy - uv) = 0$, but S and T cannot be zero simultaneously.

- Blowing up $(xy - uv = 0)$ along the plane $(x = u = 0)$:

$$\begin{aligned} X^+ = \mathrm{Bl}_{(x,u)}Y &= \underline{\mathrm{Proj}}_Y \bigoplus_{d\geq 0}(x,u)^d \\ &= \underline{\mathrm{Proj}}_Y (A[S',T']/\ker\varphi') \\ &= \left\{\left((x,y,u,v),[S':T']\right) \in \mathbb{A}^4 \times \mathbb{P}^1 \;\middle|\; xT' = uS', yS' = vT'\right\} \\ &\subseteq \mathbb{A}^1 \times \mathbb{P}^1_{[S':T']}, \end{aligned}$$

  where $\ker\varphi' = (xT' - uS', yS' - vT')$ coming from the surjection

$$\begin{aligned} \varphi' : A[S',T'] &\longrightarrow \bigoplus_{d\geq 0}(x,u)^d \\ S' &\longmapsto x \\ T' &\longmapsto u. \end{aligned}$$

Now

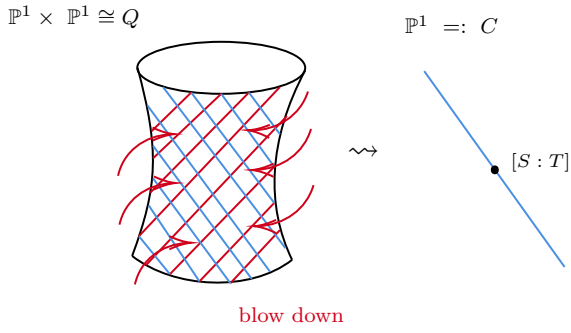
(1) at the family $\begin{cases} X = cV \\ Y = c^{-1}U \end{cases}$ as c varies, we have $\frac{X}{V} = \frac{U}{Y} = c$. So for

$$\begin{aligned} \tilde{X} &\longrightarrow X \longrightarrow Y \\ Q &\longrightarrow C \longrightarrow \{0\}. \end{aligned}$$

For $X = \mathrm{Bl}_{(x,v)}Y = V(xT - vS, yS - uT) \subseteq \mathbb{A}^4 \times \mathbb{P}^1$,we have

$$\begin{aligned} \text{on } \{S \neq 0\} \subseteq \mathbb{P}^1 : \frac{T}{S} = \frac{v}{x} = \frac{y}{u} (= \frac{V}{X} = \frac{Y}{U} \text{ on } \tilde{X}) = c^{-1} \\ \text{on } \{T \neq 0\} \subseteq \mathbb{P}^1 : \frac{S}{T} = \frac{x}{v} = \frac{u}{y} (= \frac{X}{V} = \frac{U}{Y} \text{ on } \tilde{X}) = c \end{aligned}$$

which determines a point in $\mathbb{P}^1$.



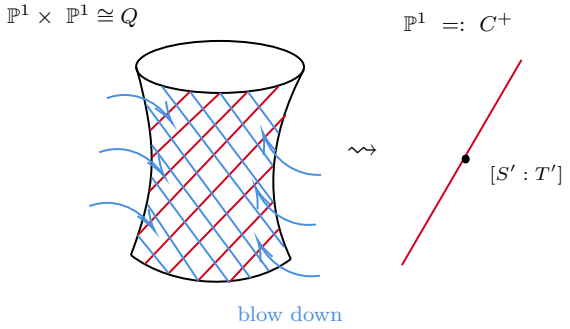
(2) at the family $\begin{cases} X = cU \\ Y = c^{-1}V \end{cases}$ as c varies, we have $\frac{X}{U} = \frac{V}{Y} = c$. So for

$$\begin{aligned} \tilde{X} &\longrightarrow X^+ \longrightarrow Y \\ Q &\longmapsto C^+ \longrightarrow \{0\}. \end{aligned}$$

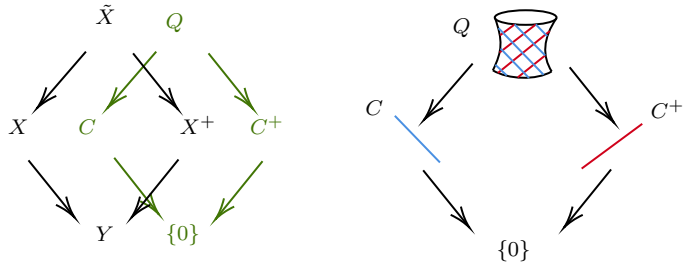
For $X^+ = \mathrm{Bl}_{(x,u)} Y = V(xT' - uS', yS' - vT') \subseteq \mathbb{A}^4 \times \mathbb{P}^1$, we have

$$\begin{aligned} \text{on } \{S' \neq 0\} \subseteq \mathbb{P}^1 : \frac{T'}{S'} = \frac{u}{x} = \frac{y}{v} (= \frac{U}{X} = \frac{Y}{V} \text{ on } \tilde{X}) = c^{-1} \\ \text{on } \{T' \neq 0\} \subseteq \mathbb{P}^1 : \frac{S'}{T'} = \frac{x}{u} = \frac{v}{y} (= \frac{X}{U} = \frac{V}{Y} \text{ on } \tilde{X}) = c \end{aligned}$$

which determines a point in $\mathbb{P}^1$.

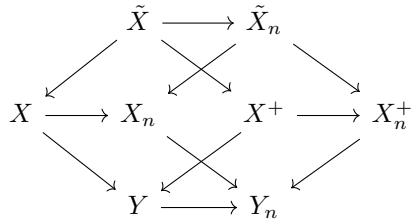


Therefore we have the following diagram



Now we consider the action of the cyclic group $\mu_n : (x, y, u, v) \mapsto (\zeta x, y, \zeta u, v)$, for which $\zeta = \exp(2\pi i/n)$ such that $\zeta^n = 1$. This defines an action on $\tilde{X}$, X, X^+ and Y acting on affine part coordinates. And we denote $\tilde{X}_n := \tilde{X}/\mu_n$, $X_n := X/\mu_n$,

$X_n^+ := X^+/\mu_n$ and $Y_n := Y/\mu_n$, equipped with natural morphisms



On Y, since we have $\zeta x = x$ implies $(\zeta - 1)x = 0$, which gives $x = 0$ and similarly $\zeta u = u$ implies that $u = 0$, so the fixed point set of the action on Y is the 2-plane $(x = u = 0)$.

On quadric $Q \subseteq \tilde{X}$, we have two cases

- $[X : Y : U : V] \mapsto [\zeta X : Y : \zeta U : V] = [X : Y : U : V]$, implies that the fixed point set is the line $(X = U = 0)$.
- $[X : Y : U : V] \mapsto [\zeta X : Y : \zeta U : V] = [\zeta X : \zeta Y : \zeta U : \zeta V]$, implies that the fixed point set is the line $(Y = V = 0)$.

On X, by above, the fixed point set has two components:

- birational transform of the line $(X = U = 0)$;
- image of the line $(Y = V = 0)$, for which is an isolated fixed point. We give local coordinates $(x, v' = vx^{-1}, u)$, which under this action is

$$(x, v', u) \longmapsto (\zeta x, v(\zeta x)^{-1}, \zeta u) = (\zeta x, \zeta^{-1} v', \zeta u).$$

On X^+, as it is the blow-up along the ideal (x, u), that is we introduce a relation

$$xT' = uS' \text{ for } [S' : T'] \in \mathbb{P}^1,$$

where on $\{S' \neq 0\}$, $T' = u/x$ is invariant under the group action. So we give the local coordinate $(x, u' = ux^{-1}, v)$, which sending to $(\zeta x, u', v)$. So the fixed point set is $(x = 0)$ which is irreducible. And on $\{T' \neq 0\}$, $S' = x/u$, the local coordinate is $(x, x' = xu^{-1}, v)$ sending to $(\zeta x, x', v)$ is similar. Therefore, for $C^+ = (x = u = 0 \text{ in } X^+)$, it is contained in the fixed point set $(x = 0 \text{ in } X^+)$. Moreover, X_n^+ is **smooth**, as it locally of the form $\operatorname{Spec} \mathbb{C}[x^n, u', v]$, corresponding to the invariant subring under the action on $\mathbb{C}[x, u', v]$.

Now, we compute the intersection number of the canonical classes with exceptional curves. Since X_n has an isolated singularity, so K_{X_n} is not a Cartier divisor, we

compute the intersection number of nK_{X_n}. Now on piece $\{S = 1\}$ the local coordinate is $(x, v' = vx^{-1}, u)$, and C has coordinate $[S : T] = [1 : T]$ so at this time C is the v'-axis. And for volume form $\omega_1 = dx \wedge dv' \wedge du$ is sent to

$$\omega_1 = dx \wedge dv' \wedge du \longmapsto (\zeta dx) \wedge (\zeta^{-1} dv') \wedge (\zeta du) = \zeta(dx \wedge dv' \wedge du) = \zeta\omega_1,$$

then taking the n-power tensor:

$$\omega_1^{\otimes n} = (dx \wedge dv' \wedge du)^{\otimes n} \longrightarrow \zeta^n (dx \wedge dv' \wedge du)^{\otimes n} = \omega_1^{\otimes n}.$$

So it is a local μ_n-invariant, naturally can descends as a volume form in nK_{X_n}. Then on $\{T = 1\}$ has the local coordinates $(y, u' = uy^{-1}, v)$, with C is the $\frac{1}{v'} = u'$-axis. It also has a volume form $\omega_2 = dy \wedge du' \wedge dv$ which will sends to

$$\omega_2 = dy \wedge du' \wedge dv \longmapsto \zeta(dy \wedge du' \wedge dv) = \zeta\omega_2,$$

is a local μ_n-invariant, thus $\omega_2^{\otimes}$ descends in nK_{X_n}.

Strictly inside the chart $\{T = 1\}$, we compute the local expression on the quotient space. On X_n since the local coordinate $(y, u', v) \mapsto (y, \zeta u', v)$, local coordinate of X_n is of the form $(y, (u')^n, v) := (y, w, v)$, and this time the quotient curve C_n is the w-axis. A natural volume form in nK_{X_n} is

$$\begin{aligned}(dy \wedge dw \wedge dv)^{\otimes n} &= (n(u')^{n-1}(dy \wedge du' \wedge dv)^{\otimes n} \\ &= n^n (u')^{n(n-1)} \omega_2^{\otimes n}.\end{aligned}$$

That is,

$$\omega_2^{\otimes n} = \frac{1}{n^n}(u')^{-n(n-1)}(dy \wedge dw \wedge dv)^{\otimes n}.$$

Now we check if this section extends globally. On the overlap $\{S \neq 0\} \cap \{T \neq 0\}$, we have the transition, for $v' = \frac{v}{x} = (\frac{u}{y})^{-1} = (u')^{-1}$ by $xy = uv$, and $x = u'v$, $u = u'y$. So taking differentials

$$\begin{aligned}\omega_1 = dx \wedge dv' \wedge du &= d(u'v) \wedge d((u')^{-1}) \wedge d(u'y) \\ &= -dv \wedge du' \wedge dy \\ &= dy \wedge du' \wedge dv \\ &= \omega_2.\end{aligned}$$

Gluing them together on the overlap, we see the global section restricts to the chart $\{T = 1\}$, as

$$\begin{aligned}\omega_1^{\otimes n} = \omega_2^{\otimes n} &= \frac{1}{n^n}(u')^{-n(n-1)}(dy \wedge dw \wedge dv)^{\otimes n} \\ &= \frac{1}{n^n}(w)^{-(n-1)}(dy \wedge dw \wedge dv)^{\otimes n}\end{aligned}$$

says that the rational section has a pole of degree $n-1$ along the C_n, so

$$(K_{X_n} \cdot C_n) = \frac{-(n-1)}{n}.$$

Similarly for X_n^+ which is smooth, on piece $\{S' = 1\}$ we have local coordinate $(x^n, u' = ux^{-1}, v)$ with the volume form

$$\omega_1 = d(x^n) \wedge du' \wedge dv.$$

And on piece $\{T' = 1\}$ we have local coordinate $(u^n, x' = xu^{-1}, y)$ with the volume form

$$\omega_2 = d(u^n) \wedge dx' \wedge dy.$$

Therefore on the overlap $x^n = u^n(x')^n$, $u' = (x')^{-1}$ and $v = x'y$, we have the relation

$$\omega_1 = -(x')^{n-1}\omega_2,$$

while as C^+ has the local coordinate u' or x' which is μ_n-invariant, so is C_n^+. Then above shows that

$$(K_{X_n^+} \cdot C_n^+) = n - 1.$$

Thus $X_n^+ \to Y_n$ is the **flip** of $X_n \to Y_n$ for $n \geq 2$.

The operation that happens in the above example can be formalized as follows:

Definition 5.1.8. (Flip) Let $f : X \longrightarrow Y$ be a proper birational morphism such that the exceptional set $\mathrm{Ex}(f)$ has codimension at least two in X. Assume furthermore that K_X is $\mathbb{Q}$-Cartier and $-K_X$ is f-ample. A variety X^+ together with a proper birational morphism $f^+ : X^+ \longrightarrow Y$ is called a *flip* of f if

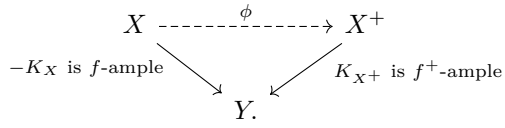
(1) K_{X^+} is $\mathbb{Q}$-Cartier,

(2) K_{X^+} is f^+-ample, and

(3) the exceptional set $\mathrm{Ex}(f^+)$ has codimension at least two in X^+.

By a slight abuse of terminology, the rational map $\phi : X \dashrightarrow X^+$ is also called a *flip*.

We will see that a flip is unique and the main question is its existence. A flip gives the following diagram:



If we perform a flip, it is not clear that X^+ is any "simpler" than X. In the example above this happens since X_n is singular but X_n^+ is smooth.

Before we formalize the above discussions, it is necessary to settle on a class of singularities where Minimal Model program actually works. Such a class was already studied in the 19-th century, see [Die85, VI, 23], in the following setting: Let $Y \subseteq \mathbb{P}^n$ be a non-singular hypersurface with resolution $f : X \longrightarrow Y$. The *adjunction formula* predicts that

$$H^0(X, \mathscr{O}_X(K_X)) \stackrel{?}{=} H^0(Y, \mathscr{O}_Y(K_Y)) = H^0(Y, \mathscr{O}_Y(d-n-1))$$

has the dimension $\binom{d-1}{n}$ via the exact sequence $0 \to \mathscr{O}_{\mathbb{P}^n}(-d) \to \mathscr{O}_{\mathbb{P}^n} \to \mathscr{O}_Y \to 0$. But in general we do not have equality for Y_d singular, because the singularities of Y pose "adjunction conditions". Thus it is of interest to understand which singularities pose no adjunction conditions. In modern terminology these are *canonical singularities*:

Definition 5.1.9. (Canonical singularities) Let Y be a normal variety. We say that Y has *canonical* singularities if

(1) K_Y is $\mathbb{Q}$-Cartier, and

(2) $f_*\mathscr{O}_X(mK_X) \cong \mathscr{O}_Y(mK_Y)$ for every resolution of singularities $f : X \longrightarrow Y$.

The following observation leads to an even smaller class. If Y is smooth, $\gamma \in H^0\big(Y, \mathscr{O}_Y(mK_Y)\big)$ then $f^*\gamma$, as a section of $\mathscr{O}_X(mK_X)$, vanishes along the exceptional divisor $\mathrm{Ex}(f)$. This is stronger than condition (2) of above, which requires only that $f^*\gamma$ have **no poles** along $\mathrm{Ex}(f)$:

$$H^0\big(X, \mathscr{O}_X(mK_X)\big) = H^0\big(Y, f_*\mathscr{O}_X(mK_X)\big) \cong H^0\big(Y, \mathscr{O}_Y(mK_Y)\big)$$
$$f^*\gamma \mapsto \gamma.$$

This stronger condition gives us the next definition, which is the smallest class where Minimal Model program can work.

Definition 5.1.10. (Terminal singularities) Let Y be a normal variety. We say that Y has *terminal* singularities if

(1) K_Y is $\mathbb{Q}$-Cartier, and

(2) $f_*\mathscr{O}_X(mK_X - E) \cong \mathscr{O}_Y(mK_Y)$ for every resolution of singularities $f : X \longrightarrow Y$ where $E \subseteq X$ is the reduced exceptional divisor, i.e. the divisorial part of $\mathrm{Ex}(f)$.

We introduce the concept of higher dimensional minimal models:

Definition 5.1.11. (Minimal model) Let X be a normal and proper variety. We say that X is *minimal* or a *minimal model* if

(1) X has terminal singularities, and

(2) K_X is nef.

If Y is a smooth proper variety birational to X, then X is also called *a minimal model* of Y. It should be stressed that, unlike for surfaces (see under the Remark 1.4), higher dimensional minimal models are not unique. With these definitions at hand, we can formulate a precise step by step outline of the Minimal Model Program.

Minimal Model Program

Starting with a smooth projective variety X, we perform a sequence of understandable birational modifications, until we arrive at a variety X^*, whose global structure is simpler. The following is a more detailed list of the necessary steps.

Step 0 (Initial datum). We start with a projective variety $X = X_0$ over $\mathbb{C}$ with only $\mathbb{Q}$-factorial and terminal singularities. We set up a recursive procedure which produces intermediate varieties X_i and then stop with a final variety X^*. Assume that we have already constructed X_i.

Step 1 (Preparation). If K_{X_i} is nef, then there is nothing to do and we go directly to *Step 3*(2). If K_{X_i} is not nef, then we will establish two results:

(1) (Cone Theorem) $\overline{\mathrm{NE}}(X_i) = \overline{\mathrm{NE}}(X_i)_{K_{X_i}\geq 0} + \sum_i \mathbb{R}_{\geq 0}[C_i]$. If K_{X_i} is not nef, we can pick a K_{X_i}-negative ray $R_i \subseteq \overline{\mathrm{NE}}(X_i)$.

(2) (Contraction of an extremal ray) Let $\mathrm{cont}_{R_i} : X_i \longrightarrow Y_i$ denote the corresponding extremal contraction.

Step 2 (Birational transformations). By Lemma 1.6, we have three possible types of contractions $\mathrm{cont}_{R_i} : X_i \longrightarrow Y_i$. Two of them are used to produce a new variety X_{i+1} as follows.

(1) (Divisorial contraction) If $\mathrm{cont}_{R_i} : X_i \longrightarrow Y_i$ is a divisorial contraction, then set $X_{i+1} = Y_i$. We will prove that X_{i+1} again has $\mathbb{Q}$-factorial and terminal singularities, so we can go back to *Step 0* with X_{i+1} and start anew.

(2) (Flipping contraction) If $\mathrm{cont}_{R_i} : X_i \longrightarrow Y_i$ is a flipping contraction (small contraction), then set $X_{i+1} = X_i^+$, the flip of $\mathrm{cont}_{R_i} : X_i \longrightarrow Y_i$. We will prove that X_{i+1} again has $\mathbb{Q}$-factorial and terminal singularities, so we can go back to *Step 0* with X_{i+1} and start anew.

Step 3 (Final outcome). We hope that eventually the procedure stops, and we get one of the following two possibilities:

(1) (Mori fibre space) If $\mathrm{cont}_{R_i} : X_i \longrightarrow Y_i$ is a Fano contraction (fibre type contraction), then set $X^* := X_i$. The hope is that the investigation of the lower dimensional variety Y_i and of the fibres provides new methods to study X^*.

(2) (Minimal model) If K_{X_i} is nef then again set $X^* := X_i$. We hope to be able to exploit the semi-positivity of the canonical class in further attempts to understand X^*.

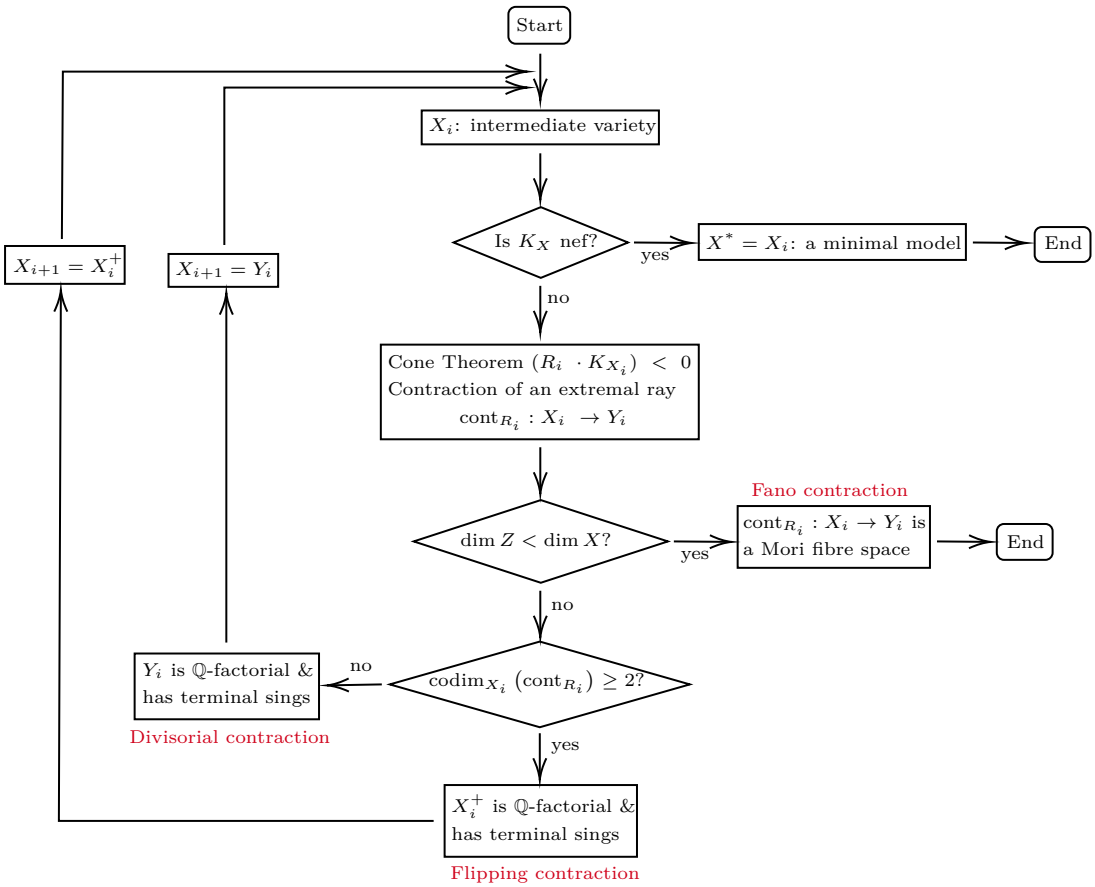


Figure 5.4.: Minimal Model Program

5.2 Singularities in the Minimal Model Program

As we saw in Theorem 1.5, contractions of extremal rays lead to singular varieties, like in (E3)-(E5). In order to continue, we have to understand the singularities that occur in the process and we need a new way to "measure" how singular a variety is. This new measure is called *discrepancy*.

Definition 5.2.1. (Canonical divisor) Let X be a normal variety. The *canonical divisor* K_X on X is a divisor such that $\mathscr{O}_{X_{\mathrm{reg}}}(K_X) \cong \omega_{X_{\mathrm{reg}}}$, where X_{reg} is the smooth locus of X. Note that the canonical divisor K_X is well-defined up to linear equivalence.

For a (Weil) divisor D, its corresponding sheaf $\mathscr{O}_X(D)$ may not be an invertible sheaf (i.e. line bundle) anymore. It is called the *divisorial sheaf*, which locally defined as below: for any open subset $U \subseteq X$,

$$U \longmapsto H^0\big(U, \mathscr{O}_X(D)\big) = \big\{h \in \mathbb{C}(X)^\times \,\big|\, \mathrm{div}(h)|_U + D|_U \geq 0\big\} \cup \{0\}.$$

Similar as in Chapter 1 of Cartier divisor case, a nonzero global section s of $\mathscr{O}_X(D)$ corresponds to a rational function h, we define the *divisor of zeros* of s by

$$\mathrm{div}(s) = \mathrm{div}(h) + D,$$

which is effective. Generally we can also define the divisor $\mathrm{div}(s)$ of a rational section s of $\mathscr{O}_X(D)$ by the corresponding rational function h as in the above equation, but in this case $\mathrm{div}(s)$ is not necessarily effective. For example like in Example 1.1.11, if we take s to be the rational section corresponding to the rational function $h = 1$, then the corresponding divisor is just D.

There is an isomorphism $\mathscr{O}_X(D)_\eta \cong \mathscr{O}_{X,\eta} = \mathbb{C}(X)$, since in the process

$$\mathscr{O}_X(D)_\eta = \varinjlim_{\eta \in U} H^0(X, \mathscr{O}_X(D)),$$

we can always shrink U to avoid of the negative part of the divisors D and $\mathrm{div}(h) = \sum v_E(h)E$. Moreover, since at the generic point the morphism $\mathscr{O}_X(D) \to \mathscr{O}_X$ will be $\mathbb{C}(X) \to \mathbb{C}(X)$, so locally over the open subset $U \subseteq X$ the morphism is defined by

$$\begin{aligned} H^0(U, \mathscr{O}_X(D)) &\longrightarrow H^0(U, \mathscr{O}_X) \\ s &\longmapsto s \cdot f \end{aligned}$$

with $f \in \mathbb{C}(X)^\times$, and we have that

$$\begin{aligned} \mathrm{div}(s \cdot f)|_U &= \mathrm{div}(s)|_U + \mathrm{div}(f)|_U \\ &= \mathrm{div}(h)|_U + D|_U + \mathrm{div}(f)|_U \\ &\geq 0, \end{aligned}$$

that is f allowing to have pole along D. Therefore we see

$$\mathscr{O}_X(D)^\vee := \mathscr{H}om_{\mathscr{O}_X}(\mathscr{O}_X(D), \mathscr{O}_X) \cong \mathscr{O}_X(-D),$$

hence the divisorial sheaf $\mathscr{O}_X(D)$ is a *reflexive sheaf of rank 1.* Here a *reflexive sheaf* is a **coherent** sheaf which is isomorphic to its double dual: $(\mathscr{F}^\vee)^\vee \cong \mathscr{F}$. Furthermore, can see [Sch]. Also two linearly equivalent divisors have the same divisorial sheaf.

5.2A Definition of Discrepancies

Definition 5.2.2. (Discrepancy) Let X be a normal variety such that mK_X is Cartier for some $m > 0$. Suppose $f : Y \longrightarrow X$ is a (not necessarily proper) birational morphism from a normal variety Y. Let $E \subseteq Y$ be a prime exceptioanl divisor, picking $e \in E$ is a general point of E and $\{y_i\}$ is a regular system of parameters at $e \in Y$ such that E is locally cut out by $y_1 = 0$. Then locally near the point e,

$$f^*\big(\text{local generator of } \mathscr{O}_X(mK_X)\big) = y_1^{m\cdot a(E,X)}(\text{unit})(dy_1 \wedge \cdots \wedge dy_n)^{\otimes m}$$

for some rational number $a(E, X)$ such that $m \cdot a(E, X)$ is an integer. Here $a(E, X)$ is called the *discrepancy* of E with respect to X.

Proposition 5.2.3. *$a(E, X)$ is independent of the choice of m.*

Proof. For general point $e \in E$, choose two different positive integers m and m' such that mK_X and $m'K_X$ are both Cartier. Obviously for $mm'K_X$ is a Cartier divisor. Now for local generators ω_X and ω'_X respect to mK_X and $m'K_X$, we have

$$\begin{aligned} f^*\omega_X &= y_1^{m\cdot a(E,X)}(\text{unit})(dy_1 \wedge \cdots \wedge dy_n)^{\otimes m}, \\ f^*\omega'_X &= y_1^{m'\cdot a'(E,X)}(\text{unit})(dy_1 \wedge \cdots \wedge dy_n)^{\otimes m'}. \end{aligned}$$

But $\omega_X^{\otimes m'}$ and ${\omega'}_X^{\otimes m}$ are both local generators of $mm'K_X$, which only differ by a unit that we omit:

$$\begin{aligned} y_1^{m'm\cdot a(E,X)}(\text{unit})(dy_1 \wedge \cdots \wedge dy_n)^{\otimes mm'} &= (f^*\omega_X)^{\otimes m'} \\ &= (f^*\omega'_X)^{\otimes m} \\ &= y_1^{mm'\cdot a'(E,X)}(\text{unit})(dy_1 \wedge \cdots \wedge dy_n)^{\otimes m'm}. \end{aligned}$$

This implies that $m'ma(E, X) = mm'a'(E, X)$, so $a(E, X) = a'(E, X)$. □

Proposition 5.2.4. *In above setting, if $f : Y \longrightarrow X$ is a proper birational morphism of normal varieties such that mK_Y is Cartier (for instance, Y is smooth), then mK_Y*

is linearly equivalent to

$$f^*(mK_X) + \sum_i \big(m \cdot a(E_i, X)\big)E_i,$$

where the E_i are the f-exceptional divisors.

Proof. Over the open subset $U \subseteq X$ they are isomorphic, we have an isomorphism $\mathscr{O}_Y(mK_Y) \cong f^*\mathscr{O}_X(mK_X)$, so the difference can only happen on the f-exceptional part. We write

$$mK_Y \sim_{lin} f^*(mK_X) + \sum c_i E_i$$

where E_i are the f-exceptional divisors. Locally at the generic point $\eta_{E_i} \in E_i$, since Y is a normal variety, the one-dimensional local ring $\mathscr{O}_{Y,\eta_{E_i}}$ is a regular local ring. That is we can pick regular system of parameters $\{y_j\}$ such that locally near η_{E_i},

$$f^*(\text{local generator of } \mathscr{O}_X(mK_X)) = y_1^{m\cdot a(E_i,X)}(\text{unit})(dy_1 \wedge \cdots \wedge dy_n)^{\otimes m} \tag{5.1}$$

where E_i is locally cut out by $y_1 = 0$, i.e. $m_{\eta_{E_i}} = (y_1)$ in $\mathscr{O}_{Y,\eta_{E_i}}$. But this is equivalent to say, locally $\mathscr{O}_Y(mK_Y)$ is a subsheaf of $f^*\mathscr{O}_X(mK_X)$, where the injective map is given by multiplication by a local section of $\mathcal{O}_Y(-m \cdot a(E, X, \Delta)E_i)$.

Therefore globally

$$mK_Y \sim_{lin} f^*(mK_X) + \sum_{E_i : f-\text{exceptional}} \big(m \cdot a(E_i, X)\big)E_i.$$

Also, it can be written as a strict equation after a suitable change from (5.1). □

Using numerical equivalence, we can divide by m and write

$$K_Y \equiv_{num} f^*K_X + \sum_{E_i : f-\text{exceptional}} a(E_i, X)E_i$$

Remark 5.2.5. On a normal variety Y with function field $\mathbb{C}(Y)$, the one-dimensional local ring $\mathscr{O}_{Y,\eta_E} \subseteq \mathbb{C}(Y)$ at the generic point of the f-exceptional divisor E, is a discrete valuation ring which corresponds to a valuation $v(E, Y)$ of $\mathbb{C}(Y) \cong \mathbb{C}(X)$. Such valuations of $\mathbb{C}(X)$ are called *algebraic valuations*.

Let $f' : Y' \longrightarrow X$ be another birational morphism and $E' \subseteq Y'$ a prime divisor such that $v(E', Y') = v(E, Y)$. This holds if and only if the rational map $Y \to X \dashrightarrow Y'$ is an isomorphism at the generic point $\eta_E \in E$ and $\eta_{E'} \in E'$. Then $a(E, X) = a(E', X)$, as one can see from the definition.

Thus $a(E, X)$ depends only on the valuation $v(E, Y)$ but not on the particular choice of f and Y. This is why f and Y do not appear in the notation $a(E, X)$.

Definition 5.2.6. Let X be a variety, with $f : Y \longrightarrow X$ is a (not necessarily proper) birational morphism from a normal variety Y, and $E \subseteq Y$ is a prime divisor. Any such E is called a *divisor over* X. The closure of $f(E)$ in X is called the *center* of E on X. It is denoted by $\operatorname{center}_X E$.

As above, the center depends only on the local isomorphism at the generic point $\eta_E \in E$, that is, it depends only on the valuation $v(E, Y)$.

This can be generalized to pairs (X, Δ) such that the divisor $K_X + \Delta$ is $\mathbb{Q}$-Cartier.

Definition 5.2.7. (Generalized discrepancy) Let (X, Δ) be a pair where X is a normal variety and $\Delta = \sum a_i D_i$ is a sum of distinct prime divisors, with each $a_i \in \mathbb{Q}$. Assume that $m(K_X + \Delta)$ is Cartier for some $m > 0$. Suppose that $f : Y \longrightarrow X$ is a birational morphism from a normal variety Y. Let $E = \operatorname{Ex}(f) \subseteq Y$ with $E_i \subseteq E$ denotes the prime divisors. Let

$$f_*^{-1}\Delta := \sum a_i f_*^{-1} D_i$$

denote the strict transform of Δ.

As in Definition 2.2, picking $e_i \in E_i$ to be a general point of E_i and $\{y_i\}$ is a regular system of parameters at $e_i \in Y$ such that E_i is locally cut out by $y_1 = 0$. Then locally near the point e_i:

$$f^*\Big(\text{local gen. of } \mathscr{O}_X\big(m(K_X+\Delta)\big)\Big) = y_1^{m\cdot a(E_i,X,\Delta)}(\text{unit})\Big(\text{local gen. of } \mathscr{O}_Y\big(m(K_Y+f_*^{-1}\Delta)\big)\Big)$$

for some rational number $a(E_i, X, \Delta)$ such that $m \cdot a(E_i, X, \Delta)$ is an integer.

Proposition 5.2.8. *With above situation, the two line bundles*

$$\mathscr{O}_X\big(m(K_Y + f_*^{-1}\Delta)\big)\big|_{Y\setminus E} \quad \textit{and} \quad f^*\mathscr{O}_X\big(m(K_X + \Delta)\big)\big|_{Y\setminus E}$$

are naturally isomorphic.

Proof. Since f is an isomorphism on $Y\setminus E$. □

Thus there is a linearly equivalent relation

$$m(K_Y + f_*^{-1}\Delta) \sim_{lin} f^*\big(m(K_X + \Delta)\big) + \sum_{E_i : f-\text{exceptional}} \big(m \cdot a(E_i, X, \Delta)\big)E_i.$$

We call $a(E, X, \Delta)$ the *discrepancy* of E with respect to pair (X, Δ). We frequently write $a(E)$ if no confusion is likely.

Recall that twisting a line bundle by $\mathscr{O}_X(m\Delta)$ allows the local sections of that line bundle to have poles along $m\Delta$, so by definition we have

$$a(D_i, X, \Delta) = -a_i$$

and $a(D, X, \Delta) = 0$ for any divisor $D \subseteq X$ which is different from the D_i.

As in the $\Delta = 0$ case, $a(E_i, X, \Delta)$ depends only on E_i but not on f.

Remark 5.2.9. Using numerical equivalence, when X is proper ensures us to define the intersection number, we can divide by m and write

$$K_Y + f_*^{-1}\Delta \equiv_{num} f^*(K_X + \Delta) + \sum_{E_i : f-\text{exceptional}} a(E_i, X, \Delta)E_i, \text{ or}$$
$$K_Y \equiv_{num} f^*(K_X + \Delta) + \sum_{E_i : \text{arbitrary}} a(E_i, X, \Delta)E_i,$$

where the second line the $-f_*^{-1}\Delta = \sum -a_i f_*^{-1} D_i$ becomes $\sum_{E_i : \text{non-exceptional}} a(E_i, X, \Delta)E_i$ since we have $a(D_i, X, \Delta) = -a_i$ and $a(f_*^{-1}D_i, X, \Delta) = a(D_i, X, \Delta)$.

Moreover, for general case when X is not necessarily proper, we often choose the birational morphism $f : Y \longrightarrow X$ is proper, such that we can define the intersection number on the proper relative curve C, and relation to be $\equiv_f$ or $\equiv_X$.

Remark 5.2.10. We can change the representative of the canonical divisor K_Y, writing down the equation of $\mathbb{Q}$-**Cartier divisors**:

$$K_Y + f_*^{-1}\Delta = f^*(K_X + \Delta) + \sum_{E_i : f-\text{exceptional}} a(E_i, X, \Delta)E_i.$$

This has been used in the proof of the generalized Kawamata-Viehweg vanishing.

Lemma 5.2.11. *Notation as above. Assume that the divisor Δ' is effective and $\mathbb{Q}$-Cartier. Then*

$$a(E, X, \Delta) \geq a(E, X, \Delta + \Delta')$$

for every divisor E over X, and $>$ holds if and only if $\operatorname{center}_X E \subseteq \operatorname{Supp} \Delta'$.

Proof. Since the divisor Δ' is $\mathbb{Q}$-Cartier, we can choose a common multiple $m > 0$ such that $m(K_X + \Delta)$ and $m\Delta'$ are both be Cartier. Also, we pick a general point $e_i \in E_i$ with a regular system of parameters $\{y_j\}$ such that E_i is locally cut out by $y_1 = 0$. Then locally near the point e_i, we have

$$f^*\Big(\text{l.g. of } \mathscr{O}_X\big(m(K_X + \Delta)\big)\Big) = y_1^{m \cdot a(E,X,\Delta)} u\Big(\text{l.g. of } \mathscr{O}_Y\big(m(K_Y + f_*^{-1}\Delta)\big)\Big),$$
$$f^*\Big(\text{l.g. of } \mathscr{O}_X\big(m(K_X + \Delta + \Delta')\big)\Big) = y_1^{m \cdot a(E,X,\Delta+\Delta')} u\Big(\text{l.g. of } \mathscr{O}_Y\big(m(K_Y + f_*^{-1}\Delta + f_*^{-1}\Delta')\big)\Big),$$

where u is a unit. So globally we write down the strict equation

$$mK_Y = mf^*(K_X + \Delta) + \sum_{E_i : \text{arbitrary}} m \cdot a(E_i, X, \Delta)E_i,$$
$$mK_Y = mf^*(K_X + \Delta + \Delta') + \sum_{E_i : \text{arbitrary}} m \cdot a(E_i, X, \Delta + \Delta')E_i,$$

whose difference is

$$\sum_{E_i:\text{arbitrary}} \big(a(E_i, X, \Delta) - a(E_i, X, \Delta + \Delta')\big)E_i = f^*\Delta' \geq 0,$$

so $a(E_i, X, \Delta) \geq a(E_i, X, \Delta + \Delta') = a(E_i, X, \Delta) - \operatorname{mult}_{E_i} f^*\Delta'$.

Moreover, $\operatorname{center}_X E \subseteq \operatorname{Supp} \Delta'$ holds if and only if the coefficient of E on $f^*\Delta'$ is strictly positive. □

For us the most important values are the minima of $a(E, X, \Delta)$ as E runs through various sets of divisors. We use several versions:

Definition 5.2.12. (Discrepancy of pair) The *discrepancy* of the pair (X, Δ) is given by

$$\operatorname{discrep}(X, \Delta) := \inf_E \{a(E, X, \Delta) \;\big|\; E \text{ is an exceptional divisor over } X\},$$

that is, $E \subseteq Y$ runs through all the irreducible exceptional divisors of all birational morphisms $f : Y \longrightarrow X$. **Equivalently**, $\operatorname{discrep}(X, \Delta) := \inf_v \{a(v, X, \Delta)\}$ where v runs through all algebraic valuations of $\mathbb{C}(X)$ such that $\operatorname{center}_X v$ is non-empty and has codimension at least 2 in X. When $\Delta = 0$ this is for X.

The *total discrepancy* of the pair (X, Δ) is defined as

$$\operatorname{totaldiscrep}(X, \Delta) := \inf_E \{a(E, X, \Delta) \;\big|\; E \text{ is a divisor over } X\},$$

that is, $E \subseteq Y$ runs through all the irreducible exceptional divisors for all birational morphisms $f : Y \longrightarrow X$ and through all the irreducible divisors of X. **Equivalently**, $\operatorname{totaldiscrep}(X, \Delta) := \inf_v \{a(v, X, \Delta)\}$ where v runs through all algebraic valuations of $\mathbb{C}(X)$ such that $\operatorname{center}_X v$ is non-empty. When $\Delta = 0$ this is for X.

5.2B Computation of Discrepancies

First we recall a result of Zariski [Zar39, 19.10], which shows that every divisor can be reached by a sequence of blow-ups. The proof is taken from [Art86, 5.2].

Lemma 5.2.13. *Let X be an algebraic variety over a arbitrary field (not necessarily closed). Let (R, m_R) be a D.V.R of the functional field $k(X)$ of X with*

$$\operatorname{tr.deg}_k R/m_R = \dim X - 1.$$

Let $Y = \operatorname{Spec} R$ be affine, with $y \in Y$ the closed point and $f : Y \longrightarrow X$ the induced birational morphism.

We define a sequence of varieties and morphisms as follows:

- *Set $X_0 = X$ and $f_0 = f$.*
- *If $f_i : Y \longrightarrow X_i$ is already defined, then let $Z_i \subseteq X_i$ be the closure of $x_i = f_i(y)$. Let $X_{i+1} = \mathrm{Bl}_{Z_i} X_i$ and $f_{i+1} : Y \longrightarrow X_{i+1}$ the induced morphism.*

Then $f_n : Y \longrightarrow X_n$ induces an isomorphism $\mathscr{O}_{X_n,x_n} \cong R$ for some $n \geq 0$.

Proof. We recall that the D.V.R defines a valuation

$$\begin{aligned} v_R(x) &= \left\{ n \in \mathbb{Z} \middle| \ x = t^n u \text{ where } u \text{ is unit and } t \in m_R \right\} \\ &= \max \left\{ n \in \mathbb{Z} \middle| \ x \in m_R^n \right\} \text{ for } x \neq 0 \text{ in } k(X). \end{aligned}$$

Step 1: Assume that the morphism $f_n : Y \longrightarrow X_n$ is already defined, sending the closed point y to x_n (possibly non-closed since R is not finite type), which gives a injective ring map $\mathscr{O}_{X_n,x_n} \hookrightarrow R$ by f_n is dominant.

The following is the ring theoretic version of the sequence of blow-ups. Set $\mathscr{O}_0 := \mathscr{O}_{X_0,x_0}$ with maximal ideal m_0, Let $z_1, \ldots, z_r$ be the generators of m_n such that

$$v_R(z_1) \leq v_R(z_2) \leq \cdots \leq v_R(z_r).$$

We set $\mathscr{O}'_n := \mathscr{O}_n[\frac{z_2}{z_1}, \ldots, \frac{z_r}{z_1}]$, which is corresponds to an affine subset $D_+(z_1)$ in $X_{n+1} = \mathrm{Bl}_{Z_n} X_n$ over $\operatorname{Spec} \mathscr{O}_n$, where $Z_n = \overline{\{x_n\}}$. Concretely speaking, for a natural flat morphism $g : \operatorname{Spec} \mathscr{O}_n \longrightarrow \operatorname{Spec} A \subseteq X_n$ where we regard $\mathscr{O}_n$ as the localization of A, we have the following Cartesian diagram

$$\begin{array}{ccc} X_{n+1} \underset{X_n}{\times} \operatorname{Spec} \mathscr{O}_n & \longrightarrow & X_{n+1} \\ \downarrow & & \downarrow \\ \operatorname{Spec} \mathscr{O}_n & \xrightarrow[g]{} & X_n \end{array}$$

where $X_{n+1} = \mathrm{Bl}_{Z_n} X_n = \underline{\mathrm{Proj}}_{X_n} \bigoplus_{d \geq 0} \mathcal{I}_{Z_n}^d$. Following from [EGA II, 3.5.3] or [GW20, 13.27], we have

$$X_{n+1} \underset{X_n}{\times} \operatorname{Spec} \mathscr{O}_n \cong \underline{\mathrm{Proj}}_{\mathscr{O}_n} g^* \left(\bigoplus_{d \geq 0} \mathcal{I}_{Z_n}^d \right) \cong \underline{\mathrm{Proj}}_{\mathscr{O}_n} \bigoplus_{d \geq 0} g^* \mathcal{I}_{Z_n}^d.$$

But since g is flat, the pullback functor is exact; so we have that

$$g^* \mathcal{I}_{Z_n}^d \cong g^{-1} \mathcal{I}_{Z_n}^d \cdot \mathscr{O}_n \cong m_n^d.$$

That is, $X_{n+1} \underset{X_n}{\times} \operatorname{Spec} \mathscr{O}_n \cong \underline{\mathrm{Proj}}_{\mathscr{O}_n} \bigoplus_{d \geq 0} m_n^d = \mathrm{Bl}_{m_n} \operatorname{Spec} \mathscr{O}_n$. As the base variety is affine, $\underline{\mathrm{Proj}}_{\mathscr{O}_n} \bigoplus_{d \geq 0} m_n^d$ is a subvariety of $\mathbb{P}^N_{\mathscr{O}_n}$ whose affine coordinate ring on $D_+(z_1)$ is

isomorphic to

$$\deg 0 \text{ of } \left(\bigoplus_{d\geq 0} m_n^d\right)_{z_1} \cong \mathscr{O}_n\left[\frac{z_2}{z_1}, \ldots, \frac{z_r}{z_1}\right].$$

Moreover since in D.V.R the every ideal is principal, the inverse image ideal sheaf $f_n^{-1}\mathcal{I}_{Z_n} \cdot \mathscr{O}_Y$ is free of rank 1 which is invertible. So by the universal property of blowing-up, the morphism f_n will factor into $f_{n+1} : Y \longrightarrow X_{n+1}$.

Also, by our choice of order of valuations, we have

$$v_R(z_i/z_1) = v_R(z_i) - v_R(z_1) \geq 0,$$

so each elements $\frac{z_i}{z_1}$ lie in R which gives the injective ring map $\mathscr{O}'_n \hookrightarrow R$. This is equivalent to say the point $x_{n+1} = f_{n+1}(y)$ lies in the affine subset $\operatorname{Spec} \mathscr{O}'_n$, so the local ring of X_{n+1} at the point x_{n+1}, $\mathscr{O}_{n+1}$ is the localization of $\mathscr{O}'_n$ at $\mathscr{O}'_n \cap m_R$ and m_{n+1} is the maximal ideal of $\mathscr{O}_{n+1}$.

Step 2: Now we claim that $R = \bigcup_{n\geq 0} \mathscr{O}_n$. By our construction we have $\mathscr{O}_n \subseteq R$ for each n, so we need to prove the converse inclusion. To see this, we take an arbitrary $u \in R \backslash \mathscr{O}_n$. Since in each inductive step is birational, they share the same function field

$$R \subseteq k(X) \cong k(X_n) \cong \operatorname{Frac} \mathscr{O}_n$$

and same valuation v_R on each subring $\mathscr{O}_n$. So elements u can be written as the from y_1/y_2 where $y_i \in \mathscr{O}_n$ for $i = 1, 2$. Here we choose that $v_R(y_2)$ is the smallest possible. Then each $y_i \in m_n$ by the following:

$$u \notin \mathscr{O}_n \Leftrightarrow y_2 \in m_n \text{ is not a unit} \Leftrightarrow v_R(y_2) > 0,$$

therefore we have

$$u \in R \Rightarrow v_R(u) = v_R(y_1) - v_R(y_2) \geq 0 \Rightarrow v_R(y_1) \geq v_R(y_2) > 0,$$

which implies that $y_1 \in m_n$. Now, we can write

$$y_i = \sum_{j=1}^{r} z_j y_{ij}$$

where $y_{ij} \in \mathscr{O}_n$ and z_j are generators of m_n as above. Let

$$y'_i = \sum_{j=1}^{r} \frac{z_j}{z_1} y_{ij} = y_{i1} + \frac{z_2}{z_1} y_{i2} + \cdots + \frac{z_r}{z_1} y_{ir}$$

which lies in $\mathscr{O}'_n = \mathscr{O}_n[\frac{z_2}{z_1}, \ldots, \frac{z_r}{z_1}]$, so furthermore in $\mathscr{O}_{n+1}$ by we have inclusion $\mathscr{O}'_n \subseteq \mathscr{O}_{n+1}$. Then $y_i = z_1 y'_i$ and $u = y'_1/y'_2$. By construction, we have

$$v_R(y'_2) = v_R(y_2) - v_R(z_1) < v_R(y_2).$$

If $v_R(y'_2) \neq 0$, then $y'_2 \in m_{n+1}$ and y'_1 also in m_{n+1} by $u \in R$. So we can continue this procedure, and eventually can obtain that $u = y_1^{(s)}/y_2^{(s)}$ and $v_R(y_2^{(s)})$ after finite steps. This implies that $u \in \mathscr{O}_{n+s}$ for some s. Hence $R = \bigcup_{n \geq 0} \mathscr{O}_n$.

Step 3: Picking $t_1, \ldots, t_r \in R$ which give a transcendence basis of R/m_R over k. Then $t_1, \ldots, t_r \in \mathscr{O}_n$ for some $n \gg 1$, by *Step 2*. Hence for residue field $k(x_n) = \mathscr{O}_n/m_n$, we have $k(x_n) \subseteq R/m_R$ as a subfield. So

$$\mathrm{tr.deg}_k\, k(x_n) \leq \mathrm{tr.deg}_k\, R/m_R.$$

And the algebraic relation of $t_1, \ldots, t_r$ in $k(x_n)$ can be extended in R/m_R, so

$$\mathrm{tr.deg}_k\, k(x_n) \geq r = \mathrm{tr.deg}_k\, R/m_R.$$

Therefore, we can get

$$\mathrm{tr.deg}_k\, k(x_n) = \mathrm{tr.deg}_k\, R/m_R = \dim X - 1 \text{ for } n \gg 1. \tag{5.2}$$

Now, since $\mathscr{O}_n$ is a localization of a finite type k-algebra, by construction in *Step 1*, we have

$$\dim_{x_n} X_n = \dim X_n - \dim \overline{\{x_n\}}$$

which is nothing but

$$\dim \mathscr{O}_n = \mathrm{tr.deg}_k\, k(X) - \mathrm{tr.deg}_k\, k(x_n).$$

Combing this with (5.2) we have

$$\begin{aligned} \mathrm{tr.deg}_{k(x_n)}\, R/m_R &= \mathrm{tr.deg}_k\, R/m_R - \mathrm{tr.deg}_k\, k(x_n) \\ &= \dim X - 1 - (\dim X - \dim \mathscr{O}_n) \\ &= \dim \mathscr{O}_n - 1 \\ &= 0 \text{ for } n \gg 1. \end{aligned}$$

Thus R/m_R is an algebraic extension of $k(x_n)$ and $\dim \mathscr{O}_n = 1$ for $n \gg 1$. Since every valuation ring is integrally closed, we have the relation

$$\mathscr{O}_n \subseteq R' \subseteq R \subseteq k(X)$$

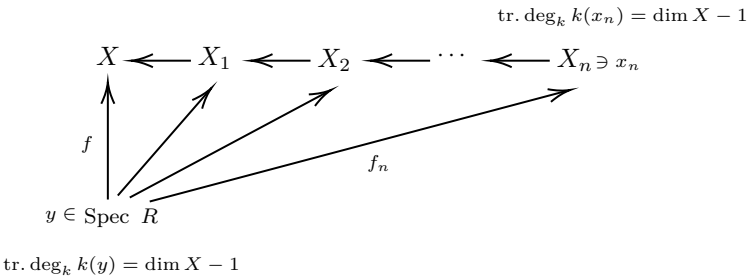
where R' is the integral closure of $\mathscr{O}_n$ in $k(X)$; and by Noether's finiteness theorem R' is a finite $\mathscr{O}_n$-module, so $\dim \mathscr{O}_n = \dim R' = 1$, see [Eis95, 13.13]. Taking localization we can get R'_p is a D.V.R since it is integrally closed and dimension one; and $R'_p \subseteq R$ gives that $R'_p = R$, since valuation ring is the maximal element among the relation of domination of local rings contained in $k(X)$, see [Har77, I, 6.1-2A]. Now for finitely many generators $v_1, \ldots, v_s \in R' \subseteq R$ of R' over $\mathscr{O}_n$, for $k \gg n$ we have $v_1, \ldots, v_s \in \mathscr{O}_k$, i.e. $R' \subseteq \mathscr{O}_k \subseteq R$. Taking localization gives $\mathscr{O}_k = R$, for $k \gg 1$. □

We explain the geometrical meaning of this lemma.

Remark 5.2.14. After finitely many blow-ups, we eventually can get a local ring $\mathscr{O}_{X_n, x_n}$ such that $\dim \mathscr{O}_{X_n, x_n} = 1$. This says

$$\dim_{x_n} X_n = \operatorname{codim}_{X_n} \overline{\{x_n\}} = 1.$$

This is our desired divisor $Z_n = \overline{\{x_n\}}$, which is a prime divisor.



The following two lemmas make it possible to compute discrepancies in many cases. The proofs are straightforward. The first one shows what happens under one blow up, and the second shows how to use this step inductively. The fact that all divisors are covered by this method follows from Lemma 2.13.

Lemma 5.2.15. *Let X be a normal variety and $\Delta = \sum a_i D_i$ be a sum of distinct prime divisors such that $K_X + \Delta$ is $\mathbb{Q}$-Cartier, and $Z \subseteq X$ be a closed subvariety of codimension $k \geq 2$ which is not contained in* $\operatorname{Sing}(X)$. *Let $p : \operatorname{Bl}_Z X \longrightarrow X$ be the blowing-up of Z and $E \subseteq \operatorname{Bl}_Z X$ the irreducible component of the exceptional divisor which dominates Z. (If X and Z are smooth, then this is the only component by [Har77, II, 8.24].) Then*

$$a(E, X, \Delta) = k - 1 - \sum_i a_i \cdot \operatorname{mult}_Z D_i.$$

Proof. **Step 1:** When the blowing-up center Z has singularity, $\mathrm{Bl}_Z X$ might not be a normal variety even when X is smooth. But we can pick an open dense subset $U := X_{reg} \backslash \mathrm{Sing}(Z)$ and we set $Z_U := Z \cap U$ is a smooth closed subvariety of U. Consider the following Cartesian diagram

$$\begin{array}{ccc} \mathrm{Bl}_Z X \underset{X}{\times} U & \longrightarrow & \mathrm{Bl}_Z X \\ \downarrow & & \downarrow \\ U & \xrightarrow[j]{} & X \end{array}$$

where we denote the ideal sheaf of Z in $\mathscr{O}_X$ by $\mathcal{I}_Z$, then we have

$$\mathrm{Bl}_Z X \underset{X}{\times} U = \left(\underline{\mathrm{Proj}}_X \bigoplus_{d \geq 0} \mathcal{I}_Z^d \right) \underset{X}{\times} U \cong \underline{\mathrm{Proj}}_U j^* \left(\bigoplus_{d \geq 0} \mathcal{I}_Z^d \right),$$

by [EGA II, 3.5.3] or [GW20, 13.27]. But $j^* \mathcal{I}_Z = \mathcal{I}_Z|_U \cong \mathcal{I}_{Z_U}$ is an ideal sheaf in $\mathscr{O}_U$, so the left side term is isomorphic to $\mathrm{Bl}_{Z_U} U$ which is the blowing-up along a smooth subvariety Z_U in a smooth variety U, hence is smooth. In this time, since E dominates Z, we have $p(\eta_E) = \eta_Z$ between generic points. But Z_U as an open subset of Z we have $\eta_Z \in Z_U$, therefore $\eta_E \in \mathrm{Bl}_{Z_U} U$. Thus the local ring

$$\mathscr{O}_{\mathrm{Bl}_Z X, \eta_E} \cong \mathscr{O}_{\mathrm{Bl}_{Z_U} U, \eta_E}$$

is a one-dimensional regular local ring, has algebraic valuation.

Step 2: Now we compute the discrepancy $a(E_U, U, \Delta|_U)$ where $E_U = E \cap \mathrm{Bl}_{Z_U} U$. For the local blowing-up $p : \mathrm{Bl}_{Z_U} U \longrightarrow U$, we have

$$K_{\mathrm{Bl}_{Z_U} U} + p_*^{-1} \Delta = p^* (K_U + \Delta|_U) + a(E_U, U, \Delta|_U) \cdot E,$$

here by [Har77, II, 8.24] the p-exceptional part is irreducible, so there is only one term. Therefore,

$$\begin{aligned} a(E_U, U, \Delta|_U) \cdot E &= (K_{\mathrm{Bl}_{Z_U} U} - p^* K_U) - (p^* \Delta|_U - p_*^{-1} \Delta) \\ &= (k-1) \cdot E - (p^* \Delta|_U - p_*^{-1} \Delta|_U) \end{aligned}$$

by [Har77, II, ex.8.5(b)], here we assume $k \geq 2$. And the right hand side part is

$$\begin{aligned} p^* \left(\sum a_i D_i|_U \right) - p_*^{-1} \left(\sum a_i D_i|_U \right) &= \sum a_i p^* (D_i|_U) - \sum a_i p_*^{-1} D_i|_U \\ &= \sum a_i \cdot (p_*^{-1} D_i|_U + \mathrm{mult}_{Z_U} D_i|_U \cdot E) - \sum a_i p_*^{-1} D_i|_U \\ &= \sum a_i \cdot \mathrm{mult}_Z D_i \cdot E \end{aligned}$$

by $\mathrm{mult}_{Z_U} D_i|_U = \mathrm{mult}_Z D_i$. Thus, we get

$$a(E_U, U, \Delta|_U) = k - 1 - \sum_i a_i \cdot \mathrm{mult}_Z D_i;$$

and $a(E_U, U, \Delta|_U) = a(E, X, \Delta)$ is directly from the definition. □

Lemma 5.2.16. *Let $f : Y \longrightarrow X$ be a proper birational morphism between normal varieties. Let Δ_Y (resp. Δ_X) be $\mathbb{Q}$-divisors on Y (resp. X) such that*

$$K_Y + \Delta_Y \equiv_f f^*(K_X + \Delta_X) \quad \text{and} \quad f_*\Delta_Y = \Delta_X.$$

Then, for any divisor F over X,

$$a(F, Y, \Delta_Y) = a(F, X, \Delta_X).$$

Before giving a proof, we need a Negativity lemma in [Bir12, 4.15].

Lemma 5.2.17. (Negativity lemma) *Let $\psi : Y \longrightarrow X$ be a proper birational morphism of normal varieties. Let D be a $\mathbb{Q}$-Cartier divisor on Y such that*

(1) $-D$ is ψ-nef over X, and

(2) $\psi_* D \geq 0$ is effective.

Then $D \geq 0$ is an effective divisor.

Proof. **Step 1:** By Chow's lemma, there is an algebraic variety Y' with a birational morphism $\mu : Y' \longrightarrow Y$, such that

$$\psi' = \psi \circ \mu : Y' \longrightarrow X$$

is a projective birational morphism. We set $D' = \mu^* D$, by projection formula we have

$$(-D' \cdot C)_{Y'} = (-D \cdot \mu_* C)_Y \geq 0$$

for every ψ'-relative curve $C \subseteq Y'$. Hence $-D'$ is ψ'-nef over X. And we have

$$\psi'_* D' = (\psi \circ \mu)_*(\mu^* D) = \psi_*(\mu_* \mu^* D) = \psi_* D \geq 0,$$

because $\mu_* \mu^* D = D$ by

$$\mu_* \mu^* D = \mu_* \left(\tilde{D} + \sum a_i E_i \right) = \mu_* \tilde{D} + \sum a_i \mu_* E_i = D + 0 = D.$$

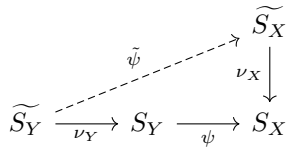
Then if we prove that $D' \geq 0$, we have $D = \mu_* \mu^* D = \mu_* D' \geq 0$. So we assume that $\psi : Y \longrightarrow X$ is a projective birational morphism.

Step 2: We reduce the case on normal surfaces. Picking $\dim Y - 2$ general hyperplanes, cut out a surface S_Y. We denote the image of S_Y under ψ by S_X. It has dimension two and birational to S_Y, because in Y the exceptional locus E has dimension $\leq \dim Y - 1$, then after taking $\dim Y - 2$ intersections, we have

$$\dim(E \cap S_Y) \leq \dim Y - 1 - (\dim Y - 2) = 1.$$

That is ψ is an isomorphism on $S_Y \cap (Y \backslash E)$, which is an open dense subset of S_Y has dimension two.

Now, we take normalizations $\nu_Y : \widetilde{S_Y} \longrightarrow S_Y$ and $\nu_X : \widetilde{S_X} \longrightarrow S_X$.



By the universal property of normalization, we have a birational morphism $\tilde{\psi} : \widetilde{S_Y} \longrightarrow \widetilde{S_X}$. Note that by closed immersion is projective, we have

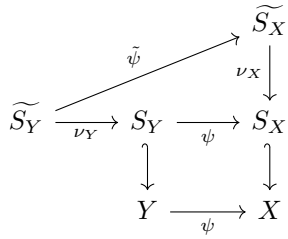
$$S_Y \hookrightarrow Y \to X$$

is projective. And it factor as $S_Y \to S_X \hookrightarrow X$ which is projective with $S_X \hookrightarrow X$ is separated, so $S_Y \to S_X$ is projective. Moreover, birational morphism $\widetilde{S_Y} \to S_Y \to S_X$ is projective because composite a finite morphism, so

$$\tilde{\psi} : \widetilde{S_Y} \longrightarrow \widetilde{S_X}$$

is projective due to $\widetilde{S_X} \to S_X$ is separated.

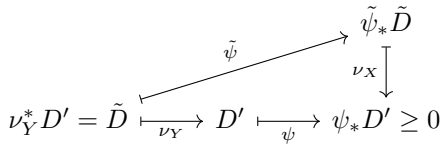
We draw the following commutative diagram:



we have that $-D|_{S_Y}$ is ψ-nef over S_X, and $\psi_*(D|_{S_Y}) \geq 0$ by diagram commutes, it is the pullback of $\psi_* D \geq 0$. We denote $D|_{S_Y}$ by D'. Furthermore, we have

$$(-\nu_Y^* D' \cdot C)_{\widetilde{S_Y}} = (-D' \cdot \nu_{Y*} C)_{S_Y} \geq 0$$

where C is contracted by $\psi \circ \nu_Y$. This C is also contracted by $\tilde{\psi}$ due to ν_X is fibre finite. So we see that $-\nu_Y^* D' =: -\tilde{D}$ is $\tilde{\psi}$-nef over $\widetilde{S_X}$. And $\tilde{\psi}_* \tilde{D} \geq 0$ by diagram commutes and ν_X does not contract anything.



At this time, if we prove that $\tilde{D} \geq 0$, then we can get $D' = (D|_{S_Y}) \geq 0$. But by our construction S_Y is cut out by general hyperplanes, so if D has negative part then one of these hyperplanes has intersection with them, which gives that D' has negative part.

We now can assume that Y and X are normal surfaces, with ψ a projective birational morphism.

Step 3: Taking a resolution of singularities of Y, we find a projective birational morphism $f : Y' \longrightarrow Y$. Also, we set $D' = f^* D$ and by projection formula

$$(-D' \cdot C)_{Y'} = (-D \cdot f_* C)_Y \geq 0$$

for every $\psi \circ f$-relative curve $C \subseteq Y'$. Hence $-D'$ is $\psi \circ f$-nef over X. And we have

$$(\psi \circ f)_* D' = (\psi \circ f)_* (f^* D) = \psi_* D \geq 0.$$

Then if we prove that $D' \geq 0$, we have $D = f_* f^* D = f_* D' \geq 0$.

So we can assume that Y is a smooth surface, and moreover with an effective exceptional divisor E which is anti-nef but not ψ-numerically trivial and $\operatorname{Supp} E = \operatorname{Ex}(\psi)$. To find such E, we could take a non-zero effective Cartier divisor H on X whose support contains the image of $\operatorname{Ex}(\psi)$. Then

$$\psi^* H = \psi_*^{-1} H + E.$$

Obviously we have

$$(\psi^* H \cdot C) = (H \cdot \psi_* C) = 0$$

for $C \subseteq Y$ is relative curve over X, and

$$(\psi_*^{-1} H \cdot C) = -(E \cdot C) \geq 0 \tag{5.3}$$

implies that E is anti-nef over X and not ψ-numerically trivial.

Step 4: Now we turn to prove this lemma. Assume that D is not effective, we prove it by contradiction. Since we have $\psi_* D \geq 0$, the negative part can only appears

in ψ-exceptional support. So we can choose e be the minimal non-negative rational number for which $D + eE \geq 0$. Then $D + eE$ has coefficient zero at some exceptional curve C, while these curves have $-e = \frac{\text{mult}_D C}{\text{mult}_E C} \in \mathbb{Q}_{<0}$.

On the other hand, by Zariski's Main Theorem every fibre is connected, so we can choose the closed point $x \in \psi(E)$ and consider the divisor $(D + eE)|_{Y_x}$. Then

$$(D + eE)|_{Y_x} = D|_{Y_x} + e \sum C_i$$

where $C_i \subseteq Y$ are exceptional curves contracted to x. Let x runs through $\psi(E)$, if we have $D + eE \neq 0$, there must exists an exceptional curve C such that it intersects some component of $D_{Y_x} + e \sum C_i$, otherwise there is no curve C has coefficient zero in $D + eE$.

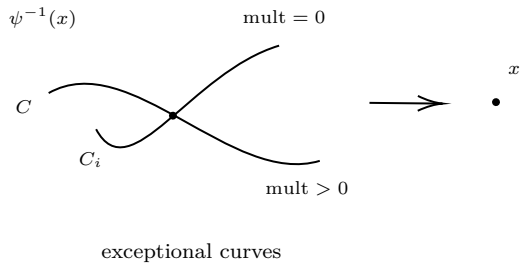


Figure 5.5.: Exceptional curve C intersects some component of $D_{Y_x} + e \sum C_i$

But in that case $\big((D+eE)\cdot C\big) > 0$, which contradicts the assumption $(D \cdot C) \leq 0$ and the choice in (5.3) of E which ensures $(E \cdot C) \leq 0$. Therefore, $D + eE = 0$ which is impossible, otherwise $D = -eE$ will gives that

$$(D \cdot C) = -e(E \cdot C) \leq 0$$

by $-D$ is ψ-nef. But by our construction of E is anti-nef over X, so $-e(E \cdot C) \geq 0$. This implies that E is ψ-numerically trivial. □

We now turn to prove the Lemma 2.16.

Proof. Our strategy is use Negativity lemma to prove some divisor is zero, ensures that coefficients are equal.

Step 1: For our two pairs (Y, Δ_Y) and (X, Δ_X), we have $K_Y + \Delta_Y$ and $K_X + \Delta_X$ are $\mathbb{Q}$-Cartier divisors. Set a $\mathbb{Q}$-Cartier divisor $D = (K_Y + \Delta_Y) - f^*(K_X + \Delta_X)$. We write down a strict equation of divisors

$$K_Y + f_*^{-1}\Delta_X = f^*(K_X + \Delta_X) + \sum_{E_i : f-\text{exceptional}} a(E_i, X, \Delta_X)E_i,$$

and applying pushout f_* on both sides can get

$$f_*(K_Y + \Delta_Y) = K_X + \Delta_X,$$

by $f_*\Delta_Y = \Delta_X$. So we pushout the divisor D:

$$\begin{aligned} f_*D &= f_*(K_Y + \Delta_Y) - f_*f^*(K_X + \Delta_X) \\ &= K_X + \Delta_X - (K_X + \Delta_X) \\ &= 0. \end{aligned}$$

Moreover, by assumption $K_Y + \Delta_Y \equiv_{num} f^*(K_X + \Delta_X)$, we have $-D$ is f-nef. So by Negativity lemma $D \geq 0$. Similarly, applying Negativity lemma on $-(-D)$ which is also f-nef and $f_*(-D) = 0$, we have $-D \geq 0$. Therefore,

$$D = 0.$$

Step 2: We choose a resolution $g : W \longrightarrow Y$ from a normal variety W, and set $h = f \circ g : W \longrightarrow X$. Then we have

$$\begin{aligned} K_W &= g^*(K_Y + \Delta_Y) + \sum_{E_i:\text{arbitrary}} a(E_i, Y, \Delta_Y)E_i \\ K_W &= h^*(K_X + \Delta_X) + \sum_{E_i:\text{arbitrary}} a(E_i, X, \Delta_X)E_i, \end{aligned}$$

where E_i are arbitrary divisor over X. Applying g^* on D, we have

$$g^*(K_Y + \Delta_Y) - g^*f^*(K_X + \Delta_X) = g^*(K_Y + \Delta_Y) - h^*(K_X + \Delta_X) = 0.$$

Thus,

$$a(E_i, Y, \Delta_Y) = a(E_i, X, \Delta_X).$$

□

Remark 5.2.18. Without using Negativity lemma, we can only get

$$\sum_{E_i:\text{arbitrary}} \Big(a(E_i, Y, \Delta_Y) - a(E_i, X, \Delta_X)\Big)E_i \equiv_{num} 0,$$

by $K_Y + \Delta_Y \equiv_{num} f^*(K_X + \Delta_X)$ combining with Remark 2.9. Therefore, we can not obtain the equality

$$a(E_i, Y, \Delta_Y) = a(E_i, X, \Delta_X)$$

by "straightforward" verification.

Example 5.2.19. For the proper birational morphism $f : Y \longrightarrow X$ between normal varieties, and we recall that in Remark 2.9, we have the numerically equivalent relation

$$K_Y \equiv_f f^*(K_X + \Delta_X) + \sum_{E_i:\text{arbitrary}} a(E_i, X, \Delta)E_i.$$

If we set

$$\begin{aligned}\Delta_Y &= \sum_{E_i:\text{arbitrary}} \Big(- a(E_i, X, \Delta_X)\Big)E_i \\ &= \sum_{D_i \subset \operatorname{Supp} \Delta} a_i f_*^{-1} D_i + \sum_{E_j \subset \operatorname{Ex}(f)} \Big(- a(E_j, X, \Delta)\Big)E_j\end{aligned}$$

such that $K_X + \Delta_X$ and $K_Y + \Delta_Y$ are $\mathbb{Q}$-Cartier divisor, then automatically we have $f_*\Delta_Y = \Delta_X$. This is the trivial case of Lemma 2.16.

The following is the first example computing discrepancies.

Corollary 5.2.20. *Consider pair (X, Δ) where X is a normal variety and $\Delta = \sum a_i D_i$ is a $\mathbb{Q}$-divisor on X, such that $K_X + \Delta$ is $\mathbb{Q}$-Cartier. Then we have the following:*

(1) Either $\operatorname{discrep}(X, \Delta) = -\infty$, *or*

$$-1 \leq \operatorname{totaldiscrep}(X, \Delta) \leq \operatorname{discrep}(X, \Delta) \leq 1.$$

(2) If X is smooth then $\operatorname{discrep}(X, 0) = 1$.

(3) Assume that X is smooth, $\sum D_i$ is an snc divisor and $a_i \leq 1$ for every i. Then

$$\operatorname{discrep}(X, \Delta) = \min\left\{ \min_{i\neq j, D_i \cap D_j \neq \emptyset} \{1 - a_i - a_j\}, \min_i\{1 - a_i\}, 1 \right\}.$$

Proof. For (1): Since X is normal, its singularities has codimension at least 2. We can find a subvariety Z whose codimension is exactly 2, lying in the smooth locus of X and disjoint with the divisor Δ. Taking a blow-up of X along the smooth center Z, we get a proper birational morphism $p : \operatorname{Bl}_Z X \longrightarrow X$ gives a irreducible exceptional divisor, which we denote it by E_0. Moreover, after shrink to smooth locus, the blowing-up $p : \operatorname{Bl}_Z X_{reg} \longrightarrow X_{reg}$ doesn't change too much (see the proof of Lemma 2.15). And we can use Lemma 2.15 to get

$$a(E_0, X, \Delta) = 2 - 1 = 1,$$

since $\operatorname{mult}_Z D_i = 0$ by our choice of Z. Therefore, we have

$$\operatorname{discrep}(X, \Delta) = \inf_E\{a(E, X, \Delta)\} \leq a(E_0, X, \Delta) = 1.$$

Due to the inclusion relative between sets

$$\{\text{exceptional divisors over } X\} \subseteq \{\text{any divisor over } X\},$$

we have

$$\operatorname{totaldiscrep}(X, \Delta) \leq \operatorname{discrep}(X, \Delta) \leq 1.$$

Now we turn to prove the lower bound is -1. Suppose that $\operatorname{totaldiscrep}(X, \Delta) < -1$, that is we assume there is a divisor E over X such that $a(E, X, \Delta) = -1 - c$ with $c > 0$. Taking a proper birational morphism $f : Y \longrightarrow X$ from a normal variety and we let $K_Y + \Delta_Y \equiv_f f^*(K_X + \Delta)$ as in Example 2.19.

Note that since $a(E, X, \Delta) = -1 - c \neq 0$, either in case $E \subseteq \operatorname{Ex}(f)$ or $f_*E \subseteq \Delta$, otherwise for $\operatorname{Supp} E \nsubseteq \operatorname{Supp} \Delta$ we have

$$a(E, X, \Delta) = -\operatorname{mult}_{\Delta} E = 0,$$

see Remark 2.9. So we let Z_0 be a codimension 2 locus of Y, which is contained in E but not in any other exceptional divisor of f or in any other component of $f_*^{-1}\Delta$. In particular, since $\operatorname{Sing}(Y)$ and $\operatorname{Sing}(E)$ are proper closed subsets, we can choose an arbitrary Z_0 such that $Z_0 \nsubseteq \operatorname{Sing}(Y)$ and $Z_0 \nsubseteq \operatorname{Sing}(E)$. Since $Z_0 \nsubseteq \operatorname{Sing}(Y)$, the generic point η_{Z_0} is in the smooth locus of Y. Thus the local ring $\mathcal{O}_{Y,\eta_{Z_0}}$ is a regular local ring with maximal ideal $\mathfrak{m}_{Z_0}$. Furthermore, since $Z_0 \nsubseteq \operatorname{Sing}(E)$, the divisor E is smooth at η_{Z_0}, the local defining equation of E lies in $\mathfrak{m}_{Z_0} \setminus \mathfrak{m}_{Z_0}^2$. This directly yields:

$$\operatorname{mult}_{Z_0} E = 1.$$

Let $g_1 : Y_1 = \operatorname{Bl}_{Z_0} Y \longrightarrow Y$ be the blow-up along the subvariety Z_0, with excpetional divisor $E_1 \subseteq Y_1$ which dominates Z_0. Then follows from Lemma 2.15, we have

$$\begin{aligned} a(E_1, Y, \Delta_Y) &= 2 - 1 - \sum_{E_i:\text{arbitrary}} \Big(- a(E_i, X, \Delta) \Big) \operatorname{mult}_{Z_0} E_i \\ &= 1 - (1 + c) \\ &= -c. \end{aligned}$$

Moreover, by Lemma 2.16 we have

$$a(E_1, Y, \Delta_Y) = a(E_1, X, \Delta) = -c.$$

Let $Z_1 \subseteq Y_1$ be the intersection of E_1 and of the strict transform of E, such that it has codimension 2. And it is not contained in $\operatorname{Sing}(Y_1)$, that is because by our choice

of $Z_0 \subseteq Y_{reg}$, we have the exceptional divisor E_1 lies in the smooth variety $\mathrm{Bl}_{Z_0} Y_{reg}$. Now we set

$$\Delta_{Y_1} = \sum_{D_i' \subseteq \mathrm{Supp}\, \Delta_Y} g_{1*}^{-1} D_i' + \sum_{E_i \subseteq \mathrm{Ex}(g_1)} \Big(- a(E_i, Y, \Delta_Y) \Big) E_i$$

such that $K_{Y_1} + \Delta_{Y_1} \equiv_{g_1} g_1^*(K_Y + \Delta_Y)$ and $g_1 \Delta_{Y_1} = \Delta_Y$. Let $g_2 : \mathrm{Bl}_{Z_1} Y_1 \longrightarrow Y_1$ be the blowing-up with exceptional divisor $E_2 \subseteq Y_2$ dominates Z_1. Then by Lemma 2.15 again, we have

$$\begin{aligned} a(E_2, Y_1, \Delta_{Y_1}) &= 2 - 1 - \sum_{E_i : \text{arbitrary}} \Big(- a(E_1, Y, \Delta_Y) \Big) \cdot \mathrm{mult}_{Z_1} E_i \\ &= 1 - \Big(- a(E_1, Y, \Delta_Y) \Big) \cdot \mathrm{mult}_{Z_1} E_1 - \Big(- a(g_{1*}^{-1} E, Y, \Delta_Y) \Big) \cdot \mathrm{mult}_{Z_1} g_{1*}^{-1} E \\ &= 1 - \big(- (-c) \big) - \big(- (-1 - c) \big) \\ &= -2c \end{aligned}$$

where $a(g_{1*}^{-1} E, Y, \Delta_Y) = a(E, X, \Delta)$ by the isomorphism $\mathscr{O}_{Y_1, g_{1*}^{-1} E} \cong \mathscr{O}_{Y,E}$ and see Remark 2.5. Now apply Lemma 2.6 on $g_1 : Y_1 \longrightarrow Y$ and $f : Y \longrightarrow X$ respectively, we have

$$a(E_2, Y_1, \Delta_{Y_1}) = a(E_2, Y, \Delta_Y) = a(E_2, X, \Delta) = -2c.$$

Repeat the blowing-up, this time along the intersection of the strict transform of E and E_2, which also not contained in the singular locus, then get a divisor with discrepancy $-3c$, etc. This shows that

$$\mathrm{discrep}(X, \Delta) = -\infty$$

when $\mathrm{totaldiscrep}(X, \Delta) < -1$.

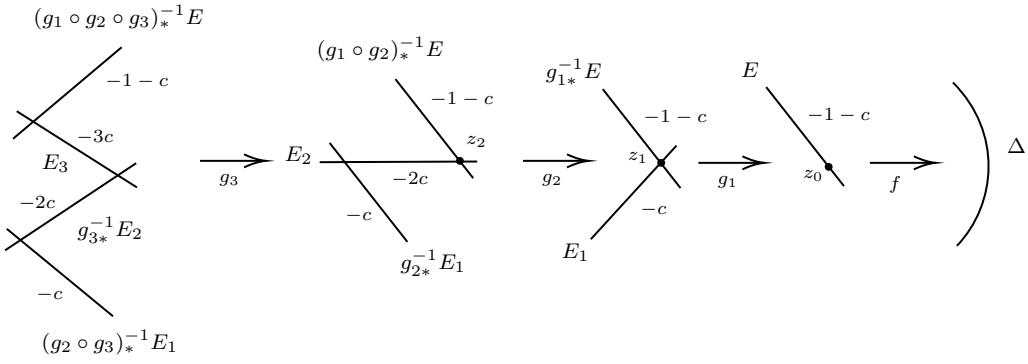


Figure 5.6.: Values of discrepancy of exceptional divisors.

For (2): it is the special case of (3), but also we can easily get it by Lemma 2.15.

For (3): Let

$$r(X,\Delta) = \min\left\{ \min_{i\neq j, D_i\cap D_j\neq\emptyset}\{1-a_i-a_j\}, \min_i\{1-a_i\}, 1 \right\}.$$

First we construct the special blow-ups to compute the discrepancies of exceptional divisor.

(1) We blow-up the codimension 2 subvariety Z of $D_i \cap D_j$, then by Lemma 2.15 we have

$$\begin{aligned} a(E,X,\Delta) &= 2-1-a_i\cdot \operatorname{mult}_Z D_i - a_j\cdot \operatorname{mult}_Z D_j \\ &= 1-a_i-a_j, \end{aligned}$$

where E is the exceptional divisor dominates Z, and since Δ has snc support $\operatorname{mult}_Z D_i = \operatorname{mult}_Z D_j = 1$.

(2) We blow-up the general codimension 2 subvariety Z of D_i, not contained in other D_j's, then by Lemma 2.15 we have

$$\begin{aligned} a(E,X,\Delta) &= 2-1-a_i\cdot \operatorname{mult}_Z D_i \\ &= 1-a_i, \end{aligned}$$

where E is the exceptional divisor dominates Z, and also $\operatorname{mult}_Z D_i = 1$ by D_i is smooth, the snc property.

(3) We blow-up the smooth center Z of codimensional 2 which is not contained in $\operatorname{Supp}\Delta$, then by Lemma 2.15 we have

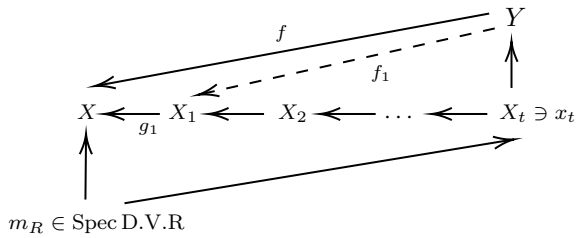
$$\begin{aligned} a(E,X,\Delta) &= 2-1 \\ &= 1, \end{aligned}$$

where E is the exceptional divisor dominates Z, and all $\operatorname{mult}_Z D_i = 0$.

Since $\operatorname{discrep}(X,\Delta)$ is the infimum of all possible choice of $a(E,X,\Delta)$, which include above cases. So we have

$$\operatorname{discrep}(X,\Delta) \leq r(X,\Delta).$$

Now we prove the converse direction. Let $E \subseteq Y$ be the exceptional divisor for some birational morphism $f : Y \longrightarrow X$. We need to prove $a(E,X,\Delta) \geq r(X,\Delta)$. By Lemma 2.13, we can assume E is obtained by a succession of (say t) blow-ups, that is, Y is locally isomorphic to X_t in Remark 2.14:



where for $Z_t = \overline{\{x_t\}}$, we have $\mathscr{O}_{Y,\eta_E} \cong \mathscr{O}_{X_t,\eta_{Z_t}}$. Moreover, in each steps we can make some shrink which is not change the local rings, so in each blow-ups we can assume that it is along a smooth center. We prove the inequality by induction on t. Also, note that $r(X,\Delta)$ does not decrease if we shrink X, because after shrink we may lost some $\operatorname{mult}_Z D_i$, the terms $1-a_i-a_j$ or $1-a_i$ might goes big.

We assume that $f(E)$ is a smooth closed subvariety of X. Let $g_1 : X_1 \longrightarrow X$ be the blowing-up along smooth center $f(E)$, and $E_1 \subseteq X_1$ be the exceptional divisor dominantes $f(E)$, By shrinking X around a general point of $f(E)$, we may assume that $E_1 \cup {g_1}_*^{-1}\Delta$ has snc support, By shrinking Y around a general point of E, we may assume that $f_1 : Y \longrightarrow X_1$ is a morphism.

Now, after renumbering that $D_i \subseteq \operatorname{Supp}\Delta$, we may assume that $\operatorname{codim}_X f(E) = k \geq 2$, and $f(E) \subseteq D_i$ if and only if $i \leq b$ for some b. Note that $b \leq k$, since for $b > k$ we have

$$D_1 \cap \cdots \cap D_b$$

will has codimension $< k$. The blow-ups formula in Lemma 2.15 says that

$$a(E_1, X, \Delta) = k - 1 - \sum_{1 \leq l \leq b} a_l,$$

where $\operatorname{mult}_{f(E)} D_i = 1$ due to smoothness. Then we treat three cases:

(1) If $b = 0$, then $a(E_1, X, \Delta) = k - 1 \geq 1 \geq r(X, \Delta)$.

(2) If $b = 1$, then $a(E_1, X, \Delta) = k - 1 - a_i \geq 1 - a_i \geq r(X, \Delta)$.

(3) If $b \geq 2$, then

$$\begin{aligned} a(E_1, X, \Delta) &= k - 1 - \sum_{1 \leq l \leq b} a_l \\ &= (k - b - 1) + \sum_{1 \leq l \leq b} (1 - a_l) \\ &\geq -1 + (1 - a_1) + (1 - a_2) \geq r(X, \Delta) \text{ by each } a_i \leq 1. \end{aligned}$$

Thus the case $t = 1$ is settled. On the other hand, if we define Δ_1 on X_1 by $K_{X_1} + \Delta_1 \equiv_{g_1} g_1^*(K_X + \Delta)$ as in Example 2.19, that is

$$\Delta_1 = \sum_{D_i \subseteq \mathrm{Supp}\,\Delta} a_i {g_1}_*^{-1} D_i + \Big(- a(E_1, X, \Delta)\Big) E_i =: \sum b_j D_j^{(1)}$$

by our assumption of the blowing-up, E_i is the unique exceptional divisor. Moreover, since $a(E_1, X, \Delta) \geq r(X, \Delta) \geq -1$, we have the coefficient of E_1 is ≤ 1, does not affect the assumption of (3). Then

$$\begin{aligned} r(X_1, \Delta_1) &= \min \left\{ \min_{i \neq j, D_i^{(1)} \cap D_j^{(1)} \neq \emptyset} \{1 - a_i - a_j\}, \min_i \{1 - a_i\}, 1 \right\} \\ &\geq \min \left\{ r(X, \Delta), 1 + a(E_1, X, \Delta) - \max_{D_i \cap f(E) \neq \emptyset} a_i \right\} \\ &\geq \min \Big\{ r(X, \Delta), a(E_1, X, \Delta) \Big\} \text{ by all } a_i \leq 1 \\ &\geq r(X, \Delta). \end{aligned}$$

We suppose that what we want is holds for $t - 1$ blow-ups, this is the induction hypothesis, that is $a(E, X_1, \Delta_1) \geq r(X_1, \Delta_1)$. But by our choice of Δ_1, use Lemma 2.19, we have

$$a(E, X_1, \Delta_1) = a(E, X, \Delta).$$

Therefore,

$$a(E, X, \Delta) \geq r(X_1, \Delta_1) \geq r(X, \Delta)$$

which completes our proof. □

Under some conditions, the discrepancy can be computed from the exceptional divisors occurring on a given resolution:

Corollary 5.2.21. *Given a pair (X, Δ) where X is a normal variety and we assume that $K_X + \Delta$ is a $\mathbb{Q}$-Cartier divisor. Let $f : Y \longrightarrow X$ be any resolution of singularities with $E \subseteq Y$ the exceptional set and $E_i \subseteq E$ all the irreducible exceptional divisors.*

(1) Assume that $1 \geq \min_i\{a(E_i, X)\} \geq 0$. Then

$$\mathrm{discrep}(X) = \min_i \{a(E_i, X)\}.$$

(2) Let $\Delta = \sum a_j D_j$ with $a_j \leq 1$. Then there is a log resolution f for (X, Δ) such that $\sum f_*^{-1} D_j$ is smooth. Let f be any such.

If $a(E_i, X, \Delta) \geq -1$ for every i, then

$$\mathrm{discrep}(X, \Delta) = \min \left\{ \min_i \{a(E_i, X, \Delta)\}, \min_j \{1 - a_j\}, 1 \right\}.$$

Proof. For (1) which is $\Delta = 0$, let Δ_Y be the $\mathbb{Q}$-divisor on Y such that

$$K_Y + \Delta_Y \equiv_f f^*K_X,$$

that is Δ_Y is completely determined by the f-exceptional divisors. By our assumption, we have

$$\Delta_Y = \sum_i \Big(- a(E_i, X)\Big) E_i \leq 0$$

and therefore

$$\operatorname{discrep}(Y, \Delta_Y) \geq \operatorname{discrep}(Y, 0) = 1$$

by Lemma 2.11 and Corollary 2.20(2). Thus

$$\begin{aligned} \operatorname{discrep}(X) &= \min\Big\{\operatorname{discrep}(Y, \Delta_Y), \min_i\{a(E_i, X)\}\Big\} \\ &= \min_i\{a(E_i, X)\}. \end{aligned}$$

For the existence in (2), we take any log resolution $g : Z \longrightarrow X$ for (X, Δ), such that $g^*\Delta$ has snc support. Then $\operatorname{Sing}(\sum_i g_*^{-1}D_i)$ is a union of finitely many (say $n(g)$) non-singular subvarieties S of codimension 2, by the simple normal crossing property. If $h : Z' \to Z \to X$ is the blowing-up of Z along one irreducible component S, then

$$n(h) = n(g) - 1.$$

So we obtain $f : Y \longrightarrow X$ with $n(f) = 0$, that is, $\sum f_*^{-1}D_i$ smooth by induction on $n(g)$. Now for the second statement, let Δ_Y be the $\mathbb{Q}$-divisor on Y as in Example 2.19, such that $K_Y + \Delta_Y \equiv_f f^*(K_X + \Delta)$. Set $b_i = -a(E_i, X, \Delta) \leq 1$. Then by Corollary 2.20, for which every a_i and $b_j \leq 1$ in Δ_Y,

$$\operatorname{discrep}(Y, \Delta_Y) = \min\left\{\min\{1 - b_i - a_j\}, \min\{1 - b_i - b_{i'}\}, \min_j\{1 - a_j\}, \min_i\{1 - b_i\}, 1\right\},$$

here we do not consider $1 - a_j - a_{j'}$ since $f_*^{-1}D_j$ and $f_*^{-1}D_{j'}$ is disjoint. Furthermore,

$$\begin{aligned} \operatorname{discrep}(X, \Delta) &= \min\Big\{\operatorname{discrep}(Y, \Delta_Y), \min_i\{a(E_i, X, \Delta)\}\Big\} \\ &= \min\Big\{\operatorname{discrep}(Y, \Delta_Y), \min_i\{-b_i\}\Big\}, \end{aligned}$$

where the first equality is due to Lemma 2.16. Note that each b_i and $a_j \leq 1$ by our assumption, so

$$-b_i \leq 1 - b_i - b_{i'} \quad \text{and} \quad -b_i \leq 1 - b_i - a_j,$$

thus

$$\operatorname{discrep}(X, \Delta) = \min\left\{\min_j\{1 - a_j\}, \min_i\{-b_i\}, 1\right\}.$$

□

5.2C Singularities for the Pair

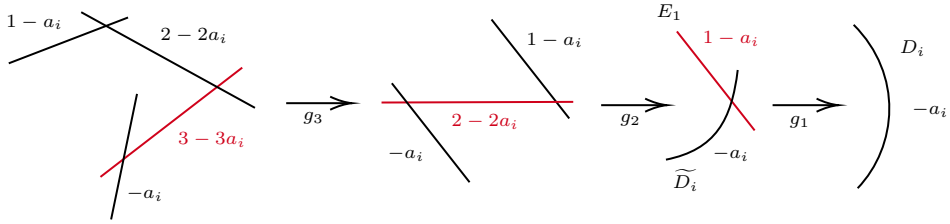
Next we define the six classes of singularities that are most important for the minimal model program.

Definition 5.2.22. (Singularities for pairs) Let (X, Δ) be a pair where X is a normal variety and $\Delta = \sum a_i D_i$ is a sum of distinct prime divisors, where $a_i \in \mathbb{Q}$. Assume that $K_X + \Delta$ is a $\mathbb{Q}$-Cartier divisor. We say that (X, Δ) is

$$\left.\begin{array}{c} \textit{terminal} \\ \textit{canonical} \\ \textit{klt} \\ \textit{plt} \\ \textit{lc} \end{array}\right\} \text{ if } \operatorname{discrep}(X,\Delta) \left\{\begin{array}{ll} > 0, & \\ \geq 0, & \\ > -1 & \text{and } \lfloor \Delta \rfloor \leq 0, \\ > -1, & \\ \geq -1. & \end{array}\right.$$

Here klt is short for "Kawamata log terminal", plt for "purely log terminal" and lc for "log canonical".

If $(X, \Delta = \sum a_i D_i)$ is lc (and the D_i are distinct), then $a_i \leq 1$ for every i. Otherwise, as in Corollary 2.20(1), we construct a blow-up sequence such that



then after k steps we will get an exceptional divisor E_k over X, such that

$$a(E_k, X, \Delta) = k - ka_i = k(1 - a_i).$$

But since $1 - a_i \leq 0$, we can finally arrive a value $k(1 - a_i) \leq -1$.

Definition 5.2.23. (Divisorial log terminal) Let (X, Δ) be a pair where X is a normal variety and $\Delta = \sum a_i D_i$ is a sum of distinct prime divisors, with rational numbers $0 \leq a_i \leq 1$. Assume that $K_X + \Delta$ is $\mathbb{Q}$-Cartier. We say that (X, Δ) is *dlt* or *divisorial log terminal* if and only if there is a closed subset $Z \subseteq X$ such that

(1) $X \backslash Z$ is smooth and $\Delta|_{X \backslash Z}$ is a snc divisor.

(2) If $f : Y \longrightarrow X$ is birational and $E \subseteq Y$ is an irreducible divisor such that $\operatorname{center}_X E \subseteq Z$, then $a(E, X, \Delta) > -1$.

Each of these notions has an important place in the theory of minimal models:

1. **Terminal: The Natural Foundation of the MMP**

   Core Definition: For a pair (X, Δ), we require $\operatorname{discrep}(X, \Delta) > 0$. Assuming $\Delta = 0$, this is the smallest class necessary to run the MMP for smooth varieties.

   Geometric Motivation: Suppose we start running the minimal model program on an absolutely smooth projective variety X. When we contract a K_X-negative extremal ray, whether it results in a divisorial contraction or a flip, the newly produced variety X' will almost inevitably acquire singularities. However, if X is smooth, all of its exceptional divisors have a discrepancy $a(E, X) > 0$. The core operations of the MMP exhibit a certain "monotonicity" - they do not make the singularities too pathological. More precisely, they preserve the property that discrepancies remain strictly greater than 0. Therefore, terminal singularities constitute the smallest category of algebraic varieties that contains all smooth varieties and remains closed under MMP operations.

2. **Canonical: Rigid Requirements for General Type and Moduli**

   Core Definition: For a pair (X, Δ), we require $\operatorname{discrep}(X, \Delta) \geq 0$. These are precisely the singularities that appear on the canonical models of varieties of general type.

   Geometric Motivation: For a variety of general type, we wish to construct its canonical ring:

   $$R(X) = \bigoplus_{m \geq 0} H^0(X, \mathcal{O}_X(mK_X))$$

   If this ring is finitely generated, we can define the canonical model $X_{can} = \operatorname{Proj} R(X)$. The algebraic essence of canonical singularities is that on any log resolution $f : Y \to X$, the pluricanonical forms defined on X, when pulled back to Y, will absolutely never acquire poles along the exceptional divisors. This guarantees the isomorphism:

   $$H^0\big(X, \mathcal{O}_X(mK_X)\big) \cong H^0\big(Y, \mathcal{O}_Y(mK_Y)\big)$$

   Because the space of sections of differential forms is perfectly protected in this way, canonical singularities become the "worst" yet most "natural" singular structures permitted when constructing moduli spaces of higher-dimensional varieties (e.g., KSBA compactifications).

3. **klt (Kawamata Log Terminal): The Domain of Vanishing Theorems**

Core Definition: Requires $\text{discrep}(X, \Delta) > -1$ and $\lfloor \Delta \rfloor \leq 0$.

Geometric Motivation: The modern cornerstones of birational geometry are the Kawamata-Viehweg (KV) Vanishing Theorem and the Basepoint Free Theorem. In the logarithmic framework, KV vanishing requires the existence of a $\mathbb{Q}$-divisor whose fractional part has coefficients strictly less than 1. In the language of singularities, this translates exactly to the discrepancy being strictly greater than -1. As soon as the discrepancy drops below or even equals -1, log poles emerge and destroy the vanishing properties of the cohomology.

Why is klt unsuitable for inductive proofs? In algebraic geometry, we frequently use the adjunction formula to perform induction on dimension (restricting to a divisor):

$$(K_X + S + \Delta)|_S = K_S + \Delta_S$$

If we require the entire pair $(X, S + \Delta)$ to be klt, the coefficients of all divisors must be strictly less than 1. This means the coefficient of S cannot be 1. In other words, the boundary of a klt pair cannot contain a full prime divisor S to restrict onto, making it impossible to complete the inductive step.

4. **plt (Purely Log Terminal): The Scalpel for Induction**

   Core Definition: Allows divisors with a coefficient of 1 to appear in the boundary, but strictly requires $a(E, X, \Delta) > -1$ for all **exceptional divisors** E.

   Geometric Motivation: To overcome the inability of klt singularities to support induction via divisors, plt was introduced. If the boundary Δ contains a prime divisor S with a coefficient of 1, and $(X, S + B)$ is a plt pair, then by adjunction to S, the resulting pair (S, B_S) **is strictly a klt pair**. This forms a rigorous logical chain: because (S, B_S) is klt, we can apply the KV vanishing theorem on S. Then, by utilizing the short exact sequence:

   $$0 \to \mathcal{O}_X(-S) \to \mathcal{O}_X \to \mathcal{O}_S \to 0$$

   combined with the properties on X, we can lift cohomological information from dimension $n - 1$ to n, successfully completing the inductive step in higher-dimensional algebraic geometry.

5. **lc (Log Canonical) and dlt: The Edge of Rationality**

   Core Definition: lc requires $\text{discrep}(X, \Delta) \geq -1$. This is the largest class where the concept of discrepancy still makes geometric sense.

Geometric Motivation: As proven previously, once an exceptional divisor with a discrepancy less than -1 appears, blowing up along intersections can yield a sequence of exceptional divisors whose discrepancies diverge to $-\infty$. Therefore, ≥ -1 is the absolute bottom line for maintaining geometric rationality. However, on the boundary of lc (where there exist exceptional divisors with discrepancies exactly equal to -1), KV vanishing fails. This makes it incredibly difficult to prove foundational results, such as the contractibility of extremal rays.

To overcome the chaotic configurations contained within the lc class, **dlt (Divisorial Log Terminal)** is introduced. dlt is a well-behaved subset of lc that acknowledges the existence of -1 discrepancies, but strictly mandates that these -1 values can only occur on the original boundary divisors of the base space X, and never on any hidden exceptional divisors. Furthermore, it requires the existence of a good resolution satisfying simple normal crossing (SNC) properties. By approximating with dlt pairs, we can significantly repair the local geometric deficiencies of lc singularities.

Remark 5.2.24. (Discrepancy for $\mathbb{R}$-Cartier case) In the definition of discrepancy for pair (X, Δ) where $K_X + \Delta$ is $\mathbb{Q}$-Cartier, our strategy is to locally check the vanishing order of corresponding exceptional divisor. But this requires us to clear the denominators and then invoke line bundles, which only applies to the $\mathbb{Q}$-Cartier case. In complex geometry, local sections are used to compute integrals.

But in algebraic geometry, we can deal with the case where $K_X + \Delta$ is $\mathbb{R}$-Cartier, we can define the discrepancies by writing down the strict equation

$$K_Y + f_*^{-1}\Delta = f^*(K_X + \Delta) + \sum_{E_i : f-\text{exceptional}} a(E_i, X, \Delta)E_i.$$

This formula means that $\sum_i a(E_i, X, \Delta)E_i$ is defined by

$$\sum_{E_i : \text{arbitrary}} a(E_i, X, \Delta)E_i = K_Y - f^*(K_X + \Delta)$$

for arbitrary divisor E_i. The real number $a(E, X, \Delta)$ is called the *discrepancy* of E with respect to the pair (X, Δ). Also, we can define

$$\text{discrep}(X, \Delta) = \inf_E \{a(E, X, \Delta) \mid E \text{ is an exceptional divisor over } X\},$$
$$\text{totaldiscrep}(X, \Delta) = \inf_E \{a(E, X, \Delta) \mid E \text{ is a divisor over } X\}.$$

Moreover, for lemmas and corollaries we have proven, if their proofs do not rely on clearing denominators to invoke line bundle properties, they remain valid for the pair

(X, Δ) such that $K_X + \Delta$ is $\mathbb{R}$-Cartier. For example, Lemma 2.11, 2.14-2.17 and Corollary 2.20-2.22. Also, we can extend the definition of terminal, canonical, klt, etc.

The following is the first example showing that the discrepancy is a lower semi-continuous function.

Corollary 5.2.25. *Let (X, Δ) be a pair with X is a proper normal variety and $K_X + \Delta$ is $\mathbb{R}$-Cartier, and $|L|$ is a linear series on X. Moreover, we assume that $\Delta \geq 0$. Let $L_0 \in |L|$ be a member and $L_g \in |L|$ a general member. Then, for every $c \geq 0$,*

$$\operatorname{discrep}(X, \Delta + cL_0) \leq \operatorname{discrep}(X, \Delta + cL_g).$$

Proof. **Step 1:** Choose a resolution of singularities $f : Y \longrightarrow X$ such that

(1) $f^*|L| = B + |F|$, where $|F|$ is free (cf. Example 3.1.34);

(2) $f_*^{-1} \operatorname{Supp} \Delta$ is smooth (cf. Corollary 2.21);

(3) $B + f_*^{-1} \operatorname{Supp} \Delta + \operatorname{Ex}(f)$ is a snc divisor.

Here we recall that L is a finite dimensional vector space.

For any Cartier divisor $L_\lambda \in |L|$, we can write

$$f^*(K_X + \Delta + cL_\lambda) \equiv_{num} K_Y + \Delta_Y + cB + cF_\lambda,$$

where $F_\lambda \in |F|$. Moreover, since

$$\begin{aligned} f_*(\Delta_Y + cB + cF_\lambda) &= f_*\Delta_Y + f_*f^*(cL_\lambda) \\ &= \Delta + cL_\lambda, \end{aligned}$$

by Lemma 2.16 of $\mathbb{R}$-version we have

$$\operatorname{discrep}(Y, \Delta_Y + cB + cF_\lambda) = \operatorname{discrep}(X, \Delta + cL_\lambda) \tag{5.4}$$

for any $L_\lambda \in |L|$. We set $\Gamma = \Delta_Y + cB$, it remains to prove

$$\operatorname{discrep}(Y, \Gamma + cF_0) \leq \operatorname{discrep}(Y, \Gamma + cF_g). \tag{5.5}$$

Step 2: If $|F| = \{0\}$, then $F_0 = F_g = 0$ and (5.5) is equality. Hence we assume that $|F| \neq \{0\}$. Note that any effetive f-exceptional divisor E_i can not be f-nef, otherwise by Negativity lemma for $\mathbb{R}$-divisors applying on $-E_i$ we can get $-E_i \geq 0$. Since $|F|$ is base-point-free, every member F_λ is f-nef over X, therefore no non-zero member of $|F|$ is supported entirely on f-exceptional divisors.

Choose general member $F_g \in |F|$. By Bertini's theorem for Y is non-singular, and (3), $\Gamma + cF_g$ has snc support for every $c \geq 0$. Write

$$\Gamma = \sum b_i G_i$$

where G_i are distinct prime divisors. We need some observations. Since $\Gamma + cF_g$ has snc support for a generam member F_g. If $G_i \cap F_g \neq \emptyset$, then the restricted line bundle $\mathscr{O}_Y(F)|_{G_i}$ has a non-zero global section $s_g|_{G_i}$, where its divisor of zeros

$$\left(s_g|_{G_i}\right)_0 = (s_g)_0 \cap G_i = F_g \cap G_i \neq \emptyset.$$

By our assumption X is **proper**, while the log resolution is projective, we have G_i is a proper variety as a subvariety of the proper variety Y. Note that a line bundle admitting a nowhere-vanishing global section, would be a trivial line bundle via the multiplication

$$\cdot s_x : \mathcal{L}_x \xrightarrow{\cong} \mathscr{O}_x.$$

So for G is proper, some member $G_i \cap F_0 = \emptyset$ which corresponds that $\mathscr{O}_Y(F)|_{G_i}$ has a nowhere-vanishing global section, then every global section in $H^0(G_i, \mathscr{O}_Y(F)|_{G_i}) = \mathbb{C}$ would be either zero or nowhere vanishing, contradicts with assumption $G_i \cap F_g \neq \emptyset$. Hence, if $F_g \cap G_i \neq \emptyset$, then for any member $F_\lambda \in |F|$ either $G_i \subseteq \operatorname{Supp} F_\lambda$ or $F_\lambda \cap G_i$ contains a codimension 2 subvariety of Y.

Step 3: Assume first that

$$\operatorname{discrep}(Y, \Gamma + cF_g) = -\infty.$$

By Corollary 2.20(1) for $\mathbb{R}$-divisors, we have

$$\operatorname{totaldiscrep}(Y, \Gamma + cF_g) \leq -1.$$

This is corresponds to four possibilities:

(1) On G_i: $-a(G_i, Y, \Gamma + cF_g) = b_i \operatorname{mult}_{G_i} F_g = b_i > 1$.

(2) On F_g: $-a(F_g, Y, \Gamma + cF_g) = \operatorname{mult}_{F_g} \Gamma + c = c > 1$.

(3) On $G_i \cap G_j \neq \emptyset$:

$$\begin{aligned} a(E, Y, \gamma + cF_g) &= 2 - 1 - b_i - b_j \\ &= 1 - (b_i + b_j) < -1, \end{aligned}$$

by applying Lemma 2.15 for $\mathbb{R}$-divisor for blowing-up along the codimension 2 center inside $G_i \cap G_j$.

(4) On $G_i \cap F_g \neq \emptyset$:

$$\begin{aligned} a(E, Y, \Gamma + cF_g) &= 2 - 1 - b_i - c \\ &= 1 - (b_i + c) < -1, \end{aligned}$$

similar as above.

These are correspond to

$$b_i > 1, c > 1, b_i + b_j > 1, b_i + c > 2.$$

Now for divisor $\Gamma + cF_0$ where F_0 is any member,

(1) If $b_i > 1$, then

$$-a(G_i, Y, \Gamma + cF_0) = b_i + \operatorname{mult}_{G_i} F_0 \geq b_i > 1.$$

(2) If $c > 1$, then

$$a(F_0, Y, \Gamma + cF_0) = \operatorname{mult}_{F_0} \Gamma + c \geq c > 1.$$

(3) If $b_i + b_j > 2$ provided by $G_i \cap G_j \neq \emptyset$, then we take blowing-up along a codimension 2 subvariety of $G_i \cap G_j$

$$a(E, Y, \Gamma + cF_0) = 2 - 1 - b_i - b_j > -1.$$

(4) If $b_i + c > 2$ provided by $G_i \cap F_g \neq \emptyset$, then by our analysis in *Step 2* we have two sub-cases

- $G_i \subseteq \operatorname{Supp} F_0$: We blow-up a codimension 2 subvariety Z inside G_i, then

$$\begin{aligned} a(E, Y, \Gamma + cF_0) &= 2 - 1 - b_i \cdot \operatorname{mult}_Z G_i - c \cdot \operatorname{mult}_Z F_0 \\ &\leq 1 - b_i - c \\ &< -1. \end{aligned}$$

- $G_i \cap F_0$ contains a codimension 2 subvariety, then similar as above

$$a(E, Y, \Gamma + cF_0) \leq 2 - 1 - b_i - c < -1.$$

In each cases, we can get $\operatorname{totaldiscrep}(Y, \Gamma + cF_0) < -1$. Then similar as the construction in Corollary 2.20(1), we have

$$\operatorname{discrep}(Y, \Gamma + cF_0) = -\infty \leq \operatorname{discrep}(Y, \Gamma + cF_g).$$

Step 4: Now we deal with the case $\operatorname{discrep}(Y, \Gamma + cF_g) > -\infty$. By our analysis in *Step 3*, we have each coefficients of $\Gamma + cF_0$ is lower than 1, thus can apply Corollary 2.20(3) for $\mathbb{R}$-divisors:

$$\operatorname{discrep}(Y, \Gamma+cF_g) = \min\left\{ \min_{G_i \cap G_j \neq \emptyset}\{1-b_i-b_j\}, \min_{G_i \cap F_g \neq \emptyset}\{1-b_i-c\}, \min_i\{1-b_i\}, 1-c, 1 \right\},$$

where recall that $\Gamma + cF_g = \sum b_i G_i + cF_g$ has snc support for general member F_g. Similarly as in *3*, we analysis the following cases for pair $(Y, \Gamma + cF_0)$:

(1) When $G_i \cap G_j \neq \emptyset$, we blow-up Y along a codimension 2 subvariety of $G_i \cap G_i$ which gives

$$a(E, Y, \Gamma + cF_0) \leq 2 - 1 - b_i - b_j = 1 - b_i - b_j.$$

(2) For $1 - b_i$, we blow-up a general codimension 2 subvariety inside G_i which gives

$$a(E, Y, \Gamma + cF_0) \leq 2 - 1 - b_i = 1 - b_i.$$

(3) For entry 1, we blow-up a codimension 2 subvariety disjoints with every irreducible components of $\Gamma + cF_0$, which gives

$$a(E, Y, \Gamma + cF_0) = 2 - 1 = 1.$$

(4) For $1 - c$, we choose a component H of F_0 which is not f-exceptional. Along H, we have the coefficient $\operatorname{mult}_H \Gamma$ is non-negative, since $f_*^{-1}\Delta$ is effective by Δ is, and B is effective. We blow-up a general codimension 2 subvariety inside H, which gives

$$a(E, Y, \Gamma + cF_0) = 2 - 1 - c - \sum b_i \operatorname{mult}_Z G_i - c \operatorname{mult}_Z B \leq 1 - c.$$

(5) Finally we suppose that $G_i \cap F_g \neq \emptyset$, then by our observation in *Step 2* we have either $G_i \subseteq \operatorname{Supp} F_0$ or $G_i \cap F_0$ has a codimension 2 center of Y, each of them after blowing-up will give

$$a(E, Y, \Gamma + cF_0) \leq 2 - 1 - b_i - c = 1 - b_i - c.$$

Thus, we have

$$\operatorname{discrep}(Y, \Gamma + cF_0) \leq \operatorname{discrep}(Y, \Gamma + cF_g).$$

Together with formula (5.4), this is

$$\operatorname{discrep}(X, \Delta + cL_0) \leq \operatorname{discrep}(X, \Delta + cL_g).$$

□

Example 5.2.26. (The assumption of properness is crucial) In the proof of Corollary 2.25, we assume that X is proper to exclude the possibility of some number $F_\lambda \in |F|$ may disjoint with divisors G_i. Since if this case arisen, we can not apply the blow-up formula to get discrepancy is lower than -1. For example, we choose $X = \mathbb{A}^2$ to be affine plane, and $L = \mathbb{C} \oplus \mathbb{C}\langle x \rangle$ is a vector subspace of $H^0(X, \mathscr{O}_X) = \mathbb{C}[x, y]$. Then $|L| = |F| = \mathbb{P}^1$ with coordinates $(1 : x)$. The special member $F_0 = L_0$ is $(1 : 0)$ corresponds the divisor 0, and the general member $F_g = L_g$ is $(a : bx)$ corresponds to the line $a + bx = 0$ in $\mathbb{A}^2$. We assume that $\Delta = 0$. Then

$$\operatorname{discrep}(X, \Delta + cL_0) = \operatorname{discrep}(X, 0) = 1$$

by X is non-singular. For general member $L_g \in |F|$, we take a blow-up $f : Y \longrightarrow X$ along some point p inside L_g, we have

$$\begin{aligned} a(E, X, cL_g) &= 2 - 1 - c \cdot \operatorname{mult}_p L_g \\ &= 1 - c. \end{aligned}$$

But $1 = \operatorname{discrep}(X, \Delta + cL_0) \leq \operatorname{discrep}(X, \Delta + cL_g) \leq 1 - c$ is absurd.

Example 5.2.27. (The assumption of effective is crucial) In many reference a *pair* (X, Δ) include the data that X is a normal variety, Δ is an effective $\mathbb{Q}$-divisor with coefficients lower than 1, and $K_X + \Delta$ is $\mathbb{Q}$-Cartier. Or furthermore, $\mathbb{R}$-divisor $\Delta \geq 0$ and $K_X + \Delta$ is $\mathbb{R}$-Cartier. For example see [Bir12]. But in [KM98], we allowed Δ to be arbitrary, not necessarily to be effective and coefficients bounded by 1, sometimes it is called by *sub-pair*. Hence, Corollary 2.25 failed when Δ is not effective.

We choose $X = \mathbb{P}^2$ be the projective plane, and $\Delta = -dH$ to be a line with multiplicity $-d$ with $d > 0$. And we choose $|L| = |F| = |\mathscr{O}_{\mathbb{P}^2}(1)|$. For special member $F_0 = H \in |F|$ and general member F_g is a line intersect transversally with H. Applying Corollary 2.20(3), for $\Delta + cL_g = -dH + cF_g$ we have

$$\operatorname{discrep}(X, \Delta + cL_g) = \min\{1 + d - c, 1 - c, 1 + d, 1\} = \min\{1 - c, 1\}.$$

And for $\Delta + cL_0 = -dH + cH = (c - d)H$ we have

$$\operatorname{discrep}(X, \Delta + cL_0) = \min\{1 - c + d, 1\}.$$

When $c = 1$, we have

$$\operatorname{discrep}(X, \Delta + cL_0) = \min\{d, 1\} \leq 0 = \operatorname{discrep}(X, \Delta + cL_g)$$

which is absurd.

Definition 5.2.28. (Alternative definition for canonical pairs) A pair (X, Δ) where X is a normal variety and Δ is effective $\mathbb{R}$-divisor, we say it has *canonical* singularities if it satisfies the following conditions

(1) $K_X + \Delta$ is $\mathbb{R}$-Cartier, and

(2) For any resolution of singularities $f : Y \longrightarrow X$, if we write

$$K_Y + C = f^*(K_X + \Delta),$$

then $-C + f_*^{-1}\Delta$ is effective.

Note that from

$$\begin{aligned} K_Y + f_*^{-1}\Delta &= f^*(K_X + \Delta) + \sum_{E_i : f\text{-exceptional}} a(E, X, \Delta)E_i \\ &= K_Y + C + \sum_{E_i : f\text{-exceptional}} a(E, X, \Delta)E_i, \end{aligned}$$

we can see every $a(E, X, \Delta) \geq 0$, this is nothing but $\operatorname{discrep}(X, \Delta) \geq 0$.

Proposition 5.2.29. *When $\Delta = 0$ and K_X is $\mathbb{Q}$-Cartier, this definition is compatible with Definition 1.9.*

Proof. We need to show that condition (2) is equivalent to $f_*\mathscr{O}_Y(mK_Y) \cong \mathscr{O}_X(mK_X)$ for $m > 0$. For resolution $f : Y \longrightarrow X$, we write

$$mK_Y = f^*(mK_X) + \sum ma_iE_i$$

where E_i are f-exceptional divisor, we denote it by $-mC$. If $-C$ is an effective divisor, that is each $a_i \geq 0$. Then we have

$$\begin{aligned} f_*\mathscr{O}_Y(mK_Y) &= f_*\mathscr{O}_Y\big(f^*(mK_X) - mC\big) \\ &\cong \mathscr{O}_X(mK_X) \otimes f_*\mathscr{O}_Y(-mC) \end{aligned}$$

thanks to the projection formula. Since X is normal, by Zariski's Main Theorem, we have $f_*\mathscr{O}_Y = \mathscr{O}_X$, that is we can get

$$\mathscr{O}_X \subseteq f_*\mathscr{O}_Y(-mC).$$

And by $-mC$ is effective, locally the rational sections in $f_*\mathscr{O}_Y(-mC)$ allowed have poles along the image of $\operatorname{Ex}(f)$, which is codimension at least 2 by X is normal and Chapter 2. Therefore, by algebraic Hartogs we have converse inclusion.

Now for the inverse direction, for $f_*\mathscr{O}_Y(-mC) = \mathscr{O}_X$, locally it gives

$$H^0\big(f^{-1}U, \mathscr{O}_Y(-mC)\big) = H^0(U, \mathscr{O}_X) \ni 1$$

which implies that $\operatorname{div}(1) + (-mC) = -mC \geq 0$, so $-C$ is effective. □

Definition 5.2.30. (Alternative definition for terminal pairs) A pair (X, Δ) where X is a normal variety and Δ is effective $\mathbb{R}$-divisor, we say it has *terminal* singularities if it satisfies the following conditions

(1) $K_X + \Delta$ is $\mathbb{R}$-Cartier;

(2) For any resolution of singularities $f : Y \longrightarrow X$, if we write

$$K_Y + C = f^*(K_X + \Delta)$$

then $-C + f_*^{-1}\Delta$ is effective.

(3) The support of $-C + f_*^{-1}\Delta$ coincides with the divisorial part of $\mathrm{Ex}(f)$.

Note that from

$$\begin{aligned} K_Y + f_*^{-1}\Delta &= f^*(K_X + \Delta) + \sum_{E_i : f\text{-exceptional}} a(E, X, \Delta)E_i \\ &= K_Y + C + \sum_{E_i : f\text{-exceptional}} a(E, X, \Delta)E_i, \end{aligned}$$

$-C + f_*^{-1}\Delta$ coincides with the divisorial part of $\mathrm{Ex}(f)$ means that there is no coefficients $a(E_i, X, \Delta)$ equals to zero, hence that $\mathrm{discrep}(X, \Delta) > 0$.

Proposition 5.2.31. *When $\Delta = 0$ and K_X is $\mathbb{Q}$-Cartier, this definition is compatible with Definition 1.10.*

Proof. We show that condition (2) is equivalent to $f_*\mathscr{O}_Y(mK_Y - E) \cong \mathscr{O}_X(mK_X)$ for $m > 0$, where $E \subseteq Y$ is the reduced exceptional divisor. For resolution $f : Y \longrightarrow X$, we write

$$mK_Y = f^*(mK_X) + (-mC)$$

where $-mC = \sum ma_iE_i$ and E_i run through all exceptional divisors inside $\mathrm{Ex}(f)$ with no a_i to be zero. We denote $\sum E_i$ by E. Since we have all $ma_i \geq 1$,

$$\begin{aligned} f_*\mathscr{O}_Y(mK_Y - E) &= f_*\mathscr{O}_Y\big(f^*(mK_X) + \sum(ma_i - 1)E_i\big) \\ &\cong \mathscr{O}_X(mK_X) \otimes f_*\mathscr{O}_Y\big(\sum(ma_i - 1)E_i\big) \end{aligned}$$

thanks to the projection formula. Since $\sum(ma_i - 1)E_i$ is an effective divisor, similarly using Zariski's Main Theorem and algebraic Hartogs, we can get

$$f_*\mathscr{O}_Y\big(\sum(ma_i - 1)E_i\big) \cong \mathscr{O}_X.$$

Now for the inverse direction, for $f_*\mathscr{O}_Y\big(\sum(ma_i - 1)E_i\big) \cong \mathscr{O}_X$, locally it gives

$$\sum(ma_i - 1)E_i \geq 0,$$

thus each $a_i > 1$, i.e. $-C$ coincides with divisorial part of $\mathrm{Ex}(f)$. □

Definition 5.2.32. (Alternative definition for klt pairs) A pair (X, Δ) where X is a normal variety and Δ is effective $\mathbb{R}$-divisor, we say it has *klt* singularities if it satisfies the following conditions:

(1) $K_X + \Delta$ is $\mathbb{R}$-Cartier;

(2) The coefficients of Δ are contained in the open interval $(0, 1)$;

(3) There exists a log resolution $f : Y \longrightarrow X$ such that if we write

$$K_Y + C = f^*(K_X + \Delta),$$

then the coefficients c_j of $C = \sum_j c_j C_j$ are contained in $(-\infty, 1)$. Here C_j are distinct prime divisors.

The equivalence of this klt and Definition 2.23 when $\Delta \geq 0$ is by applying the Corollary 2.21(2) on (3). Sometimes the klt in Definition 2.23 may be called *sub-klt*.

Definition 5.2.33. (Alternative definition for plt pairs) A pair (X, Δ) where X is a normal variety and Δ is effective $\mathbb{R}$-divisor, we say it has *lc* singularities if it satisfies the following conditions:

(1) $K_X + \Delta$ is $\mathbb{R}$-Cartier;

(2) The coefficients of Δ are contained in the open interval $(0, 1]$;

(3) For any log resolution $f : Y \longrightarrow X$, if we write

$$K_Y + C = f^*(K_X + \Delta),$$

then coefficients c_j of $C = \sum c_j C_j$ are contained in $(-\infty, 1)$ for those C_j contained in the exceptional set of f.

The equivalence of this plt and Definition 2.23 when $\Delta \geq 0$ is due to $-c_j = a(E_j, X, \Delta)$.

Definition 5.2.34. (Alternative definition for lc pairs) A pair (X, Δ) where X is a normal variety and Δ is effective $\mathbb{R}$-divisor, we say it has *lc* singularities if it satisfies the following conditions:

(1) $K_X + \Delta$ is $\mathbb{R}$-Cartier;

(2) The coefficients of Δ are contained in the open interval $(0, 1]$;

(3) There exists a log resolution $f : Y \longrightarrow X$ such that if we write

$$K_Y + C = f^*(K_X + \Delta),$$

then the coefficients c_j of $C = \sum_j c_j C_j$ are contained in $(-\infty, 1]$. Here C_j are distinct prime divisors.

Also, use Corollary 2.21(2) to deduce the equivalence when $\Delta \geq 0$. Sometimes the lc in Definition 2.23 may be call *sub-lc*.

Definition 5.2.35. (Alternative definition for dlt pairs) A pair (X, Δ) where X is a normal variety and Δ is effective $\mathbb{R}$-divisor, we say it has *lc* singularities if it satisfies the following conditions:

(1) $K_X + \Delta$ is $\mathbb{R}$-Cartier;

(2) The coefficients of Δ are contained in the open interval $(0, 1]$;

(3) There exists a log resolution $f : Y \longrightarrow X$, if we write

$$K_Y + C = f^*(K_X + \Delta),$$

then coefficients c_j of $C = \sum c_j C_j$ are contained in $(-\infty, 1)$ for those C_j contained in the exceptional set of f.

The equivalence of this dlt and Definition 2.23 when $\Delta \geq 0$ is Szabó's work, see [Sza94].

Example 5.2.36. Take the affine space $X = \mathbb{A}^n$ and coordinate hyperplanes $D_1, \ldots, D_n$, denote $\Delta = \sum a_i D_i$ is an effective $\mathbb{R}$-divisor. Then (X, Δ) is klt (resp. plt, lc) if and only if $0 \leq a_i < 1$ for all i (resp. $0 \leq a_i \leq 1$ for all i and $a_i < 1$ except for at most one i, $0 \leq a_i \leq 1$ for all i). Furthermore, dlt and lc coincide since we can choose the log resolution to be id_X, due to Szabó's result.

Example 5.2.37. Take $X = \mathbb{A}^2$ to be the affine plane with coordinates x, y, and prime divisor $D = \mathrm{div}(x^2 - y^3)$. We determine the necessary and sufficient condition for the pair (X, dD) to be klt or lc for real number $d \leq 1$.

We construct a log resolution like in [Har77, V,3.9.1]. First, we take the blow-up $f_1 : Y_1 \longrightarrow X$ along the smooth center $p_0 = (0, 0)$, the exceptional set E_1 is a prime divisor isomorphic to $\mathbb{P}^1$. The strict transform $D_1 = {f_1}_*^{-1} D$ is smooth, which intersect E_1 at one point p_1 with multiplicity 2.

Take the blow-up $f_2 : Y_2 \longrightarrow Y_1$ along the smooth center p_1, the exceptional set E_2 is a prime divisor isomorphic to $\mathbb{P}^1$. Three smooth prime divisors E_2, $D_2 = {f_2}_*^{-1} D_1$ and $E_1' = {f_2}_*^{-1} E_1$ intersect at one point p_2 with multiplicity 3.

Take the blow-up $f_3 : Y \longrightarrow Y_2$ along the smooth point p_2, the exceptional set E_3 is a prime divisor isomorphic to $\mathbb{P}^1$. The union of 4 prime divisors E_3, $D_3 = {f_3}_*^{-1} D_2$, $E_1'' = {f_3}_*^{-1} E_1'$ and $E_2' = {f_3}_*^{-1} E_2$ is a snc divisor.

The composition $f : Y \longrightarrow X$ is a log resolution of (X, dD). We have

$$K_Y + dD_3 = f^*(K_X + dD) + \sum_{E_i : f\text{-exceptional}} a(E, X, dD) \cdot E_i,$$

where $K_Y = f^* K_X + E_1'' + 2E_2' + 4E_3$ and $f^* D = D_3 + 2E_1'' + 3E_2' + 6E_3$. Therefore,

$$\sum_{E_i : f\text{-exceptional}} a(E, X, dD) \cdot E_i = (1 - 2d)E_1'' + (2 - 3d)E_2'' + (4 - 6d)E_3,$$

the pair (X, dD) is klt (resp. lc) if and only if $0 \leq d < \frac{5}{6}$ (resp. $0 \leq d \leq \frac{5}{6}$).

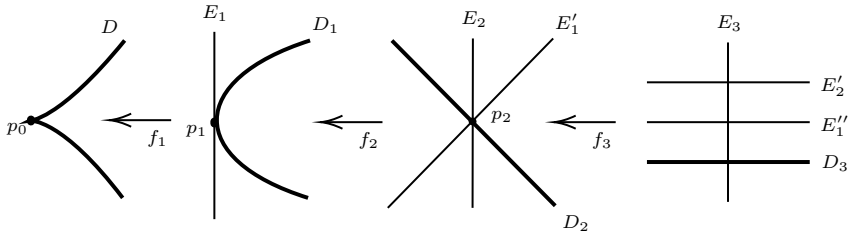


Figure 5.7.: Log resolution of (X, dD).

Lemma 2.11 and Corollary 2.21 imply the following continuity properties.

Corollary 5.2.38. *Let (X, Δ) be a pair and Δ' an effective $\mathbb{R}$-Cartier divisor. Then*

(1) If $(X, \Delta + \Delta')$ is terminal (resp. canonical, klt, plt, lc, dlt with $\Delta \geq 0$), then (X, Δ) is terminal (resp. canonical, klt, plt, lc, dlt with $\Delta \geq 0$).

(2) If (X, Δ) is terminal (resp. klt), then $(X, \Delta + \varepsilon\Delta')$ is also terminal (resp. klt) for $0 \leq \varepsilon \ll 1$.

(3) If (X, Δ) is plt, then $(X, \Delta + \varepsilon\Delta')$ is also plt for $0 \leq \varepsilon \ll 1$, assuming that Δ and Δ' have no common irreducible components.

(4) For Δ is effective, if (X, Δ) is dlt, then $(X, \Delta + \varepsilon\Delta')$ is also dlt for $0 \leq \varepsilon \ll 1$, assuming that $\operatorname{Supp} \Delta' \subseteq \operatorname{Supp}(\Delta - \lfloor \Delta \rfloor)$.

(5) If (X, Δ) is terminal, then $(X, \Delta + \Delta')$ is canonical if and only if $(X, \Delta + c\Delta')$ is terminal for every $c < 1$.

(6) If (X, Δ) is klt (resp. plt), then $(X, \Delta + \Delta')$ is lc if and only if $(X, \Delta + c\Delta')$ is klt (resp. plt) for every $c < 1$.

Proof. Note that we do not assumed Δ is effective, only with X is a normal variety and $K_X + \Delta$ is $\mathbb{R}$-Cartier.

For (1), since we have $a(E, X, \Delta + \Delta') \leq a(E, X, \Delta)$ for every divisor E over X. This means

$$\operatorname{discrep}(X, \Delta + \Delta') \leq \operatorname{discrep}(X, \Delta).$$

In particular, for dlt pair $(X, \Delta + \Delta')$, the $(\Delta + \Delta')|_{X\setminus Z}$ has snc support implies that so is $\Delta|_{X\setminus Z}$.

For (2), we write $\Delta = \sum a_j D_j$ and $\Delta' = \sum a'_k D'_k$, and we choose a log resolution $f : Y \longrightarrow X$ such that $\sum f_*^{-1} D_j + \sum f_*^{-1} D'_k$ is smooth. While we have

$$a(E, X, \Delta + \varepsilon\Delta') = a(E, X, \Delta) - \varepsilon \cdot \operatorname{mult}_E f^*\Delta'.$$

Using Corollary 2.21, we can write

$$\begin{aligned}
\operatorname{discrep}(X, \Delta + \varepsilon\Delta') &= \min\Bigg\{ \min_i\{a(E_i, X, \Delta + \varepsilon\Delta')\}, \min_j\{1 - a_j\}, \\
&\qquad \min_k\{1 - \varepsilon a'_k\}, \min_{D_j = D'_k}\{1 - a_j - \varepsilon a'_k\}, 1 \Bigg\} \\
&= \min\Bigg\{ \min_i\{a(E, X, \Delta) - \varepsilon \cdot \operatorname{mult}_E f^*\Delta'\}, \min_j\{1 - a_j\}, \\
&\qquad \min_k\{1 - \varepsilon a'_k\}, \min_{D_j = D'_k}\{1 - a_j - \varepsilon a'_k\}, 1 \Bigg\} \\
\operatorname{discrep}(X, \Delta) &= \min\left\{ \min_i\{a(E_i, X, \Delta)\}, \min_j\{1 - a_j\}, 1 \right\},
\end{aligned}$$

with our choice of ε lies in $[0, \varepsilon_0)$, where

$$\varepsilon_0 = \min\left\{ \min_i \left\{ \frac{a(E_i, X, \Delta)}{\operatorname{mult}_{E_i} f^*\Delta'} \;\middle|\; \operatorname{mult}_{E_i} f^*\Delta' > 0 \right\}, \min_{D_j = D'_k} \left\{ \frac{1 - a_j}{a'_k} \right\} \right\},$$

to ensures that each terms $a(E_i, X, \Delta + \varepsilon\Delta')$ and $1 - a_j - \varepsilon a'_k > 0$ (or > -1). Compare two equations, easy to find the conclusion.

For (3), in (2) we deal with (X, Δ) is terminal or klt, we do not assume that Δ and Δ' have no common components, because the condition of terminal or $\lfloor \Delta \rfloor \leq 0$ have already ensured that coefficients of Δ is strictly lesser than 1. But for (X, Δ) is plt, some coefficients of Δ may equal to 1, we must need this assumption to ensure that the coefficients will not larger than 1. Now, similar as above, can find that

$$\begin{aligned}
\operatorname{discrep}(X, \Delta + \varepsilon\Delta') &= \min\Bigg\{ \min_i\{a(E_i, X, \Delta + \varepsilon\Delta')\}, \min_j\{1 - a_j\}, \min_k\{1 - \varepsilon a'_k\}, 1 \Bigg\} \\
&> -1.
\end{aligned}$$

For (4), if (X, Δ) is a dlt pair, some of coefficients of Δ may equal to 1. So the assumption $\operatorname{Supp} \Delta' \subseteq \operatorname{Supp}(\Delta - \lfloor \Delta \rfloor)$ ensures that $\Delta + \varepsilon\Delta'$ is effective with coefficients lie in $[0, 1]$. Moreover, it ensures that $\Delta + \varepsilon\Delta'|_{X \setminus Z}$ has snc support for some closed subset $Z \subseteq X$ such that $X \setminus Z$ is smooth. And for a birational morphism $f : Y \longrightarrow X$ with $E \subseteq Y$ is an irreducible divisor such that $\operatorname{center}_X E \subseteq Z$, we have

$$a(E, X, \Delta + \varepsilon\Delta') = a(E, X, \Delta) - \varepsilon \cdot \operatorname{mult}_E f^*\Delta' > 1$$

for ε sufficiently small.

For (5), also write $\Delta = \sum a_j D_j$ and $\Delta' = \sum a'_k D'_k$, and use Corollary 2.21 we have

$$\begin{aligned}\operatorname{discrep}(X, \Delta + \varepsilon\Delta') = \min\Bigg\{ & \min_i\{a(E, X, \Delta) - c \cdot \operatorname{mult}_E f^*\Delta'\}, \min_j\{1 - a_j\}, \\ & \min_k\{1 - ca'_k\}, \min_{D_j = D'_k}\{1 - a_j - ca'_k\}, 1\Bigg\} \\ \operatorname{discrep}(X, \Delta) = \min\Big\{ & \min_i\{a(E_i, X, \Delta)\}, \min_j\{1 - a_j\}, 1\Big\} > 0,\end{aligned}$$

where $f : Y \longrightarrow X$ is a log resolution such that $\sum f_*^{-1} D_j + \sum f_*^{-1} D'_k$ is smooth. Easy to see that

$$a(E_i, X, \Delta) - \operatorname{mult}_{E_i} f^*\Delta', 1 - a_j, 1 - a'_k, 1 - a_j - a'_k \geq 0$$

is equivalent to

$$a(E_i, X, \Delta) - c \cdot \operatorname{mult}_{E_i} f^*\Delta', 1 - a_j, 1 - ca'_k, 1 - a_j - ca'_k > 0$$

for every $c < 1$. Where $a(E_i, X, \Delta)$ and $1 - a_j > 0$ is provided by (X, Δ) is terminal.

For (6), similar as above

$$a(E_i, X, \Delta) - \operatorname{mult}_{E_i} f^*\Delta', 1 - a_j, 1 - a'_k, 1 - a_j - a'_k \geq -1$$

is equivalent to

$$a(E_i, X, \Delta) - c \cdot \operatorname{mult}_{E_i} f^*\Delta', 1 - a_j, 1 - ca'_k, 1 - a_j - ca'_k > -1$$

for every $c < 1$. Where $\lfloor \Delta + c\Delta' \rfloor = \lfloor \Delta \rfloor$, $a(E_i, X, \Delta)$ and $1 - a_j > -1$ is provided by (X, Δ) is klt or plt. □

Proposition 5.2.39. *Let (X, Δ) be a klt pair, where $K_X + \Delta$ is $\mathbb{R}$-Cartier.*

(1) There exists a log resolution $f : X' \longrightarrow X$ such that if we write

$$f^*(K_X + \Delta) = K_{X'} + A_{X'} - B_{X'}$$

using effective $\mathbb{R}$-divisors $A_{X'}$ and $B_{X'}$ without common components, then $\operatorname{Supp} A_{X'}$ *is smooth.*

(2) If $a(E, X, \Delta) < 1 + \operatorname{totaldiscrep}(X, \Delta)$, then $\operatorname{center}_{X'} E$ *is a divisor. In particular, there are only finitely many exceptional divisors E over X such that*

$$a(E, X, \Delta) < 1 + \operatorname{totaldiscrep}(X, \Delta).$$

Proof. Since the pair (X, Δ) is klt, each coefficients of Δ is strictly lower than 1.

For (1), let us start with an arbitrary log resolution $f : Y \longrightarrow X$ and for effective divisor A_Y, we write

$$A_Y = \sum_{i \in I} (1 - \alpha_i) A_i$$

with each $0 < \alpha_i < 1$, and A_i are distinct irreducible components. Note that, by

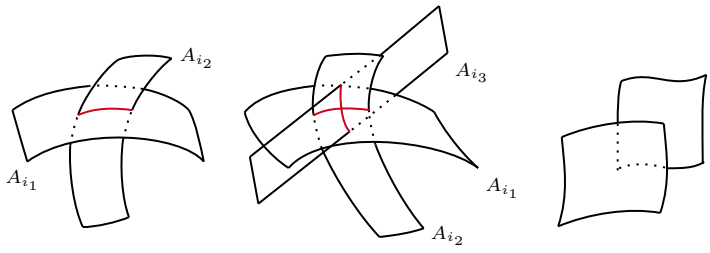
$$f^*(K_X + \Delta) = K_Y + \sum_{E_i : f\text{-exceptional}} \Big(- a(E_i, X, \Delta) \Big) E_i + f^* \Delta,$$

the coefficients of the effective part A_Y are also < 1 since $a(E_i, X, \Delta) > -1$.

Let $S(Y) = \sum_{i \in I} \mathbb{Z}_{\geq 0} \alpha_i \subseteq \mathbb{R}_{\geq 0}$. We also set

$$w(Y, J) = \begin{cases} \sum_{i \in J} \alpha_i & \text{if } \bigcap_{i \in J} A_i \neq \emptyset \text{ and } \#J \geq 2, \text{ where } J \subseteq I \\ \infty & \text{otherwise.} \end{cases}$$

which characterizes the situation of intersection of the snc divisor $\sum_{i \in I} A_i$.



$\rightsquigarrow \alpha_{i_1} + \alpha_{i_2}$ $\quad$ $\rightsquigarrow \alpha_{i_1} + \alpha_{i_2} + \alpha_{i_3}$ $\quad$ $\rightsquigarrow \infty$

And set its minimum associated to Y is

$$M(Y) = \min\{w(J) \;\big|\; J \subseteq I\},$$

which lies in $(0, \dim X) \cap S(Y)$. When $M(Y) = \infty$, it means every component of A_Y is disjoint with others, which is equivalent to $\operatorname{Supp} A_Y$ is smooth. Moreover, we set a quantity associated to Y

$$r(Y) = \#\{J \subseteq I \;\big|\; w(J) = M(Y)\}.$$

Assume that $M(Y) < \infty$. Let $J \subseteq I$ be such that $w(J) = M(Y)$. Set $Z = \bigcap_{i\in J} A_i$. Let $p : Y' \longrightarrow Y$ be the blowing-up of Y along the smooth center Z, and E is the exceptional divisor. By [Har77, II, ex8.5], we have

$$K_{Y'} = p^* K_Y + (c-1)E$$

where $c = \#J$ is the $\operatorname{codim}_Y E$. Furthermore, we pull back the divisor A_Y via p, will get

$$\begin{aligned} p^* A_Y &= p^* \left(\sum_{i\in I} (1-\alpha_i) A_i \right) \\ &= \sum_{i\in I} (1-\alpha_i) p_*^{-1} A_i + \sum_{i\in J} (1-\alpha_i) E \end{aligned}$$

for which $E \not\subseteq A_i$ the pullback gives the multiplicity is 0. Therefore, the coefficient of E in

$$\begin{aligned} &(f\circ p)^*(K_X + \Delta) \\ &= p^* \left(K_Y + \sum_{E_i : f\text{-exceptional}} \Big(- a(E_i, X, \Delta) \Big) E_i + f^* \Delta \right) \\ &= K_{Y'} + (1 - \#J)E + \sum_{i\in I} (1-\alpha_i) p_*^{-1} A_i + \sum_{i\in J} (1-\alpha_i) E + \text{non-effective part} \end{aligned}$$

is equal to $1 - \sum_{i\in J} \alpha_i$. Moreover, the new $A_{Y'}$ in Y' is determined by

$$A_{Y'} = \sum_{i\in I} (1-\alpha_i) p_*^{-1} A_i + \begin{cases} (1 - \sum_{i\in J} \alpha_i) E & \text{if } \sum_{i\in J} \alpha_i < 1 \\ 0 & \text{if } \sum_{i\in J} \alpha_i \geq 1 \end{cases}$$

thus we have $S(Y') = S(Y)$. And in each case we have $M(Y') > M(Y)$ by

$$w(Y', J') = \sum_{i\in J} \alpha_i + \sum \alpha_{i'} = M(Y) + \sum \alpha_{i'} > M(Y),$$

or $M(Y') = M(Y)$ by the coefficient of E is negative. Moreover, $r(Y') < r(Y)$ when $M(Y') = M(Y)$, since one of the choice J has removed.

Now we get an ascending sequence $\big(M(Y), -r(Y)\big)$ in lexicographical order. Since $M(Y)$ and $-r(Y)$ can take only finitely many values, we cannot keep blowing up indefinitely. Hence we get $M(Y) = \infty$ after a finite number of blow-ups.

For (2), by write $f^*(K_X + \Delta) = K_{X'} + A_{X'} - B_{X'}$ and clearly we have $f_*(A_{X'} - B_{X'}) = \Delta$, we can use Lemma 2.16 and Lemma 2.11 to get

$$a(E, X, \Delta) = a(E, X', A_{X'} - B_{X'}) \geq a(E, X', A_{X'}) \tag{5.6}$$

for every divisor E over X. Furthermore, if E is an exceptional divisor over X', by $\operatorname{Supp} A_{X'}$ is smooth where $A_{X'} = \sum_{i \in I}(1 - \alpha_i)A_i$ with $0 < \alpha_i < 1$, we have

$$a(E, X', A_{X'}) \geq \operatorname{discrep}(X, A_{X'}) = \min\left\{\min_{i \in I}\{1 - (1 - \alpha_i)\}, 1\right\} \tag{5.7}$$

thanks to Corollary 2.20(3). Moreover, we have

$$\begin{aligned}\operatorname{totaldiscrep}(X, \Delta) &\leq \min_{i \in I}\{a(A_i, X, \Delta)\} \\ &= \min_{i \in I}\{\alpha_i - 1\} \\ &= \min_{i \in I}\{\alpha_i\} - 1.\end{aligned}$$

Thus

$$\min_{i \in I}\{\alpha_i\} \geq 1 + \operatorname{totaldiscrep}(X, \Delta).$$

Combing this with (5.6) and (5.7), we have

$$a(E, X, \Delta) \geq 1 + \operatorname{totaldiscrep}(X, \Delta)$$

for which E is an exceptional divisor over X'. So if $a(E, X, \Delta)$ is not greater than it, E would not be an exceptional divisor over X', in other words $\operatorname{center}_{X'} E$ is a divisor on X'. Furthermore, by construction of X', there are only finitely many divisors on X' which are exceptional over X. This proves (2). □

Remark 5.2.40. For any pair (X, Δ) with X is a normal variety and $K_X + \Delta$ is $\mathbb{R}$-Cartier, we always have

$$\operatorname{totaldiscrep}(X, \Delta) \leq 0.$$

Since there is a divisor D have coefficient zero, i.e. $a(D, X, \Delta) = -\operatorname{mult}_D \Delta = 0$.

Proposition 5.2.41. *Let (X, Δ) be a pair where X is a normal variety and $\Delta = \sum a_i D_i$ is a sum of distinct prime divisors, $0 \leq a_i \leq 1$. Assume that $K_X + \Delta$ is $\mathbb{R}$-Cartier. Then (X, Δ) is dlt if and only if there is a closed subset $Z \subseteq X$ such that*

(1) $X \backslash Z$ is smooth and $\Delta|_{X \backslash Z}$ has snc support.

(2) There is a log resolution $f : Y \longrightarrow X$ of (X, Δ) such that $f^{-1}(Z) \subseteq Y$ has pure of dimension 1, and $a(E.X, \Delta) > -1$ for every irreducible divisor $E \subseteq f^{-1}(Z)$.

Proof. One direction is trivial, we prove the converse direction.

Assume that $f : Y \longrightarrow X$ exists as above. Write

$$K_Y + \Delta_Y = f^*(K_X + \Delta)$$

and let Δ' be an effective divisor whose support equals $\operatorname{Supp} f^{-1}(Z)$. By (2), every irreducible component of Δ' has coefficient < 1 in Δ_Y. Thus every irreducible component of $\Delta_Y + \varepsilon\Delta'$ has coefficient ≤ 1 for $0 < \varepsilon \ll 1$, by our choice of Δ. Hence, $(Y, \Delta_Y + \varepsilon\Delta')$ has lc singularities by Corollary 2.20(3).

If E is any divisor over X whose center is contained in Z, then $\operatorname{center}_Y E \subseteq f^{-1}(Z)$, thus by Lemma 2.16 and 2.11

$$a(E, X, \Delta) = a(E, Y, \Delta) > a(E, Y, \Delta_Y + \varepsilon\Delta') \geq -1.$$

In particular for divisor over X which lies between Y and X, we compute the discrepancy of its strict transform. This shows Definition 2.23(2). □

Proposition 5.2.42. *Assuming that $\Delta = \sum a_i D_i$ is an effective $\mathbb{R}$-divisor. A pair (X, Δ) is klt if and only if it is dlt and $\lfloor \Delta \rfloor = 0$.*

Proof. "⇒": If (X, Δ) is klt pair, then we can choose $Z = X$ such that the condition (1) holds. Moreover, since for non-exceptional divisors X their coefficient a_i from Δ is < 1 which gives discrepancy is $-a_i > -1$, and for exceptional divisors have $a(E, X, \Delta) > -1$, we see it is dlt.

"⇐": Assume that (X, Δ) is dlt and $\lfloor \Delta \rfloor = 0$. Let $f : Y \longrightarrow X$ be a birational morphism and $E \subseteq Y$ an exceptional divisor over X, let $Z \subseteq X$ be the closed subset witness (X, Δ) is dlt. If $\operatorname{center}_X E \subseteq Z$ then $a(E, X, \Delta) > -1$ by the definition of dlt. If $\operatorname{center}_X E \not\subseteq Z$, then we can choose $U = X \backslash Z$ such that

$$\operatorname{discrep}(U, \Delta|_U) = \min \left\{ \min_{i \neq j \ D_i|_U \cap D_j|_U \neq \emptyset} \{1 - a_i - a_j\}, \min_i \{1 - a_i\}, 1 \right\}$$

thanks to Corollary 2.20(3), to find $a(E|_U, U, \Delta_U) > -1$. But it is equal to $a(E, X, \Delta)$. Thus (X, Δ) is klt. □

Recall that for Weil divisor D, its corresponding rank 1 reflexive sheaf $\mathscr{O}_X(D)$ is a coherent sheaf, has global section

$$H^0\big(X, \mathscr{O}_X(D)\big) = \{f \in \mathbb{C}(X)^\times \mid \operatorname{div}(f) + D \geq 0\} \cup \{0\}.$$

Furthermore, we can define its linear series by set

$$\begin{aligned} |D| &= \{D_f \text{ is an effetive divisor} \mid D_f \sim_{lin} D\} \\ &= \big\{ \operatorname{div}(f) + D \;\big|\; f \in H^0\big(X, \mathscr{O}_X(D)\big)\backslash\{0\}\big\}. \end{aligned}$$

And we define its (set-theoretic) base locus to be $\operatorname{Bs}(|D|) = \bigcap_{D_f \in |D|} \operatorname{Supp} D_f$. For a finite dimensional vector subspace $V \subseteq H^0\big(X, \mathscr{O}_X(D)\big)$, its corresponding projective space $\mathbb{P}(V)$ sends each point to an element of $|D|$:

$$\begin{aligned} \Phi : \mathbb{P}(V) &\longrightarrow |D| \\ [s] &\longmapsto \operatorname{div}(s) + D, \end{aligned}$$

which is well-defined but not necessarily injective. Therefore, we can define the *linear sub-series* of $|D|$ to be $\Phi(\mathbb{P}(V)) \subseteq |D|$.

Recall that for divisors D_1 and D_2, we write $D_1 \sim_{\mathbb{R}} D_2$ (resp. $\sim_{\mathbb{Q}}$) if and only if $D_1 - D_2$ is the $\mathbb{R}$-linear (resp. $\mathbb{Q}$-linear) combination of (Weil) principal divisors.

A dlt pair is always a limit of klt pairs:

Proposition 5.2.43. *Assume that (X, Δ) is dlt and the normal variety X is quasi-projective with an ample divisor H. Let Δ_1 be an effective $\mathbb{R}$-divisor (not necessarily $\mathbb{R}$-Cartier) such that $\Delta - \Delta_1$ is effective. Then there exist a real number $c > 0$ and an effective $\mathbb{R}$-divisor $D \sim_{\mathbb{R}} \Delta_1 + cH$ such that $(X, \Delta - \varepsilon\Delta_1 + \varepsilon D)$ is dlt for all $0 < \varepsilon \ll 1$.*

Moreover, if $\operatorname{Supp}\Delta_1 = \operatorname{Supp}\Delta$, then $(X, \Delta - \varepsilon\Delta_1 + \varepsilon D)$ is klt for all sufficiently small real numbers $\varepsilon > 0$.

Before prove this result, we need a lemma of rational approximation.

Lemma 5.2.44. *Assume that (X, Δ) is a dlt pair and the normal variety X is quasi-projective with an ample divisor H. Let Θ be an effective $\mathbb{Q}$-divisor (not necessarily $\mathbb{Q}$-Cartier) such that $\Delta - \Theta$ is effective. Then there is a rational number $c > 0$ and an effective $\mathbb{Q}$-divisor $B \sim_{\mathbb{Q}} \Theta + cH$ such that $(X, \Delta - \varepsilon\Theta + \varepsilon B)$ is dlt for all $0 < \varepsilon \ll 1$.*

Moreover, if $\operatorname{Supp}\Theta = \operatorname{Supp}\Delta$, then $(X, \Delta - \varepsilon\Theta + \varepsilon B)$ is klt for all sufficiently small real numbers $\varepsilon > 0$.

Proof. Observe that $\Delta = \Delta - \Theta + \Theta$ is an effective $\mathbb{R}$-divisor.

Let $Z \subseteq X$ be the closed subset witness (X, Δ) is dlt. Choose $m, m' \in \mathbb{Z}_{>0}$ such that $m\Theta$ is an integral divisor and the sheaf $\mathscr{O}_X(m\Theta + m'H)$ is generated by global sections. Here we no need that $m\Theta$ is Cartier. Note that X is quasi-projective, its global sections $H^0(X, m\Theta + m'H)$ may not be a finite dimensional vector space. Since $\mathscr{O}_X(m\Theta + m'H)$ is coherent sheaf, on every affine open subset it is finitely generated as a module, so we can choose finitely many such neighborhood - X is a noetherian scheme it is quasi-compact - equipped with finitely many global sections form a vector subspace $V \subseteq H^0(X, m\Theta + m'H)$ generate the sheaf $\mathscr{O}_X(m\Theta + m'H)$.

Now for $U = X \backslash Z$, U is smooth so every divisor of X on U is Cartier. We can see $|V| = \mathbb{P}(V)$ is base-point-free on U, and can apply Bertini's Theorem in [Har77, III, 10.9] on this linear series to find a general member $[s] \in \mathbb{P}(V)$ whose corresponding divisor $A \sim_{lin} m\Theta + m'H$ is non-singular on U, and hence that $A|_U + \Delta|_U$ has snc support on U.

We set $B = 1/mA$ and $c = m'/m$, such that B is an effective $\mathbb{Q}$-divisor and

$$B \sim_{\mathbb{Q}} \Theta + cH.$$

Moreover, we have $B - \Theta$ is $\mathbb{Q}$-Cartier. Since $K_X + \Delta$ is $\mathbb{R}$-Cartier, for arbitrary real number ε, we have

$$K_X + \Delta - \varepsilon\Theta + \varepsilon B = K_X + \Delta + \varepsilon(B - \Theta)$$

is $\mathbb{R}$-Cartier, thus the pair $(X, \Delta - \varepsilon\Theta + \varepsilon B)$ is well-defined.

Take $f : Y \longrightarrow X$ be any log resolution of $(X, \Delta + A)$, also of $(X, \Delta + B)$. As U is smooth and $\Delta|_U + B|_U$ has snc support, we can choose this log resolution to be an isomorphism over U; thus every f-exceptional divisors have center contained in Z, i.e. $\operatorname{center}_X E \subseteq Z$. But (X, Δ) is a dlt pair, we have $a(E, X, \Delta) > -1$. And

$$a(E, X, \Delta - \varepsilon\Theta + \varepsilon B) = a(E, X, \Delta) - \varepsilon \cdot \operatorname{mult}_E f^*(B - \Theta).$$

Since this log resolution is obtained by finite step of blow-ups, there are only finitely many f-exceptional divisors on Y. Hence we can arrange a sufficiently small $0 < \varepsilon_0 \ll 1$, such that for all $0 < \varepsilon < \varepsilon_0$ we have

$$a(E, X, \Delta - \varepsilon\Theta + \varepsilon B) > -1.$$

Thus, $(X, \Delta - \varepsilon\Theta + \varepsilon B)$ is dlt for real numbers $0 < \varepsilon \ll 1$. Here we use the Szabó's result which mentioned in Definition 2.35.

For the last statement, if $\operatorname{Supp}\Theta = \operatorname{Supp}\Delta$, then we write

$$\Delta = \sum a_i D_i \quad \text{and} \quad \Theta = \sum b_i D_i$$

and set

$$\varepsilon_0 = \min\left\{\min_i\left\{\frac{a_i}{b_i}\right\}, m\right\}$$

to ensure that for all real numbers $0 < \varepsilon < \varepsilon_0$, we have $\Delta - \varepsilon\Theta$ is an effective $\mathbb{R}$-divisor. Moreover, by our choice of A, the divisor B is also an effective divisor. And

$$\lfloor \Delta - \varepsilon\Theta + \varepsilon B \rfloor = 0$$

by each coefficients $a_i - \varepsilon b_i < 1$, so by Proposition 2.42, the pair $(X, \Delta - \varepsilon\Theta + \varepsilon B)$ is klt. □

Now we turn to prove Proposition 2.43:

Proof. Our strategy is use Lemma 2.44 to approximate the proposition by rational numbers.

We write $\Delta_1 = \sum r_i\Gamma_i$ for real numbers $r_i > 0$. For each i, can take a rational number q_i such that $0 < q_i < r_i$, and set $\Theta = \sum q_i\Gamma_i$ which is an effective $\mathbb{Q}$-divisor and $\operatorname{Supp}\Theta = \operatorname{Supp}\Delta_1$. Furthermore, since $\Delta - \Delta_1 \geq 0$, we have

$$\Delta - \Theta = (\Delta - \Delta_1) + (\Delta_1 - \Theta) \geq 0.$$

So we can apply Lemma 2.44 for $\mathbb{Q}$-divisor Θ, to get a rational number $c > 0$ and an effective $\mathbb{Q}$-divisor B such that

$$B \sim_{\mathbb{Q}} \Theta + cH, \tag{5.8}$$

and $(X, \Delta - \varepsilon\Theta + \varepsilon B)$ is dlt for all $0 < \varepsilon \ll 1$.

Now, we set $D = B + (\Delta_1 - \Theta)$. Since B and $\Delta_1 - \Theta$ are effective divisors, so is D. By formula in (5.8), we have

$$D - (\Delta_1 + cH) = B + (\Delta_1 - \Theta) - (\Delta_1 + cH) = B - \Theta - cH$$

is a $\mathbb{Q}$-linear combination of principal divisors, also can regard as an $\mathbb{R}$-linear combination. Hence, we have

$$D \sim_{\mathbb{R}} \Delta_1 + cH.$$

Finally, for arbitrary real number ε, we have

$$\begin{aligned}\Delta - \varepsilon\Delta_1 + \varepsilon D &= \Delta - \varepsilon\Delta_1 + \varepsilon(B + \Delta_1 - \Theta)\\ &= \Delta - \varepsilon\Theta + \varepsilon B,\end{aligned}$$

so $(X, \Delta - \varepsilon\Delta_1 + \varepsilon D)$ is dlt for every real numbers $0 < \varepsilon \ll 1$. Similarly for the second statement. □

We need tow extra result which will be used in Chapter 6.

Lemma 5.2.45. *Fix an n-dimensional pair (X, Δ) of a normal algebraic variety and an effective $\mathbb{R}$-divisor such that $K_X + \Delta$ is $\mathbb{R}$-Cartier, and P be a closed point on X. Take effective Cartier divisor $D_1, \dots, D_n$ passing through P such that P is an irreducible component of $\bigcap_i D_i$. Then there exists a log resolution $f : Y \longrightarrow X$ of the pair $(X, \Delta + \sum D_i)$ such that if we write the discrepancy formula*

$$K_Y + C = f^*(K_X + \Delta + \sum_{i=1}^{n} D_i),$$

then there exists an irreducible component C_1 of C with coefficient at least 1 and $f(C_1) = \{P\}$.

Proof. The assertion is local, so we may possible shrink X and assume X is affine. Then, we can write $D_i = \operatorname{div}(h_i)$, where h_i are regular functions on X. Define the morphism

$$\begin{aligned} h : X &\longrightarrow Z = \mathbb{A}^n \\ x &\longmapsto (h_1(x), \dots, h_n(x)). \end{aligned}$$

And we have $h(P) = (h_1(P), \dots, h_n(P)) = 0 \in \mathbb{A}^n$, so $h^{-1}(0) = V(h_1, \dots, h_n) = \bigcap_i D_i$. By assumption, P is the irreducible component of $\bigcap_i D_i$, so that the point P is isolated in $h^{-1}(0)$; by [SP25, Tag 01TH], we see h is fibre-finite in an open neighborhood of P. Take $E_1, \dots, E_n$ to be coordinate hyperplanes of Z, and $h^* E_i = D_i$ by construction. Take $g : Z' \longrightarrow Z$ be the blowing-up at the origin and denote F by its exceptional divisor, we get

$$g^*(K_Z + \sum E_i) = K_{Z'} + F + \sum g_*^{-1} E_i$$

by the blow-up formula in [Har77, II, ex8.5]. We denote the coordinates on Z by $z_1, \dots, z_n$, and take

$$\begin{aligned} \omega_Z = dz_1 \wedge \cdots \wedge dz_n \in \Omega_Z^n, \quad K_Z = \operatorname{div}_Z(\omega_Z) = 0, \\ h^* \omega_Z = dh_1 \wedge \cdots \wedge dh_n \text{[1]} = \det \left(\frac{\partial h_i}{\partial x_j} \right) dx_1 \wedge \cdots \wedge dx_n, \end{aligned}$$

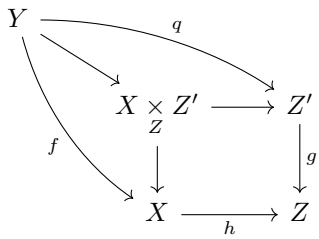
where $x_1, \dots, x_n$ is the regular system of parameters of X_{reg}, and $h^* \omega_Z$ is a n-form on X; therefore we can denote $K_X = \operatorname{div}_X(h^* \omega_Z)$, up to extend the Cartier divisor from X_{reg} to X uniquely. Note that since h_i are regular functions, we have $K_X \geq 0$. Thus,

$$h^*(K_Z + \sum E_i) = h^*(\sum E_i) = \sum D_i \leq K_X + \sum D_i \leq K_X + \Delta + \sum D_i,$$

[1] See Remark 2.46.

where the last inequality by Δ is an effective $\mathbb{R}$-divisor.

By take a log resolution $f : Y \longrightarrow X$ of the pair $(X, \Delta + \sum D_i)$ which factor through $X \underset{Z}{\times} Z'$, we get a diagram



such that $h \circ f = g \circ q$. We may choose the divisor C_1 as an irreducible component of $g^{-1}(F)$.

Now, we denote $c = \operatorname{mult}_{C_1} C$ and $e = \operatorname{mult}_{C_1} g^*F$. Combining above results,

$$\begin{aligned} K_Y + C &= f^*(K_X + \Delta + \sum D_i) \\ &\geq f^*h^*(K_Z + \sum E_i) \\ &= q^*g^*(K_Z + \sum E_i) \\ &= q^*(K_{Z'} + F + \sum g_*^{-1}E_i) \\ &\geq g^*K_{Z'} = g^*F, \end{aligned}$$

by $\sum g_*^{-1}E_i$ is effective. That is,

$$K_Y - g^*K_{Z'} + C \geq g^*F. \tag{5.9}$$

At the general point of F we take the local equation $(t = 0)$, and the general point of C_1 with the local equation $(s = 0)$. By the definition of multiplicities, we have

$$q^*t = us^e, \quad u \text{ is a unit.}$$

Therefore,

$$q^*(dt) = {}^{2}d(q^*t) = d(us^e) = s^e du + eus^{e-1}ds.$$

More precisely, we write the corresponding local section of $K_{Z'}$ to be

$$t^m v dt \wedge dz_2' \wedge \cdots \wedge dz_n',$$

[2]This is trivial in differential geometry, but in algebraic geometry this is ensured by the canonical homomorphism $q^*\Omega_{Z'} \to \Omega_Y$. See [SP25, Tag 01UV].

where v is a unit and $m = \operatorname{mult}_F K_{Z'}$. Note that $q^*(t^m v) = s^{em} \cdot u^m q^* v$ contributes the divisor $q^* K_{Z'}$, and the term

$$q^*\big(t^m v dt \wedge (n-1 \text{ form})\big) = s^{em} \cdot u^m q^* v \cdot (eus^{e-1} ds + s^e du) \wedge (n-1 \text{ form})$$

contributes the divisor K_Y. So, we have

$$\operatorname{mult}_{C_1} K_Y = em + e - 1, \quad \operatorname{mult}_{C_1} q^* K_{Z'} = em.$$

Substitute this result into (5.9), can obtain that

$$em + e - 1 - em + \operatorname{mult}_{C_1} C \geq e$$

which gives that $\operatorname{mult}_{C_1} C \geq 1$. This completes the proof. □

Remark 5.2.46. Where $dh_1 \wedge \cdots \wedge dh_n \not\equiv 0$ is due to the following argument. Consider the dominant morphism $h'X \to \overline{h(X)}$, and denote $e = \dim X - \dim \overline{h(X)}$. By [Har77, II, ex3.22(e)] we know the generic point $\eta \in \overline{h(X)}$ lies in C_e, i.e.

$$\dim X_\eta = \dim h^{-1}(\eta) = \dim X - \dim \overline{h(X)}.$$

But h is fibre-finite in an open neighborhood of P, and clearly η is in it. Thus, we see $\dim \overline{h(X)} = \dim X = \dim Z$, and obtain that $\overline{h(X)} = Z$, i.e. h is dominant.

The dominance of the morphism h implies that $h_1, \ldots, h_n$ is algebraically independent; the field extension

$$\begin{aligned} \mathbb{C}(Z) = \mathbb{C}(z_1, \ldots, z_n) &\hookrightarrow \mathbb{C}(X) \\ z_i &\mapsto h_i \end{aligned}$$

makes $h_1, \ldots, h_n$ form a transcendental basis of a sub-field of $\mathbb{C}(X)$. Moreover, since we are over characteristic zero, the extension $\mathbb{C}(X)/\mathbb{C}(Z)$ is finitely separable. So by [Eis95, 16.15], we have an isomorphism

$$\begin{aligned} \mathbb{C}(X) \underset{\mathbb{C}(Z)}{\otimes} \Omega_{\mathbb{C}(Z)/\mathbb{C}} &\xrightarrow{\sim} \Omega_{\mathbb{C}(X)/\mathbb{C}} \\ 1 \otimes dz_i &\longmapsto dh_i \end{aligned}$$

where the left-hand-side $\{1 \otimes dz_i\}$ forms a basis, so $\{dh_i\}$ also form a basis such that $dh_1 \wedge \cdots \wedge dh_n \not\equiv 0$ in $\Omega_{\mathbb{C}(X)/\mathbb{C}} = \Omega_{X_\xi}$. Therefore, globally $dh_1 \wedge \cdots \wedge dh_n \not\equiv 0$ in Ω_X.

It has an application:

Corollary 5.2.47. *Let (X, Δ) be an n-dimensional klt pair with Δ is effective $\mathbb{R}$-divisor such that $K_X + \Delta$ is $\mathbb{R}$-Cartier, and let P be a closed point of X. Take general effective Cartier divisors $D_1, \ldots, D_n, E$ passing through the point P and a positive number $0 < \varepsilon < 1$. Then there exists a sufficiently small number $\delta > 0$ such that the pair $(X, \Delta + \sum(1-\delta)D_i + \varepsilon E)$ has klt singularities in a punctured neighborhood of P, but not lc at P.*

Proof. As $D_1, \ldots, D_n, E$ are general outside point P, take a log resolution $\bar{f} : Y \longrightarrow X$ of the pair (X, Δ), and we write the discrepancy formula

$$K_Y + \bar{C} = \bar{f}^*(K_X + \Delta).$$

Use Bertini's theorem, we may assume that $\bar{C} + \bar{f}^*(\sum D_i + E)$ has simple normal crossing support outside $\bar{f}^{-1}(P)$.

The coefficients of $D_1, \ldots, D_n, E$ in the pair $(X, \Delta + \sum(1-\delta)D_i + \varepsilon E)$ are strictly smaller than 1 for $\delta > 0$, hence the pair is klt in a punctured neighborhood of P by Corollary 5.2.20.[3]

On the other hand, take the log resolution f and a prime divisor C_1 as in the above lemma, then the coefficient of C_1 in $f^*(K_X + \Delta + \sum D_i)$ is at least 1. Moreover, since E passing through the point P, the coefficient of C_1 in f^*E is at least 1. So the coefficient of C_1 in $f^*(K_X + \Delta + \sum D_i + \varepsilon E)$ is strictly larger than 1. Hence $(X, \Delta + \sum(1-\delta)D_i + \varepsilon E)$ is not lc at P for sufficiently small $\delta > 0$. □

[3] Note that this punctured neighborhood is contained in X_{reg}.

Chapter 6

Cone Theorems

In Chapter 2 we proved the Cone Theorem for smooth projective varieties using the method of deformation theory, and we noted that the lower bound in Theorem 2.1.23 did not work for singular varieties. For the minimal model program, certain singularities are unavoidable and it is essential to have Cone Theorem for pair (X, Δ).

6.1 Introduction to the Proof of the Cone Theorem

This proof is built on a very different set of ideas, which applied even in the smooth case it gives result not accessible by the previous method; namely it proves that extremal rays can always be contracted. On the other hand, it gives little information about the curves that span an extremal ray. Also, this proof works only in characteristic 0. We always assume that X is a normal variety.

We prove four basic theorems finishing with the Cone Theorem.

Theorem 6.1.1. (Basepoint-free Theorem) *Let (X, Δ) be a proper klt pair with $K_X + \Delta$ is $\mathbb{Q}$-Cartier and Δ is effective. Let D be a nef Cartier divisor such that*

$$aD - K_X - \Delta$$

is nef and big for some $a > 0$. Then $|bD|$ has no basepoints for all $b \gg 0$.

Theorem 6.1.2. (Shokurov's Non-vanishing Theorem) *Let X be a proper variety, D is a nef Cartier divisor and G is a $\mathbb{Q}$-divisor. Suppose*

(1) $aD + G - K_X$ is $\mathbb{Q}$-Cartier, nef and big for some $a > 0$, and

(2) $(X, -G)$ is klt.

Then, for all $m \gg 0$, we have $H^0(X, mD + \lceil G \rceil) \neq 0$.

Theorem 6.1.3. (Kawamata's Rationality Theorem) *Let (X, Δ) be a proper klt pair with Δ effective such that the $\mathbb{Q}$-Cartier divisor $K_X + \Delta$ is not nef. Let $a(X) > 0$ be an integer such that $a(X) \cdot (K_X + \Delta)$ is Cartier. Let H be a nef and big Cartier divisor, and define the threshold*

$$r = r(H) := \max\{t \in \mathbb{R} \big| H + t(K_X + \Delta) \text{ is nef } \}.$$

Then r is a rational number of the from u/v where

$$0 < v \leq a(X) \cdot (\dim X + 1).$$

Remark 6.1.4. Notation as above. Then there is an extremal ray R such that $\big(R\cdot(K_X+\Delta)\big)<0$ and $\big(R\cdot(H+r(K_X+\Delta))\big)=0$.

Theorem 6.1.5. (Cone Theorem) *Let (X,Δ) be a projective klt pair with Δ is effective such that $K_X+\Delta$ is $\mathbb{Q}$-Cartier. Then:*

(1) There are (countably many) rational curves $C_j\subseteq X$ such that intersection occurs $0<\big(-(K_X+\Delta)\cdot C_j\big)\le 2\dim X$, and

$$\overline{\mathrm{NE}}(X)=\overline{\mathrm{NE}}(X)_{(K_X+\Delta)\ge 0}+\sum\mathbb{R}_{\ge 0}[C_j].$$

(2) For any $\varepsilon>0$ and ample $\mathbb{Q}$-Cartier divisor H,

$$\overline{\mathrm{NE}}(X)=\overline{\mathrm{NE}}(X)_{(K_X+\Delta+\varepsilon H)\ge 0}+\sum_{\text{finite}}\mathbb{R}_{\ge 0}[C_j].$$

(3) Let $F\subseteq\overline{\mathrm{NE}}(X)$ be a $(K_X+\Delta)$-negative extremal face. Then there is a unique morphism $\mathrm{cont}_F:X\longrightarrow Z$ to a projective variety such that $(\mathrm{cont}_F)_*\mathscr{O}_X=\mathscr{O}_Z$ and an irreducible curve $C\subseteq X$ is mapped to a point by cont_F if and only if $[C]\in F$. cont_F is called the contraction of F.

(4) Let F and $\mathrm{cont}_F:X\longrightarrow Z$ be as in (3). Let L be a line bundle on X such that $(L\cdot C)=0$ for every curve C with $[C]\in F$. Then there is a line bundle L_Z on Z such that $L\cong\mathrm{cont}_F^*L_Z$.

Note. Part (3) *is frequently called the Contraction Theorem.*

Remark 6.1.6. The logical order of proof of these theorems is the following:

$$\text{Non-vanishing}^{\mathrm{D}}\ \Rightarrow\ \text{Basepoint-free}^{\mathrm{A}}\ \Rightarrow\ \text{Rationality}^{\mathrm{C}}\ \Rightarrow\ \text{Cone}^{\mathrm{B}}.$$

However, for better understanding we prove first Basepoint freeness, because its proof shows the basic underlying ideas the best.

Our main idea for proving the Basepoint-free Theorem (as well as the Non-vanishing and Rationality Theorems) is the following.

Assume for simplicity X is smooth. M is an ample Cartier divisor, $F\subseteq X$ is a smooth irreducible divisor. We have an exact sequence

$$0\longrightarrow\mathscr{O}_X(K_X+M)\longrightarrow\mathscr{O}_X(K_X+M+F)\longrightarrow\mathscr{O}_F\big((K_X+M+F)|_F\big)\longrightarrow 0.$$

By adjunction formula, we have

$$(K_X+F)|_F=K_F,$$

so the last term is

$$\mathscr{O}_F(K_F + M|_F).$$

Since by our choice X is smooth, Kodaira vanishing gives that

$$H^1\big(X, \mathscr{O}_X(K_X + M)\big) = 0,$$

that is we have a surjection

$$H^0\big(X, \mathscr{O}_X(K_X + M + F)\big) \twoheadrightarrow H^0\big(F, \mathscr{O}_F(K_F + M|_F)\big). \tag{6.1}$$

Moreover, if we have some vanishing theorem can applies to $K_X + M$, we can no need that M be ample. The surjectivity at (6.1) means that sections on F can be lifted on X, thus if we can write $M = bD - K_X - F$ as above such that

$$K_F + M|_F = K_F + bD|_F - (K_X + F)|_F = bD|_F,$$

the surjective map (6.1) will gives a section in $\mathscr{O}_X(bD)$, provided by right hand side is non-trivial, in other words is not identically vanishing on F. It is implies that F is not the fixed part of the series $|bD|$, since for non-zero sections on F it can be lifted as a non-zero global section which is not identically zero on F.

So our strategy is pick an smooth effective divisor inside basepoint part or fixed part, and construct a suitable M such that vanishing theorem can applies on $K_X + M$ (for example $M + \Delta$ is nef and big), then one can lift the sections on F to X for $K_X + M + F = bD$ to show that F is not the fixed part, and we repeat this process.

But our issue is that $bD - K_X$ might not have the ideal form, so we need some birational modifications, even for X might not be non-singular.

Taking a proper birational morphism $f : Y \longrightarrow X$ such that Y is non-singular. Set A is an effective exceptional divisor. Hence we assume that X is normal, such that the image of A has codimension at least two in X, so by algebraic Hartogs

$$f_*\mathscr{O}_Y(A) \cong \mathscr{O}_X.$$

Then we have

$$H^0\big(Y, \mathscr{O}_Y(bf^*D + A)\big) \cong H^0\big(X, \mathscr{O}_X(bD)\big)$$

thanks to the projection formula. The above sequence in (6.1) can be replaced by

$$0 \longrightarrow \mathscr{O}_Y(K_Y + M) \longrightarrow \mathscr{O}_Y(K_Y + M + F) \longrightarrow \mathscr{O}_F(K_F + M|_F) \longrightarrow 0$$

where $M = bf^*D + A - K_Y - F$, here we use the same notation F to indicate an effective smooth prime divisor. We hope our vanishing theorems can apply on

$K_Y + M = bf^*D + A - F$, for example the Kawamata-Viehweg vanishing applied on

$$M - \Delta = bf^*D - K_Y + A - F - \Delta = (\text{nef and big}), \tag{6.2}$$

where Δ has coefficients in $[0, 1)$ with snc support, then we have

$$H^1\big(Y, \mathscr{O}_Y(K_Y + \lceil M - \Delta \rceil)\big) = H^1\big(Y, \mathscr{O}_Y(K_Y + M)\big) = 0$$

by applying Theorem 4.4.14 on the surjective proper morphism $f : Y \longrightarrow \operatorname{Spec} \mathbb{C}$.

But, how can we get such b and F? We should use some linear combination of divisors given by the problem. That is, we have at out disposal $aD - K_X$ and D. Suppose that $|mD|$ is not basepoint-free for some large m, then we can take a resolution $f : Y \longrightarrow X$ to assume that $f^*|mD| = |L| + \sum r_j F_j$, where $|L|$ is Basepoint-free and $\sum r_j F_j$ is the fixed part. Let $K_Y = f^*K_X + \sum a_j F_j$, where in $\sum a_j F_j$ we just let F_j run through the exceptional divisors only, while $\sum r_j F_j$ may involve non-exceptional divisors as well. We have to pay attention to this.

Thus we can write

$$\begin{aligned} bf^*D - K_Y = (b - cm - a)f^*D + c(mf^*D - \sum r_j F_j) \\ + f^*(aD - K_X) - \sum (a_j - cr_j)F_j. \end{aligned}$$

The first term $(b - cm - a)f^*D$ is nef provided by $b \geq cm + a$, because our assumption D is nef. Second term $c(mf^*D - \sum r_j F_j)$ also is nef, since it is linearly equivalent some divisor of free linear series. Third term $f^*(aD - K_X)$ is nef and big, see Proposition 3.2.36 and recall that even for X may be singular we still assume that its K_X is $\mathbb{Q}$-Cartier. Moreover, these three summands still nef and big, because its top self-intersection is strictly positive.

Therefore, we need to choose c in such a way that

$$\sum (a_j - cr_j)F_j = A - \Delta - F,$$

to fit in the formula (6.2), where A is an effective f-exceptional divisor and F is a smooth prime divisor.

By our choice of the boundary divisor Δ has coefficients lie in $[0, 1)$, we need choose c such that $\min\limits_j \{a_j - cr_j\} = -1$. Then F is the union of these F_j such that $a_j - cr_j = -1$. But unfortunately, these may be several such components making F no longer a smooth prime divisor. In order to eliminate this possibility, we perturb everything by a very small linear combination $\sum p_j F_j$. This is a small but useful technical point, which we call it the *tie-breaking method*.

Remark 6.1.7. In above discussion we construct a nef and big $\mathbb{Q}$-Cartier divisor

$$bf^*D + \sum(a_j - cr_j)F_j - K_Y = bf^*D + (A - \Delta - F) - K_Y. \tag{6.3}$$

Before we use the Kawamata-Viehweg vanishing theorem on $K_Y + M$, we need that the vector space $H^0(F, K_F + M|_F)$ is non-zero ensures us can lift sections to Y. For this purpose, we consider the nef and big divisor in (6.3) restrict to F:

$$b(f^*D)|_F + (A - \Delta)|_F - K_F$$

and use the non-vanishing theorem to make

$$H^0\big(F, b(f^*D)|_F + \lceil (A - \Delta)|_F \rceil\big) \overset{?}{=} H^0\big(F, K_F + M|_F\big) \neq 0.$$

Our problem is, for

$$\begin{aligned} K_F + M|_F &= b(f^*D)|_F + A|_F \\ &= b(f^*D)|_F + \lceil (A - \Delta) \rceil|_F, \end{aligned}$$

we may not have $\lceil (A - \Delta) \rceil|_F = \lceil (A - \Delta)|_F \rceil$.

Therefore for this smooth hypersurface F of a smooth Y, F should not be among the components of Δ. In this case, we write

$$A - \Delta = A - \sum a_i \Delta_i \text{ where } a_i \in [0, 1).$$

Then

$$\lceil (A - \Delta)|_F \rceil = \lceil A|_F - \sum a_i \Delta|_F \rceil = A|_F + \sum \lceil -a_i \rceil \Delta|_F = A|_F.$$

6.1A Basepoint-free Theorem

The following proof is taken almost verbatim from [Rei83].

Proof. **Step 1:** In this step, we establish that $|mD| \neq \emptyset$ for every $m \gg 0$. We choose a (projective) log resolution $f : Y \longrightarrow X$ such that

(1) $f^*|mD| = |L| + \sum r_j F_j$ for some fixed m used in *Step 2*, where $|L|$ is free;

(2) $\sum r_j F_j + f_*^{-1} \operatorname{Supp} \Delta + \operatorname{Ex}(f)$ has snc support.

And we also write the strict transform of components of Δ be F_j. Furthermore, this resolution gives that

(3) $K_Y = f^*(K_X + \Delta) + \sum a_j F_j$ with all $a_j > -1$,

(4) $f^*\big(aD-(K_X+\Delta)\big)-\sum p_jF_j$ is ample for some $a>0$ and for suitable $0<p_j\ll 1$.

Where the construction of (4) see remark below, and these F_j need not all be f-exceptional. Note that these F_j need not be f-exceptional. More concretely, in (3) the coefficients a_j correspond to

$$a_j=\begin{cases} a(F_j,X,\Delta), & F_j \text{ is } f\text{-exceptional;}\\ -d_i, & F_j=f_*^{-1}\Delta_i \text{ where } \Delta=\sum d_i\Delta_i\geq 0;\\ 0, & F_j \text{ is other non-exceptional component.}\end{cases}$$

On Y we write

$$\begin{aligned} &f^*\big(aD-(K_X+\Delta)\big)-\sum p_jF_j\\ =\ &af^*D+\sum(a_j-p_j)F_j-\big(f^*(K_X+\Delta)+\sum a_jF_j\big)\\ =\ &af^*D+G-K_Y, \end{aligned}$$

where $G=\sum(a_j-p_j)F_j$. By assumption, $\lceil G\rceil$ is an effective f-exceptional divisor, since $a_j>0$ only when F_j is f-exceptional by Δ is effective. Moreover, we have af^*D+G-K_Y is ample and

$$\begin{aligned} H^0\big(Y,\mathscr{O}_Y(mf^*D+\lceil G\rceil)\big)&=H^0\big(X,f_*\mathscr{O}_Y(mf^*D+\lceil G\rceil)\big)\\ &\cong H^0\big(X,\mathscr{O}_X(mD)\big), \end{aligned}$$

thanks to the projection formula and algebraic Hartogs.

We can now apply Non-vanishing theorem to get

$$H^0(X,mD)\neq 0 \text{ for all } m\gg 0.$$

Note that this result is independent of the choice of some fixed m in condition (1), and here we use Corollary 5.2.20 to ensures $(Y,-G)$ is a klt pair.

Step 2: For $s\in\mathbb{Z}_{>0}$, let $B(s)=\mathrm{Bs}\,|sD|$ denote the set-theoretic base locus of $|sD|$, i.e. it is defined as $\bigcap_{D'\in|sD|}\mathrm{Supp}\,D'$. Clearly there is a set-theoretic inclusion $B(s^u)\subseteq B(s^v)$ for any positive integers $u>v$, thanks to the reverse inclusion $\mathfrak{b}(|s^vD|)^{s^{u-v}}\subseteq\mathfrak{b}(|s^uD|)$ on the base ideals, see Example 1.1.15. Since X is a noetherian topological space, says that descending chain condition holds on $\{B(s^i)\big|i\in\mathbb{Z}_{>0}\}$, so there is a minimal element $B(s^i)$ which we call the limit B_s. Note that this is not the *stable base locus* we have defined in 3.1.18, since in general we only have $\mathbf{B}(D)\subsetneq B_s$.

So either B_s is non-empty for some s, or B_s is empty for **arbitrary** s. In the latter case, we denote two relative prime integers s and s' such that $B_s=B_{s'}=\emptyset$.

This tells us we can take u and v such that $B(s^u)$ and $B(s'^v)$ are empty, and use the fact that every sufficiently large integer is a linear combination of s^u and s'^v with non-negative coefficients, to conclude that

$$\begin{aligned}\operatorname{Bs}|mD| &= \operatorname{Bs}\left(|(as^u + bs'^v)D|\right)\\ &\subseteq \operatorname{Bs}|as^uD| \cup \operatorname{Bs}|bs'^vD|\\ &\subseteq B(s^u) \cup B(s'^v) = \emptyset,\end{aligned}$$

where the inclusion comes from the map

$$H^0(X, as^uD) \otimes H^0(X, bs'^vD) \longrightarrow H^0(X, mD).$$

This says that $|mD|$ is Basepoint-free for all $m \gg 0$. So we must show that the assumption some B_s is non-empty leads to a contradiction.

Now, for some B_s is non-empty, we let $m = s^u$ such that $B_s = B(m)$, By our choice of log resolution in *Step 1*, we resolve the series $|mD|$ into

$$f^*|mD| = |L|(\text{moving part}) + \sum r_jF_j(\text{fixed part})$$

with $|L|$ is Basepoint-free. Therefore $\bigcup_{r_j>0} f(F_j)$ is the base locus of $|mD|$, that is each F_j with $r_j > 0$ we have $F_j \subseteq f^{-1}\operatorname{Bs}|mD|$. Note that by Zariski's Main Theorem, f is a fibre space that is $f_*\mathscr{O}_Y = \mathscr{O}_X$. So by the projection formula

$$H^0(Y, mf^*D) = H^0(X, mD) \tag{6.4}$$

that is every section at the left is of the form f^*s for some $s \in H^0(X, mD)$. For any $x = f(y) \notin \operatorname{Bs}|mD|$, there is a section $s \in H^0(X, mD)$ such that $s(x) \neq 0$, so the pullback f^*s is non-vanishing along $f^{-1}(x)$, especially at y which gives that $y \notin \operatorname{Bs}|mf^*D|$. Conversely, suppose that $x \in \operatorname{Bs}|mD|$, then every section $s \in H^0(X, mD)$ satisfies $s(x) = 0$. Hence for all $y \in f^{-1}(x)$ we have $(f^*s)(y) = 0$, condition (6.4) gives that $f^{-1}(x) \subseteq \operatorname{Bs}(|mf^*D|)$. Thus, we see $f^{-1}\operatorname{Bs}|mD| = \operatorname{Bs}|mf^*D|$.

We are going to choose $b = s^v$ for $v \in \mathbb{Z}_{>0}$, then for every $b \gg 0$ we have $\operatorname{Bs}|bf^*D| = \operatorname{Bs}|mf^*D|$. Then, our next goal is to obtain the desired contradiction by finding some F_j with $r_j > 0$ such that F_j is not contained in $\operatorname{Bs}|bf^*D|$.

Step 3: For an arbitrary integer $b > 0$ and a rational number $c > 0$ such that $b > cm + a$, here $m = s^u$ same as in *Step 2*. We can define divisors:

$$\begin{aligned}N(b, c) &:= bf^*D - K_Y + \sum(-cr_j + a_j - p_j)F_j\\ &= (b - cm - a)f^*D \quad (\text{nef})\\ &\quad + c(mf^*D - \sum r_jF_j) \quad (\text{basepoint-free})\\ &\quad + f^*(aD - K_X - \Delta) - \sum p_jF_j \quad (\text{ample}).\end{aligned}$$

Thus, $N(b,c)$ is ample provided by $b \geq cm + a$. If that is the case then, by the Kawamata-Viehweg vanishing theorem, $H^1(Y, K_Y + \lceil N(b,c) \rceil) = 0$, while

$$\lceil N(b,c) \rceil = bf^*D + \sum \lceil -cr_j + a_j - p_j \rceil F_j - K_Y$$

the fractional part $\sum F_j$ has snc support by condition (2).

Step 4: c and the p_j can be chosen so that

$$\sum (-cr_j + a_j - p_j)F_j = A - F$$

for some $F = F_j$, where $\lceil A \rceil$ is effective and does not have F as a component. This ensures that $\lceil A \rceil|_F = \lceil A|_F \rceil$, since $\operatorname{Supp} F \cap \operatorname{Supp} A$ has snc support.

In fact, we can choose $c > 0$ such that

$$\min_j \{ -cr_j + a_j - p_j \} = -1.$$

If this last condition does not single out a unique j, then we can wiggle the p_j slightly to achieve the desired uniqueness. This j satisfies $r_j > 0$ and $\lceil N(b,c) \rceil + K_Y = bf^*D + \lceil A \rceil - F$.

Now, *Step 3* implies that

$$H^0(Y, bf^*D + \lceil A \rceil) \longrightarrow H^0\big(F, (bf^*D + \lceil A \rceil)|_F\big)$$

is surjective for all $b \geq cm + a$.

Note that if some others F_j appears in $\lceil A \rceil$, then we must have $a_j > 0$ which corresponds the f-exceptional divisor. Thus the effective divisor $\lceil A \rceil$ is f-exceptional.

Step 5: Notice that

$$\begin{aligned} N(b,c)|_F &= (bf^*D + A - F - K_Y)|_F \\ &= (bf^*D + A)|_F - K_F \end{aligned}$$

by adjunction formula, and it is ample because the ampleness is preserved under finite pullbacks. And $f^*D|_F$ is also nef for the same reason. So we can apply the Non-vanishing Theorem on F to get

$$H^0\big(F, (bf^*D + \lceil A \rceil)|_F\big) \neq 0.$$

Here we use Corollary 5.2.20 to ensures that $(F, -A|_F)$ is a klt pair. So $H^0(bf^*D + \lceil A \rceil)$ has a section not vanishing on F. Since $\lceil A \rceil$ is f-exceptional and effective,

$$\begin{aligned} H^0(Y, bf^*D + \lceil A \rceil) &= H^0\big(X, f_*\mathscr{O}_Y(bf^*D + \lceil A \rceil)\big) \\ &= H^0(X, bD) \end{aligned}$$

thanks to the projection formula and algebraic Hartogs. Therefore, $f(F)$ is not contained in $\mathrm{Bs}\,|bD|$ for all $b \gg 0$, especially of the form $b = s^v$ for $v \in \mathbb{Z}_{>0}$. We arrive our desired contradiction which stated in *Step 2*, this completes the proof of the Basepoint-free Theorem. □

Remark 6.1.8. (Construction of the ample perturbation) The ampleness of $f^*\big(aD-(K_X+\Delta)\big)-\sum p_jF_j$ for some $a>0$ and suitable $0<p_j\ll 1$, is comes from the Kodaira's lemma.

Let $H := aD-(K_X+\Delta)$ be that nef and big divisor. Since X is a normal proper variety, by Chow's lemma there is a proper birational morphism $\mu : X' \longrightarrow X$ such that X' is a projective variety. For $\mathbb{Q}$-Cartier divisor μ^*H is still nef and big, so by Kodaira's lemma we can write

$$\mu^*H \sim_{\mathbb{Q}} A' + N'$$

where A' is an ample $\mathbb{Q}$-Cartier divisor on X', and N' effective. Use resolution of singularities there is a projective birational morphism $g : Y \longrightarrow X'$ such that the composition $f = \mu \circ g : Y \longrightarrow X$ is the log resolution of the proper klt pair (X,Δ), such that $\mathrm{Ex}(f)+f_*^{-1}\Delta$ has simple normal crossing support.

Moreover, Y is a smooth projective variety and we have the form

$$f^*H \sim_{\mathbb{Q}} g^*A' + g^*N'$$

where f^*H is nef and big but g^*A' may not be ample anymore. Since g is a sequence of blow-ups, we can find an effective g-exceptional divisor F such that $-F$ is g-ample. Then $\tilde{A} := g^*A'-\varepsilon F$ is ample for $0<\varepsilon\ll 1$ by Proposition 1.5.8, and $\tilde{N} := g^*N'+\varepsilon F$ is an effective divisor. We can set

$$\tilde{A}_k := \frac{1}{k}\big(\tilde{A}+(k-1)f^*H\big)$$

which is an ample divisor for every $k \geq 1$ by Corollary 1.4.7, then we can write

$$\frac{1}{k}\tilde{N}+\tilde{A}_k = \frac{1}{k}\big(g^*N'+\varepsilon F+g^*A'-\varepsilon F+(k-1)f^*H\big) \sim_{\mathbb{Q}} f^*H.$$

Thus, we can fix some k sufficiently large and by the open nature of ampleness we have

$$f^*H-\frac{1}{k}\tilde{N}-\sum_j p_jF_j \sim_{\mathbb{Q}} \underbrace{\tilde{A}_k}_{\text{ample}} - \underbrace{\left(\sum_j p_jF_j\right)}_{\text{sufficiently small}}$$

is ample, where these p_j may dependent on k. See Proposition 1.3.7. Then we can denote $1/k\tilde{N}$ as one of the summands of $\sum p_jF_j$.

One of the most important applications of the Basepoint-free Theorem is to the finite generation of canonical rings:

Theorem 6.1.9. *Let (X, Δ) be a proper klt pair with $K_X + \Delta$ is $\mathbb{Q}$-Cartier and Δ is effective. Moreover assume that $K_X + \Delta$ is nef and big. Then the canonical ring, defined as*

$$\bigoplus_{m=0}^{\infty} H^0\big(X, \mathscr{O}_X(mK_X + \lfloor m\Delta \rfloor)\big),$$

is finitely generated over $\mathbb{C}$.

Remark 6.1.10. The canonical ring is indeed a graded ring, since we have

$$\lfloor m_1\Delta \rfloor + \lfloor m_2\Delta \rfloor \leq \lfloor (m_1 + m_2)\Delta \rfloor,$$

then the graded structure comes from

$$\begin{aligned} \operatorname{div}(f_1) + m_1K_X + \lfloor m_1\Delta \rfloor &\geq 0 \\ \operatorname{div}(f_2) + m_2K_X + \lfloor m_2\Delta \rfloor &\geq 0, \end{aligned}$$

then we have

$$\begin{aligned} &\operatorname{div}(f_1 \cdot f_2) + (m_1 + m_2)K_X + \lfloor (m_1 + m_2)\Delta \rfloor \\ \geq\ &\operatorname{div}(f_1) + \operatorname{div}(f_2) + (m_1 + m_2)K_X + \lfloor m_1\Delta \rfloor + \lfloor m_2\Delta \rfloor \\ \geq\ &0. \end{aligned}$$

The inequality is reversed for $\lceil m\Delta \rceil$, and so we would not get a ring using $\lceil m\Delta \rceil$.

Proof. By Basepoint-free theorem we choose D to be $K_X + \Delta$ can conclude that, there is $r > 0$ such that $rK_X + r\Delta$ is an integral divisor and $\mathscr{O}_X(rK_X + r\Delta)$ is generated by global sections. These sections define a morphism $f : X \longrightarrow Z$ to a projective variety and there is a natural very ample line bundle $\mathscr{O}(1)$ on Z such that $f^*\mathscr{O}(1) = \mathscr{O}_X(rK_X + r\Delta)$. We denote $G_m = f_*\mathscr{O}_X(mK_X + \lfloor m\Delta \rfloor)$, then

$$\bigoplus_{m=0}^{\infty} H^0\big(X, \mathscr{O}_X(mK_X + \lfloor m\Delta \rfloor)\big) = \bigoplus_{m=0}^{\infty} H^0(Z, G_m).$$

The G_m are coherent sheaves and

$$\begin{aligned} G_{m+r} &= f_*\mathscr{O}_X\big((m + r)K_X + \lfloor m\Delta \rfloor + r\Delta\big) \\ &= f_*\big(\mathscr{O}_X(mK_X + \lfloor m\Delta \rfloor) \otimes f^*\mathscr{O}(1)\big) \\ &\cong G_m \otimes \mathscr{O}(1) \end{aligned}$$

thanks to the projection formula. Since $\mathscr{O}(1)$ is ample, its section ring $R = \bigoplus_{m=0}^{\infty} H^0\big(Z, \mathscr{O}(m)\big)$ is a finite type algebra over $\mathbb{C}$. Note that $Z = \operatorname{Proj} R$. Moreover, for every $0 \leq j < r$ we have

$$G_{j+rm} \cong G_j \otimes \mathscr{O}(m),$$

so the section ring

$$\bigoplus_{m=0}^{\infty} H^0(Z, G_{j+rm}) \cong \bigoplus_{m=0}^{\infty} H^0\big(Z, G_j(m)\big)$$

is a finitely generated R-module. This is because of that there is a finitely generated R-module M such that all its large enough degrees part have an isomorphism

$$M_{\geq d_0} \cong \bigoplus_{d \geq d_0}^{\infty} H^0\big(Z, G_j(d)\big)$$

see [Har77, II, ex5.9(b)], and the finitely man lower degree terms are indeed finitely generated R-module by coherence.

Thus,

$$\bigoplus_{m=0}^{\infty} H^0(Z, G_m) = \bigoplus_{j=0}^{r-1} \left(\bigoplus_{m=0}^{\infty} H^0(Z, G_{j+rm}) \right)$$

is finite type over $\mathbb{C}$. □

The above result is a special case of the following:

Conjecture 6.1.11. (Abundance conjecture) Let (X, Δ) be a proper log canonical pair with $K_X + \Delta$ is $\mathbb{Q}$-Cartier and Δ is effective. Then:

(1) $\bigoplus_{m=0}^{\infty} H^0\big(X, \mathscr{O}_X(mK_X + \lfloor m\Delta \rfloor)\big)$ is a finitely generated ring.

(2) If $K_X + \Delta$ is nef then $|m(K_X + \Delta)|$ is basepoint free for some $m > 0$.

Remark 6.1.12. Abundance conjecture on curve is trivial, since normal curve is non-singular. For surface it is proved by Kawamata in [Kaw92], and the modern reference can see [Fuj12]. The conjecture is also known to be true in dimension 3, see [KMM94] and [KMM04] of its correction.

Until today, abundance conjecture still unsolved in dimension larger than 4.

6.1B The Cone Theorem

We need some preparations to give the proof. First we prove the vanishing theorem on the pair has klt singularities:

Theorem 6.1.13. ([KMM87, 1.2.5]) *Let (X, Δ) be a klt pair with X is a normal variety and $K_X + \Delta$ is $\mathbb{R}$-Cartier. Let $\pi : X \longrightarrow S$ be a proper surjective morphism onto a variety S, and let D be an integral divisor which is $\mathbb{Q}$-Cartier. Assume that $D - (K_X + \Delta)$ is a π-ample $\mathbb{R}$-Cartier divisor. Then*

$$R^i\pi_*\mathscr{O}_X(D) = 0 \text{ for } i > 0.$$

Note that here $\mathscr{O}_X(D)$ is a reflexive sheaf, not a line bundle. Moreover, here π actually is a projective morphism, see Remark 1.5.2.

Proof. We take a resolution of singularities $f : Y \longrightarrow X$ such that

(1) $f_*^{-1}\Delta + \mathrm{Ex}(f)$ has snc support;

(2) $f^*\big(D - (K_X + \Delta)\big) - \sum p_j F_j$ is $\pi \circ f$-ample for suitable $0 < p_j \ll 1$, where F_j are f-exceptional divisors;

(3) Write the discrepancy formula

$$K_Y + f_*^{-1}\Delta = f^*(K_X + \Delta) + E,$$

that is $\lceil E \rceil \geq 0$ and $f_*\lceil E \rceil = 0$ by our assumption (X, Δ) is klt.

Step 1: We claim that $f_*\mathscr{O}_Y(\lceil f^*D + E \rceil) = \mathscr{O}_X(D)$. Notice that we can not use the projection formula here, so we give a tedious analysis below, one can choose to skip. In fact, we choose a rational function

$$\varphi \in H^0\big(f^{-1}U, \mathscr{O}_Y(\lceil f^*D + E \rceil)\big) \subseteq k(X)$$

which means on $f^{-1}U$ we have

$$\mathrm{div}_Y(\varphi) + \lceil f^*D + E \rceil \geq 0. \tag{6.5}$$

For any prime divisor P on X, its strict transform $\tilde{P}$ has zero multiplicity on $\lceil E \rceil$, i.e. $\mathrm{mult}_{\tilde{P}} \lceil E \rceil = 0$, since $f_*\lceil E \rceil = 0$. Then

$$-1 < \mathrm{mult}_{\tilde{P}} E \leq 0.$$

On the other hand,

$$\begin{aligned} \mathrm{mult}_{\tilde{P}} f^*D &= \mathrm{mult}_{\tilde{P}} f_*^{-1}D + \mathrm{mult}_{\tilde{P}} E \\ &= \mathrm{mult}_P D. \end{aligned}$$

Hence

$$\begin{aligned}\operatorname{mult}_{\tilde{P}}\lceil f^*D+E\rceil &= \lceil \operatorname{mult}_{\tilde{P}} f^*D + \operatorname{mult}_{\tilde{P}} E\rceil \\ &= \operatorname{mult}_P D.\end{aligned}$$

We see

$$\begin{aligned}&\operatorname{ord}_Y(\varphi) + \operatorname{mult}_{\tilde{P}}\lceil f^*D+E\rceil \\ &= \operatorname{ord}_{P_Y}(\varphi) + \operatorname{mult}_P D \\ &\geq 0\end{aligned}$$

by formula (6.5). But since f is birational, it is an isomorphism over the generic point of P, so $\operatorname{ord}_{P_Y}(\varphi) = \operatorname{ord}_P(\varphi)$. Then

$$\operatorname{ord}_P(\varphi) + \operatorname{mult}_P D \geq 0$$

for arbitrary prime divisor P on X, that is, on U we have

$$\operatorname{div}_X(\varphi) + D \geq 0.$$

So $f_*\mathscr{O}_Y(\lceil f^*D+E\rceil) \subseteq \mathscr{O}_X(D)$.

Conversely, we choose $\varphi \in \mathscr{O}_X(D)(U)$ such that

$$\operatorname{div}_X(\varphi) + D \geq 0$$

on U. Since D is $\mathbb{Q}$-Cartier, we have

$$f^*\big(\operatorname{div}_X(\varphi) + D\big) = \operatorname{div}_Y(\varphi) + f^*D \geq 0.$$

Since $\lceil E\rceil \geq 0$ by (X,Δ) is a klt pair, each coefficients of E is greater than -1, so for the divisor $\operatorname{div}_Y(\varphi) + f^*D + E$. Then,

$$\begin{aligned}&\operatorname{div}_Y(\varphi) + \lceil f^*D+E\rceil \\ &= \lceil \operatorname{div}_Y(\varphi) + f^*D + E\rceil \qquad \big(\operatorname{div}_Y(\varphi) \text{ is an integral divisor}\big) \\ &\geq 0\end{aligned}$$

on $f^{-1}U$, which gives that $\mathscr{O}_X(D) \subseteq f_*\mathscr{O}_Y(\lceil f^*D+E\rceil)$.

Step 2: Now, we have

$$\begin{aligned}&R^i f_*\mathscr{O}_Y(\lceil f^*D+E\rceil) \\ &= R^i f_*\mathscr{O}_Y\big(\lceil f^*D + K_Y - f^*(K_X+\Delta)\rceil\big) \\ &= R^i f_*\mathscr{O}_Y\Big(K_Y + \lceil f^*\big(D-(K_X+\Delta)\big) - \sum p_j F_j\rceil\Big) \\ &= 0\end{aligned}$$

for every $i > 0$ by Kawamata-Viehweg vanishing theorem, where

$$\operatorname{Supp}\left\{f^*\big(D-(K_X+\Delta)\big)-\sum p_jF_j\right\}$$

is contained in $\operatorname{Supp}(f_*^{-1}\Delta)\cup\operatorname{Ex}(f)$. Since for any prime divisor P on Y disjoint with $\operatorname{Ex}(f)$, we have

$$\begin{aligned}\operatorname{mult}_P f^*\big(D-(K_X+\Delta)\big) &= \operatorname{mult}_{f(P)}\big(D-(K_X+\Delta)\big)\\ &= \underbrace{\operatorname{mult}_{f(P)} D-\operatorname{mult}_{f(P)} K_X}_{\in\mathbb{Z}}-\operatorname{mult}_{f(P)}\Delta,\end{aligned}$$

by f is an isomorphism over the generic point of $f(P)$; therefore the fractional part can only appears in $f_*^{-1}\Delta$ and the f-exceptional divisor.

For composition $Y\xrightarrow{f}X\xrightarrow{\pi}S$, we have Grothendieck spectral sequence (D.6.3)

$$E_2^{p,q}=R^p\pi_*\big(R^qf_*\mathscr{O}_Y(\lceil f^*D+E\rceil)\big)\underset{p}{\Rightarrow}R^{p+q}(\pi\circ f)_*\mathscr{O}_Y(\lceil f^*D+E\rceil).$$

But $R^qf_*\mathscr{O}_Y(\lceil f^*D+E\rceil)=0$ for $q>0$ by above, so this spectral sequence collapses on the p-axis. By (D.5.11), we get

$$\begin{aligned}E_2^{i,0}&=R^i\pi_*\big(f_*\mathscr{O}_Y(\lceil f^*D+E\rceil)\big)\\ &\cong R^i\pi_*\mathscr{O}_X(D)\\ &\cong R^i(\pi\circ f)_*\mathscr{O}_Y(\lceil f^*D+E\rceil).\end{aligned}$$

We can use Kawamata-Viehweg vanishing theorem again to obtain

$$\begin{aligned}&R^i(\pi\circ f)_*\mathscr{O}_Y(\lceil f^*D+E\rceil)\\ =\;&R^i(\pi\circ f)_*\mathscr{O}_Y\Big(K_Y+\lceil f^*\big(D-(K_X+\Delta)\big)-\sum p_jF_j\rceil\Big)\\ =\;&0\end{aligned}$$

for every $i>0$. So directly $R^i\pi_*\mathscr{O}_X(D)=0$ for every $i>0$. □

Remark 6.1.14. The ampleness of $f^*\big(D-(K_X+\Delta)\big)-\sum p_jF_j$ comes from [EGA II, 4.6.13(ii)]. In fact, since resolution of singularities is a sequence of blow-ups, there are rational number $c_j>0$ such that the sum of exceptional divisors $-\sum c_jF_j$ are f-ample. This can easily deduce from the induction on blow-up steps and use [EGA II, 4.6.13(ii)]. So for π-ample divisor $D-(K_X+\Delta)$, we have

$$-\sum c_jF_j+mf^*\big(D-(K_X+\Delta)\big)$$

is $\pi\circ f$-ample for sufficiently large m, also the EGA II's result.

Corollary 6.1.15. *Let (X, Δ) be a klt pair with X is a normal variety and $K_X + \Delta$ is $\mathbb{R}$-Cartier. Let $\pi : X \longrightarrow S$ be a projective surjective morphism onto a variety S, and let D be an integral divisor which is $\mathbb{Q}$-Cartier. Assume that $D - (K_X + \Delta)$ is a π-nef and π-big $\mathbb{R}$-Cartier divisor. Then*

$$R^i\pi_*\mathscr{O}_X(D) = 0 \text{ for } i > 0.$$

Proof. By relative Kodaira's lemma, see Corollary 3.3.7, it gives a decomposition

$$D - (K_X + \Delta) \sim_{\mathbb{R},S} H + B \tag{6.6}$$

where H is π-ample and B is an effective $\mathbb{Q}$-Cartier divisor. We choose a rational number $0 < \varepsilon \ll 1$, by Corollary 5.2.38 we see $(X, \Delta + \varepsilon B)$ is also a klt pair. Using formula (6.6),

$$\begin{aligned} D - (K_X + \Delta) - \varepsilon B &\sim_{\mathbb{R},S} D - (K_X + \Delta) - \varepsilon\big(D - (K_X + \Delta)\big) \\ &= \underbrace{(1-\varepsilon)\big(D - (K_X + \Delta)\big)}_{\pi\text{-nef}} + \varepsilon \underbrace{H}_{\pi\text{-ample}}, \end{aligned}$$

then by Corollary 1.5.20 we see

$$D - (K_X + \Delta + \varepsilon B)$$

is π-ample. We may therefore apply Theorem 1.13 to the klt pair $(X, \Delta + \varepsilon B)$ to get that

$$R^i\pi_*\mathscr{O}_X(D) = 0$$

for every $i > 0$. □

The above arguments are suitable in the analytic category, while our main method in the proof of Theorem 1.13 is the Kawamata-Viehweg vanishing, it also has the analytic version:

Theorem 6.1.16. ([Nak04, II, 5.12]) *Let X be a non-singular complex analytic variety and $f : \mathsf{X} \longrightarrow \mathsf{S}$ be a projective surjective morphism onto a complex analytic variety S. Let D be a f-nef and f-big $\mathbb{R}$-divisor on X whose fractional part $\{D\}$ has only normal crossing support. Then*

$$\mathrm{R}^i f_*\mathcal{O}_{\mathsf{X}}(K_{\mathsf{X}} + \lceil D \rceil) = 0 \text{ for } i > 0.$$

So for the analytic version of the vanishing Theorem 1.15, we give the statement below, while the definition can see [Nak04, II, 4.6].

Theorem 6.1.17. *Let (X, Δ) be a klt pair with X is a normal variety and $K_{\mathsf{X}} + \Delta$ is $\mathbb{R}$-Cartier. Let $\pi : \mathsf{X} \longrightarrow \mathsf{S}$ be a projective surjective morphism onto a complex analytic variety S, and let D be an integral divisor which is $\mathbb{Q}$-Cartier. Assume that $D - (K_{\mathsf{X}} + \Delta)$ is π-nef and π-big $\mathbb{R}$-Cartier divisor. Then*

$$\mathrm{R}^i\pi_*\mathcal{O}_{\mathsf{X}}(D) = 0 \text{ for } i > 0.$$

The proof is analogue with Theorem 1.13 and Corollary 1.14, but be careful that the algebraic geometric result [EGA II, 4.6.13(ii)] does not holds globally in the analytic category, it is locally holds which can see [DHP24, 2.12(4)]. Furthermore, we need a complex-analytic version analogue with [Har77, III, 11.2]:

Lemma 6.1.18. *Let $\pi : \mathsf{X} \longrightarrow \mathsf{S}$ be a proper morphism of complex analytic spaces whose fibers have dimension n. Then, for every coherent analytic sheaf $\mathcal{F}$ on X,*

$$\operatorname{Supp} \mathrm{R}^k\pi_*\mathcal{F} \subset \{s \in S \mid \dim(\pi^{-1}(s) \cap \operatorname{Supp} \mathcal{F}) \geq k\},$$

and

$$\mathrm{R}^k\pi_*\mathcal{F} = 0 \text{ for every } k > n.$$

Proof. The statement can see [Kad24, 2.13]. Actually its proof is similar as in algebraic geometry, since we have Grauert's Direct Image Theorem gives that $\mathrm{R}^k\pi_*\mathcal{F}$ is coherent, then use holomorphic formal function method and the faithful flatness of the completion on coherent sheaf obtain the vanishing on each stalks. □

After these preparations, we discuss a adjunction lemma in [Kaw91].

Lemma 6.1.19. ([Kaw91]) *Let $f : X \longrightarrow Y$ be a projective birational morphism of normal algebraic varieties, H is an f-ample Cartier divisor on X and Δ is an effective $\mathbb{Q}$-divisor on X. We denote E is an irreducible component of the exceptional locus $\mathrm{Ex}(f)$ whose dimension is maximal, denote by n. Let $\nu : \bar{E} \longrightarrow E$ be the normalization.*

Suppose that the pair (X, Δ) is klt with $-(K_X + \Delta)$ is f-ample, $f(E)$ is a (closed) point. Then

$$\left(H^{n-1} \cdot (K_X + \Delta) \cdot E\right) > \left((\nu^* H)^{n-1} \cdot K_{\bar{E}}\right).$$

Here note that $\cdot E$ and $\cdot K_{\bar{E}}$ is take intersection on coherent sheaves $\mathscr{O}_X/\mathcal{I}_E$ and $\mathscr{O}_{\bar{E}}(K_{\bar{E}})$, while the latter term is a coherent rank-1 reflexive sheaf.

Proof. **Step 1:** Since the statement is local near the point $f(E)$, we may assume that Y is affine, while the intersection number does not change since

$$\mathscr{O}_X(H)|_E = \mathscr{O}_{X_V}(H|_{X_V})|_E$$

where $f_V : X_V = f^{-1}(V) \longrightarrow V \ni f(E)$, and other line bundles are the same so Euler number does not change.

Step 2: We may replace H by its multiple mH such that the inequality still holds. So we may assume that H is f-very ample. But by *Step 1* our base variety Y is affine, this is equivalent to say that H is very ample. At this time, X is a quasi-projective variety, since projective morphism f can be factored as $X \hookrightarrow \mathbb{P}^N_Y \to Y$, and with

$$\mathbb{P}^N_Y = \mathbb{P}^N_{\mathbb{C}} \times Y \hookrightarrow \mathbb{P}^N_{\mathbb{C}} \times \mathbb{A}^M_{\mathbb{C}} \underset{\text{open}}{\subseteq} \mathbb{P}^N_{\mathbb{C}} \times \mathbb{P}^M_{\mathbb{C}}$$

gives that X is the open subset of the closure of X in $\mathbb{P}^N_{\mathbb{C}} \times \mathbb{P}^M_{\mathbb{C}}$.

Step 3: We may also assume that $n = 1$. In fact, if $n > 1$ we choose a general member $X_1 \in |H|$ and set $E_1 := E \cap X_1$. Our choice of X_1 ensures that $E \not\subseteq X_1$ such that E_1 is a variety with $\dim E_1 = n-1$. Moreover, since X is quasi-projective and H is very ample, X_1 is the X intersect with a general hyperplane section, so by [FOV99, 3.4.14] (or can see [DW22, 2.4]) we get that X_1 is a normal algebraic variety. And we denote $\bar{E}_1$ by the normalization of E_1.

For normal varieties X_1 and $\bar{E}_1$, we have formulas

$$K_{X_1} \sim_{lin} (K_X + H)|_{X_1} \quad \text{and} \quad K_{\bar{E}_1} \sim_{lin} (K_{\bar{E}} + \nu^* H)|_{\bar{E}_1}, \tag{6.7}$$

which they are come from the adjunction formula on smooth locus and then extend on the whole space, by [Har77, II,6.5(b)]. Note that $\bar{E}_1 \sim_{lin} \nu^* H = \nu^*(H|_E)$. Since at the beginning we have $E_1 \sim_{lin} H|_E$, and $D := \nu^* E_1 = \nu^{-1}(E_1) \sim_{lin} \nu^*(H|_E)$ is a normal variety, by we use the Bertini normality on normal projective variety $\bar{E}$ with the free series $|\nu^* H|$, its general member D is normal (for example, see [Man82, II,7]). Therefore we get a morphism $\nu|_D : D \longrightarrow E_1$ which is finite and birational; the finiteness comes from the base change

$$\begin{array}{ccc} D = \bar{E} \underset{E}{\times} E_1 & \longrightarrow & \bar{E}_1 \\ {\scriptstyle \nu|_D}\downarrow & & \downarrow{\scriptstyle \nu} \\ E_1 & \longrightarrow & E, \end{array}$$

and the birational part comes from $\nu|_D$ is an isomorphism over the dense open subset $E_1 \cap E_{\text{nor}}$ of E_1, where E_{nor} denotes the normal locus of E. By the universal property of the normalization, we have a diagram

$$\begin{array}{ccc} D & & \\ {\scriptstyle \phi}\vdots & \searrow^{\nu|_D} & \\ \bar{E}_1 & \longrightarrow & E_1 \end{array}$$

implies that ϕ is a finite and birational morphism between normal varieties, so it is an isomorphism (cf. [Har77, II,11.4]). Thus, we see $\bar{E}_1 \sim_{lin} \nu^* H$ and isomorphic to an subvariety of $\bar{E}$, hence the second formula in (6.7) is well-defined.

Now, we can consider the pair (X_1, Δ_1) where $\Delta_1 := \Delta|_{X_1}$ such that $K_{X_1} + \Delta_1$ is $\mathbb{Q}$-Cartier, $H_1 := H|_{X_1}$ is very ample and normalization $v_1 : \bar{E}_1 \longrightarrow E_1$. Once we prove the inequality

$$\big(H_1^{n-2} \cdot (K_{X_1} + \Delta_1) \cdot E_1\big) > \big((\nu_1 H_1)^{n-2} \cdot K_{\bar{E}_1}\big), \tag{6.8}$$

then since the left-hand-side of (6.8) is

$$\begin{aligned}
&\big(H_1^{n-2} \cdot (K_{X_1} + \Delta_1) \cdot E_1\big) \\
&= \big((H|_{X_1})^{n-2} \cdot (K_X + H + \Delta)|_{X_1} \cdot E|_{X_1}\big) \\
&= \big(H^{n-2} \cdot (K_X + H + \Delta) \cdot H \cdot E\big) \quad \text{since } X_1 \in |H| \\
&= \big(H^{n-1} \cdot (K_X + \Delta) \cdot E\big) + (H^n \cdot E),
\end{aligned}$$

and the right-hand-side of (6.8) is

$$\begin{aligned}
&\big((\nu_1 H_1)^{n-2} \cdot K_{\bar{E}_1}\big) \\
&= \Big(\nu^*(H|_E)^{n-2} \cdot \big(K_{\bar{E}} + \nu^*(H|_E)\big)\big|_{\bar{E}_1}\Big) \\
&= \Big(\nu^*(H|_E)^{n-2} \cdot \big(K_{\bar{E}} + \nu^*(H|_E) \cdot \nu^*(H|_E)\big)\Big) \quad \text{since } \bar{E}_1 \sim_{lin} \nu^*(H|_E) \\
&= \big(\nu^*(H|_E)^{n-1} \cdot K_{\bar{E}}\big) + \big(\nu^*(H|_E)^n\big).
\end{aligned}$$

But $\nu : \bar{E} \longrightarrow E$ is a birational morphism has degree 1, so by the generalized projection formula Proposition 1.1.28 we have $(H^n \cdot E) = \big(\nu^*(H|_E)^n\big)$. Therefore, the inequality (6.8) will implies the inequality we need to prove. Repeat this process $n-1$ times, we may assume that $n = 1$. Note that in this process we may lose the f-ampleness of $-(K_X + \Delta)$, since

$$(K_X + \Delta + H)|_{X_1} \sim_{lin} K_{X_1} + \Delta_1$$

will gives that

$$-(K_{X_1} + \Delta_1) = \big(-(K_X + \Delta) - H\big)\big|_{X_1}.$$

Step 4: Suppose that $n = 1$ and $\big((K_X + \Delta) \cdot E\big) \leq \deg K_{\bar{E}}$, that is we prove the Lemma by contradiction. If we denote $q = \big((K_X + \Delta) \cdot E\big) \in \mathbb{Q}$, then there is a integer $d \geq \lceil q \rceil$ such that

$$q \leq d \leq 2g(\bar{E}) - 2 = \deg K_{\bar{E}}.$$

Our next goal is to find a Cartier divisor A_0 on E such that

$$q \leq \deg A_0 = \deg \nu^* A_0 = d \leq \deg K_{\bar{E}}$$

and $H^0(\bar{E}, K_{\bar{E}} - \nu^* A_0) \neq \{0\}$. There are several cases:

- $g(\bar{E}) \geq 2$: we can choose $d \geq 0$. By [Har77, IV,5.1], the canonical linear series $|K_{\bar{E}}|$ is basepoint-free. So we can move it and choose an effective divisor

  $$G = P_1 + \cdots + P_N \in |K_{\bar{E}}|,$$

  where $N = \deg K_{\bar{E}}$ and P_i are the points lie in $\nu^{-1}(E_{\mathrm{nor}})$, such that ν is an isomorphism near these points. We define the divisor A_0 on E to be

  $$A_0 = \nu(P_0) + \cdots + \nu(P_d)$$

  such that $\nu^* A_0 = P_1 + \cdots + P_d$ and $\deg A_0 = d \geq q$. Then

  $$\begin{aligned} G - \nu^* A_0 =& P_{d+1} + \cdots + P_N \\ \sim_{lin}& K_{\bar{E}} - \nu^* A_0 \end{aligned}$$

  is an effective divisor, so $H^0(\bar{E}, K_{\bar{E}} - \nu^* A_0) \neq \{0\}$.

- $g(\bar{E}) = 1$: at this time $\deg K_{\bar{E}} = 0$, so we can just take $A_0 = 0$. Then

  $$H^0(\bar{E}, K_{\bar{E}} - \nu^* A_0) = H^0(\bar{E}, K_{\bar{E}}) \neq \{0\}$$

  by $h^0(\bar{E}, K_{\bar{E}}) = g(\bar{E}) = 1$.

- $g(\bar{E}) = 0$: at this time $\bar{E} \cong \mathbb{P}^1$, so we can take $A_0 = dQ$ while as $K_{\bar{E}} \sim_{lin} -2P$ and $Q \in E_{\mathrm{nor}}$ is sent by P. Then

  $$K_{\bar{E}} - \nu^* A_0 \sim_{lin} (-2 - d)P$$

  is an effective divisor by $d \leq -2$, so $H^0(\bar{E}, K_{\bar{E}} - \nu^* A_0) \neq \{0\}$.

Therefore, analysis as above can obtain such Cartier divisor A_0.

Now, there is an exact sequence

$$0 \longrightarrow \mathscr{O}_E(A_0) \longrightarrow \nu_* \mathscr{O}_{\bar{E}}(\nu^* A_0) \longrightarrow \mathrm{Coker} \longrightarrow 0$$

thanks to the projection formula and the sequence in [Har77, IV, ex1.8], where the right-hand-side term is supported on finite points. Thus we get a surjection

$$H^1\big(E, \mathscr{O}_E(A_0)\big) \twoheadrightarrow H^1\big(E, \nu_* \mathscr{O}_{\bar{E}}(\nu^* A_0)\big). \tag{6.9}$$

But as ν is a normalization, it is an affine morphism, so by [Har77, III, ex8.2] we have

$$H^1\big(E, \nu_*\mathscr{O}_{\bar{E}}(\nu^* A_0)\big) = H^1\big(\bar{E}, \mathscr{O}_{\bar{E}}(\nu^* A_0)\big).$$

Combining this with surjection (6.9), and take a dual it will be

$$H^1\big(\bar{E}, \mathscr{O}_{\bar{E}}(\nu^* A_0)\big)^\vee \hookrightarrow H^1\big(E, \mathscr{O}_E(A_0)\big)^\vee. \tag{6.10}$$

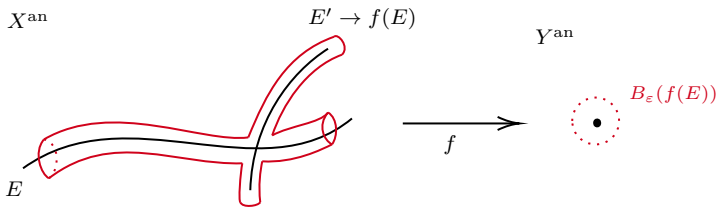
And notice that E and $\bar{E}$ are complete curves, so they are projective, we can use Serre duality to obtain that

$$\begin{aligned} H^1\big(\bar{E}, \mathscr{O}_{\bar{E}}(\nu^* A_0)\big)^\vee &\cong H^0\big(\bar{E}, \omega_{\bar{E}}(-\nu^* A_0)\big) \neq \{0\}, \\ H^1\big(E, \mathscr{O}_E(A_0)\big)^\vee &\cong H^0\big(E, \omega_E^o(-A_0)\big), \end{aligned}$$

where ω_E^o is the dualizing sheaf of E, and easy to verify that E as an integral curve, it is a Cohen-Macaulay scheme. Therefore, the injective map (6.10) gives that

$$H^0\big(E, \omega_E^o(-A_0)\big) \neq \{0\}.$$

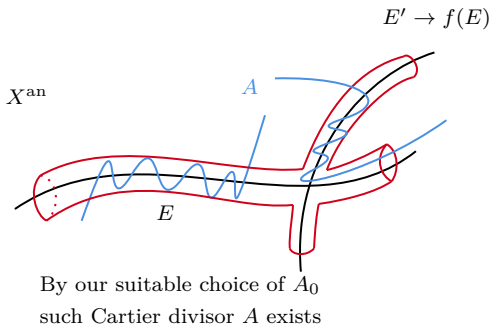
On the other hand, since E is 1-dimensional, we can take a sufficiently small analytic neighborhood $V = Y^{\mathrm{an}} \cap B_\varepsilon(f(E))$ of $f(E)$,



and we choose $U = f^{-1}(V) \cap X^{\mathrm{an}}$ such that there exists a Cartier divisor A on U where $A_0 = A|_E$ and the support of A does not **contains** with irreducible components of $\mathrm{Ex}(f|_U)$ other than E. Moreover, by our suitable choice of such Cartier divisor A, we can ensure that on other irreducible component E' it will be satisfied that

$$\big((K_X + \Delta) \cdot E'\big) \leq \deg_{E'} A = (A \cdot E').$$

Note that there are finitely many such components E' of $\mathrm{Ex}(f)$ in noetherian space, so we always can do this.



Thus we see $\Big(\big(A-(K_X+\Delta)\cdot C\big)\Big) \geq 0$ for every f-relative curve, that is $A-(K_X+\Delta)$ is f-nef for $f : U \longrightarrow V$. Now, the pair $(U, \Delta|_U)$ is a klt pair in the analytic category, with $A-(K_U+\Delta|_U)$ is f-big since f is bimeromorphic morphism over the analytic variety V. Moreover, $f : U \longrightarrow V$ is a projective surjective morphism under the complex-analytic meaning, since f is proper and there is an f-ample divisor $-(K_U+\Delta|_U)$, can see [Nak04, II, 1.10]. So by Vanishing Theorem 1.17, we have

$$\mathrm{R}^i f_* \mathcal{O}_U(A) = 0$$

for every $i > 0$, in particular for $i = 1$. We now consider $i : E \hookrightarrow U$ with $\mathcal{O}_U(A)|_E = \mathcal{O}_E(A_0)$, so we have the short exact sequence

$$0 \longrightarrow \mathcal{I}_E \otimes \mathcal{O}_U(A) \longrightarrow \mathcal{O}_U(A) \longrightarrow \mathcal{O}_E(A_0) \longrightarrow 0.$$

Take higher direct image, obtain that

$$\mathrm{R}^1 f_* \mathcal{O}_U(A) \longrightarrow \mathrm{R}^1 f_* \mathcal{O}_E(A_0) \longrightarrow \mathrm{R}^2 f_* \big(\mathcal{I}_E \otimes \mathcal{O}_U(A)\big).$$

The left-outer-term vanished by our preceding analysis, and the right-outer-term vanished because for proper bimeromorphic morphism $f : U \longrightarrow V$, our choice of E such that it has maximal dimension ensures that the fibre of f has dimension at most 1. So by Lemma 1.18, we have

$$\mathrm{R}^2 f_* \big(\mathcal{I}_E \otimes \mathcal{O}_U(A)\big) = 0.$$

This gives that the middle term $\mathrm{R}^1 f_* \mathcal{O}_E(A_0) = 0$, but as $f(E)$ is a point, this is nothing but to say $\mathrm{H}^1\big(E, \mathcal{O}_E(A_0)\big) = 0$. Use GAGA principle, we obtain that

$$H^1\big(E, \mathscr{O}_E(A_0)\big) = 0.$$

Then use Serre duality to get that $H^0\big(E, \omega_E^o(-A_0)\big) = 0$, contradiction. □

For another proof can see [Deb01, 7.47]. It has an application:

Theorem 6.1.20. (Length of an extremal rational curve) *Let $f : X \longrightarrow Y$ be a projective surjective morphism of normal algebraic varieties such that $f_*\mathscr{O}_X = \mathscr{O}_Y$, and $\Delta \geq 0$ is a $\mathbb{Q}$-divisor such that the pair (X, Δ) has klt singularities. Let E be an irreducible component of $\mathrm{Ex}(f)$ of maximal dimension, and $n = \dim E - \dim f(E)$. Assume further that $-(K_X + \Delta)$ is f-ample.*

Then the general points of E is covered by a family of rational curves $\{L_\lambda\}_{\lambda\in\Lambda}$ such that the $f(L_\lambda)$ are points and that

$$0 < \big(-(K_X + \Delta) \cdot L_\lambda \big) < {}^{1}2n.$$

Recall that in Theorem 2.3.7, when f is just a structure morphism with X is smooth and Y is a point, there is a rational curve L such that $-(K_X \cdot L) \leq \dim X + 1$. The number $-(K_X \cdot L)$ is called *length* of L.

Proof. **Step 1:** We set $r = \dim f(E)$ and $n = \dim E - r$. Take an affine open subset Y_1 of Y which intersects $f(E)$, and take Y_0 is the intersection of Y by r general hyperplane sections of $\mathbb{A}^N$. Then Y_0 is a normal variety by the quasi-projective version of [FOV99, 3.4.14]. We set

$$X_0 = f^{-1}(Y_0),\ E_0 = E \cap X_0,\ \Delta_0 = \Delta \cap X_0,\ f_0 = f|_{X_0}.$$

In other words, we cut $f(E)$ by general hyperplane section r times to reduce the dimension of $f(E)$ to 0. Pick an irreducible component of E_0 and we abuse of notation denote it by E_0 again, then

$$\dim E_0 = \dim E - \dim f(E) = n,$$

and $f_0(E_0) = \{p\}$ is a single point. Moreover, since X_0 is cut out in X by r general members of a free series $f^*|\mathscr{O}_{Y_1}(1)|$, so it is also a normal variety by [Man82, II,7]. And by [KM98, 5.17(1)] the pair (X_0, Δ_0) is also has klt singularities. Note that f_0 is also projective after base change, and $-(K_{X_0} + \Delta_0)$ is f_0-ample.

Step 2: We can take an f_0-ample Cartier divisor H on X_0, and a normalization $\nu : \bar{E}_0 \longrightarrow E_0$. Since ν is a finite morphism, the pullback $\nu^*H = \nu^*(H|_E)$ is an $(f_0 \circ \nu)$-ample Cartier divisor on $\bar{E}_0$, hence ample. By pass through to a sufficiently large multiple of H, we may assume that ν^*H is very ample. Choose $n - 1$ general

[1] This is the case $E \neq X$, i.e. f is birational; but when f is a contraction such that $E = X$, it will be $\leq 2n$.

members in $|\nu^* H|$, can cut out a smooth curve C since $\bar{E}_0$ is normal, i.e. we use Bertini's theorem on the smooth locus of $\bar{E}_0$. We also set

$$M = -\nu^*(K_{X_0} + \Delta_0)$$

which is an $(f_0 \circ \nu)$-ample $\mathbb{Q}$-Cartier divisor on $\bar{E}_0$, hence ample. Note that $\bar{E}_0$ is projective by finite morphism is projective.

Step 3: By adjunction Lemma 1.19 applied to f_0, where E_0 has the maximal dimension, we get that

$$\big(H^{n-1} \cdot (K_{X_0} + \Delta_0) \cdot E_0\big) > \big((\nu^* H)^{n-1} \cdot K_{\bar{E}_0}\big). \tag{6.11}$$

From the construction of curve C, we can get

$$\begin{aligned}
\big(H^{n-1} \cdot (K_{X_0} + \Delta_0) \cdot E_0\big)_{X_0} &= \big((H|_{E_0})^{n-1} \cdot (K_{X_0} + \Delta_0)|_{E_0}\big)_{E_0} \\
&= \big(\nu^*(H|_{E_0})^{n-1} \cdot \nu^*(K_{X_0} + \Delta_0)\big)_{\bar{E}_0} \\
&= \big(\nu^*(K_{X_0} + \Delta_0) \cdot C\big)_{\bar{E}_0} \\
&= (-M \cdot C)_{\bar{E}_0} \\
&< 0
\end{aligned}$$

by M is ample. Follows from (6.11), we have

$$\big((\nu^* H)^{n-1} \cdot K_{\bar{E}_0}\big) = (K_{\bar{E}_0} \cdot C) = \deg_C(K_{\bar{E}_0}|_C) < 0.$$

If $n = 1$, i.e. $\bar{E}_0$ is a smooth curve such that $C = \bar{E}_0$. Then

$$\deg K_{\bar{E}_0} = 2g(\bar{E}_0) - 2 < 0$$

implies that $\bar{E}_0 \cong \mathbb{P}^1$. Since (X_0, Δ_0) is a klt pair with f_0-ample divisor $-(K_{X_0} + \Delta_0)$, by vanishing Theorem 1.13 we have

$$R^1 f_{0*} \mathscr{O}_{X_0} = 0.$$

Then consider the short exact sequence

$$0 \longrightarrow \mathscr{I}_{E_0} \longrightarrow \mathscr{O}_{X_0} \longrightarrow \mathscr{O}_{E_0} \longrightarrow 0$$

which gives that

$$R^1 f_{0*} \mathscr{O}_{X_0} \longrightarrow R^1 f_{0*} \mathscr{O}_{E_0} \longrightarrow R^2 f_{0*} \mathscr{I}_{E_0}.$$

The right-hand-side term vanished by [Har77, III, 11.2] and our choice of E has the maximal dimension. So $R^1 f_{0*} \mathscr{O}_{E_0} \cong H^1(E_0, \mathscr{O}_{E_0}) = 0$. But follows from the short exact sequence

$$0 \longrightarrow \mathscr{O}_{E_0} \longrightarrow \nu_* \mathscr{O}_{\bar{E}_0} \longrightarrow \text{Coker} \longrightarrow 0,$$

we have

$$\underbrace{H^0(E_0, \mathscr{O}_{E_0})}_{\cong\mathbb{C}} \longrightarrow \underbrace{H^0(E_0, \nu_*\mathscr{O}_{E_0})}_{\cong\mathbb{C}} \longrightarrow H^0(E_0, \mathrm{Coker}) \longrightarrow 0.$$

Then we obtain that $H^0(E_0, \mathrm{Coker}) = 0$, so $\mathrm{Coker} = 0$ and $E_0 \cong \bar{E}_0 = \mathbb{P}^1$.

If $n > 1$, by our construction $\bar{E}_0$ is smooth along the curve C, so we can obtain the formula in Chapter 2:

$$\dim_{[i]} \mathrm{Mor}(C, \bar{E}_0; i|_{\{*\}}) \geq -(K_{\bar{E}_0} \cdot C) - g(C) \cdot n.$$

Then we can use the Bend and Break method, to get that for arbitrary point[2] x of C, there exists a rational curve L on $\bar{E}_0$ passing through x such that

$$(M \cdot L) \leq 2n \cdot \frac{(M \cdot C)}{(-K_{\bar{E}_0} \cdot C)}^{3} < 2n.$$

Then $\nu(L)$ is the rational curve[4] we are looking for, while $f \circ \nu(L)$ is a point and

$$\begin{aligned}
\big(-(K_X + \Delta) \cdot \nu(L)\big)_X &= \big(-(K_X + \Delta) \cdot \nu_* L\big)_X \\
&= \big(-(K_{X_0} + \Delta_0) \cdot \nu_* L)_{X_0} \quad (\text{Since } \nu(L) \subsetneq X_0 \subseteq X) \\
&= (M \cdot L) \\
&< 2n.
\end{aligned}$$

Step 4: Let such smooth C sweep out the smooth locus of $\bar{E}_0$, we see the general points of $\bar{E}_0$ is covered by rational curves. So via the projective birational morphism ν, the general points of E_0 is covered by rational curves. Moreover, since Y is covered by open affine subsets, let Y_1 runs through affine subsets intersect $f(E)$, can get that the general points of E is also covered by rational curves. □

Remark 6.1.21. (Counter-example when only assume that f is projective) In Kawamata's origin paper [Kaw91], the statement only assumes that $f : X \longrightarrow Y$ is a projective morphism of algebraic varieties. The Theorem 1.20 will fails when we drop out the assumption $f_*\mathscr{O}_X = \mathscr{O}_Y$. For example, we consider

$$\begin{aligned}
f : \mathbb{P}^1 &\longrightarrow \mathbb{P}^1 \\
[s : t] &\longmapsto [s^2 : t^2]
\end{aligned}$$

with the zero boundary divisor $\Delta = 0$. Since $X = \mathbb{P}^1$ is smooth, so the pair $(X, 0)$ has klt singularities. Note that f is finite and projective, but it is not birational since at

[2] Since every point of C will be send to a point of $\bar{E}_0$) via $i : C \hookrightarrow \bar{E}_0$.

[3] cf. [Kol96, II, 5.8] or [Deb01, 3.6]

[4] Since ν is a finite birational morphism does not contract anything.

every point it won't be an isomorphism, so $E = X$ and $n = 0$. And $-K_X = \mathscr{O}_{\mathbb{P}^1}(2)$ is an ample divisor, in particular is f-ample. Since f is finite contracts no curve, there is no rational curve L with $f(L)$ is a point; or even more

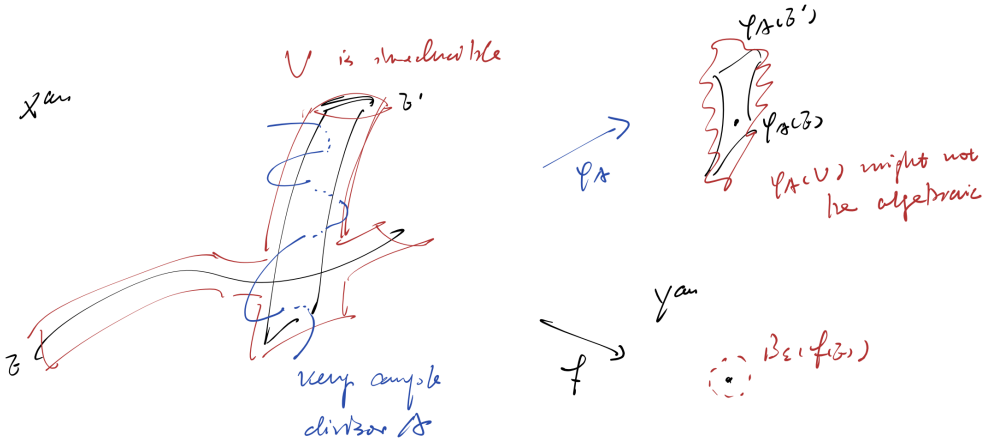
$$\big(-(K_X + \Delta) \cdot L\big) = \int_L c_1(\mathscr{O}_{\mathbb{P}^1}(2)) \leq 2n = 0$$

which is absurd. Moreover, it can be extended on arbitrary dimension $f : \mathbb{P}^n \longrightarrow \mathbb{P}^n$ which sending $[x_0 : \cdots : x_n]$ to $[x_0^2 : \cdots : x_n^2]$ to obtain the counter-example.

So in Kawamata's recent textbook [Kaw24, 2.8], we see that the assumption changed to be $f : X \longrightarrow Y$ is birational.

Remark 6.1.22. Our assumption here is the irreducible component E has the maximal dimension among $\mathrm{Ex}(f)$, to give the desired vanishing of $\mathrm{R}^2 f_* \big(\mathcal{I}_E \otimes \mathcal{O}_U(A)\big)$. But in Debarre's book he written a footnote and say that there is an another way to ensure that E can be chosen by arbitrary dimension by use some magic analytic geometry method, see [Deb01, 7.11]:

> This holds for example when E is a top-dimensional component of the exceptional locus of c, and this is enough for us. In general, one may use a neat trick of Kawamata's to reduce to the case where X_0 is actually irreducible. This goes roughly as follows (see [K], (2.3.3)): take a very ample divisor on X that meets X_0 transversely and shrink X to an analytic neighborhood of E so that this divisor has a component which meets only the other components of X_0, not E. The holomorphic map to a projective space associated with the sections of (a multiple of) this divisor contracts only E. Its image might not be algebraic, but the vanishing theorem that we use below is still valid in this context (see [N], 3.6) and the proof therefore proceeds in the same way.



This will gives that

$$\mathrm{R}^1\varphi_{A*}\mathcal{O}_U(A) \longrightarrow \mathrm{R}^1\varphi_{A*}\mathcal{O}_E(A_0) \longrightarrow \mathrm{R}^2\varphi_{A*}\big(\mathcal{I}_E \otimes \mathcal{O}_U(A)\big),$$

and the left-hand-side term vanishes by Kawamata-Viehweg vanishing in analytic version, the right-hand-side is due to all fibres are at most dimension one. So the middle term $\mathrm{R}^1\varphi_{A*}\mathcal{O}_E(A_0) = H^1(E, A_0) = 0$. But whether this is true or false, does not affect the proof of the Cone Theorem.

Now, we give a way the Rationality Theorem is used to get the information of cone of curves. Let H be an ample Cartier divisor on X, and choose $r = r(H)$ as in the Rationality Theorem such that $H + r(K_X + \Delta)$ is a nef $\mathbb{Q}$-Cartier divisor. Since r is a rational number, $m\big(H + r(K_X + \Delta)\big)$ is an integral Cartier divisor for some $m > 0$. Note that $m\big(H + r(K_X + \Delta)\big)$ is nef but not ample. Thus

$$\overline{\mathrm{NE}}(X) \cap \big(H + r(K_X + \Delta)\big)^{\perp} = \left\{ \sum_{a_i \geq 0} a_i [C_i] \middle| \Big(C_i \cdot \big(H + r(K_X + \Delta)\big)\Big) = 0 \right\}$$

is an extremal face of $\overline{\mathrm{NE}}(X)$.

Starting with various ample divisors, we get various extremal faces of $\overline{\mathrm{NE}}(X)$. The proof of the Cone Theorem follows immediately from the Rationality Theorem and the following abstract result.

Theorem 6.1.23. *Let $N_{\mathbb{Z}}$ be a free $\mathbb{Z}$-module of finite rank and $N_{\mathbb{R}} = N_{\mathbb{Z}} \underset{\mathbb{Z}}{\otimes} \mathbb{R}$ be the change of coefficients to $\mathbb{R}$. Let $\overline{\mathrm{NE}}(X) \subseteq N_{\mathbb{R}}$ be a closed convex cone not containing a straight line. Let K be an element of the dual $\mathbb{Q}$-vector space $N^{\vee}_{\mathbb{Q}}$ such that $(K \cdot C) < 0$ for some $C \in \overline{NE}(X)$.*

Assume that there exists $a(K) \in \mathbb{Z}_{>0}$ such that, for all $H \in N^{\vee}_{\mathbb{Z}}$ with $H > 0$ on $\overline{\mathrm{NE}}(X) \backslash \{0\}$,

$$r := \max\big\{ t \in \mathbb{R} \big| H + tK \geq 0 \text{ on } \overline{\mathrm{NE}}(X) \big\}$$

is a rational number of the form $u/a(K)$ $(u \in \mathbb{Z})$. Then

$$\overline{\mathrm{NE}}(X) = \overline{\mathrm{NE}}(X)_{K \geq 0} + \sum \mathbb{R}_{\geq 0}[\xi_i],$$

for a collection of $\xi_i \in N_{\mathbb{Z}}$ with $(\xi_i \cdot K) < 0$ such that the $\mathbb{R}_{\geq 0}[\xi_i]$ do not accumulate in the half space $K < 0$.

We prove the Theorem 1.12 and the Cone Theorem together:

Proof. We may assume that $K := K_X + \Delta$ is not nef.

Step 1: For a nef $\mathbb{Q}$-divisor class L, we set extremal face $F_L = L^{\perp} \cap \overline{\mathrm{NE}}(X)$. If L is not ample, then by Kleiman's ampleness criterion, we have $F_L \neq \{0\}$. Assume that $F_L \not\subseteq \overline{\mathrm{NE}}(X)_{K \geq 0}$ and let H be an arbitrary ample Cartier divisor. For $n \in \mathbb{Z}_{>0}$, let

$$r_L(n, H) := \left\{ t \in \mathbb{R} \middle| \underbrace{nL + H}_{\text{nef and big}} + \frac{t}{a(K)} K \text{ is nef} \right\}.$$

By the Rationality Theorem, $r_L(n,H)$ is a (non-negative) integer, and, since L is nef, $r_L(n,H)$ is a non-decreasing function of n by K is not nef. Now $r_L(n,H)$ stabilizes to a fixed $r_L(H)$ for every $n \geq n_0$, since if $\xi \in F_L \backslash \overline{\mathrm{NE}}(X)_{K\geq 0}$, then

$$r_L(n,H) \leq a(K)\frac{(H\cdot\xi)}{-(K\cdot\xi)}$$

for all n. Also, L and $D(nL,H) := na(K)L + a(K)H + r_L(H)K$ are both non-ample nef divisors for $n > n_0$, while notice that by our choice of the maximum of t which ensures the latter term is non-ample. So,

$$\{0\} \neq F_{D(nL,H)} \subseteq F_L$$

because we have $D\big((n-1)L,H\big) = D(nL,H) - a(K)L$ and

$$0 \leq \Big(D\big((n-1)L,H\big)\cdot\xi\Big) = D(nL,H) - a(K)(L\cdot\xi) = -a(K)(L\cdot\xi)$$

implies that $(L\cdot\xi) \leq 0$ for all $\xi \in F_{D(nL,H)}$. But L is nef so we also have $(L\cdot\xi)\geq 0$ which gives that $\xi \in F_L$. Moreover,

$$F_{D(nL,H)} \subseteq \overline{\mathrm{NE}}(X)_{K<0} \cup \{0\}$$

since L is nef and H is ample, the negative part can only arises on K.

Step 2: We claim that, if $\dim_{v.s.} F_L > 1$ and $F_L \nsubseteq \overline{\mathrm{NE}}(X)_{K\geq 0}$, then we can find an ample H with

$$\dim_{v.s.} F_{D(nL,H)} < \dim_{v.s.} F_L.$$

To see this, we choose ample divisors H_i which give a basis for $F_L^\vee$, (cf. Corollary 1.3.16). If $\dim_{v.s.} F_L > 1$, the linear functions

$$\left(nL + H_i + \frac{r_L(H_i)}{a(K)}K\right)\Big|_{F_L} = \left(H_i + \frac{r_L(H_i)}{a(K)}K\right)\Big|_{F_L}$$

cannot **all** be identically zero on F_L, otherwise all these $H_i|_{F_L}$ are some multiple of the linear form $K|_{F_L}$ contradicts the dimensional assumption. Thus we see

$$F_{D(nL,H_i)} = F_L \cap \ker\left(H_i + \frac{r_L(H_i)}{a(K)}K\right)\Big|_{F_L}$$

which means $\dim_{v.s.} F_{D(nL,H_i)} < \dim_{v.s.} F_L$ for some i, proving our claim.

Repeating the argument over successively smaller extremal faces, we obtain that for every nef L with $F_L \nsubseteq \overline{\mathrm{NE}}(X)_{K\geq 0}$ there is a nef $\mathbb{Q}$-divisor L' such that $F_L \supseteq F_{L'}$, $\dim_{v.s.} F_{L'} = 1$ and $F_L \subseteq \overline{\mathrm{NE}}(X)_{K<0} \cup \{0\}$.

Step 3: We claim that

$$\overline{\mathrm{NE}}(X) \quad \text{and} \quad \overline{\mathrm{NE}}(X)_{K\geq 0} + \sum_{\dim_{v.s.}=1} F_L$$

have the same closure.

To prove this, we assume the contrary, that is the right-hand-side

$$W := \text{the closure of} \left(\overline{\mathrm{NE}}(X)_{K\geq 0} + \sum_{\dim_{v.s.}=1} F_L \right)$$

of the claimed equality is smaller, strictly contained in the left-hand-side $\overline{\mathrm{NE}}(X)$. We pick a one-cycle class γ inside their difference $\overline{\mathrm{NE}}(X)\backslash W$, can find a divisor class $M \in N^{\vee}_{\mathbb{Z}}$ such that $(M\cdot\xi) > 0$ for all $\xi \in W\backslash\{0\}$ but $(M\cdot\gamma) < 0$ after a small rational perturbation and by the separating hyperplane theorem, see [BV04, 2.5.1]. So the hyperplane $\{M = 0\}$ misses the $W\backslash\{0\}$ but not the $\overline{\mathrm{NE}}(X)$.

Let H be an ample Cartier divisor on X and $\mu > 0$ be the largest (real) number such that $H + \mu M$ is nef but not ample. We fix a non-zero element $z \in \overline{\mathrm{NE}}(X)$ with $\big((H+\mu M)\cdot z\big) = 0$, then $(M\cdot z) < 0$ because $(H\cdot z) > 0$ by H is ample, so $z \notin W$. Moreover $(K\cdot z) < 0$ as $\overline{\mathrm{NE}}(X)_{K\geq 0} \subseteq W$. Note that here X is not smooth, we no need choose a sequence of effective one-cycles, use bend and break method and Theorem 2.3.7 to obtain rational curves. We choose to use Rationality Theorem and *Step 2*.

For any rational number $0 < \mu' < \mu$, we set $A_{\mu'} := H + \mu' M$ is a $\mathbb{Q}$-Cartier ample divisor on X, because it is the sum of an ample and a nef:

$$A_{\mu'} = \left(1 - \frac{\mu'}{\mu}\right) H + \frac{\mu'}{\mu}(H + \mu M).$$

We consider the nef threshold along K:

$$\tau_{\mu'} = \max\left\{t \geq 0 \middle| A_{\mu'} + tK \text{ is nef}\right\}.$$

Since $A_{\mu'}$ is an ample $\mathbb{Q}$-Cartier divisor, up to clear the denominator of $A_{\mu'}$, we see that $\tau_{\mu'}$ is a rational number by the Rationality Theorem, and it is positive and bounded above. We set $L_{\mu'} := A_{\mu'} + \tau_{\mu'} K$ which is a nef but not ample $\mathbb{Q}$-Cartier divisor, and we have seen $F_{L_{\mu'}} \neq \{0\}$. Any non-zero element $\xi \in F_{L_{\mu'}}$ such that $(L_{\mu'}\cdot\xi) = 0$, we can see

$$(K\cdot\xi) = -\frac{1}{\tau_{\mu'}}(A_{\mu'}\cdot\xi) < 0$$

hence that $F_{L_{\mu'}} \not\subseteq \overline{\mathrm{NE}}(X)_{K\geq 0}$.

Now we can use *Step 2*, there is a nef $\mathbb{Q}$-Cartier divisor P such that $F_{L_{\mu'}} \supseteq F_P$ and $\dim_{v.s.} F_P = 1$. By the definition of W, $F_P \subseteq W$. So for a non-zero element

$\xi_{\mu'} \in F_P \subseteq F_{L_{\mu'}} \cap W$ which is dependent on the choice of μ', we have

$$(L_{\mu'} \cdot \xi_{\mu'}) = 0 \ \text{ and } \ (M \cdot \xi_{\mu'}) > 0$$

by our choice of M. So,

$$\big((H + \mu' M + \tau_{\mu'} K) \cdot \xi_{\mu'}\big) = 0$$

will gives that

$$\tau_{\mu'} = \frac{\big((H + \mu' M) \cdot \xi_{\mu'}\big)}{-(K \cdot \xi_{\mu'})} > \frac{(H \cdot \xi_{\mu'})}{-(K \cdot \xi_{\mu'})} > \frac{1}{B},$$

where B is a sufficiently large integer such that $BH + K$ is ample, while notice that this number is independent of the choice of the rational number $0 < \mu' < \mu$.

On the other hand, our previous fixed non-zero element $z \in F_{H+\mu M}$ satisfies that

$$\big((H + \mu' M + \tau_{\mu'} K) \cdot z\big) \geq 0 \tag{6.12}$$

by $L_{\mu'}$ is nef, while z is independent of the choice of μ', and we have seen that $z \notin W$. Choose a seuqnece of rational numbers

$$\mu'_j < \mu \ \text{ and } \ \mu'_j \to \mu.$$

For each μ'_j, $L_{\mu'_j}$ is nef but not ample satisfies the formula (6.12). Hence

$$\begin{aligned} 0 \leq & \big((H + \mu'_j M + \tau_{\mu'_j} K) \cdot z\big) \\ = & \big((H + \mu M) \cdot z\big) + \big((\mu'_j - \mu) M \cdot z\big) + \big(\tau_{\mu'_j} K \cdot z\big) \\ = & (\mu'_j - \mu)(M \cdot z) + \tau_{\mu'_j} (K \cdot z). \end{aligned}$$

Since

$$(M \cdot z) < 0 \ \text{ and } \ (K \cdot z) < 0,$$

they have the same sign, so we can obtain that

$$0 \leq (\mu - \mu'_j)\big(-(M \cdot z)\big) - \tau_{\mu'_j}\big(-(K \cdot z)\big),$$

that is

$$0 \leq \tau_{\mu'_j} \leq (\mu - \mu'_j) \frac{-(M \cdot z)}{-(K \cdot z)}.$$

When μ'_j tends to μ, we have $\tau_{\mu'_j}$ will tends to 0, contradicts the fact that all $\tau_{\mu'_j}$ is bounded below by a fixed number $1/B$. Thus we complete that $W = \overline{\mathrm{NE}}(X)$.

Step 4: Next we show that the one-dimensional extremal rays F_L do not accumulate in the half space $K < 0$.

To see this, we choose ample divisors $H_1, \dots, H_d \in N^\vee_{\mathbb{Z}}$ such that they, together with K, form a basis of $N^\vee_{\mathbb{R}}$. For each one-dimensional F_L and $i = 1, \dots, d$, take $n_{L,i}$ such that

$$F_{D(n_{L,i}L, H_i)} = F_L$$

as in *Step 2*. Then, for ξ generating $F_L = \mathbb{R}_{\geq 0}[\xi]$ and for all i, we have

$$\begin{aligned}\big(\xi \cdot D(n_{L,i}L, H_i)\big) &= \big(\xi \cdot (n_{L,i}a(K)L + a(K)H_i + r_L(H_i)K)\big) \\ &= n_{L,i}a(K)\underbrace{(\xi \cdot L)}_{=0} + a(K)(\xi \cdot H_i) + r_L(H_i)(\xi \cdot K) \\ &= 0,\end{aligned}$$

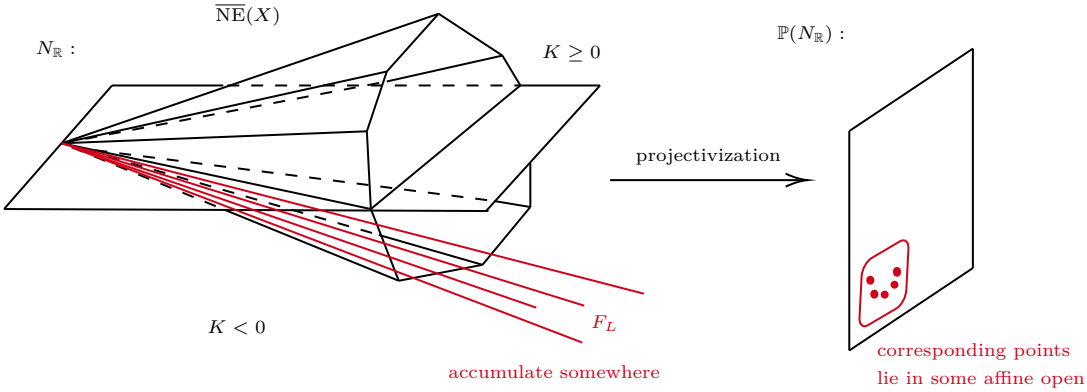
which gives that

$$\frac{(\xi \cdot H_i)}{(\xi \cdot K)} = \frac{-r_L(H_i)}{a(K)} = \frac{\text{integer}}{a(K)}. \tag{6.13}$$

If these F_L accumulated somewhere in the half space $K < 0$, then the points of the projectivization $\mathbb{P}(N_{\mathbb{R}})$:

$$\mathbb{R}_{\geq 0}[\xi] \longmapsto \text{points in } \mathbb{P}(N_{\mathbb{R}})$$

would have to accumulate somewhere in the affine subset $U \subseteq \mathbb{P}(N_{\mathbb{R}})$ given by $K < 0$.



Futhermore, we find that

$$\xi \in U \longmapsto \left(\frac{(\xi \cdot H_1)}{(\xi \cdot K)}, \dots, \frac{(\xi \cdot H_d)}{(\xi \cdot K)}\right)$$

is an affine coordinate system in $K < 0$, since it is stable under the scalar change $\xi \mapsto \lambda\xi_i$. But these points are discrete in $\mathbb{R}^d$, because the equation above (6.13) gives that they are lie in

$$\left(\frac{1}{a(K)}\mathbb{Z}\right) \subseteq \mathbb{R}^d,$$

so we get the contradiction.

This shows that there are only finitely many F_L such that $\big(F_L \cdot (K+\varepsilon H)\big) < 0$, for any $\varepsilon > 0$ and an ample $\mathbb{Q}$-divisor H. Thus,

$$\overline{\mathrm{NE}}(X)_{(K+\varepsilon H)\geq 0} + \sum_{\big(F_L\cdot(K+\varepsilon H)\big)<0}^{\text{finite}} F_L$$

is a closed cone. We have proved in *Step 3* that its closure is $\overline{\mathrm{NE}}(X)$. Thus,

$$\overline{\mathrm{NE}}(X) = \overline{\mathrm{NE}}(X)_{(K+\varepsilon H)\geq 0} + \sum_{\big(F_L\cdot(K+\varepsilon H)\big)<0}^{\text{finite}} F_L.$$

This proves Theorem 1.5(2), except that we do not yet know that each F_L is spanned by a curve in Theorem 1.5.

Remark 6.1.24. This is a convenient place to prove Remark 1.4 on a side step, while note that our setting is under the projectivity assumption, since in *Step 3* we have shown that there exist extremal rays negative on K. The proof of the Cone Theorem will continue with *Step 5*.

To avoid notation confusion, let M denote the nef and big divisor in Remark 1.4. We can write

$$M = m_0 K + \sum_{i=1}^{d} m_i H_i$$

for some $m_i \in \mathbb{Q}$. *Step 4* has shown that

$$\begin{aligned} -\frac{(\xi\cdot M)}{\big(\xi\cdot(K_X+\Delta)\big)} &= -\frac{m_0(\xi\cdot K)+\sum m_i(\xi\cdot H_i)}{(\xi\cdot K)} \\ &= -m_0 - \sum m_i \frac{(\xi\cdot H_i)}{(\xi\cdot K)} \\ &= -m_0 - \sum m_i \frac{-r_L(H_i)}{a(K)} \end{aligned}$$

are non-negative[5] rational numbers with bounded denominators as ξ runs through all $(K_X+\Delta)$-negative extremal rays. So, the infimum of the set

$$\mathfrak{S} = \left\{ -\frac{(\xi\cdot M)}{(\xi\cdot K)} \;\middle|\; \xi \text{ generating a } K\text{-negative extremal ray} \right\}$$

is a minimum, archived by an extremal ray $\mathbb{R}_{\geq 0}[\xi_M]$. Thus

$$r(M) = -\frac{(\xi_M\cdot M)}{\big(\xi_M\cdot(K_X+\Delta)\big)} \quad \text{and} \quad \Big(\xi_M\cdot\big(r(M)\cdot(K_X+\Delta)\big)\Big) = 0.$$

[5] Since M is nef and $(\xi\cdot K)\leq 0$.

Step 5: It is not hard to prove that the right hand side of Theorem 1.5(1) is a closed cone, and thus it is a formal consequence of Theorem 1.5(2) and *Step 4*, see [Kol96, P.161]. We do not need this in the sequel.

The arising of rational curves will be put in *Step 8*, after the proof of the Contraction Theorem.

Step 6: In this step we plan to prove that if $F \subseteq \overline{\mathrm{NE}}(X)$ is a $(K_X + \Delta)$-negative extremal face, then there is a nef Cartier divisor D such that $F = F_D = D^{\perp} \cap \overline{\mathrm{NE}}(X)$.

Let $\langle F \rangle$ be the linear span of F, i.e.

$$\langle F \rangle = \mathbb{R}\xi_1 + \cdots + \mathbb{R}\xi_l$$

for $F = \sum^{l} \mathbb{R}_{\geq 0}[\xi_l]$. And we let $V \subseteq N_1(X)^{\vee}$ be the set of linear divisorial forms which vanish on $\langle F \rangle$. Since F is spanned by extremal ray, and each of them is defined over $\mathbb{Q}$, i.e. we have $F_{\mathbb{Q}} = F \cap N_1(X)_{\mathbb{Q}}$, thus V can also defined on $\mathbb{Q}$, i.e. $V_{\mathbb{Q}} = V \cap N^1(X)_{\mathbb{Q}}$.

Since F is extremal, $\langle F \rangle \cap \overline{\mathrm{NE}}(X) = F$. This is because for x in the left side we can write

$$x = f_1 - f_2 \in \langle F \rangle \cap \overline{\mathrm{NE}}(X)$$

where $f_1, f_2 \in F$. So $f_1 = x + f_2 \in F$ which gives that x and f_2 in F. Now, for

$$W_F := \overline{\mathrm{NE}}(X)_{(K_X+\Delta+\varepsilon H)\geq 0} + \sum_{\dim_{v.s.} F_L=1; F_L \not\subseteq F} F_L$$

is a closed cone which intersects $\langle F \rangle$ only at the origin, because we can write

$$w = y + z_1 + \cdots + z_n \in W_F \cap \big(\langle F \rangle \cap \overline{\mathrm{NE}}(X)\big),$$

then y and each z_i is in F. So F is K-negative gives that $\big(y \cdot (K + \varepsilon H)\big) < 0$ for some fixed sufficiently small $\varepsilon > 0$[6], but this contradicts with the fact $y \in \overline{\mathrm{NE}}(X)_{(K+\varepsilon H)\geq 0}$. Thus w can only be 0, and we have $\overline{\mathrm{NE}}(X) = W_F + F$. By separating hyperplane theorem, see [BV04, 2.5.1], there is a hyperplane $\{g = 0\}$ which contains $\langle F \rangle$ but intersects W_F only at the origin. But this linear form g might not be rational.

The supporting functions of F, we mean, are exactly those elements of V which are strictly positive on $W_F \backslash \{0\}$. That is,

$$l \text{ is a supporting function } \Rightarrow \ l \geq 0 \text{ on } \overline{\mathrm{NE}}(X) \text{ and } \overline{\mathrm{NE}}(X) \cap \{l = 0\} = F.$$

This is an open condition since "positive" is such, and it is non-empty since g is such. Thus F has a supporting function which is defined over $\mathbb{Q}$. A suitable multiple of it

$$\overline{\mathrm{NE}}(X) \cap \big\{ \underbrace{ml' = 0}_{l' \text{ is } \mathbb{Q}\text{-Cartier}} \big\} = F,$$

[6] Which we also use this ε to define W_F.

gives a supporting function which corresponds to a Cartier divisor; denote it by D[7].

Step 7: The Contraction Theorem 1.5(3), is an easy consequence of the Basepoint-free Theorem now.

By our assumption, $-(K_X + \Delta)$ is strictly positive on F. This implies that $mD - (K_X + \Delta)$ is strictly positive on $\overline{\mathrm{NE}}(X)\backslash\{0\}$ for $m \gg 1$. Hence by Kleiman's ampleness criterion, $mD - (K_X + \Delta)$ is ample for $m \gg 1$. Thus $|mD|$ is base-point free for $m \gg 1$, by Theorem 1.1. Let g_F be the Stein factorization of the morphism given by $|mD|$ for some $m \gg 1$, i.e.

$$X \xrightarrow{g_F} Z \overset{\text{finite part}}{\longrightarrow} \varphi_{|mD|}(X).$$

Moreover, note that

$$\begin{aligned}\varphi_{|mD|}(C) = \text{point} &\iff \deg_C \varphi_{|mD|}^* \mathscr{O}(1) = 0 \\ &\iff [C] \in mD^\perp \cap \overline{\mathrm{NE}}(X) = F.\end{aligned}$$

So, this is the contraction morphism associated to F.

Step 8: If $F \neq \{0\}$, then the contraction morphism is not an isomorphism since mD is not very ample, by Kleiman's ampleness criterion. Thus, at least one curve $C \subseteq X$ is contracted by g_F. This shows that each $(K_X + \Delta)$-negative extremal face contains the class of a curve.

If X is smooth then we can use Bend and Break method to show that there is a rational curve in F, but at here X is just normal. So we need the method in [Kaw91] to produce rational curves.

First g_F is the projective morphism in Theorem 1.20[8]. And there exists $\mathrm{Ex}(g_F)$, while we can pick out an irreducible component E of the maximal dimension, since we are in noetherian scheme. Recall that $mD - (K_X + \Delta)$ is ample, and mD defines the morphism $\varphi_{|mD|}$, so there is a Cartier divisor A_Z such that

$$mD = g_F^* A_Z.$$

Thus, we have

$$-(K_X + \Delta) \sim_{\mathbb{Q}} \underbrace{mD - (K_X + \Delta)}_{\text{ample}} - g_F^* A_Z$$

is a g_F-ample $\mathbb{Q}$-divisor. Now, by Theorem 1.20 we see the general points of E is covered by rational curves $\{C_j\}$ such that $g_F(C_j)$ are points, i.e. $[C_j] \in F$. This also

[7] Recall that $\overline{\mathrm{NE}}(X)$ and $\mathrm{Nef}(X)$ are dual to each other, so D is a nef divisor.

[8] Recall that X is normal and $Z = \underline{\mathrm{Spec}}_Y \varphi_{|mD|*} \mathscr{O}_X$, so locally Z has coordinate ring $H^0(\varphi_{|mD|}^{-1} U, \mathscr{O}_U)$ gives that Z is normal.

shows that these $F_L = \mathbb{R}_{\geq 0}[C_j]$ for such rational curves C_j. We finish the proof of Theorem 1.2(1) and (2).

Step 9: We now prove (4). Let F be a $(K_X + \Delta)$-negative extremal face. Let D be **any** supporting Cartier divisor of F, $g_F : X \longrightarrow Z$ the contraction morphism associated to F and $W_F \subseteq \overline{\mathrm{NE}}(X)$ as in *Step 6*. As we showed in *Step 7*, g_F is given by $|mD|$ for **every** $m \gg 1$. The reason is for two Stein factorizations

$$X \xrightarrow{g_m} Z_m \longrightarrow \varphi_{|mD|}(X) \quad \text{and} \quad X \xrightarrow{g_n} Z_n \longrightarrow \varphi_{|nD|}(X),$$

Proposition 1.4.38 gives that there is a (proper) morphism $\pi : Z_m \longrightarrow Z_n$. Moreover, it is finite and birational between normal algebraic varieties, so π is an isomorphism (cf. [Har77, III, 11.4]). Thus, both mD and $(m+1)D$[9] is a pullback of the Cartier divisors A_m and A_{m+1} on Z, respectively. Then we get that $D = g_F^*(A_{m+1} - A_m) =: g_F^* D_Z$, where D_Z is a Cartier divisor on Z.

Let L be any Cartier divisor on X such that $(L \cdot C) = 0$ for every $[C] \in F$. Since D is positive on $W_F \backslash \{0\}$, $L+mD$ is also positive on $W_F \backslash \{0\}$ for $m \gg 1$. Thus $L+mD$ is also a supporting Cartier divisor of F for $m \gg 1$, hence as we just proved, there is a Cartier divisor M_Z such that $L + mD = g_F^* M_Z$[10]. We set

$$L_Z := M_Z - mD_Z$$

to obtain the Theorem 1.5(4). □

The following two corollaries examine the divisor theory of contractions.

Corollary 6.1.25. *Let (X, Δ) be a projective morphism klt pair such that $K_X + \Delta$ is $\mathbb{Q}$-Cartier, and $R \subseteq \overline{\mathrm{NE}}(X)$ a $(K_X + \Delta)$-negative extremal ray with contraction morphism $g_R : X \longrightarrow Z$. Let $C \subseteq X$ be a curve which generates R. We have an exact sequence*

$$0 \longrightarrow \mathrm{Pic}\, Z \xrightarrow{L \mapsto g_R^* L} \mathrm{Pic}\, X \xrightarrow{M \mapsto (M \cdot C)} \mathbb{Z}.$$

In particular, $\rho(Z) = \rho(X) - 1$.

Proof. Let L be a line bundle on Z. Then

$$g_{R*} g_R^* L \cong L \otimes g_{R*} \mathscr{O}_X = L$$

[9] Recall as in the proof of the Basepoint-free Theorem, m and $m+1$ can be written as the linear combination of s^u and s'^v

[10] Similar as above, can use Proposition 1.4.38 to show that the contraction morphism is only depends on the choice of extremal face, not the related free series. Thus $|mD|$ and $|k(L + mD)|$ will gives the same contraction g_F.

thanks to the projection formula and g_R is a contraction. Thus, $L \mapsto g_R^* L$ is an injection. If M is a line bundle on X and $(M \cdot C) = 0$, then there is a line bundle L_Z such that $M = g_R^* L_Z$ by Theorem 1.5(4). □

Corollary 6.1.26. *Let (X, Δ) be a projective klt pair such that $K_X + \Delta$ is $\mathbb{Q}$-Cartier, and $R \subseteq \overline{\mathrm{NE}}(X)$ is a $(K_X + \Delta)$-negative extremal ray with contraction morphism $g_R : X \longrightarrow Z$. Assume that X is $\mathbb{Q}$-factorial, and g_R is either a divisorial or fibre type contraction. Then Z is also $\mathbb{Q}$-factorial.*

Proof. Assume first that g_R is a divisorial contraction with exceptional divisor $E \subseteq X$. Note that $(E \cdot R) < 0$, since if $(E \cdot R) \geq 0$ implies that E is g_R-nef, and $g_{R*}E = 0$. So by Negativity Lemma 5.2.17, we can get that $-E \geq 0$ contradicts with $E > 0$. In particular, $(E \cdot R) \neq 0$. Let B by any Weil divisor on Z. Since X is $\mathbb{Q}$-factorial, there is a rational number s such that

$$\big((g_{R*}^{-1}B + sE) \cdot R\big) = 0,$$

where $m(g_{R*}^{-1}B + sE)$ is a Cartier divisor on X for some $m > 0$, and by Cone Theorem 1.5(4) it is the pullback of a Cartier divisor M_Z on Z. Thus

$$\begin{aligned} mB &= m\big(g_{R*}(g_{R*}^{-1}B) + s g_{R*}E\big) \\ &= g_{R*}\big(m(g_{R*}^{-1}B + sE)\big) \\ &= g_{R*}g_R^* M_Z \\ &\sim_{lin} M_Z, \end{aligned}$$

so B is $\mathbb{Q}$-Cartier.

Assume next that g_R is a fibre type contraction. Let B by any Weil divisor on Z and $Z^o \subseteq Z$ be the smooth locus. Let $D \subseteq X$ be the closure of $g_R^{-1}(B|_{Z^o})$, it is a Weil divisor on X. And D is disjoint from the general fibre of g_R, especially over the $Z \setminus \operatorname{Supp} B$. We can pick a curve C inside the general fibre, which contracted by g_R so $[C] \in R$ and $R = \mathbb{R}_{\geq 0}[C]$. Then $(D \cdot R) > 0$. mD is a Cartier divisor for some $m > 0$ and by Cone Theorem 1.5(4) it is the pullback of a Cartier divisor M_Z on Z. Moreover, we have $g_R^* \mathscr{O}_{Z^o}(mB|_{Z^o} - M_Z|_{Z^o}) \cong \mathscr{O}_{g^{-1}Z^o}$, and we have $g_{R*}\mathscr{O}_{g^{-1}Z^o} = \mathscr{O}_{Z^o}$. So the group map

$$g_R^* : \mathrm{Pic}(Z^o) \longrightarrow \mathrm{Pic}(g^{-1}Z^o)$$

is injective, thanks to the projection formula. Then we have

$$\mathscr{O}_{Z^o}(mB|_{Z^o} - M_Z|_{Z^0}) \cong \mathscr{O}_{Z^o},$$

i.e. $mB \sim_{lin} M_Z$ since $\mathrm{codim}_Z(Z \setminus Z^o) \geq 2$. So B is $\mathbb{Q}$-Cartier. □

6.1C The Rationality Theorem

The proof is taken from [Kaw84, 3.1] with simplifications and additions as in [Kol84].

The main idea is the following.

Assume for simplicity that X is smooth, H is ample and $\Delta = 0$. We want to use the *basepoint freeness* method to show that certain linear systems $|pH + qK_X|$ are basepoint free. By looking at the proof carefully, we see that a crucial part is the vanishing

$$H^1\big(X, \mathscr{O}_X(pH + qK_X)\big) = 0,$$

which holds if $pH + qK_X$ is nef and big, via Theorem 4.3.1. If we can choose p, q such that

$$\frac{q-1}{p} < r < \frac{q}{p}, \tag{6.14}$$

where $r = \max\{t \in \mathbb{R} | H + tK_X \text{ is nef}\}$. Then the above vanishing holds by we choose $D = pH + qK_X$, and

$$H^1\Big(X, \mathscr{O}_X\big(K_X + (D - K_X)\big)\Big) = 0,$$

where $D - K_X$ is ample, since it is inside $\mathrm{Nef}(X)$. But $|pH + qK_X|$ is not basepoint free, since it is not even nef. Of course we need other vanishing results as well, and so we can prove only that if there are lots of pair (p, q) satisfying (6.14), then we get a contraction. We show in *Step 3* how the existence of such pairs is connected with the arithmetic properties of r.

Now we begin to prove the Rationality Theorem:

Proof. **Step 1:** In this step we reduce the general case of Theorem 1.3 when H is basepoint free. Assume that $a(K_X + \Delta)$ is Cartier. Then, we set

$$D = c\left(H + \frac{da}{c}(K_X + \Delta)\right)$$

where c is sufficiently large such that $da/c < r(H)$, hence that D is a nef Cartier divisor. And we can choose $s = 1/da$ such that

$$\begin{aligned} sD - (K_X + \Delta) &= \frac{1}{da}\big(cH + da(K_X + \Delta)\big) - (K_X + \Delta) \\ &= \frac{c}{da}H \end{aligned}$$

is a nef and big divisor, since H is. So by Basepoint-free Theorem, we have

$$H' := m\big(cH + da(K_X + \Delta)\big)$$

is basepoint free for $m \gg c \gg d > 0$. For any real number λ,

$$\begin{aligned} H' + \lambda(K_X + \Delta) &= mcH + (mda + \lambda)(K_X + \Delta) \\ &= mc\left(H + \frac{mda + \lambda}{mc}(K_X + \Delta)\right), \end{aligned}$$

and it gives that

$$r(H) = \frac{r(H') + mda}{mc}.$$

Thus $r(H)$ is a rational number if and only if $r(H')$ is. Assume furthermore that $r(H')$ has a form u/v, then

$$r(H) = \frac{u + vmda}{mcv}.$$

So the denominator q of $r(H)$ dividing mcv, but as m and c can be chosen in sufficiently large, we can have q is relative prime with mc such that q dividing v. Therefore,

$$q \leq v \leq a(X) \cdot (\dim X + 1).$$

Now, we turn to prove the case when H is basepoint free.

Step 2: Suppose Y is a smooth proper variety, $\{D_i\}_{i=1}^k$ is a finite collection of Cartier divisors and A is an snc $\mathbb{Q}$-divisor with $\lceil A \rceil$ effective. Consider the Hilbert polynomial

$$P(u_1, \ldots, u_k) = \chi\left(Y, \sum^k u_i D_i + \lceil A \rceil\right).$$

Suppose that, for some values of the u_i, $\sum u_i D_i$ is nef and $\sum u_i D_i + A - K_Y$ is nef and big. Then, for all integers $m \gg 0$, $\sum mu_i D_i + A - K_Y$ is still nef and big by Example 3.3.12, so that

$$H^i\left(Y, \sum mu_i D_i + \lceil A \rceil\right) = H^i\left(Y, K_Y + \lceil \sum mu_i D_i + A - K_Y \rceil\right) = 0$$

for every $i > 0$, by the Kawamata-Viehweg Vanishing Theorem. By Corollary 5.2.20(3), $(Y, -A)$ has klt singularities, then we can use the Non-vanishing Theorem to get that $\mathscr{O}_Y\left(\sum mu_i D_i + \lceil A \rceil\right)$ must have a global section. Therefore, $\chi\left(\sum mu_i D_i + \lceil A \rceil\right) \neq 0$.

Thus, $P(u_1, \ldots, u_k)$ is not identically zero and its degree is $\leq \dim Y$ due to the asymptotic Riemann-Roch[11].

Step 3: We need a result about zeros of polynomials:

[11] Also, we can view this geometrically. We denote $D = \sum u_i D_i$ and $\mathscr{F} = \mathscr{O}_Y(\lceil A \rceil)$, then we have $\chi(\sum mu_i D_i + \lceil A \rceil) = \chi\big(Y, \mathscr{F}(mD)\big)$. Consider

$$Q(m) = P(mu_1, \ldots, mu_k) = \chi\big(Y, \mathscr{F}(mD)\big) = a_n m^n + O(m^{n-1})$$

by the asymptotic Riemann-Roch, we can introduce a new variable $t_1 + \cdots + t_n$ such that

$$Q(t_1 + \cdots + t_n) = \chi\Big(Y, \mathscr{F}\big((t_1 + \cdots + t_n)D\big)\Big) = a_n \cdot (t_1 + \cdots + t_n)^n + \text{lower degree terms},$$

Lemma 6.1.27. *Let $P(x,y) \in \mathbb{R}[x,y]$ be a non-trivial polynomial of total degree $\leq n$. Fix some $a \in \mathbb{Z}_{>0}$, $\varepsilon > 0$ and $r \in \mathbb{R}$. Suppose that there exists $R > 0$ such that $P(x,y) = 0$ for every $(x,y) \in \mathbb{Z}^2$ satisfying*

$$\|(x,y)\| > R \quad \text{and} \quad 0 < ay - rx < \varepsilon.$$

Then $r \in \mathbb{Q}$. More precisely, if $r = u/v$ in lowest terms with $v > 0$, then

$$v \leq \frac{a(n+1)}{\varepsilon}.$$

Proof. For the first statement, we assume that r is irrational, thus $\alpha := r/a$ is also irrational. We claim that for any $\delta > 0$, can find $(x,y) \in \mathbb{Z}^2$ such that $0 < ay - rx < \delta$. In fact, for any given large enough $N \in \mathbb{N}$, the $N+1$ number of the fractional part $0, \{\alpha\}, \{2\alpha\}, \ldots, \{N\alpha\}$ will cut the interval $[0,1)$ into N small parts each of them has $1/N$ length. So there is at least two members lie in the same small interval, which we denote it by $\{i\alpha\}$ and $\{j\alpha\}$ for $i < j$. Then

$$\Big|\{i\alpha\} - \{j\alpha\}\Big| < \frac{1}{N}. \tag{6.15}$$

Set $q = j - i$, then follows from (6.15) after we cancel out the non-fractional part, we can get that there is an integer $p \in \mathbb{Z}$ such that $|q\alpha - p| < 1/N$. Since $\alpha \notin \mathbb{Q}$, $q\alpha - p \neq 0$. Hence we can get

$$0 < q\alpha - p < \frac{1}{N} \quad \text{or} \quad 0 < (-q)\alpha - (-p) < \frac{1}{N},$$

while we can take $(x,y) = (p,q)$ or $(-p,-q)$ such that $0 < ay - rx < \delta$ by we choose $N > a/\delta$. Moreover, note that when $\delta \to 0$ we must have $\|(x,y)\| \to \infty$, otherwise there are only finite member of the pair (x,y) inside $\|(x,y)\| \leq R$ such that one of them eventually satisfies $ay - rx = 0$, contradicts with the fact $r \notin \mathbb{Q}$.

Now, we take such pair (x',y') satisfies that

$$\|(x',y')\| > R \ \text{ and } \ 0 < ay' - rx' < \varepsilon/(n+2).$$

Then for integers $j = 1, \ldots, n+1$, we have

$$0 < a(jy') - r(jx') = j(ay' - rx') < \varepsilon \ \text{ and } \ \|(jx', jy')\| > R.$$

---

where $a_n \cdot n! = (D^n \cdot \mathscr{F})$. Therefore,

$$\begin{aligned} Q(m) &= P(mu_1, \ldots, mu_k) \\ &= \frac{(D^n \cdot \mathscr{F})}{n!} m^n = O(m^{n-1}) \\ &= \frac{\big((\sum u_i D_i)^n \cdot \mathscr{F}\big)}{n!} + O(m^{n-1}). \end{aligned}$$

So by our assumption, $P(jx', jy') = 0$ for $j = 1, \ldots, n+1$. Set $Q(t) = P(tx', ty')$, we have $\deg Q \leq n$ since P is. But, $Q(t)$ has $n+1$ different roots, so $Q \equiv 0$ and implies that P is identically zero on the line $y'x - x'y = 0$, i.e. $y'x - x'y$ must divide P. Since δ can be arbitrarily small, we can repeat this process $n+1$ times, that is in each time we pick a new pair $(x^{(i)}, y^{(i)})$ produce a new line $x^{(i)}y - y^{(i)}x = 0$ such that P has $n+1$ different roots. This contradicts with that $\deg P \leq n$, so r must be a rational number.

For the second statement, we suppose that $r = u/v$ (in lowest terms with $v > 0$). For any $j \in \mathbb{Z}_{>0}$, consider the line $ay - rx = aj/v (=: l)$ which is equivalent to $avy - ux = aj$. Since $\gcd(u, av) = \gcd(u, a)$ divides a, so[12] we can find an integral solution (x_j, y_j). And for each $k \in \mathbb{Z}$, $(x_j + kav, y_j + ku)$ us also an integral solution, thus on the line l there exists integral pairs with arbitrarily large norm.

Now, if we have $aj/v < \varepsilon$, then $0 < ay - rx < \varepsilon$ implies that P is vanishing on some integral pairs on the line l, by we choose the pairs sufficiently large. But, there are infinitely many such pairs as k vary, so P is identically zero on l, i.e. $ay - rx - (aj/v)$ must divide P. Since $\deg P \leq n$, there only exists n numbers of such j. So if $v > a(n+1)/\varepsilon$, then for $j = 1, \ldots, n+1$ we have

$$0 < \frac{aj}{v} < \frac{a(n+1)}{v} < \varepsilon$$

which will gives that $n+1$ different roots of the from $ay - rx - (aj/v)$, contradicts with the fact $\deg P \leq n$. □

Step 4: Fix $\varepsilon > 0$ and H be a nef and big Cartier divisor which is basepoint free. Let $a \in \mathbb{Z}_{>0}$ be such that $a(K_X + \Delta)$ is also Cartier. Assume that $K_X + \Delta$ is not nef and let $r = \max\{t \in \mathbb{R} | H + r(K_X + \Delta) \text{ is nef}\}$.

For each $(p, q) \in \mathbb{Z}^2$, let $L(p, q)$ denote the set-theoretic base locus of the series $|pH + qa(K_X + \Delta)|$ on X. By definition,

$$L(p, q) = X \text{ if and only if } |pH + qa(K_X + \Delta)| \neq \emptyset.$$

Claim 6.1.28. For (p, q) sufficiently large and $0 < aq - rp < \varepsilon$, $L(p, q)$ will eventually stabilizes to the same subset of X. We call this subset L_{stab}.

Proof. When there is no (p, q) satisfies $0 < aq - rp < \varepsilon$, the statement is vacuously true, so we assume that such (p, q) exists.

[12] Easy to verify that the Diophantine equation $Ax + By = C$ has solution if and only if $\gcd(A, B) | C$.

Step 1: First we reduce the case that H can be ample. For this basepoint free Cartier divisor H, we take a Stein factorization of its associated morphism $\varphi_{|H|}$:

$$X \xrightarrow{\varphi} X_0 \xrightarrow{\nu} \varphi_{|H|}(X),$$

where X_0 is normal[13], φ is a birational contraction and ν is finite. We denote $K := K_X + \Delta$. Set $H_0 := \nu^*\mathscr{O}(1)$ is an ample divisor, we have $H = \varphi^* H_0$. Note that a curve C is contracted by φ if and only if $(H \cdot C) = 0$, since H_0 is ample, and at same time it will gives that $(K \cdot C) = 0$[14].

Now, we aim to prove that there is a $\mathbb{Q}$-Cartier divisor K_0 on X_0 such that $K \sim_{\mathbb{Q}} \varphi^* K_0$, and moreover K_0 has the form $K_0 = K_{X_0} + \Delta_0$. Take a sufficiently large integer N such that $a/N < r$ and $(a-1)/N < r$, and set a Cartier divisor $D_N := NH + aK$. It is nef and big, since

$$D_N = N\Big(\underbrace{(1 - \frac{a}{Nr})H}_{\text{nef and big}} + \frac{a}{Nr} \underbrace{(H + rK)}_{\text{nef}} \Big). \tag{6.16}$$

We replace a by $a-1$ can easily see that $D_N - K$ is also nef and big $\mathbb{Q}$-divisor. So by the Basepoint-free Theorem we see D_N is semi-ample, we can denote its associated fibre space by φ_N. Combine the decomposition (6.16) with footnote 14 gives that

$$(D_N \cdot C) = 0 \Longleftrightarrow (H \cdot C) = 0,$$

that is, they contract the same curve with $\mathrm{NE}(\varphi) = \mathrm{NE}(\varphi_N)$. Then by the Rigidity Lemma and Proposition 1.4.38(b) we see φ and φ_N are the same birational contraction. Thus, there is an ample divisor $D_{N,0}$ such that $D_N \sim_{lin} \varphi^* D_{N,0}$[15]. So,

$$\begin{aligned} aK &= D_N - NH \\ &\sim_{lin} \varphi^*(D_{N,0} - NH_0). \end{aligned}$$

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[13] Since X is normal, cf.[Har77, III, 11.5].

[14] Recall that $H = mc(H_{\text{old}} + \frac{da}{c}K)$ for $da/c < r_{\text{old}}$, and

$$H_{\text{old}} + \frac{da}{c}K = \left(1 - \frac{da}{r_{\text{old}}c}H_{\text{old}}\right) + \frac{da}{r_{\text{old}}c}(H_{\text{old}} + r_{\text{old}}K)$$

gives that $H_{\text{old}} + \frac{da}{c}K$ is nef. So if $(H \cdot C) = 0$, then solve the linear equation

$$\begin{cases} (H_{\text{old}} \cdot C) \geq 0, \\ \big((H_{\text{old}} + r_{\text{old}}K) \cdot C\big) \geq 0, \\ (H_{\text{old}} + \frac{da}{c}K) \cdot C) = 0, \\ 0 < \frac{da}{c} < r_{\text{old}}, \end{cases} \quad \Longrightarrow \quad (K \cdot C) = 0.$$

[15] Use the method in *Step 9* of the proof of the Cone Theorem to find that bD_N and $(b+1)D_N$ are pullback some Cartier divisors. And actually it is differ from a sign of an isomorphism between the target space of φ and φ_N, but never mind.

Moreover, since φ is a birational morphism between normal varieties, the image of φ-exceptional part has codimension at least two, thus we can choose a suitable representative such that $\varphi_* K_X = K_{X_0}$. More precisely, we push forward the

$$\begin{aligned}
\varphi_* K_X &= \varphi_* \operatorname{div}_X(\omega), \text{ where } \omega \in \Omega^{\mathrm{top}}_{k(X)} \\
&= \varphi_* \left(\sum_{P \subseteq X_0} \operatorname*{ord}_{\tilde{P}}(\omega)\tilde{P} + \sum_{E:\varphi\text{-exceptional}} \operatorname*{ord}_{E}(\omega)E \right) \\
&= \varphi_* \left(\sum_{P \subseteq X_0} \operatorname*{ord}_{P}(\omega)\tilde{P} + \varphi\text{-exceptional part} \right), \text{ since } \mathscr{O}_{X,\eta_{\tilde{P}}} \cong \mathscr{O}_{X_0,\eta_P} \text{ as D.V.R.} \\
&= \sum_{P \subseteq X_0} \operatorname*{ord}_{P}(\omega)P, \text{ since } \deg \varphi = 1 \\
&= \operatorname{div}_{X_0}(\omega) \\
&= K_{X_0}.
\end{aligned}$$

Also, we set $\Delta_0 = \varphi_* \Delta$. Then we have

$$K_X + \Delta \sim_{lin} \varphi^* \varphi_* (K_X + \Delta) = \varphi^* (K_{X_0} + \Delta_0),$$

where $K_{X_0} + \Delta_0$ is $\mathbb{Q}$-Cartier such that it can be pullback is promised by

$$K_{X_0} + \Delta_0 = \varphi_*(K_X + \Delta) \sim_{lin} \frac{1}{a}(D_{N,0} - NH_0).$$

Furthermore, we set $\Theta = K_X + \Delta - \varphi^*(K_{X_0} + \Delta_0)$, and then Θ and $-\Theta$ both be φ-nef since for every relative φ-curve C we have $\big((K_X + \Delta) \cdot C\big) = 0$. So by the Negativity Lemma 5.2.17 we see $\Theta = 0$, i.e. $K = \varphi^* K_0$.

Now we have

$$H = \varphi^* H_0, \quad K = \varphi^* K_0, \quad H_0 \text{ is ample.}$$

Consider $D(p,q) := pH + qaK$ and $D_0(p,q) := pH_0 + qaK_0$. Due to above we have $D(p,q) = \varphi^* D_0(p,q)$. Moreover,

$$\begin{aligned}
H^0\Big(X, \mathscr{O}_X\big(D(p,q)\big)\Big) &= H^0\Big(X_0, \varphi_*\varphi^* \mathscr{O}_X\big(D_0(p,q)\big)\Big) \\
&\cong H^0\Big(X_0, \mathscr{O}_X\big(D_0(p,q)\big)\Big)
\end{aligned}$$

thanks to the projection formula and $\varphi_* \mathscr{O}_X = \mathscr{O}_{X_0}$, so $L_X(p,q) = \varphi^{-1}\big(L_{X_0}(p,q)\big)$ for every p, q as sets and we only need to show that $L_{X_0}(p,q)$ will stabilizes on X_0. Furthermore, since $H + tK = \varphi^*(H_0 + tK_0)$, the threshold number r won't be changed.

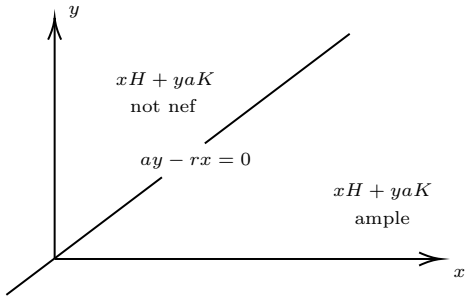
Step 2: From now on, we assume that H is an ample divisor in this claim[16], by replace X by X_0, H by H_0 and K by K_0.

[16] Of course at this time X is projective, not only proper.

When $ay - rx < 0$, then $0 < \frac{ay}{x} =: t < r$ and

$$H + tK = \underbrace{\left(1 - \frac{t}{r}\right) H}_{\text{ample}} + \frac{t}{r}(\underbrace{H + rK}_{\text{nef}})$$

implies that $H + tK$ is ample, so is $D(x, y) = x(H + tK)$. And when $0 < ay - rx$ we have $t > r$, thus $H + tK$ is not nef and same as $D(x, y)$. So we have the following diagram



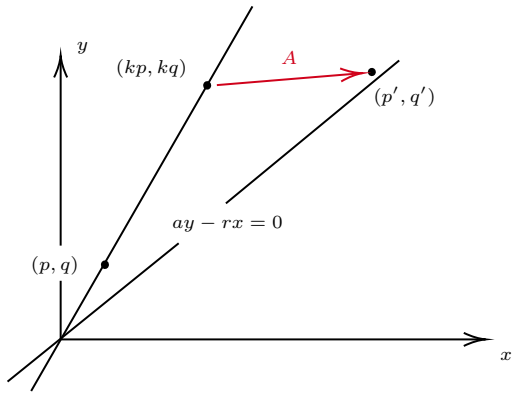
We begin with an arbitrary integral pair (p, q) satisfies that $0 < aq - rp =: \delta < \varepsilon$, and show that for any sufficiently large (p', q') which satisfies the claimed condition, there is $L(p', q') \subseteq L(p, q)$. Take any rational number $0 < t_0 < r$ and set $\gamma := r - t_0 > 0$. Since

$$H + t_0 K = \underbrace{\left(1 - \frac{t_0}{r}\right) H}_{\text{ample}} + \frac{t_0}{r}(\underbrace{H + rK}_{\text{nef}}),$$

we see $H + t_0 K$ is ample. Take any such pair (p', q') and denote $aq' - rp'$ by δ'. We define

$$k := \left\lfloor \frac{\gamma p'}{\delta + \gamma p} \right\rfloor$$

and consider the arrow from (kp, kq) to (p', q'): $(x, y) = (p' - kp, q' - kq)$.



It is corresponds to the divisor

$$A := D(p', q') - kD(p, q) = xH + yaK.$$

Note that $k \leq \frac{\gamma p'}{\delta+\gamma p}$, so

$$x = p' - kp \geq p' - \frac{\gamma p'}{\delta + \gamma p} p = p' \frac{\delta}{\delta + \gamma p} \to \infty \text{ as } p' \to \infty,$$

thus we see the length of the arrow will tend to infinity. Moreover, the direction of this arrow will be tend to $H + t_0 K$, since from

$$\begin{aligned} ay - t_0 x &= a(q' - kq) + t_0(p' - kp) \\ &= (aq' - rp') + (r - t_0)p' - k\big((aq - rp) + (r - t_0)p\big) \\ &= \delta' + \gamma p' - k(\delta + \gamma p). \end{aligned}$$

And from $k = \lfloor \frac{\gamma p'}{\delta+\gamma p} \rfloor$, we see $0 \leq \gamma p' - k(\delta + \gamma p) < \delta + \gamma p$. Combine this with $0 < \delta' < \varepsilon$, can get that $0 < ay - t_0 x < (\delta + \gamma p) + \varepsilon$. Divide x on both sides to get

$$t_0 < \frac{ay}{x} < t_0 + \frac{(\delta + \gamma p) + \varepsilon}{x}. \tag{6.17}$$

Then as $x \to \infty$, we have $\frac{ay}{x} \to t_0$. In particular, since $0 < t_0 < r$, for every sufficiently large (p', q') we have $0 < \frac{ay}{x} < r$, and hence

$$A = x\left(H + \frac{ay}{x} K\right)$$

is ample.

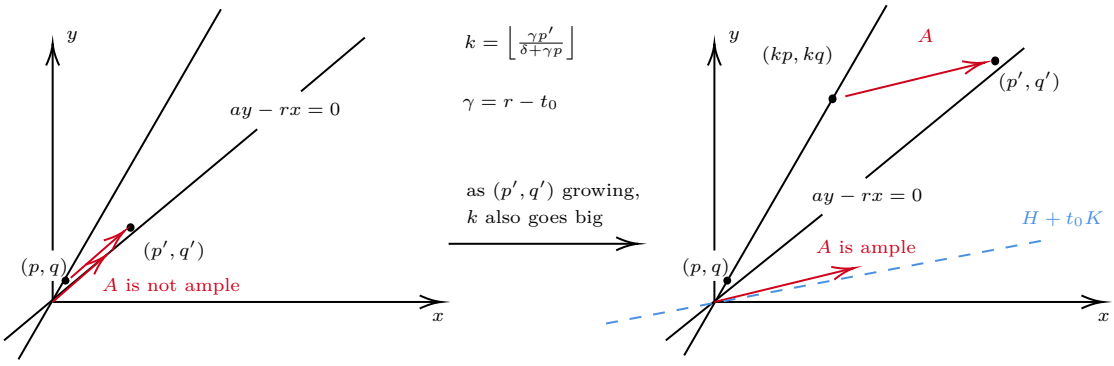


Figure 6.1.: By suitable choice of the parameters, as p' growing the corresponding divisor A are tend to an ample divisor, hence A itself will become ample.

Moreover, we need enhance the conclusion that A can be very ample. Since $t_0 \in \mathbb{Q}$, we can write $t_0 = aV/U$, Can take U and V sufficiently large simultaneously such

that $\Xi := U(H + t_0K)$ is a very ample Cartier[17] divisor. We substitute this t_0 into the above of (6.17), to get that

$$0 < \frac{a}{U}(Uy - Vx) < (\delta + \gamma p) + \varepsilon.$$

Thus the integral number $Uy - Vx$ is bounded, which can only has finitely many choice of values. Use Euclidean division to divide x by U, can get

$$x = nU + s,\ 0 \leq s < U.$$

But since s has only finitely many choice of values, and same for $Uy - Vx$, so in the form $y = nV + t$ the number $t = \frac{Uy - Vx + Vs}{U}$ also has finitely many of choice. Therefore,

$$\begin{aligned} A &= n(UH + VaK) + (sH + taK) \\ &= n\Xi + R_j \end{aligned}$$

where R_j belongs to a finite set, which corresponding to the finite choice of s and t. For each R_j, since Ξ is very ample so it is ample, we can take a large enough n such that $n\Xi + R_j$ is generated by global sections. Furthermore, by [Har77, II, ex7.5(d)] can replace n by $n + 1$ such that $n\Xi + R_j$ is very ample. Since there are only finitely many R_j, can choose a maximal n such that A is very ample (provided by we can take (p', q') sufficiently large so that n can vary).

Step 3: Now, we have the decomposition

$$D(p', q') = kD(p, q) + A$$

where A is a very ample Cartier divisor. Suppose that the point $z \notin L(p, q)$, then there is a non-trivial section $s \in H^0\Big(X, \mathscr{O}_X\big(D(p, q)\big)\Big)$ such that $s(z) \neq 0$ in $\mathscr{O}_{X,z}/m_z = \mathbb{C}$. Since $|A|$ is basepoint free, there exists $t \in H^0\big(X, \mathscr{O}_X(A)\big)$ such that $t(z) \neq 0$. Thus, via the natural multiplication

$$\begin{aligned} \mu : H^0\Big(X, \mathscr{O}_X\big(D(p, q)\big)\Big)^{\otimes k} \otimes H^0\big(X, \mathscr{O}_X(A)\big) &\longrightarrow H^0\Big(X, \mathscr{O}_X\big(D(p', q')\big)\Big) \\ s^{\otimes k} \otimes t &\longmapsto s^k t \end{aligned}$$

we can see that $(s^k t)(z) = s(z)^k t(z) \neq 0$. Therefore, $z \notin L(p', q')$. Hence we prove that $L(p', q') \subseteq L(p, q)$.

Now, we can replace (p, q) by (p', q') and repeat this process again and again to obtain a descending chain of closed subsets. Since we are in the noetherian topological space, this chain must stabilizes to a minimal element L_{Stab}. □

[17] Recall that aK is Cartier.

Now we take $\varepsilon = 1$, and for (p,q) as in claimed the series $\big|pH + qa(K_X + \Delta)\big|$ cannot be basepoint free on X, since $qa/p > r$ such that $pH + qa(K_X + \Delta)$ is not nef. We let $I \subseteq \mathbb{Z} \times \mathbb{Z}$ be the set of (p,q) for which $0 < aq - rp < 1$ and $L(p,q) = L_{\mathrm{Stab}}$. Let us emphasize that I contains all sufficiently large (p,q).

Step 5: Now we back to the proof of the Rationality Theorem.

We take a log resolution $g : Y \longrightarrow X$ of the klt pair (X, Δ), we denote that $D_1 = g^*H$, $D_2 = g^*\big(a(K_X + \Delta)\big)$ and define A by the discrepancy formula

$$K_Y + g^*(K_X + \Delta) + A.$$

Note that $\lceil A \rceil$ is effective and g-exceptional, since $\lfloor \Delta \rfloor = 0$.

Then we put $P(x,y) = \chi\big(Y, xD_1 + yD_2 + \lceil A \rceil\big)$.

Since D_1 is nef and big[18], and we can choose $u_1 = aM$ for some $M < 1/r$, $u_2 = 1$ such that

$$u_1 D_1 + u_2 D_2 = aMg^*\Big(H + \underbrace{\frac{1}{M}(K_X + \Delta)}_{\text{nef}}\Big) =: T$$

is a nef Cartier divisor. And from

$$\begin{aligned} u_1 D_1 + u_2 D_2 + A - K_Y &= aMg^*\Big(H + \frac{1}{M}(K_X + \Delta)\Big) - g^*(K_X + \Delta) \\ &= aMD_1 + \frac{a-1}{a} D_2 \\ &= \underbrace{MD_1}_{\text{nef and big}} + \underbrace{\frac{a-1}{a} T}_{\text{nef}}, \end{aligned}$$

we see it is nef and big, by Example 3.3.12. Therefore, the Hilbert polynomial P is not identically zero by *Step 2*. Moreover, since $\lceil A \rceil$ is effective and g-exceptional Cartier divisor, we have

$$\begin{aligned} H^0\big(Y, \mathscr{O}_Y(pD_1 + qD_2 + \lceil A \rceil)\big) &= H^0\Big(X, \mathscr{O}_X\big(pH + qa(K_X + \Delta)\big) \otimes g_*\mathscr{O}_Y(\lceil A \rceil)\Big) \\ &= H^0\Big(X, \mathscr{O}_X\big(pH + qa(K_X + \Delta)\big)\Big) \end{aligned}$$

thanks to the projection formula and the algebraic Hartogs such that $g_*\mathscr{O}_Y(\lceil A \rceil) \cong \mathscr{O}_X$, while as $g(\lceil A \rceil)$ has codimension at least two[19].

[18] See Example 1.4.3, Proposition 3.2.36 and Remark 3.3.10.

[19] From Zariski's Main Theorem and $\lceil A \rceil \geq 0$, we have $\mathscr{O}_X \cong g_*\mathscr{O}_Y \hookrightarrow g_*\mathscr{O}_Y(\lceil A \rceil)$. Then by the algebraic Hartogs and $\operatorname{codim}_X g(\lceil A \rceil) \geq 2$,

$$H^0\big(X, g_*\mathscr{O}_Y(\lceil A \rceil)\big) \hookrightarrow H^0\big(X \backslash g(\lceil A \rceil), \mathscr{O}_X\big) \cong H^0(X, \mathscr{O}_X)$$

to get that $g_*\mathscr{O}_Y(\lceil A \rceil) \hookrightarrow \mathscr{O}_X$.

Step 6: Suppose now that the assertion of the Rationality Theorem that r is rational is false. For values of $x = aM$ and $y = 1$ as above, or moreover the integral solutions of $0 < ay - rx < 1$, we have $xD_1 + yD_2 + A - K_Y$ is a nef and big $\mathbb{Q}$-Cartier divisor. Thus, by the Kawamata-Viehweg Vanishing Theorem,

$$H^i(Y, xD_1 + yD_2 + \lceil A \rceil) = H^i(Y, K_Y + \lceil xD_1 + yD_2 + A - K_Y \rceil) = 0$$

for every $i > 0$. And by the contrapositive of *Step 3*, there must exist arbitrary large (p, q) with $0 < aq - rp < 1$ for which $P(p, q) = h^0(Y, pD_1 + qD_2 + \lceil A \rceil) \neq 0$, since otherwise $P(x, y)$ would vanish "too often" implying that r is a rational number. Combining this result with *Step 4* and *Step 5*, we see for all $(p, q) \in I$ we have

$$\big|pH + qa(K_X + \Delta)\big| = |pD_1 + qD_2 + \lceil A \rceil| \neq \emptyset,$$

that is $L(p, q) = L_{\text{Stab}} \neq X$.

Step 7: For some $(p, q) \in I$, i.e. $0 < aq - rp < 1$ and (p, q) sufficiently large, we choose a log resolution $f : Y \longrightarrow X$ such that

(1) $K_Y = f^*(K_X + \Delta) + \sum a_j F_j$, where $a_j > -1$ by (X, Δ) has klt singularities;

(2) $f^*\big(pH + (qa - 1)(K_X + \Delta)\big) - \sum p_j F_j$ is ample[20] for some rationals $0 < p_j \ll 1$;

(3) $f^*\big|pH + qa(K_X + \Delta)\big| = |L| + \sum r_j F_j$ for some non-negative integers r_j, where L is basepoint free and $\sum r_j F_j$ is the fixed part.

Step 8: Let $(p, q) \in I$ be as chosen in *Step 7*. As before, we can take

$$c = \min_j \left\{ \frac{1 + a_j - p_j}{r_j} \middle| \, r_j > 0 \right\} > 0$$

and suitable permutation of p_i such that such c is unique, hence that

$$\sum \underbrace{(-cr_j + a_j - p_j)}_{\geq -1} F_j = A' - F$$

[20] Note that $pH + (qa - 1)(K_X + \Delta)$ is nef and big. Since when $0 < aq - rp < 1$ we have $\frac{qa-1}{p} < r$ and hence $H + \frac{qa-1}{p}(K_X + \Delta)$ is nef. And moreover, it is equal to

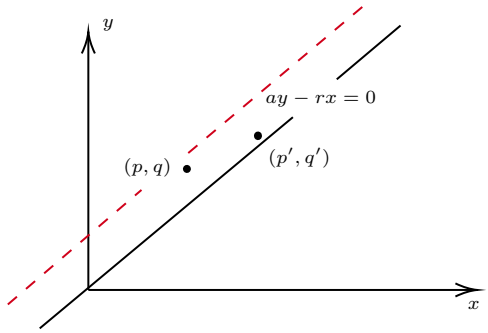
$$\underbrace{(p-1)H}_{\text{nef and big}} + \underbrace{\left(H + \frac{qa-1}{p}(K_X + \Delta)\right)}_{\text{nef}},$$

so by Example 3.3.12 it is also big. f is birational so its pullback is also nef and big, see Example 1.4.3, Proposition 3.2.36 and Remark 3.3.10. Since each F_j are effective, follows from Example 3.2.21 and Example 1.3.15 can get the conclusion. Or just can see Remark 1.8.

where F is the one of F_j's, $\lceil A' \rceil$ is effective and A' not involving F. By our choice of c, the coefficient of F in expression (3) must be non-zero. So we can see F maps into some component B of the base locus $L(p,q) = f(\cup_{r_j>0} F_j)$ of $\big|pH + qa(K_X + \Delta)\big|$[21]. Define

$$\begin{aligned} N(p',q') &:= f^*\big(p'H + q'a(K_X+\Delta)\big) + A' - F - K_Y \\ &= f^*\Big(\big(p' - (1+c)p\big)H + \big(q' - (1+c)q\big)a(K_X+\Delta)\Big) \\ &\quad + f^*\big((1+c)pH + (1+c)qa(K_X+\Delta)\big) \\ &\quad + \sum(-cr_j + a_j - p_j)F_j - K_Y \\ &= cL \text{ (base point free)} \\ &\quad + f^*\Big(\big(p' - (1+c)p\big)H + \big(q' - (1+c)q\big)a(K_X+\Delta)\Big) \\ &\quad + f^*\big(pH + (qa-1)(K_X+\Delta)\big) - \sum p_j F_j \text{ (ample)}. \end{aligned}$$

Notice that if p' and q' are big enough and $aq' - rp' < aq - rp$,



then $\big(q' - (1+c)q\big)a < r\big(p' - (1+c)p\big)$, so the Cartier divisor

$$\big(p' - (1+c)p\big)H + \big(q' - (1+c)q\big)a(K_X+\Delta)$$

is nef. Therefore, $N(p',q')$ is ample by Corollary 1.4.7. Thus, by the Kawamata-Viehweg vanishing Theorem, the following map is surjective

$$H^0\Big(Y, f^*\big(p'H + q'a(K_X+\Delta)\big) + \lceil A' \rceil\Big) \twoheadrightarrow H^0\Big(F, \Big(f^*\big(p'H + q'a(K_X+\Delta)\big) + \lceil A' \rceil\Big)\Big|_F\Big),$$

while notice that by our construction, $\lceil A'|_F \rceil = \lceil A' \rceil|_F$.

Step 9: By the adjunction formula, for $0 < aq' - rp' < aq - rp$, the restriction of the ample divisor $N(p',q')$ to F is the divisor

$$\Big(f^*\big(p'H + q'a(K_X+\Delta)\big) + A'\Big)|_F - K_F,$$

[21] See the proof of Example 3.1.34, the fixed part is comes from the blowing-up of the base ideal sheaf.

which is also ample. So as in *Step 2*, we repeat the process on F to get that the Hilbert polynomial

$$\chi\Big(F, \Big(f^*\big(p'H + q'a(K_X + \Delta)\big) + \lceil A' \rceil\Big)\Big|_F\Big)$$

is not identically zero, since by Corollary 5.2.20(3) the pair $(F, -A'|_F)$ has klt singularities. Furthermore, it is nothing but

$$\chi\Big(F, \Big(f^*\big(p'H+q'a(K_X+\Delta)\big)+\lceil A' \rceil\Big)\Big|_F\Big) = h^0\Big(F, \Big(f^*\big(p'H+q'a(K_X+\Delta)\big)+\lceil A' \rceil\Big)\Big|_F\Big).$$

So, by *Step 3* applied to the Hilbert polynomial on F with $\varepsilon = aq - rp$, there must be arbitrarily large (p', q') such that $0 < aq' - rp' < aq - rp$ and

$$h^0\Big(F, \Big(f^*\big(p'H + q'a(K_X + \Delta)\big) + \lceil A' \rceil\Big)\Big|_F\Big) \neq 0.$$

Step 10: We are now ready to derive a contradiction. By our previous choice, we have $L(p, q) = L_{\mathrm{Stab}}$. For (p', q') as in *Step 9*,

$$H^0\Big(Y, f^*\big(p'H+q'a(K_X+\Delta)\big)+\lceil A' \rceil\Big) \to H^0\Big(F, \Big(f^*\big(p'H+q'a(K_X+\Delta)\big)+\lceil A' \rceil\Big)\Big|_F\Big) \neq 0$$

is surjective. Thus F is not a component of the $\mathrm{Bs}\Big|f^*\big(p'H + q'a(K_X + \Delta)\big) + \lceil A' \rceil\Big|$, that is there is a global section s such that $s|_F \neq 0$. Since $\lceil A' \rceil \geq 0$ is f-exceptional,

$$H^0\Big(Y, f^*\big(p'H + q'a(K_X + \Delta)\big) + \lceil A' \rceil\Big) \overset{f^*}{\cong} H^0\big(X, p'H + q'a(K_X + \Delta)\big)$$

thanks to the projection formula and algebraic Hartogs. And so $f(F)$ is not contained in $L(p', q')$, since otherwise there is a global section t in the right-hand-side such that $s = f^*t \cdot s_{\lceil A' \rceil}$, where $s_{\lceil A' \rceil}$ is the canonical section of $\lceil A' \rceil$. Since A' does not involve F, we can pick a closed point $\xi \in F - \mathrm{Supp}\lceil A' \rceil$ such that $s(\xi) \neq 0$ and $s_{\lceil A' \rceil}(\xi) \neq 0$. Then $(f^*t)(\xi) = t(f(\xi)) \neq 0$, gives the conclusion. But in *Step 8* we have already shown that $f(F) \subseteq L(p, q) = L_{\mathrm{Stab}}$, so this implies that $L(p', q')$ is a proper subset of L_{Stab}, giving the desired contraction with $(p', q') \in I$.

Step 11: So now we know that r is a rational number. We next suppose that the assertion the Rationality Theorem concerning the denominator of r is false. We proceed to a contradiction in much the same way.

Using *Step 3* with $\varepsilon = 1$, conclude like in *Step 6* that there exists arbitrarily large (p_0, q_0) with $0 < aq - rp < 1$ such that

$$P(p_0, q_0) = h^0(Y, p_0D_1 + q_0D_2 + \lceil A \rceil) \neq 0,$$

since otherwise $P(x, y)$ would vanish "too often". Thus

$$\big|p_0H + q_0a(K_X + \Delta)\big| \neq \emptyset,$$

and by *Step 4* we know this holds for all $(p,q) \in I$.

Now choose $(p,q) \in I$ such that $aq - rp$ is the maximum (by we take a translation $(p,q) \mapsto (p + kav, q + ku)$ to make (p,q) sufficiently large), and say it is equal to d/v since $r = u/v$. Choose a resolution f as in *Step 7*. In the strip $0 < aq' - rp' < d/v$, we have as before that

$$\chi\Big(F, \big(f^*\big(p'H+q'a(K_X+\Delta)\big)+\lceil A'\rceil\big)\Big|_F\Big) = h^0\Big(F, \big(f^*\big(p'H+q'a(K_X+\Delta)\big)+\lceil A'\rceil\big)\Big|_F\Big).$$

By the contrapositive of *Step 3*, there exists (p',q') in the strip $0 < aq' - rp' < 1$ with $\varepsilon = 1$ for which

$$h^0\Big(F, \big(f^*\big(p'H + q'a(K_X + \Delta)\big) + \lceil A'\rceil\big)\Big|_F\Big) \neq 0.$$

But then $aq' - rp' < d/v = aq - rp$ automatically, so we obtain the surjection in *Step 9*. The desired contradiction is then derived as in *Steps 7-10*. This completes the proof of the Rationality Theorem. □

6.1D The Non-vanishing Theorem

Before that, we need a lemma which is an application of the Kodaira's vanishing theorem, says the Euler characteristic of a Cartier divisor D depends only on the numerical equivalence class of D.

Lemma 6.1.29. *Let X be a complex projective variety and let D, D' be two Cartier divisors on X such that $D \equiv_{num} D'$. Then*

$$\chi\big(X, \mathscr{O}_X(D)\big) = \chi\big(X, \mathscr{O}_X(D')\big).$$

Proof. Let H be an ample divisor on X. Set

$$P(v) := \chi\big(X, \mathscr{O}_X(D + vH)\big)$$
$$\text{and } P'(u,v) := \chi\Big(X, \mathscr{O}_X\big(D + u(D' - D) + vH\big)\Big).$$

P and P' are polynomials of degree $\leq \dim X$ by asymptotic Riemann-Roch. Fix any $v \gg 1$ such that $D + vH - K_X$ is ample, see [Har77, II, ex7.5(b)]. Then for all u,

$$D + u(D' - D) + vH - K_X$$

is also ample by Kleiman's ampleness criterion and $D' - D \equiv_{num} 0$. Now, we have

$$\begin{aligned} H^i\big(X, \mathscr{O}_X(D + vH)\big) &= 0 \\ &= H^i\Big(X, \mathscr{O}_X\big(D + u(D' - D) + vH\big)\Big) \end{aligned}$$

for all u and $i > 0$, by Kodaira's vanishing theorem. Hence

$$P'(u,v) = h^0\Big(X, \mathscr{O}_X\big(D + u(D'-D) + vH\big)\Big)$$

for every u. We would like to conclude from this, that for every $v \gg 1$, $P'(u,v)$ is bounded as a function of u.

We can prove this by induction on $n = \dim X$ as follows. Pick $m \gg 1$ such that

$$\Big(\big(D + u(D'-D) + vH - mH\big)\cdot(H^{n-1})\Big) = \big((D + vH - mH)\cdot(H^{n-1})\big) = 0,$$

and let $Y \in |mH|$ be a smooth divisor, by Bertini's theorem. Then we must have

$$H^0\Big(X, \mathscr{O}_X\big(D + u(D'-D) + vH - mH\big)\Big) = 0,$$

since otherwise we can pick a canonical section s_E of some effective divisor E such that $\big(E\cdot(H^{n-1})\big) > 0$ due to the Nakai-Moishezon theorem. Hence, there is an injection

$$H^0\Big(X, \mathscr{O}_X\big(D + u(D'-D) + vH\big)\Big) \hookrightarrow H^0\Big(Y, \mathscr{O}_Y\Big(\big(D + u(D'-D) + vH\big)\big|_Y\Big)\Big).$$

A bounded polynomial is constant, thus $P'(u,v) = P(v)$ for $v \gg 1$, hence for every v. Therefore,

$$\chi\big(X, \mathscr{O}_X(D)\big) = P(0) = P'(1,0) = \chi\big(X, \mathscr{O}_X(D')\big).$$

□

The proof of the Non-vanishing Theorem is taken from [Sho86].

Proof. **Step 0:** Let $f : X' \to \tilde{X} \to X$ be a (projective) log resolution of X, that is we use Chow's lemma and take a log resolution. Write down the discrepancy formula

$$K_{X'} - G' = f^*(K_X - G),$$

where G' has snc support. Then $(X', -G')$ has klt singularities, by $(X, -G)$ is, and

$$af^*D + G' - K_{X'} = f^*(\underbrace{aD + G - K_X}_{\text{nef and big}}).$$

As Remark 1.8, we choose small effective divisor F with $\operatorname{Supp} G' \cup \operatorname{Supp} F$ has snc support[22], such that $af^*D + G' - K_{X'} - F$ is ample and $(X', -G' + F)$ is also klt by Corollary 5.2.38.

[22] In the projective resolution $f : X' \xrightarrow{\mu} \tilde{X} \xrightarrow{\varepsilon} X$ where $\tilde{X}$ is the projective variety, we can resolve the pair $(\tilde{X}, -\varepsilon^*G + \mu(F))$ to ensure this intersection property.

Set $G'' := G' - F$ which has snc support. Then $f_*G'' = G - f_*F \leq G$ and so

$$f_*\lceil G'' \rceil = \lceil f_*G'' \rceil \leq \lceil G \rceil. \tag{6.18}$$

Identify the function fields $\mathbb{C}(X) = \mathbb{C}(X')$, and pick a non-zero rational function

$$\varphi \in H^0(X', mf^*D + \lceil G'' \rceil),$$

that is

$$\operatorname{div}_{X'}(\varphi) + mf^*D + \lceil G'' \rceil \geq 0.$$

But since f is birational, it is an isomorphism at generic points, so after the pushforward f_* we have

$$\operatorname{div}_X(\varphi) + mD + f_*\lceil G'' \rceil \geq 0.$$

By (6.18) we have $\varphi \in H^0(X, mD + \lceil G \rceil)$. Therefore,

$$h^0(X', mf^*D + \lceil G'' \rceil) \leq h^0(X, mD + \lceil G \rceil).$$

This shows that it is sufficient to prove the Non-vanishing Theorem under the following additional conditions, which we assume for the rest of the proof.

> X is smooth and projective, $(X, -G)$ has klt singularities where the $\mathbb{Q}$-divisor G has snc support, and $aD + G - K_X$ is ample for some $a > 0$.

Step 1: Since $(X, -G)$ has klt singularities, the coefficient of every summands of G is greater than -1, that is, $\lceil G \rceil$ is effective.

If D is numerically trivial and hence $G - K_X$ is ample, then

$$h^0(X, mD + \lceil G \rceil) = \chi(X, mD + \lceil G \rceil) = \chi(X, \lceil G \rceil) = h^0(X, \lceil G \rceil) \neq 0,$$

where the middle equality holds by Lemma 1.29, and the two others by Kawamata-Viehweg[23] vanishing theorem, where the fractional part of $mD + G - K_X$ and $G - K_X$ has snc support is by our assumption. So the assertion of the theorem is trivially satisfied, and for the rest of the section we assume that D is not numerically trivial.

Step 2: Pick a point $x \in X$ which $x \notin \operatorname{Supp} G$. (We blow up this point first in the construction of f below).

[23] Be careful that, the round up of an ample divisor might not be ample anymore, so we cannot use Kodaira's vanishing here. For example $(\mathrm{Bl}_p\mathbb{P}^2, -\frac{1}{2}E)$ has klt singularities, where E is the exceptional curve; and

$$\frac{1}{2}E - K_{\mathrm{Bl}_p\mathbb{P}^2} = \frac{1}{2}E - (-3H - E) = 3H - \frac{1}{2}E$$

is ample. But its round up $3H$ has $(3H \cdot E) = 0$ due to the projection formula.

We claim that, we can pick a positive integer q_0 such that for every $q \geq q_0$ there is a $\mathbb{Q}$-divisor

$$M(q) \sim_{\mathbb{Q}} qD + G - K_X$$

with $\operatorname{mult}_x M(q) > 2 \dim X$. This is called the *concentration method*.

To see that this possible, let $d = \dim X$. Since D is nef, $(D^e \cdot A^{d-e}) \geq 0$ for any ample divisor A, by Kleiman's Theorem 1.4.6. Thus,

$$\begin{aligned}\big((qD + G - K_X)^d\big) &= \Big(\big((q-a)D + aD + G - K_X\big)^d\Big) \\ &= \sum_{i=0}^{d} \binom{d}{i} (q-a)^i \big(D^i \cdot (aD + G - K_X)^{d-i}\big) \\ &\geq d(q-a)\big(D \cdot (aD + G - K_X)^{d-1}\big).\end{aligned}$$

There is some curve C so that $(D \cdot C) > 0$ by $D \not\equiv_{num} 0$, and some p such that

$$\Big(\big(p(aD + G - K_X)\big)^{d-1}\Big)$$

is represented by C plus an effective one-cycle[24]. So,

$$\begin{aligned}\Big(\big(D \cdot (aD + G - K_X)\big)^{d-1}\Big) &= \frac{(\text{const.})}{p^{d-1}}\big(\underbrace{(D \cdot C)}_{>0} + \underbrace{(D \cdot \text{one-cycle})}_{\geq 0}\big) \\ &> 0.\end{aligned}$$

Thus, the right-hand side quantity in the inequality goes to infinity with q. Moreover, by asymptotic Riemann-Roch and the Serre's vanishing, we have

$$h^0\big(m(qD + G - K_X)\big) \geq \frac{m^d}{d!}\big((qD + G - K_X)^d\big) + O(m^{d-1}).$$

On the other hand, the number of condition on $M(q,m) \in \big|m(qD + G - K_X)\big|$ that x be a point of multiplicity $> 2dm$ on $M(q,m)$ is at most

$$\binom{2dm + d}{d} = \frac{m^d}{d!}(2d)^d + O(m^{d-1})^{25}.$$

Since $\big((qD + G - K_X)^d\big) \to \infty$ as $q \to \infty$, we have more sections than condition. Thus, we can find such $M(q,m)$ such that $\operatorname{mult}_x M(q,m) > 2dm$. Set $M(q) := M(q,m)/m$. This prove the claim.

[24]This is because that $\mathscr{I}_C \otimes \mathscr{O}_X\big(p(aD + G - K_X)\big)$ is generated by global sections, and we have

$$V := H^0\Big(X, \mathscr{I}_C \otimes \mathscr{O}_X\big(p(aD + G - K_X)\big)\Big) \subseteq H^0\Big(X, \mathscr{O}_X\big(p(aD + G - K_X)\big)\Big),$$

where $\operatorname{Bs}|V| = C$. Thus we can make a suitable intersection by the sub-series $|V|$ to cut out a $C \cup$ one-cycle.

[25]Locally at the point x, after take a local trivialization the section $s \in H^0\big(X, m(qD + G - K_X)\big)$

Step 3: Let $f = f(q) : Y \longrightarrow X$ be some log resolution of the pair $\big(X, G + M(q)\big)$, such that f can be decomposed into

$$Y \longrightarrow \mathrm{Bl}_x X \longrightarrow X.$$

Set

(1) $K_Y + f^*G = f^*K_X + \sum b_j F_j$, where $b_j > -1$ by assumption,

(2) $\frac{1}{2} f^*(aD + G - K_X) - \sum p_j F_j$ is ample for suitable $0 < p_j \ll 1$,

(3) $f^*M(q) = \sum r_j F_j$ with F_0 corresponding to the exceptional divisor of the blow-up of x.

Step 4: We define

$$N(b, c) := bf^*D + \sum(-cr_j + b_j - p_j)F_j - K_Y.$$

As before, we want to make $N(b, c)$ ample. We calculate:

$$\begin{aligned}
N(b,c) &= bf^*D + \sum(-cr_j + b_j - p_j)F_j - K_Y \\
&= bf^*D - cf^*(qD + G - K_X) - \sum p_j F_j + f^*G - f^*K_X \\
&= \big(b - a - c(q-a)\big)f^*D + (1-c)f^*(aD + G - K_X) - \sum p_j F_j \\
&= \big(b - a - c(q-a)\big)f^*D \quad (\text{nef if } b - a - c(q-a) \geq 0) \\
&\quad + (\frac{1}{2} - c)f^*(aD + G - K_X) \quad (\text{nef if } \frac{1}{2} \geq c) \\
&\quad + \frac{1}{2} f^*(aD + G - K_X) - \sum p_j F_j \quad (\text{ample}).
\end{aligned}$$

Thus as long as $1/2 \geq c$ and $b \geq a + c(q-a)$, $N(b, c)$ is ample by Corollary 1.4.7.

Step 5: As before, we pick

$$c = \min_j \left\{ \frac{1 + b_j - p_j}{r_j} \Big| r_j > 0 \right\} > 0$$

---

can be expressed as a regular function at x of the form

$$f = \sum_{\alpha \in \mathbb{Z}^d_{\geq 0}} A_\alpha z_1^{\alpha_1} \cdots z_d^{\alpha_d}, \ |\alpha| = \alpha_1 + \cdots + \alpha_d.$$

The condition $\mathrm{mult}_x M(q, m) > 2dm$ is equivalent to $A_\alpha = 0$ for all $|\alpha| \leq 2dm$. Therefore,

$$\#\{(\alpha_1, \ldots, \alpha_d) \in \mathbb{Z}^d_{\geq 0} | \alpha_1 + \cdots + \alpha_d \leq 2dm\} = \binom{2dm + d}{d}.$$

and suitable perturbing of p_i such that such c is unique corresponding to the divisor $F = F_{j'}$, hence that

$$\sum(-cr_j + b_j - p - j)F_j = A - F. \tag{6.19}$$

Since $x \notin \operatorname{Supp} G$, by [Har77, II, ex8.5] $b_0 = d - 1$, and $r_0 > 2d$ by $\operatorname{mult}_x M(q) > 2d$. Therefore,

$$c \leq \frac{1 + b_0 - p_0}{r_0} < \frac{1 + (d-1) - p_0}{2d} < \frac{1}{2}.$$

Thus, $c < 1/2$, and if we take $b \geq a + c(q - a)$ sufficiently large then $N(b, c)$ is ample.

Step 6: The rest of the story is as in the proofs of the Basepoint-free and Rationality Theorems. We write

$$N(b, c) = bf^*D + A - F - K_Y.$$

Let $f^*G = \sum g_j F_j$, so from (1) in *Step 3* we have

$$\begin{aligned} K_Y - f^*K_X + \sum g_j F_j &= \sum_{f\text{-excpetional}} a_j F_j + \sum g_j F_j \\ &= \sum b_j F_j, \end{aligned}$$

where a_j comes from the blow-up formula in [Har77, II, ex8.5]. If F_j is not f-exceptional, i.e. f is an isomorphism at the generic point of F_j, then $b_j = g_j$. The coefficients of F_j in A is $(-cr_j + b_j - p_j) < b_j$, and therefore

$$\lceil A \rceil \leq f^*\lceil G \rceil + (f\text{-excpetional divisors}).$$

Thanks to the projection formula and algebraic Hartogs, adding exceptional divisors to a pull-back does not increase H^0. Thus

$$H^0(Y, bf^*D + \lceil A \rceil) \subseteq H^0(Y, bf^*D + f^*\lceil G \rceil + f\text{-exceptional divisors}) \cong H^0(X, bD + \lceil G \rceil).$$

Since $N(b, c)$ is ample, so by Kawamata-Viehweg vanishing theorem,

$$H^1(Y, bf^*D + \lceil A \rceil - F) = H^1(Y, bf^*D + \lceil A - F \rceil) = 0,$$

where the snc condition is obtain by the choice of log resolution. Thus, $H^0(X, bD + \lceil G \rceil) \neq 0$ if we show that

$$H^0\big(F, (bf^*D + \lceil A \rceil)|_F\big) \neq 0.$$

This last inequality can be achieved by making an induction on $\dim X$. We can assume that we have already proved the Non-vanishing Theorem for varieties of dimension $< \dim X$. Applying the induction assumption to F, we complete the proof of the theorem. □

6.2 Relative versions

In this section, we will discuss the relative versions of Basepoint-free Theorem, Non-vanishing Theorem, Rationality Theorem and Cone Theorem. In particular, there is an effective version of the Basepoint-free Theorem proved by Kollár in [Kol93], and we will formulate it.

Further more, we will extend these result in the general case where the boundary divisor Δ is $\mathbb{R}$-divisor, prepare for learning [BCHM10].

Remark 6.2.1. The logical order of proof of these theorems is the following:

Rel. Non-vanishing^{A} $\Rightarrow$ Rel. Basepoint-free^{B} $\Rightarrow$ Rel. Rationality^{C} $\Rightarrow$ Rel. Cone^{D}.

6.2A The Relative Non-vanishing Theorem

Theorem 6.2.2. (Relative Non-vanishing) *Let (X, Δ) be a klt pair with $\Delta \geq 0$ such that $K_X + \Delta$ is $\mathbb{R}$-Cartier, let $f : X \longrightarrow S$ be a projective [26] morphism, and let D, E be Cartier divisors on X. Assume the following conditions:*

(1) D is f-nef.

(2) There is a positive integer m_1 such that $m_1 D + E - (K_X + \Delta)$ is f-nef and f-big.

(3) E is effective and there exists a positive integer m_2 such that for any positive integer, the natural homomorphism

$$f_* \mathscr{O}_X(mD) \longrightarrow f_* \mathscr{O}_X(mD + E)$$

is an isomorphism. (Trivial when $E = 0$).

Then there exists a positive integer m_3 such that for any integer $m \geq m_3$, we have

$$f_* \mathscr{O}_X(mD) \neq 0.$$

Proof. This is a bookkeeping proof, and the only issue is how to approximate the $\mathbb{R}$-Cartier version from the $\mathbb{Q}$-Cartier version, and how to apply the absolute version of the Non-vanishing Theorem. We will reduce to the geometric generic fibre, where the properties of the divisors and condition (3) remains by the flat base change.

[26] By replace S by its image which is reduced and irreducible, we can always assume that f is surjective. Notice here f might not has irreducible geometric generic fibre, so the bigness is defined on generic fibre, which must be irreducible since locally it is of the form $B \underset{A}{\otimes} \mathrm{Frac}(A)$.

Step 0: It is suffices to show the theorem on the geometric generic fibre of f, that is by $f_*\mathscr{O}_X(mD)$ is coherent and $k(\eta) \hookrightarrow \overline{k(\eta)}$ is faithfully flat,

$$f_*\mathscr{O}_X(mD)_\eta \underset{k(\eta)}{\otimes} \overline{k(\eta)} = H^0(X_{\bar{\eta}}, \mathscr{O}_X(mD)|_{X_{\bar{\eta}}}) \neq 0 \Rightarrow f_*\mathscr{O}_X(mD)_\eta \neq 0,$$

cf.[Har77, III, 9.3]. Where $X_{\bar{\eta}}$ is a projective normal scheme[27] over an algebraically closed field of characteristic 0, so we can replace X by its connected component.

Step 1: We set $L = m_1D + E - (K_X + \Delta)$ is a $\mathbb{R}$-Cartier divisor, which is nef and big. So by Kodaira's lemma, we have $L \sim_{\mathbb{R}} A + N$ where A is ample and N effective, both be $\mathbb{R}$-Cartier. Taking sufficiently $\varepsilon > 0$, then by Corollary 5.2.18, $(X, \Delta + \varepsilon N)$ also has klt singularities. And

$$L - \varepsilon N = m_1D + E - (K_X + \Delta + \varepsilon N) \sim_{\mathbb{R}} (1-\varepsilon)L + \varepsilon A$$

is ample by Corollary 1.4.7 and $1 - \varepsilon > 0$. We denote $\Delta' = \Delta + \varepsilon N$. Now, since $K_X + \Delta' = K_X + \sum_i a_i\Delta_i' =: \sum_j b_jB_j$ is $\mathbb{R}$-Cartier, the linear equation

$$K_X + \sum_i x_i\Delta_i' = \sum_j y_jB_j$$

already has a real solution, therefore it has rational solutions since rational solutions are dense. Therefore, for the rational solution $\Delta_{\mathbb{Q}}$ which is greater than and sufficiently closed to the given real solution, we have

$$(X, \Delta_{\mathbb{Q}}) \text{ is klt, } K_X + \Delta_{\mathbb{Q}} \text{ is } \mathbb{Q}\text{-Cartier, } m_1D + E - (K_X + \Delta_{\mathbb{Q}}) \text{ is ample,}$$

where for klt singularities we can apply Corollary 5.2.38 by finite steps, and the ampleness comes from the open nature. Thus We may replace Δ by $\Delta_{\mathbb{Q}}$.

Step 2: In Shokurov's Non-vanishing Theorem, we can take $G = E - \Delta$, such that it is a $\mathbb{Q}$-divisor and

$$m_1D + G - K_X = m_1D + E - (K_X + \Delta)$$

is ample automatically nef and big. Moreover, $(X, -G) = (X, \Delta - E)$ has klt singularities, since (X, Δ) has and also by Corollary 5.2.38. Therefore, we have

$$\begin{aligned} H^0(X, mD + \lceil G \rceil) &= H^0(X, mD + \lceil E - \Delta \rceil) \\ &= H^0(X, mD + E + \lfloor \Delta \rfloor) \\ &= H^0(X, mD + E) \neq 0. \end{aligned}$$

But by condition (3), we have $H^0(X, mD) \cong H^0(X, mD + E) \neq 0$. □

[27] See [SP25, Tag 0C3M] and [SP25, Tag 033M].

Remark 6.2.3. If above proof we take an irreducible component U of the geometric generic fibre of $f : X \longrightarrow S$, and now we explain that why the pair (U, Δ_U) also has klt singularities.

Indeed, for log resolution $\mu : Y \longrightarrow X$ where Y is a non-singular variety, after base change on the field $\operatorname{Spec} \overline{k(\eta)}$ which we denote by $\operatorname{Spec} \bar{K}$, the scheme $X_{\bar{\eta}}$ and $Y_{\bar{\eta}}$ might be reducible[28]. But we have the following commutative algebra fact:

> If $A \to B$ is a flat ring map, then $\operatorname{Spec} B \to \operatorname{Spec} A$ maps minimal primes to minimal primes by going-down property.

This is easily to check, therefore the flat morphism $X_{\bar{\eta}} \longrightarrow X$ must sends generic points of each irreducible component to the unique generic point of X. Thus, for the open dense subset $W \subseteq X$ where μ is an isomorphism, and the decomposition

$$X_{\bar{\eta}} = U_1 \bigsqcup \cdots \bigsqcup U_k$$

we have each $U_i \cap W_{\bar{\eta}} \neq \emptyset$ contains the generic point. Use the same argument to the flat morphism $Y_{\bar{\eta}} \longrightarrow Y$ and open dense subset $\mu^{-1}(W) \subseteq Y$, where $Y_{\bar{\eta}}$ decomposed into the connected components

$$Y_{\bar{\eta}} = V_1 \bigsqcup \cdots \bigsqcup V_l$$

with each component is a non-singular variety since the smoothness is stable under base change, see [Har77, III, 10.1]. So we also have each $V_j \cap \mu_{\bar{\eta}}^{-1}(W_{\bar{\eta}}) \neq \emptyset$, which gives an isomorphism $V_j \cap \mu_{\bar{\eta}}^{-1}(W_{\bar{\eta}}) \longrightarrow U_i \cap W_{\bar{\eta}}$, so that $\mu_{V_j} : V_j \longrightarrow U_{\sigma(j)}$ is a birational morphism from a non-singular variety to a normal variety. We omit the index.

On the other hand, we denote

$$A := \mathscr{O}_{Y,y} = \mathscr{O}_{Y_\eta,y} \text{ and } B := \mathscr{O}_{Y_{\bar{\eta}},\bar{y}} = \left(A \underset{K}{\otimes} \bar{K}\right)_{q_{\bar{y}} =: q},$$

[28] Actually they are non-connected, since for $X_{\bar{\eta}}$ the local rings are normal domain and we have

$$\mathscr{O}_{X_{\bar{\eta}},\bar{x}} = \left(\mathscr{O}_{X_\eta,x} \underset{k(\eta)}{\otimes} \overline{k(\eta)}\right)_{q_{\bar{x}}} = \left((\mathscr{O}_{X,x}/m_\eta \mathscr{O}_{X,x}) \underset{k(\eta)}{\otimes} \overline{k(\eta)}\right)_{q_{\bar{x}}} = \left(\mathscr{O}_{X,x} \underset{k(\eta)}{\otimes} \overline{k(\eta)}\right)_{q_{\bar{x}}},$$

then use [SP25, Tag 0C30] and [SP25, Tag 06DF] can get that it is normal domain. For $Y_{\bar{\eta}}$ is similar, by [SP25, Tag 07EQ] we see $k(\eta) \hookrightarrow \overline{k(\eta)}$ is a regular ring map, so for finite type map $k(\eta) \to \mathscr{O}_{Y_\eta,y}$ the base change $\mathscr{O}_{Y_\eta,y} \to \mathscr{O}_{Y_\eta,y} \underset{k(\eta)}{\otimes} \overline{k(\eta)}$ is a regular ring map due to [SP25, Tag 07C1]. Then, $\left(\mathscr{O}_{Y_\eta,y} \underset{k(\eta)}{\otimes} \overline{k(\eta)}\right)_{q_{\bar{y}}} = \mathscr{O}_{Y_{\bar{\eta}},\bar{y}}$ is a regular local ring by [SP25, Tag 0H7S]. But regular local ring is U.F.D., the intersection part cannot have more than one minimal primes which implies that they must be disjoint.

and the ring map $A \to B$ is flat and local, while the flatness comes from the $K \hookrightarrow \bar{K}$ and the localization. Then we consider

$$\begin{aligned} B/m_A B &= \left(A \underset{K}{\otimes} \bar{K}\right)_q \Big/ m_A(A \underset{K}{\otimes} \bar{K}) \cdot \left(A \underset{K}{\otimes} \bar{K}\right)_q \\ &\cong \left((A \underset{K}{\otimes} \bar{K}\right) \Big/ m_A \left(A \underset{K}{\otimes} \bar{K}\right)_{\bar{q}} \\ &\cong \left(k(y) \underset{K}{\otimes} \bar{K}\right)_{\bar{q}}. \end{aligned}$$

This is a field. Since $\bar{K}/K$ is separably generated so $k(y) \underset{K}{\otimes} \bar{K}$ is reduced by [SP25, Tag 030U]. And $k(y) \underset{K}{\otimes} \bar{K}$ is integral over $k(y)$, since for any element $r = \sum^{m} k_i \otimes a_i \in k(y) \underset{K}{\otimes} \bar{K}$, we have $K(a_1, \ldots, a_m)/K$ is a finite field extension, so $r \in k(y) \underset{K}{\otimes} K(a_1, \ldots, a_m)$ which lies in a finite dimensional $k(y)$-vector space. Thus, $1, r, r^2, \ldots$ must dependent over $k(y)$, i.e. we have

$$r^n + c_{n-1} r^{n-1} + \cdots + c_0 = 0, \ c_i \in k(y),$$

which implies that r is integral over $k(y)$. This gives that $\dim k(y) \underset{K}{\otimes} \bar{K} = 0$, since for any prime ideal its quotient ring still integral over $k(y)$ implies that it is a field. But at this time, a zero dimensional reduced local ring must be a field, so we see $B/m_A B$ is a field, in other words, $m_A B = m_B$. This is nothing but to say the image of the regular system of parameters in $\mathscr{O}_{Y,y}$ will generated m_B, and by famous formula

$$\dim B = \dim A + \dim(B/m_A B) = \dim A$$

in [Eis95, 10.10], we see their image is exactly be the regular system of parameters in $\mathscr{O}_{Y_{\bar{\eta}}, \bar{y}}$. Hence, the simple normal crossing condition still holds on V. So, the birational morphism $U \longrightarrow V$ is a log resolution of the normal variety U.

Next we need to check that $(U, \Delta|_U)$ still has klt singularities, by examining the coefficients on the discrepancies formula.

Since we are only care about the generic fibre, we may replace S by its non-singular dense open locus. That is, the canonical divisor K_S is Cartier. So we can define the relative canonical divisors of the morphisms $f : X \longrightarrow S$ and $f \circ \mu : Y \longrightarrow S$, to be

$$K_{X/S} = K_X - f^* K_X \text{[29]} \quad \text{and} \quad K_{Y/S} = K_Y - (f \circ \mu)^* K_S.$$

[29] This is a Weil divisors, and in the sheaf theory it is corresponding to the

$$\omega_{X/S} := \mathscr{O}_X(K_{X/S}) = j_* \Omega^{\mathrm{top}}_{X_{\mathrm{reg}}} \otimes f^* \omega_S^{\vee}$$

which is a reflexive sheaf.

Then for the discrepancy formula

$$K_Y + \Theta = \mu^*(K_X + \Delta),$$

we minus both sides by $(f \circ \mu)^* K_S$, can get that

$$K_{Y/S} + \Theta = \mu^*(K_{X/S} + \Delta),$$

while notice that the multiplicities does not change. By our previous analysis in the footnote and [SP25, Tag 0B8X], we see $Y_{\bar{\eta}}$ is smooth over $\overline{k(\eta)}$, same for its connected component V. By [SP25, Tag 02K4] we have the following exact sequence on V:

$$0 \longrightarrow \mathscr{O}_V \underset{\bar{K}}{\otimes} \Omega_{\bar{K}/\mathbb{C}} \longrightarrow \Omega_{V/\mathbb{C}} \longrightarrow \Omega_{V/\bar{K}} \longrightarrow 0. \tag{6.20}$$

We **claim** that the middle term has the form $\Omega_{V/\mathbb{C}} \cong p_V^* \Omega_{Y/\mathbb{C}}$ and the first term has the form $\mathscr{O}_V \underset{\bar{K}}{\otimes} \Omega_{\bar{K}/\mathbb{C}} \cong p_V^* \big((f \circ \mu)^* \Omega_{S/\mathbb{C}}\big)$, where $p_V : V \longrightarrow Y$ is the natural projection.

Proof. For the first statement, we consider the natural projection

$$p_V : V \overset{i}{\hookrightarrow} Y_{\bar{\eta}} \overset{\pi}{\to} Y.$$

Then we have $\pi^* \Omega_Y \cong \Omega_{Y_{\bar{\eta}}}$, and we can see it on affine opens. Denote $K = k(\eta)$, and choose the notations

$$\operatorname{Spec} \bar{B} \subseteq Y_{\bar{\eta}} \xrightarrow{\pi} \operatorname{Spec} B \subseteq Y \xrightarrow{f \circ \mu} \eta \in \operatorname{Spec} A \subseteq S.$$

We denote $C = S^{-1}B = B \underset{A}{\otimes} K$ for $S = A \backslash \{0\}$. Then $\bar{B} = B \underset{A}{\otimes} \bar{K} = C \underset{K}{\otimes} \bar{K}$. So the formula above is equivalent to prove that

$$\bar{B} \underset{B}{\otimes} \Omega_B \cong \Omega_{\bar{B}}.$$

For the localization ring map $B \to C$, we have $\Omega_C \cong C \underset{B}{\otimes} \Omega_B$ by [Har77, II, 8.2A], and then for $C \to C \underset{K}{\otimes} \bar{K} = \bar{B}$, we need to check that $\Omega_{\bar{B}} \cong \bar{B} \underset{C}{\otimes} \Omega_C$.

Taking any $\bar{B}$-module M with a $\mathbb{C}$-derivation $\delta : C \to M$, and restrict on $K \subseteq C$[30] it will gives a $\mathbb{C}$-derivation $K \to M \in \operatorname{Hom}_K(\Omega_K, M)$. Since $\bar{K}/K$ is separably generated, [Eis95, 16.15] says that $\bar{K} \underset{K}{\otimes} \Omega_K = \Omega_{\bar{K}}$. Thus, that $\mathbb{C}$-derivation can be uniquely extended to $\delta_{\bar{K}} : \bar{K} \to M$ by the universal property of the Kähler differentials, where M regard as $\bar{K}$-module via $\bar{K} \to \bar{B}$. Hence can define that

$$\begin{aligned} \tilde{\delta} : C \underset{K}{\otimes} \bar{K} &\longrightarrow M \\ c \otimes \lambda &\longmapsto \lambda \delta(c) + c \delta_{\bar{K}}(\lambda). \end{aligned}$$

[30] Recall that K is flat over A with $A \hookrightarrow B$ by the surjectivity and A reduced.

Since δ and $\delta_{\bar{K}}$ coincide on K, this definition is compatible with the multiplication $(kc)\otimes\lambda = c\otimes(k\lambda)$ for $k\in K$.

Conversely, for every such extension $D:\bar{B}\to M$ of $\delta: C\to M$ we must have $D(1\otimes\lambda)=\delta_{\bar{K}}(\lambda)$ for any $\lambda\in\bar{K}$. Since on K, we have

$$D(1\otimes a)=D(a\otimes 1)=\delta(a)=\delta_{\bar{K}}(a),$$

that is the derivations D and $\delta_{\bar{K}}$ coincide on K as δ_K. And by $\bar{K}/K$ is separably generated, the extension is unique

$$\mathrm{Der}_{\mathbb{C}}(\bar{K},M)\cong \mathrm{Hom}_{\bar{K}}(\Omega_{\bar{K}},M)\cong \mathrm{Hom}_{\bar{K}}(\bar{K}\underset{K}{\otimes}\Omega_K,M)=\mathrm{Hom}_K(\Omega_K,M)\cong \mathrm{Der}_{\mathbb{C}}(K,M),$$

where the second isomorphism is due to [Eis95, 16.15], so D and $\delta_{\bar{K}}$ coincide on $\bar{K}$. Now, every element $c\otimes\lambda\in\bar{B}$ can be written as the form $(c\otimes 1)(1\otimes\lambda)$, then by Leibniz rule

$$\begin{aligned} D(c\otimes\lambda) &= (1\otimes\lambda)D(c\otimes 1)+(c\otimes 1)D(1\otimes\lambda)\\ &= (1\otimes\lambda)\delta(c)+(c\otimes 1)\delta_{\overline{K}}(\lambda). \end{aligned}$$

gives that $D=\tilde{\delta}$.

Therefore, we have a natural bijection $\mathrm{Der}_{\mathbb{C}}(\bar{B},M)\cong\mathrm{Der}_{\mathbb{C}}(C,M)$, which gives that our desired $\Omega_{\bar{B}}\cong\bar{B}\underset{C}{\otimes}\Omega_C$.

Combining above results, we found that

$$\Omega_{\bar{B}}\cong\bar{B}\underset{C}{\otimes}\Omega_C\cong\bar{B}\underset{C}{\otimes}\left(C\underset{B}{\otimes}\Omega_B\right)\cong\bar{B}\underset{B}{\otimes}\Omega_B.$$

And then glue them together to get that $\Omega_{Y_{\bar{\eta}}}\cong\pi^*\Omega_Y$. Now, since V is a connected component of $Y_{\bar{\eta}}$, the closed immersion $i:V\hookrightarrow Y_{\bar{\eta}}$ is also an opem immersion. Thus,

$$\begin{aligned} p_V^*\Omega_Y &= i^*\pi^*\Omega_Y\\ &\cong i^*\Omega_{Y_{\bar{\eta}}}\\ &\cong \Omega_V. \end{aligned}$$

Alternatively, we can use some "fancy" method to deduce this result. In fact, as $\bar{K}/K$ is a separably generated field extension, we have

$$K\cong\varinjlim_{L\in I}L,$$

where $I=\left\{L \text{ field } \middle| K\subseteq L\subseteq\bar{K} \text{ and } [L:K]<\infty\right\}$ is the set contains the data about fields that finitely separable over K. Then each L is formally étale over K by [SP25,

Tag 090W]. So by [SP25, Tag 031N] this property commutes with direct limit, we can see $\bar{K}$ is formally étale over K. Now, the morphism

$$\operatorname{Spec}\bar{K} \longrightarrow \operatorname{Spec}K \longrightarrow \operatorname{Spec}A \subseteq S$$

is formally étale, since this property is stable under the composition and the last one is due to [SP25, Tag 04EG]. After base change, we see

$$\pi : Y_{\bar{\eta}} = Y \underset{S}{\times} \operatorname{Spec}\bar{K} \longrightarrow Y$$

is formally étale, since this property is stable under base change, while one can see [SP25, Tag 0H92]. Therefore, by [EGA IV$_4$, 17.2.4] we know that

$$\pi^*\Omega_Y \cong \Omega_{Y_{\bar{\eta}}}.$$

For the second statement, it is deduce from the following diagram

$$\begin{array}{ccc} V & \xrightarrow{p_V} & Y \\ \downarrow & & \downarrow{\scriptstyle f\circ\mu} \\ \operatorname{Spec}\overline{k(\eta)} & \xrightarrow[b]{} & S \end{array}$$

and $b^*\Omega_S \cong \bar{K}\underset{K}{\otimes}(\Omega_S)_\eta \cong \bar{K}\underset{K}{\otimes}\Omega_K \cong \Omega_{\bar{K}}$, where the last comes from [Eis95, 16.15]. □

Thus, sequence (6.20) can be rewritten as

$$0 \longrightarrow p_V^*(f\circ\mu)^*\Omega_{S/\mathbb{C}} \longrightarrow p_V^*\Omega_{Y/\mathbb{C}} \longrightarrow \Omega_{V/\bar{K}} \longrightarrow 0.$$

Taking the top wedge on this sequence, can get that $p_V^*\omega_Y \cong p_V^*(f\circ\mu)^*\omega_S \otimes \omega_V$, that is, $p_V^*\big(\omega_Y \otimes (f\circ\mu)^*\omega_S^\vee\big) \cong \omega_V$. Therefore, $p_V^*K_{Y/S} = K_V$.

Now. for normal variety X, we denote

$$X^o = X_{\mathrm{reg}},\ U^o = p_U^{-1}(X^o),\ p_U^o : U^o \longrightarrow X^o,\ f^o : X^o \longrightarrow S.$$

We claim that U^o contains all codimension one points in U. In fact, p_U is a flat morphism, since it comes from the (open) restriction and base change of the flat morphism

$$\operatorname{Spec}\bar{K} \longrightarrow \operatorname{Spec}K \longrightarrow \operatorname{Spec}A \subseteq S.$$

Taking any codimension one point $u \in U$, set $x = p_U(u)$. Then for the local ring map $\mathscr{O}_{X,x} \to \mathscr{O}_{U,u}$ which is flat, we have

$$\dim \mathscr{O}_{X,x} \leq \dim \mathscr{O}_{U,u} = 1$$

by the going-down property, see [Mat86, 9.5] or [SP25, Tag 00HS]. But since X is normal, $\mathscr{O}_{X,x}$ is a regular local ring gives that $x \in X_{\mathrm{reg}}$, so $u \in U^o$.

On the other hand, the generic fibre X^o_η has the local ring

$$\mathscr{O}_{X^o_\eta,x} \cong \mathscr{O}_{X_\eta,x}/m_\eta \mathscr{O}_{X^o,x} \cong \mathscr{O}_{X^o,x}$$

which is a regular local ring. So by [SP25, Tag 0B8X] again we see X^o_η is smooth over $\operatorname{Spec} K$, and since smoothness is stable under base change, $X^o_{\bar\eta}$ is smooth over $\operatorname{Spec} \bar K$. This gives that $U^o \longrightarrow \operatorname{Spec} \bar K$ is smooth. Again, by [SP25, Tag 02K4] we have the following exact sequence

$$0 \longrightarrow \mathscr{O}_{U^o} \underset{\bar K}{\otimes} \Omega_{\bar K/\mathbb{C}} \longrightarrow \Omega_{U^o/\mathbb{C}} \longrightarrow \Omega_{U^o/\bar K} \longrightarrow 0.$$

Using the similar formally étale argument or Kähler differential computations, we can rewrite this sequence to be

$$0 \longrightarrow (p^o_U)^*(f^o)^*\Omega_S \longrightarrow (p^o_U)^*\Omega_{X^o} \longrightarrow \Omega_{U^o/\bar K} \longrightarrow 0.$$

So, take the top wedge can obtain that $(p^o_U)Y^*\big(\omega_{X^o} \otimes (f^o)^*\omega_S^\vee\big) \cong \omega_{U^o}$. Therefore,

$$(p^o_U)^*(K_{X/S}|_{X^o}) = K_U|_{U^o}. \tag{6.21}$$

Since X^o is non-singular, $\Delta|_{X^o}$ is $\mathbb{R}$-Cartier, we can define $\Delta_U|_{U^o} := (p^o_U)^*(\Delta|_{X^o})$ and then take closure in U to get Δ_U. Equipping this on (6.21), we have

$$\big(p_U^*(K_{X/S}+\Delta)\big)\big|_{U^o} = (p^o_U)^*\big((K_{X/S}+\Delta)|_{X^o}\big) = (K_U+\Delta_U)|_{U^o}.$$

But this is the equation between Weil divisors, and U^o contains all codimension one points of U, so the equality can be extended on the whole of U:

$$p_U^*(K_{X/S}+\Delta) = K_U + \Delta_U$$

which also implies that $K_U + \Delta_U$ is $\mathbb{R}$-Cartier.

Finally, combining the above results into the discrepancy formula

$$K_{Y/S} + \Theta = \mu^*(K_{X/S}+\Delta),$$

then via the following commutative diagram, we have

$$\begin{array}{ccc} V & \xrightarrow{p_V} & Y \\ {\scriptstyle \mu_V}\downarrow & & \downarrow{\scriptstyle \mu} \\ U & \xrightarrow[p_U]{} & X \end{array}$$

$$\begin{aligned} p_V^*(K_{Y/S}+\Theta) &= p_V^*K_{Y/S} + \Theta_V \\ &= K_V + \Theta_V \\ &= (\mu\circ p_V)^*(K_{X/S}+\Delta) \\ &= (p_U\circ\mu_V)^*(K_{X/S}+\Delta) \\ &= \mu_V^*(K_U+\Delta_U). \end{aligned}$$

But the multiplicities does not change, so the pair (U, Δ_U) has klt singularities.

Remark 6.2.4. My advisor tell me that can replace f by its Stein factorization, so that can assume that f has connected geometric fibres, see [SP25, Tag 0A1B], not stated as in [Har77] which only has connected fibres.

At this time, the geometric generic fibre is automatically a normal projective variety over an algebraically closed field of characteristic zero. And then, a finite pushforward of a non-zero sheaf must be non-zero.

6.2B The Relative Basepoint-free & Effective Basepoint-free Theorem

Theorem 6.2.5. (Relative Basepoint-free) *Let (X, Δ) be a klt pair with $\Delta \geq 0$ such that $K_X + \Delta$ is $\mathbb{R}$-Cartier, let $f : X \longrightarrow S$ be a projective morphism, and let D, E be Cartier divisors on X. Assume the following conditions:*

(1) D is f-nef.

(2) There is a positive integer m_1 such that $m_1 D + E - (K_X + \Delta)$ is f-nef and f-big.

(3) E is effective and there exists a positive integer m_2 such that for any positive integer, the natural homomorphism

$$f_* \mathscr{O}_X(mD) \longrightarrow f_* \mathscr{O}_X(mD + E)$$

is an isomorphism.

Then there exists a positive integer m_3 such that for any integer $m \geq m_3$, mD is f-free. That is, the natural homomorphism

$$f^* f_* \mathscr{O}_X(mD) \longrightarrow \mathscr{O}_X(mD)$$

is surjective.

Proof. **Step 0:** As the assertion is local over S, we may assume that S is affine, and in this case X is quasi-projective. Moreover as S is quasi-compact, the maximum of the resulting bounds give the assertion. Then the theorem says that the natural homomorphism

$$H^0(X, mD) \otimes \mathscr{O}_X \longrightarrow \mathscr{O}_X(mD)$$

is surjective, in other words, $\operatorname{Bs} |mD| = \emptyset$. Note that $H^0(X, mD)$ may not be finite dimensional, so the linear series $|mD|$ is defined by

$$|mD| = \big\{ D' \big| mD \sim_{lin} D' \geq 0 \big\}.$$

Step 1: We may assume that the $\mathbb{R}$-Cartier divisor $m_1D+E-(K_X+\Delta)$ is f-ample and that Δ is a $\mathbb{Q}$-divisor. Indeed, by the relative Kodaira's lemma, we can write

$$L := m_1D + E - (K_X + \Delta) \sim_{\mathbb{R},S} A + N$$

where A is f-ample and N effective. Then for a real number $0 < \varepsilon \ll 1$,

$$L - \varepsilon N \sim_{R,S} (1-\varepsilon)L + \varepsilon A$$

which is f-ample by Corollary 1.5.20 and $1-\varepsilon \geq 0$. If ε is sufficiently small, $(X, \Delta+\varepsilon N)$ will has klt singularities by Corollary 5.2.38. We may replace Δ by $\Delta + \varepsilon N$, can therefore assume that

$$m_1D + E - (K_X + \Delta)$$

is f-ample. And similarly with the proof of the relative Non-vanishing, solve the linear equation by rational solution we may assume that Δ is a $\mathbb{Q}$-divisor and won't change anything by the open nature of the ampleness.

Step 2: Note that $H^0(X, mD) \neq \{0\}$ for all sufficiently large $m \geq m_3'$ is by the relative version of the Non-vanishing Theorem, and S is affine. We fix an integer $m \geq m_3'$ and suppose that $|mD|$ has a basepoint. We take a member $M \in |mD|$ and a log resolution $g : Y \longrightarrow X$ of the pair $(X, \Delta + E + M)$ such that also resolving the base ideal of $|mD|$[31]. Then, we have

$$K_Y + C = g^*(K_X + \Delta), \quad g^*M = M_1 + M_2$$

where $|M_1|$ is a free series, $M_2 \geq 0$ is the fixed part of $g^*|mD|$. In particular, by our assumptions, $M_2 \neq 0$ and all coefficients of C all lesser than 1 as (X, Δ) is a klt pair.

Note that our log resolution make that

$$\Sigma = \mathrm{Ex}(g) \cup \mathrm{Supp}(g_*^{-1}\Delta + g_*^{-1}E + g_*^{-1}M)$$

[31] Note that even $H^0(X, mD)$ may not be a finite dimensional $\mathbb{C}$-vector space, we can choose its finite dimensional subspace W such that $\mathfrak{b}(|W|) = \mathfrak{b}(|mD|)$.

In fact, for $S = \mathrm{Spec}\, A$, since f is a projective morphism we see $f_*\mathscr{O}_X(mD)$ is a coherent sheaf. So

$$H^0\big(X, \mathscr{O}_X(mD)\big) = H^0\big(S, f_*\mathscr{O}_X(mD)\big)$$

is a finite A-module, not necessary finite over $\mathbb{C}$. Take its generators and set

$$W = \mathsf{Span}_{\mathbb{C}}\{s_1, \ldots, s_r\} \subseteq H^0(X, mD),$$

then two maps

$$W \underset{\mathbb{C}}{\otimes} \mathscr{O}_X(mD)^\vee \to \mathscr{O}_X, \quad H^0(X, mD) \underset{\mathbb{C}}{\otimes} \mathscr{O}_X(mD)^\vee \to \mathscr{O}_X$$

have the same image, which gives that $\mathfrak{b}(|W|) = \mathfrak{b}(|mD|)$ by the definition of the base ideal sheaf.

has snc support. We can construct an effective $\mathbb{Q}$-divisor C' on Y such that

$$\operatorname{Supp} C' = \operatorname{Ex}(g) \cup \operatorname{Supp}(C + g^*E + M_2) \subseteq \Sigma$$

and $-C'$ is g-ample. In fact, resolution of singularities is a sequence of blow-ups, so admits an effective $\mathbb{Q}$-divisor C'' with support $\operatorname{Ex}(g)$ such that $-C''$ is g-ample. Adding sufficiently small positive coefficients on the remaining components gives C', by the open nature of the relative ampleness, see Proposition 1.5.16.

We denote $h = f \circ g$, then by [EGA II, 4.6.13(ii)] we can choose a sufficiently small positive rational number $0 < \varepsilon \ll 1$ such that

$$g^*\big(m_1 D + E - (K_X + \Delta)\big) - \varepsilon C'$$

is a h-ample $\mathbb{Q}$-divisor on Y, and $\lceil -C - \varepsilon C' \rceil = \lceil -C \rceil \geq 0$ by coefficients of C are less than 1 and $C' \geq 0$. Consider the threshold

$$r = \sup\big\{ t \in \mathbb{R} \big| \lfloor tM_2 - g^*E + C + \varepsilon C' \rfloor \leq 0 \big\}.$$

Since $M_2 \neq 0$ and the coefficients of $C + \varepsilon C'$ are lesser than 1, the threshold r is positive. Moreover, it is rational; since if we expand $M_2 = \sum a_i P_i$ and $-g^*E + C + \varepsilon C' = \sum b_i P_i$, the threshold condition is $ta_i + b_i < 1$ and therefore

$$r = \min_{a_i > 0} \left\{ \frac{1 - b_i}{a_i} \right\}.$$

The maximal coefficient of $rM_2 - g^*E + C + \varepsilon C'$ is exactly 1, while be careful that there may $\lfloor rM_2 - g^*E + C + \varepsilon C' \rfloor \leq 0$ no more. Use the same method before, we can perturbing the coefficients of C' while preserving the preceding properties, we may assume that there exactly one prime divisor has coefficient 1. We denote this (smooth) prime divisor by F. Since $g^*E \geq 0$ as E effective, and coefficients of $C + \varepsilon C'$ are less than 1, we must have $F \subseteq \operatorname{Supp} M_2$. Consequently, $g(F) \subseteq \operatorname{Bs} |mD|$. We write

$$rM_2 - g^*E + C + \varepsilon C' = N + F,$$

where F is not the component of N, every coefficient of N is strictly less than 1 and $N + F$ has snc support. Using $g^*(mD) \sim_{lin} M_1 + M_2$, we obtain that

$$\begin{aligned} m'g^*D - N - F - K_Y &\sim_{\mathbb{Q}} (m' - rm)g^*D + rM_1 + g^*E - (K_Y + C + \varepsilon C') \\ &= (m' - rm - m_1)g^*D \quad (\text{nef if } m' \geq m_1 + rm) \\ &\quad + rM_1 \quad (\text{nef since basepoint free}) \\ &\quad + g^*\big(m_1 D + E - (K_X + \Delta)\big) - \varepsilon C' \quad (h\text{-ample}). \end{aligned}$$

So if $m' \geq m_1 + rm$, the right hand side is h-ample, by Corollary 1.5.20. Using Kawamata-Viehweg vanishing theorem, can obtain that

$$H^1(Y, m'g^*D + \lceil -N \rceil - F) = H^1(Y, K_Y + \lceil m'g^*D - N - F - K_Y \rceil) = 0, \quad (6.22)$$

by the base variety S is affine. The associated long exact sequence with F therefore yields a surjection

$$H^0(Y, m'g^*D + \lceil -N \rceil) \twoheadrightarrow H^0(F, m'g^*D|_F + \lceil -N \rceil|_F). \quad (6.23)$$

On the other hand, the negative part of $N + g^*E$ is supported in $\mathrm{Ex}(g)$, because any such component comes from the negative coefficient part of C in the discrepancy formula. Hence $0 \leq g_*\lceil -N \rceil \leq g_*g^*E = E$ by

$$\lceil -N \rceil = -\lfloor N \rfloor \geq 0^{32} \quad \text{and} \quad g_*\lfloor N + g^*E \rfloor = g_*\lfloor N \rfloor + g_*g^*E \geq 0.$$

Therefore, there are natural inclusions

$$H^0(X, m'D) \overset{\cdot\lceil -N \rceil}{\hookrightarrow} H^0(Y, m'g^*D + \lceil -N \rceil) \hookrightarrow H^0(X, m'D + E), \quad (6.24)$$

where the first inclusion comes from that g is birational and X normal, so $g_*\mathscr{O}_Y = \mathscr{O}_X$; and $\lceil -N \rceil$ is effective. The second inclusion, is because of that

$$H^0\big(Y, \mathscr{O}_Y(m'g^*D + \lceil -N \rceil\big) \cong H^0\big(X, \mathscr{O}_X(m'D) \otimes g_*\mathscr{O}_Y(\lceil -N \rceil)\big),$$

and $g_*\mathscr{O}_Y(\lceil -N \rceil) \hookrightarrow \mathscr{O}_Y(g_*\lceil -N \rceil)$ due to

$$\begin{aligned} f \in H^0\big(U, g_*\mathscr{O}_Y(\lceil -N \rceil)\big) = H^0(g^{-1}U, \mathscr{O}_Y(\lceil -N \rceil)) \\ \Rightarrow \mathrm{div}_Y(f) + \lceil -N \rceil \geq 0 \\ \Rightarrow g_*\,\mathrm{div}_Y(f) + g_*\lceil -N \rceil = \mathrm{div}_X(f)^{33} + g_*\lceil -N \rceil \geq 0 \\ \Rightarrow f \in H^0\big(V, \mathscr{O}_X(g_*\lceil -N \rceil\big). \end{aligned}$$

Combining these with $H^0(X, m'D + g_*\lceil -N \rceil) \hookrightarrow H^0(X, m'D + E)$, which is comes from $\mathscr{O}_X(g_*\lceil -N \rceil) \hookrightarrow \mathscr{O}_X(E)$[34], to get the desired inclusions.

Step 3: Now, for $m' \geq m_2$ the composition (6.24) is an isomorphism by condition (3), so both inclusions are isomorphism. In particular, we have

$$H^0(Y, m'g^*D) \cong H^0(X, m'D) \cong H^0(Y, m'g^*D + \lceil -N \rceil). \quad (6.25)$$

[32] Since we drop the coefficient 1 divisor F, we have $\lfloor N \rfloor \leq 0$.

[33] This is because $\mathbb{C}(X) \cong \mathbb{C}(Y)$. More precisely, we write $\mathrm{div}_Y(f) = \sum a_iD_i + \sum b_iE_i$, where D_i are divisors on Y and E_i are g-exceptional part. Then $g_*\,\mathrm{div}_Y(f) = \sum a_ig(D_i)$. But since g is birational, we have the isomorphisms between D.V.Rs: $\mathscr{O}_{Y,D_i} \cong \mathscr{O}_{X,g(D_i)}$. So the valuations in $\mathrm{div}_X(f)$ must coincide with these a_i, and won't arise other components.

[34] Be careful that, this injectivity is not like the Cartier divisor case which is obtained by the ideal

Consider the projective surjective morphism $\pi = h|_F : F \longrightarrow W = h(F)_{\text{red}} \subseteq S$, where W is an affine variety by [Har77, II, ex4.4]. Set $L = g^*D|_F$ and $N_F = N|_F$. The variety F is non-singular, N_F has snc support and its coefficients are less than 1. Thus by Corollary 5.2.20, $(F, \{N_F\})$ has klt singularities. Moreover, by our construction we have $\lceil -N_F \rceil = \lceil -N|_F \rceil = \lceil -N \rceil|_F$, and divisor L is π-nef.

For a fixed integer $a \geq m_1 + rm$, the adjunction formula and the preceding relative ampleness give that

$$aL + \lceil -N_F \rceil - (K_F + \{N_F\}) = aL - N_F - K_F = (ag^*D - N - F - K_Y)|_F$$

is π-ample. Now, the relative Non-vanishing Theorem applies on π, where the isomorphism of

$$H^0(F, m'L) \overset{\cdot \lceil -N_F \rceil}{\hookrightarrow} H^0(F, m'L + \lceil -N_F \rceil)$$

is comes from (6.25), (6.21) and the following commutative diagram

$$\begin{array}{ccc} H^0(Y, m'g^*D) & \xrightarrow{\cong} & H^0(Y, m'g^*D + \lceil -N \rceil) \\ {\scriptstyle \text{res}_{F,Y}}\downarrow & & \downarrow{\scriptstyle \text{res}_{F,Y}} \\ H^0(F, m'L) & \hookrightarrow & H^0(F, m'L + \lceil -N_F \rceil). \end{array}$$

Therefore, $H^0(F, m'L + \lceil -N_F \rceil) \neq 0$ for $m' \gg 0$.

We choose m' sufficiently large so that the non-vanishing holds and $m' \geq \max\{m_2.m_1 + rm\}$. The surjective composite

$$\begin{array}{ccccc} H^0(X, m'D) \cong H^0(Y, m'g^*D) & \xrightarrow{\cong} & H^0(Y, m'g^*D + \lceil -N \rceil) & \twoheadrightarrow & H^0(F, m'L + \lceil -N_F \rceil) \\ s & \longmapsto & s \cdot s_{\lceil -N \rceil} & & \end{array}$$

gives that $(s \cdot s_{\lceil -N \rceil})|_F \neq 0$, and hence $s|_F \neq 0$. But since $g^{-1} \operatorname{Bs} |m'D| = \operatorname{Bs} |m'g^*D|$[35], we have $g(F) \not\subseteq \operatorname{Bs} |m'D|$. If m' is a multiple of m, then $\operatorname{Bs} |m'D| \subseteq \operatorname{Bs} |mD|$. Since $g(F) \subseteq \operatorname{Bs} |mD|$, the inclusion is strict: $\operatorname{Bs} |m'D| \subsetneq \operatorname{Bs} |mD|$.

Step 4: Fix a prime number p, and take m', m to be powers of p. The descending sequence of closed subsets

$$\cdots \subseteq \operatorname{Bs} |p^{t+1}D| \subseteq \operatorname{Bs} |p^tD| \subseteq \operatorname{Bs} |p^{t-1}D| \subseteq \cdots$$

---

sheaf. Since $E \geq g_* \lceil -N \rceil$, then by

$$\begin{aligned} f \in H^0(X, g_* \lceil -N \rceil) &\Rightarrow \operatorname{div}_X(f) + g_* \lceil -N \rceil \geq 0 \\ &\Rightarrow \operatorname{div}_X(f) + E \geq 0 \\ &\Rightarrow f \in H^0(X, E). \end{aligned}$$

This gives that $\mathscr{O}_X(g_* \lceil -N \rceil) \hookrightarrow \mathscr{O}_X(E)$ between the reflexive sheaves.

[35] See the proof of *Step 2* in Basepoint-free Theorem.

will stabilizes, since X is a noetherian topological space. If its stable value were non-empty, the argument above would produce a strict inclusion between two terms after stabilization, arise a contradiction. Thus, $\mathrm{Bs}\,|p^t D| = \emptyset$ for all sufficiently large t.

For another prime number q, the same argument gives an integer s with $\mathrm{Bs}\,|q^s D| = \emptyset$. We fic such exponent t and s. Since p^t and q^s are coprime, every sufficiently large integer m can be written as a combination $m = ap^t + bq^s$ for some $a, b \in \mathbb{Z}_{\geq 0}$. But

$$\mathrm{Bs}\,|mD| \subseteq \mathrm{Bs}\,|p^t D| \cup \mathrm{Bs}\,|q^s D| = \emptyset,$$

see *Step 2* in the proof of the Basepoint-free Theorem. This implies that $|mD|$ is basepoint-free, and we take the maximum of the bounds over a finite number of affine open covering of S, to complete the proof. □

The following corollary is an equivalent statement of the relative Basepoint-free Theorem:

Corollary 6.2.6. *Let (X, Δ) be a klt pair with $\Delta \geq 0$ such that $K_X + \Delta$ is $\mathbb{R}$-Cartier, let $f : X \longrightarrow S$ be a projective morphism, and let D, E be Cartier divisors on X. Assume the following conditions:*

(1) D is f-nef.

(2) There is a positive integer m_1 such that $m_1 D + E - (K_X + \Delta)$ is f-nef and f-big.

(3) E is effective and there exists a positive integer m_2 such that for any positive integer, the natural homomorphism

$$f_* \mathscr{O}_X(mD) \longrightarrow f_* \mathscr{O}_X(mD + E)$$

is an isomorphism.

Then there exists a projective morphism $g : Z \longrightarrow S$ from a normal algebraic variety, a surjective projective morphism $h : X \longrightarrow Z$ with connected geometric fibre such that $f = g \circ h$, and a g-ample Cartier divisor H such that $h^* H \sim_{lin} D$.

Proof. By the relative Basepoint-free Theorem, there exists a positive integer m_3 such that mD is f-free for $m \geq m_3$. For morphism $X \longrightarrow \mathbb{P}\big(f_* \mathscr{O}_X(mD)\big)$, we take a Stein factorization

$$\phi_m = h'_m \circ h_m : X \longrightarrow Z \longrightarrow \mathbb{P}\big(f_* \mathscr{O}_X(mD)\big),$$

where h'_m is the finite part and h_m is the connected geometric fibres part. We denote the S-structure morphism of Z_m by g_m. By our construction, there exists[36] a g_m-ample Cartier divisor H_m on Z_m such that $mD \sim_{lin} h_m^* H_m$.

[36] Can use the argument of Theorem 1.5.6 on every closed point of S.

We can use a similar method in the proof of the Cone Theorem. In fact, for a f-relative curve C on X, C is contracted by h_m if and only if $(D \cdot C) = 0$, thanks to the projection formula and the relative ampleness of H_m. Therefore, $\mathrm{NE}(h_m)$ is unique determined. That is, there is an isomorphism $k_m : Z_m \longrightarrow Z_{m+1}$ and a commutative diagram

$$\begin{array}{ccccc} X & \xrightarrow{h_m} & Z_m & \longrightarrow & \mathbb{P}\big(f_*\mathscr{O}_X(mD)\big) \\ \| & & \downarrow{\scriptstyle k_m} & & \\ X & \xrightarrow[h_{m+1}]{} & Z_{m+1} & \longrightarrow & \mathbb{P}\big(f_*\mathscr{O}_X((m+1)D)\big) \end{array}$$

such that $k_m \circ h_m = h_{m+1}$. So, we can take $H := k_m^* H_{m+1} - H_m$, then so that $h_m^* H = (k_m \circ h_m)^* H_{m+1} - H_m \sim_{lin} (m+1)D - mD = D$. □

Corollary 6.2.7. *Let (X, Δ) be a klt pair with $\Delta \geq 0$ such that $K_X + \Delta$ is $\mathbb{Q}$-Cartier. Let $f : X \longrightarrow S$ be a projective morphism, Assume that $K_X + \Delta$ is f-nef and Δ is an f-big $\mathbb{Q}$-Cartier divisor. Then there exists a projective morphism $g : Z \longrightarrow S$ from a normal algebraic variety, a surjective projective morphism $h : X \longrightarrow Z$ with connected geometric fibres such that $f = g \circ h$, and a g-ample $\mathbb{Q}$-Cartier divisor H such that $h^* H \sim_{lin} K_X + \Delta$.*

Proof. The assertion is local, so we may assume that S is affine. As $K_X + \Delta$ is $\mathbb{Q}$-Cartier, there exists a positive integer m_1 such that $D = m_1(K_X + \Delta)$ is an f-nef Cartier divisor. As Δ is f-big, by relative Kodaira's lemma we have $\Delta \sim_{\mathbb{Q}} A + N$, where A is f-ample and N effective. take a sufficiently small positive rational number ε, $\big(X, (1-\varepsilon)\Delta + \varepsilon N\big)$ also has klt singularities by Corollary 5.2.38. Also,

$$D - \big(K_X + (1-\varepsilon)\Delta + \varepsilon N\big) = (m_1 - 1)(K_X + \Delta) + \varepsilon A$$

is f-ample, and of course, is f-nef and f-big. We therefore get the conclusion by the above corollary. □

Remark 6.2.8. The condition that the coefficients of B are rational numbers can be removed by using the relative Cone Theorem.

The following lemma is useful when generalizing statements for klt pairs to dlt pairs, which is the relative version of Proposition 5.2.43.

Lemma 6.2.9. *Assume that (X, Δ) is a dlt pair with $\Delta \geq 0$, let $f : X \longrightarrow S$ be a projective morphism with S is quasi-projective. Let H be an f-ample Cartier divisor on X, and let $c > 0$ be a real number. Then there exists an ample Cartier divisor A on S and an effective $\mathbb{R}$-divisor $B \sim_{\mathbb{R}} \Delta + c(H + f^* A)$ such that (X, B) is klt.*

Proof. Choose an ample Cartier divisor A on S such that $H + f^*A$ is ample on X, and apply Proposition 5.2.43 to $D = B$. Easy to verify that X is quasi-projective. □

Now we can show the relative Basepoint-free Theorem for dlt pairs:

Corollary 6.2.10. *Let (X, Δ) be a dlt pair with $\Delta \geq 0$ such that $K_X + \Delta$ is $\mathbb{R}$-Cartier, let $f : X \longrightarrow S$ be a projective morphism with S is quasi-projective, and let D, E be Cartier divisors on X. Assume the following conditions:*

(1) D is f-nef.

(2) There is a positive integer m_1 such that $m_1 D + E - (K_X + \Delta)$ is f-ample.

(3) E is effective and there exists a positive integer m_2 such that for any positive integer, the natural homomorphism

$$f_*\mathscr{O}_X(mD) \longrightarrow f_*\mathscr{O}_X(mD + E)$$

is an isomorphism.

Then there exists a positive integer m_3 such that for any integer $m \geq m_3$, mD is f-free. That is, the natural homomorphism

$$f^* f_*\mathscr{O}_X(mD) \longrightarrow \mathscr{O}_X(mD)$$

is surjective.

Proof. As the assertion is local, we may assume that S is affine. Take an effective $\mathbb{R}$-divisor $B \sim_{\mathbb{R}} \Delta + c(H + f^*A)$ as in above lemma, such that (X, B) has klt singularities. If the real number c can be sufficiently small, then $m_1 D + E - (K_X + \Delta)$ is still f-ample, by the open nature of relative ampleness. The corollary follows. □

The Basepoint-free Theorem states that a multiple of a certain Cartier divisor is free. Its effective version shows how large this multiple can be taken in practice.

Theorem 6.2.11. (Effective Basepoint-free [Kol93]) *Let (X, Δ) be a klt pair consisting of an n-dimensional normal algebraic variety and an effective $\mathbb{R}$-divisor, such that $K_X + \Delta$ is $\mathbb{R}$-Cartier. Let E be an effective Cartier divisor on X, let D be a Cartier divisor on X, and let $f : X \longrightarrow S$ be a projective morphism. Assume that the following conditions holds:*

(1) D is f-nef and $D + E - (K_X + \Delta)$ is f-nef and f-big.

(2) The natural homomorphism

$$f_*\mathscr{O}_X(mD) \longrightarrow f_*\mathscr{O}_X(mD+E)$$

is bijective for any positive integer m.

Then for any $m \geq 2n+3$, $|m^{n+1}D|$ is f-free, that is, the natural homomorphism

$$f^*f_*\mathscr{O}_X(m^{n+1}D) \longrightarrow \mathscr{O}_X(m^{n+1}D)$$

is surjective. Therefore, there exist a positive integer m_0 depending only on the dimension n such that $|mD|$ is f-free for any $m \geq m_0$.

Proof. **Step 0:** Since the assertion is local, we may assume that S is affine. So the statement "f-ample" and "f-free" is simply the same as "ample" and "free". As *Step 1* of the proof of the Theorem 2.5, we may assume that $m_1D+E-(K_X+\Delta)$ is ample.

Step 1: By Corollary 2.6, there exists a normal algebraic variety Y, a surjective projective morphism $g : X \longrightarrow Y$ over S with connected geometric fibres, and a g-ample Cartier divisor H on Y such that $D \sim_{lin} g^*H$. Denote the S-structure morphism of Y by h such that $f = h \circ g$ and $d = \dim Y$. Take X_s, Y_s to be the fibres over a general point s in $h(Y)$ and we denote

$$d_s = \dim Y_s \leq d \leq n.$$

Step 2: First, we show the effective version of the Non-vanishing theorem. The generic flatness says that there is an open dense subset $U \subseteq S$, such that for the flat morphism $f_U : X_U \longrightarrow U$ we have $\mathscr{O}_X(mD+E)|_{X_U}$ and $\mathscr{O}_X(mD)|_{X_U}$ are flat over U. See [FGA05, 5.12] or [SP25, Tag 052A]. But notice that, by the Vanishing Theorem 1.15 and

$$(mD+E)-(K_X+\Delta) = \underbrace{(m-1)D}_{\text{nef}} + \underbrace{D+E-(K_X+\Delta)}_{\text{ample}}$$

is an ample $\mathbb{R}$-Cartier divisor, we have

$$R^if_*\mathscr{O}_X(mD+E) = 0^{37} \tag{6.26}$$

for every $i > 0$. Consider a general point $s \in S$ lies in U, and the natural homomorphism

$$\varphi^i(s) : R^if_*\mathscr{O}_X(mD+E) \otimes k(s)^{38} \longrightarrow H^i(X_s, mD+E).$$

[37] Since can replace S by its image, we may assume that f is surjective.

For $i = N > \dim X_s$, we have $\varphi^N(s) : 0 \to 0$ is a surjective map. So by the *Cohomology and Base Change* in [Har77, III, 12.11]. we see $\varphi^{N-1}(s)$ is also surjective. Repeating this argument with the vanishing (6.26), in each step we can obtain that

$$H^i(X_s, mD + E) = 0$$

for every $i > 0$. By condition (2) we may replace $R^i f_* \mathscr{O}_X(mD + E)$ by $R^i f_* \mathscr{O}_X(mD)$ in (6.26), and repeat the above argument can obtain that

$$H^i(X_s, mD) = 0$$

for every $i > 0$. Therefore,

$$h^0(X_s, mD) = h^0(X_s, mD + E) = \chi(X_s, mD + E).$$

Applying the relative Non-vanishing Theorem for base variety is $\operatorname{Spec} k(s) = \operatorname{Spec} \mathbb{C}$, while $m_1 D_s + E_s - (K_{X_s} + \Delta_s)$ is ample by ampleness is stable under the finite pullbacks, we have $h^0(X_s, mD) \neq 0$. Moreover, since $D \sim_{lin} g^* H$, we have

$$H^0\big(X_s, \mathscr{O}_{X_s}(mD)\big) = H^0\big(X_s, g^* \mathscr{O}_{Y_s}(mH)\big) \cong H^0\big(Y_s, (g_s)_* \mathscr{O}_{X_s} \otimes \mathscr{O}_{Y_s}(mH)\big)$$

thank to the projection formula, and where $(g_s)_* \mathscr{O}_{X_s}$ is a coherent sheaf. Sy by the asymptotic Riemann-Roch, we know that $\chi(X_s, mD + E)$ is a (non-zero) polynomial of degree $\dim \operatorname{Supp}(g_s)_* \mathscr{O}_{X_s} = \dim Y_s = d_s$, and it has at most d_s distinct roots.

Now, we claim that for $m \geq 2d_2 + 3$, $H^0(X_s, mD) \neq 0$. Take $d_s + 1$ pairs $\{(i, m-i) | 1 \leq i \leq d_s + 1\}$[39]. If $H^0(X_s, mD) = 0$, then from the natural multiplication map

$$\begin{aligned} H^0(X_s, iD) \otimes H^0\big(X_s, (m-i)D\big) &\longrightarrow H^0(X_s, mD) \\ s \otimes t &\longmapsto s \cdot t, \end{aligned}$$

we know that either $H^0(X_s, iD) = 0$ or $H^0\big(X_s, (m-i)D\big) = 0$, which implies that $\chi(X_s, mD + E)$ has at least $d_s + 1$ distinct roots. This is a contradiction.

Step 3: We need prove the following lemma to complete the theorem:

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[38] Note that

$$R^i f_* \mathscr{O}_X(mD + E) \underset{\mathscr{O}_S}{\otimes} k(s) \cong R^i f_* \mathscr{O}_X(mD + E)_s \underset{\mathscr{O}_{S,s}}{\otimes} k(s) \cong R^i (f_U)_* \mathscr{O}_X(mD + E)_s \underset{\mathscr{O}_{U,s}}{\otimes} k(s).$$

[39] The pairs $\{1, m-1\}, \{2, m-2\}, \ldots, \{d_s + 1, m - d_s - 1\}$ are mutually disjoint: $i = m - i \Rightarrow m = 2i$ for $1 \leq i \leq d_s + 1$.

Lemma 6.2.12. *For any $m \geq 2d+3$ and take an irreducible component $\bar{Z}$ of the base locus*

$$\operatorname{Bs}|mH| = \operatorname{Supp}\Big(\operatorname{coker}\big(h^*h_*\mathscr{O}_Y(mH) \longrightarrow \mathscr{O}_Y(mH)\big)\Big),$$

where recall that $d = \dim Y$. Then for any $k \geq 2d+3$, $\bar{Z} \not\subseteq \operatorname{Bs}|kmH|$.

Proof. Note that $h(\bar{Z})$ is a subset of $h(Y)$, and they may not coincide. Denote $Z = g^{-1}(\bar{Z}) \subseteq X$.

First, we construct singularities in a neighborhood of Z. Denote $\bar{d} = \dim \bar{Z}$. Take $d - \bar{d} + 1$ Cartier divisors corresponding to general global sections of $\mathscr{O}_Y(mH)$, say, $\bar{M}_i (1 \leq i \leq d - \bar{d})$ and $\bar{N}$. note that the supports of $\bar{M}_i$ and $\bar{N}$ all contain $\operatorname{Bs}|mH|$. For positive numbers $\varepsilon, \delta > 0$, we set

$$M_{\varepsilon,\delta} = \sum_i (1-\delta) g^* \bar{M}_i + \varepsilon g^* \bar{N},$$

and take a neighborhood $\bar{U}$ of the generic point of $\bar{Z}$ and denote $U = g^{-1}(\bar{U})$. By shrinking $\bar{U}$, we may assume that it does not intersect irreducible components of $\operatorname{Bs}|mH|$ other than $\bar{Z}$. Here note that $\bar{Z}$ is not necessarily contained in $\bar{U}$. Moreover, since g is projective, every irreducible component of Z has closed image in $\bar{Z}$; so we may further shrink $\bar{U}$ and assume that $\bar{U} \cap g(W) = \emptyset$ for components W of Z not dominates $\bar{Z}$. Therefore,

$$Z \cap U = \bigcup_{W_i \subseteq Z} W_i \cap U$$

where $W_i \subseteq Z$ are irreducible components dominate $\bar{Z}$.

Fix $0 < \varepsilon < 1/2$, we **claim** that for sufficiently small $\delta > 0$, the pair

$$(X, \Delta + M_{\varepsilon,\delta}) \text{ is } \begin{cases} \text{klt on } U \setminus \operatorname{Bs}|mD| = U \setminus Z, \\ \text{non-lc on } U \cap \operatorname{Bs}|mD| = U \cap Z. \end{cases}$$

Proof. **Step 1:** We take W an irreducible component of Z which dominates $\bar{Z}$. Set

$$q = \operatorname{codim}_X W, \quad s = \dim W, \quad \dim X = q + s = n.$$

Near the generic point $\eta_{\bar{Z}}$ of $\bar{Z}$, the base locus $\operatorname{Bs}|mH|$ only consist of the component $\bar{Z}$ after we shrink the neighborhood $\bar{U}$. Since $\operatorname{codim}_Y \bar{Z} = d - \bar{d}$, for $d - \bar{d}$ general Cartier divisor $\bar{M}_i$, we have the set-theoretic equality

$$\bar{M}_1 \cap \cdots \cap \bar{M}_{d-\bar{d}} \cap \bar{U} = \bar{Z} \cap \bar{U}.$$

Therefore, by each $\bar{M}_i$ effective, we have

$$g^*\bar{M}_1 \cap \cdots \cap g^*\bar{M}_{d-\bar{d}} \cap U = Z \cap U.$$

By our choice of W dominates $\bar{Z}$, $\eta_W \in Z \cap U$, i.e. $W \cap U \neq \emptyset$. Since $\dim W \cap U = \dim W$ and $W \cap U$ still irreducible, we have $q \leq d - \bar{d}$ by Krull's height theorem. We may take an affine open neighborhood V of the generic point η_W, disjoint with other irreducible components of Z, such that $\operatorname{Bs}|mD| \cap V = W \cap V$[40].

Step 2: As our choice of $g^*\bar{M}_i$ is general, there only need q members of them make W to be an irreducible component of $g^*\bar{M}_1 \cap \cdots \cap g^*\bar{M}_q$; the reason is before when we cut q times, any larger irreducible component which contains W will not be the component of $\operatorname{Bs}|mD|$, so the next general member does not contains it and can continue cut down the dimension. Possibly shrinking V, we may arrange that

$$\operatorname{Supp} g^*\bar{M}_1 \cap \cdots \cap \operatorname{Supp} g^*\bar{M}_q \cap V = W \cap V, \tag{6.27}$$

where V is still affine. Immerse V into some projective space, and we take s general hyperplane sections $H_1, \ldots, H_s$. Set $T := V \cap H_1 \cap \cdots \cap H_s$, which satisfies the following conditions:

(1) T is a normal algebraic variety, by Bertini's normality theorem. See [FOV99, 3.4.14] or [Man82, II, 7].

(2) $\dim T = q$.

(3) $T \cap W$ is non-empty and consists of finite points. This is possible by the projective compactification of $W \cap V$ has dimension s.

Take a log resolution $\mu : \tilde{V} \longrightarrow V$ of the pair $(V, \Delta|_V + \sum g^*\bar{M}_i|_V + g^*\bar{N}|_V)$ such that

$$\operatorname{Supp} \mu_*^{-1}\Delta \bigcup \operatorname{Supp} \mu_*^{-1}(\sum g^*\bar{M}_i) \bigcup \operatorname{Supp} \mu_*^{-1} g^*\bar{N} \bigcup \operatorname{Ex}(\mu)$$

has snc support. Moreover, we denote all strict transforms and μ-exceptional divisors by $E_1, \ldots, E_N$, and their center by $C_j = \mu(E_j) \subseteq V$. By using Bertini-type theorem on non-singular variety $\tilde{V}$, we may assume that

$$\tilde{T} = \tilde{V} \cap \mu^*H_1 \cap \cdots \cap \mu^*H_s$$

is a non-singular algebraic variety. See [FOV99, 3.4.14] and [Man82, II, 7]. Furthermore, we can re-choose these μ^*H_j intersect with all strata of the snc divisor $E_1 + \cdots + E_N$ transversally, such that every non-empty $E_j|_{\tilde{T}}$ is reduced and different of them have no common components; this is possible by the series $|\mu^*H_j|$ is free. Note that for every center $C_j \not\supseteq W$, we have $T \cap W \cap C_j = \emptyset$; this is possible since

[40] Note that by our construction in Corollary 6.2.6, we have $g_*\mathscr{O}_X = \mathscr{O}_Y$; so $g^{-1}\operatorname{Bs}|mH| = \operatorname{Bs}|mD|$ by $D \sim_{lin} g^*H$.

$\dim(W \cap C_j) < s$ for these centers, we can choose s general hyperplane sections avoids it and there are only finitely many such centers.

Step 3: Now, we claim that (T, Δ_T) has klt singularities, where

$$\Delta_T = \Delta|_V \cap H_1 \cap \cdots \cap H_s = \Delta|_T.$$

Repeat the adjunction formula, can obtain that

$$K_T + \Delta_T = (K_V + \Delta|_V + H_1 + \cdots + H_s)|_T,$$

so the $K_T + \Delta_T$ is $\mathbb{R}$-Cartier. For log resolution $\mu : \tilde{V} \longrightarrow V$, we write

$$K_{\tilde{V}} + \Theta_0 = \mu^*(K_V + \Delta|_V).$$

Since $(V, \Delta|_V)$ is klt pair, the coefficients of Θ_0 are strictly lesser than 1. By our construction and the choice of H_i's, its restriction $\mu_T := \mu|_{\tilde{T}} : \tilde{T} \longrightarrow T$ is the corresponding log resolution on T. Therefore,

$$\begin{aligned} K_{\tilde{T}} + \Theta_0|_{\tilde{T}} &= \left(K_{\tilde{V}} + \Theta_0 + \sum_{i=1}^{s} \mu^* H_i \right) \Big|_{\tilde{T}} \\ &= \mu^* \left((K_V + \Delta|_V + \sum_{i=1}^{s} H_i)|_T \right) \\ &= \mu^*(K_T + \Delta_T), \end{aligned}$$

thanks to the adjunction formula again. By intersecting transversally, $\Theta_0|_T$ will not change the coefficients since each $E_j|_{\tilde{T}} = E_j|_{\bigcap_{i=1}^{s} g^* H_i}$ either be empty or will be a reduced divisor. Thus, (T, Δ_T) has klt singularities.

The above argument of adjunction formula also suitable for any variable $M_{\varepsilon,\delta}$. That is, if we denote

$$K_{\tilde{V}} + \Theta_{\varepsilon,\delta} = \mu^*(K_V + \Delta|_V + M_{\varepsilon,\delta}|_V),$$

then

$$\begin{aligned} K_{\tilde{T}} + \Theta_{\varepsilon,\delta}|_{\tilde{T}} &= \left(K_{\tilde{V}} + \Theta_{\varepsilon,\delta} + \sum_{i=1}^{s} \mu^* H_i \right) \Big|_{\tilde{T}} \\ &= \mu^* \left((K_V + \Delta|_V + M_{\varepsilon,\delta}|_V + \sum_{i=1}^{s} H_i)|_T \right) \\ &= \mu^*(K_T + \Delta_T + M_{\varepsilon,\delta}|_T). \end{aligned}$$

Step 4: Take a closed point $P \in T \cap W$, where the latter term consists of finite points. Take an open neighborhood T^o of P which exclude other points in $T \cap W$. We denote

$$D_i = g^*\bar{M}_i|_T (1 \leq i \leq q), \quad N_T = g^*\bar{N}|_T.$$

From formula (6.27), we see P is the isolated point in $D_1 \cap \cdots \cap D_q$, where N_T passing through P by $P \in \mathrm{Bs}\,|mD| \cap T$ and $N_T \in |mD|_T|$. So using Corollary 5.2.47 on the q-dimensional klt pair (T, Δ_T), can obtain that there exists a sufficient small number $\delta > 0$ depend on P, such that the pair $\left(T, \Delta_T + \sum(1-\delta)D_i + \varepsilon N_T\right)$ is klt in a punctured neighborhood $T^o \backslash \{P\}$ of P, but not lc at P. After adding effective divisor $\sum_{i=q+1}^{d-\bar{d}} (1-\delta) g^*\bar{M}_i|_T$, the klt statement may be false, but the pair $(T, \Delta_T + M_{\varepsilon,\delta}|_T)$ still not lc at P.

But this implies that in $\Theta_{\varepsilon,\delta}|_{\tilde{T}}$ for such ε and δ, there exist a prime divisor E_i such that $\mathrm{mult}_{E_i|_T} \Theta_{\varepsilon,\delta}|_{\tilde{T}} > 1$. By intersecting transversally, it is nothing but to say $\mathrm{mult}_{E_i} \Theta_{\varepsilon,\delta} > 1$, and we have

$$P \in \mu_T(E_i \cap \tilde{T}) \subseteq \mu(E_i) = C_i.$$

Then $W \subseteq C_i$, otherwise for $W \nsubseteq C_i$ we have $T \cap W \cap C_i = \emptyset$ by our construction in *Step 2*. Thus, the pair $(V, \Delta|_V + M_{\varepsilon,\delta}|_V)$ is not lc along $W \cap V$. We write it as $W \cap V \subseteq \mathrm{Nlc}(V, \Delta|_V + M_{\varepsilon,\delta}|_V)$. Note that the right-hand-side is a closed subset, since

$$\mathrm{Nlc}(V, \Delta|_V + M_{\varepsilon,\delta}|_V) = \bigcup_{\mathrm{mult}>1} \mu(E_j).$$

Step 5: In noetherian topological space $Z = g^{-1}(\bar{Z})$, there are only finitely many irreducible components. So we can repeat the above steps for each components of $Z \cap U = \mathrm{Bs}\,|mD| \cap U$. The finiteness of components give a common sufficiently small $\delta > 0$ satisfies the desired non-lc condition, and we have

$$W \cap V_W \subseteq \mathrm{Nlc}(X, \Delta + M_{\varepsilon,\delta}) \cap U$$

for every component W of Z dominates $\bar{Z}$. As non-lc locus is closed, and the closure of $W \cap V_W$ in U is $W \cap U$, we have

$$W \subseteq \mathrm{Nlc}(X, \Delta + M_{\varepsilon,\delta}) \cap U.$$

Taking the union,

$$Z \cap U \subseteq \mathrm{Nlc}(X, \Delta + M_{\varepsilon,\delta}).$$

And in $U \backslash Z = U \backslash \mathrm{Bs}\,|mD|$, $(X, \Delta + M_{\varepsilon,\delta})$ has klt singularities. This is because the coefficients of $M_{\varepsilon,\delta}$ is strictly lesser than 1, and we can take a log resolution of the klt

pair (X, Δ); the pullbacks of $g^*\bar{M}_i$ and $g^*\bar{N}$ are basepoint free over $U\backslash Z$, so we can arrange that they have snc support by Bertini's theorem and they do not contribute the exceptional part. This completes our claim. □

We denote this $M_{\varepsilon,\delta}$ by M, and continue to prove the lemma. Use the same notation, we take a log resolution $\mu : X' \longrightarrow X$ of the pair $(X, \Delta + M)$. We write the discrepancy formula by

$$K_{X'} + \Delta' = \mu^*(K_X + \Delta),$$

where the coefficients of Δ' are all strictly lesser than 1 by (X, Δ) is klt. Since the resolution of singularities is a sequence of blow-ups, we can take an effective divisor F supported on the μ-exceptional part such that $-F$ is μ-ample. Take a sufficiently small positive rational number ε' such that coefficients of $\Delta' + \varepsilon' F$ are all strictly lesser than 1, and $\mu^*\big(D + E - (K_X + \Delta)\big) - \varepsilon' F$ is ample by Proposition 1.5.8.

Consider the log-canonical threshold r as the following: on $\mu^{-1}(U)$, we set

$$r = \sup\big\{t \in \mathbb{R}\big|\lfloor \Delta' + t\mu^* M + \varepsilon' F\rfloor \leq 0\big\},$$

while the coefficients of $\Delta' + r\mu^* M + \varepsilon' F$ are all no greater than 1 and at least one coefficient is exactly 1. Here $r < 1$ as $(X, \Delta + M)$ is non-lc on $U \cap Z$, and note that outside $\mu^{-1}(U)$ the coefficients may be greater than 1. Write

$$K_{X'} + F_0 + \Delta'' = \mu^*(K_X + \Delta + rM).$$

Here F_0 is the sum of all irreducible components with coefficient 1 intersecting $\mu^{-1}(U)$. By perturbing the coefficients of F and shrinking $\bar{U}$, we may assume that F_0 is irreducible, $g \circ \mu|_{\mu^{-1}(U)}(F_0) = \bar{Z}$ and $\lfloor \Delta''|_{\mu^{-1}(U)}\rfloor \leq 0$ on $\mu^{-1}(U)$.

For a positive integer m', consider the following exact sequence

$$\begin{aligned} 0 \longrightarrow \mathscr{O}_{X'}\big(\mu^*(m'D + E) - \lfloor \Delta''\rfloor - F_0\big) &\longrightarrow \mathscr{O}_{X'}\big(\mu^*(m'D + E) - \lfloor \Delta''\rfloor\big) \\ \longrightarrow \mathscr{O}_{F_0}\Big(\big(\mu^*(m'D + E) - \lfloor \Delta''\rfloor\big)\big|_{F_0}\Big) &\longrightarrow 0. \end{aligned}$$

If we write

$$\begin{aligned} &\mu^*(m'D + E) - \Delta'' - F_0 - K_{X'} \\ &= \mu^*(m'D + E) - \mu^*(K_X + \Delta + rM) - \varepsilon' F \\ &\sim_{lin} \Big(m' - r\big((d - \bar{d})(1 - \delta) + \varepsilon\big)m - 1\Big)\mu^* D \text{ (nef if the coefficient is non-negative)} \\ &\quad + \mu^*\big(D + E - (K_X + \Delta)\big) - \varepsilon' F \text{ (ample)}, \end{aligned}$$

by $\mu^*(rM) \sim_{lin} r\big((d-\bar{d})(1-\delta)+\varepsilon\big)m\mu^*g^*H$ and $D \sim_{lin} g^*H$. Therefore, if $m' \geq r\big((d-\bar{d})(1-\delta)+\varepsilon\big)m+1$, the above divisor is ample. By Kawamata-Viehweg vanishing theorem, we have

$$\begin{aligned}
&H^i\big(X', K_{X'} + \lceil \mu^*(m'D+E) - \Delta'' - F_0 - K_{X'} \rceil\big)\\
&= H^i\big(X', \mu^*(m'D+E) - \lfloor \Delta'' \rfloor - F_0\big) = 0 \text{ for every } i > 0;\\
&H^i\Big(F_0, K_{F_0} + \lceil \big(\mu^*(m'D+E) - \Delta'' - F_0 - K_{X'}\big)|_{F_0} \rceil\Big)\\
&= H^i\Big(F_0, K_{F_0} + \lceil \big(\mu^*(m'D-\Delta'')|_{F_0}\big) - K_{F_0} \rceil\Big)\\
&= H^i\Big(F_0, \big(\mu^*(m'D+E) - \lfloor \Delta'' \rfloor\big)\big|_{F_0}\Big) = 0 \text{ for every } i > 0,
\end{aligned}$$

where notice that $\lceil -\Delta''|_{F_0} \rceil = \lceil -\Delta'' \rceil|_{F_0}$ by the snc property. Therefore, the following natural homomorphism

$$H^0\big(X, \mu^*(m'D+E) - \lfloor \Delta'' \rfloor\big) \longrightarrow H^0\Big(F_0, \big(\mu^*(m'D+E) - \lfloor \Delta'' \rfloor\big)\big|_{F_0}\Big)$$

is surjective. On the other hand, since

$$\Delta'' = \Delta' + r\mu^*M + \varepsilon' F - F_0, \quad \operatorname{mult}_{F_0} \Delta'' = 0,$$

the positive part of $-\Delta''$ is supported on $\operatorname{Ex}(\mu)$, while it comes from the μ-exceptional part in Δ'. Thus, we have the following homomorphism

$$H^0\big(X', \mu^*(m'D+E) - \lfloor \Delta'' \rfloor\big) \longrightarrow H^0(X, m'D+E)$$

is injective, thanks to the projection formula and algebraic Hartogs applying on $\mu_*\mathscr{O}_{X'}(\lceil -\Delta'' \rceil)$ to get $\mu_*\mathscr{O}_{X'}(\lceil -\Delta'' \rceil) \subseteq \mathscr{O}_X$. Moreover, we have a bijection homomorphism

$$H^o(Y, m'H) \longrightarrow H^0(X, m'D+E)$$

by $g_*\mathscr{O}_X = \mathscr{O}_Y$, $D \sim_{lin} g^*H$ and condition (2). Therefore, for $m' \geq (d-\bar{d}+1)m$[41], follows from the following commutative diagram

$$\begin{array}{ccccc}
H^0\big(X', \mu^*(m'D+E) - \lfloor \Delta'' \rfloor\big) & \hookrightarrow & H^0(X, m'D+E) & \xrightarrow{\sim} & H^0(Y, m'H)\\
\downarrow & & & \searrow^{\operatorname{res}\circ\mu^*} & \downarrow{\scriptstyle \operatorname{res}}\\
 & & & & H^0(\bar{Z}, m'H|_{\bar{Z}})\\
 & & & & \downarrow{\scriptstyle (\dagger)}\\
H^0\Big(F_0, \big(\mu^*(m'D+E) - \lfloor \Delta'' \rfloor\big)|_{F_0}\Big) & & \xhookrightarrow{\cdot s_{\lfloor \Delta'' \rfloor}|_{F_0}} & & H^0\big(F_0, \mu^*(m'D+E)|_{F_0}\big)
\end{array}$$

[41] Since by our preceding choice, we have

$$(d-\bar{d}+1)m \geq (d-\bar{d})m + m/2 + 1 \geq (d-\bar{d}+\varepsilon)m + 1 \geq r\big((d-\bar{d})(1-\delta)+\varepsilon\big)m + 1.$$

the image of the right vertical arrow contains $H^0\Big(F_0, \big(\mu^*(m'D+E) - \lfloor \Delta'' \rfloor\big)|_{F_0}\Big)$.

Note that the natural homomorphism (†) is comes from

$$\begin{aligned} H^0(\bar{Z}, m'H|_{\bar{Z}}) &\xrightarrow{(g\circ\mu)^*} H^0\big(F_0, (g\circ\mu)^*(m'H|_{\bar{Z}})\big) \\ &\longrightarrow H^0\big(F_0, \mu^*(m'D)|_{F_0}\big) \\ &\xrightarrow{\mu^* s_E|_{F_0}} H^0\big(F_0, \mu^*(m'D+E)|_{F_0}\big), \end{aligned}$$

where $s_E \in H^0(X, E)$ is the canonical section, and first homomorphism is due to the isomorphism $(g\circ\mu)^*\big(\mathscr{O}_Y(m'H)|_{\bar{Z}}\big) \cong \mathscr{O}_{X'}\big(\mu^*(m'D)\big)|_{F_0}$. And the bottom horizontal injectivity is follows from the following argument. Since $\operatorname{mult}_{F_0}\Delta'' = 0$, $F_0^o := F_0 \setminus \operatorname{Supp}\lfloor \Delta'' \rfloor$ is a dense open subset of F_0, and therefore we have $\mathscr{O}_{X'}(\lfloor \Delta'' \rfloor)|_{F_0^o} \cong \mathscr{O}_{F_0^o}$ is the trivial line bundle. Its canonical section $s_{\lfloor \Delta'' \rfloor}$ is regular and non-trivial on F_0^0, which is invertible. Take a section σ in lower left corner, and lift it up to be $\tilde{\sigma}$ such that $\tilde{\sigma}|_{F_0} = \sigma$. Inject it into $H^0(X, m'D+E) \cong H^0\big(X', \mu^*(m'D+E)\big)$, and restrict it on F_0 we can obtain a section $\mu^*(\tilde{\sigma}\cdot s_{\lfloor \Delta'' \rfloor})|_{F_0}$. On F_0, it is equals to $\sigma \cdot s_{\lfloor \Delta'' \rfloor}|_{F_0}$, so this homomorphism is well-define because of that a non-zero global section of an invertible sheaf cannot vanish on a dense open subset, cf.[Har77, II, 8.19]. Furthermore, the injectivity is due to the same reason: if $(\sigma \cdot s_{\lfloor \Delta'' \rfloor})|_{F_0^o} = 0$, then $\sigma|_{F_0^o} = 0$ by $s_{\lfloor \Delta'' \rfloor}|_{F_0^o}$ is a unit and hence $\sigma = 0$.

Recall that $\lfloor \Delta'' \rfloor \leq 0$ on $\mu^{-1}(U)$. This means that there might be some irreducible component of Δ'' with coefficient greater than 1, but it does not intersect $\mu^{-1}(U)$. □

□

Remark 6.2.13. (Alternative way to prove the claim in above lemma) Thanks to Prof. Chen Jiang, he gave me the following guidance. The Lemma 5.2.45 and Corollary 5.2.47 can be generalized on non-closed points, and use the argument of the intersection of lc center is still an lc center to derived the conclusion.

6.2C The Relative Rationality Theorem

In this section we discuss the relative version of the Kawamata's Rationality Theorem, while if we suppose that the source and target spaces are projective variety then the relative version can easily deduced from the absolute version. But unfortunately, we want generalize it on the case of just algebraic varieties.

Theorem 6.2.14. (Relative Rationality Theorem) *Let (X, Δ) be a klt pair with X is a normal algebraic variety, and $\Delta \geq 0$ is a $\mathbb{Q}$-divisor such that the $\mathbb{Q}$-Cartier divisor $K_X + \Delta$ is not f-nef, where $f : X \longrightarrow S$ is a projective morphism. Let $a = a(f)$ be an integer such that $a \cdot (K_X + \Delta)$ is Cartier. Let A be an f-nef and f-big Cartier divisor, and define the threshold*

$$r = r(A) = \max\left\{t \in \mathbb{R} \middle| A + t(K_X + \Delta) \text{ is } f\text{-nef}\right\}.$$

Then r is a rational number, and if we write $r/a = p/q$ as an irreducible fraction then[42]

$$0 < q < a(f) \cdot \left(\max_{\substack{s \in S \\ \text{closed point}}} \left\{ \dim X_s \right\} + 1 \right).$$

Proof. Replace S by the image of f, we may assume that f is surjective. Note that

$$\max_{\substack{s \in S \\ \text{any point}}} \left\{ \dim X_s \right\} = \max_{\substack{s \in S \\ \text{closed point}}} \left\{ \dim X_s \right\} =: b.$$

One direction of the inequality is clear, the other direction is by the upper semi-continuous of the fibre dimension. The subset $\{s \in S | \dim X_s \geq d\} = f(E_d)$ is closed as f is proper, see [Har77, II, ex.3.22]. So for any (possibly non-closed) point $s \in S$, if $\dim X_s = d$ then the closed set above also contains $\overline{\{s\}}$. Then for any closed point $t \in \overline{\{s\}}$, we have $\dim X_t \geq \dim X_s$.

Step 0: Since $K_X + \Delta =: K$ is negative on some curve contracted by f, the threshold r is finite. This is because of that $t \leq (A \cdot C)/(-K \cdot C) < \infty$ for every relative curve C. And note that r is attained since the nef cone is closed, see Theorem 1.5.22, that is the choices of $r = \sup\{r \in \mathbb{R}\}$ and $r = \max\{t \in \mathbb{R}\}$ are coincide, by we can consider a number sequence $\{t_n\}_n = 1$ converges to r such that

$$\big((A + rK) \cdot C\big) = \lim_{n \to \infty} \big((A + rK) \cdot C\big) \geq 0$$

[42] Note that this bounded result is stronger than the absolute version: if S is a point then for $\frac{r}{a} = \frac{u}{av} = \frac{u/g}{av/g}$, where $g = \gcd(u, a)$; then

$$\frac{av}{g} \leq a \cdot (\dim X + 1) \Rightarrow v \leq \gcd(u, a) \cdot (\dim X + 1).$$

for every relative curve C. The closed cone implies that $A + rK$ is f-nef.

Step 1: We first reduce to the case where S is affine and A is f-ample. Since S is quasi-compact, for finite affine covering $S = \cup S_i$, let r_i denote the threshold over S_i with $r_i = +\infty$ when L is relatively nef over S_i. Then $r = \min_i\{r_i\}$. The same $a(f)$ is a Cartier index[43] on each open subset, and the fibre dimension won't be greater. Therefore, we may assume that S is affine.

If $r = 0$, there is nothing to prove. We may assume that $r > 0$. We may assume that $r > 0$. To obtain a contradiction, we need consider either r/a is irrational or $q > a(b+1)$. Choose integers $m, n \gg 0$ such that $mr > a$ and, in the rational case, $\gcd(mn, q) = 1$. Set $D_0 = mA + aK$, $A' = nD_0$. Both D_0 and $D_0 - K$ are f-nef and f-big. In fact, they can be decomposed into

$$D_0 = (m - \frac{a}{r})A + \frac{a}{r}(A + rL),$$
$$D_0 - L = (m - \frac{a-1}{r})A + \frac{a-1}{r}(A + rL).$$

The coefficients of A are positive, and $A + rK$ are f-nef, so we can use the relative version of Example 3.3.12[44].So by the relative Basepoint-free Theorem 2.5 applied on D_0, we see $A' = nD_0$ is f-free for every sufficiently large n. Using the same argument as *Step 2* in the proof absolute version on A', we may assume that A is basepoint free.

Moreover, applying Corollary 2.6 on the free series $|A|$, there is a projective birational morphism $h : X \longrightarrow Z$ such that

$$A \sim_{lin} h^*H, \quad K_X + \Delta \sim_{\mathbb{Q}} h^*(K_Z + \Delta_Z),$$

where H is a relatively ample Cartier divisor on Z, $\Delta_Z = h_*\Delta$ and $a(K_Z + \Delta_Z)$ is Cartier. Note that (Z, Δ_Z) also has klt singularities by Lemma 5.2.16. Then after replacing X, Δ, A by Z, Δ_Z, H, may assume that A is f-ample, so ample by S affine.

Step 2: We prove this by contradiction. Choose a real number $0 < \varepsilon < 1$ such that $\frac{a(b+1)}{q} < \varepsilon < 1$ in the rational case; for irrational case ε is arbitrary. For integral x and y, we put

$$D(x, y) = xA + ayK, \quad \delta(x, y) = ay - rx.$$

Consider the bounded lattice

$$\Gamma = \big\{(x, y) \in \mathbb{Z}^2 \big| 0 < ay - rx < \varepsilon\big\}.$$

[43]The small positive integer make the multiple of a divisor to be Cartier.

[44]The proof is tedious and bookkeeping for passing the absolute version to relative version, by reduce on generic fibre. We left it to readers as an exercise

Our next goal is to apply the Lemma 1.27, while we choose the integer parameter to be 1, its real slope to be r/a and the upper bound to be ε/a. Then, every non-zero polynomial $P(x,y)$ of total degree $\leq b$ is non-vanished for all sufficient large integral solutions $(x,y) \in \Gamma$. Otherwise r/a is rational, and its denominator $q \leq \frac{b+1}{\varepsilon/a}$ which contradicts our assumption[45].

Now, fix an integral vector $v \in \Gamma$ and choose an integral pair $w = (w_1, w_2)$ with positive coordinates such that $0 < aw_2 < rw_1$. Then

$$D(w) = w_1 A + aw_2 K = (w_1 - \frac{aw_2}{r})A + \frac{aw_2}{r}(A + rK)$$

is an ample divisor, by Corollary 1.5.20. By our choice of v, we have $av_2 > rv_1$; so together we have

$$\frac{w_2}{w_1} < \frac{r}{a} < \frac{v_2}{v_1},$$

that is the line $ay = rx$ lies strictly between the rays through v and w.

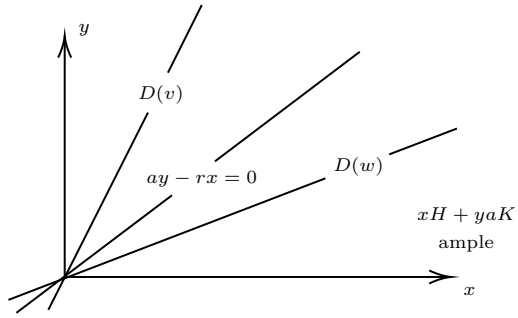


Figure 6.2.: Position of the line $ay = rx$.

As u and w are not parallel, they generates a full rank sub-lattice $\Lambda = \mathbb{Z}v + \mathbb{Z}w \subseteq \mathbb{Z}^2$. So the quotient group $\mathbb{Z}^2/\Lambda$ is finite which consisting elements $\{e_i\}$, such that every integral pair z can be written as

$$z = kv + lw + e_i, \quad k, l \in \mathbb{Z}.$$

Set $u = (1, r/a)$ which lies on the line $ay = rx$. Due to the reason in figure 6.2, there exist real number $\alpha, \beta > 0$ such that $u = \alpha v + \beta w$. Then for any $z \in \Gamma$, we have

$$z = xu + (0, \frac{\delta(x,y)}{a}).$$

[45] Note that we cannot apply Lemma 6.1.27 for $0 < ay - rx < \varepsilon$, since otherwise we will get $v \leq \frac{a(b+1)}{\varepsilon}$ and hence $q \leq \frac{a^2(b+1)}{g\varepsilon}$ where $g = \gcd(u, a)$.

Since $0 < \delta(z) < \varepsilon$, we can write

$$z = \alpha x v + \beta x w + O(1),$$

and as e_i has only finitely many choices, its subtraction will also have a bounded difference:

$$z - e_i = \alpha x v + \beta x w + O(1),$$

that is

$$z - e_i = kv + lw, \quad k = \alpha x + O(1), \quad l = \beta x + O(1).$$

Since $\alpha, \beta > 0$, when $x \to \infty$ we must have $k \to \infty$ and $l \to \infty$. Therefore, for every sufficiently large integral pairs $z \in \Gamma$ we have an expression

$$z = kv + lw + e, \tag{6.28}$$

where e is a representative in $\mathbb{Z}^2/\Lambda$ and both k, l be sufficiently large.

For each e, the Cartier divisor $D(e) + lD(w)$ is free, since $D(w)$ is ample. Via the natural multiplication homomorphism

$$\begin{aligned} H^0\big(X, D(v)\big)^{\otimes k} \otimes H^0\big(X, D(e) + lD(w)\big) &\longrightarrow H^0\big(X, D(z)\big) \\ s^{\otimes k} \otimes t &\longmapsto s^k \cdot t \end{aligned}$$

gives that $\operatorname{Bs}|D(z)| \subseteq \operatorname{Bs}|D(v)|$ for $z \in \Gamma$ sufficiently large.

Since we are in a noetherian topological space, the descending chain

$$\operatorname{Bs}|D(v_1)| \supseteq \operatorname{Bs}|D(v_1)| \cap \operatorname{Bs}|D(v_2)| \supseteq \cdots \bigcap \operatorname{Bs}|D(v_i)| \supseteq \cdots$$

will stabilizes to a closed subset $Z_0 \subseteq X$. But by our previous analysis, for each v_i appear in the intersection Z_0, there is a sufficiently large integral pair $z_i \in \Gamma$ such that $\operatorname{Bs}|D(z_i)| \subseteq \operatorname{Bs}|D(v_i)|$. Therefore, there is a large enough $(x, y) \in \Gamma$ such that $\operatorname{Bs}|D(x, y)| = Z_0$.

Step 3: In this step we show that Z_0 is non-empty, and not is X. For $(x, y) \in \Gamma$ with $ay > 1$, we have

$$D(x, y) - K = \frac{ay - 1}{r}(A + rK) + \frac{1 - \delta(x, y)}{r}A$$

to be a positive linear combination of an f-nef and an f-ample, so it is f-ample. Applying the vanishing Theorem 1.13 on klt pair (X, Δ), we obtain that

$$R^i f_* \mathscr{O}_X\big(D(x, y)\big) = 0$$

for every $i > 0$. Let $\eta \in S$ be the generic point. Since $\operatorname{Spec} k(S) \longrightarrow S$ is flat, the flat base change gives

$$H^i\Big(X_\eta, \mathscr{O}_X\big(D(x,y)\big)\big|_{X_\eta}\Big) \cong \Big(R^i f_* \mathscr{O}_X\big(D(x,y)\big)\Big)_\eta = 0 \tag{6.29}$$

for every $i > 0$.

Now, for arbitrary integers x, y we put

$$P_X(x,y) = \chi\Big(X_\eta, \mathscr{O}_X\big(D(x,y)\big)\big|_{X_\eta}\Big).$$

The asymptotic Riemann-Roch gives that the numerical polynomial P_X has $\deg P_X \leq \dim X_\eta \leq b$. This polynomial is non-zero because $P_X(x,0) = h^0(X_\eta, xA|_{X_\eta})$ and $A|_{X_\eta}$ is ample. By (6.29),

$$P_X(x,y) = h^0\Big(X_\eta, \mathscr{O}_X\big(D(x,y)\big)\big|_{X_\eta}\Big)$$

for all sufficiently large $(x,y) \in \Gamma$. By *Step 2* and our assumption applying on this polynomial, $P_X(x,y) \neq 0$ for such (x,y). So the global section space of $\mathscr{O}_X\big(D(x,y)\big)|_{X_\eta}$ is non-trivial and same for $f_*\mathscr{O}_X\big(D(x,y)\big)$; therefore $H^0\big(X, D(x,y)\big) \neq \{0\}$ by S affine. Thus $Z_0 \neq X$.

On the other hand, $ay/x > r$ on Γ, so $D(x,y)$ is not f-nef. It cannot be f-free. Therefore $Z_0 \neq \emptyset$.

Step 4: Fix a sufficiently large $(x_0, y_0) \in \Gamma$ such that achieves the closed subset Z_0. Recover our notation in *Step 1* to put $D_0 = D(x_0, y_0)$ and $\delta_0 = \delta(x_0, y_0) > 0$. We take a member $M \in |D_0|$, a log resolution $g : Y \longrightarrow X$ of the pair $(X, \Delta + M)$ and of the base ideal of $|D_0|$[46], and write $h = f \circ g$. We may arrange that

$$K_Y + C = g^*(K_X + \Delta), \quad g^*M = M_1 + M_2$$

where $|M_1|$ is a free series, $M_2 \geq 0$ is the fixed part of $g^*|D_0|$. In particular, by Z_0 is non-empty, $M_2 \neq 0$ and $g(\operatorname{Supp} M_2) = Z_0$. All coefficients of C are all lesser than 1 as (X, Δ) is a klt pair.

Note that our log resolution make that

$$\Sigma = \operatorname{Ex}(g) \cup \operatorname{Supp}(g_*^{-1}\Delta + g_*^{-1}M)$$

has simple normal crossing support. We can construct an effective $\mathbb{Q}$-divisor C' such that

$$\operatorname{Supp} C' = \operatorname{Ex}(g) \cup \operatorname{Supp}(C + M_2) \subseteq \Sigma$$

[46] Note that $H^0(X, D_0)$ may not be finite dimensional, but the choice of base ideal is by take a finite dimensional sub-vector space consisting all generators over A, where $S = \operatorname{Spec} A$. The method can see footnote 31.

and $-C'$ is g-ample. In fact, resolution of singularities is a sequence of blow-ups, so admits an effective $\mathbb{Q}$-divisor C'' with support $\mathrm{Ex}(g)$ such that $-C''$ is g-ample. Adding sufficiently small positive coefficients on the remaining components gives C', by the open nature of the relative ampleness, see Proposition 1.5.16.

Then by [EGA II, 4.6.13(ii)], we can choose a sufficiently small positive rational number $0 < \lambda \ll 1$ such that

$$\frac{1-\varepsilon}{r} g^* A - \lambda C'$$

is a h-ample $\mathbb{R}$-divisor on Y, where we may arrange that the coefficients of $C + \lambda C'$ are all lesser than 1. Consider the threshold

$$c = \sup\left\{ t \in \mathbb{R} \middle| \lfloor tM_2 + C + \lambda C' \rfloor \leq 0 \right\}.$$

Since $M_2 \neq 0$ and the coefficients of $C + \lambda C'$ are lesser than 1, the threshold c is positive. Moreover, it is rational; since if we expand $M_2 = \sum a_i P_i$ and $C + \lambda C' = \sum b_i P_i$, the threshold condition is $ta_i + b_i < 1$ and therefore

$$c = \min_{a_i > 0} \left\{ \frac{1-b_i}{a_i} \right\}.$$

The maximal coefficient of $cM_2 + C + \lambda C'$ is exactly 1, while be careful that there may $\lfloor cM_2 + C + \lambda C' \rfloor \leq 0$ no more. Use the same tie-breaking method before, we can perturbing the coefficients of C' while preserving the preceding properties, we may assume that there exactly one prime divisor has coefficient 1. We denote this (smooth) prime divisor by F. Note that we must have $F \subseteq \operatorname{Supp} M_2$, so $g(F) \subseteq \operatorname{Bs} |D_0| = Z_0$. We write

$$cM_2 + C + \lambda C' = N + F,$$

where F is not the component of N, every coefficient of N is strictly lesser than 1 and $N + F$ has snc support. Using $g^* D_0 \sim_{lin} M_1 + M_2$, for integers x and y we have

$$\begin{aligned}
&g^* D(x,y) - N - F - K_Y \\
&= g^*\big(cD_0 + D(x - cx_0, y - cy_0)\big) - N - F - K_Y \\
&\sim_{\mathbb{Q}} (x - cx_0) g^* A + (ay - acy_0 - 1) g^* K - cM_2 - \lambda C' \\
&= \frac{ay - acy_0 - 1}{r} g^*(A + rK) + \frac{\varepsilon - \delta(x,y) + c\delta_0}{r} g^* A \quad \big(\text{nef if } (x,y) \in \Gamma \text{ sufficiently large}\big) \\
&\quad + cM_1 \quad (\text{nef since basepoint-free}) \\
&\quad + \frac{1-\varepsilon}{r} g^* A - \lambda C' \quad (h\text{-ample}).
\end{aligned}$$

So if $(x,y) \in \Gamma$ sufficiently large, the right hand side is h-ample by Corollary 1.5.20. Using Kawamata-Viehweg vanishing theorem, can obtain that

$$H^1\big(Y, g^* D(x,y) + \lceil -N \rceil - F\big) = H^1\big(Y, K_Y + \lceil g^* D(x,y) - N - F - K_Y \rceil\big) = 0,$$

by the base variety S is affine. The associated long exact sequence with F therefore yields a surjection

$$H^0\big(Y, g^*D(x,y) + \lceil -N \rceil\big) \twoheadrightarrow H^0\big(F, g^*D(x,y)|_F + \lceil -N \rceil|_F\big), \tag{6.30}$$

where by our construction $\lceil -N|_F \rceil = \lceil -N \rceil|_F$.

On the other hand, the negative part of N is supported in $\mathrm{Ex}(g)$, because any such component comes from the negative coefficient part of C in the discrepancy formula. Therefore, $\lceil -N \rceil = -\lfloor N \rfloor \geq 0$ by we drop the coefficient 1 divisor F, and $g_*\mathscr{O}_Y(\lceil -N \rceil) = \mathscr{O}_X$ by algraic Hartogs. It gives that

$$H^0\big(Y, g^*D(x,y) + \lceil -N \rceil\big) \cong H^0\big(X, D(x,y)\big),$$

thanks to the projective formula. This identification sends global section σ on X to the global section $g^*\sigma \cdot s_{\lceil -N \rceil}$ on Y.

Step 5: We denote $W = g(F)$ and $T = h(W)$, and consider the morphism

$$\rho = g|_F : F \longrightarrow W, \quad \nu = f|_{g(F)} : W \longrightarrow h(W), \pi = \nu \circ \rho : F \longrightarrow T.$$

And consider $\mathscr{E} = \mathscr{O}_F(\lceil -N \rceil|_F)$,

$$\mathscr{M}(x,y) = \mathscr{O}_F\Big(\big(g^*D(x,y) + \lceil -N \rceil\big)\big|_F\Big) \quad \text{and} \quad \mathscr{L}(x,y) = \mathscr{O}_X\big(D(x,y)\big)\big|_W.$$

Since F is not a component of $\lceil -N \rceil$, the restriction $\lceil -N \rceil|_F$ is an effective Cartier divisor. In particular, $\mathscr{O}_F \hookrightarrow \mathscr{E}$. By construction, $\mathscr{M}(x,y) = \mathscr{E} \otimes \rho^*\mathscr{L}(x,y)$. Apply the adjunction formula on smooth subvariety $F \subseteq Y$ gives that

$$\big(g^*D(x,y) - N - F - K_Y\big)\big|_F = g^*D(x,y)|_F - N|_F - K_F.$$

For all sufficiently large $(x,y) \in \Gamma$, the right hand side is h-ample over T. Since T is also an affine variety by h is proper, so it is an ample divisor on F, consequently ρ-ample. Thus, we can apply the Kawamata-Viehweg vanishing theorem to get

$$R^i\pi_*\mathscr{M}(x,y) = 0, \quad R^i\rho_*\mathscr{M}(x,y) = 0 \tag{6.31}$$

for every $i > 0$. Fix one sufficiently large vector of Γ, we have

$$\begin{aligned} R^i\rho_*\mathscr{M}(x,y) &= R^i\rho_*\big(\mathscr{E} \otimes \rho^*\mathscr{L}(x,y)\big) \\ &\cong R^i\rho_*\mathscr{E} \otimes \mathscr{L}(x,y) \\ &= 0 \end{aligned}$$

for every $i > 0$ thanks to the projection formula. But as $\mathscr{L}(x, y)$ is a line bundle, it gives that $R^i\rho_*\mathscr{E} = 0$ for $i > 0$. We set $\mathscr{G} = \rho_*\mathscr{E}$, dominance of the morphism ρ and the injection $\mathscr{O}_F \hookrightarrow \mathscr{E}$ give an injection $\mathscr{O}_W \hookrightarrow \rho_*\mathscr{O}_F \hookrightarrow \rho_*\mathscr{E} = \mathscr{G}$, while recall that the functor ρ_* is left exact.

Consider the generic point η_T of T, the morphism $\operatorname{Spec} k(\eta_T) \longrightarrow T$ is flat. And consider the flat base change

$$\rho_{\eta_T} : F \underset{T}{\times} k(\eta_T) = F_{\eta_T} \longrightarrow W \underset{T}{\times} k(\eta_T) = W_{\eta_T}.$$

The schemes F_{η_T} and W_{η_T} are projective over k_{η_T}, since π and ν are projective. We use subscript η_T on sheaves to denote their natural pullbacks. Flat base change for ρ gives

$$(\rho_{\eta_T})_*\mathscr{E}_{\eta_T} \cong (\rho_*\mathscr{E})_{\eta_T}{}^{47} = \mathscr{G}_{\eta_T}, \quad R^i(\rho_{\eta_T})_*\mathscr{E}_{\eta_T} \cong (R^i\rho_*\mathscr{E})_{\eta_T} = 0 \text{ for } i > 0, \tag{6.32}$$

see [Har77, III, 9.3]. Flatness also preserves the injection, so $\mathscr{O}_{W_{\eta_T}} \hookrightarrow \mathscr{G}_{\eta_T}$ and $\mathscr{G}_{\eta_T} \neq 0$.

We denote the restrictions of $\mathscr{O}_X(A)$ and $\mathscr{O}_X(aK)$ on W_{η_T} by A_{η_T} and aK_{η_T}, while the former is ample by immersion into a projective space is stable under base change. By formula in (6.32), for all integers x, y we have

$$\begin{aligned} P_F(x, y) &= \chi\big(F_{\eta_T}, M(x, y)_{\eta_T}\big) \\ &= \chi\Big(F_{\eta_T}, \mathscr{E}_{\eta_T} \otimes \big(\rho^*\mathscr{L}(x, y)\big)_{\eta_T}\Big) \\ &= \chi\big(F_{\eta_T}, \mathscr{E}_{\eta_T} \otimes (\rho_{\eta_T})^*\mathscr{L}(x, y)_{\eta_T}\big) \\ &= \chi\big(W_{\eta_T}, \mathscr{G}_{\eta_T} \otimes \mathscr{O}_{W_{\eta_T}}(xA_{\eta_T} + ayK_{\eta_T})\big), \end{aligned}$$

thanks to the projection formula, the vanishing in (6.32) and [Har77, III, ex8.1]. The asymptotic Riemann-Roch gives

$$\deg P_F \leq \dim \operatorname{Supp} \mathscr{G}_{\eta_T} \leq \dim W_{\eta_T} \leq b,$$

where the last equality follows from $W_{\eta_T} \subseteq X_{\eta_T}$. This polynomial is non-zero because $P_F(x, 0) = h^0\big(W_{\eta_T}, \mathscr{G}_{\eta_T} \otimes \mathscr{O}_{W_{\eta_T}}(xA_{\eta_T})\big)$ and A_{η_T} is ample. By (6.31),

$$P_F(x, y) = h^0\big(F_{\eta_T}, M(x, y)_{\eta_T}\big)$$

for all sufficiently large $(x, y) \in \Gamma$, while the flat base change gives

$$H^i\big(F_{\eta_T}, \widetilde{M(x, y)_{\eta_T}}\big) \cong \big(R^i\pi_*M(x, y)\big)_{\eta_T} = 0$$

[47] Recall that on the generic point the pullback coincides with taking stalk.

for every $i > 0$. Like in *Step 3*, by *Step 2* and our assumption applying on this polynomial, $P_F(x,y) \neq 0$ for such (x,y). So the global section space of $M(x,y)_{\eta_T}$ is non-trivial and same for $\pi_* M(x,y)$; therefore $H^0\big(F, M(x,y)\big) \neq \{0\}$ by T affine.

Step 6: Choose $(x,y) \in \Gamma$ large enough such that it achieves the stable base locus Z_0 and the surjection (6.30) holds. Via previous step $H^0\big(F, \mathscr{M}(x,y)\big) \neq \{0\}$ and the surjection (6.30), we may pick a non-zero section on F and lift it to Y, then identify it on X to some $\sigma \in H^0\big(X, D(x,y)\big)$. Its restriction to F is $(g^*\sigma \cdot s_{\lceil -N \rceil})|_F \neq 0$. Since F is not a component of $\lceil -N \rceil$, $s_{\lceil -N \rceil}|_F \neq 0$, and therefore $g^*\sigma|_F \neq 0$. But this means that F is not contained in $g^{-1}\operatorname{Bs}|D(x,y)| = g^{-1}(Z_0)$, contradicts our construction that $g(F) \subseteq \operatorname{Bs}|D(x,y)|$ in *Step 4*. This completes the proof. □

6.2D The Relative Cone Theorem

In this subsection, we will give the relative Contraction Theorem and the relative Cone Theorem separately. The following is directly comes from [Kaw24].

The following relative Contraction Theorem is a consequence of the relative Basepoint-free Theorem. But be careful that, in the proof of the (absolute) Cone Theorem, we first prove that the extremal face F has a supporting function which is defined over $\mathbb{Q}$ in *Step 6* then began to prove the (absolute) Contraction Theorem. So in the assumption we need count this in.

Theorem 6.2.15. (Relative Contraction Theorem) *Let (X, Δ) be a klt pair with $\Delta \geq 0$ such that $K_X + \Delta$ is $\mathbb{R}$-Cartier, let $f : X \longrightarrow S$ be a projective morphism and let F be a $K_X + \Delta$-negative extremal face of $\overline{\mathrm{NE}}(X/S)$. Assume that there is an f-nef $\mathbb{Q}$-Cartier divisor D on X such that*

$$F = F_D = D^{\perp} \cap \overline{\mathrm{NE}}(X/S).$$

Then the following assertions hold:

(1) The contraction morphism $\operatorname{cont}_F : X \longrightarrow Z$ associated to F exists.

(2) The smallest linear subspace containing F coincides with the image of the injective map $N_1(X/Z) \hookrightarrow N_1(X/S)$, and F coincides with the image of $\overline{\mathrm{NE}}(X/Z)$.

(3) $-(K_X + \Delta)$ is cont_F-ample.

(4) Let L be a Cartier divisor on X such that $(L \cdot C) = 0$ for every curve C with $[C] \in F$. Then there is a Cartier divisor L_Z on Z such that $L \sim_{lin} \operatorname{cont}_F^* L_Z$.

(5) $\rho(X/S) = \rho(Z/S) + \rho(X/Z)$.

Proof. (1) By taking a multiple of D, we may assume that D is an f-nef Cartier divisor.

As $K_X + \Delta$ is negative on $F\backslash\{0\}$, we have

$$\overline{\mathrm{NE}}(X/S)_{K_X+\Delta\geq 0} \cap F = \{0\}. \tag{6.33}$$

We consider the dimension $\rho(X/S) - 1$ unit sphere $\big(N_1(X/S)\backslash\{0\}\big)/\mathbb{R}_{>0}$ which is isomorphic to $\big\{\gamma \in N_1(X/S)\big|\|\gamma\| = 1\big\}$, by we identify γ to $\gamma/\|\gamma\|$. On the other hand,

$$\frac{\overline{\mathrm{NE}}(X/S)_{K_X+\Delta\geq 0}\backslash\{0\}}{\mathbb{R}_{>0}} \cong \begin{pmatrix}\text{closed subset of the } \rho(X/S)-1\\ \text{dimensional unit sphere}\end{pmatrix}$$

which is a compact subset, and the quotient function $(K_X + \Delta)/D$ is well-defined on the quotient space. By (6.33), the function $(D\cdot -)$ is strictly positive on that compact subset, therefore the values of $(K_X+\Delta)/D$ is bounded. There exists a sufficiently small positive real number ε such that $D-\varepsilon(K_X+\Delta)$ is strictly positive on $\overline{\mathrm{NE}}(X/S)\backslash\{0\}$. By Kleiman's ampleness criterion, $D-\varepsilon(K_X+\Delta)$ is f-ample, see Theorem 1.5.26.

Applying the relative Basepoint-free Theorem, while there is a positive integer m_1 such that

$$m_1 D - (K_X + \Delta) = \frac{1}{\varepsilon}\big(D - \varepsilon(K_X + \Delta)\big) + (m_1 - \frac{1}{\varepsilon})D$$

is an f-nef and f-big $\mathbb{R}$-Cartier divisor, then the natural homomorphism

$$f^* f_* \mathscr{O}_X(mD) \longrightarrow \mathscr{O}_X(mD)$$

is surjective. It induces a projective morphism $\varphi_{|mD|} : X \longrightarrow \mathbb{P}\big(f_*\mathscr{O}_X(mD)\big) = \underline{\mathrm{Proj}}_S \mathrm{Sym}\big(f_*\mathscr{O}_X(mD)\big)$ over S. And by definition, we have $\varphi^*_{|mD|}\mathscr{O}(1) \cong \mathscr{O}_X(mD)$.

Take the *Stein factorization* of $\varphi_{|mD|}$, we get a surjective morphism $h : X \longrightarrow Z = \underline{\mathrm{Spec}}_S f_*\mathscr{O}_X$ to a normal algebraic variety and a finite morphism $Z \longrightarrow \mathbb{P}\big(f_*\mathscr{O}_X(mD)\big)$. We denote the S-structure morphism of Z by g.

We claim that h is the contraction morphism cont_F associated to F. First, for a curve C on X which is contracted by h, then we have $\deg_C \varphi^*_{|mD|}\mathscr{O}(1) = 0$. That is, $(mD \cdot C) = 0$ and hence $C \in mD^\perp \cap \overline{\mathrm{NE}}(X/S) = F$. Second, take F' to be the closed convex cone spanned by equivalence classes of curves contracted by h, then $F' = \overline{\mathrm{NE}}(X/Z) \subseteq F$.

Now, we assume the contrary that $F' \neq F$, then there exists a Cartier divisor D' on X which is strictly positive on $F'\backslash\{0\}$ but negative on somewhere of F, by the separating hyperplane theorem which can see [BV04, 2.5.1]. Note that $mD \sim_{lin} h^*D''$ for some g-ample Cartier divisor D''. Since D' is h-ample by our construction and Theorem 1.5.26, for any sufficiently large n, $D' + nh^*D''$ is f-ample by [EGA II,

4.6.13(ii)]. In this case, $D'+nh^*D''$ is strictly positive on $\overline{\mathrm{NE}}(X/S)\backslash\{0\}$ and hence on $F\backslash\{0\}$; but this is a contradiction since $h^*D'' \sim_{lin} mD$ is identically zero on $F\backslash\{0\}$.

(2) is comes the proof of (1).

(3) is follows from the relative Kleiman's ampleness criterion testing on $F\backslash\{0\}$.

(4) Since $N_1(X/Z) \cong \mathsf{Span}_{\mathbb{R}}F$ by (2), the Cartier divisor L is cont_F-nef, and hence $L-(K_X+\Delta)$ is cont_F-ample. We apply the relative Basepoint-free Theorem on L and cont_F, which implies that $m'L$ is cont_F-free for $m' \gg 0$. It induces a projective morphism $X \longrightarrow \mathbb{P}\big((\mathrm{cont}_F)_*\mathscr{O}_X(m'L)\big)$ over Z. Take a *Stein factorization*

$$\begin{array}{ccccc} X & \xrightarrow{\psi} & Z' & \longrightarrow & \mathbb{P}\big((\mathrm{cont}_F)_*\mathscr{O}_X(m'L)\big) \\ & \searrow^{\mathrm{cont}_F} & \downarrow p & \swarrow & \\ & & Z & & \end{array}$$

where $\mathrm{cont}_F = p\circ\psi$ and $\psi_*\mathscr{O}_X = \mathscr{O}_{Z'}$. Note that $m'L \sim_{lin} \psi^*A$ for some p-ample Cartier divisor A. Now we prove that p is an isomorphism. Suppose that p has positive dimension fibres, then we pick a curve Γ in it and some curve $C \in F$ on X which dominates Γ. Then

$$(m'L\cdot C) = (\psi^*A\cdot C) = (A\cdot\psi_*C) > 0$$

by the p-ampleness of A, and this contradicts the fact that $(L\cdot C)=0$ for every $[C]\in F$. Since p is fibre-finite and projective, we see it is a finite morphism. On the other hand, since we have

$$p_*\mathscr{O}_{Z'} = p_*\psi_*\mathscr{O}_X = (\mathrm{cont}_F)_*\mathscr{O}_X = \mathscr{O}_Z. \tag{6.34}$$

By finite morphism is affine, the equality (6.33) gives the isomorphism between Z' and Z on each affine cover. Thus, $Z' \cong Z$ and then $m'L \sim_{lin} \mathrm{cont}_F^* E_{m'}$ via this isomorphism pulls back A.

The above argument holds for every $m' \gg 0$, so can set $L \sim_{lin} \mathrm{cont}_F^*(E_{m'+1} - E_{m'})$.

(5) is the relative version of Corollary 1.25. □

6.3 Running the MMP

Appendix A

Relatively Linearly Equivalent

All varieties are assumed to be normal to establish the theory of Weil divisors. In [Kaw24, P.8] gives two definitions of *relatively linearly equivalent*:

Definition A.0.1. (Version I) Given a morphism $f : X \longrightarrow S$ between two normal algebraic varieties, two Weil divisors are *relatively linearly equivalent over* S, which we denote by $D \sim_S D'$, if there exists an open covering $\{S_i\}$ of S s.t.

$$D|_{S_i} \sim D'|_{S_i}$$

after restriction over each S_i.

Definition A.0.2. (Version II) Under same condition, we say D and D' are *relatively linearly equivalent over* S if there exists a Cartier divisor B on S such that

$$D \sim D' + f^*B$$

In general, these two definitions are not the same, and the version I is much weaker, but under the condition that f is proper surjective, and has connected geometric fibres, they are equivalent.

Remark A.0.3. I think Kawamata's setting is too strong as my advisor has this similar consideration. We might just need that f is proper dominant, and has a connected generic fibre.

Since version II is the stronger version, it trivially implies version I. Now we discuss how version I implies version II under the above suitable condition.

Since $\{S_i\}$ covers S, this is just saying $D|_{f^{-1}S_i} \sim D'|_{f^{-1}S_i}$, so

$$D|_{f^{-1}S_i} - D'|_{f^{-1}S_i} = \mathrm{div}(g_i)$$

for some $g_i \in k(X)^*$.

On overlaps $f^{-1}S_i \cap f^{-1}S_j = f^{-1}(S_i \cap S_j)$, we have

$$\phi_{ij} := g_i/g_j \in \mathscr{O}_X(f^{-1}(S_i \cap S_j))^\times.$$

To show $\mathscr{O}_X(D - D') \cong f^*L$ for some line bundle on S, we construct it as follows:

- Want units $v_{ij} \in \mathscr{O}_S(S_j \cap S_j)^\times$ such that
$$f^*(v_{ij}) = \phi_{ij}$$
- f is proper dominant, and has a connected generic fibre, [SP25, Tag 0AY8] implies that
$$f_*\mathscr{O}_X = \mathscr{O}_S$$
- For any open subsets $U \subseteq S$, the natural map
$$\mathscr{O}_S(U) \longrightarrow \mathscr{O}_X(f^{-1}U) = f_*\mathscr{O}_X(U)$$
is an isomorphism.
- Taking $U = S_i \cap S_j$, via this isomorphism we obtain units using $v_{ij} \in \mathscr{O}_S(S_i \cap S_j)^\times$ s.t.
$$f^*(v_{ij}) = \phi_{ij}$$
- These v_{ij} satisfy the cocycle condition, so they can be glued together to form a line bundle L on S such that
$$f^*L \cong \mathscr{O}_X(D - D')$$
i.e. $D - D' \sim f^*B$ as we are working on integral schemes.

Appendix B

Local Properties of the Algebraic Scheme

B.1 Bertini's Theorem

Theorem B.1.1. (Bertini) *Let X be a non-singular closed subvariety in $\mathbb{P}^n_k$, where k is an algebraically closed field. Then there exists a hyperplane $H \subseteq \mathbb{P}^n_k$, not containg X, and such that the scheme $H \cap X$ is regular at every point.*

Furthermore, the set of hyperplanes with this property forms an open dense subset of the complete linear series $|H|$, considered as a projective space. We say that the ***general hyperplane section*** *satisfies the theorem.*

Proof. We fix a closed point $x \in X$ and consider the set of hyperplanes

$$B_x = \{H \mid H \supseteq X \text{ or } H \not\supseteq X \text{ but } x \in H \cap X \text{ with } x \text{ is not regular in it.}\}$$

These are the **bad** hyperplanes with respect to the point x.

Now a hyperplane H is cut by a non-zero global section $f \in V = H^0(\mathbb{P}^n, \mathscr{O}_{\mathbb{P}^n}(1))$. Let us fix an $f_0 \in V$ such that $x \notin H_0$, where H_0 is the hyperplane defined by f_0. Then we can define a map of k-vector spaces

$$\begin{aligned} \varphi_x : V &\longrightarrow \mathscr{O}_{X,x}/m_x^2 \cong \mathscr{O}_{X,x}/m_x \oplus m_x/m_x^2 \\ f &\longmapsto f/f_0 \text{ in } \mathscr{O}_{X,x} \text{ modulo } m_x^2. \end{aligned}$$

Given $f \in V$, then f/f_0 is a regular function on $\mathbb{P}^n - H_0$, which induces a regular function on $X - X \cap H_0$. Now the scheme $H \cap X$ is defined at x by the ideal generated

by f/f_0 in $\mathscr{O}_x$. So $x \in H \cap X$ if and only if $\varphi_x(f) \in m_x$, and x is non-regular on $H \cap X$ if and only if $\varphi_x(f) \in m_x^2$, because in that case:

$$\begin{aligned}\dim_k \frac{m_x/\varphi_x(f)}{(m_x/\varphi_x(f))^2} \underset{\varphi_x(f)\in m_x^2}{=} \dim_k \frac{m_x/\varphi_x(f)}{m_x^2/\varphi_x(f)} &= \dim_k m_x/m_x^2 \\ &= r \\ &= \dim \mathscr{O}_x\end{aligned}$$

but $\dim \mathscr{O}_x/\varphi_x(f) = r-1$ by Hauptidealsatz, i.e. $\mathscr{O}_x/\varphi_x(f)$ won't be a regular local ring. Thus we see that the hyperplanes $H \in B_x$ correspond exactly to those $f \in \ker\varphi_x$ (note also that $\varphi_x(f) = 0$ if and only if $H \supseteq X$.)

Since x is a closed point and k is algebraically closed, m_x is generated by linear forms in the coordinates, so we see that φ_x is surjective, as in $\mathscr{O}_{X,x}/m_x^2$ the higher degree terms vanish.

If $\dim X = r$, then $\dim_k \mathscr{O}_x/m_x^2 = \dim_k \mathscr{O}_{X,x}/m_x \oplus m_x/m_x^2 = r+1$. We have $\dim V = n+1$, so $\dim \ker \varphi_x = n+1-(r+1) = n-r$. This shows that B_x is a linear system of hyperplanes of dimension $n-r-1$.

Now, considering the complete linear system $|H|$ as a projective space, consider the subset $B \subseteq X \times |H|$ consisting of all pairs $\langle x, H\rangle$ such that $x \in X$ is a closed point and $H \in B_x$. Clearly B is the set of closed points of a closed subset of $X \times |H|$ since by the above construction, it is locally the zero locus of $\varphi_x(f)$, for H is determined by such f. We also denote this subset by B, and give it a reduced induced scheme structure.

We have just seen that the first projection

$$p_1 : B \longrightarrow X$$

is surjective, hence B is irreducible because X is. And the fibres of this p_1 are projective spaces of dimension $n-r-1$, so B has the dimension $(n-r-1)+r = n-1$.

Therefore, we considering the second projection

$$p_2 : B \longrightarrow |H|,$$

we have $\dim p_2(B) \leq n-1 = \dim B$. Since $\dim |H| = n$, we conclude that $p_2(B) \subseteq |H|$. If $H \in |H| \backslash p_2(B)$, then $H \not\supseteq X$ and every point of $H \cap X$ is regular, so that H satisfies the requirements of the theorem.

Finally note that since X is projective variety, $p_2 : X \times |H| \longrightarrow |H|$ is a proper morphism by properness is stable under base change. B is closed in $X \times |H|$, so by [Har77, II, ex.4.4] the $p_2(B)$ is also closed in $|H|$. Thus $|H| - p_2(B)$ is an open dense subset of $|H|$ which is what we desired. □

B.2 Differential Forms in Algebraic Geometry

We fix our ground field k to be always algebraically closed.

Recall that, for X a variety over k and $x \in X$ a closed point, we have the *Zariski cotangent space* $T^{\vee}_{X,x}$ to be m_x/m_x^2 which is a $k(x) = k$-vector space, and the *Zariski tangent space* $T_{X,x} = \mathrm{Hom}_{k(x)}(m_x/m_x^2, k(x))$. When we say X is *smooth* or *non-singular* over k at closed point $x \in X$ we mean that

$$\dim_k m_x/m_x^2 = \dim X,$$

and furthermore it is an open condition. So for arbitrary variety we always have an open dense subset $U \subseteq X$ such that X is smooth at any closed point $x \in U$.

Example B.2.1. (Differential forms in $\mathbb{A}^n$) For $X = \mathbb{A}^n$, the differential forms on X look like df for $f \in \mathscr{O}_{\mathbb{A}^n}$, i.e. for any closed point $x \in X$, there is an open neighborhood U_x of x equipped with a map

$$\begin{aligned} f_x : U_x &\longrightarrow k(x) = k \\ x &\longmapsto \sum \frac{\partial f}{\partial x_i}(x) dx_i \end{aligned}$$

satisfying that

(1) k-linear: $\lambda_1 df_1 + \lambda_2 df_2 = d(\lambda_1 f_1 + \lambda_2 f_2)$;

(2) $d\alpha = 0$ if $\alpha \in k$ is a constant function;

(3) $d(f_1 f_2) = f_1 df_2 + f_2 df_1$.

Definition B.2.2. (Module of Kähler Differential) Let $\phi : A \to B$ be a ring homomorphism, which makes B into an A-algebra. Then we define a B-module $\Omega_{B/A}$ to be the quotient of the free B-module generated by $\{d\beta \mid \beta \in B\}$, while the quotient relations are

(1) $d\beta_1 + d\beta_2 = d(\beta_1 + \beta_2)$;

(2) $d\alpha = 0$ if $\alpha \in \phi(A)$;

(3) $d(\beta_1 \beta_2) = \beta_1 d\beta_2 + \beta_2 d\beta_1$.

Note that $\Omega_{B/A}$ has a canonical map

$$\begin{aligned} d : B &\to \Omega_{B/A} \\ \beta &\mapsto d\beta. \end{aligned}$$

Example B.2.3. If $A = \mathbb{C}$, and $B = \mathscr{O}_{\mathbb{C}^n} \cong \mathbb{C}[x_1, \ldots, x_n]$. Then

$$\Omega_{B/A} = \{df \mid f \in \mathscr{O}_{\mathbb{C}^n} \text{ , } df = \text{its regular differentials}\}$$

Proposition B.2.4. (Universal property of $\Omega_{B/A}$) *Let $\phi : A \to B$ be a ring homomorphism. For any B-module M, we define the module of A-derivations denoted by $\mathrm{Der}_A(B, M)$, to be the following data*

- *an A-derivation is a B-module homomorphism $B \xrightarrow{D} M$ such that*
  - *(1) $D(ab) = aD(b)$ for any $a \in \phi(A)$ and $b \in B$;*
  - *(2) $D(b_1b_2) = b_1 \cdot D(b_2) + b_2 \cdot D(b_1)$ for any $b_1, b_2 \in B$.*
- *$B \to \Omega_{B/A}$ is a A-derivation, i.e. $(d : B \to \Omega_{B/A}) \in \mathrm{Der}_A(B, \Omega_{B/A})$.*

We have the result that $\Omega_{B/A}$ represents the $\mathrm{Der}_A(B, -)$ functor, i.e. for any B-module M and for any $D : B \to M \in \mathrm{Der}_A(B, M)$, there is a unique B-module homomorphism $\widetilde{D} : \Omega_{B/A} \to M$ making the following diagram commute

$$\begin{array}{ccc} B & \xrightarrow{\quad D \quad} & M \\ {\scriptstyle d}\downarrow & \nearrow_{\exists!\ \widetilde{D}} & \\ \Omega_{B/A} & & \end{array}$$

and we say that $\Omega_{B/A}$ is the universal object with this factorization property, and hence that $\mathrm{Der}_A(B, M) \cong \mathrm{Hom}_{\mathrm{Mod}_B}(\Omega_{B/A}, M)$.

Example B.2.5. Take $A = \mathbb{C}$ and $B = \mathbb{C}[x] = M$, corresponding to the affine line.

- $\mathrm{Der}_{\mathbb{C}}(\mathbb{C}[x], \mathbb{C}[x]) \cong \mathbb{C}[x]\langle\partial_x\rangle$, which is a free B-module generated by the 1st order differential operator

$$\begin{aligned} \partial_x : \mathbb{C}[x] &\longrightarrow \mathbb{C}[x] \\ f &\longmapsto f'. \end{aligned}$$

- $\Omega_{\mathbb{C}[x]/\mathbb{C}} \cong \mathbb{C}[x]\langle dx\rangle$, where

$$\begin{aligned} d : \mathbb{C}[x] &\longrightarrow \Omega_{\mathbb{C}[x]/\mathbb{C}} \\ f &\longmapsto df = f'(x)dx. \end{aligned}$$

- $\mathrm{Hom}_{\mathbb{C}[x]}(\mathbb{C}[x]\langle dx\rangle, \mathbb{C}[x]) \cong \mathbb{C}[x]\langle\partial_x\rangle = \mathrm{Der}_{\mathbb{C}}(\mathbb{C}[x], \mathbb{C}[x])$, here ∂_x is viewed as the dual of dx, which coincides with differential geometry.

Example B.2.6. (1) If $M = B$, then we have

$$\begin{aligned}\mathrm{Der}_A(B, B) &\cong \mathrm{Hom}_B(\Omega_{B/A}, B)\\ &\cong \Omega^{\vee}_{B/A},\end{aligned}$$

that is the tangent vectors are the dual of differentials.

(2) If $A \to B \to 0$ exact, then for any $\beta \in B$ we can write $\beta = \bar{a}$ for some $a \in A$. We have $d\beta = 0$ so $\Omega_{B/A} = 0$, meaning there are no relative differentials.

(3) Take $A = \mathbb{C}$, $B = \mathbb{C}[x_1, \ldots, x_n]$, then

$$\mathrm{Der}_A(B, B) \cong \mathbb{C}[x_1, \ldots, x_n]\langle \partial_{x_1}, \ldots, \partial_{x_n}\rangle.$$

These are the 1st order differential operators, and then the canonical map

$$\begin{aligned} d : B &\longrightarrow \Omega_{B/A} \cong \mathbb{C}[x_1, \ldots, x_n]\langle dx_1, \ldots, dx_n\rangle\\ f &\longmapsto df\end{aligned}$$

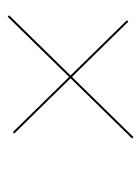
gives the 1st order differential forms.

Proof. Write $f = xg_1 + g_2$ for $g_2 \in \mathbb{C}[x_2, \ldots, x_n]$, then

$$\begin{aligned} df &= d(xg_1) + dg_2\\ &= g_1 dx_1 + dg_2 + x_1 dg_1 \text{ repeating this process for } g_1 \text{ i.e. drop degree on } x_1\\ &= \frac{\partial f}{\partial x_1} \cdot dx_1 + dg_2 \text{ repeat for } g_2.\end{aligned}$$

□

(4) Take $A = \mathbb{C}$ and $B = \mathbb{C}[x, y]/(xy)$ corresponding to the crossing lines, which is singular at the origin. Then we have $\Omega_{B/A} = \frac{\mathbb{C}[x,y]}{(xy)}\langle dx, dy\rangle/(xdy + ydx)$:



- Since $d(xy) = d0 = 0$, we have $w := xdy = -ydx$ and $y \cdot w = x \cdot w = 0$, thus

$$B \cdot w = \mathbb{C}\langle w\rangle \subseteq \Omega_{B/A}$$

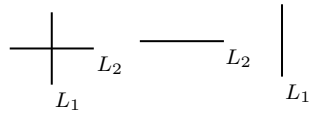
is a submodule and a one dimensional $\mathbb{C}$-vector space.

- Furthermore, we have

$$\begin{aligned}\Omega_{B/A}/B\cdot w &\cong \mathbb{C}[x]\langle dy\rangle \oplus \mathbb{C}[y]\langle dx\rangle \\ &\cong \frac{\mathbb{C}[x,y]}{(y)}\langle dy\rangle \oplus \frac{\mathbb{C}[x,y]}{(x)}\langle dx\rangle \\ &= \Omega_{B_2/\mathbb{C}} \oplus \Omega_{B_1/\mathbb{C}}.\end{aligned}$$

  Therefore we have the short exact sequence where $B\cdot w$ is a torsion module

$$0 \longrightarrow B\cdot w \longrightarrow \Omega_{B/\mathbb{C}} \longrightarrow \Omega_{B_2/\mathbb{C}} \oplus \Omega_{B_1/\mathbb{C}} \longrightarrow 0$$



  by $xw = yw = 0$, so it is supported on the origin $(0,0)$. Taking dual we see that taking dual will lose information such as killing the torsion.

$$0 \longleftarrow 0 \longleftarrow T_{X,p} \xleftarrow{\cong} T_{L_1,p} \oplus T_{L_2,p} \longleftarrow 0$$

  That is, differentials contain more information than tangents, and are more canonical.

Example B.2.7. Let $\mathbb{C} \hookrightarrow K$ be a field extension and K be a finitely generated $\mathbb{C}$-algebra.

- $\mathrm{Der}_{\mathbb{C}}(K,K)$ has a basis $\{df_1,\dots,df_n\}$ forming a K-vector space if and only if $\{f_1,\dots,f_n\}$ is a transcendental basis of K over $\mathbb{C}$.
- The relations of $df_1,\dots,df_n$ are correspond one-to-one to $P(f_1,\dots,f_n)=0$ for some $P \in \mathbb{C}[x_1,\dots,x_n]$.
- $\dim_K(\mathrm{Der}_{\mathbb{C}}(K,K)) = \mathrm{tr.deg}_{\mathbb{C}} K = \dim X$ for $K = k(X)$ being the rational function field of a variety X. Note that the left side is equal to

$$\begin{aligned}\dim_K \mathrm{Hom}_K(\Omega_{K/\mathbb{C}}, K) &= \dim_K \Omega_{\mathscr{O}_{X,\eta}/\mathbb{C}} \\ &= \dim_K (\Omega_{X/\mathbb{C}})_\eta\end{aligned}$$

  while the last equality follows from [Har77, II, 8.6A].

Proposition B.2.8. (Canonical way) *Let $\phi : A \to B$ be a ring homomorphism, together with a multiplicative map*

$$\begin{aligned} B \underset{A}{\otimes} B &\xrightarrow{\delta} B \\ f \otimes g &\longmapsto f\cdot g\end{aligned}$$

We take $I := \ker \delta$ *which is a* $B \underset{A}{\otimes} B$*-module, making* $B \cong B \underset{A}{\otimes} B/I$*.*

Then $I/I^2 \cong \Omega_{B/A}$ *as a* B*-module, while the* B*-module structure comes from*

$$I/I^2 \cong I \otimes (B \underset{A}{\otimes} B/I).$$

Proof. For I/I^2 as a B-module, we define the $B \cong B \underset{A}{\otimes} B/I$ action as

$$\beta \cdot \bar{f} := \overline{(\beta \otimes 1)} \cdot \bar{f} = \overline{(1 \otimes \beta)} \cdot \bar{f} \in I/I^2$$

for $\beta \in B$ and $f \in I$, while as $\beta \otimes 1 - 1 \otimes \beta \mapsto \beta \cdot 1 - 1 \cdot \beta = 0$ for any $\beta \in B$.

Then we define

$$\begin{aligned} \varphi : \Omega_{B/A} &\longrightarrow I/I^2 \\ d\beta &\longmapsto \overline{1 \otimes \beta - \beta \otimes 1} \in I/I^2 \end{aligned}$$

where satisfies

- φ is A-linear and $d\alpha = 0$ for any $\alpha \in A$.
- By computation we have

$$\begin{aligned} d(\beta_1\beta_2) \longmapsto \overline{1 \otimes \beta_1\beta_2 - \beta_1\beta_2 \otimes 1} &= \overline{1 \otimes \beta_1\beta_2 - \beta_1 \otimes \beta_2 + \beta_1 \otimes \beta_2 - \beta_1\beta_2 \otimes 1} \\ &= \overline{(1 \otimes \beta_2)(1 \otimes \beta_1 - \beta_1 \otimes 1) + (\beta_1 \otimes 1)(1 \otimes \beta_2 - \beta_2 \otimes 1)} \\ &= \beta_1 \cdot \overline{(1 \otimes \beta_2 - \beta_2 \otimes 1)} + \beta_2 \cdot \overline{(1 \otimes \beta_1 - \beta_2 \otimes 1)} \end{aligned}$$

  which is what $\beta_1 d\beta_2 + \beta_2 + d\beta_1$ maps to.

Now we define $\psi : I/I^2 \to \Omega_{B/A}$ as follows. To get this, first we consider the module

$$B \oplus I/I^2 \cong B \underset{A}{\otimes} B/I \oplus I/I^2 = B \underset{A}{\otimes} B/I^2$$

which is a ring, and the module $B \oplus \Omega_{B/A}$ defines a product

$$(\beta_1 \oplus d\beta_2) \cdot (\beta_1' \oplus d\beta_2') = \beta_1 \cdot \beta_1' \oplus (\beta_1 d\beta_1' + \beta_1' d\beta_2)$$

making it into a ring. Then we get a ring homomorphism

$$\begin{aligned} \psi' : B \underset{A}{\otimes} B/I^2 &\longrightarrow B \oplus \Omega_{B/A} \\ \overline{\beta_1 \otimes \beta_2} &\longmapsto \beta_1\beta_2 \oplus \beta_1 d\beta_2 \end{aligned}$$

and since we mod out the I^2, we have that for $b, c \in I$ the form

$$(1 \otimes b - b \otimes 1)(1 \otimes c - c \otimes 1) = 1 \otimes bc - c \otimes b - b \otimes c + bc \otimes$$

is mapped to

$$bc \oplus d(bc) - cb \oplus cdb - bc \oplus dbc + bc \oplus bcd(1) = 0,$$

thus this ψ' is well-defined. Moreover for $B \hookrightarrow B \underset{A}{\otimes} B/I^2$ maps β to $\overline{\beta \otimes 1}$ the ψ' will send it to $\beta \oplus \beta d(1) = \beta \oplus 0$, which is the identity map on B. And its restriction to I/I^2 is

$$\psi : \overline{\beta_1 \otimes \beta_2} \longmapsto \beta_1\beta_2 \oplus \beta_1 d\beta_2 = 0 \oplus \beta_1 d\beta_2,$$

so ψ sends $\overline{1 \otimes \beta - \beta \otimes 1}$ to be $d\beta - \beta d(1) = d\beta$, which is the inverse of φ. □

Definition B.2.9. For any closed point x in a variety X, let m_x be the maximal ideal in the local ring $\mathscr{O}_{X,x}$. We define

$$(\Omega_{X/k})_x := (\Omega_{X/k})_{m_x} \cong \Omega_{\mathscr{O}_{X,x}/k}$$

is an $\mathscr{O}_{X,x}$-module. And

$$\Omega_{X/k}(x) := (\Omega_{X/k})_{m_x} \underset{\mathscr{O}_{X,x}}{\otimes} k(x)$$

which is a $k(x)$-vector space. We often omit the subscript k.

In fact, we also use non-closed points. For example, pick η is a generic point, then

$$\Omega_X(\eta) = \Omega_{k(X)} \underset{k(X)}{\otimes} k(X) = \Omega_{k(X)} = (\Omega_X)_\eta,$$

that is at the generic point the rank gives the dimension of X.

Theorem B.2.10. *For any $x \in X$ closed points, the following are isomorphic as k-vector space.*

(1) Zariski tangent space $\mathrm{Hom}_k(m_x/m_x^2, k(x)) = T_{X,x}$.

(2) The space of point differentials $D_x : \mathscr{O}_{X,x} \to k(x)$, i.e. a k-linear map with

$$D_x(f \cdot g) = f(x) \cdot D_x(g) + g(x) \cdot D_x(f)$$

for $D_x \in \mathrm{Der}_{k(x)}(\mathscr{O}_{X,x}, k(x))$.

(3) $\mathrm{Hom}_{\mathscr{O}_{X,x}}((\Omega_X)_x, k(x)) \cong (\Omega_X)_x^\vee \underset{\mathscr{O}_{X,x}}{\otimes} k(x)$.

(4) $\{f : I \to X \big| f((\varepsilon)) = x \text{ for } I = \mathrm{Spec}\, k[\varepsilon]/(\varepsilon^2)\}$ *as a set.*

We omit the subscript $k(x)$ by as x is closed point so $k(x) \cong k$ with k algebraically closed.

Proof. "(2) $\Leftrightarrow$ (3)": Follows from

$$\begin{aligned}\{\text{point differentials}\} &= \mathrm{Der}_{k(x)}(\mathscr{O}_{X,x}, k(x)) \\ &\cong \mathrm{Hom}_{\mathscr{O}_{X,x}}(\Omega_{\mathscr{O}_{X,x}}, k(x)) \\ &= \mathrm{Hom}_{\mathscr{O}_{X,x}}((\Omega_X)_x, k(x)).\end{aligned}$$

"(1) $\Leftrightarrow$ (2)": For $D_x : \mathscr{O}_{X,x} \to k(x)$, it will sends m_x^2 to zero. The reason is by taking value in residue field, we have

$$D_x(f \cdot g) = f(x) \cdot D_x(g) + g(x) \cdot D_x(f) = 0.$$

So it is induces a k-linear map $\bar{D}_x : m_x/m_x^2 \to k(x)$. This is one direction.

For another, for any k-linear map $\varphi : m_x/m_x^2 \to k(x)$ we can define

$$\begin{aligned}D_{x,\varphi} : \mathscr{O}_{X,x} &\longrightarrow k(x) \\ f &\longmapsto \varphi(\overline{f - f(x)}) \in m_x.\end{aligned}$$

Easy to verify that it is a point differential. That is the 1-1 correspondence.

Before completing this proof, we need more discussion about the *dual number*. □

Corollary B.2.11. *For any $x \in X$ closed points we have the isomorphism between $k(x)$-vector spaces*

$$\begin{aligned}\Omega_X(x) = (\Omega_X)_x \underset{\mathscr{O}_{X,x}}{\otimes} k(x) &\cong m_x/m_x^2 \\ df_x \otimes 1 &\mapsto \overline{f_x - f_x(x)}.\end{aligned}$$

So $\Omega_X(x)$ is a Zariski cotangent space.

Proof. By previous theorem we have

$$\mathrm{Hom}_{k(x)}(m_x/m_x^2, k(x)) \cong \mathrm{Hom}_{\mathscr{O}_{X,x}}((\Omega_X)_x, k(x)) = (\Omega_X)^\vee \underset{\mathscr{O}_{X,x}}{\otimes} k(x).$$

Then we take the dual to get

$$m_x/m_x^2 \cong (\Omega_X)_x \otimes k(x)^\vee \cong (\Omega_X)_x \otimes k(x) = \Omega_X(x).$$

□

Now we explain (4) in (B.2.10). For *dual number* $I = \mathrm{Spec}\, k[\varepsilon]/(\varepsilon^2) = \{(\varepsilon)\}$ it is a one point set but needs counting multiplicity, which we can regard it as a single closed point plus a tangent direction (since it keeps the linear terms!). Then for p denote the unique point, we have $m_p/m_p^2 = (\varepsilon)/(\varepsilon^2) = k\langle\varepsilon\rangle$, is the tangent direction part of the dual number, which we can regard it as the information of $T_{\mathbb{A}^1,0}$. By taking the dual, we have $T_{I,p} = k\langle\mathscr{L}\rangle$ where $\mathscr{L}$ is a linear form dual to ε, sending ε to 1 in k.

Proposition B.2.12. *Let X be a variety, and consider $V \in T_{X,x}$ be any tangent vector for $x \in X$ a closed point. Then there is a unique schematic morphism $f : I \to X$ such that $f((\varepsilon)) = x$ and $df(\mathscr{L}) = V$.*

Proof. The existence is trivial so we only need to verify the uniqueness. For any such morphism $f : I \to X$, it induces a local ring homomorphism

$$\begin{aligned} \mathscr{O}_{X,x} &\longrightarrow \mathscr{O}_{I,(\varepsilon)} = k[\varepsilon]/(\varepsilon^2) \\ m_x &\longmapsto (\varepsilon). \end{aligned}$$

Then it will induce a k-linear map $m_x/m_x^2 \xrightarrow{f_x} (\varepsilon)/(\varepsilon^2) = k\langle\varepsilon\rangle$. Taking the dual, we have

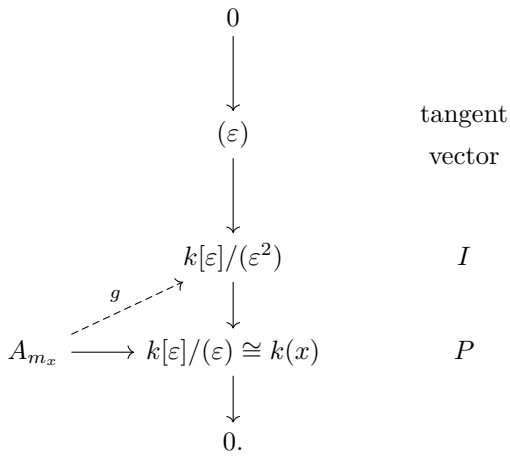
$$\begin{aligned} k\langle\mathscr{L}\rangle = T_{I,p} &\xrightarrow{df} T_{X,x} = (m_x/m_x^2)^\vee \\ \mathscr{L} &\longmapsto (m_x/m_x^2 \xrightarrow{f_x} k\langle\varepsilon\rangle \xrightarrow{\mathscr{L}} k). \end{aligned}$$

As $\mathscr{L}$ is an isomorphism we see that f is 1-1 correspondence with $\mathscr{L}$. □

Now we go back to theorem (B.2.10) for "(4) ⇔ (1)" as sets: since we are dealing with the local property, without loss of generality we can assume that $X = \operatorname{Spec} A$ is affine, and the closed point $x = m_x$ with the exact sequence

$$A \to A_{m_x} \to A_{m_x}/m_x A_{m_x} = k(x) \to 0$$

giving rise to the closed immersion $\operatorname{Spec} k(x) \hookrightarrow \operatorname{Spec} A = X$. For the middle term $A_{m_x} \to k(x)$ we can fill it into the diagram



Notice that such an extension always exists as this short exact sequence splits, that is, $k(x) \to k[\epsilon]/(\epsilon^2)$ has a right inverse, so composing it with the canonical valuation

map can yield our desired extension g, though it might have different coefficients on ϵ.

Proposition B.2.13. *The set of such lifts g is in bijective correspondence with $\Omega_A(x)^\vee = T_{X,x}$ as a set, where x is a closed point in the affine variety $X = \operatorname{Spec} A$.*

Proof. We prove this in the following several steps. Remember that we always have the isomorphisms

$$k(x) \cong k \cong k\langle\varepsilon\rangle \cong k[\varepsilon]/(\varepsilon).$$

Step 1: If g_1 and g_2 are two different lifts, then $g_1 - g_2 =: \theta \in \operatorname{Der}_{k(x)}(A_{m_x}, k(x))$. Since (ε) is a $k[\varepsilon]/(\varepsilon^2)$-module as it is an ideal, but the ε-part acting on it will kill the generator, so we see it is also a $k[\varepsilon]/(\varepsilon)$-module isomorphic to $k\langle\varepsilon\rangle$. Via the ring homomorphism $f : A_{m_x} \to k(x) \cong k[\varepsilon]/(\varepsilon)$ we see it can be viewed as an A_{m_x}-module, i.e. $a \cdot \varepsilon$ will be mapped to $a(x) \cdot \varepsilon$ where $a \in A_{m_x}$ takes its value at closed point x in the residue field. Now for any $a \in A_{m_x}$, we have

$$\begin{aligned} g_i : A_{m_x} &\longrightarrow k[\varepsilon]/(\varepsilon^2) \cong k \oplus k\langle\varepsilon\rangle \\ a &\longmapsto a(x) + h_i(x) \cdot \varepsilon \end{aligned}$$

then

$$\begin{aligned} (g_1 - g_2)(a) &= g_1(a) - g_2(a) \\ &= a(x) + h_1(x) \cdot \varepsilon - a(x) - h_2(x) \cdot \varepsilon \\ &= (h_1(x) - h_2(x)) \cdot \varepsilon \in (\varepsilon). \end{aligned}$$

Therefore we see that $\theta : A_{m_x} \to (\varepsilon)$ is a module homomorphism. Now we verify that θ is a *derivation*. Since for each lift g we have that

$$\begin{aligned} g(ab) = g(a)g(b) &= (a(x) + h(x)\varepsilon)(b(x) + h(x)\varepsilon) \\ &= a(x)b(x) + (a(x) \cdot h(x) + b(x) \cdot h(x)) \cdot \varepsilon, \end{aligned}$$

then directly

$$\begin{aligned} \theta(ab) &= (g_1 - g_2)(ab) \\ &= g_1(ab) - g_2(ab) \\ &= a(x) \cdot (g_1 - g_2)(b) + b(x) \cdot (g_1 - g_2)(a), \end{aligned}$$

that means $\theta \in \operatorname{Der}_{\mathscr{O}_{X,x}}(\mathscr{O}_{X,x}, k(x))$ is a point differential.

Step 2: Given any $\theta \in \mathrm{Der}_{k(x)}(\mathscr{O}_{X,x}, k(x))$, the

$$\begin{aligned} f + \theta : A_{m_x} &\longrightarrow k[\varepsilon]/(\varepsilon^2) \\ a &\longmapsto f(a) + \theta(a) \cdot \varepsilon \end{aligned}$$

is a lift of $f : A_{m_x} \to k$. This is obvious by straightforward verification.

Step 3: Now we have that $\mathrm{Der}_{k(x)}(\mathscr{O}_{X,x}, k(x))$ parametrizes all such lifts, and it is isomorphic to $\mathrm{Hom}_{\mathscr{O}_{X,x}}((\Omega_X)_x, k(x)) \cong (\Omega_X^\vee)_x \otimes k(x) = \Omega_X(x)^\vee$, is our desired. □

Now we give some computations on Ω_X.

Proposition B.2.14. *Let $Y \hookrightarrow X$ be closed affine subvarieties, and $x \in Y$ a closed point. Then for short exact sequence*

$$0 \to \mathscr{I}_X \to \mathscr{O}_{X,x} \to \mathscr{O}_{Y,x} \to 0$$

we have the short exact sequence

$$0 \to \sum_{f \in \mathscr{I}_x} k(x) \cdot df \to \Omega_X(x) \to \Omega_Y(x) \to 0.$$

Before giving the proof, let us make some analysis. The statement is local so we can regard everything as happening over an affine neighborhood, while as closed immersion $Y \hookrightarrow X$ corresponding to surjective morphism on the structure sheaf, so we have $\Omega_{Y/X} = 0$. Hence the right exactness part in the proposition is given by

$$(\Omega_X)_x \to (\Omega_Y)_x \to (\Omega_{Y/X})_x = 0 \to 0$$

and then twisted by the flat module $k(x)$. After taking the dual, we have

$$0 \to T_{Y,x} \to T_{X,x} \to \left(\sum_{f \in \mathscr{I}_x} k(x) \cdot df \right)^\vee \to 0,$$

which geometrically speaking is that $T_{Y,x}$ is the *subspace* of $T_{X,x}$ killed by these linear forms $\sum\limits_{f \in \mathscr{I}_x} k(x) \cdot df$.

Proof. The canonical inclusion $T_{Y,x} \hookrightarrow T_{X,x}$ is given as above, by taking the dual of $\Omega_X(x) \to \Omega_Y(x) \to 0$.

Give a set-wise description via the dual number, here we abuse of notations

$$\begin{aligned} T_{X,x} &= \{\tilde{f} : I \to X \mid I = \operatorname{Spec} k[\varepsilon]/(\varepsilon^2) \text{ such that } \tilde{f}((\varepsilon)) = x\} \\ &= \{\tilde{f} : \mathscr{O}_{X,x} \to k[\varepsilon]/(\varepsilon^2) \mid \tilde{f}((\varepsilon)) = m_x\} \\ T_{Y.x} &= \{f : \mathscr{O}_{Y,x} \to k[\varepsilon]/(\varepsilon^2) \mid f((\varepsilon)) = m_x\} \end{aligned}$$

The canonical inclusion $T_{Y,x} \hookrightarrow T_{X,x}$ is given by the extension:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{I}_x & \longrightarrow & \mathscr{O}_{X,x} & \longrightarrow & \mathscr{O}_{Y,x} & \longrightarrow & 0 \\ & & & & & \searrow^{\tilde{f}} & \downarrow{\scriptstyle f} & & \\ & & & & \text{a canonical} & & k[\varepsilon]/(\varepsilon^2). & & \\ & & & & \text{lift} & & & & \end{array}$$

So we need to determine which $\tilde{f} \in T_{X,x}$ can be factored through in this diagram, but this is equivalent to $\mathcal{I}_x \subseteq \ker \tilde{f}$. Now for

$$\begin{aligned} \tilde{f} : \mathscr{O}_{X,x} &\longrightarrow k[\varepsilon]/(\varepsilon^2) \cong k \oplus k\langle\varepsilon\rangle \\ \varphi &\longmapsto \varphi(x) + \langle \bar{f}, d\varphi \rangle \cdot \varepsilon \end{aligned}$$

where the second term comes from evaluating the tangent vector $\bar{f}$ on the canonical differential via the pairing $\langle -, - \rangle$. Explicitly, the homomorphism $\tilde{f}$ induces a k-linear map $\bar{f} : m_x/m_x^2 \to k\langle\varepsilon\rangle \cong k(x)$, and via the canonical isomorphism

$$\begin{aligned} m_x/m_x^2 &\cong \Omega_{\mathscr{O}_{X,x}} \otimes k(x) \to k\langle\varepsilon\rangle \\ \overline{\varphi - \varphi(x)} &\longmapsto d\varphi \otimes 1 \longmapsto \langle \bar{f}, d\varphi \otimes 1 \rangle \cdot \varepsilon \end{aligned}$$

while notice here we omit the $\otimes 1$ since by the natural isomorphism

$$\operatorname{Hom}_{\mathscr{O}_{X,x}}(\Omega_{\mathscr{O}_{X,x}}, k(x)) \cong \operatorname{Hom}_{k(x)}(\Omega_{\mathscr{O}_{X,x}}, k(x)) \ni \bar{f}.$$

So $\mathcal{I}_x \subseteq \ker \tilde{f}$ if and only if for all $\varphi \in \mathcal{I}_x$ we have $\varphi(x) = 0$ and $\langle \bar{f}, d\varphi \rangle = 0$. But the condition $\varphi(x) = 0$ requires that $\mathcal{I}_x \subseteq m_x$ in $\mathscr{O}_{X,x}$, which holds automatically. And the condition $\langle \bar{f}, d\varphi \rangle = 0$ with fixed $\bar{f}$ requires that $d\varphi = 0$ for all $\varphi \in \mathcal{I}_x$, which is nothing but saying that $T_{Y.x} \hookrightarrow T_{X,x}$ is killed by $\sum_{\varphi \in \mathcal{I}_x} k(x) \cdot d\varphi$. □

Corollary B.2.15. *Let $X \subseteq \mathbb{A}_k^n$ be an affine variety. $I = (g_1, g_2, \ldots, g_m)$ is the ideal of X in $k[x_1, \ldots, x_n]$. We take $x = 0$ to be the origin.*

Then $T_{X,x}$ is the kernel of the differential map $(\partial g_i/\partial x_j)(0)$, just like in differential geometry.

Proof. By previous theorem we have the short exact sequence

$$0 \longrightarrow T_{X,0} \hookrightarrow T_{\mathbb{A}^n,0} \xrightarrow{\varphi} \left(\sum_{i=1}^{m} k(0) \cdot dg_i \right)^{\vee} \longrightarrow 0$$

thus by exactness $T_{X,0} \cong \ker \varphi$. Furthermore since we have

$$\begin{aligned}(\Omega_{\mathbb{A}^n})(0) &= (\Omega_{\mathbb{A}^n})_0 \underset{\mathscr{O}_{\mathbb{A}^n,0}}{\otimes} k(0) \\ &\cong \left(\Omega_{\mathbb{A}^n} \underset{\mathscr{O}_{\mathbb{A}^n}}{\otimes} \mathscr{O}_{\mathbb{A}^n,0}\right) \underset{\mathscr{O}_{\mathbb{A}^n,0}}{\otimes} k(0) \\ &\cong \Omega_{\mathbb{A}^n} \otimes k(0) \\ &= \mathsf{Span}_{k(0)}\{dx_1 \otimes 1, \ldots, dx_n \otimes 1\} \text{ for } 1 \in k(0) = k \\ &=: \mathsf{Span}_k\{dx_1|_0, \ldots, dx_n|_0\}.\end{aligned}$$

Therefore

$$T_{\mathbb{A}^n,0} \cong \Omega_{\mathbb{A}^n}(0)^\vee = \mathsf{Span}_k \left\{ \left.\frac{\partial}{\partial x_1}\right|_0, \ldots, \left.\frac{\partial}{\partial x_n}\right|_0 \right\}$$

where $\frac{\partial}{\partial x_j}|_0$ is dual to $dx_j|_0$. We pick arbitrary $v \in \ker \varphi$, write it as

$$v = \sum_{j=1}^n \left.\frac{\partial}{\partial x_j}\right|_0 \cdot v^j,$$

then letting it act on the generators, we get

$$\begin{aligned}v(dg_i|_0) &= v\left(\sum_{j=1}^n \frac{\partial g_i}{\partial x_j}(0) \cdot dx_j\right) \\ &= \sum_{j=1}^n \frac{\partial g_i}{\partial x_j}(0) \cdot v(dx_j) \\ &= \sum_{j=1}^n \frac{\partial g_i}{\partial x_j}(0) \cdot v^j = 0.\end{aligned}$$

That is nothing but

$$\begin{pmatrix} \frac{\partial g_1}{\partial x_1}(0) & \ldots & \frac{\partial g_1}{\partial x_n}(0) \\ \vdots & \ddots & \vdots \\ \frac{\partial g_m}{\partial x_1}(0) & \ldots & \frac{\partial g_m}{\partial x_n}(0) \end{pmatrix} \begin{pmatrix} v^1 \\ \vdots \\ v^n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

So $T_{X,0} \cong \ker(\partial g_i/\partial x_j)(0)$, whose values are taken in the residue field $k(0)$. (But since we are over an algebraically closed field here, we actually never need to emphasize this thing.) □

Proposition B.2.16. *For $B = A[x_1, \ldots, x_n]/I = R/I$ is a finitely generated A-algebra with $I = (f_1, \ldots, f_m)$ be generators. Then $\Omega_{B/A}$ is a B-module generated by*

$\langle dx_1, \ldots, dx_n \rangle$ *modulo out by relations*

$$\begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{pmatrix} \begin{pmatrix} dx_1 \\ \vdots \\ dx_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

where the left hand side is the Jacobian matrix with elements in B*. Equivalently speaking, it is the cokernel of*

$$\Omega_{A/k} \underset{A}{\otimes} B \xrightarrow{\left(\frac{\partial f_i}{\partial x_j}\right)} \Omega_{B/k} \longrightarrow \Omega_{B/A} \to 0.$$

Proof. By the second exact sequence of [Eis95, 16.12], we have that

$$\begin{aligned} I/I^2 &\xrightarrow{\delta} \Omega_{R/A} \underset{R}{\otimes} B \longrightarrow \Omega_{B/A} \longrightarrow 0 \\ \bar{b} &\longmapsto db \otimes 1 \end{aligned}$$

where $I/I^2 = B\langle \bar{f}_1, \ldots, \bar{f}_m \rangle$ and $\Omega_{R/A} \underset{R}{\otimes} B = \mathsf{Span}\{dx_1 \otimes 1, \ldots, dx_n \otimes 1\}$. Then on each generators

$$\delta(\bar{f}_i) = df_i \otimes 1 = \left(\sum_{j=1}^{n} \frac{\partial f_i}{\partial x_j} dx_j \right) \otimes 1,$$

so we see

$$\Omega_{B/A} \cong \frac{\bigoplus_{j=1}^{n} B dx_j}{\left\langle \sum_{j=1}^{n} \frac{\partial f_i}{\partial x_j} dx_j \right\rangle_{i=1}^{m}}.$$

□

B.3 Implicit Function Theorem in Algebraic Geometry

Recall that in analytic geometry, for $X \subseteq \mathbb{C}^n$ a sub-manifold with a smooth of dimension $n-k$. Then there is a analytic coordinate change such that $X = V(z_1, \ldots, z_k)$. This is the Implicit Function Theorem in [GH78, P.20].

In algebraic geometry, we have

Theorem B.3.1. *X is a non-singular variety with $Y \overset{i}{\hookrightarrow} X$ closed subvariety. Then for $x \in Y$ any closed point, we have Y is non-singular at x if and only if i is a regular immersion at x and $\Omega_X|_Y$ contains a free subsheaf of rank $r = \operatorname{codim}_X Y$.*

Explicitly, there is an open affine neighborhood U of x with $f_1, \ldots, f_r \in \mathscr{O}_X(U)$ such that

(1) $Y \cap U = \operatorname{Spec} \mathscr{O}_X(U)/(f_1, \ldots, f_r)$;

(2) $\{df_1, \ldots, df_r\}$ are linearly independent, i.e. $\{f_i\}_i$ is a regular sequence in $\mathscr{O}_{X,x}$.

Corollary B.3.2. *In this situation for Y non-singular, we have the short exact sequence of locally free sheaves on Y*

$$0 \to \mathcal{I}/\mathcal{I}^2 \to \Omega_X|_Y \to \Omega_Y \to 0.$$

After taking dual we have

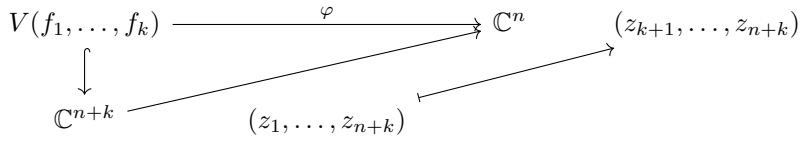
$$0 \to T_Y \to T_X|_Y \to N_X Y \to 0$$

where the right term is called the normal bundle of Y in X.

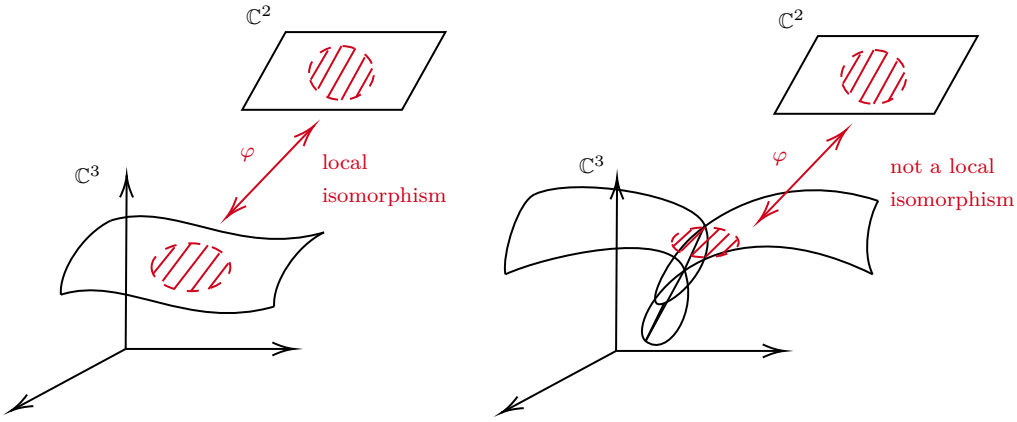
Now we go back to the Implicit Function Theorem in analytic geometry, let $f_1, \ldots, f_k$ be analytic functions on x in $\mathbb{C}^{n+k} \to \mathbb{C}$ such that

$$\det \left| \frac{\partial f_i}{\partial x_j}(x) \right|_{1 \le i,j \le k} \neq 0.$$

Then



such that φ is local isomorphism.



But this will "fail" in algebraic geometry, since a Zariski open subset is too big and we need some modifications:

(1) Add the right condition on the maps about "local isomorphism";

(2) "Take smaller neighborhood".

Thus we turn to consider *Étale morphisms*. Let R be a finite type k-algebra.

Definition B.3.3. (Étale morphism) Let $Y = \operatorname{Spec} R$ be affine scheme, and we let $X = \operatorname{Spec} R[x_1, \ldots, x_n]/(f_1, \ldots, f_n) \hookrightarrow Y \times \mathbb{A}^n_k$ such that $f : X \to Y$ is given by the

$$R \hookrightarrow R[x_1, \ldots, x_n]/(f_1, \ldots, f_n).$$

We say f is *étale* at point $x \in X$, if

$$\det\left(\frac{\partial f_i}{\partial x_j}(x)\right) \neq 0,$$

where the determinant takes values in the residue field $k(x)$ with respect to localization with $R[x_1, \ldots, x_n]/(f_1, \ldots, f_n)$. By [SP25, Tag 02GS] we see f is flat at x.

In general, for $f : X \to Y$ is a morphism of algebraic schemes, we say f is *étale* at a point $x \in X$ if there is an open affine neighborhood $U \subseteq X$ of x and $V \subseteq Y$ of $y = f(x)$ such that for $f|_U : U \to V$ we have

$$\begin{array}{ccc} U & \hookrightarrow & \operatorname{Spec}(R[x_1, \ldots, x_n]/(f_1, \ldots, f_n) \\ {\scriptstyle f|_U}\downarrow & \nearrow_{\text{open immersions}} & \downarrow \\ V & \hookrightarrow & \operatorname{Spec} R \end{array} \qquad \text{and } \det\left(\tfrac{\partial f_i}{\partial x_j}(x)\right) \neq 0 \text{ in } k(x).$$

In short words, étale morphism is "standard étale" with respect to an affine cover.

Remark B.3.4. We denote $S = R[x_1, \dots, x_n]/(f_1, \dots, f_n)$ which is finite type over k. Then the ring homomorphism $R \to S$ is locally characterizes the morphism $X \to Y$, we have the short exact sequence

$$\Omega_R \to \Omega_S \to \Omega_{S/R} \to 0,$$

where $\Omega_{S/R}$ is S-module spanned by $\{dx_1, \dots, dx_n\}$ modulo out the relations

$$\begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n} \end{pmatrix} \begin{pmatrix} dx_1 \\ \vdots \\ dx_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

But we have the determinant on the left is non-zero at point $x \in X$. So in this setting,

$$\begin{aligned} \text{étale at } x \in X &\iff \Omega_{X/Y}(x) = \Omega_{S/R}(x) = 0 \\ &\iff \det\left(\frac{\partial f_i}{\partial x_j}(x)\right) \neq 0. \end{aligned}$$

Moreover as $\Omega_{X/Y}$ is coherent, by Nakayama's lemma $\Omega_{X/Y}(x) \neq 0$ implies that $\Omega_{X/Y} \neq 0$ in a neighborhood of x.

Furthermore in this setting, we say f is *étale* on X if and only if f is étale at any point $x \in X$, if and only if $\Omega_{X/Y} = 0$.

Now we will not restrict on algebraically closed fields.

Proposition B.3.5. *Let X be an algebraic scheme. Then $\pi : X \to \operatorname{Spec} k$ is étale on X if and only if X is the union of finite set of points, and each point is of the form $\operatorname{Spec} k_i$ with $k \hookrightarrow k_i$ is a finite separable extension.*

In particular if $\bar{k} = k$, then $X \to \operatorname{Spec} k$ is étale if and only if X is union of finite set of $\operatorname{Spec} k$.

Proof. "⇒": Take $U = \operatorname{Spec} R$ be an affine piece of X. By above remark we see $(\Omega_{R/k})_x = 0$ for any point $x \in U$ and $R = k[x_1, \dots, k_n]/(f_1, \dots, f_n)$. But by [Eis95, 16.16] we see $R = \bigoplus_{i=k}^{k} k_i$ where each k_i is a finite separable extension of k.

"⇐": Let $X = \coprod_{\text{finite}} \operatorname{Spec} k_i$. Then $X \to \operatorname{Spec} k$ is étale if and only if for each i the $\operatorname{Spec} k_i \to \operatorname{Spec} k$ is étale. But $\Omega_{k_i/k} = 0$ by [Har77, II, 8.6A] for $k \hookrightarrow k_i$ is finite separable extension.

Moreover, at this time X is reduced. □

Proposition B.3.6. *Let $f : X \to Y$ be étale defined as above.*

(1) For any point $y \in Y$, the fibre $f^{-1}(y)$ as a scheme over $k(y)$ is the finite union of $\operatorname{Spec} k_i$ with $k \hookrightarrow k_i$ is a finite separable extension.

(2) For any point $x \in X$, take $y = f(x)$ which induces a ring homomorphism $\mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$. Then $m_y \cdot \mathscr{O}_{X,x} = m_x$ where the left is extension of m_y in $\mathscr{O}_{X,x}$, and $k(y) \hookrightarrow k(x)$ is finite separable.

Proof. (1) Easily to see that an étale morphism is stable under base change, so for the fibre product diagram

$$\begin{array}{ccc} f^{-1}(y) & \hookrightarrow & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ \{y\} & \hookrightarrow & Y \end{array}$$

we see $f^{-1}(y) \xrightarrow{f'} \{y\}$ is étale. Then we apply the previous proposition.

(2) Take a local picture, with f is étale for $R \hookrightarrow S = R[x_1, \ldots, x_n]/(f_1, \ldots, f_n)$. And we denote $y = p_y \subseteq R$, $x = p_x \subseteq S$ such that the contraction $p_x^c = p_x \cap R = p_y$. Then by straightforward computation we have

$$\begin{array}{ccccccc} & & S' & \longleftarrow & S_{p_x} & & \\ & \swarrow & \uparrow & & \uparrow & & \\ k(x) & \longleftarrow & S_{p_y{}^e}/p_y^e S_{p_y{}^e} & \longleftarrow & S_{p_y{}^e} & \longleftarrow & S \\ \uparrow & & \uparrow & & \uparrow & & \uparrow \\ k(y) & = & R_{p_y}/p_y R_{p_y} & \longleftarrow & R_{p_y} & \longleftarrow & R \end{array}$$

such that

$$\mathscr{O}_{f^{-1}(y),x} = (S_{p_y{}^e}/p_y^e S_{p_y{}^e})_{\overline{p_x}} \cong S_{p_x}/p_y{}^e S_{p_x} = \mathscr{O}_{X,x}/(m_y \cdot \mathscr{O}_{X,x}).$$

So by (1) telling us that $f^{-1}(y) \cong \coprod_{\text{finite}} \operatorname{Spec} k_i$, we directly have

$$\mathscr{O}_{f^{-1}(y),x} \cong \text{ one of } k_i = k(x)$$

implies that $m_y \cdot \mathscr{O}_{X,x} = m_x$.

In this case we say f is *unramified* at point $x \in X$. The finite separable part is also by (1).

□

We go back to an algebraically closed field k.

Proposition B.3.7. *Let $f : X \to Y$ be morphism of varieties over k with f is étale. Then*

(1) f is dominant;

(2) $k(Y) \hookrightarrow k(X)$ is finite extension;

(3) for any $y \in Y$ closed point, $f^{-1}(y)$ is a finite set of reduced points $\operatorname{Spec} k$.

Proof. (3) is trivial, since there is no finite separable extension of $l = k(y)$.

(1) by definition, locally see we have

$$\begin{array}{ccc} X & \hookrightarrow & Y \times \mathbb{A}_n^r \\ {\scriptstyle f}\downarrow & \swarrow_{\pi} & \\ Y & & \end{array}$$

where X is cut out by $(f_1, \ldots, f_n)$ in $Y \times \mathbb{A}^n$, so we have $\dim X \geq \dim Y$. Assume $\overline{f(X)} \subseteq Y$ is a proper subvariety [SP25, Tag 03WT], i.e. $\dim \overline{f(X)} \leq \dim Y - 1$. Then we take $\eta \in \overline{f(X)}$ a generic point, so $f^{-1}(\eta)$ has dimension at least 1 by [Har77, II, ex3.22(b)]. But this contradicts that $f^{-1}(\eta)$ is a finite set, so we see $\dim \overline{f(X)} = \dim Y$, and hence $\overline{f(X)} = Y$ as Y is a variety. Notice here we use the result that an étale morphism is an open morphism ensure the irreducibility of the image.

(2) we take $y = \eta_Y$ and $x = \eta_X$, then $k(\eta_Y) = k(Y) \hookrightarrow k(\eta_X) = k(X)$ is a finite (separable) extension. □

Example B.3.8. Open immersion is always étale, since locally we have

$$\begin{array}{ccc} U & \xhookrightarrow{\cong} & \operatorname{Spec} R[x]/(x) \\ {\scriptstyle i}\downarrow & & \downarrow \\ i(U) & = & \operatorname{Spec} R. \end{array}$$

Moreover this means étale morphism may not be surjective, and étale morphism may not be finite as open immersion is not affine.

Remark B.3.9. Over algebraic scheme for finite type morphism f, we have

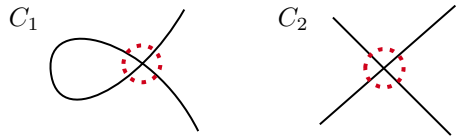
$$\begin{aligned} f \text{ is étale} &\iff f \text{ is flat and } \Omega_{X/Y} = 0 \\ &\iff f \text{ is flat and unramified} \end{aligned}$$

Definition B.3.10. (Étale neighborhood) Let $x \in X$ be a point of an algebraic scheme X, by an *étale neighborhood* we mean a morphism

$$\varphi : (U', x') \longrightarrow (U, x)$$

such that $U \subseteq X$ is Zariski open neighborhood and φ is étale.

To find a "smaller neighborhood", we consider the *formal neighborhood* of an algebraic scheme.



nodal curve cross line

$y^2 - x^2(x+1) = 0$ $x^2 - y^2 = 0$

These two cases of curves are locally isomorphic near the origin x in analytic geometry, but not in algebraic geometry, since the concept of "near" is not sufficiently near as a Zariski open neighborhood is too big. For example, $\mathscr{O}_{C_1,x} \cong \mathbb{C}[x,y]_{(x,y)}/(y^2 - x^2(x+1))$ is an integral domain, but $\mathscr{O}_{C_2,x} \cong \mathbb{C}[x,y]_{(x,y)}/(xy)$ is not an integral domain, thus they are not isomorphic near the origin. This property is still "Zariski local".

But what if we consider that there is an element u which is a unit and $u^2 = x+1$, that is, we can get a well-defined map

$$\frac{?}{(y+ux)(y-ux)} \xrightarrow{x\mapsto 1/u\cdot x} \frac{?}{(y+x)(y-x)}$$

to make everything work? In analytic geometry, for $\mathscr{O}^{\mathrm{an}}_{X,x}$ the analytic germ this thing can happening, as the analytic branch of $\sqrt{x+1}$ exists. So in algebraic geometry, we need to consider about formal completion.

To give a local picture, let A be a finite type k-algebra with maximal ideal m, we define $A_n := A/m^n$ for each $n \in \mathbb{Z}_{>0}$. Then for arbitrary $n_1 < n_2 < n_3$ we have a commutative diagram

$$\begin{array}{ccc} A_{n_1} & \longleftarrow & A_{n_2} \\ \uparrow & \nearrow & \\ A_{n_3} & & \end{array}$$

which gives an inverse system which is surjective and satisfies the *Mittag-Leffler* condition. So by taking the limit, we get the *m-adic completion* of A which is $\widehat{A} := \varprojlim A_n$. Informally speaking, $\widehat{A}$ is the ring of *convergent* power series, in the sense of m-adic topology.

Example B.3.11. $A = k[x,y]$ with $m = (x,y)$. Then $\widehat{A} = k[[x,y]]$ is the formal power series ring.

Proposition B.3.12. *Let m be its maximal ideal and let $(A_m, \tilde{m})$ be the local ring, we have*

$$\begin{array}{ccc} A & \longrightarrow & A_m \\ \downarrow & & \downarrow \\ \widehat{A} & \xrightarrow[\cong]{} & \widehat{A}_m \end{array}$$

where the right side arrow is the completion of A_m along $\tilde{m}$.

Informally speaking, "formal completion" is very "local", only depends on local property.

Example B.3.13. For $C_1 = V(y^2 - x^2(x+1))$ and $C_2 = V(x^2 - y^2)$. We have

$$\begin{array}{ccc} \mathscr{O}_{C_1,x} & \hookrightarrow & \widehat{\mathscr{O}}_{C_1,x} \\ \wr\downarrow & & \| \wr \\ \mathscr{O}_{C_2,x} & \hookrightarrow & \widehat{\mathscr{O}}_{C_2,x} \end{array}$$

as $\sqrt{x+1} \in \widehat{\mathscr{O}}_{C_1,x}$ is a unit.

Remark B.3.14. $\widehat{\mathscr{O}}_{X,x}$ is the "limit" that preserves the property, which is much smaller than on Zariski open neighborhood.

$$\mathscr{O}_{X,x} \hookrightarrow \mathscr{O}_{X,x}^{\mathrm{an}} \hookrightarrow \widehat{\mathscr{O}}_{X,x} = \widehat{\mathscr{O}}_{X,x}^{\mathrm{an}}$$

Definition B.3.15. (Formal neighborhood) Let X be a scheme with a point $x \in X$. A *formal neighborhood* of X at x is the following data

$$\begin{cases} (1) \text{ a single point,} \\ (2) \text{ a ring } \widehat{\mathscr{O}}_{X,x} = \varprojlim \mathscr{O}_{X,x}/m_x^n \text{ as the structure sheaf.} \end{cases}$$

Which we denote by $\mathrm{Spf}(\widehat{\mathscr{O}}_{X,x})$.

Recall that for the natural map $A \to \widehat{A}$, we have that $\widehat{A}$ is a *faithfully flat* A-algebra.

Proposition B.3.16. *Let $\mathscr{F} \subseteq \mathscr{G}$ be coherent sheaves on X, and let $i : \widehat{X} \to X$ be the formal neighborhood of X at x. If $i^*\mathscr{F} = i^*\mathscr{G}$ on $\widehat{X}$, then $\mathscr{F} \cong \mathscr{G}$ on a Zariski open neighborhood of X around x.*

Proof. For $x \in X$ we take an affine open $U \subseteq X$ such that $x = p \in U = \operatorname{Spec} A$. Then

the diagram

$$\begin{array}{ccccc} \widehat{A} \cong \widehat{A}_p & \longleftarrow & A_p & \longleftarrow & A \\ \downarrow & & \downarrow & & \\ \widehat{A}_p/\widehat{p} & = & A_p/pA_p = k(x) & & \\ \| & & & & \\ k(x) & & & & \end{array}$$

gives the formal neighborhood

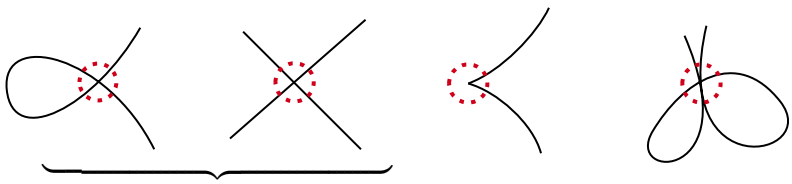
$$\begin{array}{ccc} \widehat{X} & \xrightarrow{i} & X \\ & \nwarrow & \uparrow \\ & & \{x\}. \end{array}$$

We can think U is small enough (under the Zariski meaning) to make $\mathscr{F} \subseteq \mathscr{G}$ can be written as $\widetilde{M} \subseteq \widetilde{N}$, and then at the stalk $M_p \hookrightarrow N_p$ as A_p-modules. Then

$$\begin{aligned} i^*\mathscr{F} \cong i^*\mathscr{G} &\iff M_p \underset{A_p}{\otimes} \widehat{A}_p \cong N_p \underset{A_p}{\otimes} \widehat{A}_p \\ &\iff M_p \cong N_p \text{ as } A \to \widehat{A} = \widehat{A}_p \text{ is faithfully flat} \end{aligned}$$

i.e. $\mathscr{F}_x \cong \mathscr{G}_x$. So by Nakayama's lemma, we get that $\mathscr{F} = \mathscr{G}$ on a Zariski open neighborhood of x, since its cokernel is 0 over this neighborhood. □

We can classify the singularity type of curves by the local rings $\widehat{\mathscr{O}}_{X,x}$.



"same" singularity

Example B.3.17. Over $\mathbb{C}$ up to an isomorphism given by multiplying by i, we say that a curve C has A_n-*singularity* at x if $\widehat{\mathscr{O}}_{C,x} \cong \mathbb{C}[[x,y]]/(y^2 - x^{n+1})$.

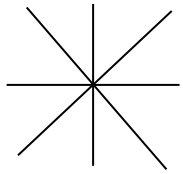
In above picture, the two left side curves have A_1-singularity at the origin, since

$$\widehat{\mathscr{O}}_{C,x} \cong \mathbb{C}[[x,y]]/(y^2 - x^2) \cong \mathbb{C}[[x,y]]/(xy)$$

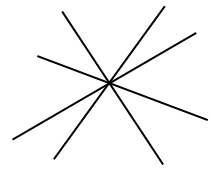
by change of coordinates $x \mapsto x + y, y \mapsto x - y$.

But the completion has very rigid structure, for example even if two curves' structures have the same local topology, their formal neighborhoods may not be the same.

Example B.3.18. Let X and Y be the union of four distinct lines in $\mathbb{C}^2$, but with different coefficients.



$X = V((x + a_1y) \cdots (x + x_4y))$



$Y = V((x + b_1y) \cdots (x + b_4y))$

We have $\widehat{\mathscr{O}}_{X,0} \cong \mathbb{C}[[x, y]]/(\prod_{i=1}^{4}(x + a_iy))$ and $\widehat{\mathscr{O}}_{Y,0} \cong \mathbb{C}[[x, y]]/(\prod_{i=1}^{4}(x + b_iy))$ respectively. Topologically speaking, X and Y should have the "same" singularity type, but in fact they are not isomorphic.

Suppose they are isomorphic. Expanding them into

$$\begin{aligned}\mathrm{gr}_m \widehat{\mathscr{O}}_{X,0} &= \mathbb{C} \oplus m/m^2 \oplus m^2/m^3 \oplus m^3/m^4 \oplus m^4/m^5 \oplus \cdots \\ \mathrm{gr}_{\bar{m}} \widehat{\mathscr{O}}_{Y,0} &= \mathbb{C} \oplus \bar{m}/\bar{m}^2 \oplus \bar{m}^2/\bar{m}^3 \oplus \bar{m}^3/\bar{m}^4 \oplus \bar{m}^4/\bar{m}^5 \oplus \cdots\end{aligned}$$

and suppose they're isomorphic via φ, so this must induce an isomorphism on each degree of the associated graded ring expansion. Therefore at degree 1 we have an isomorphism φ_1 between the 2-dimensional $\mathbb{C}$-vector spaces

$$\varphi_1 : m/m^2 \cong \bar{m}/\bar{m}^2.$$

Geometrically, φ_1 acts as the linear change of coordinates that must map the tangent cone of X to the tangent cone of Y, corresponding to the Möbius transformation over $\mathbb{P}^1_{\mathbb{C}}$ sending a_i to b_i. But this can only happen when their **cross ratios** are the same.

Before give our main theorem, we need two lemmas of commutative algebra:

Lemma B.3.19. *Let (A, m_A) and (B, m_B) be noetherian local rings with same residue field, let $\mathrm{gr}_{m_A} A$ and $\mathrm{gr}_{m_B} B$ be their associated graded algebra. Then $\mathrm{gr}_{m_A} A \longrightarrow \mathrm{gr}_{m_B} B$ is an isomorphism induced by the local homomorphism $f : A \longrightarrow B$ if and only if $\hat{f} : \widehat{A} \longrightarrow \widehat{B}$ is an isomorphism.*

Proof. "⇒": From short exact sequence

$$0 \longrightarrow m^k/m^{k+1} \longrightarrow A/m^{k+1} \longrightarrow A/m^k \longrightarrow 0,$$

can use induction and short five lemma to get the isomorphism $A/m^{k+1} \cong B/m^{k+1}$. So is after take the inverse limit.

"$\Leftarrow$": Since we have $\mathrm{gr}_{m_A} A \cong \mathrm{gr}_{m_{\widehat{A}}} \widehat{A}$, which follows from [Eis95, 7.13]. □

Lemma B.3.20. *Let (A, m_A) and (B, m_B) be noetherian local rings. Given local homomorphisms $f : A \longrightarrow B$ and $g : B \longrightarrow \widehat{A}$, fill up the commutative diagram*

$$\begin{array}{ccc} A & \xrightarrow{\quad f \quad} & B \\ & {\scriptstyle\varphi}\searrow \quad \swarrow{\scriptstyle g} & \\ & \widehat{A} & \end{array}$$

where $\varphi : A \longrightarrow \widehat{A}$ is the natural inclusion. If m_B is the extension of m_A. then we have an isomorphism $\widehat{B} \cong \widehat{A}$.

Proof. We apply the completion on f and g, to get local homomorphisms

$$\hat{f} : \widehat{A} \longrightarrow \widehat{B} \quad \text{and} \quad \hat{g} : \widehat{B} \longrightarrow \widehat{\widehat{A}} = \widehat{A}.$$

And $\hat{\varphi} : \widehat{A} \longrightarrow \widehat{\widehat{A}} = \widehat{A}$ is the identity map. So we have $\hat{g} \circ \hat{f} = id_{\widehat{A}}$, that is $\hat{f}$ is injective and $\hat{g}$ is surjective,

$$\begin{array}{ccc} \widehat{A} & \xrightarrow{\quad \hat{f} \quad} & \widehat{B} \\ & {\scriptstyle id_{\widehat{A}}}\searrow \quad \swarrow{\scriptstyle \hat{g}} & \\ & \widehat{A}. & \end{array}$$

Therefore, we get a short exact sequence

$$\begin{aligned} 0 \longrightarrow \widehat{A} \xrightarrow{\hat{f}} \widehat{B} &\longrightarrow \ker \hat{g} =: I \longrightarrow 0 \\ a &\longmapsto \hat{f}(a) \\ b &\longmapsto b - \hat{f}(\hat{g}(b)) \end{aligned}$$

which is split, so $\widehat{B} \cong \widehat{A} \oplus I$.

The assumption $m_A B = m_B$ will gives that $m_{\widehat{A}} \widehat{B} = m_{\widehat{B}}$, since via the equations

$$m_{\widehat{A}} \widehat{B} = (m_A \widehat{A}) \cdot \widehat{B} = m_A (\widehat{A} \cdot \widehat{B}) = m_A \widehat{B} = (m_A B) \cdot \widehat{B} = m_B \widehat{B} = m_{\widehat{B}},$$

where B (resp. $\widehat{A}$) acting on $\widehat{B}$ as B (resp. $\widehat{A}$)-algebra. So under decomposition $\widehat{B} \cong \widehat{A} \oplus I$, the maximal ideal will becomes $m_{\widehat{B}} = m_{\widehat{A}} \oplus I$, since

$$\frac{\widehat{B}}{m_{\widehat{A}} \oplus I} \cong \frac{\widehat{A} \oplus I}{m_{\widehat{A}} \oplus I} \cong \widehat{A}/m_{\widehat{A}} \cong A/m_A$$

is a field. At same time, we have

$$\begin{aligned} m_{\widehat{A}}\widehat{B} = m_{\widehat{A}}(\widehat{A} \oplus I) &= m_{\widehat{A}} \oplus m_{\widehat{A}} I \\ &\cong m_{\widehat{B}} \\ &= m_{\widehat{A}} \oplus I, \end{aligned}$$

that is we have $m_{\widehat{A}} I = I$. Combining this with $m_{\widehat{A}} I \subseteq m_{\widehat{B}} I$, we can get that $I \subseteq m_{\widehat{B}} I$, so $I = m_{\widehat{B}} I$ where I is a finite $\widehat{B}$-module. By Nakayama's lemma, we have $I = 0$ hence that $\hat{g}$ is injective, which gives $\widehat{B} \cong \widehat{A}$. □

Our main theorem is the following, which says an étale morphism is the algebraic analog of a local diffeomorphism:

Theorem B.3.21. *Let $f : X \longrightarrow Y$ be a morphism of finite type between noetherian schemes. Assume that $x \in X$ and $y = f(x) \in Y$ are points such that $k(x) \cong k(y)$. Then f is étale at x if and only if $f_x^* : \widehat{\mathscr{O}}_{Y,y} \longrightarrow \widehat{\mathscr{O}}_{X,x}$ is an isomorphism.*

On algebraically closed field k we always have $k(x) \cong k(y) \cong k$ (where y is a closed point we will prove it later), so can get:

Corollary B.3.22. *Let k be an algebraically closed field, X and Y are schemes of finite type over k. Let $f : X \longrightarrow Y$ be a morphism. Then f is étale if and only if for every closed point $x \in X$, $f_x^* : \widehat{\mathscr{O}}_{Y,y} \longrightarrow \widehat{\mathscr{O}}_{X,x}$ is an isomorphism.*

Also, this can be used on the analysis on normal cones at a point:

Corollary B.3.23. *Let $f : X \longrightarrow Y$ be a morphism between algebraic varieties, and for any point $x \in X$, $y = f(x) \in Y$ let we draw the following commutative diagram*

$$\begin{array}{ccc} T_{X,x} & \hookleftarrow & C_x X \\ \downarrow & & \downarrow \\ Y_{Y,y} & \hookleftarrow & C_y Y \end{array}$$

of Zariski tangent spaces and normal cones. Then f is an étale morphism if and only if the two normal cones are isomorphic for all closed points $x \in X$.

Proof. Note that their tangent space are isomorphic when f is étale, since étale is equivalent to smooth of dimension 0.

"⇒": When f is étale, then by our main theorem, we have $\widehat{\mathscr{O}}_{Y,y} \cong \widehat{\mathscr{O}}_{X,x}$, while note that algebraic variety is defined over an algebraically closed field. Then we get

an isomorphism $\mathrm{gr}(\widehat{\mathscr{O}}_{Y,y}) \cong \mathrm{gr}(\widehat{\mathscr{O}}_{X,x})$ which states that $\mathrm{gr}(\mathscr{O}_{Y,y}) \cong \mathrm{gr}(\mathscr{O}_{X,x})$, due to [Eis95, 7.13]. And we recall that the normal cone is defined by

$$\begin{aligned} C_x X &= \underline{\mathrm{Spec}}_{\{x\}} \left(\bigoplus_{n=0}^{\infty} m_x^n / m_x^{n+1} \right) \\ &= \mathrm{Spec}\, \mathrm{gr}(\mathscr{O}_{X,x}), \end{aligned}$$

so we have the isomorphism.

"$\Leftarrow$": For such f such that $C_x X \cong C_y Y$, we verify that the image point $y = f(x)$ is closed for which $x \in X$ is a closed point. Since we are deal with finite type schemes, if $x \in X$ is a closed point then locally see $x = m \in \mathrm{Spec}\, B$ and $y = p \in \mathrm{Spec}\, A$, where m is a maximal ideal in B and p is a prime ideal in B is the contraction of m. This morphism f induces a injective ring map

$$\varphi : A/p \longrightarrow B/m = k(x)$$

which factor as $A/p \hookrightarrow k(y) \hookrightarrow k(x)$, and the right hand side is a finite dimensional vector space over k. So the left term is a finite dimension vector space k which is also an integral domain. Consider the endomorphism $\cdot a$ on A/p for any $a \in A/p$, it is injective by A/p is a domain, and is surjective by A/p is a finite dimensional vector space. Thus we see A/p is a field, since a has inverse. (Note that here we not use k is algebraically closed.)

Now we have a fields map $k(y) \hookrightarrow k(x) \cong k$, so they are isomorphic by y is a closed point. Since we have $C_x X \cong C_y Y$, it is nothing but $\mathrm{gr}(\mathscr{O}_{X,x}) \cong \mathrm{gr}(\mathscr{O}_{Y,y})$. So by lemma we have $\widehat{\mathscr{O}}_{Y,y} \cong \widehat{\mathscr{O}}_{X,x}$, which implies that f is étale at x. □

Remark B.3.24. This can not imply $A \longrightarrow B$ is isomorphic, that is fails to obtain the information on the level of "Zariski local". For example, set $A = B = \mathbb{C}[X]_{(x)}$, and $f : A \longrightarrow B$ to be $x \longmapsto x + x^2$. We see f is not isomorphism, but $\hat{f} : \widehat{A} \longrightarrow \widehat{B}$ is isomorphic. Since $x = 0$ is a regular point in affine line, by [Eis95, 10.16] we see $\widehat{A} = \widehat{B} = \mathbb{C}[[x]]$, which make

$$\begin{aligned} \hat{f} : \mathbb{C}[[x]] &\longrightarrow \mathbb{C}[[x]] \\ x &\longmapsto x + x^2 = f(x) \end{aligned}$$

has inverse sending x to $h(x)$, to find this inverse is equivalent to solve the coefficients in formal power series $h(x + x^2) = \sum a_n (x + x^2)^n = x$.

We say a finite étale morphism is the algebraic analogue of covering map in analytic topology.

B.3A Proof of the Main Theorem

B.4 Blowing-up Property and Infinitesimal Lifting

Definition B.4.1. (Regular sequence and depth) Let A be a ring, and M an A-module. A sequence $x_1, \ldots, x_r$ of elements of A is called a *regular sequence for* M if

- x_1 is not a zero divisor in M.
- x_2 is not a zero divisor in M/x_1M.
  ...
- X_i is not a zero divisor in $M/\sum^{i-1} x_iM$ for all i.

Moreover, if (A, m) is a local ring, the *depth of* M is the maximal length of a regular sequence $x_1, \ldots, x_r$ of M with all $x_i \in m$, which is denoted by depth M.

Definition B.4.2. (Cohen-Macaulay ring) A noetherian local ring A is *Cohen-Macaulay* if depth $A = \dim A$.

Example B.4.3.

- $k[[x_1, \ldots, x_n]]$ is Cohen-Macaulay ring.
- For any maximal ideal $m \subseteq k[x_1, \ldots, x_n]$, the $k[x_1, \ldots, x_n]_m$ is Cohen-Macaulay ring.
- $\mathbb{Z}_p$ is Cohen-Macaulay ring.
- $k[\varepsilon]/(\varepsilon^2)$ is Cohen-Macaulay ring.

Theorem B.4.4. *Let (A, m) be a noetherian local ring.*

(a) If A is regular, then it is Cohen-Macaulay.

(b) If A is Cohen-Macaulay, then so is A_p for all $p \in$ Spec A.

(c) If A is Cohen-Macaulay, then $x_1, \ldots, x_r$ with $x_i \in m$ forms a regular sequence for A if and only if $\dim\ A/(x_1, \ldots, x_r) = \dim\ A - r$.

(d) If A is Cohen-Macaulay, and $x_1, \ldots, x_r$ with $x_i \in m$ is a regular sequence for A, then $A/(x_1, \ldots, x_r)$ is Cohen-Macaulay.

(e) If A is Cohen-Macaulay, and $x_1, \ldots, x_r$ with $x_i \in m$ is a regular sequence for A, let $I = (x_1, \ldots, x_r)$ be the ideal. Then the natural map

$$(A/I)[t_1, \ldots, t_r] \longrightarrow \mathrm{gr}_I A = \bigoplus_{n \geq 0} I^n/I^{n+1}$$

$$t_i \longmapsto x_i$$

is an isomorphism, that is, I/I^2 is a free A/I-module of rank r, and for each $n \geq 1$ the natural map

$$\mathrm{Sym}^n(I/I^2) \longrightarrow I^n/I^{n+1}$$

is an isomorphism.

Theorem B.4.5. *Let X be a nonsingular variety over $\mathbb{C}$, and let $Y \subseteq X$ be a nonsingular closed subvariety, with ideal sheaf $\mathscr{I}$.*

Let $\pi : \widetilde{X} = \mathrm{Bl}_Y X \longrightarrow X$ be the blowing-up along Y, and let $Y' \subseteq \widetilde{X}$ be the subscheme defined by the inverse image ideal sheaf

$$\mathscr{I}' = \pi^{-1}\mathscr{I} \cdot \mathscr{O}_{\widetilde{X}}.$$

Then

(a) $\widetilde{X}$ is also nonsingular.

(b) Y', together with the induced projection map $\pi : Y' \longrightarrow Y$, is isomorphic to $\mathbb{P}(\mathscr{I}/\mathscr{I}^2)$, the projective space bundle associated to the locally free sheaf $\mathscr{I}/\mathscr{I}^2$ on Y.

(c) under this isomorphism, the normal sheaf

$$\mathscr{N}_{Y'/\widetilde{X}} = \mathscr{H}om_{\mathscr{O}_{Y'}}(\mathscr{I}'/\mathscr{I}'^2, \mathscr{O}_{Y'})$$

corresponds to $\mathscr{O}_{\mathbb{P}(\mathscr{I}/\mathscr{I}^2)}(-1)$.

Proof. (b): Since $\widetilde{X} = \mathrm{Proj.}\bigoplus_{n\geq 0}\mathscr{I}^n$, we have

$$\begin{aligned} Y' &\cong \mathrm{Proj.}\bigoplus_{n\geq 0}(\mathscr{I}^n \underset{\mathscr{O}_X}{\otimes} \mathscr{O}_X/\mathscr{I}) \\ &\cong \mathrm{Proj.}\bigoplus_{n\geq 0}\mathscr{I}^n/\mathscr{I}^{n+1}. \end{aligned}$$

But as Y is nonsingular, so $\mathscr{I}$ is locally generated by $\mathrm{codim}_X Y$ elements which is a regular sequence in $\mathscr{O}_X$. Then using (B.4.4)(e), saying that $\mathscr{I}/\mathscr{I}^2$ is a locally free $\mathscr{O}_X/\mathscr{I}$-module of rank $\mathrm{codim}_X Y$, and for each $n \geq 1$,

$$\mathscr{I}^n/\mathscr{I}^{n+1} \cong \mathrm{Sym}^n(\mathscr{I}/\mathscr{I}^2).$$

Thus

$$\begin{aligned} Y' &\cong \mathrm{Proj.}\bigoplus_{n\geq 0}\mathrm{Sym}^n(\mathscr{I}/\mathscr{I}^2) \\ &\cong \mathrm{Proj.}\mathrm{Sym}(\mathscr{I}/\mathscr{I}^2) \\ &\cong \mathbb{P}(\mathscr{I}/\mathscr{I}^2). \end{aligned}$$

(a): By the above we see that Y' is locally isomorphic to $Y \times \mathbb{P}^{r-1}$, where $r = \operatorname{codim}_X Y$, by $\operatorname{Sym}(\mathscr{I}/\mathscr{I}^2) \cong \mathscr{O}_Y[t_1, \ldots, t_r]$. So Y' is also non-singular.

Since for $\widetilde{X}$ is defined as $\operatorname{Proj.} \bigoplus_{n\geq 0} \mathscr{I}^n$, equipped with a natural line bundle $\mathscr{O}_{\widetilde{X}}(1)$, picking any open affines $U \subseteq X$, the sheaf $\mathscr{O}(1)$ on $\operatorname{Proj.} \bigoplus_{n\geq 0} \mathscr{I}^n(U)$ is the sheaf associated to the graded $\bigoplus_{n\geq 0} \mathscr{I}^n(U)$-module $(\bigoplus_{n\geq 0} \mathscr{I}^n(U))(1) = \bigoplus_{n\geq 0} \mathscr{I}^{n+1}(U)$, equal to the ideal $\mathscr{I}(U) \cdot (\bigoplus_{n\geq 0} \mathscr{I}^n(U))$. This implies that $\mathscr{O}_{\widetilde{X}}(1)$ is equal to the inverse image ideal sheaf $\mathscr{I}' = \pi^{-1}\mathscr{I} \cdot \mathscr{O}_{\widetilde{X}}$. So $\mathscr{I}'$ is an invertible sheaf, where the local equation defines the locally principal subvariety Y'.

But at every point p in $\widetilde{X}$, the fact that Y' is non-singular says that the local ring $\mathscr{O}_{\widetilde{X},p}$ quotiented by an element which is not a zero-divisor, is regular. So the $\mathscr{O}_{\widetilde{X},p}$ itself is indeed regular, hence $\widetilde{X}$ is non-singular.

(c): By $\mathscr{I}' = \pi^{-1}\mathscr{I} \cdot \mathscr{O}_{\widetilde{X}} \cong \mathscr{O}_{\widetilde{X}}(1)$, the

$$\begin{aligned}
\mathscr{I}'/\mathscr{I}'^2 &\cong \mathscr{I}' \underset{\mathscr{O}_{\widetilde{X}}}{\otimes} \mathscr{O}_{\widetilde{X}}/\mathscr{I}' \\
&\cong \mathscr{O}_{\widetilde{X}}(1) \underset{\mathscr{O}_{\widetilde{X}}}{\otimes} \mathscr{O}_{Y'} \\
&= \mathscr{O}_{Y'}(1).
\end{aligned}$$

Hence $\mathscr{N}_{Y'/\widetilde{X}} = \mathscr{H}om(\mathscr{I}'/\mathscr{I}'^2, \mathscr{O}_{Y'}) \cong \mathscr{O}_{Y'}(-1) = \mathscr{O}_{\mathbb{P}(\mathscr{I}/\mathscr{I}^2)}(-1)$. □

Corollary B.4.6. (Blowing up along nonsingular variety) *Let X be a nonsingular variety, and let $Y \subseteq X$ be a nonsingular closed subvariety of $r = \operatorname{codim}_X Y \geq 2$. Let $\pi : \widetilde{X} = \operatorname{Bl}_Y X \longrightarrow X$ be the blowing-up, and let $Y' = \pi^{-1}(Y)$.*

Then

(a) $\operatorname{Pic} \widetilde{X} \cong \operatorname{Pic} X \oplus \mathbb{Z}$.

(b) $\omega_{\widetilde{X}} \cong \pi^*\omega_X \otimes \mathscr{O}\big((r-1)Y'\big) \cong \pi^*\omega_X \otimes \mathscr{O}_{\mathbb{P}(\mathscr{I}/\mathscr{I}^2)}(r-1)$.

Proof. (a): As the above theorem, X and $\widetilde{X}$ are both non-singular, so we can sat the Weil divisor class group instead of line bundles.

Since Y is of codimension at least two, we have

$$\operatorname{Pic}(X - Y) \cong \operatorname{Pic} X$$

by [Har77, II,6.5(b)]. And as $\widetilde{X} - Y' \cong X - Y$, so compositing the natural restriction map

$$\begin{aligned}
\operatorname{Pic} \widetilde{X} &\longrightarrow \operatorname{Pic}(\widetilde{X} - Y') \cong \operatorname{Pic}(X_Y) \\
L &\longmapsto L|_{\widetilde{X}-Y'}
\end{aligned}$$

we have a map

$$\text{Pic } \widetilde{X} \longrightarrow \text{Pic } X.$$

As in the proof of previous theorem, we see that the ideal sheaf $\mathscr{I}'$ corresponding to Y' is equal to $\mathscr{O}_{\widetilde{X}}(1)$, which implies that the map

$$\begin{aligned} \mathbb{Z} &\longrightarrow \text{Pic } \widetilde{X} \\ n &\longmapsto [nY'] = \mathscr{O}_{\widetilde{X}}(1)^{\otimes n} = \mathscr{O}_{\widetilde{X}}(n) \end{aligned}$$

is injective, so we have a short exact sequence

$$\begin{aligned} 0 \longrightarrow \mathbb{Z} &\longrightarrow \text{Pic } \widetilde{X} \longrightarrow \text{Pic } X \longrightarrow 0 \\ 1 &\longmapsto \mathscr{O}_{\widetilde{X}}(1) \end{aligned}$$

But there is a splitting pullback $\pi^* : \text{Pic } X \longrightarrow \text{Pic } \widetilde{X}$, therefore we have

$$\text{Pic } \widetilde{X} \cong \text{Pic } X \oplus \mathbb{Z}.$$

(b): As a consequence of (a), for line bundle $\omega_{\widetilde{X}} \in \text{Pic } \widetilde{X}$, we can write

$$\begin{aligned} \omega_{\widetilde{X}} &\cong \pi^* L \otimes \mathscr{O}_{\widetilde{X}}(qY') \\ &\cong \pi^* L \otimes \mathscr{O}_{\widetilde{X}}(q) \end{aligned}$$

for $L \in \text{Pic } X$.

Then by pushing out the $\omega_{\widetilde{X}}$ into Pic X via the restriction map

$$\begin{array}{ccc} \text{Pic } \widetilde{X} \longrightarrow \text{Pic}(\widetilde{X} - Y') \xrightarrow{\cong} & & \text{Pic}(X - Y) \cong \text{Pic } X \\ \omega_{\widetilde{X}} \longmapsto \omega_{\widetilde{X}}|_{\widetilde{X}-Y'} \cong \pi^* L|_{\widetilde{X}-Y'} \longmapsto & & L|_{X-Y} \longmapsto L \end{array}$$

where the $\mathscr{O}_{\widetilde{X}}(qY')$ part becomes trivial on $\widetilde{X} - Y'$, we see that

$$\omega_{\widetilde{X}}|_{\widetilde{X}-Y'} \cong \otimes_X |_{X-Y}.$$

But there is only one way to extending the line bundle outside the open subset $X - Y$, which is just ω_X. So $L \cong \omega_X$.

Now to determine the integer q, we pullback both side to Y' - the codimension 1 subvariety of $\widetilde{X}$ - and can use the adjunction formula:

$$\begin{aligned} \omega_{Y'} &\cong \omega_{\widetilde{X}} \otimes \mathscr{O}_{\widetilde{X}}(Y') \otimes \mathscr{O}_{Y'} \\ &\cong \pi^* \omega_X \otimes \mathscr{O}_{\widetilde{X}}((q+1)Y')|_{Y'} \\ &= \pi^* \omega_X \otimes (\mathscr{O}_{\widetilde{X}}(-Y')|_{Y'})^{\otimes(-q-1)} \\ [\text{Har77}, \text{II}, 6.18] &= \pi^* \omega_X \otimes (\mathscr{I}' \otimes \mathscr{O}_{\widetilde{X}}/\mathscr{I}')^{\otimes(-q-1)} \\ &\cong \pi^* \omega_X \otimes (\mathscr{N}_{Y'/\widetilde{X}}^{-1})^{\otimes(-q-1)} \\ &= \pi^* \omega_X \otimes \mathscr{O}_{Y'}(-q-1) \end{aligned}$$

by previous theorem.

Now we take a closed point $y \in Y$ and consider the fibre $\pi^{-1}(y) = Y' \underset{Y}{\times} \operatorname{Spec} k(y)$, then with respect to the projections $p_1 : \pi^{-1}(y) \longrightarrow Y'$, $p_2 : \pi^{-1}(y) \longrightarrow \operatorname{Spec} k(y)$, we have

$$\begin{aligned}
\omega_{\pi^{-1}(y)} &\cong p_1^*\omega_{Y'} \otimes p_2^*\omega_{k(y)} \\
&\cong p_1^*(\pi^*\omega_X \otimes \mathscr{O}_{Y'}(-q-1)) \otimes p_2^*\mathscr{O}_{Y,y} \\
&= p_1^*\pi^*\omega_X \otimes p_1^*\mathscr{O}_{Y'}(-q-1) \otimes \mathscr{O}_{\pi^-(y)} \\
&= (\pi \circ p_1)^*\omega_X \otimes \mathscr{O}_{\pi^{-1}(y)}(-q-1) \\
&\cong (\pi \circ p_1)^*\omega_X|_{\{y\}} \otimes \mathscr{O}_{\pi^{-1}(y)}(-q-1) \\
&\cong \mathscr{O}_{\pi^{-1}(y)} \otimes \mathscr{O}_{\pi^{-1}(y)}(-q-1) \\
&= \mathscr{O}_{\pi^{-1}(y)}(-q-1)
\end{aligned}$$

as $\pi \circ p_1 : \pi^{-1}(y) \to Y' \to \{y\}$ sending to a point. But by the previous theorem,

$$\pi^{-1}(y) \cong \mathbb{P}^{r-1}_{k(y)},$$

so we have $\omega_{\pi^{-1}(y)} \cong \mathscr{O}_{\pi^{-1}(y)}(-r)$, that is $-r = -q-1$, i.e. $q = r-1$. □

There is also a useful proposition, by generalize the (B.2.13):

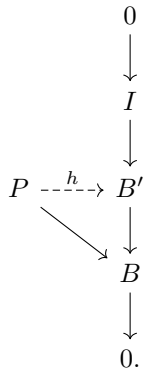
Lemma B.4.7. *Let $B' \longrightarrow B$ be a surjective homomorphism of k-algebras, with kernel I s.t. $I^2 = 0$. Let $P \longrightarrow B$ be a homomorphism of k-algebras.*

(a) If h, $h' : P \longrightarrow B'$ are two liftings of the map $P \longrightarrow B$ to B', then $\theta = h - h'$ is a k-derivation from P to I.

(b) Conversely, if $h : P \longrightarrow B'$ is one lifting, and $\theta : P \longrightarrow I$ is a k-derivation, then $h' = h + \theta$ is another homomorphism of P to B' which lifts the given map $P \longrightarrow B$.

Note that since $I^2 = 0$, I has a natural structure of B-module and hence of P-module, then we can regard θ as an element of $\operatorname{Hom}_P(\Omega_P, I)$ by the universal property.

Proof. Suppose that $h : P \longrightarrow B'$ is a given homomorphism lifting $P \longrightarrow B$.



If $h' : P \longrightarrow B'$ is another lifting, we have that $\theta = h - h'$ composed with $B' \longrightarrow B$ is zero, so θ maps P into I. Then

- θ is additive: OK.
- θ satisfies Leibniz rule:

$$\begin{aligned}\theta(a_1a_2) &= h(a_1a_2) - h'(a_1a_2)\\ &= h(a_1)h(a_2) - h(a_1)h'(a_2) + h(a_1)h'(a_2) - h'(a_1)h'(a_2)\\ &= h(a_1)\theta(a_2) + \theta(a_1)h'(a_2) \text{ , for } a_i \in P.\end{aligned}$$

- θ is k-linear: $\forall c \in k$, $\theta(c) = h(c) - h'(c) = c(h(1) - h'(1)) = c(1-1) = 0$.

So θ is a k-linear derivation of P into I.

Conversely, if θ is a k-linear derivation from P to I, and h be a lifting of $P \longrightarrow B$. Then $h' = h + \theta$ is

- k-linear: OK since h and θ are.
- algebra homomorphism:

$$\begin{aligned}h'(a_1a_2) &= h(a_1a_2) + \theta(a_1a_2)\\ &= h(a_1)h(a_2) + h(a_1)\theta(a_2) + h(a_2)\theta(a_1)\\ (\text{As } \theta(a_1)\cdot\theta(a_2) \in I^2 = 0) &= (h(a_1) + \theta(a_1))\cdot(h(a_2) + \theta(a_2))\\ &= h'(a_1)'(a_2).\end{aligned}$$

□

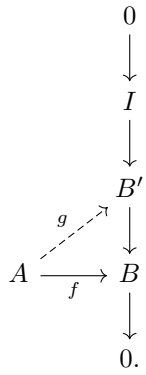
Proposition B.4.8. (Infinitesimal lifting property) *Let k be an algebraically closed field, and let A be a finitely generated k-algebra such that* Spec *A is non-singular variety over k.*

Let

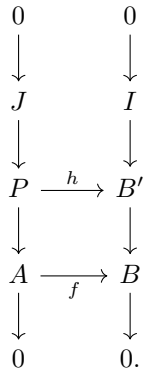
$$0 \longrightarrow I \longrightarrow B' \longrightarrow B \longrightarrow 0$$

be an exact sequence, where B' is a k-algebra and I is an ideal of B' with $I^2 = 0$.

Finally suppose given a k-algebra homomorphism $f : A \longrightarrow B$. Then there exists a k-algebra homomorphism $g : A \longrightarrow B'$ making the diagram commute:



Proof. Let $P = k[x_1, \ldots, x_n]$ be a polynomial ring, and A be its quotient with ideal J. As $B' \longrightarrow B$ is onto, there exists a homomorphism $h : P \longrightarrow B'$ making the diagram commute



Then we see h sends the ideal J into I, hence it sends J^2 into $I^2 = 0$, inducing a map $\bar{h} : J/J^2 \longrightarrow I$. And

$$\bar{h}(\underbrace{(p+J)}_{\in A}(j+J^2)) = f(p+J)h(j+J^2)$$

says that the $\bar{h}$ is A-linear.

Now as $\operatorname{Spec} A$ is non-singular, there is an exact sequence

$$0 \longrightarrow J/J^2 \longrightarrow \Omega_P \otimes A \longrightarrow \Omega_A \longrightarrow 0$$

then we apply the functor $\operatorname{Hom}_A(-, I)$:

$$0 \to \operatorname{Hom}_A(\Omega_A, I) \to \operatorname{Hom}_P(\Omega_P, I) \to \operatorname{Hom}_A(J/J^2, I) \to \operatorname{Ext}^1_A(\Omega_A, I) \to \cdots$$

Also by $\operatorname{Spec} A$ is non-singular, so Ω_A is finite locally free by [Har77, II, 8.15] which **the algebraically closed field condition** used here, hence finite projective see [SP25, Tag 00NX], consequently $\operatorname{Ext}^1_A(\Omega_A, I) = 0$.

Then we get an exact sequence

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \operatorname{Hom}_A(\Omega_A, I) & \longrightarrow & \operatorname{Hom}_P(\Omega_P, I) & \longrightarrow & \operatorname{Hom}_A(J/J^2, I) & \longrightarrow & 0 \\ & & & & \theta & \underset{\text{lifting}}{\longmapsto} & \bar{h} & & \end{array}$$

for $\theta \in \operatorname{Hom}_P(\Omega_P, I)$ can be regarded as a k-derivation of P to the module I. Then we define a new map $h' : P \longrightarrow B'$ by $h' = h - \theta$. The lemma above says that h' is a homomorphism lifting f, and $h'(J) = 0$ because by picking each $y \in J$, then $h'(y) = h(y) - \theta(y)$ just only need to consider y mod J^2 which maps to zero by our choice of θ.

$$\begin{array}{ccc} 0 & & 0 \\ \downarrow & & \downarrow \\ J & & I \\ \downarrow & & \downarrow \\ P & \xrightarrow{h'} & B' \\ \downarrow & \overset{g}{\nearrow} & \downarrow \\ A & \xrightarrow[f]{} & B \\ \downarrow & & \downarrow \\ 0 & & 0. \end{array}$$

Then h' descends to give a desired homomorphism $g : A \longrightarrow B'$ lifting f. □

Appendix C

Chevalley's Nullstellensatz

Definition C.0.1. For a scheme X, a *constructible subset* $S \subseteq X$ is a set of the form

$$S = S_1 \cup S_2 \cup \cdots \cup S_t$$

where S_i is locally closed, meaning that it's an intersection of an open subset and a closed subset.

Theorem C.0.2. (Chevalley's Nullstellensatz) *Let $f : X \longrightarrow Y$ be a morphism of finite type with Y is a noetherian scheme. Then for every constructible set $S \subseteq X$, the $f(S) \subseteq Y$ is also constructible.*

Proof. First as $f : X \longrightarrow Y$ is of finite type, so we know X also be a noetherian scheme, that is they are both covered by finitely many affine opens. Now if $\{U_{ij}\}$ is the affine cover of X sending into $\{V_i\}$ the affine cover of Y, we write

$$f(S) = \bigcup_{i,j} f(S \cap U_{ij}).$$

Since S is constructib;e, so is $S \cap U_{ij} \subseteq U_{ij}$, thus we enough to reduced the case X and Y are both affine.

Secondly we write $X = \operatorname{Spec} R$ and $Y = \operatorname{Spec} S$ where R is a finitely generated S-algebra. Since we can factor f as

$$\begin{aligned} X = \operatorname{Spec} S[a_1, \ldots, a_n] \to \operatorname{Spec} S[a_1, \ldots, a_{n-1}] \to \cdots \\ \cdots \to \operatorname{Spec} S[a_1] \to \operatorname{Spec} S = Y, \end{aligned}$$

we can reduce the theorem to the case $R = S[a]$, and we can write it as $S[X]/\mathfrak{a}$ mapping into $\operatorname{Spec} S[x]$ is a closed immersion, keeps constructiblity. Thus we enough

to reduce to the case

$$f : \operatorname{Spec} S[x] \longrightarrow \operatorname{Spec} S.$$

Now we make so-called "noetherian induction", that the closed subsets $V(-) \subseteq \operatorname{Spec} S$ satisfy the descending chain condition, if the theorem is false, then the set of closed subsets of $\operatorname{Spec} S$ which does not take constructibles to constructibles is non-empty, wo can find a minimal element $V(I) \subseteq \operatorname{Spec} S$ s.t.

$$f|_{f^{-1}V(I)} : f^{-1}(V(I)) \longrightarrow V(I)$$

fails to send constructibles to constructibles. Since $f^{-1}V(I) = V(IS[x]) \cong \operatorname{Spec}(S/I)[x]$, we can replace $\operatorname{Spec} S$ by $\operatorname{Spec} S/I$ and $\operatorname{Spec} S[x]$ by $\operatorname{Spec}(S/I)[x]$, and of course assume that $I = \sqrt{I}$ so that new S is nilpotent. So we reduce to the case which outside the minimal element:

for all constructible sets $C \subseteq \operatorname{Spec} S[x]$, if $\overline{f(C)} \subsetneq \operatorname{Spec} S$ then$f(C)$ is constructible.

which is the **noetherian hypothesis**.

Moreover $\operatorname{Spec} S$ must be irreducible too, since if not, we write

$$\operatorname{Spec} S = Z_1 \cup Z_2 \text{ for } Z_i \subsetneq \operatorname{Spec} S \text{ and } Z_i \text{ closed.}$$

As S is the minimal counter-example, the above noetherian hypothesis must holds on Z_i, then if $C \subseteq \operatorname{Spec} S[x]$ is constructible, so are $C \cap f^{-1}(Z_i)$. By noetherian induction on

$$f(C \cap f^{-1}(Z_i)) = f(C) \cap Z_i$$

which the closure lies in $Z_i \subsetneq \operatorname{Spec} S$, it is constructible, hence

$$f(C) = f(C \cap f^{-1}(Z_1)) \cap f(C) = f(C \cap f^{-1}(Z_2))$$

is also constructible. In particular we can choose $C = \operatorname{Spec} S[x]$, contradicts the minimality of our choice of $\operatorname{Spec} S$ is not constructible. So S must be an integral domain.

Now we need the following lemma to complete the proof:

Lemma C.0.3. *Let S be an integral domain and $\eta \in \operatorname{Spec} S$ be its generic point. Let $C \subseteq \operatorname{Spec} S[x]$ be an irreducible closed subset and $C_0 \subseteq C$ an open subset. Consider the morphism*

$$f : \operatorname{Spec} S[x] \longrightarrow \operatorname{Spec} S.$$

Then there is a non-empty open subset $U \subseteq \operatorname{Spec} S$ s.t. either $U \subseteq f(C_0)$ or $U \cap f(C_0)$ is empty.

Using this lemma, we have two cases for locally closed set C_0:

- if $U \subseteq f(C_0)$, then

$$f(C_0) = U \cap (f(C_0) \cap (\operatorname{Spec} S - U)).$$

  As U is open so constructible, and as $\operatorname{Spec} S - U$ is closed so the set $C_0 \cap f^{-1}(\operatorname{Spec} S - U)$ is also constructible. By out noetherian hypothesis, the image $f(C_0) \cap (\operatorname{Spec} S - U)$ is constructible, so $f(C_0)$ is constructible.

- if $U \cap f(C_0)$ is empty, then $f(C_0) \subseteq \operatorname{Spec} S - U$ so constructible by noetherian hypothesis.

Hence we proved that for arbitrary C_0 the open subset of an irreducible closed set, its image is constructible. But notice that every constructible subset of $\operatorname{Spec} S[x]$, even $\operatorname{Spec} S[x]$ itself, is the finite union of such C_0's and has the constructible image $\operatorname{Spec} S$. So we get the contradiction, the theorem holds.

To complete the proof, we now proving the lemma

Proof. Let K be the quotient field of S. Note that the generic fibre

$$f^{-1}(\eta) \cong \operatorname{Spec} S[x] \underset{S}{\times} k(\eta) \cong \operatorname{Spec} K[X] = \mathbb{A}^1_K,$$

which consists only of a generic point η^* and its closed points. And $C \cap f^{-1}(\eta)$ is a closed irreducible subset of $f^{-1}(\eta)$ has three possibilities:

Case i) $C \supseteq f^{-1}(\eta)$, then C contains generic point so $C = \operatorname{Spec} S[x]$.

Case ii) $C \cap f^{-1}(\eta) = \{\zeta\}$, where ζ is a closed point of $f^{-1}(\eta)$, and

Case iii) $C \cap f^{-1}(\eta) = \varnothing$.

In *case (i)*, C_0 contains some distinguished open subset $\operatorname{Spec} S[x]_g$, where we write

$$g = a_0 x^n + a_1 x^{n-1} + \cdots + a_n,\ a_0 \neq 0.$$

Let $U = \operatorname{Spec} S_{a_0}$. For all $s \in \operatorname{Spec} S$, $f^{-1}(s) \cong \operatorname{Spec} k(s)[x] = \mathbb{A}^1_{k(s)}$ and

$$C_0 \cap f^{-1}(s) \supseteq \left\{ t \in \mathbb{A}^1_{k(s)} \;\middle|\; \begin{array}{l} \bar{g}(t) \neq 0, \text{ where } \bar{g} = \bar{a}_0 x^n + \cdots + \bar{a}_n \\ \text{and } \bar{a}_i = a_i(s) \text{ the image of } a_i \text{ in } k(s) \end{array} \right\}.$$

So if $s \in U$ which means $a_0 \notin \mathfrak{p}_s$, then $\bar{a}_0 \neq 0$, hence $\bar{g} \not\equiv 0$, hence the generic point of $f^{-1}(s)$ is in $D(\bar{g}) \subseteq \mathbb{A}^1_{k(s)}$. Moreover

$$C_0 \cap f^{-1}(s) \supseteq D(g) \cap f^{-1}(s) = D(\bar{g})$$

so the generic point of $f^{-1}(s)$ lies in $C_0 \cap f^{-1}(s)$, hence $s \in f(C_0)$. So $U \subseteq f(C_0)$.

In *case (ii)*, let $C = V(\mathfrak{p})$. Then

$$\mathfrak{p} \cdot K[x] = g \cdot K[x] \text{ for } g \text{ irreducible.}$$

We may assume that $g = a_0x^n + a_1x^{n-1} + \cdots + a_n \in \mathfrak{p}$ because $\mathfrak{p} \cap S = (0)$, and $a_i \in S$. Then

$$V(g) \supseteq C \supseteq C_0,$$

but all three sets intersect the generic fibre $f^{-1}(\eta)$ in only one point ζ, since on $f^{-1}(\eta) \cong \operatorname{Spec} K[x]$ the irreducible g defines a closed point which is exactly ζ, and if $C_0 \cap f^{-1}(\eta) = \emptyset$ the closure of $f(C_0)$ is proper closed subset hence constructible by our noetherian hypothesis. Thus

$$\begin{aligned} V(g) - C_0 &= (V(g) - C) \cup (V(g) - W) \\ &= (V(g) \cap (\operatorname{Spec} S[x] - C)) \cup (V(g) \cap (\operatorname{Spec} S[x] - W)) \end{aligned}$$

where $C_0 = C \cap W$ for W open, hence $V(g) - C_0$ is a constructible set disjoint from $f^{-1}(\eta)$. Let

$$\overline{V(g) - C_0} = W_1 \cup \cdots \cup W_t \text{ for } W_i \text{ irreducible with generic points } w_i \notin f^{-1}(\eta).$$

Then $f(W_i) \subseteq \overline{\{f(w_i\}}$ and $\overline{\{f(w_i\}}$ is a proper closed subset of $\operatorname{Spec} S$. Thus

$$f(V(g) - C_0) \subseteq \bigcup_{i=1}^{t} \overline{\{f(w_i\}} \subseteq \text{ some subset } V(\alpha) \text{ of } \operatorname{Spec} S$$

with $\alpha \in S$ is non-zero. Now let $U = \operatorname{Spec} S_{a_0\alpha}$. Then if $s \in U$ and $s \notin V(\alpha)$,

$$\begin{aligned} \alpha(s) \neq 0 \Rightarrow f^{-1}(s) \cap C_0 &= f^{-1}(s) \cap V(g) \\ &= \left\{ t \in \mathbb{A}^1_{k(s)} \;\middle|\; \bar{g}(t) = 0 \right\}. \end{aligned}$$

Since $a_0(s) = \bar{a}_0 \neq 0$ as $s \in U$, and $\bar{g} \not\equiv 0$, hence $\bar{g}$ has an irreducible factor $\bar{g}_1$ and the prime ideal $\bar{g}_1 \cdot k(s)[x]$ defines a point of $f^{-1}(s) \cong \mathbb{A}^1_{k(s)}$ where g is zero. Thus we have $f^{-1}(s) \cap V(g)$ is non-empty implies $s \in f(C_0)$ which proves $U \subseteq f(C_0)$.

In *case (iii)*, let ζ be the generic point of C, then $f(C) \subseteq \overline{f(\zeta)} \not\ni \eta$, hence we pick the non-empty open subset $U = \operatorname{Spec} S - \overline{f(\zeta)}$ which is disjoint from $f(C_0)$. □

□

Appendix D

Spectral Sequence towards to Algebraic Geometry

D.1 Bicomplexes

Almost all spectral sequence arise from bicomplexes, so we begin with discussion of them now. We will over associated ring R, and almost all settings are related with cohomology.

Definition D.1.1. (Graded module) A *graded module* is $\mathbb{Z}$-indexed family $M = (M^i)_{i\in\mathbb{Z}}$ of R-modules. Graded modules M are often denoted by $M^\bullet$.

Example D.1.2. (i) If $(C^\bullet, d) = \cdots \to C^{n-1} \xrightarrow{d} C^n \xrightarrow{d} C^{n+1} \to \cdots$ is a cochain complex, then $C^\bullet = (C^n)_{n\in\mathbb{Z}}$ is a graded module.

(ii) The family $H^\bullet(C^\bullet) = (H^p(C^\bullet))_{p\in\mathbb{Z}}$ of its cohomology modules is a graded module.

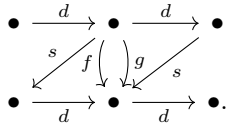
Definition D.1.3. (Graded maps) Let $M^\bullet$ and $N^\bullet$ be graded modules, and let $a \in \mathbb{Z}$. A *graded map of degree* a, which denoted by $f : M^\bullet \to N^\bullet$ is a family of homomorphisms $f = (f^p : M^p \to N^{p+a})_{p\in\mathbb{Z}}$. The *degree* of f is a, and denote it by $\deg(f) = a$.

Example D.1.4.

(i) If $(C^\bullet, d)$ is a cochain complex, then the differential $d : C^\bullet \to C^\bullet$ given by $d = (d^n : C^n \to C^{n+1})_{n\in\mathbb{Z}}$ is a map of degree $+1$

(ii) If $\phi : C^\bullet \to C'^\bullet$ is a cochain complex morphism, then $f = (f^n : C^n \to C'^n)_{n\in\mathbb{Z}}$ is a graded map of degree 0.

(iii) If $f, g : C^\bullet \to C'^\bullet$ are homotopic cochain maps, then a homotopy $s = (s^n : C^n \to C'^{n-1})_{n\in\mathbb{Z}}$ is a graded map of degree -1.



Proposition D.1.5. *If $M^\bullet \xrightarrow{f} N^\bullet \xrightarrow{g} P^\bullet$ are graded maps with degree a and b, respectively, then* $\deg(g \circ f) = a + b$.

So easily to see the graded modules over R equipped with graded maps form a category $\mathcal{A}$. If $f : M^\bullet \to N^\bullet$ is a graded map of degree n, then $f \in \prod_p \mathrm{Hom}(M^p, N^{p+n})$, and so

$$\mathrm{Hom}(M^\bullet, N^\bullet) = \bigcup_{a\in\mathbb{Z}} \left(\prod_p \mathrm{Hom}(M^p, N^{p+a}) \right).$$

Now let us define the exactness in $\mathcal{A}$.

Definition D.1.6. (Exactness of graded modules) If $M^\bullet$ and $M'^\bullet$ are graded modules, then M' is a *submodule* of M if $M'^p \subseteq M^p$ for all p, i.e. inclusions are graded maps of degree 0. If $M'^\bullet \subseteq M^\bullet$, then the *quotient* is $M^\bullet/M'^\bullet = (M^p/M'^p)_{p\in\mathbb{Z}}$, i.e. the natural map $M^\bullet \to M^\bullet/M'^\bullet$ is a graded map of degree 0.

If $f : M^\bullet \to N^\bullet$ is a graded map of degree a, then there is an inclusion

$$\ker f^\bullet = (\ker f^p) \hookrightarrow (M^p) = M^\bullet.$$

On the other hand, we have

$$\mathrm{im}\, f^\bullet = ((\mathrm{im}\, f)^p) = (\mathrm{im}\, f^{p-a}) \hookrightarrow (N^p) = N^\bullet.$$

So directly the *exactness* of $A^\bullet \xrightarrow{f} B^\bullet \xrightarrow{g} C^\bullet$ is defined by

$$\ker g^\bullet = \mathrm{im}\, f^\bullet,$$

that is $\ker g^p = \mathrm{im}\, f^{p-a}$ for all $p \in \mathbb{Z}$.

Given a short exact sequence of cochain complexes

$$0 \to C'^\bullet \xrightarrow{i} C^\bullet \xrightarrow{\pi} C''^\bullet \to 0.$$

The long exact sequence of cohomology modules is sometimes called an *exact triangle*:

$$\begin{array}{ccc} H^\bullet(C'^\bullet) & \xrightarrow{H^\bullet(i)} & H^\bullet(C^\bullet) \\ & {\scriptstyle\delta}\nwarrow \quad \swarrow{\scriptstyle H^\bullet(\pi)} & \\ & H^\bullet(C''^\bullet). & \end{array}$$

If we regard each vertex as a graded module, then the maps $H^\bullet(i)$ and $H^\bullet(\pi)$ have degree 0, and the connecting homomorphism δ has degree $+1$.

More generally, given an exact triangle of graded modules and graded maps,

$$\begin{array}{ccc} A^\bullet & \xrightarrow{\alpha} & B^\bullet \\ & {\scriptstyle\gamma}\nwarrow \quad \swarrow{\scriptstyle\beta} & \\ & C^\bullet & \end{array}$$

with α, β, γ having degrees a, b, c respectively, we can reconstruct the long exact sequence from which it comes:

$$\cdots B^{p-b-c} \xrightarrow{\beta} C^{p-c} \xrightarrow{\gamma} A^p \xrightarrow{\alpha} B^{p+a} \xrightarrow{\beta} \cdots$$

Definition D.1.7. (Bigraded modules) A *bigraded module* is a doubly indexed family of R-modules

$$M^{\bullet,\bullet} = (M^{p,q})_{(p,q)\in\mathbb{Z}\times\mathbb{Z}}.$$

We will view $p \in \mathbb{Z}$ on the x-axis with a module sitting on each integer point.

Definition D.1.8. (Bigraded maps) Let $M^{\bullet,\bullet}$ and $N^{\bullet,\bullet}$ be bigraded modules, and let $(a, b) \in \mathbb{Z} \times \mathbb{Z}$. A *bigraded map of bidegree* (a, b), denoted by $f : M^{\bullet,\bullet} \to N^{\bullet,\bullet}$ is a family of homomorphisms $f = (f^{p,q} : M^{p,q} \to M^{p+a,q+b})_{(p,q)\in\mathbb{Z}\times\mathbb{Z}}$, while we denote it by $\deg f = (a, b)$.

Similarly, for $\deg f = (a, b)$ and $\deg g = (a', b')$, then $\deg g \circ f = (a + a', b + b')$.

All bigraded modules over R equipped with the bigraded maps form a category. If $f : M^{\bullet,\bullet} \to N^{\bullet,\bullet}$ of degree (a, b), then $f \in \prod_{(p,q)} \mathrm{Hom}(M^{p,q}, N^{p+a,q+b})$, and so

$$\mathrm{Hom}(M, N) = \bigcup_{(a,b)} \left(\prod_{(p,q)} \mathrm{Hom}(M^{p,q}, N^{p+a,q+b}) \right).$$

And similarly, *submodule* are the data $(M'^{p,q} \subseteq M^{p,q})$ for all (p, q) equipping with bidegree $(0, 0)$ inclusions, and *quotient* are the data $(M^{p,q}/M'^{p,q})$ equipping with the natural bigraded map $M \to M/M'$ of bidegree $(0, 0)$. Moreover for $f : M^{\bullet,\bullet} \to N^{\bullet,\bullet}$ of bidegree (a, b), then

$$\ker f^{\bullet,\bullet} = (\ker f^{p,q}) \hookrightarrow (M^{p,q}) = M^{\bullet,\bullet}.$$

On the other hand,

$$\operatorname{im} f^{\bullet,\bullet} = ((\operatorname{im} f)^{p,q}) = (\operatorname{im} f^{p-a,q-b}) \hookrightarrow (N^{p,q}) = N^{\bullet,\bullet}.$$

So the *exactness* of $A^{\bullet,\bullet} \xrightarrow{f} B^{\bullet,\bullet} \xrightarrow{g} C^{\bullet,\bullet}$ means that $\ker g^{\bullet,\bullet} = \operatorname{im} f^{\bullet,\bullet}$, that is $\operatorname{im} f^{p-a,q-b} = \ker g^{p,q}$ for all $(p,q) \in \mathbb{Z} \times \mathbb{Z}$.

Generally we consider a triangle $(A, B, C; \alpha, \beta, \gamma)$ of bigraded modules and bidegree maps

$$\begin{array}{ccc} A^{\bullet,\bullet} & \xrightarrow{\quad\alpha\quad} & B^{\bullet,\bullet} \\ & {\scriptstyle\gamma}\nwarrow \quad \swarrow{\scriptstyle\beta} & \\ & C^{\bullet,\bullet} & \end{array}$$

which is exact in each vertex. If we set α, β, γ have bidegree $(a,a'), (b,b'), (c,c')$ respectively, we obtain a long exact sequence

$$\cdots \to C^{p-c,q-c'} \xrightarrow{\gamma} A^{p,q} \xrightarrow{\alpha} B^{p+a,q+a'} \xrightarrow{\beta} C^{p+a+b,q+a'+b'} \to \cdots$$

Conversely, given a family of such long exact sequence, we can define an exact triangle from which they come.

Definition D.1.9. (Double complex) A *bicomplex* (or *double complex*) is an ordered triple $(M^{\bullet,\bullet}, d_{\prime}, d_{\prime\prime})$, where $M^{\bullet,\bullet} = (M^{p,q})$ is a bigraded module, and $d_{\prime}, d_{\prime\prime} : M^{\bullet,\bullet} \to M^{\bullet,\bullet}$ are differentials of $(1,0)$ and $(0,1)$ respectively, such that $d_{\prime}^2 = 0 = d_{\prime\prime}^2$, and with anti-commutativity

$$d_{\prime}^{p,q+1} \circ d_{\prime\prime}^{p,q} + d_{\prime\prime}^{p+1,q} \circ d_{\prime}^{p,q} = 0.$$

Viewing it as

$$\begin{array}{ccc} M^{p,q+1} & \xrightarrow{\quad d_{\prime}^{p,q+1}\quad} & M^{p+1,q+1} \\ {\scriptstyle d_{\prime}^{p,q}}\uparrow & & \uparrow{\scriptstyle d_{\prime\prime}^{p+1,q}} \\ M^{p,q} & \xrightarrow[\quad d_{\prime}^{p,q}\quad]{} & M^{p+1,q}. \end{array}$$

Example D.1.10. Let $M^{\bullet,\bullet} = (M^{p,q})$ be a bigraded module, and assume that there are bidegree maps $d_{\prime} : M^{\bullet,\bullet} \to M^{\bullet,\bullet}$ of degree $(1,0)$ and $d_{\prime\prime} : M^{\bullet,\bullet} \to M^{\bullet,\bullet}$ of degree $(0,1)$, with commutative diagram

$$\begin{array}{ccc} \bullet & \xrightarrow{\quad d_{\prime}\quad} & \bullet \\ {\scriptstyle d_{\prime\prime}}\uparrow & & \uparrow{\scriptstyle d_{\prime\prime}} \\ \bullet & \xrightarrow[\quad d_{\prime}\quad]{} & \bullet. \end{array}$$

Clearly it is not a double complex by the lack of anti-commutativity. But we can make it into a double complex, Define $\Delta_{''}^{p,q} = (-1)^p d_{''}^{p,q}$, obviously $\Delta_{''}^2 = 0$ and

$$\begin{aligned} d_{'}^{p,q+1}\Delta_{''}^{p,q} + \Delta_{''}^{p+1,q} d_{'}^{p,q} &= d_{'}^{p,q+1}(-1)^p d_{''}^{p,q} + (-1)^{p+1} d_{''}^{p+1,q} d_{'}^{p,q} \\ &= 0, \end{aligned}$$

because the diagram commutes. This is called the *sign change*. Therefore $(M^{\bullet,\bullet}, d_{'}, \Delta_{''})$ is a double complex.

Now let us explain the significance of the anti-commutativity equation $d_{'}d_{''} + d_{''}d_{'} = 0$.

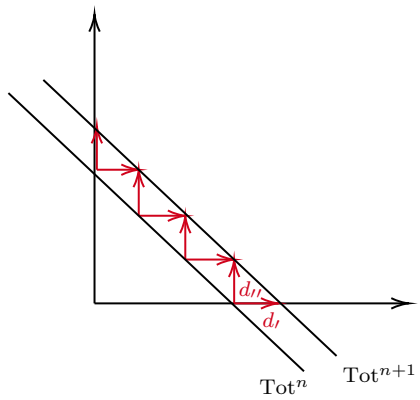
Definition D.1.11. (Total complex) Given a bicomplex $M^{\bullet,\bullet}$, then its *total complex* which denoted by $\mathrm{Tot}(M)$, is the complex with nth term

$$\mathrm{Tot}(M)^n = \bigoplus_{p+q=n} M^{p,q},$$

and with differentials $D^n : \mathrm{Tot}(M)^n \to \mathrm{Tot}(M)^{n+1}$ given by

$$D^n = \sum_{p+q=n} (d_{'}^{p,q} + d_{''}^{p,q}).$$

We can view it as



Lemma D.1.12. *If $M^{\bullet,\bullet}$ is a double complex, then $(\mathrm{Tot}(M), D)$ is a complex.*

Proof. Clearly $\mathrm{im}\, D \subseteq \mathrm{Tot}(M)^{n+1}$ by $\mathrm{im}\, d_{'}^{p,q} \subseteq M^{p+1,q}$ and $\mathrm{im}\, d_{''}^{p,q} \subseteq M^{p,q+1}$ with $p+q+1 = n+1$. Moreover,

$$\begin{aligned} D^2 &= \sum_{p,q} (d_{'} + d_{''})(d_{'} + d_{''}) \\ &= \sum d_{'}d_{'} + \sum (d_{'}d_{''} + d_{''}d_{'}) + \sum d_{''}d_{''} \\ &= 0, \end{aligned}$$

which the middle term is 0 by anti-commutivity. □

Example D.1.13. (i) Let

$$A_\bullet = \cdots \to A_p \overset{\Delta'}{\to} A_{p-1} \overset{\Delta'}{\to} A_{p-2} \to \cdots \to A_0 \to 0$$

and

$$B_\bullet = \cdots \to B_q \overset{\Delta''}{\to} B_{q-1} \overset{\Delta''}{\to} B_{q-2} \to \cdots \to B_0 \to 0$$

be *positive complexes* of R-modules, that is $A_p = \{0\}$ and $B_q = \{0\}$ for $p < 0$ and $q < 0$. Define $(M^{\bullet,\bullet}, d', d'')$ by

$$M^{p,q} = A_p \underset{R}{\otimes} B_q \ , \ d'_{p,q} = \Delta'_p \otimes id_B \ , \ d''_{p,q} = (-1)^p id_A \otimes \Delta''_q.$$

This bicomplex is concentrated in the first quadrant.

Definition D.1.14. (First quadrant bicomplex) A *first quadrant bicomplex* is a bi-complex $M^{\bullet,\bullet}$ (resp. $M_{\bullet,\bullet}$) for which $M^{p,q} = \{0\}$ (resp. $M_{p,q} = \{0\}$) whenever p or q is negative.

And the total complex here is called the *tensor product of complexes*, denoted by $\mathrm{Tot}(M) = A_\bullet \underset{R}{\otimes} B_\bullet$, with

$$(A_\bullet \underset{R}{\otimes} B_\bullet)_n = \bigoplus_{p+q=n} A_p \underset{R}{\otimes} B_q,$$

and $D_n = \Delta'_p \otimes id_B + (-1)^p id_A \otimes \Delta''_q$.

(ii) Let

$$I_A = 0 \to I^0 \to \cdots \to I^p \overset{\Delta'_p}{\to} I^{p+1} \to \cdots$$
$$J_B = 0 \to J^0 \to \cdots \to J^q \overset{\Delta''_q}{\to} J^{q+1} \to \cdots$$

be deleted injective resolutions. A special double complex is $I_A \otimes J_B$.

D.2 Filtrations and Exact Couples

Definition D.2.1. (Filtration) A *filtration* of a module M is a family $(M^p)_{p\in\mathbb{Z}}$ of submodules of M such that

$$\cdots \subseteq M^{p+1} \subseteq M^p \subseteq M^{p-1} \subseteq \cdots .$$

The *factor modules* of this filtration from the graded module $(M^p/M^{p+1})_{p\in\mathbb{Z}}$.

Similarly, in any abelian category one can define a filtration. For example for cochain complex $C^\bullet$, its filtration is a family of subcomplexes $(F^pC^\bullet)_{p\in\mathbb{Z}}$ with

$$\cdots \subseteq F^{p+1}C^\bullet \subseteq F^pC^\bullet \subseteq F^{p-1}C^\bullet \subseteq \cdots$$

And the factors are $(F^pC^\bullet/F^{p+1}C^\bullet)_{p\in\mathbb{Z}}$. In more detail, a filtration of a complex $C^\bullet$ is a commutative diagram such that, for each n, the n-th column is a filtration of C^n.

$$\begin{array}{ccccccc}
\longrightarrow & C^{n-1} & \longrightarrow & C^n & \longrightarrow & C^{n+1} & \longrightarrow \\
 & \uparrow & & \uparrow & & \uparrow & \\
\longrightarrow & F^pC^{n-1} & \longrightarrow & F^pC^n & \longrightarrow & F^pC^{n+1} & \longrightarrow \\
 & \uparrow & & \uparrow & & \uparrow & \\
\longrightarrow & F^{p+1}C^{n-1} & \longrightarrow & F^{p+1}C^n & \longrightarrow & F^{p+1}C^{n+1} & \longrightarrow
\end{array}$$

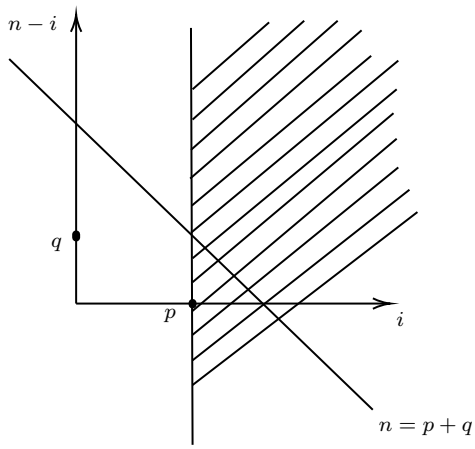
Filtrations may have only finitely many terms, like composition series of finite dimensional vector spaces. But moreover, the "end point" of a filtration of M need not be $\{0\}$ or M, that is, neither $\{0\}$ or M must be equal to M^p for some $p \in \mathbb{Z}$.

Now we give two very important filtrations of the total complex. For $(M, d_{\prime}, d_{\prime\prime})$ be a double cochain complex.

Definition D.2.2. (First filtration) The *first filtration* of $\mathrm{Tot}(M)$ is given by

$$\begin{aligned}
({}_{\mathrm{I}}F^p\,\mathrm{Tot}(M))^n &= \bigoplus_{i\geq p} M^{i,n-i} \\
&= \cdots \oplus M^{p+2,q-2} \oplus M^{p+1,q-1} \oplus M^{p,q},
\end{aligned}$$

for $p + q = n$.



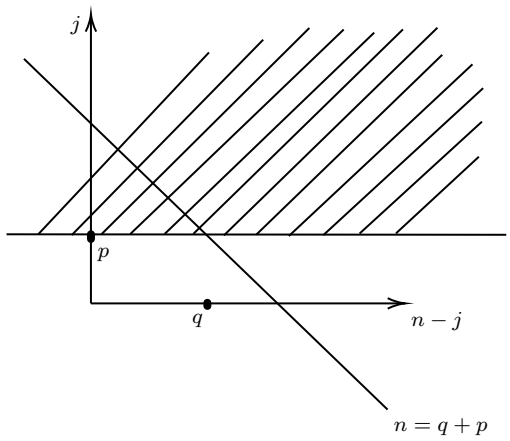
Let us check that ${}_{\mathrm{I}}F^p\,\mathrm{Tot}(M)$ is a subcomplex of $\mathrm{Tot}(M)$.

$$\begin{aligned} D^{i,n-i}M^{i,n-i} = (d_{\prime}^{i,n-i} + d_{\prime\prime}^{i,n-i})M^{i,n-i} &\subseteq d_{\prime}M^{i,n-i} + d_{\prime\prime}M^{i,n-i} \\ &\subseteq M^{i+1,n-i} \oplus M^{i,n-i+1} \\ &\subseteq ({}_{\mathrm{I}}F^p\,\mathrm{Tot}(M))^{n+1}. \end{aligned}$$

Definition D.2.3. (Second filtration) The *second filtration* of $\mathrm{Tot}(M)$ is given by

$$\begin{aligned} ({}_{\mathrm{II}}F^p\,\mathrm{Tot}(M))^n &= \bigoplus_{j\geq p} M^{n-j,j} \\ &= \cdots \oplus M^{q-2,p+2} \oplus M^{q-1,p+1} \oplus M^{q,p}, \end{aligned}$$

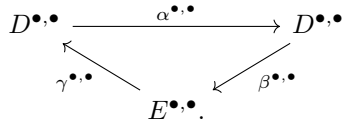
for $q + p = n$.



Let us check ${}_{\mathrm{II}}F^p\,\mathrm{Tot}(M)$ is a subcomplex of $\mathrm{Tot}(M)$.

$$\begin{aligned} D^{n-j,j}M^{n-j,j} = (d_{\prime}^{n-j,j} + d_{\prime\prime}^{n-j,j})M^{n-j,j} &\subseteq d_{\prime}M^{n-j,j} + d_{\prime\prime}M^{n-j,j} \\ &\subseteq M^{n-j+1,j} \oplus M^{n-j,j+1} \\ &\subseteq ({}_{\mathrm{II}}F^p\,\mathrm{Tot}(M))^{n+1}. \end{aligned}$$

Definition D.2.4. (Exact couple) An *exact couple* is a 5-tuple $(D, E; \alpha, \beta, \gamma)$ where $D^{\bullet,\bullet}$ and $E^{\bullet,\bullet}$ are bigraded modules, and α, β, γ are bigraded maps, such that the triangle is exact at each vertex



Proposition D.2.5. (Filtration determines exact couple) *Every filtration $(F^pC^\bullet)_{p\in\mathbb{Z}}$ of a cochain complex $C^\bullet$ determines an exact couple*

$$\begin{array}{ccc} D & \xrightarrow{\alpha \quad (1,-1)} & D \\ & {\scriptstyle\gamma \quad (1,0)} \nwarrow \quad \swarrow {\scriptstyle \beta \quad (0,0)} & \\ & E. & \end{array}$$

Proof. For each fixed p, there is a short exact sequence of complexes

$$0 \to F^{p+1}C^\bullet \overset{j^{p+1}}{\hookrightarrow} F^pC^\bullet \overset{\nu^p}{\twoheadrightarrow} F^pC^\bullet/F^{p+1}C^\bullet \to 0,$$

this gives rise to the long exact sequence

$$\cdots \to H^n(F^{p+1}C^\bullet) \xrightarrow{\alpha} H^n(F^pC^\bullet) \xrightarrow{\beta} H^n(F^pC^\bullet/F^{p+1}C^\bullet) \xrightarrow{\gamma}$$
$$H^{n+1}(F^{p+1}C^\bullet) \xrightarrow{\alpha} H^{n+1}(F^pC^\bullet) \xrightarrow{\beta} H^{n+1}(F^pC^\bullet/F^{p+1}C^\bullet) \xrightarrow{\gamma} \cdots,$$

where $\alpha = H(j^{p+1}), \beta = H(\nu^p)$ and $\gamma = \delta$ the connecting homomorphism.

Introduce q by writing $n = p + q$, we have

$$\cdots \to H^{p+q}(F^{p+1}C^\bullet) \xrightarrow{\alpha} H^{p+q}(F^pC^\bullet) \xrightarrow{\beta} H^{p+q}(F^pC^\bullet/F^{p+1}C^\bullet) \xrightarrow{\gamma}$$
$$H^{p+q+1}(F^{p+1}C^\bullet) \xrightarrow{\alpha} H^{p+q+1}(F^pC^\bullet) \xrightarrow{\beta} H^{p+q+1}(F^pC^\bullet/F^{p+1}C^\bullet) \xrightarrow{\gamma} \cdots.$$

So we define

$$D^{\bullet,\bullet} = (D^{p,q}) \text{ where } D^{p,q} = H^{p+q}(F^pC^\bullet),$$
$$E^{\bullet,\bullet} = (E^{p,q}) \text{ where } E^{p,q} = H^{p+q}(F^pC^\bullet/F^{p+1}C^\bullet),$$

which we notice that $H^{p+q}(F^{p+1}C^\bullet) = H^{p+1+(q-1)}(F^{p+1}C^\bullet) = D^{p+1,q-1}$. Thus with this notation, for each fixed $p \in \mathbb{Z}$ the long exact sequence is

$$\cdots \to D^{p+1,q-1} \underset{(-1,1)}{\xrightarrow{\alpha}} D^{p,q} \underset{(0,0)}{\xrightarrow{\beta}} E^{p,q} \underset{(1,0)}{\xrightarrow{\gamma}} D^{p+1,q} \xrightarrow{\alpha} D^{p,q+1} \xrightarrow{\beta} E^{p,q+1} \to \cdots$$

Therefore, $(D, E; \alpha, \beta, \gamma)$ is an exact couple with the displayed bidegree. □

Every exact couple determines another exact couple.

Definition D.2.6. (Differential graded module) A *differential bigraded module* is an ordered pair (M, d), where $M^{\bullet,\bullet}$ is bigraded and $d^{\bullet,\bullet} : M^{\bullet,\bullet} \to M^{\bullet,\bullet}$ is a bigraded map with $d^2 = 0$.

Given (M, d) which $\deg d = (a, b)$, then its cohomology $H(M, d)^{\bullet,\bullet}$ is a bigraded module whose p, q term is

$$H(M, d)^{p,q} = \frac{\ker d^{p,q}}{\operatorname{im} d^{p-a,q-b}}.$$

A double complex $(M, d_{\prime}, d_{\prime\prime})$ gives rise to two differential bigraded modules, namely $(M, d_{\prime})$ and $(M, d_{\prime\prime})$. But notice that $(M, d_{\prime} + d_{\prime\prime})$ is not a differential bigraded module, i.e. $d_{\prime} + d_{\prime\prime} : M^{\bullet,\bullet} \to M^{\bullet,\bullet}$ is not a bigraded map.

Proposition D.2.7. (Derived couple) *If $(D, E; \alpha, \beta, \gamma)$ is an exact couple*

$$\begin{array}{ccc} D & \xrightarrow{\alpha} & D \\ & {\scriptstyle\gamma}\nwarrow \quad \swarrow{\scriptstyle\beta} & \\ & E. & \end{array}$$

Then $d_1 = \beta\gamma$ is a differential $d_1 : E^{\bullet,\bullet} \to E^{\bullet,\bullet}$, and there is an exact couple $(D_2, E_2; \alpha_2, \beta_2, \gamma_2)$ called the derived couple, with $E_2 = H(E, d_1)$.

$$\begin{array}{ccc} D_2 & \xrightarrow{\alpha_2} & D_2 \\ & {\scriptstyle\gamma_2}\nwarrow \quad \swarrow{\scriptstyle\beta_2} & \\ & E_2. & \end{array}$$

Proof. We will give a concrete construction. Let α, β, γ have respective bidegree $(-1, 1), (0, 0)$ and $(1, 0)$, while we always write $p + q = n$.

Step 1: Define the differential $d_1^{\bullet,\bullet} : E^{\bullet,\bullet} \to E^{\bullet,\bullet}$ by $d_1 = \beta\gamma$. Note that for

$$\begin{array}{ccccccccccccc} \cdots \to & D^{p+1,q-1} & \xrightarrow{\alpha} & D^{p,q} & \xrightarrow{g} & E^{p,q} & \xrightarrow{\gamma} & D^{p+1,q} & \xrightarrow{\alpha} & D^{p,q+1} & \xrightarrow{\beta} & E^{p,q+1} & \longrightarrow \cdots \\ & & & & & & x \longmapsto & \gamma^{p,q}(x) & & & & & & \end{array}$$

$$\begin{array}{ccccccccccc} \cdots \to & D^{p+1,q-1} & \xrightarrow{\beta} & E^{p+1,q-1} & \xrightarrow{\gamma} & D^{p+2,q-1} & \xrightarrow{\alpha} & D^{p+1,q} & \xrightarrow{\beta} & E^{p+1,q} & \longrightarrow \cdots \\ & & & & & & & \gamma(x) & \longmapsto & \beta^{p+1,q}(\gamma(x)) & \end{array}$$

the composition $\beta\gamma$ is in the "wrong order", but makes sense:

$$\begin{array}{ccccc} E^{p,q} & \longrightarrow & E^{p+1,q} & \longrightarrow & E^{p+1,q} \\ x & \longmapsto & \gamma(x) & \longmapsto & \beta(\gamma(x)) \end{array}$$

i.e. $d_1 = (d_1^{p,q} := \beta^{p+1,q} \circ \gamma^{p,q})$. Moreover,

$$\begin{aligned} d_1^{p+1,q} d_1^{p,q} &= (\beta^{p+2,1} \circ \gamma^{p+1,q}) \circ (\beta^{p+1,q} \circ \gamma^{p,q}) \\ &= \beta^{p+2,q} \circ (\gamma^{p+1,q} \circ \beta^{p+1,q}) \circ \gamma^{p,q} \\ &= 0, \end{aligned}$$

so it is indeed a differential with degree $(1, 0)$.

Step 2: We define $E_2^{\bullet,\bullet} = H(E, d_1)$. Thus, $E_2^{p,q} = \ker d_1^{p,q} / \operatorname{im} d_1^{p,q}$.

Step 3: We define $D_2^{\bullet,\bullet} = \operatorname{im} \alpha \subseteq D$, i.e.

$$D_2 = ((\operatorname{im} \alpha)^{p,q}) = (\operatorname{im} \alpha^{p+1,q-1}) \subseteq (D^{p,q}) = D^{\bullet,\bullet}.$$

And construct the bigraded maps. Define $\alpha_2^{\bullet,\bullet} : D_2 \to D_2$ to be restriction $\alpha|_{D_2}$, i.e. $\alpha_2 = \alpha i$ for $i : \operatorname{im}\alpha \hookrightarrow D$ is the inclusion. So $\deg\alpha_2 = \deg\alpha + \deg i = (-1,1)$, the same bidegree as of α. If $x \in D_2^{p,q}$, then $x = \alpha u$ for some $u \in D^{p+1,q-1}$, and

$$\alpha_2^{p,q} : x = \alpha u \mapsto \alpha i(\alpha u) = \alpha\alpha u.$$

Step 4: We define $\beta_2^{\bullet,\bullet} : D_2 \to E_2$ as follows. Picking $y \in D_2^{p,q}$, then $y = \alpha v$ for some $v \in D^{p+1,q-1}$, and βv is a cycle since $d_1\beta v = \beta\gamma(\beta v) = 0$, which is independent of the choice of $v \in \alpha^{-1}\{y\}$ (see later). So we choose arbitrary representation v in $\alpha^{-1}\{y\}$, and set

$$\beta_2(y) = [\beta\alpha^{-1}y],$$

where $\alpha^{-1}y$ means an arbitrary representation in $\alpha^{-1}\{y\}$.

Now to see β_2 is well-defined, suppose that $\alpha v = y = \alpha v'$, then we have $v - v' \in \ker\alpha = \operatorname{im}\gamma$, such that $v - v' = \gamma w$ for some $w \in E$, and hence

$$\beta(v - v') = \beta\gamma w = d_1 w$$

is a boundary, i.e. $[\beta v] = [\beta v'] = \beta_2(y)$. Note that β_2 has bidegree $(1,-1)$.

Step 5: We now define $\gamma_2^{\bullet,\bullet} : E_2 \to D_2$. Pick $[z] \in E_2^{p,q}$ so that $z \in E^{p,q}$ is a cycle, i.e. $d_1 z = \beta\gamma z = 0$. Hence $\gamma z \in \ker\beta = \operatorname{im}\alpha$, such that $\gamma z \in D_2$ and $\gamma^{p,q} z \in D^{p+1,q}$. We define γ_2 by

$$\begin{aligned} \gamma_2 : E_2 &\longrightarrow E_2 \\ [z] &\longmapsto \gamma z \end{aligned}$$

which is well-defined. Since for a boundary $w \in \operatorname{im} d_1$, we have $w = \beta\gamma x$ and so $\gamma w = \gamma\beta\gamma x = 0$, that is $0 = \gamma_2 0 = \gamma(z - z') = \gamma w = 0$.

Step 6: It remains to prove the exactness. First of all,

$$\begin{aligned} &\beta_2\alpha_2 : x = \alpha u \overset{\alpha_2}{\longmapsto} \alpha\alpha u \overset{\beta_2}{\longmapsto} [\beta\alpha^{-1}(\alpha\alpha u)] = [\beta\alpha u] = 0, \text{ i.e. } \operatorname{im}\alpha_2 \subseteq \ker\beta_2; \\ &\gamma_2\beta_2 : x = \alpha u \overset{\beta_2}{\longmapsto} [\beta\alpha^{-1}\alpha u] = [\beta u] \overset{\gamma_2}{\longmapsto} \gamma\beta u = 0, \text{ i.e. } \operatorname{im}\beta_2 \subseteq \ker\gamma_2; \\ &\alpha_2\gamma_2 : [z] \overset{\gamma_2}{\longmapsto} \gamma z \overset{\alpha_2}{\longmapsto} \alpha\gamma z = 0, \text{ i.e. } \operatorname{im}\gamma_2 \subseteq \ker\alpha_2. \end{aligned}$$

So for reverse inclusions, we verify:

"$\ker\alpha_2 \subseteq \operatorname{im}\gamma_2$": For $x \in \ker\alpha_2$, we have $\alpha_2 x = \alpha x = 0$, so $x \in \ker\alpha = \operatorname{im}\gamma$, i.e. $x = \gamma y$ for some $y \in E$. But as $x \in D_2 = \operatorname{im}\alpha = \ker\beta$, we have

$$0 = \beta x = \beta\gamma y = d_1\gamma.$$

Thus y is a cycle, and $x = \gamma y = \gamma_2[y] \in \operatorname{im} y_2$.

"$\ker \beta_2 \subseteq \operatorname{im} \alpha_2$": For $x \in \ker \beta_2$, then $x \in D_2 = \operatorname{im} \alpha$ and $\beta_2 x = 0$. Thus, we have $x = \alpha u$ for some $u \in D$, and $0 = \beta_2 x = [\beta\alpha^{-1}(\alpha u)] = [\beta u]$, so $\beta u \in \operatorname{im} d_1$, i.e. $\beta u = d_1 w = \beta\gamma w$ for some $w \in E$. Now we have

$$u - \gamma w \in \ker \beta = \operatorname{im} \alpha = D_2,$$

and so $\alpha_2(u - \gamma w) = \alpha u - \alpha\gamma w = \alpha u = x \in \operatorname{im} \alpha_2$.

"$\ker \gamma_2 \subseteq \operatorname{im} \beta_2$": For $[z] \in \ker \gamma_2$, we have $\gamma_2[z] = \gamma z = 0$. Thus $z \in \ker \gamma = \operatorname{im} \beta$, i.e. $z = \beta v$ for some $v \in D$. But then

$$\beta_2(\alpha v) = [\beta\alpha^{-1}\alpha v] = [\beta v] = [z],$$

that is $[z] \in \operatorname{im} \beta_2$. □

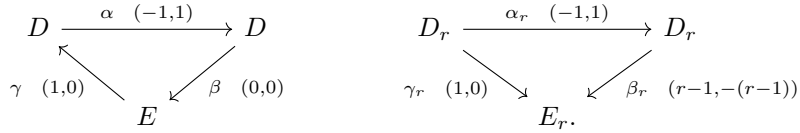
Remark D.2.8. By construction, easy to check that if the maps α, β, γ have bidegrees $(a, a'), (b, b')$ and (c, c') respectively. Thus, only the bidegree of β changes.

Definition D.2.9. (Higher derived couple) The *r-th derived couple* of an exact couple $(D, E; \alpha, \beta, \gamma)$ is constructed inductively: its $(r+1)$st derived couple

$$(D_{r+1}, E_{r+1}; \alpha_{r+1}, \beta_{r+1}, \gamma_{r+1})$$

is the derived couple of $(D_r, E_r; \alpha_r, \beta_r, \gamma_r)$, the r-th derived couple.

Corollary D.2.10. *Let $(D, E; \alpha, \beta, \gamma)$ be the exact couple arising from a filtration $(F^pC^\bullet)$ associated to cochain complex $C^\bullet$:*



Then the r-th derived couple has the following properties:

(i) *$\alpha_r, \beta_r, \gamma_r$ have bidegrees $(-1, 1), (r-1, 1-r), (1, 0)$ respectively;*

(ii) *d_r has bidegree $(r, 1-r)$, which is induced by $\beta\alpha^{-r+1}\gamma$;*

(iii) $E_{r+1}^{p,q} = \ker d_r^{p,q} / \operatorname{im} d_r^{p-r,q+r-1}$;

(iv) $D_r^{p,q} = \operatorname{im}(\alpha^{p+1,q-1})(\alpha^{p+2,q-2}) \cdots (\alpha^{p+r-1,q-r+1})$.

Definition D.2.11. (Spectral sequence) A *spectral sequence* is a sequence $(E_r, d_r)_{r \geq 1}$ of differential bigraded modules such that $E_{r+1} = H(E_r, d_r)$ for all r.

Easily to get:

Theorem D.2.12. *Every filtration of a cochain complex yields a spectral sequence as described above.*

D.3 Convergence

Definition D.3.1. (Subquotient) For M a module, a *subquotient* of M is a module of the form M'/M'', where $M'' \subseteq M' \subseteq M$.

Example D.3.2. (i) If M has a filtration (F^pM), then each factor module $F^pM/F^{p+1}M$ is a subquotient of M.

(ii) The n-th cohomology $H^n(C^\bullet)$ of a cochain complex $C^\bullet$ is a subquotient of C^n : $\operatorname{im} d^{n-1} \subseteq \ker d^n \subseteq C^n$.

(iii) If $\{E_r, d_r\}$ is a spectral sequence, then $E_2 = H^\bullet(E_1, d_1)$ is a subquotient of E_1, since $E_2 = Z_2/B_2 =$ cycles/boundaries, where

$$B_2 \subseteq Z_2 \subseteq E_1.$$

Moreover, every submodule of E_2 is isomorphic to S/B_2 for a unique submodule $S \subseteq E_1$ with $B_2 \subseteq S$. In particular, the derived cycle Z_3 and boundaries B_3 can be regarded as quotients $B_3/B_2 \subseteq Z_3/B_2 \subseteq Z_2/B_2 = E_2$, so that up to isomorphism

$$B_2 \subseteq B_3 \subseteq Z_3 \subseteq Z_2 \subseteq E_1,$$

that is E_3 is a subquotient of E_2. Furthermore, for every $r \geq 1$, there is a chain

$$B_2 \subseteq \cdots \subseteq B_r \subseteq Z_r \subseteq \cdots \subseteq Z_2 \subseteq E_1,$$

i.e. E_r is a subquotient of E_{r-1}.

Definition D.3.3. (Limit of spectral sequence) For a given spectral sequence $\{E_r, d_r\}_{r\geq 1}$, define $Z_\infty = \bigcap_r Z_r$ and $B_\infty = \bigcup_r B_r$ (an ascending union of submodules is a submodule). Then $B_\infty \subseteq Z_\infty$, and the *limit term* of the spectral sequence is the bigraded module E_∞ defined by

$$E_\infty^{p,q} = Z_\infty^{p,q}/B_\infty^{p,q}.$$

Lemma D.3.4. *Let $\{E_r, d_r\}$ be a spectral sequence.*

(i) $E_{r+1} = E_r$ if and only if $Z_{r+1} = Z_r$ and $B_{r+1} = B_r$.

(ii) If $E_{r+1} = E_r$ for all $r \geq s$, then $E_s = E_\infty$.

Proof. (i) If X/Y is a subquotient of Z, i.e. $Y \subseteq X \subseteq Z$, then $X/Y = Z$ if and only if $Y = \{0\}$ and $X = Z$. So for $E_{r+1} = Z_{r+1}/B_{r+1} = E_r$, and for

$$B_{r+1} \subseteq Z_{r+1} \subseteq E_r$$

we have $B_{r+1} = \{0\}$ in $E_r = Z_r/B_r$, i.e. $B_{r+1} = B_r$. Hence $E_{r+1} = Z_{r+1}/B_r = E_r = Z_r/B_r$ implies also that $Z_{r+1} = Z_r$.

The converse is trivial.

(ii) If $E_r = E_{r+1}$ for some $r \geq s$, then $Z_s = Z_r$ for all $r \geq s$ via (i), hence $Z_s \bigcap\limits_{r\geq s} Z_r = Z_\infty$. Also by (i), $B_s = B_r$ for all $r \geq s$, hence $B_s = \bigcup\limits_{r\geq s} B_r = B_\infty$. Therefore

$$E_s = Z_s/B_s = Z_\infty/B_\infty = E_\infty.$$

□

Now for a given filtration $(F^pC^\bullet)_p$ of a cochain complex $C^\bullet$ with inclusions $i^p : F^pC^\bullet \hookrightarrow C^\bullet$, then we get $H(i^p) : H^\bullet(F^pC^\bullet) \to H^\bullet(C^\bullet)$. And since $F^pC^\bullet \subseteq F^{p-1}C^\bullet$, we have $\operatorname{im} H(i^p) \subseteq \operatorname{im} H(i^{p-1})$:

$$H^\bullet(F^pC^\bullet) \xrightarrow{\quad} H^\bullet(F^{p-1}C^\bullet) \xrightarrow{H(i^{p-1})} H^\bullet(C^\bullet)$$
$$\underbrace{\qquad\qquad\qquad\qquad\qquad\qquad}_{H(i^p)}$$

that is, $\operatorname{im} H(i^p)$ is a filtration of $H^\bullet(C^\bullet)$.

Definition D.3.5. (Induced filtration) If $(F^pC^\bullet)$ is a filtration of a cochain complex $C^\bullet$ and $\{i^p : F^pC^\bullet \to C^\bullet\}$ are inclusions, define

$$\Phi^pH^n(C^\bullet) = \operatorname{im} H^n(i^p).$$

We call $(\Phi^pH^n(C^\bullet))_p$ the *induced filtration* of $H^n(C^\bullet)$.

For a given filtration of $C^\bullet$, the induced filtration $(\Phi^pH^n(C^\bullet))_{p,n}$ is obviously a filtration on the cohomology $H^\bullet(C^\bullet)$.

$$\begin{array}{ccc}
H^{n-1}(C^\bullet) & H^n(C^\bullet) & H^{n+1}(C^\bullet) \\
\uparrow & \uparrow & \uparrow \\
\Phi^pH^{n-1}(C^\bullet) & \Phi^pH^n(C^\bullet) & \Phi^pH^{n+1}(C^\bullet) \\
\uparrow & \uparrow & \uparrow \\
\Phi^{p+1}H^{n-1}(C^\bullet) & \Phi^{p+1}H^n(C^\bullet) & \Phi^{p+1}H^{n+1}(C^\bullet)
\end{array}$$

The spectral sequence $\{E_r, d_r\}$ of a filtration of $C^\bullet$ obviously determines E_∞, thus, if

$$E_\infty^{p,q} \cong \Phi^pH^{p+q}(C^\bullet)/\Phi^{p-1}H^{p+q}(C^\bullet)$$

for all p, q, then the spectral sequence determines the factor module of $H^\bullet(C^\bullet)$.

Definition D.3.6. (Bounded filtration) A filtration $(F^pM^\bullet)$ of a graded module $M^\bullet = (M^n)$ is *bounded*, if for each $n \in \mathbb{Z}$, there exist integers $s = s(n)$ and $t = t(n)$ such that

$$F^sM^n = \{0\} \text{ and } F^tM^n = M^n,$$

that is

$$\cdots = 0 = \cdots = 0 = F^sM^n \subseteq F^{s-1}M^n \subseteq \cdots \subseteq F^{t-1}M^n \subseteq F^tM^n = M^n.$$

Example D.3.7. (Bounded filtration induced bounded cohomology) If $(F^pC^\bullet)_p$ is a bounded filtration for cochain complex $C^\bullet$, then the induced filtration on chomology is also bounded and with same bounds.

Proof. For $i^p : F^pC^\bullet \hookrightarrow C^\bullet$ the inclusion, then $\Phi^pH^n(C^\bullet) = \operatorname{im} H^n(i^p)$. Since $(F^pC^\bullet)_p$ is bounded, there are integers $s = s(n)$ and $t = t(n)$ with $F^sC^\bullet = 0$ and $F^tC^\bullet = C^\bullet$, so we have for each fixed n,

$$\begin{aligned}\Phi^pH^n(C^\bullet) &= \operatorname{im}(H^n(i^p) : H^n(F^pC^\bullet) \to H^n(C^\bullet)) \\ &= \begin{cases} 0 & \text{if } p = s \\ H^n(C^\bullet) & \text{if } p = t \end{cases}\end{aligned}$$

Thus for each fixed n, there is a finite ascending chain

$$\{0\} = \Phi^sH^n(C^\bullet) \subseteq \Phi^{s-1}H^n(C^\bullet) \subseteq \cdots \subseteq \Phi^{t+1}H^n(C^\bullet) \subseteq \Phi^tH^n(C^\bullet) = H^n(C^\bullet).$$

Of course, $\Phi^iH^n(C^\bullet) = \{0\}$ for all $i \geq s$, and $\Phi^jH^n(C^\bullet) = H^n(C^\bullet)$ for $j \leq t$. □

Definition D.3.8. (Convergent spectral sequence) A spectral sequence $\{E_r, d_r\}_{r\geq 1}$ is said to *converges* to a graded module $H^\bullet$, denoted by

$$E_2^{p,q} \underset{p}{\Rightarrow} H^n$$

if there is some bounded filtration (Φ^pH^n) of $H^\bullet$ with

$$E_\infty^{p,q} \cong \Phi^pH^n/\Phi^{p+1}H^n$$

for all n (with notation $n = p + q$).

Here the superscript p is called the *filtration degree*, and notice that in notation $E_2^{p,q} \underset{p}{\Rightarrow} H^n = H^{p+q}$, we have $E_2^{p,q}$ may **not** be isomorphic to $E_2^{q,p}$, for example in algebraic geometry the p-th cohomologies of q-th higher direct image sheaves may not be equal with the indexes reversed.

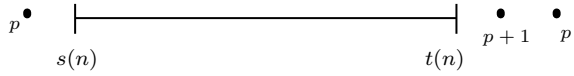
Theorem D.3.9. *Let $(F^pC^\bullet)_p$ be a bounded filtration of a cochain complex $C^\bullet$, and let $\{E_r, d_r\}_{r\geq 1}$ be the spectral sequence of (D.2.12), by the way so that the induced filtration $(\Phi^p H^n(C^\bullet))_{p,n}$ is bounded with the same bounds $s(n)$ and $t(n)$ as $(F^pC^\bullet)_p$.*

Then

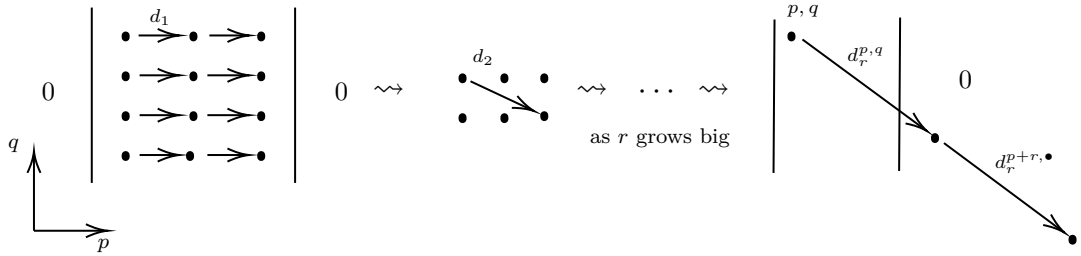
(i) for each p, q we have $E_\infty^{p,q} = E_r^{p,q}$ for large r which depending on p and q.

(ii) $E_2^{p,q} \underset{p}{\Rightarrow} H^n(C^\bullet) = H^{p+q}(C^\bullet)$.

Proof. (i) For p "small" enough such that $p < t(n)$, then $F^{p+1}C^\bullet = F^pC^\bullet$ and therefore $F^pC^\bullet/F^{p+1}C^\bullet = \{0\}$. By definition, $E^{p,q} = H^{p+q}(F^pC^\bullet/F^{p+1}C^\bullet)$ in (D.2.5), so $E^{p,q} = \{0\}$. Since $E_r^{p,q}$ is a subquotient of $E^{p,q}$, we have $E_r^{p,q} = \{0\}$ for all $r \geq 1$ under this choice of p. Similarly for p "large" enough to be $p > s(n)$, then $F^pC^\bullet = \{0\}$ and $E_r^{p,q} = \{0\}$ for all $r \geq 1$.



Now the differential d_r has bidegree $(r, 1-r)$ by (D.2.10). Focus on first superscript, for any fixed bi-integer (p, q), we have $d_r(E_r^{p,q}) \subseteq E_r^{p+r,\bullet}$. So for r grows big, the first index $p + r$ also goes big such that $E_r^{p+r,\bullet} = \{0\}$. Hence $\ker d_r^{p,q} = E_r^{p,q}$.



Let us compute $E_{r+1}^{p,q} = \ker d_r^{p,q}/\operatorname{im} d_r^{p+r,\bullet}$. Now, as r grows big, $E_r^{p+r,\bullet} = \{0\}$ so $\operatorname{im} d_r^{p+r,\bullet} = \{0\}$. Therefore, $E_{r+1}^{p,q} = \ker d_r^{p,q}/0 = \ker d_r^{p,q} = E_r^{p,q}$ for large enough r which depending on (p, q). Thus, the (p, q) term is constant for large r, which says that $E_\infty^{p,q} = E_r^{p,q}$ by (D.3.4).

(ii) Continue focusing on the first index in the superscript p. Consider the exact sequence obtained from the r-th derived couple:

$$D_r^{p-r+2,\bullet} \xrightarrow{\alpha_r} D_r^{p-r+1,\bullet} \xrightarrow{\beta_r} E_r^{p,q} \xrightarrow{\gamma_r} D_r^{p+1,q}.$$

These indices arise from the bidegrees in (D.2.10). Also as in (D.2.10)(iv), the module

$$\begin{aligned} D_r^{p,q} &= \operatorname{im}(\alpha^{p+1,\bullet}\alpha^{p+2,\bullet}\cdots\alpha^{p+r-1,\bullet}) \\ &= \operatorname{im}(H(j^{p+1})H(j^{p+2})\cdots H(j^{p+r-1})) : H^n(F^{p+r-1}C^\bullet) \to H^n(F^pC^\bullet), \end{aligned}$$

as the notation in (D.2.5). Replacing p first by $p-r+1$ and then by $p-r+2$, we have

$$D_r^{p-r+1,\bullet} = \mathrm{im}(H^n(j^{p-r+2})\cdots H^n(j^p)) : H^n(F^pC^\bullet) \to H^n(F^{p-r+1}C^\bullet)$$

and

$$D_r^{p-r+2,\bullet} = \mathrm{im}(H^n(j^{p-r+3})\cdots H^n(j^{p+1})) : H^n(F^{p+1}C^\bullet) \to H^n(F^{p-r+2}C^\bullet).$$

For large n, we have $F^{p-r+1}C^\bullet = F^tC^\bullet = C^\bullet$, and the composition $j^{p-r+2}\cdots j^p$ of inclusions are just nothing but the inclusion $i^p : F^pC^\bullet \hookrightarrow F^{p-r+1}C^\bullet = C^\bullet$. Therefore, $D_r^{p-r+1,\bullet} = \mathrm{im}\, H^n(i^p) = \Phi^pH^n(C^\bullet)$. Similarly, $D_r^{p-r+2,\bullet} = \mathrm{im}\, H^n(i^{p+1}) = \Phi^{p+1}H^n(C^\bullet)$ for large r. So, when r grows big we may rewrite above exact sequence as

$$\Phi^{p+1}H^n(C^\bullet) \xrightarrow{\alpha_r} \Phi^pH^n(C^\bullet) \xrightarrow{\beta_r} E_r^{p,q} \longrightarrow D_r^{p+1,q}$$

for large enough$r$ with $n = p+q$. Here α_r becomes an inclusion. Moreover if we prove $D_r^{p,q} = \{0\}$ for large r, then by exactness

$$\Phi^pH^n(C^\bullet)/\Phi^{p+1}H^n(C^\bullet) \cong E_r^{p,q} = E_\infty^{p,q},$$

that is $E_2^{p,q} \underset{p}{\Rightarrow} H^n(C^\bullet)$. But this is obviously since

$$D_r^{p+1,q} = \mathrm{im}(H^n(F^{p+r}C^\bullet) \to H^n(F^{p+1}C^\bullet)),$$

which is $\{0\}$ because $F^{p+r}C^\bullet = \{0\}$ for large r. □

D.4 Cartan-Eilenberg Resolution

For a given abelian category $\mathcal{A}$ with enough injectives (or enough projectives), then its associated category of all (co)chain complexes over $\mathcal{A}$ is an abelian category with enough injectives (or projectives). For example, sheaves of $\mathscr{O}_X$-modules over a ringed space X is an abelian category with enough injectives (but without enough projectives, see [Har77, III, ex6.2]), so its associated category $\mathbf{Comp}(\mathscr{O}_X\text{-Mod})$ is an abelian category with enough injectives. In particular we're mainly interested in $\mathrm{Coh}(X)$, the category of coherents sheaves of $\mathscr{O}_X$-modules over a noetherian scheme X.

A *Cartan-Eilenberg resolution* is a special kind of injective (or projective) resolution of a complex in $\mathbf{Comp}(\mathcal{A})$.

Definition D.4.1. (Fundamental exact sequence) Let $\mathcal{A}$ be an abelian category, and $(C^\bullet, d)$ be a cochain complex in $\mathcal{A}$. The *fundamental exact sequences* of $C^\bullet$ are, for all $n \in \mathbb{Z}$:

$$0 \longrightarrow Z^n \longrightarrow C^n \xrightarrow{d'} B^{n+1} \longrightarrow 0$$
$$0 \longrightarrow B^n \longrightarrow Z^n \longrightarrow H^n(C^\bullet) \to 0,$$

where the morphism $d' : C^n \to B^{n+1} \subseteq C^{n+1}$ is obtained from d, i.e. $d = id'$ for $i : B^\bullet \hookrightarrow C^\bullet$ the inclusions. We can do this since in abelian category we always have **epi-monic factorization**.

Definition D.4.2. (Split complex) A cochain complex $(C^\bullet, d)$ in $\mathbf{Comp}(\mathcal{A})$ is said to be *split* if all its fundamental exact sequences are split.

Behold the difference with the *split exact sequence*.

Lemma D.4.3. *$\mathcal{A}$ is an abelian category, then for any elements $A, B \in \mathrm{Ob}(\mathcal{A})$ every morphism $f : A \to B$ is a differential.*

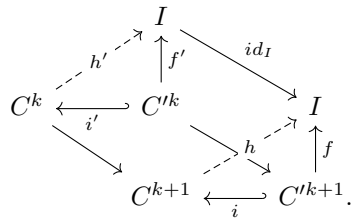
Proof. Consider $\Sigma^k(f) \in \mathbf{Comp}(\mathcal{A})$ whose k-th term is A and $(k+1)$-th term is B, while others are 0:

$$\Sigma^k(f) = \cdots \to 0 \to 0 \to A \xrightarrow{f} B \to 0 \to 0 \to \cdots$$

We call $\Sigma^k(f)$ is the complex having *f concentrated in degree $(k, k+1)$*. □

Lemma D.4.4. *If I is an injective object in an abelian category $\mathcal{A}$ and for $k \in \mathbb{Z}$, then $\Sigma^k(id_I)$ is an injective object in $\boldsymbol{Comp}(\mathcal{A})$.*

Proof. Consider the diagram in $\mathbf{Comp}(\mathcal{A})$



Here i and i' are parts of an monic cochain map $C'^\bullet \to C^\bullet$, so that each of them is monic in $\mathcal{A}$. Since I is injective, there exist $h : C^k \to I$ such that $hi = f$. Define $h' : C^{k+1} \to I$ by $h' = hd$. All the faces of the prism are commutative, with the possible exception of the triangle on the left. In particular, $(h', h) : C^\bullet \to \Sigma^k(id_I)$ is a cochain map. It remains to prove that $h'i' = f'$. But $h'i' = hdi' = hid' = fd' = f'$. □

Proposition D.4.5. *If $(C^\bullet, d)$ is split, then*

$$C^\bullet \cong \bigoplus_{k\in\mathbb{Z}} \Sigma^k(\delta^k)$$

where $\delta^k : B^k(C^\bullet) \hookrightarrow Z^k(C^\bullet)$ is the inclusion, and each $\Sigma^k(\delta^k)$ is the cochain complex having δ^k concentrated in degrees $(k, k+1)$.

Proof. By Freyd-Mitchell embedding we may assme $\mathcal{A}$ is R-modules, but notice here R may not necessarily commutative which just an associative ring.

Write $B^k = B^k(C^\bullet)$ and $Z^k = Z^k(C^\bullet)$. Since $(C^\bullet, d)$ is split, there are morphisms

$$0 \longrightarrow Z^k \underset{q}{\overset{i}{\rightleftarrows}} C^k \underset{j}{\overset{p}{\rightleftarrows}} B^{k+1} \longrightarrow 0$$

where p is merely d which is surjective, and

$$qi = id_Z \ , \ pj = id_B \ , \ pi = 0 \text{ and } iq + jp = id_C.$$

Now we have

$$\begin{aligned} \Sigma^k(i) &: 0 \to Z^k \to C^k \cong Z^k \oplus B^{k+1} \to 0, \\ \Sigma^k(id_C) &: 0 \to C^k \to C^k \to 0, \end{aligned}$$

we see $\Sigma^k(i)$ is a direct summand of $\Sigma^k(id_C)$ for all $k \in \mathbb{Z}$.

The n-th term of $\bigoplus_{k\in\mathbb{Z}} \Sigma^k(\delta^k)$ is $Z^n \oplus B^{n+1} \cong C^n$, and the n-th differential is

$$\begin{aligned} \Delta^n : Z^n \oplus B^{n+1} &\longrightarrow Z^{n+1} \oplus B^{n+2} \\ (z_n, b_{n+1}) &\longmapsto (\delta^{n+1} b_{n+1}, 0). \end{aligned}$$

Define $\theta^n : C^n \to Z^n \oplus B^{n+1}$ by $c_n \mapsto (qc_n, pc_n)$ which is an isomorphism, so trivially a cochain map. So by commutative diagram

$$\begin{array}{ccc} C^n & \xrightarrow{\quad d \quad} & C^{n+1} \\ \downarrow{\scriptstyle\theta} & & \downarrow{\scriptstyle\theta} \\ Z^n \oplus B^{n+1} & \xrightarrow[\Delta]{} & Z^{n+1} \oplus B^{n+2} \end{array}$$

we see Δ is just d. □

Corollary D.4.6. *If $\mathcal{A}$ is an abelian category and $C^\bullet$ is split complex with each terms is an injective object in $\mathcal{A}$, then $C^\bullet$ itself is an injective object in $\boldsymbol{Comp}(\mathcal{A})$.*

Proof. For $C^\bullet \cong \bigoplus_{k\in\mathbb{Z}} \Sigma^k(\delta^k)$ for $\delta^k : B^k \to Z^k$ are the inclusions. Since $C^k$are injective, so $\Sigma^k(id_C)$ is an injective object in $\mathbf{Comp}(\mathcal{A})$ by (D.4.4), moreover its direct summand $\Sigma^k(i^k)$ is also an injective object in $\mathbf{Comp}(\mathcal{A})$.

Similarly, the splitting of $0 \to B^k \xrightarrow{\delta^k} Z^k \to H^k \to 0$ gives $\Sigma^k(\delta^k)$ is a direct summand of $\Sigma^k(id_Z)$, while the latter is a direct summand of $\Sigma^k(i^k)$, this implies the injectivity of $\Sigma^k(\delta^k)$, hence so is $C^\bullet$ by it being a direct sum of injectives. □

Now we are going to prove the converse of (D.4.6).

Definition D.4.7. (Shifted complex) If $p \in \mathbb{Z}$ and $(C^\bullet, d)$ is a cochain complex, we define $C^\bullet[p]$ to be the *shifted complex* whose indices have been shifted by p:

$$C^n[p] = C^{n+p}$$

and the n-th differential is $(-1)^p d^n$. Obviously we have $H^n(C^\bullet) = H^{n+p}(C^\bullet)$.

Example D.4.8. If $(C^\bullet, d)$ is a cochain complex with cocycles $Z^\bullet$ and coboundaries $B^\bullet$, then

$$0 \longrightarrow Z^\bullet \longrightarrow C^\bullet \xrightarrow{d} B^\bullet[1] \longrightarrow 0$$

is an exact sequence of complexes.

Definition D.4.9. (Mapping cone) If $f : (C^\bullet, d) \to (D^\bullet, d')$ is a cochain map, then its *mapping cone* is the complex $\mathrm{Cone}(f) = C^\bullet[1] \oplus D^\bullet$, the n-th term is

$$\mathrm{Cone}(f)^n = C^{n+1} \oplus D^n,$$

and whose differential $\delta^n : \mathrm{Cone}(f)^n \to \mathrm{Cone}(f)^{n+1}$ is

$$\delta^n : (x^{n+1}, y^n) \mapsto (-d^{n+1}x^{n+1}, d'^n y^n - f^{n+1}x^{n+1}).$$

That is in column vector form:

$$\begin{aligned} \delta^n : C^{n+1} \oplus D^n &\longrightarrow C^{n+2} \oplus D^{n+1} \\ \begin{bmatrix} x^{n+1} \\ y^n \end{bmatrix} &\longmapsto \begin{bmatrix} -d^{n+1} & 0 \\ -f^{n+1} & d'^n \end{bmatrix} \begin{bmatrix} x^{n+1} \\ y^n \end{bmatrix}. \end{aligned}$$

And easily check that $(\mathrm{Cone}(f)^\bullet, \delta)$ is a complex.

Lemma D.4.10. *Let $f : (C^\bullet, d) \to (D^\bullet, d')$ be a cochain map. There is an exact sequence of complexes*

$$\begin{aligned} 0 \longrightarrow D^\bullet \xrightarrow{i} \mathrm{Cone}(f)^\bullet \xrightarrow{j} C^\bullet[1] \longrightarrow 0 \\ y \longmapsto (0, y) \\ (x, y) \longmapsto -x \end{aligned}$$

The connecting homomorphism $H^n(C^\bullet[1]) = H^{n+1}(C^\bullet) \to H^{n+1}(D^\bullet)$ in the corresponding long exact sequence is the induced map $H^{n+1}(f)$.

Proof. Fixed $n \in \mathbb{Z}$, consider the sequence

$$0 \longrightarrow D^n \xrightarrow{i^n} \mathrm{Cone}(f)^n \xrightarrow{j^n} C^{n+1} \longrightarrow 0$$

which is obviously exact. Moreover, the connecting homomorphism ∂ is induced by $j^{-1}\delta i^{-1}$, that is sending

$$x \overset{j^{-1}}{\longmapsto} (-x, 0) \overset{\delta}{\longmapsto} (-d(-x), -f(-x)) \overset{i^{-1}}{\longmapsto} fx$$

this is nothing but $H^n(f)$. □

Lemma D.4.11. *Let $\mathcal{A}$ be an abelian category, and let $(C^\bullet, d)$ be a complex in $\mathcal{A}$.*

(i) $\mathrm{Cone}(id_C)$ *is an acyclic complex.*

(ii) If there exists a morphism $s : C^\bullet \to C^\bullet$ of degree -1, with $d^n = d^n s^{n+1} d^n$ for all $n \in \mathbb{Z}$, then $C^\bullet$ is split.

(iii) $\mathrm{Cone}(id_C)$ *is a split complex.*

Proof. (i) By (D.4.10), the connecting homomorphism arising from

$$0 \to C^\bullet \to \mathrm{Cone}(id_C) \to C^\bullet[1] \to 0$$

are identities. Therefore $H^n(\mathrm{Cone}(id_C)) = 0$ for all n, that is $\mathrm{Cone}(id_C)$ is acyclic.

(ii) Again we assume $\mathcal{A} = \mathrm{Mod}_R$ for some associated ring R, so we can concretely consider elements. If $b \in B^{n+1}$, then $b = dc$ for some $c \in C^n$. If we define

$$t = s|_{B^{n+1}} : B^{n+1} \to C^n,$$

then the sequence $0 \to Z^\bullet \to C^\bullet \xrightarrow{d} B^\bullet[1] \to 0$ splits, since for $dtb = dsb = dsdc = dc = b$, i.e. $dt = id_{B^\bullet[1]}$. To see that $0 \to B^\bullet \to Z^\bullet \xrightarrow{\nu} H^\bullet \to 0$ splits, we define

$$\begin{aligned} u : H^n &\longrightarrow Z^n \\ [z] &\longmapsto z - dsz. \end{aligned}$$

We check that u is well-defined. If $b \in B$, then $b = dc$ for some $c \in C^{n-1}$, and $b - dsb = dc - dsdc = dc - dc = 0$. Hence, for $[z] = [z']$, that is $z = z' + b$ for some $b \in B^n$, we have $u[z] = u[z']$. Now, the map ν is splitting, because

$$\nu u : [z] \mapsto z - dsz \mapsto [z - dsz] = [z],$$

i.e. $\nu u = id_H$.

(iii) Define $s : \mathrm{Cone}(id_C) \to \mathrm{Cone}(id_C)$ by sending (x^{n+1}, x^n) to $(-x^n, 0)$. Now we verify:

$$\delta s \delta : \begin{bmatrix} x^{n+1} \\ x^n \end{bmatrix} \overset{\delta}{\longmapsto} \begin{bmatrix} -d^{n+1}x^{n+1} \\ d^n x^n - x^{n+1} \end{bmatrix} \overset{s}{\longmapsto} \begin{bmatrix} x^{n+1} - d^n x^n \\ 0 \end{bmatrix} \overset{\delta}{\longmapsto} \begin{bmatrix} -d^{n+1}x^{n+1} \\ d^n x^n - x^{n+1} \end{bmatrix}$$

implies that $\delta = \delta s \delta$, so (ii) shows that $\mathrm{Cone}(id_C)$ is split. □

Definition D.4.12. (Quasi-isomorphism) A cochain map $f : (C^\bullet, d) \to (D^\bullet, d')$ is called a *quasi-isomorphism* (or abbreviate *qis*) if all of its induced maps $H^n(f) : H^n(C^\bullet) \to H^n(D^\bullet)$ are isomorphisms.

Corollary D.4.13. *A cochain map f is qis if and only if* $\mathrm{Cone}(f)$ *is acyclic.*

Theorem D.4.14. (Characterizing injective objects) *If $\mathcal{A}$ is an abelian category, then a complex $(C^\bullet, d)$ is an injective object in $\boldsymbol{Comp}(\mathcal{A})$ if and only if it is splits with every term C^n is an injective object in $\mathcal{A}$.*

Proof. Sufficiency is (D.4.6). Conversely, if $C^\bullet$ is injective, then by universal property

$$\begin{array}{ccc} & & \mathrm{Cone}(id_C) \\ & \swarrow & \uparrow \\ C^\bullet & = & C^\bullet \end{array}$$

we see the short exact sequence of complexes

$$0 \to C^\bullet \to \mathrm{Cone}(id_C)^\bullet \to C^\bullet[1] \to 0$$

is split. And by (D.4.11), $\mathrm{Cone}(id_C)$ is a split complex, hence its direct summand complex $C^\bullet$ also split.

Now we need to show each term C^n is injective in $\mathcal{A}$. Given an object A' in $\mathcal{A}$, let $Q^n(A')$ be the complex with A' concentrated in degree n:

$$\cdots \to 0 \to 0 \to A \to 0 \to 0 \to \cdots .$$

Given a morphism $f : A' \to C^n$, define a cochain map $F : Q^n(A') \to C^\bullet$

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & 0 & \longrightarrow & A' & \longrightarrow & 0 & \longrightarrow & \cdots \\ & & \downarrow{\scriptstyle 0} & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle 0} & & \\ \cdots & \longrightarrow & C^{n-1} & \longrightarrow & C^n & \longrightarrow & C^{n+1} & \longrightarrow & \cdots \end{array}$$

where $F^n = f$ and all other $F_i = 0$. Similarly, if $g : A" \hookrightarrow A$ is a monomorphism and $Q^n(A)$ is the concentrated complex, there is a cochain map $G : Q^n(A') \to Q^n(A)$

extending g by zeros outside degree n. Since $C^\bullet$ is injective in $\mathbf{Comp}(\mathcal{A})$, there is a cochain map $H : Q^n(A) \to C^\bullet$ with $HG = F$,

$$\begin{array}{ccc} & & Q^n(A) \\ & \overset{\exists H}{\swarrow} & \uparrow{\scriptstyle G} \\ C^\bullet & \xleftarrow[F]{} & Q^n(A') \end{array}$$

but it follows that $hg = f$, and so C^n is injective in $\mathcal{A}$. □

Theorem D.4.15. *If an abelian category $\mathcal{A}$ has enough injectives, then so does $\boldsymbol{Comp}(\mathcal{A})$.*

Proof. Let $C^\bullet$ be a complex in $\mathcal{A}$. For each n, there exists an injective I^n and a monic $j^n : C^n \to I^n$. Consider the following cochain map $J^n : C^\bullet \to \Sigma^n(id_{I^n})$:

$$\begin{array}{ccccccccccc} \longrightarrow & C^{n-2} & \longrightarrow & C^{n-1} & \xrightarrow{d^{n-1}} & C^n & \longrightarrow & C^{n+1} & \longrightarrow \\ & \downarrow & & \downarrow{\scriptstyle i^n d^{n-1}} & & \downarrow{\scriptstyle i^n} & & \downarrow & \\ \longrightarrow & 0 & \longrightarrow & I^n & \xrightarrow[id_{I^n}]{} & I^n & \longrightarrow & 0 & \longrightarrow \end{array}$$

Now $\Sigma^\bullet := \bigoplus_{k\in\mathbb{Z}} \Sigma^n(id_{I^n})$ is injective in $\mathbf{Comp}(\mathcal{A})$ by (D.4.4), and $J = \bigoplus J^n : C^\bullet \to \Sigma^\bullet$ is a monomorphism. □

Remark D.4.16. All of these discussions can be dualizes on projectives.

The full subcategories $\mathbf{Comp}_{\geq 0}(\mathcal{A})$ and $\mathbf{Comp}_{\leq 0}(\mathcal{A})$ also have enough injectives when $\mathcal{A}$ does.

Definition D.4.17. (Cartan-Eilenberg resolution) Let $C^\bullet$ be a cochain complex in $\mathbf{Comp}(\mathcal{A})$, where $\mathcal{A}$ is an abelian category. A *Cartan-Eilenberg injective resolution* of $C^\bullet$ is an exact sequence in $\mathbf{Comp}(\mathcal{A})$,

$$0 \to C^\bullet \to M^{\bullet,0} \to M^{\bullet,1} \to \cdots \to M^{\bullet,q} \to \cdots$$

such that the following sequences in $\mathcal{A}$ are injective resolutions for each p:

(i) $0 \to C^p \to M^{p,0} \to M^{p,1} \to \cdots$;

(ii) $0 \to Z^p(C^\bullet) \to Z^{p,0} \to Z^{p,1} \to \cdots$;

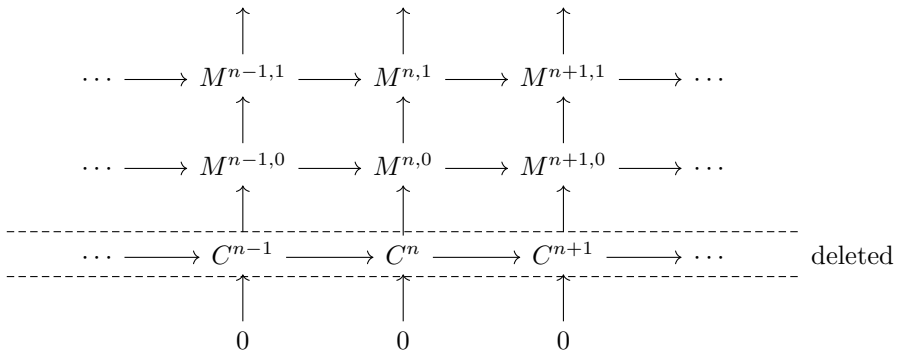
(iii) $0 \to B^p(C^\bullet) \to B^{p,0} \to B^{p,1} \to \cdots$;

(iv) $0 \to H^p(C^\bullet) \to H^{p,0} \to H^{p,1}$.

Every term with a double superscript is an injective object in $\mathcal{A}$.

There is a dual notion of *Cartan-Eilenberg projective resolution*.

A Cartan-Eilenberg injective resolution can be viewed as a large commutative diagram $M^{\bullet,\bullet}$ in $\mathcal{A}$, and if we invoked the sign change trick it would be a double complex. For each $p \in \mathbb{Z}$, the p-th column $M^{p,\bullet}$ is a deleted injective resolution of C^p, and for each $q \in \mathbb{Z}$, the q-th row $M^{\bullet,q}$ is a complex each of whose terms is injective:



Now the fundamental exact sequences of the q-th row $M^{\bullet,q}$

$$0 \to Z^{p,q} \to M^{p,q} \to B^{p+1,q} \to 0$$
$$0 \to B^{p,q} \to Z^{p,q} \to H^{p,q} \to 0$$

are split, since $Z^{p,q}$ and $B^{p,q}$ are injective. Therefore, each row $M^{\bullet,q}$ is injective in $\mathbf{Comp}(\mathcal{A})$ by (D.4.15), and so Cartan-Eilenberg injective resolution of a complex $C^\bullet$ in the category $\mathbf{Comp}(\mathcal{A})$.

Theorem D.4.18. (Construction of CE resolution) *If $\mathcal{A}$ an abelian category with enough injectives (or projectives), then every $C^\bullet$ in $\boldsymbol{Comp}(\mathcal{A})$ has a Cartan-Eilenberg injective (or projective) resolution.*

Proof. Let $C^\bullet =\to C^{-1} \xrightarrow{d} C^0 \xrightarrow{d} C^1 \xrightarrow{d} C^2 \to \cdots$ be a complex. For each $p \in \mathbb{Z}$, there are fundamental exact sequences

$$0 \to B^p \to Z^p \to H^p \to 0 \text{ and } 0 \to Z^p \to C^p \to B^{p+1} \to 0.$$

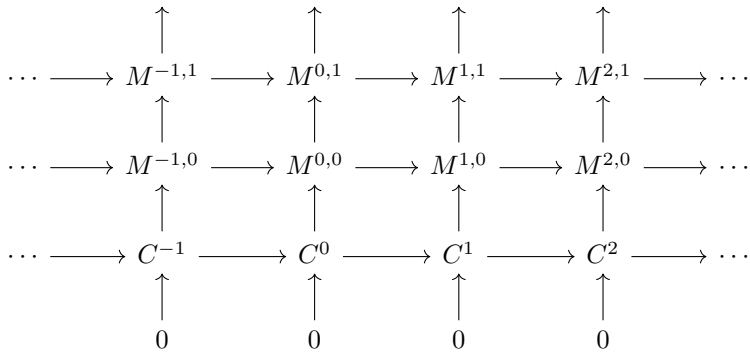
Now choose an injective resolution $B^{p,\bullet}$ and $H^{p,\bullet}$ of B^p and H^p, respectively, by $\mathcal{A}$ has enough injectives. Then by Horseshoe lemma can construct an injective resolution $Z^{p,\bullet}$ of Z^p, then an injective resolution $M^{p,\bullet}$ of C^p, such that there are exact sequences of complexes

$$0 \to B^{p,\bullet} \to Z^{p,\bullet} \to H^{p,\bullet} \to 0$$
$$0 \to Z^{p,\bullet} \to M^{p,\bullet} \to B^{p+1,\bullet} \to 0.$$

For each $p \in \mathbb{Z}$, we define cochain maps $d^{p,q} : M^{p,q} \to M^{p+1,q}$ to be the composite

$$d^{p,q} : M^{p,q} \to B^{p+1,q} \hookrightarrow Z^{p+1,q} \hookrightarrow M^{p+1,q}.$$

Therefore can draw a two dimensional commutative diagram whose columns are injective resolutions $M^{p,\bullet}$ of C^p.



This is the construction of Cartan-Eilenberg injective resolution. The dual statement is similar. □

D.5 Cohomology of The Total Complex

Recall that, given a double complex $(M^{\bullet,\bullet}, d_{\prime}, d_{\prime\prime})$, the *first filtration* of $\mathrm{Tot}(M)$ is $({}_{\mathrm{I}}F^p)$, where

$$\begin{aligned} {}_{\mathrm{I}}F^p \,\mathrm{Tot}(M)^n &= \bigoplus_{i \geq p} M^{i,n-i} \\ &= \cdots \oplus M^{p+2,q-2} \oplus M^{p+1,q-1} \oplus M^{p,q}, \end{aligned}$$

for $p+q=n$, is the subcomplex of $\mathrm{Tot}(M)$ whose n-th term is the direct sum of those $M^{i,n-i}$ lying to the right of the vertical line.

The *second filtration* of $\mathrm{Tot}(M)$ is $({}_{\mathrm{II}}F^p)$, where

$$\begin{aligned} {}_{\mathrm{II}}F^p \,\mathrm{Tot}(M)^n &= \bigoplus_{j \geq p} M^{n-j,j} \\ &= \cdots \oplus M^{q-2,p+2} \oplus M^{q-1,p+1} \oplus M^{q,p}, \end{aligned}$$

for $q+p=n$, is the subcomplex of $\mathrm{Tot}(M)$ whose n-th term is the direct sum of those $M^{n-j,j}$ lying over the horizontal line.

Lemma D.5.1. *If* $\mathrm{Tot}(M)$ *os the total complex of a double complex* $(M^{\bullet,\bullet}, d_{\prime}, d_{\prime\prime})$, *then we have* ${}_{\mathrm{II}}F^p \,\mathrm{Tot}(M) = {}_{\mathrm{I}}F^p \,\mathrm{Tot}(M^t)$, *where* $\mathrm{Tot}(M^t)$ *arises from the transposed double complex* $(M^t, d_{\prime\prime}, d_{\prime})$.

Proof. As $(M^t)^{p,q} = M^{q,p}$, and so $\bigoplus_{j\geq p} M^{n-j,j} = \bigoplus_{j\geq p} (M^t)^{j,n-j}$. □

Theorem D.5.2. *Let $M^{\bullet,\bullet}$ be a first quadrant double complex, and let ${}_{\mathrm{I}}E_r$ and ${}_{\mathrm{II}}E_r$ be the spectral sequences determined by ${}_{\mathrm{I}}F^p\,\mathrm{Tot}(M)$ and ${}_{\mathrm{II}}F^p\,\mathrm{Tot}(M)$. Then*

(i) ${}_{\mathrm{I}}F^p\,\mathrm{Tot}(M)$ and ${}_{\mathrm{II}}F^p\,\mathrm{Tot}(M)$ are bounded.

(ii) For all $p, q \in \mathbb{Z}_{\geq 0}$, we have ${}_{\mathrm{I}}E_\infty^{p,q} = {}_{\mathrm{I}}E_r^{p,q}$ and ${}_{\mathrm{II}}E_\infty^{p,q} = {}_{\mathrm{II}}E_r^{p,q}$ for large enough r which depending on p and q.

(iii) ${}_{\mathrm{I}}E_2^{p,q} \underset{p}{\Rightarrow} H^n(\mathrm{Tot}(M))$ and ${}_{\mathrm{II}}E_2^{p,q} \underset{p}{\Rightarrow} H^n(\mathrm{Tot}(M))$.

Proof. Since ${}_{\mathrm{I}}F^p\,\mathrm{Tot}(M)^n = \bigoplus_{i\geq p} M^{i,n-i}$. As $M^{\bullet,\bullet}$ is first quadrant we have for $p = 0$, ${}_{\mathrm{I}}F^p\,\mathrm{Tot}(M)^n = \mathrm{Tot}(M)^n$; and for $p \geq n+1$, $i \geq n+1$ means $n - i \leq -1$, so when $p = n+1$ we have all things become zero. Therefore we have the bounds $s(n) = n+1$, $t(n) = 0$. Similarly for ${}_{\mathrm{II}}F^p\,\mathrm{Tot}(M)^n$. This is (i). The statement (ii) and (iii) for ${}_{\mathrm{I}}E$ follows from (D.3.9), by $\mathrm{Tot}(M)^\bullet = \mathrm{Tot}(M^t)^\bullet$. □

Example D.5.3. (Computation of ${}_{\mathrm{I}}E_2$) Now let us compute ${}_{\mathrm{I}}E_2$, the E_2-term of the spectral sequence arising from the first filtration ${}_{\mathrm{I}}F^p$ of $\mathrm{Tot}(M)^\bullet$. For convenience we drop the subscript I in the discussion. As in the proof of (D.2.7), $E^{p,q} = H^n(F^p/F^{p+1})$ for $p + q = n$, where

$$(F^p)^n = F^p\,\mathrm{Tot}(M)^n = \cdots \oplus M^{p+2,q-2} \oplus M^{p+1,q-1} \oplus M^{p,q}$$
$$(F^{p+1})^n = F^{p+1}\,\mathrm{Tot}(M)^n = \cdots \oplus M^{p+2,q-2} \oplus M^{p+1,q-1}.$$

Hence, the n-th term of F^p/F^{p+1} is $M^{p,q}$, and the differential $(F^p/F^{p+1})^n \to (F^p/F^{p+1})^{n+1}$ is

$$\bar{D}^n : a_n + (F^{p+1})^n \longmapsto D^n a_n + (F^{p+1})^{n+1},$$

where $D^n a_n \in (F^p)^{n+1}$ for $a_n \in (F^p)^n$. We can just assume that $a_n \in M^{p,q}$, and then

$$D^n a_n = (d_{\prime}^{p,q} + d_{\prime\prime}^{p,q}) a_n \in M^{p+1,q} \oplus M^{p,q+1},$$

but $M^{p+1,q} \subseteq (F^{p+1})^{n+1}$ so that $D^n a_n \equiv d_{\prime\prime}^{p,q} a_n \bmod (F^{p+1})^{n+1}$. Thus only $d_{\prime\prime}$ survives in F^p/F^{p+1}. More precisely,

$$H^n(F^p/F^{p+1}) = \frac{\ker \bar{D}^n}{\mathrm{im}\, \bar{D}^{n+1}} \cong \frac{\ker d_{\prime\prime}^{p,q}}{\mathrm{im}\, d_{\prime\prime}^{p,q+1}} = H^q(M^{p,\bullet}) =: H_{\prime\prime}(M)^{p,q},$$

where $M^{p,\bullet}$ is the p-th column of $M^{\bullet,\bullet}$. We have constructed a new bigraded module $H_{\prime\prime}(M)^{\bullet,\bullet}$ whose (p,q) term is $H^q(M^{p,\bullet})$, and the elements have the form $[z]$ where $z \in M^{p,q}$ and $d_{\prime\prime}^{p,q} z = 0$.

Now for each fixed q, the q-th row $H_{\prime\prime}(M)^{\bullet,q}$ of $H_{\prime\prime}(M)^{\bullet,\bullet}$

$$\cdots H^q(M^{p-1,\bullet}), H^q(M^{p,\bullet}), H^q(M^{p+1,\bullet}), \cdots$$

can be made into a complex by define $\bar{d}_{\prime}^{\,p} : H^q(M^{p,\bullet}) \to H^q(M^{p+1,\bullet})$ by

$$\bar{d}_{\prime}^{\,p} : [z] \longmapsto [d_{\prime}^{p,q} z],$$

where $z \in \ker d_{\prime}^{p,q}$. So can induce a new bigraded module whose (p,q) term is denoted by $H_{\prime}^{p} H_{\prime\prime}^{q}(M)$, is the p-th cohomology of the q-th row

$$\cdots \longrightarrow H^q(M^{p-1,\bullet}) \xrightarrow{\bar{d}_{\prime}^{\,p-1}} H^q(M^{p,\bullet}) \xrightarrow{\bar{d}_{\prime}^{\,p}} H^q(M^{p+1,\bullet}) \longrightarrow \cdots$$

Note that elements of $H_{\prime}^{p} H_{\prime\prime}^{q}(M)$ have the form $[\bar{d}_{\prime} z]$, where $z \in \ker d_{\prime\prime} \subseteq M^{p,q}$.

$$\begin{array}{ccccc}
M^{p-1,q+1} & \longrightarrow & M^{p,q+1} & \longrightarrow & M^{p+1,q+1} \\
\uparrow & & \uparrow & & \uparrow \\
M^{p-1,q} & \longrightarrow & M^{p,q} & \longrightarrow & M^{p+1,q} \\
\uparrow & & \uparrow & & \uparrow \\
M^{p-1,q-1} & \longrightarrow & M^{p,q-1} & \longrightarrow & M^{p+1,q-1}
\end{array}$$

$$\rightsquigarrow H_{\prime\prime}(M)^{\bullet,q} : H^q(M^{p-1,\bullet}) \xrightarrow{\bar{d}_{\prime}^{\,p-1}} H^q(M^{p,\bullet}) \xrightarrow{\bar{d}_{\prime}^{\,p}} H^q(M^{p+1,\bullet});$$

$$\rightsquigarrow H^p(H_{\prime\prime}(M)^{\bullet,q}) = \frac{\ker \bar{d}_{\prime}^{\,p}}{\operatorname{im} \bar{d}_{\prime}^{\,p-1}} =: H_{\prime}^{p} H_{\prime\prime}^{q}(M).$$

Definition D.5.4. (First iterated cohomology) If $(M^{\bullet,\bullet}, d_{\prime}, d_{\prime\prime})$ is a double complex, its *first iterated cohomology* is the bigraded module whose (p,q) term is $H_{\prime}^{p} H_{\prime\prime}^{q}(M)$.

The first iterated cohomology of a double complex $(M^{\bullet,\bullet}, d_{\prime}, d_{\prime\prime})$ can be computed in above way. Taking cohomology of p-th columns gives a bigraded module $\{H^q(M^{p,\bullet})\}_{(p,q)}$, then taking cohomology of the q-th rows gives the first iterated cohomology.

We prove that $H_{\prime}^{p} H_{\prime\prime}^{q}(M)$ is the $E_2^{p,q}$-term of the first spectral sequence.

Proposition D.5.5. *If $M^{\bullet,\bullet}$ is a first quadrant double complex, then*

$$_{\mathrm{I}}E_1^{p,q} = H^q(M^{p,\bullet})$$

and

$$_{\mathrm{I}}E_2^{p,q} = H_{\prime}^{p} H_{\prime\prime}^{q}(M) \underset{p}{\Rightarrow} H^n(\mathrm{Tot}(M)).$$

Proof. In (D.5.3) we have seen that F^p/F^{p+1} is the cochain complex $(M^{p,\bullet}, d_{//})$. As the construction in (D.2.5), we claim that $d_1^{p,q} : E^{p,q} \to E^{p+1,q}$ is just the connecting homomorphism of $H^n(F^p/F^{p+1}) \to H^{n+1}(F^{p+1}/F^{p+2})$ arising from

$$0 \to F^{p+1}/F^{p+2} \to F^p/F^{p+2} \to F^p/F^{p+1} \to 0.$$

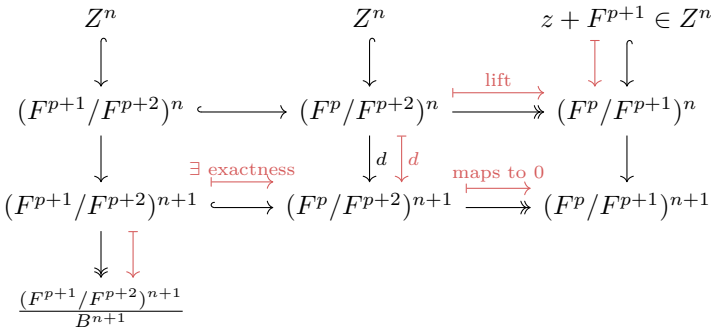
In fact, for $d_1^{p,q} = \beta^{p+1,q}\gamma^{p,q}$, we have

$$\begin{aligned} \beta^{p+1,q} \circ \gamma^{p,q} : H^n(F^p/F^{p+1}) &\xrightarrow{\gamma} H^{n+1}(F^{p+1}) \xrightarrow{\beta} H^{n+1}(F^{p+1}/F^{p+2}) \\ [z] &\longmapsto [dz] \longmapsto [dz + F^{p+2}] \end{aligned}$$

for z is a cocycle so $dz = 0$ is in F^{p+1}. Furthermore,

$$\begin{aligned} \partial : H^n(F^p/F^{p+1}) &\longrightarrow H^{n+1}(F^{p+1}/F^{p+2}) \\ [z + F^{p+1}] &\longmapsto [d(z + F^{p+2})] = [dz + F^{p+2}] \end{aligned}$$

by diagram chasing. Therefore we verify our claim.



Moreover by (D.2.5) we see ${}_{\mathrm{I}}E_1^{p,q} = H^n(F^p/F^{p+1})$, while our discussion above shows that $H^n(F^p/F^{p+1}) = H^q(M^{p,\bullet})$. Therefore ${}_{\mathrm{I}}E_1^{p,q} = H^q(M^{p,\bullet})$, and the elements of ${}_{\mathrm{I}}E_1^{p,q}$ have the form $[z]$ with $d_{//}z = 0$.

By (D.5.2) we are already know that ${}_{\mathrm{I}}E_2^{p,q} \underset{p}{\Rightarrow} H^n(\mathrm{Tot}(M))$. Now we omit the prescipt I in the rest of this proof. It only remains to prove that the first iterated cohomology is the E_2-term of the spectral sequence arising from the first filtration. So we show that

$$d_1^{p,q} : E_1^{p,q} \to E_1^{p+1,q}$$

will take $[z]$ to $[\bar{d}_{/}z] \in H_{/}^p H_{//}^q(M)$, hence that the differentials $d_1^{p,q}$ on $H^q(M^{p,\bullet})$ are coincide with $\bar{d}_{/}^p$.

As $d_1 : H^n(F^p/F^{p+1}) \to H^{n+1}(F^{p+1}/F^{p+2})$ is the connecting homomorphism, it

arises from the diagram like above, while we write it concretely up to isomorphism.

$$\begin{array}{ccccccc} & & & & (F^p/F^{p+2})^n = M^{p+1,q-1} \oplus M^{p,q} & \xrightarrow{\pi} & M^{p,q} \longrightarrow 0 \\ & & & & \downarrow \bar{D} & & \\ 0 & \longrightarrow & M^{p+1,q} & \xrightarrow{i} & (F^p/F^{p+2})^{n+1} = M^{p,q+1} \oplus M^{p+1,q} & & \end{array}$$

where $\bar{D} : (a_{p+1,q}, a_{p,q}) \mapsto (d_{\prime\prime} a_{p+1,q} + d_{\prime} a_{p,q}, d_{\prime\prime} a_{p,q})$ as $D = d_{\prime} + d_{\prime\prime}$. Let $z \in M^{p,q}$ be a cocycle respect to $d_{\prime\prime}^{p,q}$. Choose a representation of $\pi^{-1} z = (0, z)$, so that $\bar{D}(0, z) = (d_{\prime}^{p,q} z, 0)$ projects to 0. Then

$$d_1[z] = [i^{-1} \bar{D} \pi^{-1} z] = [d_{\prime} z] = \bar{d}_{\prime}[z] \in H_{\prime}^p H_{\prime\prime}^q(M).$$

□

We have seen that both the first and second spectral sequences are converge to $H^\bullet(\mathrm{Tot}(M))$, then let us now compute the E_1-term and E_2-terms from the second filtration.

Example D.5.6. (Computation of ${}_{\mathrm{II}}E_2$) Since ${}_{\mathrm{II}}F^p \,\mathrm{Tot}(M)$ is ${}_{\mathrm{I}}F^p \,\mathrm{Tot}(M^t)$, so we denote $d_{\prime\prime}$ by $\delta_{\prime}$, $d_{\prime}$ by $\delta_{\prime\prime}$, and denote transposed double complex by $(M^t, \delta_{\prime}, \delta_{\prime\prime})$. Thus, taking cohomology of p-th columns of m^t gives a bigraded module whose (p, q) terms is $H^q((M^t)^{p,\bullet})$; then taking cohomology of q-th rows gives the second iterated cohomology, by similar calculus as (D.5.3), let us rewrite $H_{\prime}^p H_{\prime\prime}^q(M^t)$ in terms of M. Of course, the terms $(M^t)^{p,q} = M^{q,p}$, thus the p-th column of M^t is the p-th row of M, and the (p, q) term is $H^q(M^{\bullet,p})$ the cohomology of $M^{\bullet,p}$ with respect to $\delta_{\prime\prime} = d_{\prime}$.

$${}_{\mathrm{II}}E_1^{p,q} = H^q(M^{\bullet,p}).$$

For each fixed q, the q-th row of this double complex are the cohomologies with respect to $\delta_{\prime} = d_{\prime\prime}$ of the q-th column:

$$\cdots, H^q(M^{\bullet,p-1}), H^q(M^{\bullet,p}), H^q(M^{\bullet,p+1}), \cdots$$

Taking cohomology gives a new bigraded module with (p, q) term is $H_{\prime\prime}^p H_{\prime}^q(M)$, the p-th cohomology of the q-th column.

$$\begin{array}{ccccc} M^{q-1,p+1} & \longrightarrow & M^{q,p+1} & \longrightarrow & M^{q+1,p+1} \\ \uparrow & & \uparrow{\scriptstyle \delta_{\prime}=d_{\prime\prime}} & & \uparrow \\ M^{q-1,p} & \longrightarrow & M^{q,p} & \xrightarrow{\delta_{\prime\prime}=d_{\prime}} & M^{q+1,p} \\ \uparrow & & \uparrow & & \uparrow \\ M^{q-1,p-1} & \longrightarrow & M^{q,p-1} & \longrightarrow & M^{q+1,p-1} \end{array}$$

$$\rightsquigarrow H_{\prime}(M)^{q,\bullet}: \quad \begin{array}{c} H^q(M^{\bullet,p+1}) \\ \uparrow{\scriptstyle \bar{\delta}_{\prime}^{p}} \\ H^q(M^{\bullet,p}) \\ \uparrow{\scriptstyle \bar{\delta}_{\prime}^{p-1}} \\ H^q(M^{\bullet,p-1}) \end{array}$$

$$\rightsquigarrow H^p(H_{\prime}(M)^{q,\bullet}) = \frac{\ker \bar{\delta}_{\prime}^{p}}{\operatorname{im} \bar{\delta}_{\prime}^{p-1}} = \frac{\ker \bar{d}_{\prime\prime}^{p}}{\operatorname{im} \bar{d}_{\prime\prime}^{p-1}} =: H_{\prime\prime}^{p} H_{\prime}^{q}(M).$$

Definition D.5.7. (Second iterated cohomology) If $(M^{\bullet,\bullet}, d_{\prime}, d_{\prime\prime})$ is a double complex, its *second iterated cohomology* is the bigraded module whose (p,q) term is $H_{\prime}^{p} H_{\prime\prime}^{q}(M)$.

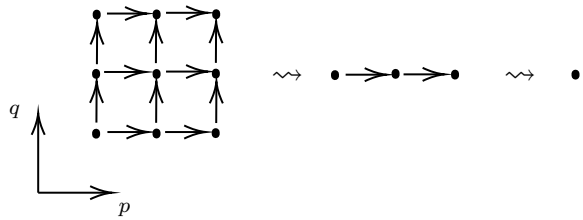
Proposition D.5.8. *If $M^{\bullet,\bullet}$ is a first quadrant double complex, then*

$${}_{\mathrm{II}}E_1^{p,q} = H^q(M^{\bullet,p})$$

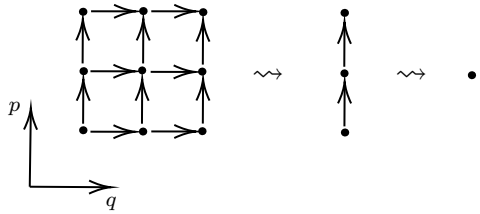
and

$${}_{\mathrm{II}}E_2^{p,q} = H_{\prime\prime}^{p} H_{\prime}^{q}(M) \underset{p}{\Rightarrow} H^n(\mathrm{Tot}(M)).$$

Remark D.5.9. Informally speaking, the first iterated cohomology computes ${}_{\mathrm{I}}E_2$ by taking cohomology of the columns and then taking cohomology of the rows: The



second iterated cohomology computes ${}_{\mathrm{II}}E_2$ by taking cohomology of the rows then taking cohomology of the columns:



Even though both the first and second spectral sequences converge to $H^n(\mathrm{Tot}(M))$, we may not have that ${}_{\mathrm{I}}E_{\infty}^{p,q} \cong {}_{\mathrm{II}}E_{\infty}^{p,q}$. The induced filtration on H^n from $({}_{\mathrm{I}}F^p)$ and $({}_{\mathrm{II}}F^p)$ need not be the same, and so factor modules of H^n may be different.

Definition D.5.10. (Collapsed spectral sequence) A spectral sequence (E_r, d_r) *collapses* on the p-axis if $E_2^{p,q} = \{0\}$ for all $q \neq 0$. A spectral sequence (E_r, d_r) *collapses* on the q-axis if $E_2^{p,q} = \{0\}$ for all $p \neq 0$.

Proposition D.5.11. *Let (E_r, d_r) be a first quadrant spectral sequence, and let $E_2^{p,q} \underset{p}{\Rightarrow} H^n(\mathrm{Tot}(M))$. Then*

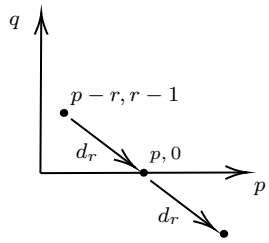
(i) If (E_r, d_r) collapses on either axis, then $E_\infty^{p,q} = E_2^{p,q}$ for all p, q.

(ii) If (E_r, d_r) collapses on the p-axis, then $H^n(\mathrm{Tot}(M)) \cong E_2^{n,0}$;
if (E_r, d_r) collapses on the q-axis, then $H^n(\mathrm{Tot}(M)) \cong E_2^{0,n}$.

Proof. (i) Assume that (E_r, d_r) collapses on the p-axis, and choose $r \geq 2$. First of all, $E_r^{p,q} = \{0\}$ for all $r \geq 2$ with $q \neq 0$, because $E_r^{p,q}$ is a subquotient of $E_2^{p,q} = \{0\}$. Now $E_{r+1}^{p,0} = \ker d_r^{p,0} / \operatorname{im} d_r^{p-r,r-1}$. Regard E_r as the r-th page of the spectral sequence, the differential

$$d_r : E_r \to E_r$$

is a family of arrows each having non-zero slope $\frac{1-r}{r}$. Now $d_r^{p,0} = 0$ since its target is off the p-axis, hence is $\{0\}$, thus $\ker d_r^{p,0} = E_r^{p,0}$. Also, $\operatorname{im} d_r^{p-r,r-1} = \{0\}$ since its domain is off the p-axis. Therefore, $E_{r+1}^{p,0} = E_r^{p,0}/0 = E_r^{p,0}$, and (D.3.4) gives $E_2 = E_\infty$.



For collapses on the q-axis is similar.

(ii) The induced filtration on $H^n = H^n\big(\mathrm{Tot}(M)\big)$ is

$$\{0\} = \Phi^{n+1}H^n \subseteq \Phi^n H^n \subseteq \cdots \subseteq \Phi^1 H^n \subseteq \Phi^0 H^n = H^n$$

by in (D.5.2) we have seen that $s(n) = n + 1$ and $t(n) = 0$.

If the spectral sequence collapses on the p-axis, then $E_2^{p,q} = \{0\}$ for all $p \leq n - 1$, that is for all $q = n - p \geq 1$. As $M^{\bullet,\bullet}$ is first quadrant we no need care about $q \leq -1$ case. But by part (i) $E_2^{p,q} = E_\infty^{p,q}$, so that $\{0\} = E_\infty^{p,q} = \Phi^p H^n / \Phi^{p+1} H^n$ for all $p \leq n - 1$. Hence

$$H^n = \Phi^0 H^n = \cdots = \Phi^n H^n$$

and so $E_2^{p,q} \cong \Phi^n H^n / \Phi^{n+1} H^n = H^n$. A similar argument can be given when the spectral sequence collapses on the q-axis. □

D.6 Grothendieck Spectral Sequence

This section we discuss the spectral sequence due to Grothendieck in his Tohoku paper.

Definition D.6.1. (Right/Left acyclic) Let $\mathcal{B}$ be an abelian category with enough injectives (or with enough projectives), and let $F : \mathcal{B} \to \mathrm{Mod}_R$ be an additive functor. And object B is called *right F-acyclic* if $(R^iF)B = 0$ for all $i \geq 1$. An object B is called *left F-acyclic* if $(L_iF)B = 0$ for all $i \geq 1$.

Example D.6.2. (i) The covariant additive left exact functor

$$\mathrm{Hom}_R(M, -) : \mathrm{Mod}_R \to \mathrm{Mod}_R$$

making every injective R-modules I to be right acyclic, i.e. $\mathrm{Ext}^i_R(M, I) = 0$ for all $i \geq 1$.

(ii) The covariant additive right exact functor

$$M \underset{R}{\otimes} - : \mathrm{Mod}_R \to \mathrm{Mod}_R$$

making every flat R-modules N to be left acyclic, i.e. $\mathrm{Tor}^R_i(M, N) = 0$ for all $i \geq 1$. In fact as every projective module is flat, it is also left acyclic.

(iii) The covariant additive left exact functor

$$\Gamma : \mathrm{Mod}_{\mathscr{O}_X} \to \mathrm{Mod}_{\Gamma(X,\mathscr{O}_X)}$$

making every flasque sheaf $\mathscr{F}$ to be right acyclic, i.e. $H^i(X, \mathscr{F}) = 0$ for all $i \geq 1$. In fact as every injective $\mathscr{O}_X$-module is flasque, it is also left acyclic.

(iv) The covariant additive left exact functor

$$f_* : \mathrm{Qcoh}(X) \to \mathrm{Qcoh}(Y)$$

making every flasque sheaf $\mathscr{F}$ to be right acyclic, i.e. $R^if_*(\mathscr{F}) = 0$ for all $i \geq 1$. Here $f : X \to Y$ are morphism of schemes with X noetherian.

Theorem D.6.3. (Grothendieck) *Let $\mathcal{A} \xrightarrow{G} \mathcal{B} \xrightarrow{F} \mathcal{C}$ be covariant additive functors, where $\mathcal{A}, \mathcal{B}$ and $\mathcal{C}$ are abelian categories with enough injectives. Assume that F is left exact and that GI is right F-acyclic for every injective object I in $\mathcal{A}$. Then for every object A in $\mathcal{A}$, there is a first quadrant spectral sequence with*

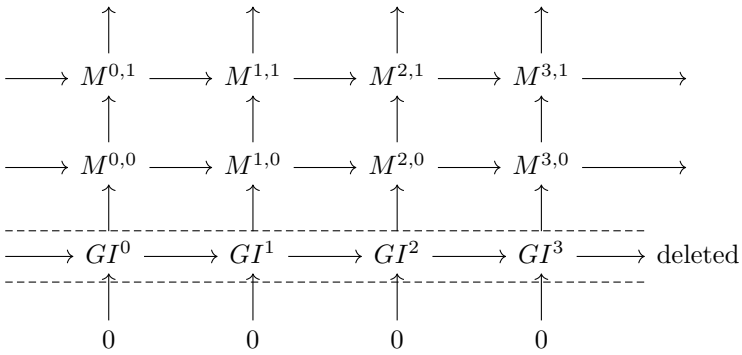
$$E_2^{p,q} = (R^pF)(R^qG)A \underset{p}{\Rightarrow} R^n(FG)A$$

for $n = p + q$.

Proof. Choosing an injective resolution $\mathbf{I} = 0 \to A \to I^0 \to I^1 \to \cdots$, and apply G to its deletion $\mathbf{I}^A$ to obtain the complex

$$G\mathbf{I}^A = 0 \to GI^0 \to GI^1 \to GI^2 \to \cdots .$$

By (D.4.18), there exists a Cartan-Eilenberg injective resolution of $G\mathbf{I}^A$, a first quadrant complex $M^{\bullet,\bullet}$ whose rows are complexes and whose columns are deleted resolution of $G\mathbf{I}^A$.



Applying left exact functor F on $M^{\bullet,\bullet}$ and consider the double complex $FM^{\bullet,\bullet}$ and its total complex $\mathrm{Tot}(FM)$. We compute its first iterated cohomology, For fixed p, notice that $M^{\bullet,\bullet}$ is first quadrant, and the p-th column $FM^{p,\bullet}$ is corresponding to a deleted injective resolution of GI^p. Hence $(R^qF)(GI^p)$ is just the q-th cohomology of this p-th column:

$$H^q(FM^{p,\bullet}) = (R^qF)(GI^p).$$

Now as I^p is injective, so that GI^p is right F-acyclic; that is $(R^qF)(GI^p) = \{0\}$ for all $q \geq 1$. Hence,

$$H^q(FM^{p,\bullet}) = \begin{cases} (R^0F)(GI^p) & \text{if } q = 0, \\ \{0\} & \text{if } q \geq 0. \end{cases}$$

But F is left exact, so $R^0F = F$, such that survives is on the p-axis,

$$0 \to FG(I^0) \to FG(I^1) \to FG(I^2) \to \cdots$$

and this is FG applied to the deleted injective resolution $\mathbf{I}^A$. Hence, its p-th cohomology is $R^p(FG)A$:

$${}_{\mathrm{I}}E_2^{p,q} = \begin{cases} R^p(FG)A & \text{if } q = 0, \\ \{0\} & \text{if } q \geq 0. \end{cases}$$

Thus, the first spectral sequence of $FM^{\bullet,\bullet}$ collapses on the p-axis, by (D.5.11) we have

$$H^n(\mathrm{Tot}(FM)) \cong {}_{\mathrm{I}}E_2^{n,0} = R^n(FG)A.$$

To compute the second iterated cohomology, we first transpose the indices p, q in the double complex $FM^{\bullet,\bullet}$, noting that

$$H^q(FM^{\bullet,p}) = \frac{\ker Fd^{q,p}}{\operatorname{im} Fd^{q-1,p}},$$

where d is the differential in the (D.4.18): $d^{p,q} : M^{p,q} \to M^{p+1,q}$. Applying F to the commutative diagram in which $j : B \hookrightarrow Z$ and $i : Z \hookrightarrow M$ are inclusions, and $\delta : M \to B$ by restrict the target arising from d, i.e. $d = ij\delta$.

$$\begin{array}{ccccccccc} & & & & & & M^{q+1,p} & & \\ & & & & & \overset{d}{\nearrow} & \uparrow{\scriptstyle ij} & & \\ 0 & \longrightarrow & Z^{q,p} & \overset{i}{\hookrightarrow} & M^{q,p} & \overset{\delta}{\twoheadrightarrow} & B^{q+1,p} & \longrightarrow & 0 \end{array}$$

Since by our construction of the Cartan-Eilenberg injective resolution, we see that $Z^{\bullet,p}$ is the injective resolution of $Z(GI^p)$, so each $Z^{q,p}$ is injective, the short exact sequence

$$0 \longrightarrow Z^{q,p} \xrightarrow{i} M^{q,p} \xrightarrow{\delta} B^{q+1,p} \longrightarrow 0$$

splits. Moreover, as F is an additive functor sending $M^{q,p}$ to $F(Z^{q,p}) \oplus F(B^{q+1,p})$, the sequence remains exact after applying F, such that Fi is monic $F\delta$ is epic and $\ker F\delta = \operatorname{im} Fi$. Similarly, the short exact sequence

$$0 \longrightarrow B^{q,p} \xrightarrow{j} Z^{q,p} \longrightarrow H^{q,p} \longrightarrow 0$$

splits by $B^{q,p}$ is injective, also remains exact after F. Hence Fj is monic.

We use Freyd-Mitchell embedding making everything into Mod_R. Now compute $\ker Fd/\operatorname{im} Fd$. As we already seen, $Fd = F(ij\delta) = (Fi)(Fj)(F\delta)$ where both Fi and Fj are injections, the numerator

$$\ker Fd = \ker F\delta = \operatorname{im} Fi = (Fi)(FZ).$$

The denominator

$$\operatorname{im} Fd = (Fd)(FM) = (Fi)\big[(Fj)(F\delta)(FM)\big] = (Fi)\big[(Fj)(FB)\big]$$

as $F\delta$ is surjective. In particular we have a linear algebra level fact: the homomorphism $Fi : FZ \hookrightarrow FM$ and the submodule $(Fj)(FB) \subseteq FZ$ give a natural surjective map

$$\frac{FZ}{(Fj)(FB)} \overset{Fi}{\twoheadrightarrow} \frac{(Fi)(FZ)}{(Fi)\big[(Fj)(FB)\big]},$$

moreover as Fi is injective, we have the isomorphism

$$\frac{\ker Fd}{\operatorname{im} Fd} = \frac{(Fi)(FZ)}{(Fi)\big[(Fj)(FB)\big]} \cong \frac{FZ}{(Fj)(FB)}.$$

But the right hand side is $\operatorname{coker} Fj \cong FH$, so restoring indices we conclude that

$$H^q(FM^{\bullet,p}) = \frac{\ker Fd^{q,p}}{\operatorname{im} Fd^{q-1,p}} \cong FH^{q,p} = FH^q(M^{\bullet,p});$$

that is, F commutes with H^q. By Cartan-Eilberg injective resolution, each

$$0 \to H^q(G\mathbf{I}^A) \to H^{q,0} = H^q(M^{\bullet,0}) \to H^{q,1} = H^q(M^{\bullet,1}) \to \cdots \to H^{q,p} = H^q(M^{\bullet,p}) \to \cdots$$

is an injective resolution. By definition, $H^q(G\mathbf{I}^A) = (R^qG)A$, so that the modules $H^q(M^{\bullet,p})$ form an injective resolution of $(R^qG)A$. Hence

$${}_{\mathrm{II}}E_2^{p,q} = H^pH^q(FM) = H^p\big(FH^q(M)\big) = H^p\big(F(R^qG)A\big) = (R^pF)(R^qG)A.$$

But by (D.5.8), we have

$$(R^pF)(R^qG)A = {}_{\mathrm{II}}E_2^{p,q} \underset{p}{\Rightarrow} H^n\big(\operatorname{Tot}(FM)\big) \cong {}_{\mathrm{I}}E_2^{p,q} = R^n(FG)A$$

where $n = p + q$. So can drop the subscripts I and II. □

Now we give an application of Grothendieck spectral sequence.

Example D.6.4. (Generalized projection formula) Let $f : X \longrightarrow Y$ be a surjective morphism between proper varieties. Let $D_1, \ldots, D_n$ be Cartier divisors on Y with $n \geq \dim X$. Then we have

$$(f^*D_1 \cdots f^*D_n) = \deg(f)(D_1 \cdots D_n).$$

Proof. For any coherent sheaf $\mathscr{F}$ on X, the higher direct image sheaf $R^qf_*\mathscr{F}$ are coherent, see [Har77, III, 8.8.1]. And by Grothendieck spectral sequence applying on $\operatorname{Coh}(X) \xrightarrow{f_*} \operatorname{Coh}(Y) \xrightarrow{\Gamma} \operatorname{Vect}_{\mathbb{C}}$ between abelian categories, we have a spectral sequence

$$E_2^{p,q} = (R^p\Gamma)(R^qf_*)\mathscr{F} = H^p(Y, R^qf_*\mathscr{F}) \underset{p}{\Rightarrow} R^{p+q}\big(H^0(Y, f_*\mathscr{F})\big) = H^n(X, \mathscr{F}).$$

Therefore there is a filtration $\Phi^\bullet H^n(X, \mathscr{F})$ such that

$$E_\infty^{p,q} \cong \Phi^pH^n/\Phi^{p+1}H^n,$$

but as we're in finite dimensional vector spaces, there are short exact sequences

$$0 \to \Phi^{p+1}H^n \to \Phi^pH^n \to E_\infty^{p,q} \to 0$$

equipping with an additive function $\dim_{\mathbb{C}}$ for each p. So respect to the filtration

$$\{0\} = \Phi^sH^n \subseteq \Phi^{s-1}H^n \subseteq \cdots \subseteq \Phi^{t+1}H^n \subseteq \Phi^tH^n = H^n(X, \mathscr{F}),$$

we have

$$\begin{aligned}\dim_{\mathbb{C}} H^n(X,\mathscr{F}) &= \dim_{\mathbb{C}} \Phi^t H^n - \dim_{\mathbb{C}} \Phi^s H^n \\ &= \dim_{\mathbb{C}} \Phi^t H^n - \dim_{\mathbb{C}} \Phi^{t+1} H^n + \dim_{\mathbb{C}} \Phi^{t+1} H^n - \dim_{\mathbb{C}} \Phi^{t+2} H^n \\ &+ \cdots + \dim_{\mathbb{C}} \Phi^{s-1} H^n - \dim_{\mathbb{C}} \Phi^s H^n \\ &= \dim_{\mathbb{C}} E_\infty^{t,n-t} + \dim_{\mathbb{C}} E_\infty^{t+1,n-(t+1)} + \cdots + \dim_{\mathbb{C}} E_\infty^{s-1,n-(s-1)}.\end{aligned}$$

But as our spectral sequence is first quadrant, so $s = s(n) = n+1$ and $t = t(n) = 0$ (by (D.5.2) and (D.3.7) for M is FM and H^n is $H^n\big(\mathrm{Tot}(FM)\big)$); that is

$$\dim_{\mathbb{C}} H^n(X,\mathscr{F}) = \sum_{i=0}^{n} \dim_{\mathbb{C}} E_\infty^{i,n-i} = \sum_{p+q=n} \dim_{\mathbb{C}} E_\infty^{p,q}.$$

And then

$$\begin{aligned}\chi(X,\mathscr{F}) &= \sum_n \left((-1)^n \sum_{p+q=n} \dim_{\mathbb{C}} E_\infty^{p,q} \right) \\ &= \sum_{p,q} (-1)^{p+q} \dim_{\mathbb{C}} E_\infty^{p,q} \\ (*) &= \sum_{p,q} (-1)^{p+q} \dim_{\mathbb{C}} E_2^{p,q} \\ &= \sum_{p,q} (-1)^p (-1)^q \dim_{\mathbb{C}} H^p(Y, R^q f_* \mathscr{F}) \\ &= \sum_{q \geq 0} (-1)^q \chi(Y, R^q f_* \mathscr{F}),\end{aligned}$$

while the equation $(*)$ is by, for finite term complex, its Euler characteristic is equal to its cohomology's Euler characteristic, i.e.

$$\chi(C^\bullet) = \chi(H^\bullet(C)),$$

since we have two split exact sequences in $\mathrm{Vect}_{\mathbb{C}}$:

$$\begin{aligned}0 \to B^n \to & Z^n \to H^n(C^\bullet) \to 0 \\ 0 \to Z^n \to & C^n \to B^{n+1} \to 0\end{aligned}$$

and $E_{r+1} = H^\bullet(E_r)$ for spectral sequences.

Now applying our result to $\mathscr{F} = \mathscr{O}_X(m_1 D_1 + \cdots m_n D_n)$, and using the projection formula

$$R^q f_* \mathscr{F} = R^q f_* \big(\mathscr{O}_X \otimes f^* \mathscr{O}_Y(m_1 D_1 + \cdots + m_n D_n) \big) \cong R^q f_* \mathscr{O}_X \otimes \mathscr{O}_Y(m_1 D_1 + \cdots + m_n D_n),$$

we get that $(f^*D_1 \cdots f^*D_n)$ is equal to the coefficent of $m_1 \cdots m_n$ in

$$\sum_{q\geq 0}(-1)^q \chi\big(Y, R^q f_* \mathscr{O}_X \otimes \mathscr{O}_Y(m_1 D_1 + \cdots + m_n D_n)\big).$$

If f is not generically finite, i.e. $n \geq \dim X \geq \dim Y$, then we have the coefficient of $m_1 \cdots m_n$ in each term of the sum vanishes, as the degree must at most the dimension of $\operatorname{Supp} R^q f_* \mathscr{O}_X \otimes \mathscr{O}_Y(\sum^n m_i D_i)$.

Otherwise, we suppose f is finite of degree d over an open dense subset U of Y, i.e. it is fibre finite over U. So by [Har77, III, 11.2] we see

$$R^q f_* \mathscr{O}_X = 0 \text{ for all } q > 0$$

over this open dense subset $U \subseteq Y$; that is, coherent sheaves $R^q f_* \mathscr{O}_X$ have support outside of U for $q > 0$, and the coefficient of $m_1 \cdots m_n$ in the corresponding term vanishes for the same reason: $n > \dim Y - U$. Now as f is generically finite of degree d, we have for generic point $\eta \in Y$,

$$(f_* \mathscr{O}_X)_\eta \cong \mathbb{C}(X)$$

which is a d-dimensional $\mathbb{C}(Y)$-vector space, so by [Har77, II, ex5.7(a)] after possibly refinement, we have $f_* \mathscr{O}_X \cong \mathscr{O}_Y^d$ over the dense open subset $U \subseteq Y$. Consider two short exact sequences

$$0 \to \mathscr{G} \to f_* \mathscr{O}_X \to \mathscr{G}_1 \to 0$$
$$0 \to \mathscr{G} \to \mathscr{O}_Y^d \to \mathscr{G}_2 \to 0$$

where $\mathscr{G} = f_* \mathscr{O}_X \cap \mathscr{O}_Y^d$ taking intersection in $\mathscr{K}_Y^d$. Then support of $\mathscr{G}_1$ and $\mathscr{G}_2$ do not meet U. Then we have

$$\chi\left(Y, f_* \mathscr{O}_X \otimes \mathscr{O}_Y(\sum^n m_i D_i)\right) = \chi\left(Y, \mathscr{G}(\sum^n m_i D_i)\right) + \chi\left(Y, \mathscr{G}_1(\sum^n m_i D_i)\right);$$
$$\chi\left(Y, \mathscr{O}_Y^d(\sum^n m_i D_i)\right) = \chi\left(Y, \mathscr{G}(\sum^n m_i D_i)\right) + \chi\left(Y, \mathscr{G}_2(\sum^n m_i D_i)\right),$$

to get

$$\chi\left(Y, f_* \mathscr{O}_X(\sum^n m_i D_i)\right) - \chi\left(Y, \mathscr{O}_Y^d(\sum^n m_i D_i)\right) = \chi\left(Y, \mathscr{G}_1(\sum^n m_i D_i)\right) - \chi\left(Y, \mathscr{G}_2(\sum^n m_i D_i)\right).$$

In right hand side as their supports outside U, so coefficents the $m_1 \cdots m_n$ vanishes by reason of dimension of support. So at left hand side the coefficients are the same, i.e.

$$(f^*D_1 \cdots f^*D_n) = d \cdot (D_1 \cdots D_n).$$

□

Index

$\mathbb{Q}$-Cartier divisor, 44
$\mathbb{Q}$-factorial, 100
$\mathbb{R}$-Cartier divisor, 48
k-cycle, 8

Abhyankar lemma, 101
Algebraic fibre space, 163
Ample and nef cone, 62
Ample Cartier divisor, 26
Ample classes, 37
Ample cone, 62
Ampleness for a map, 84
Ampleness in families, 31
Asymptotic Riemann-Roch, 16
Asymptotic Riemann-Roch on Ample case, 35

Base ideal, 12
Base scheme, 12
Bend and Break I, 124
Bend and Break II, 127
Bertini's Theorem, 431
big $\mathbb{Q}$-Cartier divisor, 195
big $\mathbb{R}$-Cartier divisors, 198
big cone, 199
Big line bundle/Cartier divisor, 190
Bloch–Gieseker covering, 233

Canonical Cartier divisor, 11
Canonical line bundle, 11
canonical singularities, 283, 309, 318
Cartier divisor, 7
Castelnuovo's Contraction, 69
Chevalley's Nullstellensatz, 467
Chow's lemma, 51
Cone of curves, 64
Constructible subset, 467
contraction of morphism, 150
Curves on a surface, 68
Cyclic Coverings, 228

Deform morphisms, 111
Deform rational curves, 109
Deform the curve, 122
deformation space, 111
Deformations with fixed part, 121
degree, 16
Differential forms with logarithmic poles, 243
Discrepancy, 287
divisor of zeros, 10
divisorial contraction, 150, 274
Divisorial log terminal, 309
divisorial sheaf, 10, 286
dlt singularities, 321

Dolbeault theorem, 241
Dual number, 113

Euler Characteristics number, 16
Evaluation map, 113
Exceptional locus, 99
exponent of a line bundle/Cartier divisor, 159
extremal face, 68
extremal ray, 68
Extremals, 68

Fano variety, 132
fibre space, 163
Finding rational curves on Fano variety, 136
Fine moduli space, 113
Finite pullbacks I, 28
Finite pullbacks II, 38
fractional part, 254
Free linear series, 13
Fujita's vanishing theorem, 78

Geauert-Riemenschneider vanishing, 253
Generalized Kawamata-Viehweg vanshing, 263
Generalized Kodaira vanishing, 260
Globally generated, 13
Grothendieck Spectral Sequence, 502
Grothendieck's Hilbert scheme theorem, 111
Growth of cohomology, 43

Hilbert polynomial, 15
Hodge decomposition, 241
Hodge Index Theorem, 153

Iitaka dimension, 160
Iitaka fibration, 176
Infinitesimal lifting property, 465
Integral Cartier divisor, 44
Intersection number, 19

Kawamata log terminal, 309
Kawamata-Viehweg vanishing, 255
Kleiman's criterion, 66
Kleiman's theorem, 53
Kleiman-Mori Cone, 65
klt singulairites, 309
klt singularities, 320
Kodaira dimension, 161
Kodaira dimension of a singular variety, 161
Kodaira Vanishing Theorem, 11
Kodaira vanishing theorem, 242
Kodaira's lemma, 192

lc singularities, 309, 320
Lefschetz hyperplane theorem, 241
length of an extremal rational curve, 358
Line Bundle, 10
Linear series, 12
Linearly equivalent, 8
local equation, 7
Local structure of $\mathrm{Mor}(Y, X)$, 119
Local structure of $\mathrm{Mor}(Y, X; g)$, 122
log canonical, 309

Minimal surface, 61
MMP for Surfaces, 155, 270
Mori's Cone theorem, 146

Néron-Severi group, 25
Nakai-Moishezon-Kleiman criterion, 36

Nef cone, 62
Nef line bundle and Cartier divisor, 51
Nefness in families, 58
Negativity Lemma, 105
Negativity lemma, 297
Numerical equivalence, 25
Numerical equivalence classes of curves, 64
Numerical nature of bigness, 194
Numerically equivalent, 25

of fibre type, 151, 274

Picard number, 25
plt singularities, 309, 320
Principal Cartier divisor, 8
Projection formula, 23
pseudo-effective cone, 199
Pullback of Cartier divisors, 8
Purely log terminal, 309

Ramification divisor, 100
relative big, 213
relative cone NE(π), 74
Relatively linearly equivalent, 429
Resolution of singularities, 224
Rigidity lemma I, 74
Rigidity lemma II, 75
round down, 254
round up, 254

Scheme-theoretic base locus, 12
Section of line bundle, 11
Section ring associated to a line bundle, 167
Semi-ample Cartier divisor, 169
Semiample fibrations, 170
semigroup of a line bundle/Cartier divisor, 159
Seshadri's criterion, 56
Set-theoretic base locus, 167
simple normal crossing divisor, 222
small contraction, 150, 274
Stable base locus, 167

Tangent space of Mor(Y, X), 115
terminal singularities, 283, 309, 319

Very ample Cartier divisor, 26
very general, 181
Volume of a line bundle, 200

Weil $\mathbb{Q}$-divisor, 47
Weil divisor, 8

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